HL7 FHIR R4 — Chapter 8: Administration Module (sections 8 – 8.32.14)

Source: HL7 FHIR R4 (v4.0.1) specification, https://hl7.org/fhir/R4/ — text-only rendition for AI ingestion; page order follows toc.html.

8 Administration Module

URL: https://hl7.org/fhir/R4/administration-module.html
Work Group Patient Administration | Standards Status: Informative

8.0 Administration Module

8.0.1 Introduction

The Administrative module covers the base data that is then linked into the other modules for clinical content, finance/billing, workflow, etc. It is built on the FHIR technology platform modules.

Before any clinical data can be recorded, the basic information of the patient must be recorded, and then often the basis of the interaction (such as an encounter).

8.0.2 Index

Patient RelatedPerson Person Group Practitioner PractitionerRole | Organization Location HealthcareService Endpoint Schedule Slot | EpisodeOfCare Encounter Appointment AppointmentResponse Account Flag | Device DeviceDefinition DeviceMetric Substance

8.0.2.1 Patient Registers

Track people involved in receiving healthcare, the basics nearly everything else references back to

Name | Aliases | Description
Patient | SubjectOfCare Client Resident | Demographics and other administrative information about an individual or animal receiving care or other health-related services.
RelatedPerson | | Information about a person that is involved in the care for a patient, but who is not the target of healthcare, nor has a formal responsibility in the care process.
Person | | Demographics and administrative information about a person independent of a specific health-related context.
Group | | Represents a defined collection of entities that may be discussed or acted upon collectively but which are not expected to act collectively, and are not formally or legally recognized; i.e. a collection of entities that isn't an Organization.

Implementation Note: Patient linking should also be considered when evaluating searches with references to other resources. e.g. searching for a patients' conditions for a patient. At present the specification does not define if the links should be also followed to include conditions that reference the linked patients too. We are currently seeking feedback on this.

Implementation Note: The Person resource may be used as a centralized register of people that may eventually be involved in healthcare, and could be used as the central core demographics register. However, the fields/values in Person are duplicated in the other resources, and in many cases the Person resource will be hosted on external systems.

8.0.2.2 Clinical Categorization Resources

Most clinical activities occur grouped in some way. Long term care is typically covered by an EpisodeOfCare, whereas short term care is covered by encounters. Account associates the tracking of transactions back to a Patient (or other resource). Flag is just used to highlight a warning or other notification about a patient (or other resource)

Name | Aliases | Description
EpisodeOfCare | Case Program Problem | An association between a patient and an organization / healthcare provider(s) during which time encounters may occur. The managing organization assumes a level of responsibility for the patient during this time.
Encounter | Visit | An interaction between a patient and healthcare provider(s) for the purpose of providing healthcare service(s) or assessing the health status of a patient.
Account | Cost center | A financial tool for tracking value accrued for a particular purpose. In the healthcare field, used to track charges for a patient, cost centers, etc.
Flag | Barriers to Care, Alert | Prospective warnings of potential issues when providing care to the patient.

Implementation Note: Resources shown with a dotted box are described in other sections of the specification: Coverage and Claim are from the section on Finance.

8.0.2.3 Service Provider Directory Resources

Service Provider Directory resources are usually stored in the administration section of applications, and may even be synchronized from external systems.

Name | Aliases | Description
Organization | | A formally or informally recognized grouping of people or organizations formed for the purpose of achieving some form of collective action. Includes companies, institutions, corporations, departments, community groups, healthcare practice groups, payer/insurer, etc.

Location | | Details and position information for a physical place where services are provided and resources and participants may be stored, found, contained, or accommodated.
Practitioner | | A person who is directly or indirectly involved in the provisioning of healthcare.
PractitionerRole | | A specific set of Roles/Locations/specialties/services that a practitioner may perform at an organization for a period of time.
HealthcareService | | The details of a healthcare service available at a location.
Endpoint | | The technical details of an endpoint that can be used for electronic services, such as for web services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context information.

8.0.2.4 Scheduling and Appointments

The Scheduling/Appointment resources permit the planning of encounters to occur and follow on with other clinical activities.

Name | Aliases | Description
Schedule | Availability | A container for slots of time that may be available for booking appointments.
Slot | | A slot of time on a schedule that may be available for booking appointments.
Appointment | | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).
AppointmentResponse | | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection.

8.0.2.5 Devices and Substances

Other assets are often registered in the administration system, and maintained as master files.

Name | Aliases | Description
Device | | A type of a manufactured item that is used in the provision of healthcare without being substantially changed through that activity. The device may be a medical or non-medical device.
DeviceDefinition | | The characteristics, operational status and capabilities of a medical-related component of a medical device.
DeviceMetric | | Describes a measurement, calculation or setting capability of a medical device.
Substance | | A homogeneous material with a definite composition.

8.0.3 Security and Privacy

Patient privacy is handled with security labels and tags in the Resource Meta property. This is the standard way in which that the FHIR specification provides this supporting information to a sub-system that implements it (which is not defined by FHIR).

One of the more common use cases is for marking a patient as being a celebrity.

Note that privacy considerations apply to Person, Practitioner and RelatedPerson records in addition to Patient's.

While Organization, Location, Device and other non-person-identifying records are generally subject to less stringent security precautions, such data must still be protected to avoid safety issues (e.g. someone maliciously changing the ingredients associated with a drug to cause/fail to cause alerts)

Devices can be linked to Patients. If this occurs, they must be protected as any other patient-linked element

For more general considerations, see the Security and Privacy module.

8.0.4 Common Use Cases

Administration Resources are cornerstone resources that are used by clinical and other domains of the FHIR Standard.

• Managing a Master Record of a Patient and a Person (e.g. MPI) A Patient resource is used to describe patient demographic information and any updates to it. It can be used to communicate Patient information to other systems (e.g. other registries, clinical, ancillary and financial systems). Some systems distinguish the Patient Registry (or Client Registry) from the Person Registry. A Person resource is a base for the Person Registry system. The Patient/Person Management use case includes creation, update, as well as merge/unmerge and link/unlink scenarios.

• Managing a Master Record of a Provider and Service Catalogue (e.g. Provider Registry, Service Directory) A Practitioner resource is a base resource for enabling the registry of individuals, related to providing health care services. Other resources, such as Organization, Location, HealthcareService, are creating a complete picture of where, how and by whom the care services are offered to a patient. The resources can be used for managing the master record or as a reference in clinical resources to inform about participants and places for various clinical resources.

• Managing Other Administrative Records The Administration domain of the FHIR standard includes creation and update of Device and Substance records. Resources can be used for managing a master record or communicating its information to other systems.

• Enabling Patient Profiles, Clinical Reporting and Connecting Clinical Records Administration Resources are referred to by almost all clinical resources. Querying systems, using the references to Administration Resources enables the creation of profiles and reports of various complexities.

• Enabling Clinical Grouping and Financial Reporting Other use cases are included in the roadmap of resources, developed by the Patient Administration group. The roadmap section lists plans and updates of the

current work.

8.0.5 Developmental Roadmap

The Patient Administration is currently working through resources that support:

• Encounters and Scheduling (enhance maturity of encounters and further develop in/outpatient scheduling)

• Service Provider Directory (in co-ordination with the Argonaut Provider Directory group)

• Financial Management interactions (account/coverage, then charge item, which links administration to billing)

Many of the administrative resources are part of the core resources that most systems use first and have formed the basis for most people's first experiences with FHIR. However this limited exposure has still to be proven in all contexts, such as veterinary, public health and clinical research.

8.1 Resource Patient - Content

URL: https://hl7.org/fhir/R4/patient.html
Patient Administration Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

| This page has been approved as part of an ANSI standard. See the Patient Package for further details.

Demographics and other administrative information about an individual or animal receiving care or other health-related services.

8.1.1 Scope and Usage

This Resource covers data about patients and animals involved in a wide range of health-related activities, including:

• Curative activities

• Psychiatric care

• Social services

• Pregnancy care

• Nursing and assisted living

• Dietary services

• Tracking of personal health and exercise data

The data in the Resource covers the "who" information about the patient: its attributes are focused on the demographic information necessary to support the administrative, financial and logistic procedures. A Patient record is generally created and maintained by each organization providing care for a patient. A patient or animal receiving care at multiple organizations may therefore have its information present in multiple Patient Resources.

Not all concepts are included within the base resource (such as race, ethnicity, organ donor status, nationality, etc.), but may be found in profiles defined for specific jurisdictions (e.g., US Meaningful Use Program) or standard extensions. Such fields vary widely between jurisdictions and often have different names and valuesets for the similar concepts, but they are not similar enough to be able to map and exchange.

This resource is referenced by Annotation, Signature, Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BiologicallyDerivedProduct, BodyStructure, CarePlan, CareTeam, ChargeItem, Claim, ClaimResponse, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, Coverage, CoverageEligibilityRequest, CoverageEligibilityResponse, DetectedIssue, Device, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, Encounter, EnrollmentRequest, EpisodeOfCare, ExplanationOfBenefit, FamilyMemberHistory, Flag, Goal, Group, GuidanceResponse, ImagingStudy, Immunization, ImmunizationEvaluation, ImmunizationRecommendation, Invoice, List, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MolecularSequence, NutritionOrder, Observation, itself, Person, Procedure, Provenance, QuestionnaireResponse, RelatedPerson, RequestGroup, ResearchSubject, RiskAssessment, Schedule, ServiceRequest, Specimen, SupplyDelivery, SupplyRequest, Task and VisionPrescription

8.1.2 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Patient | N | | DomainResource | Information about an individual or animal receiving health care services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | An identifier for this patient
active | ?!Σ | 0..1 | boolean | Whether this patient's record is in active use
name | Σ | 0..* | HumanName | A name associated with the patient
telecom | Σ | 0..* | ContactPoint | A contact detail for the individual
gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | Σ | 0..1 | date | The date of birth for the individual
deceased[x] | ?!Σ | 0..1 | | Indicates if the individual is deceased or not
deceasedBoolean | | | boolean
deceasedDateTime | | | dateTime
address | Σ | 0..* | Address | An address for the individual
maritalStatus | | 0..1 | CodeableConcept | Marital (civil) status of a patient MaritalStatus (Extensible)
multipleBirth[x] | | 0..1 | | Whether patient is part of a multiple birth
multipleBirthBoolean | | | boolean
multipleBirthInteger | | | integer
photo | | 0..* | Attachment | Image of the patient
contact | I | 0..* | BackboneElement | A contact party (e.g. guardian, partner, friend) for the patient + Rule: SHALL at least contain a contact's details or a reference to an organization
relationship | | 0..* | CodeableConcept | The kind of relationship Patient Contact Relationship (Extensible)
name | | 0..1 | HumanName | A name associated with the contact person
telecom | | 0..* | ContactPoint | A contact detail for the person
address | | 0..1 | Address | Address for the contact person
gender | | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
organization | I | 0..1 | Reference(Organization) | Organization that is associated with the contact
period | | 0..1 | Period | The period during which this contact person or organization is valid to be contacted relating to this patient
communication | | 0..* | BackboneElement | A language which may be used to communicate with the patient about his or her health
language | | 1..1 | CodeableConcept | The language which can be used to communicate with the patient about his or her health Common Languages (Preferred but limited to AllLanguages)
preferred | | 0..1 | boolean | Language preference indicator
generalPractitioner | | 0..* | Reference(Organization | Practitioner | PractitionerRole) | Patient's nominated primary care provider
managingOrganization | Σ | 0..1 | Reference(Organization) | Organization that is the custodian of the patient record
link | ?!Σ | 0..* | BackboneElement | Link to another patient resource that concerns the same actual person
other | Σ | 1..1 | Reference(Patient | RelatedPerson) | The other patient or related person resource that the link refers to
type | Σ | 1..1 | code | replaced-by | replaces | refer | seealso LinkType (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Patient xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for this patient --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this patient's record is in active use -->
 <name><!-- 0..* HumanName A name associated with the patient --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the individual --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date of birth for the individual -->
 <deceased[x]><!-- 0..1 boolean|dateTime Indicates if the individual is deceased or not --></deceased[x]>
 <address><!-- 0..* Address An address for the individual --></address>
 <maritalStatus><!-- 0..1 CodeableConcept Marital (civil) status of a patient --></maritalStatus>
 <multipleBirth[x]><!-- 0..1 boolean|integer Whether patient is part of a multiple birth --></multipleBirth[x]>
 <photo><!-- 0..* Attachment Image of the patient --></photo>
 <contact>  <!-- 0..* A contact party (e.g. guardian, partner, friend) for the patient -->
  <relationship><!-- 0..* CodeableConcept The kind of relationship --></relationship>
  <name><!-- 0..1 HumanName A name associated with the contact person --></name>
  <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
  <address><!-- 0..1 Address Address for the contact person --></address>
  <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
  <organization><!--  0..1 Reference(Organization) Organization that is associated with the contact --></organization>
  <period><!-- 0..1 Period The period during which this contact person or organization is valid to be contacted relating
     to this patient --></period>
 </contact>
 <communication>  <!-- 0..* A language which may be used to communicate with the patient about his or her health -->
  <language><!-- 1..1 CodeableConcept The language which can be used to communicate with the patient about his or her
    health --></language>
  <preferred value="[boolean]"/><!-- 0..1 Language preference indicator -->
 </communication>
 <generalPractitioner><!-- 0..* Reference(Organization|Practitioner|
   PractitionerRole) Patient's nominated primary care provider --></generalPractitioner>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that is the custodian of the patient record
    --></managingOrganization>
 <link>  <!-- 0..* Link to another patient resource that concerns the same actual person -->
```

```
  <other><!-- 1..1 Reference(Patient|RelatedPerson) The other patient or related person resource that the link refers to
     --></other>
  <type value="[code]"/><!-- 1..1 replaced-by | replaces | refer | seealso -->
 </link>
</Patient>
```

JSON Template

```
{
  "resourceType" : "Patient",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for this patient
  "active" : <boolean>, // Whether this patient's record is in active use
  "name" : [{ HumanName }], // A name associated with the patient
  "telecom" : [{ ContactPoint }], // A contact detail for the individual
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date of birth for the individual
  // deceased[x]: Indicates if the individual is deceased or not. One of these 2:
  "deceasedBoolean" : <boolean>,
  "deceasedDateTime" : "<dateTime>",
  "address" : [{ Address }], // An address for the individual
  "maritalStatus" : { CodeableConcept }, // Marital (civil) status of a patient
  // multipleBirth[x]: Whether patient is part of a multiple birth. One of these 2:
  "multipleBirthBoolean" : <boolean>,
  "multipleBirthInteger" : <integer>,
  "photo" : [{ Attachment }], // Image of the patient
  "contact" : [{ // A contact party (e.g. guardian, partner, friend) for the patient
    "relationship" : [{ CodeableConcept }], // The kind of relationship
    "name" : { HumanName }, // A name associated with the contact person
    "telecom" : [{ ContactPoint }], // A contact detail for the person
    "address" : { Address }, // Address for the contact person
    "gender" : "<code>", // male | female | other | unknown
    "organization" : { Reference(Organization) }, // C? Organization that is associated with the contact
    "period" : { Period } // The period during which this contact person or organization is valid to be contacted
    relating to this patient
  }],
  "communication" : [{ // A language which may be used to communicate with the patient about his or her health
    "language" : { CodeableConcept }, // R!  The language which can be used to communicate with the patient about his or
     her health
    "preferred" : <boolean> // Language preference indicator
  }],
  "generalPractitioner" : [{ Reference(Organization|Practitioner|
   PractitionerRole) }], // Patient's nominated primary care provider
  "managingOrganization" : { Reference(Organization) }, // Organization that is the custodian of the patient record
  "link" : [{ // Link to another patient resource that concerns the same actual person
    "other" : { Reference(Patient|RelatedPerson) }, // R!  The other patient or related person resource that the link
    refers to
    "type" : "<code>" // R!  replaced-by | replaces | refer | seealso
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Patient;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Patient.identifier [ Identifier ], ... ; # 0..* An identifier for this patient
  fhir:Patient.active [ boolean ]; # 0..1 Whether this patient's record is in active use
  fhir:Patient.name [ HumanName ], ... ; # 0..* A name associated with the patient
  fhir:Patient.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the individual
  fhir:Patient.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Patient.birthDate [ date ]; # 0..1 The date of birth for the individual
  # Patient.deceased[x] : 0..1 Indicates if the individual is deceased or not. One of these 2
    fhir:Patient.deceasedBoolean [ boolean ]
    fhir:Patient.deceasedDateTime [ dateTime ]
  fhir:Patient.address [ Address ], ... ; # 0..* An address for the individual
  fhir:Patient.maritalStatus [ CodeableConcept ]; # 0..1 Marital (civil) status of a patient
  # Patient.multipleBirth[x] : 0..1 Whether patient is part of a multiple birth. One of these 2
    fhir:Patient.multipleBirthBoolean [ boolean ]
    fhir:Patient.multipleBirthInteger [ integer ]
  fhir:Patient.photo [ Attachment ], ... ; # 0..* Image of the patient
  fhir:Patient.contact [ # 0..* A contact party (e.g. guardian, partner, friend) for the patient
    fhir:Patient.contact.relationship [ CodeableConcept ], ... ; # 0..* The kind of relationship
    fhir:Patient.contact.name [ HumanName ]; # 0..1 A name associated with the contact person
    fhir:Patient.contact.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
    fhir:Patient.contact.address [ Address ]; # 0..1 Address for the contact person
    fhir:Patient.contact.gender [ code ]; # 0..1 male | female | other | unknown
    fhir:Patient.contact.organization [ Reference(Organization) ]; # 0..1 Organization that is associated with the
    contact
    fhir:Patient.contact.period [ Period ]; # 0..1 The period during which this contact person or organization is valid
    to be contacted relating to this patient
  ], ...;
  fhir:Patient.communication [ # 0..* A language which may be used to communicate with the patient about his or her
    health
    fhir:Patient.communication.language [ CodeableConcept ]; # 1..1 The language which can be used to communicate with
```

```
    the patient about his or her health
    fhir:Patient.communication.preferred [ boolean ]; # 0..1 Language preference indicator
  ], ...;
  fhir:Patient.generalPractitioner [ Reference(Organization|Practitioner|PractitionerRole) ], ... ; # 0..* Patient's
    nominated primary care provider
  fhir:Patient.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that is the custodian of the
    patient record
  fhir:Patient.link [ # 0..* Link to another patient resource that concerns the same actual person
    fhir:Patient.link.other [ Reference(Patient|RelatedPerson) ]; # 1..1 The other patient or related person resource
    that the link refers to
    fhir:Patient.link.type [ code ]; # 1..1 replaced-by | replaces | refer | seealso
  ], ...;
]
```

Changes since R3

Patient
Patient.active | Default Value "true" removed
Patient.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
Patient.contact.relationship | Change value set from http://terminology.hl7.org/ValueSet/v2-0131 to http://hl7.org/fhir/ValueSet/patient-contactrelationship
Patient.contact.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
Patient.communication.language | Change binding strength from extensible to preferred
Patient.generalPractitioner | Type Reference: Added Target Type PractitionerRole
Patient.link.type | Change value set from http://hl7.org/fhir/ValueSet/link-type to http://hl7.org/fhir/ValueSet/link-type|4.0.1
Patient.animal | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 16 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Patient | N | | DomainResource | Information about an individual or animal receiving health care services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | An identifier for this patient
active | ?!Σ | 0..1 | boolean | Whether this patient's record is in active use
name | Σ | 0..* | HumanName | A name associated with the patient
telecom | Σ | 0..* | ContactPoint | A contact detail for the individual
gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | Σ | 0..1 | date | The date of birth for the individual
deceased[x] | ?!Σ | 0..1 | | Indicates if the individual is deceased or not
deceasedBoolean | | | boolean
deceasedDateTime | | | dateTime
address | Σ | 0..* | Address | An address for the individual
maritalStatus | | 0..1 | CodeableConcept | Marital (civil) status of a patient MaritalStatus (Extensible)
multipleBirth[x] | | 0..1 | | Whether patient is part of a multiple birth
multipleBirthBoolean | | | boolean
multipleBirthInteger | | | integer
photo | | 0..* | Attachment | Image of the patient
contact | I | 0..* | BackboneElement | A contact party (e.g. guardian, partner, friend) for the patient + Rule: SHALL at least contain a contact's details or a reference to an organization
relationship | | 0..* | CodeableConcept | The kind of relationship Patient Contact Relationship (Extensible)
name | | 0..1 | HumanName | A name associated with the contact person
telecom | | 0..* | ContactPoint | A contact detail for the person
address | | 0..1 | Address | Address for the contact person
gender | | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
organization | I | 0..1 | Reference(Organization) | Organization that is associated with the contact
period | | 0..1 | Period | The period during which this contact person or organization is valid to be contacted relating to this patient
communication | | 0..* | BackboneElement | A language which may be used to communicate with the patient about his or her health
language | | 1..1 | CodeableConcept | The language which can be used to communicate with the patient about his or her health Common Languages (Preferred but limited to AllLanguages)
preferred | | 0..1 | boolean | Language preference indicator
generalPractitioner | | 0..* | Reference(Organization | Practitioner | PractitionerRole) | Patient's nominated primary care provider
managingOrganization | Σ | 0..1 | Reference(Organization) | Organization that is the custodian of the patient record
link | ?!Σ | 0..* | BackboneElement | Link to another patient resource that concerns the same actual person
other | Σ | 1..1 | Reference(Patient | RelatedPerson) | The other patient or related person resource that the link refers to
type | Σ | 1..1 | code | replaced-by | replaces | refer | seealso LinkType (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Patient xmlns="http://hl7.org/fhir">
```

```
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for this patient --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this patient's record is in active use -->
 <name><!-- 0..* HumanName A name associated with the patient --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the individual --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date of birth for the individual -->
 <deceased[x]><!-- 0..1 boolean|dateTime Indicates if the individual is deceased or not --></deceased[x]>
 <address><!-- 0..* Address An address for the individual --></address>
 <maritalStatus><!-- 0..1 CodeableConcept Marital (civil) status of a patient --></maritalStatus>
 <multipleBirth[x]><!-- 0..1 boolean|integer Whether patient is part of a multiple birth --></multipleBirth[x]>
 <photo><!-- 0..* Attachment Image of the patient --></photo>
 <contact>  <!-- 0..* A contact party (e.g. guardian, partner, friend) for the patient -->
  <relationship><!-- 0..* CodeableConcept The kind of relationship --></relationship>
  <name><!-- 0..1 HumanName A name associated with the contact person --></name>
  <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
  <address><!-- 0..1 Address Address for the contact person --></address>
  <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
  <organization><!--  0..1 Reference(Organization) Organization that is associated with the contact --></organization>
  <period><!-- 0..1 Period The period during which this contact person or organization is valid to be contacted relating
     to this patient --></period>
 </contact>
 <communication>  <!-- 0..* A language which may be used to communicate with the patient about his or her health -->
  <language><!-- 1..1 CodeableConcept The language which can be used to communicate with the patient about his or her
    health --></language>
  <preferred value="[boolean]"/><!-- 0..1 Language preference indicator -->
 </communication>
 <generalPractitioner><!-- 0..* Reference(Organization|Practitioner|
   PractitionerRole) Patient's nominated primary care provider --></generalPractitioner>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that is the custodian of the patient record
    --></managingOrganization>
 <link>  <!-- 0..* Link to another patient resource that concerns the same actual person -->
  <other><!-- 1..1 Reference(Patient|RelatedPerson) The other patient or related person resource that the link refers to
     --></other>
  <type value="[code]"/><!-- 1..1 replaced-by | replaces | refer | seealso -->
 </link>
</Patient>
```

JSON Template

```
{
  "resourceType" : "Patient",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for this patient
  "active" : <boolean>, // Whether this patient's record is in active use
  "name" : [{ HumanName }], // A name associated with the patient
  "telecom" : [{ ContactPoint }], // A contact detail for the individual
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date of birth for the individual
  // deceased[x]: Indicates if the individual is deceased or not. One of these 2:
  "deceasedBoolean" : <boolean>,
  "deceasedDateTime" : "<dateTime>",
  "address" : [{ Address }], // An address for the individual
  "maritalStatus" : { CodeableConcept }, // Marital (civil) status of a patient
  // multipleBirth[x]: Whether patient is part of a multiple birth. One of these 2:
  "multipleBirthBoolean" : <boolean>,
  "multipleBirthInteger" : <integer>,
  "photo" : [{ Attachment }], // Image of the patient
  "contact" : [{ // A contact party (e.g. guardian, partner, friend) for the patient
    "relationship" : [{ CodeableConcept }], // The kind of relationship
    "name" : { HumanName }, // A name associated with the contact person
    "telecom" : [{ ContactPoint }], // A contact detail for the person
    "address" : { Address }, // Address for the contact person
    "gender" : "<code>", // male | female | other | unknown
    "organization" : { Reference(Organization) }, // C? Organization that is associated with the contact
    "period" : { Period } // The period during which this contact person or organization is valid to be contacted
    relating to this patient
  }],
  "communication" : [{ // A language which may be used to communicate with the patient about his or her health
    "language" : { CodeableConcept }, // R!  The language which can be used to communicate with the patient about his or
     her health
    "preferred" : <boolean> // Language preference indicator
  }],
  "generalPractitioner" : [{ Reference(Organization|Practitioner|
   PractitionerRole) }], // Patient's nominated primary care provider
  "managingOrganization" : { Reference(Organization) }, // Organization that is the custodian of the patient record
  "link" : [{ // Link to another patient resource that concerns the same actual person
    "other" : { Reference(Patient|RelatedPerson) }, // R!  The other patient or related person resource that the link
    refers to
    "type" : "<code>" // R!  replaced-by | replaces | refer | seealso
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Patient;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Patient.identifier [ Identifier ], ... ; # 0..* An identifier for this patient
  fhir:Patient.active [ boolean ]; # 0..1 Whether this patient's record is in active use
  fhir:Patient.name [ HumanName ], ... ; # 0..* A name associated with the patient
  fhir:Patient.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the individual
  fhir:Patient.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Patient.birthDate [ date ]; # 0..1 The date of birth for the individual
  # Patient.deceased[x] : 0..1 Indicates if the individual is deceased or not. One of these 2
    fhir:Patient.deceasedBoolean [ boolean ]
    fhir:Patient.deceasedDateTime [ dateTime ]
  fhir:Patient.address [ Address ], ... ; # 0..* An address for the individual
  fhir:Patient.maritalStatus [ CodeableConcept ]; # 0..1 Marital (civil) status of a patient
  # Patient.multipleBirth[x] : 0..1 Whether patient is part of a multiple birth. One of these 2
    fhir:Patient.multipleBirthBoolean [ boolean ]
    fhir:Patient.multipleBirthInteger [ integer ]
  fhir:Patient.photo [ Attachment ], ... ; # 0..* Image of the patient
  fhir:Patient.contact [ # 0..* A contact party (e.g. guardian, partner, friend) for the patient
    fhir:Patient.contact.relationship [ CodeableConcept ], ... ; # 0..* The kind of relationship
    fhir:Patient.contact.name [ HumanName ]; # 0..1 A name associated with the contact person
    fhir:Patient.contact.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
    fhir:Patient.contact.address [ Address ]; # 0..1 Address for the contact person
    fhir:Patient.contact.gender [ code ]; # 0..1 male | female | other | unknown
    fhir:Patient.contact.organization [ Reference(Organization) ]; # 0..1 Organization that is associated with the
    contact
    fhir:Patient.contact.period [ Period ]; # 0..1 The period during which this contact person or organization is valid
    to be contacted relating to this patient
  ], ...;
  fhir:Patient.communication [ # 0..* A language which may be used to communicate with the patient about his or her
    health
    fhir:Patient.communication.language [ CodeableConcept ]; # 1..1 The language which can be used to communicate with
    the patient about his or her health
    fhir:Patient.communication.preferred [ boolean ]; # 0..1 Language preference indicator
  ], ...;
  fhir:Patient.generalPractitioner [ Reference(Organization|Practitioner|PractitionerRole) ], ... ; # 0..* Patient's
    nominated primary care provider
  fhir:Patient.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that is the custodian of the
    patient record
  fhir:Patient.link [ # 0..* Link to another patient resource that concerns the same actual person
    fhir:Patient.link.other [ Reference(Patient|RelatedPerson) ]; # 1..1 The other patient or related person resource
    that the link refers to
    fhir:Patient.link.type [ code ]; # 1..1 replaced-by | replaces | refer | seealso
  ], ...;
]
```

Changes since Release 3

Patient

Patient.active | Default Value "true" removed

Patient.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Patient.contact.relationship | Change value set from http://terminology.hl7.org/ValueSet/v2-0131 to http://hl7.org/fhir/ValueSet/patient-contactrelationship

Patient.contact.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Patient.communication.language | Change binding strength from extensible to preferred

Patient.generalPractitioner | Type Reference: Added Target Type PractitionerRole

Patient.link.type | Change value set from http://hl7.org/fhir/ValueSet/link-type to http://hl7.org/fhir/ValueSet/link-type|4.0.1

Patient.animal | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 16 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.1.2.1 Terminology Bindings

Path | Definition | Type | Reference

Patient.gender Patient.contact.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender

Patient.maritalStatus | The domestic partnership status of a person. | Extensible | Marital Status Codes

Patient.contact.relationship | The nature of the relationship between a patient and a contact person for that patient. | Extensible | PatientContactRelationship

Patient.communication.language | A human language. | Preferred, but limited to AllLanguages | CommonLanguages

Patient.link.type | The type of link between this patient resource and another patient resource. | Required | LinkType

8.1.2.2 Constraints

id | Level | Location | Description | Expression

pat-1 | Rule | Patient.contact | SHALL at least contain a contact's details or a reference to an organization | name.exists() or telecom.exists() or address.exists() or organization.exists()

Notes:

• multipleBirth can be either expressed as a Boolean (just indicating whether the patient is part of a multiple birth) or as an integer, indicating the actual birth order.

• Patient records may only be in one of two statuses: in use (active=true) and not in use (active=false). A normal record is active, i.e. it is in use. Active is set to 'false' when a record is created as a duplicate or in error. A record does not need to be linked to be inactivated.

• The link element is used to assert that two or more Patient resources are both about the same actual patient. See below for further discussion

• There should be only one preferred language (Language.preference = true) per mode of expression.

• The Contact for a Patient has an element organization, this is for use with guardians or business related contacts where just the organization is relevant.

8.1.3 Patient ids and Patient resource ids

A Patient record's Resource Id can never change. For this reason, the identifiers with which humans are concerned (often called MRN - Medical Record Number, or UR - Unit Record) should not be used for the resource's id, since MRN's may change, i.e. as a result of having duplicate records of the same patient. Instead they should be represented in the Patient.identifier list where they can be managed. This is also useful for the case of institutions that have acquired multiple numbers because of mergers of patient record systems over time.

Where there is a need to implement an automated MRN Identifier created for a patient record, this could be achieved by providing an identifier in the patient with an appropriate assigner, MRN Type and/or system but with no value assigned. Internal business rules can then detect this and replace/populate this identifier with 1 or more identifiers (as required).

8.1.4 Linking Patients

The link element is used to assert that patient resources refer to the same patient. This element is used to support the following scenarios where multiple patient records exist:

8.1.4.1 Duplicate Patient records

Managing Patient registration is a well-known difficult problem. Around 2% of registrations are in error, mostly duplicate records. Sometimes the duplicate record is caught fairly quickly and retired before much data is accumulated. In other cases, substantial amounts of data may accumulate. By using a link of type 'replaced-by', the record containing such a link is marked as a duplicate and the link points forward to a record that should be used instead. Note that the record pointed to may in its turn have been identified as created in error and forward to yet another Patient resource. Records that replace another record may use a link type of 'replaces' pointing to the old record.

8.1.4.2 Patient record in a Patient index

A Patient record may be present in a system that acts as a Patient Index: it maintains a (summary of) patient data and a list of one or more servers that are known to hold a more comprehensive and/or authoritative record of the same patient. The link type 'refer' is used to denote such a link. Note that linked records may contain contradictory information. The record referred to does not point back to the referring record.

8.1.4.3 Distributed Patient record

In a distributed architecture, multiple systems keep separate patient records concerning the same patient. These records are not considered duplicates, but contain a distributed, potentially overlapping view of the patient's data. Each such record may have its own focus or maintaining organization and there need not be a sense of one record being more complete or more authoritative than another. In such cases, links of type 'see also' can be used to point to other patient records. It is not a requirement that such links are bilateral.

8.1.5 Patient vs. Person vs. Patient.Link vs. Linkage

The Person resource on the surface appears to be very similar to the Patient resource, and the usage for it is very similar to using the Patient.Link capability. The intention of the Person resource is to be able to link instances of resources together that are believed to be the same individual. This includes across resource types, such as RelatedPerson, Practitioner, Patient and even other Person resources. The Patient Link however is only intended to be used for Patient resources.

The primary use case for the Person resource is to be able to support person registries that do not necessarily have a healthcare context, and are able to identify and quantify confidence levels that this is the same person. This could include consumer portals where the maintainer of the person information is the actual person themselves. A system could use the Person entry to cross check changes to information applied to one part of a record to values in another system; e.g., when moving, a consumer updates his contact numbers and address in his person record, and then a Patient Administration system is able to see that this data is

changed and prompt the organization to follow up with the patient that was linked to the person record if they want their details updated, or if they no longer need services and they should be cancelled, as they've moved from the area.

The Linkage resource and the Patient.link property conceptually perform similar functions in FHIR, both provide an assertion of linkages between multiple resource instances that are referring to the same underlying individual. When a Patient resource is linked/merged then it needs to have an internal indication that there is another patient resource that should be considered when referencing other records, which is achieved using the patient.link property. Not detecting/checking for a potential linkage could mean that related clinical records are not discovered, potentially impacting patient safety. (which is why using the Linkage resource is not appropriate, as its use in this manner would force the use of either another query to potentially locate other patient resources to consider, or use _revinclude)

8.1.6 Patient.contact vs. RelatedPerson

The contact element on the Patient resource should be used for storing the details of people to contact. This information always travels with the Patient resource, and cannot be used as the target of a reference. Where related people need to be referenced by other resources (e.g. CarePlan.participant, Encounter.participant, DocumentReference.author, Appointment.participant), the RelatedPerson resource must be used.

It is not expected that these records will be used for recording the primary care provider; this information should be stored in the Patient.generalPractitioner field.

8.1.7 Patient Gender and Sex

Many systems and organizations only provide for a single attribute that aspires to represent all aspects of a patient's gender and sex with a single value. However, there are many considerations around sex and gender documentation and interoperability. Listed below are the various social and biological attributes that are relevant in the healthcare setting, as well as information on how each can be communicated.

• Administrative Gender - in order to interoperate with systems that use a single generic property, the basic Patient.gender property represents an administrative gender: the gender that the patient is considered to have for administration and record keeping purposes. This property is often used as an input to patient matching algorithms, for example.

In addition to this administrative gender, other kinds of gender or sex properties may be represented:

• Clinical Sex - a testable observation about a biological property of the patient. There are several different types of clinical sex, including karyotypic/genetic/chromosomal, gonadal, ductal, phenotypic, etc. Clinical sex observations should be represented using Observation, qualified with the appropriate clinical codes from LOINC and/or SNOMED.

• Clinical Gender - an observation about the patient, often collected as part of social history documentation, and represented as an Observation(example) using, for example, the LOINC code 76691-5 . Clinical gender observations can provide both history and confidentiality, where the genderIdentity extension does not.

• Gender Identity - an indication from the patient about what gender they consider themselves to be. This can influence how the patient prefers to be addressed by care providers and other individuals. The standard genderIdentity extension may be used to communicate this property. This extension is appropriate when the gender identity is openly known.

• Sex assigned at Birth - the sex assigned at birth, as documented on the birth registration. Some countries allow variations such as not yet determined, unknown, or undifferentiated, while others do not. Some countries also allow birth registration information to be updated. The US realm defines a US specific extension for this property. Alternatively, if you were representing this concept with an observation, you could use the LOINC code 76689-9 .

• Legal Sex - regional and national entities often categorize citizens using a single legal sex value. The legal sex of a patient can vary from region to region and country to country. A single patient may have multiple legal sex values at the same time in different jurisdictions. In case where the Patient.gender administrative property is not sufficient to communicate legal sex, realm specific extensions should be used.

For veterinary use, the animal extension also includes the genderStatus which indicates sterility information.

8.1.8 Mother and newborn relationships

There are several ways to represent the relationship between a mother and a child. This is due to the when it is recorded and the purpose for which it is recorded:

• To express the family relationship and legal responsibility thereof for administrative purposes: use the Patient/RelatedPerson structure. This structure is consistent over time.

• To relate the encounters of a mother and her baby in a maternity encounter, for administrative and billing purposes: use the encounter.partof property

• To collect information about the patient's relatives that might be relevant to the patient's medical condition: use the FamilyMemberHistory resource

During a maternity encounter, the Patient and Encounter resources for the mother will be present. After the child is born, new Patient, Encounter and RelatedPerson (for the mother) records will be created. The Child's encounter should reference the Mother's encounter using the partOf property. The Patient/RelatedPerson structure should also be created for ongoing usage, as shown in this example:

```
<Patient>
  <id value="child"/>
  <!-- The details of the child -->
</Patient>
<RelatedPerson>
  <id value="rp-mom"/>
  <patient>
    <reference value="Patient/child"/>
  </patient>
</RelatedPerson>
<Patient>
  <id value="pat-mom"/>
  <!-- The details of the mom -->
  <link>
    <other value="rp-mom"/>
    <type value="see-also"/>
  </link>
</Patient>
<Encounter>
  <id value="mom-enc"/>
  <status value="in-progress"/>
  <class value="inpatient"/>
  <patient>
    <reference value="Patient/mom"/>
  </patient>
</Encounter>
<Encounter>
  <id value="child-enc"/>
  <status value="in-progress"/>
  <class value="inpatient"/>
  <patient>
    <reference value="Patient/child"/>
  </patient>
  <partOf>
    <reference value="Encounter/mom-enc"/>
  </partOf>
</Encounter>
```

8.1.9 Merging records

This specification does not specify merge functionality: if multiple patient records are found to be duplicates, they can be linked together, as described above. These links merely express the relationship between records, and in the case of a replacement link, indicate a "master" record. This specification does not mandate that FHIR servers migrate information between such records on finding such a link. Note:

• Health information administrators may call the process "merging", but it is often implemented as "linking" at the record level

• Servers are allowed to implement merging/record migration even though it is not mandated.

Note: We are seeking input from the implementer community on what effect linking/merging/unlinking should have on other functionality such as the GET operation, searching, reverse includes, etc.; How should an unlink behavior be done? How will the patient compartment interact with the merge? This functionality and related behaviors is subject to ongoing experimentation and implementation testing, with a definition to be proposed in a future version of this specification.

Feedback here .

8.1.10 Patient Matching using an MPI

A Master Patient Index (MPI) is a service used to manage patient identification in a context where multiple patient databases exist. Healthcare applications and middleware use the MPI to match patients between the databases, and as new patient details are encountered. MPIs are highly specialized applications, often tailored extensively to the institution's particular mix of patients. MPIs can also be run on a regional and national basis.

To ask an MPI to match a patient, clients call the patient $match operation, which processes a parameters resource containing a complete or fragment of a patient resource, along with some other control parameters. This provided patient resource does not need to pass full validation (mandatory fields, or invariants) as the resource will not be stored, it does however need to be a parsable instance. The MPI can then use the properties of the resource as MPI inputs, and process them using an internal MPI algorithm of some kind to determine the most appropriate matches in the patient set. It does not have to use all the properties provided, and may ignore others provided quietly. A specific profile (with the required fields/invariants) can be used to define what parameters the MPI algorithm requires.

```
  POST [base]/Patient/$match
  [some headers including content-type xml or json]
  [parameters body with patient resource inside]
```

The response from an MPI $match operation is a set of patient records, ordered from most likely to least likely. If there are not patient matches, the MPI SHALL return an empty search set with no error, but may include an operation outcome with further advice. All patient records should have a score from 0 to 1, where 1 is the most certain match, along with an extension "match-grade" that indicates the MPI's position on the match quality:

```
<entry>
  <resource>
    <Patient>
      <!-- patient details -->
    </Patient>
  </resource>
  <search>
    <extension url="http://hl7.org/fhir/StructureDefinition/match-grade">
      <valueCode value="probable"/>
    </extension>
    <score value="0.80"/>
  </search>
</entry>
```

The match-grade extension has one of the following codes:

certain | This record meets the matching criteria to be automatically considered as a full match.
probable | This record is a close match, but not a certain match. Additional review (e.g. by a human) may be required before using this as a match.
possible | This record may be a matching one. Additional review (e.g. by a human) SHOULD be performed before using this as a match.
certainly-not | This record is known not to be a match. Note that usually non-matching records are not returned, but in some cases records previously or likely considered as a match may specifically be negated by the matching engine.

The purpose of using an MPI search versus a regular search is that the MPI search is really intended to target and find a specific single patient for recording information about reducing errors through incorrectly selecting the wrong patient. Often MPIs won't return data if there is insufficient search parameter data, such as a partial surname. This compares to a regular search which can be used for finding lists of patients, such as to locate a group of patients that share a property in common, such as live in a specific location, or fit within an age range for performing population analysis.

A formal definition for the MPI $match operation is published.

8.1.11 Veterinary Care

Veterinary care is very much within the scope of FHIR, and the Patient resource can be used to communicate information about animal patients. To support this, there is a standard patient-animal extension which can be used for recording details about species, breed, and gender status. This extension is not intended to cover all relevant properties for veterinary care, and the use of additional domain-relevant extensions is expected for areas such as laboratory, zoological and livestock care.

The veterinary client ("owner") is represented using the RelatedPerson resource.

8.1.12 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active TU | token | Whether the patient record is active | Patient.active
address TU | string | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Patient.address | 3 Resources
address-city TU | string | A city specified in an address | Patient.address.city | 3 Resources
address-country TU | string | A country specified in an address | Patient.address.country | 3 Resources
address-postalcode TU | string | A postalCode specified in an address | Patient.address.postalCode | 3 Resources
address-state TU | string | A state specified in an address | Patient.address.state | 3 Resources
address-use TU | token | A use code specified in an address | Patient.address.use | 3 Resources
birthdate TU | date | The patient's date of birth | Patient.birthDate | 2 Resources
death-date TU | date | The date of death has been provided and satisfies this search value | (Patient.deceased as dateTime)
deceased TU | token | This patient has been marked as deceased, or as a death date entered | Patient.deceased.exists() and Patient.deceased != false
email TU | token | A value in an email contact | Patient.telecom.where(system='email') | 4 Resources
family TU | string | A portion of the family name of the patient | Patient.name.family | 1 Resources
gender TU | token | Gender of the patient | Patient.gender | 3 Resources
general-practitioner TU | reference | Patient's nominated general practitioner, not the organization that manages the record | Patient.generalPractitioner (Practitioner, Organization, PractitionerRole)
given TU | string | A portion of the given name of the patient | Patient.name.given | 1 Resources
identifier TU | token | A patient identifier | Patient.identifier
language TU | token | Language code (irrespective of use value) | Patient.communication.language
link TU | reference | All patients linked to the given patient | Patient.link.other (Patient, RelatedPerson)
name TU | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Patient.name

organization TU | reference | The organization that is the custodian of the patient record | Patient.managingOrganization (Organization)
phone TU | token | A value in a phone contact | Patient.telecom.where(system='phone') | 4 Resources
phonetic TU | string | A portion of either family or given name using some kind of phonetic matching algorithm | Patient.name | 3 Resources
telecom TU | token | The value in any kind of telecom details of the patient | Patient.telecom | 4 Resources

8.1.13 Resource Patient - Examples

URL: https://hl7.org/fhir/R4/patient-examples.html

8.1.13 Resource Patient - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Person Example | example | XML | JSON | Turtle
Patient 1 for linking | pat1 | XML | JSON | Turtle
Patient 2 for linking | pat2 | XML | JSON | Turtle
Deceased patient (using time) | pat3 | XML | JSON | Turtle
Deceased patient (using boolean) | pat4 | XML | JSON | Turtle
Stock people (defined by HL7 publishing) | b248b1b2-1686-4b94-9936-37d7a5f94b51 | XML | JSON | Turtle
Example People from cypress project | b0a5e4277-83c4-4adb-87e2-e3efe3369b6f | XML | JSON | Turtle
2nd person example | xcda | XML | JSON | Turtle
XDS Patient | xds | XML | JSON | Turtle
An example of an animal | animal | XML | JSON | Turtle
Taken from a DICOM sample | dicom | XML | JSON | Turtle
Example from IHE-PCD example | ihe-pcd | XML | JSON | Turtle
Real-world patient example (anonymized) | f001 | XML | JSON | Turtle
Real-world patient example (anonymized) | f201 | XML | JSON | Turtle
Example for glossy | glossy | XML | JSON | Turtle
Genetic Risk Assessment Person | proband | XML | JSON | Turtle
Additional Genetics Example | genetics-example1 | XML | JSON | Turtle
Example Patient resource with Chinese content | ch-example | XML | JSON | Turtle
Newborn Patient Example | newborn | XML | JSON | Turtle
Mother of Newborn Patient Example | mom | XML | JSON | Turtle
Newborn Eldest Twin Example | infant-twin-1 | XML | JSON | Turtle
Newborn Youngest Twin Example | infant-twin-2 | XML | JSON | Turtle
Pre-birth fetal infant Example | infant-fetal | XML | JSON | Turtle
Mother of infant twins and fetal infant. | infant-mom | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.1.14 Resource Patient - Detailed Descriptions

URL: https://hl7.org/fhir/R4/patient-definitions.html

8.1.14 Resource Patient - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Patient resource.

Patient
Element Id | Patient
Definition | Demographics and other administrative information about an individual or animal receiving care or other health-related services.
Cardinality | 0..*
Type | DomainResource
Requirements | Tracking patient is the center of the healthcare process.
Alternate Names | SubjectOfCare Client Resident
Patient.identifier
Element Id | Patient.identifier
Definition | An identifier for this patient.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Patients are almost always assigned specific numerical identifiers.
Summary | true
Patient.active
Element Id | Patient.active
Definition | Whether this patient record is in active use. Many systems use this property to mark as non-current patients, such as those that have not been seen for a period of time based on an organization's business rules. It is often used to filter patient lists

to exclude inactive patients Deceased patients may also be marked as inactive for the same reasons, but may be active for some time after death.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a patient record as not to be used because it was created in error.
Summary | true
Comments | If a record is inactive, and linked to an active record, then future patient/record updates should occur on the other patient.
Patient.name
Element Id | Patient.name
Definition | A name associated with the individual.
Cardinality | 0..*
Type | HumanName
Requirements | Need to be able to track the patient by multiple names. Examples are your official name and a partner name.
Summary | true
Comments | A patient may have multiple names with different uses or applicable periods. For animals, the name is a "HumanName" in the sense that is assigned and used by humans and has the same patterns.
Patient.telecom
Element Id | Patient.telecom
Definition | A contact detail (e.g. a telephone number or an email address) by which the individual may be contacted.
Cardinality | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Summary | true
Comments | A Patient may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently and also to help with identification. The address might not go directly to the individual, but may reach another party that is able to proxy for the patient (i.e. home phone, or pet owner's phone).
Patient.gender
Element Id | Patient.gender
Definition | Administrative Gender - the gender that the patient is considered to have for administration and record keeping purposes.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed for identification of the individual, in combination with (at least) name and birth date.
Summary | true
Comments | The gender might not match the biological sex as determined by genetics or the individual's preferred identification. Note that for both humans and particularly animals, there are other legitimate possibilities than male and female, though the vast majority of systems and contexts only support male and female. Systems providing decision support or enforcing business rules should ideally do this on the basis of Observations dealing with the specific sex or gender aspect of interest (anatomical, chromosomal, social, etc.) However, because these observations are infrequently recorded, defaulting to the administrative gender is common practice. Where such defaulting occurs, rule enforcement should allow for the variation between administrative and biological, chromosomal and other gender aspects. For example, an alert about a hysterectomy on a male should be handled as a warning or overridable error, not a "hard" error. See the Patient Gender and Sex section for additional information about communicating patient gender and sex.
Patient.birthDate
Element Id | Patient.birthDate
Definition | The date of birth for the individual.
Cardinality | 0..1
Type | date
Requirements | Age of the individual drives many clinical processes.
Summary | true
Comments | At least an estimated year should be provided as a guess if the real DOB is unknown There is a standard extension "patient-birthTime" available that should be used where Time is required (such as in maternity/infant care systems).
LOINC Code | 21112-8
Patient.deceased[x]
Element Id | Patient.deceased[x]
Definition | Indicates if the individual is deceased or not.
Cardinality | 0..1
Type | boolean|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Is Modifier | true (Reason: This element is labeled as a modifier because once a patient is marked as deceased, the actions that are appropriate to perform on the patient may be significantly different.)
Requirements | The fact that a patient is deceased influences the clinical process. Also, in human communication and relation management it is necessary to know whether the person is alive.
Summary | true
Comments | If there's no value in the instance, it means there is no statement on whether or not the individual is deceased. Most systems will interpret the absence of a value as a sign of the person being alive.
Patient.address
Element Id | Patient.address
Definition | An address for the individual.

Cardinality | 0..*
Type | Address
Requirements | May need to keep track of patient addresses for contacting, billing or reporting requirements and also to help with identification.
Summary | true
Comments | Patient may have multiple addresses with different uses or applicable periods.
Patient.maritalStatus
Element Id | Patient.maritalStatus
Definition | This field contains a patient's most recent marital (civil) status.
Cardinality | 0..1
Terminology Binding | MaritalStatus (Extensible)
Type | CodeableConcept
Requirements | Most, if not all systems capture it.
Patient.multipleBirth[x]
Element Id | Patient.multipleBirth[x]
Definition | Indicates whether the patient is part of a multiple (boolean) or indicates the actual birth order (integer).
Cardinality | 0..1
Type | boolean|integer
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | For disambiguation of multiple-birth children, especially relevant where the care provider doesn't meet the patient, such as labs.
Comments | Where the valueInteger is provided, the number is the birth number in the sequence. E.g. The middle birth in triplets would be valueInteger=2 and the third born would have valueInteger=3 If a boolean value was provided for this triplets example, then all 3 patient records would have valueBoolean=true (the ordering is not indicated).
Patient.photo
Element Id | Patient.photo
Definition | Image of the patient.
Cardinality | 0..*
Type | Attachment
Requirements | Many EHR systems have the capability to capture an image of the patient. Fits with newer social media usage too.
Comments | Guidelines: Use id photos, not clinical photos. Limit dimensions to thumbnail. Keep byte count low to ease resource updates.
Patient.contact
Element Id | Patient.contact
Definition | A contact party (e.g. guardian, partner, friend) for the patient.
Cardinality | 0..*
Requirements | Need to track people you can contact about the patient.
Comments | Contact covers all kinds of contact parties: family members, business contacts, guardians, caregivers. Not applicable to register pedigree and family ties beyond use of having contact.
Invariants | Defined on this element pat-1 Rule SHALL at least contain a contact's details or a reference to an organization name.exists() or telecom.exists() or address.exists() or organization.exists()
Defined on this element
pat-1 | Rule | SHALL at least contain a contact's details or a reference to an organization | name.exists() or telecom.exists() or address.exists() or organization.exists()
Patient.contact.relationship
Element Id | Patient.contact.relationship
Definition | The nature of the relationship between the patient and the contact person.
Cardinality | 0..*
Terminology Binding | Patient Contact Relationship (Extensible)
Type | CodeableConcept
Requirements | Used to determine which contact person is the most relevant to approach, depending on circumstances.
Patient.contact.name
Element Id | Patient.contact.name
Definition | A name associated with the contact person.
Cardinality | 0..1
Type | HumanName
Requirements | Contact persons need to be identified by name, but it is uncommon to need details about multiple other names for that contact person.
Patient.contact.telecom
Element Id | Patient.contact.telecom
Definition | A contact detail for the person, e.g. a telephone number or an email address.
Cardinality | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Comments | Contact may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently, and also to help with identification.
Patient.contact.address
Element Id | Patient.contact.address
Definition | Address for the contact person.
Cardinality | 0..1
Type | Address
Requirements | Need to keep track where the contact person can be contacted per postal mail or visited.
Patient.contact.gender

Element Id | Patient.contact.gender
Definition | Administrative Gender - the gender that the contact person is considered to have for administration and record keeping purposes.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed to address the person correctly.
Patient.contact.organization
Element Id | Patient.contact.organization
Definition | Organization on behalf of which the contact is acting or for which the contact is working.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | For guardians or business related contacts, the organization is relevant.
Invariants | Affect this element pat-1 Rule SHALL at least contain a contact's details or a reference to an organization name.exists() or telecom.exists() or address.exists() or organization.exists()
Affect this element
pat-1 | Rule | SHALL at least contain a contact's details or a reference to an organization | name.exists() or telecom.exists() or address.exists() or organization.exists()
Patient.contact.period
Element Id | Patient.contact.period
Definition | The period during which this contact person or organization is valid to be contacted relating to this patient.
Cardinality | 0..1
Type | Period
Patient.communication
Element Id | Patient.communication
Definition | A language which may be used to communicate with the patient about his or her health.
Cardinality | 0..*
Requirements | If a patient does not speak the local language, interpreters may be required, so languages spoken and proficiency are important things to keep track of both for patient and other persons of interest.
Comments | If no language is specified, this implies that the default local language is spoken. If you need to convey proficiency for multiple modes, then you need multiple Patient.Communication associations. For animals, language is not a relevant field, and should be absent from the instance. If the Patient does not speak the default local language, then the Interpreter Required Standard can be used to explicitly declare that an interpreter is required.
Patient.communication.language
Element Id | Patient.communication.language
Definition | The ISO-639-1 alpha 2 code in lower case for the language, optionally followed by a hyphen and the ISO-3166-1 alpha 2 code for the region in upper case; e.g. "en" for English, or "en-US" for American English versus "en-EN" for England English.
Cardinality | 1..1
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | CodeableConcept
Requirements | Most systems in multilingual countries will want to convey language. Not all systems actually need the regional dialect.
Comments | The structure aa-BB with this exact casing is one the most widely used notations for locale. However not all systems actually code this but instead have it as free text. Hence CodeableConcept instead of code as the data type.
Patient.communication.preferred
Element Id | Patient.communication.preferred
Definition | Indicates whether or not the patient prefers this language (over other languages he masters up a certain level).
Cardinality | 0..1
Type | boolean
Requirements | People that master multiple languages up to certain level may prefer one or more, i.e. feel more confident in communicating in a particular language making other languages sort of a fall back method.
Comments | This language is specifically identified for communicating healthcare information.
Patient.generalPractitioner
Element Id | Patient.generalPractitioner
Definition | Patient's nominated care provider.
Cardinality | 0..*
Type | Reference(Organization | Practitioner | PractitionerRole)
Alternate Names | careProvider
Comments | This may be the primary care provider (in a GP context), or it may be a patient nominated care manager in a community/disability setting, or even organization that will provide people to perform the care provider roles. It is not to be used to record Care Teams, these should be in a CareTeam resource that may be linked to the CarePlan or EpisodeOfCare resources. Multiple GPs may be recorded against the patient for various reasons, such as a student that has his home GP listed along with the GP at university during the school semesters, or a "fly-in/fly-out" worker that has the onsite GP also included with his home GP to remain aware of medical issues. Jurisdictions may decide that they can profile this down to 1 if desired, or 1 per type.
Patient.managingOrganization
Element Id | Patient.managingOrganization
Definition | Organization that is the custodian of the patient record.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | Need to know who recognizes this patient record, manages and updates it.
Summary | true
Comments | There is only one managing organization for a specific patient record. Other organizations will have their own Patient record, and may use the Link property to join the records together (or a Person resource which can include confidence ratings for the association).

Patient.link
Element Id | Patient.link
Definition | Link to another patient resource that concerns the same actual patient.
Cardinality | 0..*
Is Modifier | true (Reason: This element is labeled as a modifier because it might not be the main Patient resource, and the referenced patient should be used instead of this Patient record. This is when the link.type value is 'replaced-by')
Requirements | There are multiple use cases: Duplicate patient records due to the clerical errors associated with the difficulties of identifying humans consistently, and Distribution of patient information across multiple servers.
Summary | true
Comments | There is no assumption that linked patient records have mutual links.
Patient.link.other
Element Id | Patient.link.other
Definition | The other patient resource that the link refers to.
Cardinality | 1..1
Type | Reference(Patient | RelatedPerson)
Hierarchy | This reference may point back to the same instance (including transitively)
Summary | true
Comments | Referencing a RelatedPerson here removes the need to use a Person record to associate a Patient and RelatedPerson as the same individual.
Patient.link.type
Element Id | Patient.link.type
Definition | The type of link between this patient resource and another patient resource.
Cardinality | 1..1
Terminology Binding | LinkType (Required)
Type | code
Summary | true

8.1.15 Resource Patient - Mappings

URL: https://hl7.org/fhir/R4/patient-mappings.html

8.1.15 Resource Patient - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

LOINC code for the element

CDA (R2)

RIM Mapping

Mappings for the patient resource (see Mappings to Other Standards for further information & status).

8.1.15.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Patient
identifier | FiveWs.identifier
active | FiveWs.status

8.1.15.2 HL7 v2 Mapping (http://hl7.org/v2)

Patient
identifier | PID-3
active
name | PID-5, PID-9
telecom | PID-13, PID-14, PID-40
gender | PID-8
birthDate | PID-7
deceased[x] | PID-30 (bool) and PID-29 (datetime)
address | PID-11
maritalStatus | PID-16
multipleBirth[x] | PID-24 (bool), PID-25 (integer)
photo | OBX-5 - needs a profile
contact
relationship | NK1-7, NK1-3
name | NK1-2
telecom | NK1-5, NK1-6, NK1-40
address | NK1-4
gender | NK1-15
organization | NK1-13, NK1-30, NK1-31, NK1-32, NK1-41
period
communication

language | PID-15, LAN-2
preferred | PID-15
generalPractitioner | PD1-4
managingOrganization
link
other | PID-3, MRG-1
type

8.1.15.3 LOINC code for the element (http://loinc.org)

Patient
identifier
active
name
telecom
gender
birthDate | 21112-8
deceased[x]
address
maritalStatus
multipleBirth[x]
photo
contact
relationship
name
telecom
address
gender
organization
period
communication
language
preferred
generalPractitioner
managingOrganization
link
other
type

8.1.15.4 CDA (R2) (http://hl7.org/v3/cda)

Patient | ClinicalDocument.recordTarget.patientRole
identifier | .id
active | n/a
name | .patient.name
telecom | .telecom
gender | .patient.administrativeGenderCode
birthDate | .patient.birthTime
deceased[x] | n/a
address | .addr
maritalStatus | .patient.maritalStatusCode
multipleBirth[x] | n/a
photo | n/a
contact | n/a
relationship | n/a
name | n/a
telecom | n/a
address | n/a
gender | n/a
organization | n/a
period | n/a
communication | patient.languageCommunication
language | .languageCode
preferred | .preferenceInd
generalPractitioner | n/a
managingOrganization | .providerOrganization
link | n/a
other | n/a
type | n/a

8.1.15.5 RIM Mapping (http://hl7.org/v3)

Patient | Patient[classCode=PAT]
identifier | id

active | statusCode
name | name
telecom | telecom
gender | player[classCode=PSN|ANM and determinerCode=INSTANCE]/administrativeGender
birthDate | player[classCode=PSN|ANM and determinerCode=INSTANCE]/birthTime
deceased[x] | player[classCode=PSN|ANM and determinerCode=INSTANCE]/deceasedInd, player[classCode=PSN|ANM and determinerCode=INSTANCE]/deceasedTime
address | addr
maritalStatus | player[classCode=PSN]/maritalStatusCode
multipleBirth[x] | player[classCode=PSN|ANM and determinerCode=INSTANCE]/multipleBirthInd, player[classCode=PSN|ANM and determinerCode=INSTANCE]/multipleBirthOrderNumber
photo | player[classCode=PSN|ANM and determinerCode=INSTANCE]/desc
contact | player[classCode=PSN|ANM and determinerCode=INSTANCE]/scopedRole[classCode=CON]
relationship | code
name | name
telecom | telecom
address | addr
gender | player[classCode=PSN|ANM and determinerCode=INSTANCE]/administrativeGender
organization | scoper
period | effectiveTime
communication | LanguageCommunication
language | player[classCode=PSN|ANM and determinerCode=INSTANCE]/languageCommunication/code
preferred | preferenceInd
generalPractitioner | subjectOf.CareEvent.performer.AssignedEntity
managingOrganization | scoper
link | outboundLink
other | id
type | typeCode

8.1.16 Resource Patient - Extensions & Profiles

URL: https://hl7.org/fhir/R4/patient-profiles.html

8.1.16 Resource Patient - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Patient resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.1.16.1 Profiles

No Profiles defined for this resource

8.1.16.2 Extensions

patient-adoptionInfo | Patient | adoptionInfo | for Patient HL7 Extensions
patient-animal | Patient | animal | for Patient HL7 Extensions
patient-birthPlace | Patient | birthPlace | for Patient HL7 Extensions
patient-birthTime | Patient.birthDate | birthTime | for Patient HL7 Extensions
patient-cadavericDonor | Patient | cadavericDonor | for Patient HL7 Extensions
patient-citizenship | Patient | citizenship | for Patient HL7 Extensions
patient-congregation | Patient | congregation | for Patient HL7 Extensions
patient-disability | Patient | disability | for Patient HL7 Extensions
patient-genderIdentity | Patient | genderIdentity | for Patient HL7 Extensions
patient-importance | Patient | importance | for Patient HL7 Extensions
patient-interpreterRequired | Patient | interpreterRequired | for Patient HL7 Extensions
patient-mothersMaidenName | Patient | mothersMaidenName | for Patient HL7 Extensions
patient-nationality | Patient | nationality | for Patient HL7 Extensions
patient-preferenceType | Patient.communication.preferred | preferenceType | for Patient HL7 Extensions
patient-proficiency | Patient.communication | proficiency | for Patient HL7 Extensions
patient-relatedPerson | Patient.contact | relatedPerson | for Patient HL7 Extensions
patient-religion | Patient | religion | for Patient HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

consent-Transcriber | Transcriber | for Consent HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
cqf-initiatingPerson | initiatingPerson | for Clinical Reasoning Extensions
cqf-receivingPerson | receivingPerson | for Clinical Reasoning Extensions
familymemberhistory-patient-record | patient-record | for Family Member History HL7 Extensions
goal-acceptance | acceptance | for FHIR Core Goal Profile

procedure-directedBy | directedBy | for Procedure HL7 Extensions
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.1.16.3 Search Extensions

age | number | Searches for patients based on age as calculated based on current date and date of birth. Deceased patients are excluded from the search. Expression:
birthOrderBoolean | token | Search based on whether a patient was part of a multiple birth or not. Expression:
mothersMaidenName | string | Search based on patient's mother's maiden name Expression:
Patient.extension('http://hl7.org/fhir/StructureDefinition/patient-extensions-Patient-mothersMaidenName')

8.1.17 Resource Patient - Operations

URL: https://hl7.org/fhir/R4/patient-operations.html

8.1.17 Resource Patient - Operations

Patient Administration Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

This resource has 2 operations operations associated with it:

$match | Find patient matches using MPI based logic | Trial Use
$everything | Fetch Patient Record | Trial Use

For more information about operations, including how they are invoked, see Operations.

8.1.17.1 Operation $match on Patient

URL: https://hl7.org/fhir/R4/patient-operation-match.html

8.1.17.1 Operation $match on Patient

Patient Administration Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

A Master Patient Index (MPI) is a service used to manage patient identification in a context where multiple patient databases exist. Healthcare applications and middleware use the MPI to match patients between the databases, and to store new patient details as they are encountered. MPIs are highly specialized applications, often tailored extensively to the institution's particular mix of patients. MPIs can also be run on a regional and national basis.

To ask an MPI to match a patient, clients use the "$match" operation, which accepts a patient resource which may be only partially complete. The data provided is interpreted as an MPI input and processed by an algorithm of some kind that uses the data to determine the most appropriate matches in the patient set. Note that different MPI matching algorithms have different required inputs. The generic $match operation does not specify any particular algorithm, nor a minimum set of information that must be provided when asking for an MPI match operation to be performed, but many implementations will have a set of minimum information, which may be declared in their definition of the $match operation by specifying a profile on the resource parameter, indicating which properties are required in the search. The patient resource submitted to the operation does not have to be complete, nor does it need to pass validation (i.e. mandatory fields don't need to be populated), but it does have to be a valid instance, as it is used as the reference data to match against.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Patient-match
```

Formal Definition (as a OperationDefinition).

URL: [base]/Patient/$match

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
resource | 1..1 | Resource | | | Use this to provide an entire set of patient details for the MPI to match against (e.g. POST a patient record to Patient/$match).
onlyCertainMatches | 0..1 | boolean | | | If there are multiple potential matches, then the match should not return the results with this flag set to true. When false, the server may return multiple results with each result graded accordingly.
count | 0..1 | integer | | | The maximum number of records to return. If no value is provided, the server decides how many matches to return. Note that clients should be careful when using this, as it may prevent probable - and valid - matches from being returned

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | A bundle contain a set of Patient records that represent possible matches, optionally it may also contain an OperationOutcome with further information about the search results (such as warnings or information messages, such as a count of records that were close but eliminated) If the operation was unsuccessful, then an OperationOutcome may be returned along with a BadRequest status Code (e.g. security issue, or insufficient properties in patient fragment - check against profile) Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The response from an "mpi" query is a bundle containing patient records, ordered from most likely to least likely. If there are no patient matches, the MPI SHALL return an empty search set with no error, but may include an operation outcome with further advice regarding patient selection. All patient records SHALL have a search score from 0 to 1, where 1 is the most certain match, along with an extension "match-grade" that indicates the MPI's position on the match quality.

8.1.17.1.1 Examples

Request: Patient match using patient resource:

```
POST /open/Patient/$match
[some headers]

{
  "resourceType": "Parameters",
  "id": "example",
  "parameter": [
    {
      "name": "resource",
      "resource": {
        "resourceType": "Patient",
        "identifier": [
          {
            "use": "usual",
            "type": {
              "coding": [
                {
                  "system": "http://hl7.org/fhir/v2/0203",
                  "code": "MR"
                }
              ]
            },
            "system": "urn:oid:1.2.36.146.595.217.0.1",
            "value": "12345"
          }
        ],
        "name": [
          {
            "family": [
              "Chalmers"
            ],
            "given": [
              "Peter"
            ]
          }
        ],
        "gender": "male",
        "birthDate": "1974-12-25"
      }
    },
    {
      "name": "count",
      "valueInteger": "3"
    },
    {
      "name": "onlyCertainMatches",
      "valueBoolean": "false"
    }
  ]
}
```

Response: Results from imaginary MPI algorithm

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2016-03-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 2,
  "entry": [{
    "fullUrl": "http://server/path/Patient/example",
    "resource": {
      "resourceType": "Patient",
      "id": "example",
```

```
      .. snip ...
    },
    "search": {
      "extension": [{
        "url": "http://hl7.org/fhir/StructureDefinition/match-grade",
        "valueCode": "certain"
      }],
      "mode": "match",
      "score": 0.9
    }
  },{
    "fullUrl": "http://server/path/Patient/292",
    "resource": {
      "resourceType": "Patient",
      "id": "292",
      .. snip ...
    },
    "search": {
      "extension": [{
        "url": "http://hl7.org/fhir/StructureDefinition/match-grade",
        "valueCode": "possible"
      }],
      "mode": "match",
      "score": 0.2
    }
  }]
}
```

For more information about operations, including how they are invoked, see Operations.

8.1.17.2 Operation $everything on Patient

URL: https://hl7.org/fhir/R4/patient-operation-everything.html

8.1.17.2 Operation $everything on Patient

Patient Administration Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

This operation is used to return all the information related to one or more patients described in the resource or context on which this operation is invoked. The response is a bundle of type "searchset". At a minimum, the patient resource(s) itself is returned, along with any other resources that the server has that are related to the patient(s), and that are available for the given user. The server also returns whatever resources are needed to support the records - e.g. linked practitioners, medications, locations, organizations etc.

The intended use for this operation is to provide a patient with access to their entire record (e.g. "Blue Button"), or for provider or other user to perform a bulk data download. The server SHOULD return at least all resources that it has that are in the patient compartment for the identified patient(s), and any resource referenced from those, including binaries and attachments. In the US Realm, at a minimum, the resources returned SHALL include all the data covered by the meaningful use common data elements as defined in the US Core Implementation Guide. Other applicable implementation guides may make additional rules about how much information that is returned.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Patient-everything
```

Formal Definition (as a OperationDefinition).

URL: [base]/Patient/$everything

URL: [base]/Patient/[id]/$everything

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

start | 0..1 | date | | | The date range relates to care dates, not record currency dates - e.g. all records relating to care provided in a certain date range. If no start date is provided, all records prior to the end date are in scope.

end | 0..1 | date | | | The date range relates to care dates, not record currency dates - e.g. all records relating to care provided in a certain date range. If no end date is provided, all records subsequent to the start date are in scope.

_since | 0..1 | instant | | | Resources updated after this period will be included in the response. The intent of this parameter is to allow a client to request only records that have changed since the last request, based on either the return header time, or or (for asynchronous use), the transaction time

_type | 0..* | code | | | One or more parameters, each containing one or more comma-delimited FHIR resource types to include in the return resources. In the absence of any specified types, the server returns all resource types

_count | 0..1 | integer | | | See discussion below on the utility of paging through the results of the $everything operation

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The bundle type is "searchset" Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this operation and simply searching the patient compartment are:

• unless the client requests otherwise, the server returns the entire result set in a single bundle (rather than using paging)

• the server is responsible for determining what resources to return as included resources (rather than the client specifying which ones).

This frees the client from needing to determine what it could or should ask for, particularly with regard to included resources. Servers should consider returning appropriate Provenance and AuditTrail on the returned resources, even though these are not directly part of the patient compartment.

It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single patient, or determine whether the context has the rights to the nominated patient, if there is one, or can determine an appropriate list of patients to provide data for from the context of the request. If there is no nominated patient (GET /Patient/$everything) and the context is not associated with a single patient record, the actual list of patients is all patients that the user associated with the request has access to. This may be all patients in the family that the patient has access to, or it may be all patients that a care provider has access to, or all patients on the entire record system. In such cases, the server may choose to return an error rather than all the records. Specifying the relationship between the context, a user and patient records is outside the scope of this specification (though see The Smart App Launch Implementation Guide .

When this operation is used to access multiple patient records at once, the return bundle could be rather a lot of data; servers may choose to require that such requests are made asynchronously, and associated with bulk data formats. Alternatively, clients may choose to page through the result set (or servers may require this). Paging through the results is done the same as for Searching, using the _count parameter, and Bundle links. Implementers should note that paging will be slower than simply returning all the results at once (more network traffic, multiple latency delays) but may be required in order not to exhaust available memory reading or writing the whole response in a single package. Unlike searching, there is no inherent user-display order for the $everything operation. Servers might consider sorting the returned resources in descending order of last record update, but are not required to do so.

The _since parameter is provided to support periodic queries to get additional information that has changed about the patient since the last query. This means that the _since parameter is based on record time. The value of the _since parameter should be set to the time from the server. If using direct response, this is the timestamp in the response header. If using the async interface, this is the transaction timestamp in the json response. Servers should ensure that the timestamps a managed such that the client does not miss any changes. Clients should be able to handle getting the same response more than once in the case that the transaction falls on a time boundary. Clients should ensure that the other query parameters are constant to ensure a coherent set of records when doing periodic queries.

8.1.17.2.1 Examples

Request: Get all resources linked to a single patient

```
GET /open/Patient/f001/$everything
[some headers]
```

Response: Everything linked to the nominated patient (in this case, built automatically by the publication tooling)

```
HTTP/1.1 200 OK
[other headers]

<Bundle xml="http://hl7.org/fhir">
  <id value="2039212f-3fa5-406b-b4ff-d4a71de090c9"/>
  <type value="searchset"/>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Patient/f001"/>
    <resource>
     <Patient xmlns="http://hl7.org/fhir">
       <id value="f001"/>
       <identifier>
         <use value="usual"/>
         <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!--  BSN identification system  -->
         <value value="738472983"/>
       </identifier>
       <identifier>
         <use value="usual"/>
         <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!--  BSN identification system  -->
       </identifier>
       <active value="true"/>
       <name>
         <use value="usual"/>
         <family value="van de Heuvel"/>
         <given value="Pieter"/>
         <suffix value="MSc"/>
       </name>
       <telecom>
         <system value="phone"/>
         <value value="0648352638"/>
         <use value="mobile"/>
       </telecom>
       <telecom>
```

```
        <system value="email"/>
        <value value="p.heuvel@gmail.com"/>
        <use value="home"/>
      </telecom>
      <gender value="male"/>
      <birthDate value="1944-11-17"/>
      <deceasedBoolean value="false"/>
      <address>
        <use value="home"/>
        <line value="Van Egmondkade 23"/>
        <city value="Amsterdam"/>
        <postalCode value="1024 RJ"/>
        <country value="NLD"/>
      <!--  ISO 3166 Codes (Countries)  -->
      </address>
      <maritalStatus>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-MaritalStatus"/>
          <code value="M"/>
          <display value="Married"/>
        </coding>
        <text value="Getrouwd"/>
      </maritalStatus>
      <multipleBirthBoolean value="true"/>
      <contact>
        <relationship>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v2-0131"/>
            <code value="C"/>
          </coding>
        </relationship>
        <name>
          <use value="usual"/>
          <family value="Abels"/>
          <given value="Sarah"/>
        </name>
        <telecom>
          <system value="phone"/>
          <value value="0690383372"/>
          <use value="mobile"/>
        </telecom>
      </contact>
      <communication>
        <language>
          <coding>
            <system value="urn:ietf:bcp:47"/>
          <!--  IETF language tag  -->
            <code value="nl"/>
            <display value="Dutch"/>
          </coding>
          <text value="Nederlands"/>
        </language>
        <preferred value="true"/>
      </communication>
      <managingOrganization>
        <reference value="Organization/f001"/>
        <display value="Burgers University Medical Centre"/>
      </managingOrganization>
    </Patient>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f003"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f003"/><!--   urn:oid:2.16.840.1.113883.4.642.1.7   --><!--   2.16.840.1.113883.4.642.1.118
  --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6325"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="11557-6"/>
        <display value="Carbon dioxide in blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
```

```
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="6.2"/>
      <unit value="kPa"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="kPa"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="4.8"/>
        <unit value="kPa"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kPa"/>
      </low>
      <high>
        <value value="6.0"/>
        <unit value="kPa"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kPa"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f003"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="172960003"/>
               <display value="Incision of retropharyngeal abscess"/>
           </coding>
       </code>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
    <encounter>
           <reference value="Encounter/f003"/>
       </encounter>
    <performedPeriod>
           <start value="2013-03-24T09:30:10+01:00"/>
           <end value="2013-03-24T10:30:10+01:00"/>
       </performedPeriod>
    <performer>
           <function>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
           </function>
      <actor>
               <reference value="Practitioner/f001"/>
               <display value="E.M.J.M. van den broek"/>
           </actor>
       </performer>
       <reasonCode>
         <text value="abcess in retropharyngeal area"/>
       </reasonCode>
       <bodySite>

               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="83030008"/>
                   <display value="Retropharyngeal area"/>
               </coding>

       </bodySite>
       <outcome>
         <text value="removal of the retropharyngeal abscess"/>
       </outcome>
       <report>
           <reference value="DiagnosticReport/f001"/>
           <display value="Lab results blood test"/>
```

```
      </report>
      <followUp>
        <text value="described in care plan"/>
      </followUp>
   </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notTime"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notTime"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 6. Withhold or withdraw consent for disclosure of records related published during a
  timeframe
    </p><p>
       Patient "P. van de Heuvel" wishes to withhold disclosure of all data from a timeframe to any provider.
      </p>
     </div>
    </text>
    <status value="active"/>
    <scope>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
        <code value="patient-privacy"/>
      </coding>
      </scope>
        <category>
      <coding>
        <system value="http://loinc.org"/>
        <code value="59284-0"/>
      </coding>
    </category>
    <patient>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </patient>
    <dateTime value="2015-11-18"/>
    <!-- not bound by a timeframe - Consent.period -->
    <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <period>
         <start value="2015-01-01"/>
         <end value="2015-02-01"/>
       </period>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notOrg"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notOrg"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
         or withholding consent for that purpose. In other jurisdictions, where an express consent
         model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
         "withhold" or "withdraw" consent for. <p> specific to use-case 3) Withhold or withdraw consent
           for disclosure to a specific provider organization </p><p> Patient "P. van de Heuvel"
```

```
          ex-spouse is a Physician that is employed by Good Health Clinic. P. decides that he would
          like to withdraw/withhold consent to disclose any instance of his health information to Good
          Health Clinic. </p>
        </div>
      </text>

      <issued value="2015-11-18"/>
      <!-- not bound by a timeframe - Contract.applies -->

      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>

      <authority>
        <reference value="Organization/3"/>
        <display value="Michigan Health"/>
      </authority>

      <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
    Juristiction) -->
      <domain>
        <reference value="Location/ukp"/>
        <display value="UK Pharmacies"/>
      </domain>

      <type>
        <coding>
          <system value="http://loinc.org"/>
          <code value="57016-8"/>
        </coding>
      </type>

      <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
    -->
      <subType>
        <coding>
          <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
          <code value="Opt-In"/>
          <display value="Default Authorization with exceptions."/>
        </coding>
      </subType>

      <term>
        <type>
          <coding>
            <!-- made up code system -->
            <system value="http://example.org/fhir/consent-term-type-codes"/>
            <code value="withhold-from"/>
            <display value="Withhold all data from specified actor entity."/>
          </coding>
        </type>
        <offer>
          <topic>
            <reference value="Organization/2.16.840.1.113883.19.5"/>
            <display value="Good Health Clinic"/>
          </topic>
          <!-- by not specifying term.actor I assume this means everyone? -->
          <text value="Withhold this order and any results or related objects from any provider."/>
        </offer>
      </term>

      <!-- the terms of the consent in friendly consumer speak -->
      <friendly>
        <contentAttachment>
          <title value="The terms of the consent in friendly consumer speak."/>
          <!-- likely use url pointer to common text -->
        </contentAttachment>
      </friendly>

      <!-- the legal terms of the consent in lawyer speak -->
      <legal>
        <contentAttachment>
          <title value="The terms of the consent in lawyer speak."/>
          <!-- likely use url pointer to common text -->
          <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
    Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
        </contentAttachment>
      </legal>
    </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f003"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f003"/>
     <clinicalStatus>
       <coding>
```

```
        <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
        <code value="active"/>
      </coding>
    </clinicalStatus>

    <verificationStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
        <code value="confirmed"/>
      </coding>
    </verificationStatus>

    <category>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="439401001"/>
        <display value="diagnosis"/>
      </coding>
    </category>
    <severity>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="371923003"/>
        <display value="Mild to moderate"/>
      </coding>
    </severity>
    <code>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="18099001"/>
        <display value="Retropharyngeal abscess"/>
      </coding>
    </code>
    <bodySite>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="280193007"/>
        <display value="Entire retropharyngeal area"/>
      </coding>
    </bodySite>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <encounter>
      <reference value="Encounter/f003"/>
    </encounter>
    <onsetDateTime value="2012-02-27"/>
    <recordedDate value="2012-02-20"/>
    <asserter>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </asserter>
    <evidence>
      <code>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="169068008"/>
          <display value="CT of neck"/>
        </coding>
      </code>
    </evidence>
   </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-Emergency"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-Emergency"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <p>
    Withhold Authorization for Treatment rule for Emegency Treatment
    </p>
         <p>
      Patient "P. van de Heuvel" wishes to have no data at the Good Health Psychiatric Hospital
      available rule for Emergency treatment use.
      An overall Deny consent Directive, with an exception "Permit" of purposeOfUse "ETREAT"
      at "Good Health" hospital, and exception "Deny" all other purposeOfUse from good health hospital.
      </p>
       </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
```

```
        </coding>
            </scope>
     <category>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
           <code value="EMRGONLY"/>
         </coding>
       </category>
       <patient>
         <reference value="Patient/f001"/>
         <display value="P. van de Heuvel"/>
       </patient>
       <dateTime value="2015-11-18"/>
       <!-- not bound by a timeframe - Consent.period -->
       <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
    Consent.authority element -->
       <organization>
         <reference value="Organization/f001"/>
       </organization>
       <!-- the text terms of the consent in lawyer speak -->
       <sourceAttachment>
         <title value="The terms of the consent in lawyer speak."/>
         <!-- likely use url pointer to common text -->
       </sourceAttachment>
       <!--
         this is opt-out - e.g. nothing approved unless otherwise stated.
       -->
       <policyRule>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
           <code value="OPTOUT"/>
         </coding>
       </policyRule>
       <provision>
         <actor>
           <role>
             <coding>
               <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
               <code value="CST"/>
             </coding>
           </role>
           <reference>
             <reference value="Organization/f001"/>
           </reference>
         </actor>
         <purpose>
           <system value="http://terminology.hl7.org/CodeSystem/v3-ActReason"/>
           <code value="ETREAT"/>
         </purpose>
         <!-- this rule overrides the other one -->
         <provision>
           <type value="deny"/>
           <actor>
             <role>
               <coding>
                 <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
                 <code value="CST"/>
               </coding>
             </role>
             <reference>
               <reference value="Organization/f001"/>
             </reference>
           </actor>
         </provision>
       </provision>
     </Consent>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/Procedure/f001"/>
   <resource>
    <Procedure xmlns="http://hl7.org/fhir">
      <id value="f001"/>
        <status value="completed"/>
        <code>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="34068001"/>
                <display value="Heart valve replacement"/>
            </coding>
        </code>
     <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
     <encounter>
            <reference value="Encounter/f001"/>
        </encounter>
        <performedPeriod>
```

```
            <start value="2011-06-26"/>
            <end value="2011-06-27"/>
        </performedPeriod>
      <performer>
            <function>
                <coding>
                    <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                    <code value="01.000"/>
                    <display value="Arts"/>
                </coding>
                <text value="Care role"/>
            </function>
        <actor>
                <reference value="Practitioner/f002"/>
                <display value="P. Voigt"/>
            </actor>
        </performer>
      <reasonCode>
           <text value="Heart valve disorder"/>
        </reasonCode>
        <bodySite>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="17401000"/>
                    <display value="Heart valve structure"/>
                </coding>
        </bodySite>
        <outcome>
          <text value="improved blood circulation"/>
        </outcome>
        <report>
            <reference value="DiagnosticReport/f001"/>
            <display value="Lab results blood test"/>
        </report>
        <followUp>
          <text value="described in care plan"/>
        </followUp>
     </Procedure>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notThem"/>
    <resource>
     <Consent xmlns="http://hl7.org/fhir">
       <id value="consent-example-notThem"/>
       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
     are realized in existing systems in Canada.
      The default policy is one of implied consent for the provision of care, so these scenarios all deal with
    withdrawal or withholding consent for that purpose.
      In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
    phrase "consent to" rather
      than "withhold" or "withdraw" consent for.
         <p>
      specific to use-case 4) Withhold or withdraw consent for disclosure to a specific provider agent (an individual
    within an organization)
      </p><p>
         Patient "P. van de Heuvel" ex-spouse, Bill T Lookafter is a Nurse that is employed by Good Health Hospital.
        P. realizes that she may receive treatment at GHH, however she does not want her ex-spouse to have any
        access to her IIHI.  She indicates that she would like to withdraw/withhold consent to disclose any instance
        of her health information to her ex-spouse as a result of his employment at Good Health Clinic.
        </p>
        </div>
       </text>
       <status value="active"/>
       <scope>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
           <code value="patient-privacy"/>
         </coding>
         </scope>
           <category>
         <coding>
           <system value="http://loinc.org"/>
           <code value="59284-0"/>
         </coding>
       </category>
       <patient>
         <reference value="Patient/f001"/>
         <display value="P. van de Heuvel"/>
       </patient>
       <dateTime value="2015-11-18"/>
       <!-- not bound by a timeframe - Consent.period -->
       <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
    Consent.authority element -->
       <organization>
         <reference value="Organization/f001"/>
       </organization>
```

```
    <!-- the text terms of the consent in lawyer speak -->
    <sourceAttachment>
      <title value="The terms of the consent in lawyer speak."/>
      <!-- likely use url pointer to common text -->
    </sourceAttachment>
    <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
    <policyRule>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
        <code value="OPTIN"/>
      </coding>
    </policyRule>
    <provision>
      <!-- no read or write access for this individual -->
      <actor>
        <role>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
            <code value="PRCP"/>
            <!-- PRCP: recipient of the information -->
          </coding>
        </role>
        <reference>
          <reference value="Practitioner/f204"/>
          <display value="Fictive Nurse"/>
        </reference>
      </actor>
      <action>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
          <code value="access"/>
        </coding>
      </action>
      <action>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
          <code value="correct"/>
        </coding>
      </action>
    </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notAuthor"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notAuthor"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
         or withholding consent for that purpose. In other jurisdictions, where an express consent
         model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
         "withhold" or "withdraw" consent for. <p> specific to use-case 5) Withhold or withdraw consent
           for disclosure of records that were authored by a specific organization (or service delivery
           location). </p><p> Patient "P. van de Heuvel" wishes to have all of the PHI collected at the
           Good Health Psychiatric Hospital restricted from disclosure to every provider. </p>
       </div>
     </text>
     <issued value="2015-11-18"/>
     <!-- not bound by a timeframe - Contract.applies -->

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <authority>
       <reference value="Organization/3"/>
       <display value="Michigan Health"/>
     </authority>

     <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
     <domain>
       <reference value="Location/ukp"/>
       <display value="UK Pharmacies"/>
     </domain>

     <type>
       <coding>
         <system value="http://loinc.org"/>
         <code value="57016-8"/>
       </coding>
     </type>
```

```
      <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
    -->
      <subType>
        <coding>
          <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
          <code value="Opt-In"/>
          <display value="Default Authorization with exceptions."/>
        </coding>
      </subType>

      <term>
        <type>
          <coding>
            <!-- made up code system -->
            <system value="http://example.org/fhir/consent-term-type-codes"/>
            <code value="withhold-authored-by"/>
            <display value="Withhold all data authored by specified actor entity."/>
          </coding>
        </type>
        <offer>
          <topic>
            <reference value="Organization/2.16.840.1.113883.19.5"/>
            <display value="Good Health Clinic"/>
          </topic>

          <text value="Withhold all data authored by Good Health provider."/>
        </offer>
      </term>

      <!-- the terms of the consent in friendly consumer speak -->
      <friendly>
        <contentAttachment>
          <title value="The terms of the consent in friendly consumer speak."/>
          <!-- likely use url pointer to common text -->
        </contentAttachment>
      </friendly>

      <!-- the legal terms of the consent in lawyer speak -->
      <legal>
        <contentAttachment>
          <title value="The terms of the consent in lawyer speak."/>
          <!-- likely use url pointer to common text -->
          <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
    Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
        </contentAttachment>
      </legal>
     </Contract>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Consent/consent-example-grantor"/>
    <resource>
     <Consent xmlns="http://hl7.org/fhir">
      <id value="consent-example-grantor"/>
      <text>
        <status value="generated"/>
        <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on a question asked on John Moehrke's
     blog in the
      <a href="http://healthcaresecprivacy.blogspot.com/2016/04/consent-given-to-authorized.html">Consent given to
    authorized representative</a>
        <p>
     This is a case where a Patient is giving a limited (read-only) access to an individual.
      </p><p>
        Patient "P. van de Heuvel" wishes to provide XXX with read-only access to the data at Good Health Psychiatric
    Hospital
       available for patient requested access use.
       </p>
       </div>
      </text>
      <status value="active"/>
      <scope>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
          <code value="patient-privacy"/>
        </coding>
           </scope>
     <category>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
          <code value="INFAO"/>
        </coding>
      </category>
      <patient>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </patient>
      <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->
      <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
```

```
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-out - e.g. nothing approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTOUT"/>
       </coding>
     </policyRule>
     <provision>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
           <reference value="Organization/f001"/>
         </reference>
       </actor>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="PRCP"/>
           </coding>
         </role>
         <reference>
           <reference value="Patient/example"/>
           <display value="Good Health Clinic"/>
         </reference>
       </actor>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="access"/>
         </coding>
       </action>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/unsat"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
     <id value="unsat"/>
   <!--  urn:oid:2.16.840.1.113883.4.642.1.7  -->
   <!--  2.16.840.1.113883.4.642.1.118  -->
     <identifier>
       <use value="official"/>
       <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
       <value value="6323"/>
     </identifier>
     <status value="cancelled"/>
     <code>
       <coding>
         <system value="http://loinc.org"/>
         <code value="15074-8"/>
         <display value="Glucose [Moles/volume] in Blood"/>
       </coding>
     </code>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <effectivePeriod>
       <start value="2013-04-02T09:30:10+01:00"/>
       <end value="2013-04-05T09:30:10+01:00"/>
     </effectivePeriod>
     <issued value="2013-04-03T15:30:10+01:00"/>
     <performer>
       <reference value="Practitioner/f005"/>
       <display value="A. Langeveld"/>
     </performer>
     <dataAbsentReason>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="125154007"/>
         <display value="Specimen unsatisfactory for evaluation"/>
       </coding>
```

```
      </dataAbsentReason>
      <note><text value="Tube broken in transit and sample leaked"/></note>
      <referenceRange>
        <low>
          <value value="3.1"/>
          <unit value="mmol/l"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mmol/L"/>
        </low>
        <high>
          <value value="6.2"/>
          <unit value="mmol/l"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mmol/L"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-basic"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
    <id value="consent-example-basic"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">
        <p>
    Authorize Normal access for Treatment
        </p>
        <p>
       Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
      available for normal treatment use.
        </p>
      </div>
    </text>
    <status value="active"/>
    <scope>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
        <code value="patient-privacy"/>
      </coding>
    </scope>
    <category>
      <coding>
        <system value="http://loinc.org"/>
        <code value="59284-0"/>
      </coding>
    </category>
    <patient>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </patient>
    <dateTime value="2016-05-11"/>
    <organization>
      <reference value="Organization/f001"/>
    </organization>
    <!-- the terms of the consent as signed by the patient in ink -->
    <sourceAttachment>
      <title value="The terms of the consent in lawyer speak."/>
      <!-- likely use url pointer to common text -->
    </sourceAttachment>
    <!-- this is plain out opt-in with no exceptions -->
    <policyRule>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
        <code value="OPTIN"/>
      </coding>
    </policyRule>
    <provision>
      <!-- bound by a timeframe between 1964 and 2016 - consent.applies -->
      <period>
        <start value="1964-01-01"/>
        <end value="2016-01-01"/>
      </period>
    </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-Out"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-Out"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
```

```
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    Specific to use-case 0) Withhold Authorization of sharing data for Treatment
    </p><p>
       Patient "P. van de Heuvel" wishes to have no data at the Good Health Psychiatric Hospital
      available for normal treatment use.
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-out - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTOUT"/>
       </coding>
     </policyRule>
     <provision>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
           <reference value="Organization/f001"/>
         </reference>
       </actor>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f003"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f001"/>
            <display value="E.M. van den broek"/>
            </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <lifecycleStatus value="completed"/>
      <achievementStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/goal-achievement"/>
        <code value="achieved"/>
        <display value="Achieved"/>
```

```
      </coding>
      <text value="Achieved"/>
      </achievementStatus>
         <description>
           <text value="Retropharyngeal abscess removal"/>
         </description>
         <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished without complications"/>
         </note>
       </Goal>
     </contained>
     <identifier>
       <use value="official"/>
       <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
       <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
       <value value="CP3953"/>
     </identifier>
     <status value="completed"/>
     <intent value="plan"/>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <period>
       <start value="2013-03-08T09:00:10+01:00"/>
       <end value="2013-03-08T09:30:10+01:00"/>
     </period>
     <careTeam>
       <reference value="#careteam"/>
     </careTeam>
     <addresses>
       <reference value="Condition/f201"/>
       <!--TODO Correcte referentie-->
       <display value="?????"/>
     </addresses>
     <goal>
       <reference value="#goal"/>
     </goal>
     <activity>
       <detail>
         <kind value="ServiceRequest"/>
         <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
         <code>
           <coding>
             <system value="http://snomed.info/sct"/>
             <code value="172960003"/>
             <display value="Incision of retropharyngeal abscess"/>
           </coding>
         </code>
         <status value="completed"/>
         <doNotPerform value="true"/>
         <scheduledString value="2011-06-27T09:30:10+01:00"/>
         <performer>
           <reference value="Practitioner/f001"/>
           <display value="E.M. van den broek"/>
         </performer>
       </detail>
     </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notAuthor"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notAuthor"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 5) Withhold or withdraw consent for disclosure of records that were authored by a specific
  organization (or service delivery location).
    </p><p>
       Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
      restricted from disclosure to every provider.
      </p>
      </div>
     </text>
     <status value="active"/>
```

```
      <scope>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
          <code value="patient-privacy"/>
        </coding>
        </scope>
        <category>
        <coding>
          <system value="http://loinc.org"/>
          <code value="59284-0"/>
        </coding>
      </category>
      <patient>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </patient>
      <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->
      <organization>
        <reference value="Organization/f001"/>
      </organization>
      <!-- the text terms of the consent in lawyer speak -->
      <sourceAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
      </sourceAttachment>
      <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
      <policyRule>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
          <code value="OPTIN"/>
        </coding>
      </policyRule>
      <provision>
        <actor>
          <role>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
              <code value="CST"/>
            </coding>
          </role>
          <reference>
            <reference value="Organization/f001"/>
          </reference>
        </actor>
      </provision>
    </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f002"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
     <id value="f002"/>
       <identifier>
           <use value="official"/>
           <system value="http://www.bmc.nl/zorgportal/identifiers/encounters"/>
           <value value="v3251"/>
       </identifier>
       <status value="finished"/>
       <class>
           <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
           <code value="AMB"/> <!-- outpatient -->
           <display value="ambulatory"/>
       </class>
       <type>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="270427003"/>
               <display value="Patient-initiated encounter"/>
           </coding>
       </type>
       <priority>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="103391001"/>
               <display value="Urgent"/>
           </coding>
       </priority>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
       <participant>
           <individual>
               <reference value="Practitioner/f003"/>
               <display value="M.I.M Versteegh"/>
           </individual>
       </participant>
```

```
        <length>
            <value value="140"/>
            <unit value="min"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="min"/>
        </length>
        <reasonCode>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="34068001"/>
                <display value="Partial lobectomy of lung"/>
            </coding>
        </reasonCode>
        <hospitalization>
            <preAdmissionIdentifier>
                <use value="official"/>
                <system value="http://www.bmc.nl/zorgportal/identifiers/pre-admissions"/>
                <value value="98682"/>
            </preAdmissionIdentifier>
        <!--    <preAdmissionTest>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="399208008"/>
                    <display value="Chest X-ray"/>
                </coding>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="396550006"/>
                    <display value="Blood test"/>
                </coding>
            </preAdmissionTest> -->
            <admitSource>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="305997006"/>
                    <display value="Referral by radiologist"/>
                </coding>
            </admitSource>

            <dischargeDisposition>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="306689006"/>
                    <display value="Discharge to home"/>
                </coding>
            </dischargeDisposition>
        </hospitalization>
        <serviceProvider>
            <reference value="Organization/f001"/>
            <display value="BMC"/>
        </serviceProvider>
    </Encounter>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f002"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f002"/>
        <status value="completed"/>
        <code>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="359615001"/>
                <display value="Partial lobectomy of lung"/>
            </coding>
        </code>
     <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
     <encounter>
            <reference value="Encounter/f002"/>
        </encounter>
        <performedPeriod>
            <start value="2013-03-08T09:00:10+01:00"/>
            <end value="2013-03-08T09:30:10+01:00"/>
        </performedPeriod>
     <performer>
            <function>
                <coding>
                    <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                    <code value="01.000"/>
                    <display value="Arts"/>
                </coding>
                <text value="Care role"/>
            </function>
       <actor>
                <reference value="Practitioner/f003"/>
                <display value="M.I.M. Versteegh"/>
```

```
            </actor>
        </performer>
     <reasonCode>
          <text value="Malignant tumor of lung"/>
        </reasonCode>
        <bodySite>

                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="39607008"/>
                    <display value="Lung structure"/>
                </coding>

        </bodySite>
        <outcome>
          <text value="improved blood circulation"/>
        </outcome>
        <report>
            <reference value="DiagnosticReport/f001"/>
            <display value="Lab results blood test"/>
        </report>
        <followUp>
          <text value="described in care plan"/>
        </followUp>
    </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f004"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f004"/>
        <status value="completed"/>
        <code>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="48387007"/>
                <display value="Tracheotomy"/>
            </coding>
        </code>
     <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
     <encounter>
            <reference value="Encounter/f003"/>
        </encounter>
        <performedPeriod>
            <start value="2013-03-22T09:30:10+01:00"/>
            <end value="2013-03-22T10:30:10+01:00"/>
        </performedPeriod>
        <performer>
            <function>
                <coding>
                    <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                    <code value="01.000"/>
                    <display value="Arts"/>
                </coding>
                <text value="Care role"/>
            </function>
       <actor>
                <reference value="Practitioner/f005"/>
                <display value="A. Langeveld"/>
            </actor>
        </performer>
        <reasonCode>
          <text value="ensure breathing during surgery"/>
        </reasonCode>
        <bodySite>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="83030008"/>
                <display value="Retropharyngeal area"/>
            </coding>
        </bodySite>
        <outcome>
          <text value="removal of the retropharyngeal abscess"/>
        </outcome>
        <report>
            <reference value="DiagnosticReport/f001"/>
            <display value="???????????"/>
        </report>
        <followUp>
          <text value="described in care plan"/>
        </followUp>
    </Procedure>
  </resource>
</entry>
<entry>
```

```
<fullUrl value="http://hl7.org/fhir/List/long"/>
<resource>
 <List xmlns="http://hl7.org/fhir">
   <id value="long"/>
   <status value="current"/>
   <mode value="changes"/>
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</entry>
<entry>
  <item>
    <reference value="Patient/animal"/>
  </item>
</entry>
<entry>
  <item>
    <reference value="Patient/dicom"/>
  </item>
</entry>
<entry>
  <item>
    <reference value="Patient/ihe-pcd"/>
  </item>
</entry>
<entry>
  <item>
    <reference value="Patient/f001"/>
  </item>
```

```
      </entry>
      <entry>
        <item>
          <reference value="Patient/f201"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/glossy"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/proband"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/genetics-example1"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/ch-example"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/newborn"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/mom"/>
        </item>
      </entry>
    </List>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f001"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f001"/>
     <clinicalStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
         <code value="active"/>
       </coding>
     </clinicalStatus>

     <verificationStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
         <code value="confirmed"/>
       </coding>
     </verificationStatus>
     <category>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="439401001"/>
         <display value="diagnosis"/>
       </coding>
     </category>
     <severity>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="6736007"/>
         <display value="Moderate"/>
       </coding>
     </severity>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="368009"/>
         <display value="Heart valve disorder"/>
       </coding>
     </code>
     <bodySite>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="40768004"/>
         <display value="Left thorax"/>
       </coding>
       <text value="heart structure"/>
     </bodySite>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
```

```
      </subject>
      <encounter>
        <reference value="Encounter/f001"/>
      </encounter>
      <onsetDateTime value="2011-08-05"/>
      <recordedDate value="2011-10-05"/>
      <asserter>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </asserter>
      <evidence>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="426396005"/>
            <display value="Cardiac chest pain"/>
          </coding>
        </code>
      </evidence>
    </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f001"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f001"/><!--  urn:oid:2.16.840.1.113883.4.642.1.7  --><!--  2.16.840.1.113883.4.642.1.118
  --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6323"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="15074-8"/>
        <display value="Glucose [Moles/volume] in Blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T09:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="6.3"/>
      <unit value="mmol/l"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mmol/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="3.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="6.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notOrg"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notOrg"/>
     <text>
       <status value="generated"/>
```

```
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 3) Withhold or withdraw consent for disclosure to a specific provider organization
    </p><p>
       Patient "P. van de Heuvel" ex-spouse is a Physician that is employed by Good Health Clinic.  P. decides
      that he would like to withdraw/withhold consent to disclose any instance of his health information
      to Good Health Clinic.
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <!-- no read or write access for this org -->
       <type value="deny"/>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="PRCP"/>
             <!-- PRCP: recipient of the information -->
           </coding>
         </role>
         <reference>
           <reference value="Organization/f001"/>
         </reference>
       </actor>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="access"/>
         </coding>
       </action>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="correct"/>
         </coding>
       </action>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RelatedPerson/f001"/>
  <resource>
   <RelatedPerson xmlns="http://hl7.org/fhir">
    <id value="f001"/>
     <text>
```

```
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">
       Sarah Abels
      </div>
    </text>

    <identifier>
      <use value="official"/>
      <type><text value="BSN"/></type>
      <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
      <!-- Dutch BSN identification system -->
    </identifier>

    <patient>
      <reference value="Patient/f001"/>
    </patient>

    <relationship>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
        <code value="SIGOTHR"/>
      </coding>
    </relationship>

    <name>
      <use value="usual"/>
      <family value="Abels"/>
      <given value="Sarah"/>
    </name>
    <telecom>
      <system value="phone"/>
      <value value="0690383372"/>
      <use value="mobile"/>
    </telecom>
    <telecom>
      <system value="email"/>
      <value value="s.abels@kpn.nl"/>
      <use value="home"/>
    </telecom>
    <gender value="female"/>

   </RelatedPerson>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f002"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f002"/>
     <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f003"/>
            <display value="M.I.M. Versteegh"/>
            </member>
     </participant>
      </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <lifecycleStatus value="completed"/>
      <achievementStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/goal-achievement"/>
        <code value="achieved"/>
        <display value="Achieved"/>
      </coding>
      <text value="Achieved"/>
      </achievementStatus>
         <description>
           <text value="succesful surgery on lung of patient"/>
         </description>
         <subject>
    <reference value="Patient/f001"/>
    <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished with minor complications"/>
         </note>
       </Goal>
     </contained>
     <identifier>
       <use value="official"/>
       <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
       <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
       <value value="CP2934"/>
```

```
      </identifier>
      <status value="completed"/>
      <intent value="plan"/>
      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>
      <period>
        <start value="2011-07-06"/>
        <end value="2013-07-07"/>
      </period>
      <careTeam>
        <reference value="#careteam"/>
      </careTeam>
      <addresses>
        <reference value="Condition/f201"/>
        <!--TODO Correcte referentie-->
        <display value="?????"/>
      </addresses>
      <goal>
        <reference value="#goal"/>
      </goal>
      <activity>
        <detail>
          <kind value="ServiceRequest"/>
          <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
          <code>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="359615001"/>
              <display value="Partial lobectomy of lung"/>
            </coding>
          </code>
          <status value="completed"/>
          <doNotPerform value="true"/>
          <scheduledString value="2011-07-07T09:30:10+01:00"/>
          <performer>
            <reference value="Practitioner/f003"/>
            <display value="M.I.M. Versteegh"/>
          </performer>
        </detail>
      </activity>
    </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example3"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
     <id value="example3"/>

     <status value="draft"/>
     <intent value="order"/>
     <code>
       <text value="Refill Request"/>
     </code>
     <focus>
       <reference value="MedicationRequest/medrx002"/>
     </focus>
     <for>
       <reference value="Patient/f001"/>
     </for>
     <authoredOn value="2016-03-10T22:39:32-04:00"/>
     <lastModified value="2016-03-10T22:39:32-04:00"/>
     <requester>
         <reference value="Patient/example"/>
     </requester>
     <owner>
       <reference value="Practitioner/example"/>
     </owner>
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f001"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f001"/>
     <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f002"/>
            <display value="P. Voigt"/>
               </member>
      </participant>
       </CareTeam>
     </contained>
```

```
      <contained>
        <Goal>
          <id value="goal"/>
          <lifecycleStatus value="completed"/>
       <achievementStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/goal-achievement"/>
         <code value="achieved"/>
         <display value="Achieved"/>
       </coding>
       <text value="Achieved"/>
       </achievementStatus>
          <description>
            <text value="recovery surgery on heart of patient"/>
          </description>
          <subject>
     <reference value="Patient/f001"/>
     <display value="P. van de Heuvel"/>
          </subject>
          <note>
            <text value="goal accomplished without complications"/>
          </note>
        </Goal>
      </contained>
      <identifier>
        <use value="official"/>
        <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
        <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
        <value value="CP2903"/>
      </identifier>
      <status value="completed"/>
      <intent value="plan"/>
      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>
      <period>
        <start value="2011-06-26"/>
        <end value="2011-06-27"/>
      </period>
      <careTeam>
        <reference value="#careteam"/>
      </careTeam>
      <addresses>
        <reference value="Condition/f201"/>
        <!--TODO Correcte referentie-->
        <display value="?????"/>
      </addresses>
      <goal>
        <reference value="#goal"/>
      </goal>
      <activity>
        <detail>
          <kind value="ServiceRequest"/>
          <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
          <code>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="64915003"/>
              <display value="Operation on heart"/>
            </coding>
          </code>
          <status value="completed"/>
          <doNotPerform value="true"/>
          <scheduledString value="2011-06-27T09:30:10+01:00"/>
          <performer>
            <reference value="Practitioner/f002"/>
            <display value="P. Voigt"/>
          </performer>
        </detail>
       </activity>
    </CarePlan>
   </resource>
</entry>
<entry>
   <fullUrl value="http://hl7.org/fhir/Observation/f002"/>
   <resource>
    <Observation xmlns="http://hl7.org/fhir">
     <id value="f002"/><!--  urn:oid:2.16.840.1.113883.4.642.1.7  --><!--  2.16.840.1.113883.4.642.1.118
   --><identifier>
        <use value="official"/>
        <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
        <value value="6324"/>
      </identifier>
      <status value="final"/>
      <code>
        <coding>
          <system value="http://loinc.org"/>
          <code value="11555-0"/>
```

```
        <display value="Base excess in Blood by calculation"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="12.6"/>
      <unit value="mmol/l"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mmol/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="7.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="11.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notThis"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notThis"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
         or withholding consent for that purpose. In other jurisdictions, where an express consent
         model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
         "withhold" or "withdraw" consent for. <p> specific to use-case 2. Withhold or withdraw consent
           for disclosure of a specific record (e.g. Lab Order/Result) </p><p> Patient "P. van de
           Heuvel" Primary Care Provider, Dr. Philip Primary, has ordered a set of lab test which Adam
           wishes to keep as private as possible. At the sample collection facility, he indicates that
           he would like withhold consent to disclose the order and all results associated with that
           specific order from all other providers </p>
       </div>
     </text>

     <issued value="2015-11-18"/>
     <!-- not bound by a timeframe - Contract.applies -->

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <authority>
       <reference value="Organization/3"/>
       <display value="Michigan Health"/>
     </authority>

     <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
     <domain>
       <reference value="Location/ukp"/>
       <display value="UK Pharmacies"/>
     </domain>

     <type>
```

```
      <coding>
        <system value="http://loinc.org"/>
        <code value="57016-8"/>
      </coding>
    </type>

    <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
    <subType>
      <coding>
        <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
        <code value="Opt-In"/>
        <display value="Default Authorization with exceptions."/>
      </coding>
    </subType>

    <!-- I can easily say to exclude a type of FHIR Resource, but lab-orders is not a direct Resource type. Might we
  put in FHIR based query parameters? -->
    <term>
      <identifier>
        <system value="http://example.org/fhir/term-items"/>
        <value value="3347689"/>
      </identifier>
      <issued value="2015-11-01"/>
      <applies>
        <start value="2015-11-18"/>
      </applies>
      <type>
        <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-identified-object-and-related"/>
          <display value="Withhold the identified object and any other resources that are related to this object."/>
        </coding>
      </type>
      <offer>
        <topic>
          <reference value="ServiceRequest/lipid"/>
        </topic>
        <!-- by not specifying term.actor I assume this means everyone? -->
        <text value="Withhold this order and any results or related objects from any provider."/>
      </offer>
    </term>

    <!-- the terms of the consent in friendly consumer speak -->
    <friendly>
      <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
    </friendly>

    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notLabs"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
    <id value="pcd-example-notLabs"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
        jurisdictional policy and are realized in existing systems in Canada. The default policy is
        one of implied consent for the provision of care, so these scenarios all deal with withdrawal
        or withholding consent for that purpose. In other jurisdictions, where an express consent
        model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
        "withhold" or "withdraw" consent for. <p> specific to use-case 1. Withhold or withdraw consent
          for disclosure of records related to specific domain (e.g. DI, LAB, etc.) </p><p> Patient
          "P. van de Heuvel" wishes to withhold disclosure of all of her lab orders and results to any
          provider. </p>
      </div>
    </text>

    <issued value="2014-08-17"/>
    <!-- not bound by a timeframe - Contract.applies -->

    <subject>
      <reference value="Patient/f001"/>
```

```
    <display value="P. van de Heuvel"/>
  </subject>

  <authority>
    <reference value="Organization/3"/>
    <display value="Michigan Health"/>
  </authority>

  <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
Juristiction) -->
  <domain>
    <reference value="Location/ukp"/>
    <display value="UK Pharmacies"/>
  </domain>

  <type>
    <coding>
      <system value="http://loinc.org"/>
      <code value="57016-8"/>
    </coding>
  </type>

  <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
-->
  <subType>
    <coding>
      <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
      <code value="Opt-In"/>
      <display value="Default Authorization with exceptions."/>
    </coding>
  </subType>

  <!-- I can easily say to exclude a type of FHIR Resource, but lab-orders is not a direct Resource type. Might we
put in FHIR based query parameters? -->
  <term>
    <offer>
      <text value="sample"/>
    </offer>
    <group>
      <type>
        <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-object-type"/>
        </coding>
      </type>
      <subType>
        <coding>
          <system value="http://hl7.org/fhir/resource-types"/>
          <code value="ServiceRequest"/>
        </coding>
      </subType>
      <!-- by not specifying term.actor I assume this means everyone? -->
      <offer>
        <text value="Withhold orders from any provider."/>
      </offer>
    </group>
    <group>
      <type>
        <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-object-type"/>
        </coding>
      </type>
      <subType>
        <coding>
          <system value="http://hl7.org/fhir/resource-types"/>
          <code value="DiagnosticReport"/>
        </coding>
      </subType>
      <!-- by not specifying term.actor I assume this means everyone? -->
      <offer>
        <text value="Withhold order results from any provider."/>
      </offer>
    </group>
  </term>

  <!-- the terms of the consent in friendly consumer speak -->
  <friendly>
    <contentAttachment>
      <title value="The terms of the consent in friendly consumer speak."/>
      <!-- likely use url pointer to common text -->
    </contentAttachment>
  </friendly>

  <!-- the legal terms of the consent in lawyer speak -->
  <legal>
    <contentAttachment>
```

```
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f003"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
    <id value="f003"/>
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/encounters"/>
      <value value="v6751"/>
    </identifier>
    <status value="finished"/>
    <class>
      <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
      <code value="AMB"/>
      <!-- outpatient -->
      <display value="ambulatory"/>
    </class>
    <type>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="270427003"/>
        <display value="Patient-initiated encounter"/>
      </coding>
    </type>
    <priority>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="103391001"/>
        <display value="Non-urgent ear, nose and throat admission"/>
      </coding>
    </priority>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <participant>
      <individual>
        <reference value="Practitioner/f001"/>
        <display value="E.M. van den Broek"/>
      </individual>
    </participant>
    <length>
      <value value="90"/>
      <unit value="min"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="min"/>
    </length>
    <reasonCode>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="18099001"/>
        <display value="Retropharyngeal abscess"/>
      </coding>
    </reasonCode>
    <hospitalization>
      <preAdmissionIdentifier>
        <use value="official"/>
        <system value="http://www.bmc.nl/zorgportal/identifiers/pre-admissions"/>
        <value value="93042"/>
      </preAdmissionIdentifier>
      <!--     <preAdmissionTest>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="168719007"/>
                   <display value="Neck soft tissue X-ray"/>
               </coding>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="396550006"/>
                   <display value="Blood test"/>
               </coding>
           </preAdmissionTest> -->
      <admitSource>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="305956004"/>
          <display value="Referral by physician"/>
        </coding>
      </admitSource>
      <dischargeDisposition>
```

```
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="306689006"/>
          <display value="Discharge to home"/>
        </coding>
      </dischargeDisposition>
    </hospitalization>
    <serviceProvider>
      <reference value="Organization/f001"/>
    </serviceProvider>
   </Encounter>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notThis"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notThis"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 2. Withhold or withdraw consent for disclosure of a specific record (e.g. Lab Order/Result)
    </p><p>
       Patient "P. van de Heuvel" Primary Care Provider, Dr. Philip Primary, has ordered a set of lab test which Adam
      wishes to keep as private as possible.  At the sample collection facility, he indicates that he would
      like withhold consent to disclose the order and all results associated with that specific order from
      all other providers
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <data>
         <meaning value="related"/>
         <reference>
           <reference value="Task/example3"/>
         </reference>
       </data>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notThem"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notThem"/>
```

```
   <text>
     <status value="generated"/>
     <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
       jurisdictional policy and are realized in existing systems in Canada. The default policy is
       one of implied consent for the provision of care, so these scenarios all deal with withdrawal
       or withholding consent for that purpose. In other jurisdictions, where an express consent
       model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
       "withhold" or "withdraw" consent for. <p> specific to use-case 4) Withhold or withdraw consent
         for disclosure to a specific provider agent (an individual within an organization) </p><p>
         Patient "P. van de Heuvel" ex-spouse, Bill T Lookafter is a Nurse that is employed by Good
         Health Hospital. P. realizes that she may receive treatment at GHH, however she does not
         want her ex-spouse to have any access to her IIHI. She indicates that she would like to
         withdraw/withhold consent to disclose any instance of her health information to her
         ex-spouse as a result of his employment at Good Health Clinic. </p>
     </div>
   </text>

   <issued value="2015-11-18"/>

   <subject>
     <reference value="Patient/f001"/>
     <display value="P. van de Heuvel"/>
   </subject>

   <authority>
     <reference value="Organization/3"/>
     <display value="Michigan Health"/>
   </authority>

   <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
Juristiction) -->
   <domain>
     <reference value="Location/ukp"/>
     <display value="UK Pharmacies"/>
   </domain>

   <type>
     <coding>
       <system value="http://loinc.org"/>
       <code value="57016-8"/>
     </coding>
   </type>

   <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
-->
   <subType>
     <coding>
       <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
       <code value="Opt-In"/>
       <display value="Default Authorization with exceptions."/>
     </coding>
   </subType>

   <term>
     <type>
       <coding>
         <!-- made up code system -->
         <system value="http://example.org/fhir/consent-term-type-codes"/>
         <code value="withhold-from"/>
         <display value="Withhold all data from specified actor entity."/>
       </coding>
     </type>
     <offer>
       <topic>
         <reference value="Practitioner/f204"/>
         <display value="Fictive Nurse"/>
       </topic>

       <text value="Withhold this order and any results or related objects from specified nurse provider."/>
     </offer>
   </term>

   <signer>
     <type>
       <system value="http://terminology.hl7.org/CodeSystem/contractsignertypecodes"/>
       <code value="COVPTY"/>
     </type>
     <party>
       <reference value="Patient/f001"/>
     </party>
     <signature>
       <type>
         <system value="urn:iso-astm:E1762-95:2013"/>
         <code value="1.2.840.10065.1.12.1.1"/>
       </type>
       <when value="2013-06-08T10:57:34-07:00"/>
       <who>
         <reference value="Patient/f001"/>
       </who>
```

```
      </signature>
    </signer>

    <!-- the terms of the consent in friendly consumer speak -->
    <friendly>
      <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
    </friendly>

    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
    <!-- not bound by a timeframe - Contract.applies -->
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f004"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f004"/><!--   urn:oid:2.16.840.1.113883.4.642.1.7   --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6326"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="789-8"/>
        <display value="Erythrocytes [#/volume] in Blood by Automated count"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="4.12"/><!--  Actually, 4.12E+12  --><unit value="10^12/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*12/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
    </interpretation><!-- this example demonstrates using the reference range text element for stuff that is not
  easily structured !--><referenceRange>
      <text value=" 12-14 y Male: 4.4 - 5.2  x  10^12/L ; 12-14 y Female: 4.2 - 4.8  x  10^12/L ; 15-17 y Male: 4.6 -
  5.4  x  10^12/L ; 15-17 y Female: 4.2 - 4.8  x  10^12/L ; 18-64 y Male: 4.6 - 5.4  x  10^12/L ; 18-64 y Female: 4.0
  - 4.8  x  10^12/L ; 65-74 y Male: 4.3 - 5.3  x  10^12/L ; 65-74 y Female: 4.1 - 4.9  x  10^12/L       "/>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f002"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
    <id value="f002"/>
    <clinicalStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
        <code value="active"/>
      </coding>
    </clinicalStatus>

    <verificationStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
```

```
        <code value="confirmed"/>
      </coding>
    </verificationStatus>
    <category>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="439401001"/>
        <display value="diagnosis"/>
      </coding>
    </category>
    <severity>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="24484000"/>
        <display value="Severe"/>
      </coding>
    </severity>
    <code>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="254637007"/>
        <display value="NSCLC - Non-small cell lung cancer"/>
      </coding>
    </code>
    <bodySite>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="51185008"/>
        <display value="Thorax"/>
      </coding>
    </bodySite>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <encounter>
      <reference value="Encounter/f002"/>
    </encounter>
    <onsetDateTime value="2011-05-05"/>
    <recordedDate value="2012-06-03"/>
    <asserter>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </asserter>
    <stage>
      <summary>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="258219007"/>
          <display value="stage II"/>
        </coding>
      </summary>
   <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="260998006"/>
          <display value="Clinical staging (qualifier value)"/>
        </coding>
      </type>
    </stage>
    <evidence>
      <code>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="169069000"/>
          <display value="CT of thorax"/>
        </coding>
      </code>
    </evidence>
  </Condition>
 </resource>
</entry>
<entry>
 <fullUrl value="http://hl7.org/fhir/Observation/f005"/>
 <resource>
  <Observation xmlns="http://hl7.org/fhir">
   <id value="f005"/><!--  urn:oid:2.16.840.1.113883.4.642.1.7  --><identifier>
     <use value="official"/>
     <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
     <value value="6327"/>
   </identifier>
   <status value="final"/>
   <code>
     <coding>
       <system value="http://loinc.org"/>
       <code value="718-7"/>
       <display value="Hemoglobin [Mass/volume] in Blood"/>
     </coding>
   </code>
```

```
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-05T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-05T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="7.2"/>
      <unit value="g/dl"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="g/dL"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="7.5"/>
        <unit value="g/dl"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/dL"/>
      </low>
      <high>
        <value value="10"/>
        <unit value="g/dl"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/dL"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/ekg"/>
  <resource>
   <!-- Sampled ECG potentials (Leads I, II, II). There are 2 chunks (numbered as 0 and 1) for each lead. The voltage
  of each lead is near zero. --><Observation xmlns="http://hl7.org/fhir">
    <id value="ekg"/>
    <status value="final"/>
    <category>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
        <code value="procedure"/>
        <display value="Procedure"/>
      </coding>
    </category>
    <code>
      <coding>
        <system value="urn:oid:2.16.840.1.113883.6.24"/>
        <code value="131328"/>
        <display value="MDC_ECG_ELEC_POTL"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectiveDateTime value="2015-02-19T09:30:35+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <device>
      <display value="12 lead EKG Device Metric"/>
    </device>
    <!-- Lead I data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131329"/>
          <display value="MDC_ECG_ELEC_POTL_I"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
```

```
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
   2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
   2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
   2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
   2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
   2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
    <!-- Lead II data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131330"/>
          <display value="MDC_ECG_ELEC_POTL_II"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
   2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
   2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
   2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
   2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
   2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
    <!-- Lead III data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131389"/>
          <display value="MDC_ECG_ELEC_POTL_III"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
   2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
   2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
   2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
   2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
   2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f001"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
     <id value="f001"/>
       <identifier>
```

```
            <use value="official"/>
            <system value="http://www.amc.nl/zorgportal/identifiers/visits"/>
            <value value="v1451"/>
        </identifier>
        <status value="finished"/>
        <class>
            <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
            <code value="AMB"/> <!-- outpatient -->
            <display value="ambulatory"/>
        </class>
        <type>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="270427003"/>
                <display value="Patient-initiated encounter"/>
            </coding>
        </type>
        <priority>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="310361003"/>
                <display value="Non-urgent cardiological admission"/>
            </coding>
        </priority>
      <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
        <participant>
            <individual>
                <reference value="Practitioner/f002"/>
                <display value="P. Voigt"/>
            </individual>
        </participant>
        <length>
            <value value="140"/>
            <unit value="min"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="min"/>
        </length>
        <reasonCode>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="34068001"/>
                <display value="Heart valve replacement"/>
            </coding>
        </reasonCode>
        <hospitalization>
            <preAdmissionIdentifier>
                <use value="official"/>
                <system value="http://www.amc.nl/zorgportal/identifiers/pre-admissions"/>
                <value value="93042"/>
            </preAdmissionIdentifier>
       <!--      <preAdmissionTest>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="164847006"/>
                    <display value="Standard ECG"/>
                </coding>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="396550006"/>
                    <display value="Blood test"/>
                </coding>
            </preAdmissionTest> -->
            <admitSource>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="305956004"/>
                    <display value="Referral by physician"/>
                </coding>
            </admitSource>
            <dischargeDisposition>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="306689006"/>
                    <display value="Discharge to home"/>
                </coding>
            </dischargeDisposition>
        </hospitalization>
        <serviceProvider>
            <reference value="Organization/f001"/>
            <display value="Burgers University Medical Center"/>
        </serviceProvider>
    </Encounter>
   </resource>
  </entry>
</Bundle>
```

For more information about operations, including how they are invoked, see Operations.

8.1.18 Resource Patient - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/patient-version-maps.html

No validation errors - all conversions are clean

8.1.18 Resource Patient - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Patient.

Functional status for this map: 16 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.1.18.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Patient3to4" = "R3 to R4 Conversions for Patient"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Patient" alias PatientR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Patient" alias Patient as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Patient(source src : PatientR3, target tgt : Patient) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.deceased : boolean as vs -> tgt.deceased = create('boolean') as vt then boolean(vs, vt);
  src.deceased : dateTime as vs -> tgt.deceased = create('dateTime') as vt then dateTime(vs, vt);
  src.address -> tgt.address;
  src.maritalStatus -> tgt.maritalStatus;
  src.multipleBirth : boolean as vs -> tgt.multipleBirth = create('boolean') as vt then boolean(vs, vt);
  src.multipleBirth : integer as vs -> tgt.multipleBirth = create('integer') as vt then integer(vs, vt);
  src.photo -> tgt.photo;
  src.contact as vs0 -> tgt.contact as vt0 then contact(vs0, vt0);
  src.animal as vs0 -> tgt.extension as vt0 then animal(vs0, vt0);
  src.communication as vs0 -> tgt.communication as vt0 then communication(vs0, vt0);
  src.generalPractitioner -> tgt.generalPractitioner;
  src.managingOrganization -> tgt.managingOrganization;
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
}

group contact(source src, target tgt) extends BackboneElement {
  src.relationship -> tgt.relationship;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
  src.gender -> tgt.gender;
  src.organization -> tgt.organization;
  src.period -> tgt.period;
}

group animal(source src, target tgt) extends Element {
  src -> tgt.url = 'http://hl7.org/fhir/StructureDefinition/patient-animal' "animal";
  src.species as vs ->  tgt.extension as ext,  ext.url = 'species',  ext.value = vs;
  src.breed as vs ->  tgt.extension as ext,  ext.url = 'breed',  ext.value = vs;
  src.genderStatus as vs ->  tgt.extension as ext,  ext.url = 'genderStatus',  ext.value = vs;
}

group communication(source src, target tgt) extends BackboneElement {
  src.language -> tgt.language;
  src.preferred -> tgt.preferred;
}

group link(source src, target tgt) extends BackboneElement {
  src.other -> tgt.other;
  src.type -> tgt.type;
}
```

8.1.18.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Patient4to3" = "R4 to R3 Conversion for Patient"

uses "http://hl7.org/fhir/StructureDefinition/Patient" alias Patient as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Patient" alias PatientR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Patient(source src : Patient, target tgt : PatientR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.name -> tgt.name;
```

```
  src.telecom -> tgt.telecom;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.deceased : boolean as vs -> tgt.deceased = create('boolean') as vt then boolean(vs, vt);
  src.deceased : dateTime as vs -> tgt.deceased = create('dateTime') as vt then dateTime(vs, vt);
  src.address -> tgt.address;
  src.maritalStatus -> tgt.maritalStatus;
  src.multipleBirth : boolean as vs -> tgt.multipleBirth = create('boolean') as vt then boolean(vs, vt);
  src.multipleBirth : integer as vs -> tgt.multipleBirth = create('integer') as vt then integer(vs, vt);
  src.photo -> tgt.photo;
  src.contact as vs0 -> tgt.contact as vt0 then contact(vs0, vt0);
  src.extension as vs0 where url = 'http://hl7.org/fhir/StructureDefinition/patient-animal' -> tgt.animal as vt0 then
    animal(vs0, vt0) "animal";
  src.communication as vs0 -> tgt.communication as vt0 then communication(vs0, vt0);
  src.generalPractitioner -> tgt.generalPractitioner;
  src.managingOrganization -> tgt.managingOrganization;
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
}

group contact(source src, target tgt) extends BackboneElement {
  src.relationship -> tgt.relationship;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
  src.gender -> tgt.gender;
  src.organization -> tgt.organization;
  src.period -> tgt.period;
}

group animal(source src, target tgt) {
  src.extension as ext where url = 'species' then {
    ext.value as vs0 -> tgt.species = vs0 "species2";
  } "species";
  src.extension as ext where url = 'breed' then {
    ext.value as vs0 -> tgt.breed = vs0 "breed";
  } "breed";
  src.extension as ext where url = 'genderStatus' then {
    ext.value as vs0 -> tgt.genderStatus = vs0 "genderStatus";
  } "genderStatus";
}

group communication(source src, target tgt) extends BackboneElement {
  src.language -> tgt.language;
  src.preferred -> tgt.preferred;
}

group link(source src, target tgt) extends BackboneElement {
  src.other -> tgt.other;
  src.type -> tgt.type;
}
```

8.1.18.3 R4 Validation Errors

8.1.19 Patient HL7 Extensions

URL: https://hl7.org/fhir/R4/patient-extensions.html

Defines common extensions used with or related to the Patient resource

8.1.19 Patient HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.1.19.1 Content

Extensions:

patient-mothersMaidenName | mothersMaidenName : Mother's maiden (unmarried) name, commonly collected to help verify patient identity.

patient-birthPlace | birthPlace : The registered place of birth of the patient. A sytem may use the address.text if they don't store the birthPlace address in discrete elements.

patient-birthTime | birthTime : The time of day that the Patient was born. This includes the date to ensure that the timezone information can be communicated effectively.

patient-nationality | nationality : The nationality of the patient.

patient-citizenship | citizenship : The patient's legal status as citizen of a country.

patient-cadavericDonor | cadavericDonor : Flag indicating whether the patient authorized the donation of body parts after death.

patient-congregation | congregation : A group or place of religious practice that may provide services to the patient.

patient-adoptionInfo | adoptionInfo : Code indication the adoption status of the patient.

patient-disability | disability : Value(s) identifying physical or mental condition(s) that limits a person's movements, senses, or activities.

patient-importance | importance : The importance of the patient (e.g. VIP).

patient-interpreterRequired | interpreterRequired : This Patient requires an interpreter to communicate healthcare information to the practitioner.

patient-religion | religion : The patient's professed religious affiliations.

patient-relatedPerson | relatedPerson : In some cases a Patient.contact will also be populated as a RelatedPerson resource. This linkage permits the linkage between the 2 resources to be able to accurately indicate a representation of the same individual, and updating details between could be appropriate.
patient-genderIdentity | genderIdentity : The gender the patient identifies with. The Patient's gender identity is used as guidance (e.g. for staff) about how to interact with the patient.
patient-preferenceType | preferenceType : Indicates what mode of communication the patient prefers to use for the indicated language.
patient-animal | animal : This patient is known to be an animal.
patient-proficiency | proficiency : Proficiency level of the communication.

8.1.19.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
age | number | Searches for patients based on age as calculated based on current date and date of birth. Deceased patients are excluded from the search. | f:Patient/f:birthDate | XML / JSON
birthOrderBoolean | token | Search based on whether a patient was part of a multiple birth or not. | f:Patient/f:multipleBirthBoolean | f:Patient/f:multipleBirthInteger | XML / JSON
mothersMaidenName | string | Search based on patient's mother's maiden name | | XML / JSON

8.1.19.3 Patient-extensions-Patient-age

URL: https://hl7.org/fhir/R4/patient-extensions-Patient-age.html

8.1.19.3 Patient-extensions-Patient-age

Search Parameter age (type: number) defined as part of the patient-extensions Profile.

Patient Administration | Maturity Level: 0 | Informative

Searches for patients based on age as calculated based on current date and date of birth. Deceased patients are excluded from the search.

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-age
• Paths : f:Patient/f:birthDate
• Targets :

8.1.19.4 Patient-extensions-Patient-birthOrderBoolean

URL: https://hl7.org/fhir/R4/patient-extensions-Patient-birthOrderBoolean.html

8.1.19.4 Patient-extensions-Patient-birthOrderBoolean

Search Parameter birthOrderBoolean (type: token) defined as part of the patient-extensions Profile.

Patient Administration | Maturity Level: 0 | Informative

Search based on whether a patient was part of a multiple birth or not.

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-birthOrderBoolean
• Paths : f:Patient/f:multipleBirthBoolean | f:Patient/f:multipleBirthInteger
• Targets :

8.1.19.5 Patient-extensions-Patient-mothersMaidenName

URL: https://hl7.org/fhir/R4/patient-extensions-Patient-mothersMaidenName.html

8.1.19.5 Patient-extensions-Patient-mothersMaidenName

Search Parameter mothersMaidenName (type: string) defined as part of the patient-extensions Profile.

Patient Administration | Maturity Level: 0 | Informative

Search based on patient's mother's maiden name

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-mothersMaidenName
• Paths :
• Targets :

8.1.20 JSON Schema for Patient

URL: https://hl7.org/fhir/R4/patient.schema.json.html

8.1.20 JSON Schema for Patient

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Patient, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Patient",
  "$ref": "#/definitions/Patient",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Patient": {
      "description": "Demographics and other administrative information about an individual or animal receiving care or
    other health-related services.",
      "properties": {
        "resourceType": {
          "description": "This is a Patient resource",
          "const": "Patient"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "An identifier for this patient.",
```

```
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Whether this patient record is in active use. \nMany systems use this property to mark as non-
current patients, such as those that have not been seen for a period of time based on an organization\u0027s
business rules.\n\nIt is often used to filter patient lists to exclude inactive patients\n\nDeceased patients may
also be marked as inactive for the same reasons, but may be active for some time after death.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A name associated with the individual.",
      "items": {
        "$ref": "HumanName.schema.json#/definitions/HumanName"
      },
      "type": "array"
    },
    "telecom": {
      "description": "A contact detail (e.g. a telephone number or an email address) by which the individual may be
contacted.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "gender": {
      "description": "Administrative Gender - the gender that the patient is considered to have for administration
and record keeping purposes.",
      "enum": [
        "male",
        "female",
        "other",
        "unknown"
      ]
    },
    "_gender": {
      "description": "Extensions for gender",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "birthDate": {
      "description": "The date of birth for the individual.",
      "$ref": "#/definitions/date"
    },
    "_birthDate": {
      "description": "Extensions for birthDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "deceasedBoolean": {
      "description": "Indicates if the individual is deceased or not.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_deceasedBoolean": {
      "description": "Extensions for deceasedBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "deceasedDateTime": {
      "description": "Indicates if the individual is deceased or not.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_deceasedDateTime": {
      "description": "Extensions for deceasedDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "address": {
      "description": "An address for the individual.",
      "items": {
        "$ref": "Address.schema.json#/definitions/Address"
      },
      "type": "array"
    },
    "maritalStatus": {
      "description": "This field contains a patient\u0027s most recent marital (civil) status.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "multipleBirthBoolean": {
      "description": "Indicates whether the patient is part of a multiple (boolean) or indicates the actual birth
order (integer).",
      "pattern": "^true|false$",
      "type": "boolean"
    },
```

```
    "_multipleBirthBoolean": {
      "description": "Extensions for multipleBirthBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "multipleBirthInteger": {
      "description": "Indicates whether the patient is part of a multiple (boolean) or indicates the actual birth
order (integer).",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_multipleBirthInteger": {
      "description": "Extensions for multipleBirthInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "photo": {
      "description": "Image of the patient.",
      "items": {
        "$ref": "Attachment.schema.json#/definitions/Attachment"
      },
      "type": "array"
    },
    "contact": {
      "description": "A contact party (e.g. guardian, partner, friend) for the patient.",
      "items": {
        "$ref": "#/definitions/Patient_Contact"
      },
      "type": "array"
    },
    "communication": {
      "description": "A language which may be used to communicate with the patient about his or her health.",
      "items": {
        "$ref": "#/definitions/Patient_Communication"
      },
      "type": "array"
    },
    "generalPractitioner": {
      "description": "Patient\u0027s nominated care provider.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "managingOrganization": {
      "description": "Organization that is the custodian of the patient record.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "link": {
      "description": "Link to another patient resource that concerns the same actual patient.",
      "items": {
        "$ref": "#/definitions/Patient_Link"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Patient_Contact": {
  "description": "Demographics and other administrative information about an individual or animal receiving care or
other health-related services.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "relationship": {
      "description": "The nature of the relationship between the patient and the contact person.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "name": {
      "description": "A name associated with the contact person.",
      "$ref": "HumanName.schema.json#/definitions/HumanName"
    },
    "telecom": {
      "description": "A contact detail for the person, e.g. a telephone number or an email address.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "address": {
      "description": "Address for the contact person.",
      "$ref": "Address.schema.json#/definitions/Address"
    },
    "gender": {
      "description": "Administrative Gender - the gender that the contact person is considered to have for
administration and record keeping purposes.",
      "enum": [
        "male",
        "female",
        "other",
        "unknown"
      ]
    },
    "_gender": {
      "description": "Extensions for gender",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "organization": {
      "description": "Organization on behalf of which the contact is acting or for which the contact is working.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "period": {
      "description": "The period during which this contact person or organization is valid to be contacted relating
to this patient.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
},
"Patient_Communication": {
  "description": "Demographics and other administrative information about an individual or animal receiving care or
other health-related services.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "language": {
      "description": "The ISO-639-1 alpha 2 code in lower case for the language, optionally followed by a hyphen and
 the ISO-3166-1 alpha 2 code for the region in upper case; e.g. \"en\" for English, or \"en-US\" for American
English versus \"en-EN\" for England English.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
        },
        "preferred": {
          "description": "Indicates whether or not the patient prefers this language (over other languages he masters up
     a certain level).",
          "$ref": "#/definitions/boolean"
        },
        "_preferred": {
          "description": "Extensions for preferred",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "language"
      ]
    },
    "Patient_Link": {
      "description": "Demographics and other administrative information about an individual or animal receiving care or
    other health-related services.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "other": {
          "description": "The other patient resource that the link refers to.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": {
          "description": "The type of link between this patient resource and another patient resource.",
          "enum": [
            "replaced-by",
            "replaces",
            "refer",
            "seealso"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "other"
      ]
    }
  }
}
```


8.1.21 Resource Patient - Maturity Dependency View

URL: https://hl7.org/fhir/R4/patient-dependencies.html

8.1.21 Resource Patient - Maturity Dependency View

Patient Administration Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Patient, Practitioner, RelatedPerson

| Some of the content on this page (marked clearly) has been approved as part of an ANSI standard. See the Patient Package for further details.

Dependency Graph for Patient FMM level 5

Name | Card. | Type | Dependency Analysis
Patient | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
name | 0..* | HumanName
telecom | 0..* | ContactPoint
gender | 0..1 | code
birthDate | 0..1 | date
deceased[x] | 0..1
deceasedBoolean | | boolean
deceasedDateTime | | dateTime
address | 0..* | Address
maritalStatus | 0..1 | CodeableConcept
multipleBirth[x] | 0..1
multipleBirthBoolean | | boolean
multipleBirthInteger | | integer
photo | 0..* | Attachment
contact | 0..* | BackboneElement
relationship | 0..* | CodeableConcept
name | 0..1 | HumanName
telecom | 0..* | ContactPoint
address | 0..1 | Address
gender | 0..1 | code
organization | 0..1 | Reference(Organization) | Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM5-Normative)
period | 0..1 | Period
communication | 0..* | BackboneElement
language | 1..1 | CodeableConcept
preferred | 0..1 | boolean
generalPractitioner | 0..* | Reference(Organization | Practitioner | PractitionerRole) | Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM5-Normative)
managingOrganization | 0..1 | Reference(Organization) | Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM5-Normative)
link | 0..* | BackboneElement
other | 1..1 | Reference(Patient | RelatedPerson) | Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM5-Normative)
type | 1..1 | code

8.2 Resource RelatedPerson - Content

URL: https://hl7.org/fhir/R4/relatedperson.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, RelatedPerson

Information about a person that is involved in the care for a patient, but who is not the target of healthcare, nor has a formal responsibility in the care process.

8.2.1 Scope and Usage

RelatedPersons typically have a personal or non-healthcare-specific professional relationship to the patient. A RelatedPerson resource is primarily used for attribution of information, since RelatedPersons are often a source of information about the patient. For keeping information about people for contact purposes for a patient, use a Patient's Contact element. Some individuals may serve as both a Patient's Contact and a Related Person.

Example RelatedPersons are:

• A patient's wife or husband
• A patient's relatives or friends
• A neighbor bringing a patient to the hospital
• The owner or trainer of a horse
• A patient's attorney or guardian
• A Guide Dog

8.2.2 Boundaries and Relationships

The primary distinction between a Practitioner and a RelatedPerson is based on whether:

• The person/animal operates on behalf of the care delivery organization over multiple patients (Practitioner) or,

• Where the person/animal is not associated with the organization, and instead is allocated tasks specifically for the RelatedPerson's Patient (RelatedPerson).

A standard extension animalSpecies can be used to indicate the species of a service animal.

This resource is referenced by Annotation, Signature, Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, CarePlan, CareTeam, ChargeItem, Claim, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, Coverage, DeviceRequest, DeviceUseStatement, DocumentManifest, DocumentReference, Encounter, ExplanationOfBenefit, Goal, Group, ImagingStudy, Invoice, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Observation, Patient, Person, Procedure, Provenance, QuestionnaireResponse, RequestGroup, Schedule, ServiceRequest, SupplyRequest and Task

8.2.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
RelatedPerson | TU | | DomainResource | A person that is related to a patient, but who is not a direct target of care Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | A human identifier for this person
active | ?!Σ | 0..1 | boolean | Whether this related person's record is in active use
patient | Σ | 1..1 | Reference(Patient) | The patient this person is related to
relationship | Σ | 0..* | CodeableConcept | The nature of the relationship Patient relationship type (Preferred)
name | Σ | 0..* | HumanName | A name associated with the person
telecom | Σ | 0..* | ContactPoint | A contact detail for the person
gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | Σ | 0..1 | date | The date on which the related person was born
address | Σ | 0..* | Address | Address where the related person can be contacted or visited
photo | | 0..* | Attachment | Image of the person
period | | 0..1 | Period | Period of time that this relationship is considered valid
communication | | 0..* | BackboneElement | A language which may be used to communicate with about the patient's health
language | | 1..1 | CodeableConcept | The language which can be used to communicate with the patient about his or her health Common Languages (Preferred but limited to AllLanguages)
preferred | | 0..1 | boolean | Language preference indicator
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RelatedPerson xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this related person's record is in active use -->
 <patient><!-- 1..1 Reference(Patient) The patient this person is related to --></patient>
 <relationship><!-- 0..* CodeableConcept The nature of the relationship --></relationship>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the related person was born -->
 <address><!-- 0..* Address Address where the related person can be contacted or visited --></address>
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <period><!-- 0..1 Period Period of time that this relationship is considered valid --></period>
 <communication>  <!-- 0..* A language which may be used to communicate with about the patient's health -->
  <language><!-- 1..1 CodeableConcept The language which can be used to communicate with the patient about his or her
    health --></language>
  <preferred value="[boolean]"/><!-- 0..1 Language preference indicator -->
 </communication>
</RelatedPerson>
```

JSON Template

```
{
  "resourceType" : "RelatedPerson",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "active" : <boolean>, // Whether this related person's record is in active use
  "patient" : { Reference(Patient) }, // R!  The patient this person is related to
  "relationship" : [{ CodeableConcept }], // The nature of the relationship
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the related person was born
```

```
  "address" : [{ Address }], // Address where the related person can be contacted or visited
  "photo" : [{ Attachment }], // Image of the person
  "period" : { Period }, // Period of time that this relationship is considered valid
  "communication" : [{ // A language which may be used to communicate with about the patient's health
    "language" : { CodeableConcept }, // R!  The language which can be used to communicate with the patient about his or
     her health
    "preferred" : <boolean> // Language preference indicator
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RelatedPerson;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RelatedPerson.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:RelatedPerson.active [ boolean ]; # 0..1 Whether this related person's record is in active use
  fhir:RelatedPerson.patient [ Reference(Patient) ]; # 1..1 The patient this person is related to
  fhir:RelatedPerson.relationship [ CodeableConcept ], ... ; # 0..* The nature of the relationship
  fhir:RelatedPerson.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:RelatedPerson.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:RelatedPerson.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:RelatedPerson.birthDate [ date ]; # 0..1 The date on which the related person was born
  fhir:RelatedPerson.address [ Address ], ... ; # 0..* Address where the related person can be contacted or visited
  fhir:RelatedPerson.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:RelatedPerson.period [ Period ]; # 0..1 Period of time that this relationship is considered valid
  fhir:RelatedPerson.communication [ # 0..* A language which may be used to communicate with about the patient's health
    fhir:RelatedPerson.communication.language [ CodeableConcept ]; # 1..1 The language which can be used to communicate
    with the patient about his or her health
    fhir:RelatedPerson.communication.preferred [ boolean ]; # 0..1 Language preference indicator
  ], ...;
]
```

Changes since R3

RelatedPerson
RelatedPerson.active | Default Value "true" removed
RelatedPerson.relationship | Max Cardinality changed from 1 to *
RelatedPerson.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
RelatedPerson.communication | Added Element
RelatedPerson.communication.language | Added Mandatory Element
RelatedPerson.communication.preferred | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
RelatedPerson | TU | | DomainResource | A person that is related to a patient, but who is not a direct target of care Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | A human identifier for this person
active | ?!Σ | 0..1 | boolean | Whether this related person's record is in active use
patient | Σ | 1..1 | Reference(Patient) | The patient this person is related to
relationship | Σ | 0..* | CodeableConcept | The nature of the relationship Patient relationship type (Preferred)
name | Σ | 0..* | HumanName | A name associated with the person
telecom | Σ | 0..* | ContactPoint | A contact detail for the person
gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | Σ | 0..1 | date | The date on which the related person was born
address | Σ | 0..* | Address | Address where the related person can be contacted or visited
photo | | 0..* | Attachment | Image of the person
period | | 0..1 | Period | Period of time that this relationship is considered valid
communication | | 0..* | BackboneElement | A language which may be used to communicate with about the patient's health
language | | 1..1 | CodeableConcept | The language which can be used to communicate with the patient about his or her health Common Languages (Preferred but limited to AllLanguages)
preferred | | 0..1 | boolean | Language preference indicator
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RelatedPerson xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this related person's record is in active use -->
```

```
 <patient><!-- 1..1 Reference(Patient) The patient this person is related to --></patient>
 <relationship><!-- 0..* CodeableConcept The nature of the relationship --></relationship>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the related person was born -->
 <address><!-- 0..* Address Address where the related person can be contacted or visited --></address>
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <period><!-- 0..1 Period Period of time that this relationship is considered valid --></period>
 <communication>  <!-- 0..* A language which may be used to communicate with about the patient's health -->
  <language><!-- 1..1 CodeableConcept The language which can be used to communicate with the patient about his or her
    health --></language>
  <preferred value="[boolean]"/><!-- 0..1 Language preference indicator -->
 </communication>
</RelatedPerson>
```

JSON Template

```
{
  "resourceType" : "RelatedPerson",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "active" : <boolean>, // Whether this related person's record is in active use
  "patient" : { Reference(Patient) }, // R!  The patient this person is related to
  "relationship" : [{ CodeableConcept }], // The nature of the relationship
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the related person was born
  "address" : [{ Address }], // Address where the related person can be contacted or visited
  "photo" : [{ Attachment }], // Image of the person
  "period" : { Period }, // Period of time that this relationship is considered valid
  "communication" : [{ // A language which may be used to communicate with about the patient's health
    "language" : { CodeableConcept }, // R!  The language which can be used to communicate with the patient about his or
     her health
    "preferred" : <boolean> // Language preference indicator
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RelatedPerson;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RelatedPerson.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:RelatedPerson.active [ boolean ]; # 0..1 Whether this related person's record is in active use
  fhir:RelatedPerson.patient [ Reference(Patient) ]; # 1..1 The patient this person is related to
  fhir:RelatedPerson.relationship [ CodeableConcept ], ... ; # 0..* The nature of the relationship
  fhir:RelatedPerson.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:RelatedPerson.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:RelatedPerson.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:RelatedPerson.birthDate [ date ]; # 0..1 The date on which the related person was born
  fhir:RelatedPerson.address [ Address ], ... ; # 0..* Address where the related person can be contacted or visited
  fhir:RelatedPerson.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:RelatedPerson.period [ Period ]; # 0..1 Period of time that this relationship is considered valid
  fhir:RelatedPerson.communication [ # 0..* A language which may be used to communicate with about the patient's health
    fhir:RelatedPerson.communication.language [ CodeableConcept ]; # 1..1 The language which can be used to communicate
    with the patient about his or her health
    fhir:RelatedPerson.communication.preferred [ boolean ]; # 0..1 Language preference indicator
  ], ...;
]
```

Changes since Release 3

RelatedPerson
RelatedPerson.active | Default Value "true" removed
RelatedPerson.relationship | Max Cardinality changed from 1 to *
RelatedPerson.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
RelatedPerson.communication | Added Element
RelatedPerson.communication.language | Added Mandatory Element
RelatedPerson.communication.preferred | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.2.3.1 Terminology Bindings

Path | Definition | Type | Reference
RelatedPerson.relationship | The nature of the relationship between a patient and the related person. | Preferred | PatientRelationshipType
RelatedPerson.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender
RelatedPerson.communication.language | A human language. | Preferred, but limited to AllLanguages | CommonLanguages

8.2.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Indicates if the related person record is active | RelatedPerson.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | RelatedPerson.address | 3 Resources
address-city | string | A city specified in an address | RelatedPerson.address.city | 3 Resources
address-country | string | A country specified in an address | RelatedPerson.address.country | 3 Resources
address-postalcode | string | A postal code specified in an address | RelatedPerson.address.postalCode | 3 Resources
address-state | string | A state specified in an address | RelatedPerson.address.state | 3 Resources
address-use | token | A use code specified in an address | RelatedPerson.address.use | 3 Resources
birthdate | date | The Related Person's date of birth | RelatedPerson.birthDate | 2 Resources
email | token | A value in an email contact | RelatedPerson.telecom.where(system='email') | 4 Resources
gender | token | Gender of the related person | RelatedPerson.gender | 3 Resources
identifier | token | An Identifier of the RelatedPerson | RelatedPerson.identifier
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | RelatedPerson.name
patient | reference | The patient this related person is related to | RelatedPerson.patient (Patient)
phone | token | A value in a phone contact | RelatedPerson.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of name using some kind of phonetic matching algorithm | RelatedPerson.name | 3 Resources
relationship | token | The relationship between the patient and the relatedperson | RelatedPerson.relationship
telecom | token | The value in any kind of contact | RelatedPerson.telecom | 4 Resources

8.2.5 Resource RelatedPerson - Examples

URL: https://hl7.org/fhir/R4/relatedperson-examples.html

8.2.5 Resource RelatedPerson - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, RelatedPerson

Example Name | id | Format
RelatedPerson Benedicte du Marche | benedicte | XML | JSON | Turtle
RelatedPerson Peter Chalmers | peter | XML | JSON | Turtle
Sarah Abels | f001 | XML | JSON | Turtle
Ariadne Bor-Jansma | f002 | XML | JSON | Turtle
Mother of Newborn Example for Newborn Example | newborn-mom | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.2.6 Resource RelatedPerson - Detailed Descriptions

URL: https://hl7.org/fhir/R4/relatedperson-definitions.html

8.2.6 Resource RelatedPerson - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, RelatedPerson

Detailed Descriptions for the elements in the RelatedPerson resource.

RelatedPerson
Element Id | RelatedPerson
Definition | Information about a person that is involved in the care for a patient, but who is not the target of healthcare, nor has a formal responsibility in the care process.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to track persons related to the patient or the healthcare process.
RelatedPerson.identifier
Element Id | RelatedPerson.identifier
Definition | Identifier for a person within a particular scope.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | People are known by a variety of ids. Some institutions maintain several, and most collect identifiers for exchange with other organizations concerning the patient. Examples are national person identifier and local identifier.

Summary | true
RelatedPerson.active
Element Id | RelatedPerson.active
Definition | Whether this related person record is in active use.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a related person record as not to be used, such as if it was created in error.
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
RelatedPerson.patient
Element Id | RelatedPerson.patient
Definition | The patient this person is related to.
Cardinality | 1..1
Type | Reference(Patient)
Requirements | We need to know which patient this RelatedPerson is related to.
Summary | true
RelatedPerson.relationship
Element Id | RelatedPerson.relationship
Definition | The nature of the relationship between a patient and the related person.
Cardinality | 0..*
Terminology Binding | Patient relationship type (Preferred)
Type | CodeableConcept
Requirements | We need to know the relationship with the patient since it influences the interpretation of the information attributed to this person.
Summary | true
RelatedPerson.name
Element Id | RelatedPerson.name
Definition | A name associated with the person.
Cardinality | 0..*
Type | HumanName
Requirements | Related persons need to be identified by name, but it is uncommon to need details about multiple other names for that person.
Summary | true
RelatedPerson.telecom
Element Id | RelatedPerson.telecom
Definition | A contact detail for the person, e.g. a telephone number or an email address.
Cardinality | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Summary | true
Comments | Person may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently, and also to help with identification.
RelatedPerson.gender
Element Id | RelatedPerson.gender
Definition | Administrative Gender - the gender that the person is considered to have for administration and record keeping purposes.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed for identification of the person, in combination with (at least) name and birth date.
Summary | true
RelatedPerson.birthDate
Element Id | RelatedPerson.birthDate
Definition | The date on which the related person was born.
Cardinality | 0..1
Type | date
Summary | true
RelatedPerson.address
Element Id | RelatedPerson.address
Definition | Address where the related person can be contacted or visited.
Cardinality | 0..*
Type | Address
Requirements | Need to keep track where the related person can be contacted per postal mail or visited.
Summary | true
RelatedPerson.photo
Element Id | RelatedPerson.photo
Definition | Image of the person.
Cardinality | 0..*
Type | Attachment
Requirements | Many EHR systems have the capability to capture an image of persons. Fits with newer social media usage too.

RelatedPerson.period
Element Id | RelatedPerson.period
Definition | The period of time during which this relationship is or was active. If there are no dates defined, then the interval is unknown.
Cardinality | 0..1
Type | Period
RelatedPerson.communication
Element Id | RelatedPerson.communication
Definition | A language which may be used to communicate with about the patient's health.
Cardinality | 0..*
Requirements | If a related person does not speak the local language, interpreters may be required, so languages spoken and proficiency is an important things to keep track of both for patient and other persons of interest.
Comments | If no language is specified, this implies that the default local language is spoken. If you need to convey proficiency for multiple modes, then you need multiple RelatedPerson.Communication associations. If the RelatedPerson does not speak the default local language, then the Interpreter Required Standard can be used to explicitly declare that an interpreter is required.
RelatedPerson.communication.language
Element Id | RelatedPerson.communication.language
Definition | The ISO-639-1 alpha 2 code in lower case for the language, optionally followed by a hyphen and the ISO-3166-1 alpha 2 code for the region in upper case; e.g. "en" for English, or "en-US" for American English versus "en-EN" for England English.
Cardinality | 1..1
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | CodeableConcept
Requirements | Most systems in multilingual countries will want to convey language. Not all systems actually need the regional dialect.
Comments | The structure aa-BB with this exact casing is one the most widely used notations for locale. However not all systems actually code this but instead have it as free text. Hence CodeableConcept instead of code as the data type.
RelatedPerson.communication.preferred
Element Id | RelatedPerson.communication.preferred
Definition | Indicates whether or not the patient prefers this language (over other languages he masters up a certain level).
Cardinality | 0..1
Type | boolean
Requirements | People that master multiple languages up to certain level may prefer one or more, i.e. feel more confident in communicating in a particular language making other languages sort of a fall back method.
Comments | This language is specifically identified for communicating healthcare information.

8.2.7 Resource RelatedPerson - Mappings

URL: https://hl7.org/fhir/R4/relatedperson-mappings.html

8.2.7 Resource RelatedPerson - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, RelatedPerson

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the relatedperson resource (see Mappings to Other Standards for further information & status).

8.2.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

RelatedPerson
identifier | FiveWs.identifier
active | FiveWs.status
relationship | FiveWs.class
period | FiveWs.done[x]

8.2.7.2 HL7 v2 Mapping (http://hl7.org/v2)

RelatedPerson
identifier | NK1-33
active
patient | PID-3
relationship | NK1-3
name | NK1-2
telecom | NK1-5 / NK1-6 / NK1-40
gender | NK1-15
birthDate
address | NK1-4
photo | OBX-5 - needs a profile
period
communication
language

preferred

8.2.7.3 RIM Mapping (http://hl7.org/v3)

RelatedPerson | role
identifier | .id
active | .statusCode
patient | scoper[classCode=PSN|ANM and determinerCode='INSTANCE']/playedRole[classCode='PAT']/id
relationship | code
name | name
telecom | telecom
gender | administrativeGender
birthDate | player.birthTime
address | addr
photo | player[classCode='PSN' and determinerCode='INSTANCE']/desc
period | .effectiveTime
communication | LanguageCommunication
language | player[classCode=PSN|ANM and determinerCode=INSTANCE]/languageCommunication/code
preferred | preferenceInd

8.2.8 Resource RelatedPerson - Extensions & Profiles

URL: https://hl7.org/fhir/R4/relatedperson-profiles.html

8.2.8 Resource RelatedPerson - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, RelatedPerson

This table lists profiles and extensions for the RelatedPerson resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.2.8.1 Profiles

No Profiles defined for this resource

8.2.8.2 Extensions

practitioner-animalSpecies | RelatedPerson | animalSpecies | for Practitioner HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

consent-Transcriber | Transcriber | for Consent HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
cqf-initiatingPerson | initiatingPerson | for Clinical Reasoning Extensions
cqf-receivingPerson | receivingPerson | for Clinical Reasoning Extensions
goal-acceptance | acceptance | for FHIR Core Goal Profile
patient-relatedPerson | relatedPerson | for Patient HL7 Extensions
procedure-directedBy | directedBy | for Procedure HL7 Extensions
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.2.8.3 Search Extensions

No Search Extensions defined for this resource

8.2.9 Resource RelatedPerson - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/relatedperson-version-maps.html

No validation errors - all conversions are clean

8.2.9 Resource RelatedPerson - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, RelatedPerson

R3 : R4 Conversion maps for RelatedPerson.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.2.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/RelatedPerson3to4" = "R3 to R4 Conversions for RelatedPerson"

uses "http://hl7.org/fhir/3.0/StructureDefinition/RelatedPerson" alias RelatedPersonR3 as source
uses "http://hl7.org/fhir/StructureDefinition/RelatedPerson" alias RelatedPerson as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group RelatedPerson(source src : RelatedPersonR3, target tgt : RelatedPerson) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.patient -> tgt.patient;
  src.relationship -> tgt.relationship;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.address -> tgt.address;
  src.photo -> tgt.photo;
  src.period -> tgt.period;
}
```

8.2.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/RelatedPerson4to3" = "R4 to R3 Conversion for RelatedPerson"

uses "http://hl7.org/fhir/StructureDefinition/RelatedPerson" alias RelatedPerson as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/RelatedPerson" alias RelatedPersonR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group RelatedPerson(source src : RelatedPersonR3, target tgt : RelatedPerson) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.patient -> tgt.patient;
  src.relationship -> tgt.relationship;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.address -> tgt.address;
  src.photo -> tgt.photo;
  src.period -> tgt.period;
}
```

8.2.9.3 R4 Validation Errors

8.2.10 JSON Schema for RelatedPerson

URL: https://hl7.org/fhir/R4/relatedperson.schema.json.html

8.2.10 JSON Schema for RelatedPerson

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, RelatedPerson

This is a representation of the json schema for RelatedPerson, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/RelatedPerson",
  "$ref": "#/definitions/RelatedPerson",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "RelatedPerson": {
      "description": "Information about a person that is involved in the care for a patient, but who is not the target
    of healthcare, nor has a formal responsibility in the care process.",
      "properties": {
        "resourceType": {
          "description": "This is a RelatedPerson resource",
          "const": "RelatedPerson"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
```

```
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for a person within a particular scope.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Whether this related person record is in active use.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "The patient this person is related to.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "relationship": {
      "description": "The nature of the relationship between a patient and the related person.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "name": {
      "description": "A name associated with the person.",
      "items": {
        "$ref": "HumanName.schema.json#/definitions/HumanName"
      },
      "type": "array"
    },
    "telecom": {
```

```
      "description": "A contact detail for the person, e.g. a telephone number or an email address.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "gender": {
      "description": "Administrative Gender - the gender that the person is considered to have for administration
and record keeping purposes.",
      "enum": [
        "male",
        "female",
        "other",
        "unknown"
      ]
    },
    "_gender": {
      "description": "Extensions for gender",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "birthDate": {
      "description": "The date on which the related person was born.",
      "$ref": "#/definitions/date"
    },
    "_birthDate": {
      "description": "Extensions for birthDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "address": {
      "description": "Address where the related person can be contacted or visited.",
      "items": {
        "$ref": "Address.schema.json#/definitions/Address"
      },
      "type": "array"
    },
    "photo": {
      "description": "Image of the person.",
      "items": {
        "$ref": "Attachment.schema.json#/definitions/Attachment"
      },
      "type": "array"
    },
    "period": {
      "description": "The period of time during which this relationship is or was active. If there are no dates
defined, then the interval is unknown.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "communication": {
      "description": "A language which may be used to communicate with about the patient\u0027s health.",
      "items": {
        "$ref": "#/definitions/RelatedPerson_Communication"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "patient",
    "resourceType"
  ]
},
"RelatedPerson_Communication": {
  "description": "Information about a person that is involved in the care for a patient, but who is not the target
of healthcare, nor has a formal responsibility in the care process.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
```

```
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "language": {
          "description": "The ISO-639-1 alpha 2 code in lower case for the language, optionally followed by a hyphen and
     the ISO-3166-1 alpha 2 code for the region in upper case; e.g. \"en\" for English, or \"en-US\" for American
    English versus \"en-EN\" for England English.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "preferred": {
          "description": "Indicates whether or not the patient prefers this language (over other languages he masters up
     a certain level).",
          "$ref": "#/definitions/boolean"
        },
        "_preferred": {
          "description": "Extensions for preferred",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "language"
      ]
    }
  }
}
```


8.2.11 Resource RelatedPerson - Maturity Dependency View

URL: https://hl7.org/fhir/R4/relatedperson-dependencies.html

8.2.11 Resource RelatedPerson - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, RelatedPerson

Dependency Graph for RelatedPerson FMM level 2

Name | Card. | Type | Dependency Analysis
RelatedPerson | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
patient | 1..1 | Reference(Patient)
relationship | 0..* | CodeableConcept
name | 0..* | HumanName
telecom | 0..* | ContactPoint
gender | 0..1 | code
birthDate | 0..1 | date
address | 0..* | Address
photo | 0..* | Attachment
period | 0..1 | Period
communication | 0..* | BackboneElement
language | 1..1 | CodeableConcept
preferred | 0..1 | boolean

8.3 Resource Group - Content

URL: https://hl7.org/fhir/R4/group.html
FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Business | Compartments: Device, Patient, Practitioner

Represents a defined collection of entities that may be discussed or acted upon collectively but which are not expected to act collectively, and are not formally or legally recognized; i.e. a collection of entities that isn't an Organization.

8.3.1 Scope and Usage

8.3.1.1 Use Cases

The Group resource is used in one of two ways:

• To define a group of specific people, animals, devices, etc. that is being tracked, examined or otherwise referenced as part of healthcare-related activities

• To define a set of possible people, animals, devices, etc. that are of interest for some intended future healthcare-related activities

Examples of the former could include group therapy or treatment sessions, exposed entities tracked as part of public health, etc. The latter might be used to define expected subjects for a clinical study.

Both use cases are handled by a single resource because the data elements captured tend to be similar.

Contained, bundled and remotely referenced resources convey the same meaning.

8.3.2 Boundaries and Relationships

There are a number of mechanisms in FHIR for communicating collections of resources:

• The List resource - enumerates a flat collection of resources and provides features for managing the collection. While a particular List instance may represent a "snapshot", from a business process perspective the notion of "List" is dynamic – items are added and removed over time. The List resource references other resources. Lists may be curated and have specific business meaning.

• This Group resource - defines a group of specific people, animals, devices etc. by enumerating them, or by describing qualities that group members have. The group resource refers to other resources, possibly implicitly. Groups are intended to be acted upon or observed as a whole; e.g. performing therapy on a group, calculating risk for a group, etc. This resource will commonly be used for public health (e.g. describing an at-risk population), clinical trials (e.g. defining a test subject pool) and similar purposes.

• CareTeam. Group is distinct from CareTeam. Group is patient-independent and identifies an undifferentiated set of individuals who are intended to be the target of one or more clinical activities (e.g. set of clinical trial participants, set of individuals impacted by or at risk of a public health event, a herd or flock, etc.) The CareTeam resource establishes a set of relationships and roles and is specific to a particular Patient. The actors are the individual members or organized group of individuals. CareTeam can be referenced by EpisodeOfCare, Encounter, or CarePlan to identify the set of individuals (and their respective roles) who are intended to be involved in providing the care defined by those resources.

• The Bundle resource - is an infrastructure container for a group of resources. It does not have narrative and is used to group collections of resources for transmission, persistence or processing (e.g. messages, documents, transactions, query responses, etc.) The content of bundles is typically algorithmically determined for a particular exchange or persistence purpose.

• The Composition resource - defines a set of healthcare-related information that is assembled together into a single logical document that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. The composition resource provides the basic structure of a FHIR document. The full content of the document is expressed using a bundle. Compositions will often reference Lists as the focus of particular sections.

• The DomainResource.contained element - allows multiple resources to be nested inside any DomainResource. This is a special type of grouping where the grouped resources lose independent existence - they no longer have their own identifiers, can't easily be queried independently, etc. Use of this grouping is a technical mechanism for managing the independence of resources and has no impact on meaning.

This resource is referenced by DataRequirement, UsageContext, ActivityDefinition, AdverseEvent, CarePlan, CareTeam, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, Consent, Contract, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, Encounter, EventDefinition, EvidenceVariable, Flag, Goal, itself, GuidanceResponse, ImagingStudy, Invoice, Library, List, Measure, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Observation, PlanDefinition, Procedure, RequestGroup, ResearchDefinition, ResearchElementDefinition, ResearchStudy, RiskAssessment, ServiceRequest, Specimen and Task

8.3.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints

Group | ITU | | DomainResource | Group of multiple entities + Rule: Can only have members if group is "actual" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Unique id

active | Σ | 0..1 | boolean | Whether this group's record is in active use

type | Σ | 1..1 | code | person | animal | practitioner | device | medication | substance GroupType (Required)

actual | ΣI | 1..1 | boolean | Descriptive or actual

code | Σ | 0..1 | CodeableConcept | Kind of Group members

name | Σ | 0..1 | string | Label for Group

quantity | Σ | 0..1 | unsignedInt | Number of members

managingEntity | Σ | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole) | Entity that is the custodian of the Group's definition

characteristic | | 0..* | BackboneElement | Include / Exclude group members by Trait
code | | 1..1 | CodeableConcept | Kind of characteristic
value[x] | | 1..1 | | Value held by characteristic
valueCodeableConcept | | | CodeableConcept
valueBoolean | | | boolean
valueQuantity | | | Quantity
valueRange | | | Range
valueReference | | | Reference()
exclude | | 1..1 | boolean | Group includes or excludes
period | | 0..1 | Period | Period over which characteristic is tested
member | I | 0..* | BackboneElement | Who or what is in group
entity | | 1..1 | Reference(Patient | Practitioner | PractitionerRole | Device | Medication | Substance | Group) | Reference to the group member
period | | 0..1 | Period | Period member belonged to the group
inactive | | 0..1 | boolean | If member is no longer in group
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Group xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!-- 1..1 person | animal | practitioner | device | medication | substance -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!-- 0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!-- 0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!-- 0..1 Number of members -->
 <managingEntity><!-- 0..1 Reference(Organization|RelatedPerson|Practitioner|
   PractitionerRole) Entity that is the custodian of the Group's definition --></managingEntity>
 <characteristic>  <!-- 0..* Include / Exclude group members by Trait -->
  <code><!-- 1..1 CodeableConcept Kind of characteristic --></code>
  <value[x]><!-- 1..1 CodeableConcept|boolean|Quantity|Range|Reference Value held by characteristic --></value[x]>
  <exclude value="[boolean]"/><!-- 1..1 Group includes or excludes -->
  <period><!-- 0..1 Period Period over which characteristic is tested --></period>
 </characteristic>
 <member>  <!--  0..* Who or what is in group -->
  <entity><!-- 1..1 Reference(Patient|Practitioner|PractitionerRole|Device|
    Medication|Substance|Group) Reference to the group member --></entity>
  <period><!-- 0..1 Period Period member belonged to the group --></period>
  <inactive value="[boolean]"/><!-- 0..1 If member is no longer in group -->
 </member>
</Group>
```

JSON Template

```
{
  "resourceType" : "Group",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique id
  "active" : <boolean>, // Whether this group's record is in active use
  "type" : "<code>", // R!  person | animal | practitioner | device | medication | substance
  "actual" : <boolean>, // C? R!  Descriptive or actual
  "code" : { CodeableConcept }, // Kind of Group members
  "name" : "<string>", // Label for Group
  "quantity" : "<unsignedInt>", // Number of members
  "managingEntity" : { Reference(Organization|RelatedPerson|Practitioner|
   PractitionerRole) }, // Entity that is the custodian of the Group's definition
  "characteristic" : [{ // Include / Exclude group members by Trait
    "code" : { CodeableConcept }, // R!  Kind of characteristic
    // value[x]: Value held by characteristic. One of these 5:
    "valueCodeableConcept" : { CodeableConcept },
    "valueBoolean" : <boolean>,
    "valueQuantity" : { Quantity },
    "valueRange" : { Range },
    "valueReference" : { Reference },
    "exclude" : <boolean>, // R!  Group includes or excludes
    "period" : { Period } // Period over which characteristic is tested
  }],
  "member" : [{ // C? Who or what is in group
    "entity" : { Reference(Patient|Practitioner|PractitionerRole|Device|
    Medication|Substance|Group) }, // R!  Reference to the group member
    "period" : { Period }, // Period member belonged to the group
    "inactive" : <boolean> // If member is no longer in group
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Group;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Group.identifier [ Identifier ], ... ; # 0..* Unique id
  fhir:Group.active [ boolean ]; # 0..1 Whether this group's record is in active use
  fhir:Group.type [ code ]; # 1..1 person | animal | practitioner | device | medication | substance
  fhir:Group.actual [ boolean ]; # 1..1 Descriptive or actual
  fhir:Group.code [ CodeableConcept ]; # 0..1 Kind of Group members
  fhir:Group.name [ string ]; # 0..1 Label for Group
  fhir:Group.quantity [ unsignedInt ]; # 0..1 Number of members
  fhir:Group.managingEntity [ Reference(Organization|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Entity that
    is the custodian of the Group's definition
  fhir:Group.characteristic [ # 0..* Include / Exclude group members by Trait
    fhir:Group.characteristic.code [ CodeableConcept ]; # 1..1 Kind of characteristic
    # Group.characteristic.value[x] : 1..1 Value held by characteristic. One of these 5
      fhir:Group.characteristic.valueCodeableConcept [ CodeableConcept ]
      fhir:Group.characteristic.valueBoolean [ boolean ]
      fhir:Group.characteristic.valueQuantity [ Quantity ]
      fhir:Group.characteristic.valueRange [ Range ]
      fhir:Group.characteristic.valueReference [ Reference ]
    fhir:Group.characteristic.exclude [ boolean ]; # 1..1 Group includes or excludes
    fhir:Group.characteristic.period [ Period ]; # 0..1 Period over which characteristic is tested
  ], ...;
  fhir:Group.member [ # 0..* Who or what is in group
    fhir:Group.member.entity [ Reference(Patient|Practitioner|PractitionerRole|Device|Medication|Substance|Group) ]; #
    1..1 Reference to the group member
    fhir:Group.member.period [ Period ]; # 0..1 Period member belonged to the group
    fhir:Group.member.inactive [ boolean ]; # 0..1 If member is no longer in group
  ], ...;
]
```

Changes since R3

Group
Group.active | Default Value "true" removed
Group.type | Change value set from http://hl7.org/fhir/ValueSet/group-type to http://hl7.org/fhir/ValueSet/group-type|4.0.1
Group.managingEntity | Added Element
Group.characteristic.value[x] | Add Type Reference
Group.characteristic.exclude | No longer marked as Modifier
Group.member.entity | Type Reference: Added Target Types PractitionerRole, Group
Group.member.inactive | Default Value "false" removed

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Group | ITU | | DomainResource | Group of multiple entities + Rule: Can only have members if group is "actual" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique id
active | Σ | 0..1 | boolean | Whether this group's record is in active use
type | Σ | 1..1 | code | person | animal | practitioner | device | medication | substance GroupType (Required)
actual | ΣI | 1..1 | boolean | Descriptive or actual
code | Σ | 0..1 | CodeableConcept | Kind of Group members
name | Σ | 0..1 | string | Label for Group
quantity | Σ | 0..1 | unsignedInt | Number of members
managingEntity | Σ | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole) | Entity that is the custodian of the Group's definition
characteristic | | 0..* | BackboneElement | Include / Exclude group members by Trait
code | | 1..1 | CodeableConcept | Kind of characteristic
value[x] | | 1..1 | | Value held by characteristic
valueCodeableConcept | | | CodeableConcept
valueBoolean | | | boolean
valueQuantity | | | Quantity
valueRange | | | Range
valueReference | | | Reference()
exclude | | 1..1 | boolean | Group includes or excludes
period | | 0..1 | Period | Period over which characteristic is tested
member | I | 0..* | BackboneElement | Who or what is in group
entity | | 1..1 | Reference(Patient | Practitioner | PractitionerRole | Device | Medication | Substance | Group) | Reference to the group member
period | | 0..1 | Period | Period member belonged to the group
inactive | | 0..1 | boolean | If member is no longer in group
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Group xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
```

```
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!-- 1..1 person | animal | practitioner | device | medication | substance -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!-- 0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!-- 0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!-- 0..1 Number of members -->
 <managingEntity><!-- 0..1 Reference(Organization|RelatedPerson|Practitioner|
   PractitionerRole) Entity that is the custodian of the Group's definition --></managingEntity>
 <characteristic>  <!-- 0..* Include / Exclude group members by Trait -->
  <code><!-- 1..1 CodeableConcept Kind of characteristic --></code>
  <value[x]><!-- 1..1 CodeableConcept|boolean|Quantity|Range|Reference Value held by characteristic --></value[x]>
  <exclude value="[boolean]"/><!-- 1..1 Group includes or excludes -->
  <period><!-- 0..1 Period Period over which characteristic is tested --></period>
 </characteristic>
 <member>  <!--  0..* Who or what is in group -->
  <entity><!-- 1..1 Reference(Patient|Practitioner|PractitionerRole|Device|
    Medication|Substance|Group) Reference to the group member --></entity>
  <period><!-- 0..1 Period Period member belonged to the group --></period>
  <inactive value="[boolean]"/><!-- 0..1 If member is no longer in group -->
 </member>
</Group>
```

JSON Template

```
{
  "resourceType" : "Group",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique id
  "active" : <boolean>, // Whether this group's record is in active use
  "type" : "<code>", // R!  person | animal | practitioner | device | medication | substance
  "actual" : <boolean>, // C? R!  Descriptive or actual
  "code" : { CodeableConcept }, // Kind of Group members
  "name" : "<string>", // Label for Group
  "quantity" : "<unsignedInt>", // Number of members
  "managingEntity" : { Reference(Organization|RelatedPerson|Practitioner|
   PractitionerRole) }, // Entity that is the custodian of the Group's definition
  "characteristic" : [{ // Include / Exclude group members by Trait
    "code" : { CodeableConcept }, // R!  Kind of characteristic
    // value[x]: Value held by characteristic. One of these 5:
    "valueCodeableConcept" : { CodeableConcept },
    "valueBoolean" : <boolean>,
    "valueQuantity" : { Quantity },
    "valueRange" : { Range },
    "valueReference" : { Reference },
    "exclude" : <boolean>, // R!  Group includes or excludes
    "period" : { Period } // Period over which characteristic is tested
  }],
  "member" : [{ // C? Who or what is in group
    "entity" : { Reference(Patient|Practitioner|PractitionerRole|Device|
    Medication|Substance|Group) }, // R!  Reference to the group member
    "period" : { Period }, // Period member belonged to the group
    "inactive" : <boolean> // If member is no longer in group
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Group;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Group.identifier [ Identifier ], ... ; # 0..* Unique id
  fhir:Group.active [ boolean ]; # 0..1 Whether this group's record is in active use
  fhir:Group.type [ code ]; # 1..1 person | animal | practitioner | device | medication | substance
  fhir:Group.actual [ boolean ]; # 1..1 Descriptive or actual
  fhir:Group.code [ CodeableConcept ]; # 0..1 Kind of Group members
  fhir:Group.name [ string ]; # 0..1 Label for Group
  fhir:Group.quantity [ unsignedInt ]; # 0..1 Number of members
  fhir:Group.managingEntity [ Reference(Organization|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Entity that
    is the custodian of the Group's definition
  fhir:Group.characteristic [ # 0..* Include / Exclude group members by Trait
    fhir:Group.characteristic.code [ CodeableConcept ]; # 1..1 Kind of characteristic
    # Group.characteristic.value[x] : 1..1 Value held by characteristic. One of these 5
      fhir:Group.characteristic.valueCodeableConcept [ CodeableConcept ]
      fhir:Group.characteristic.valueBoolean [ boolean ]
      fhir:Group.characteristic.valueQuantity [ Quantity ]
      fhir:Group.characteristic.valueRange [ Range ]
      fhir:Group.characteristic.valueReference [ Reference ]
    fhir:Group.characteristic.exclude [ boolean ]; # 1..1 Group includes or excludes
    fhir:Group.characteristic.period [ Period ]; # 0..1 Period over which characteristic is tested
  ], ...;
  fhir:Group.member [ # 0..* Who or what is in group
    fhir:Group.member.entity [ Reference(Patient|Practitioner|PractitionerRole|Device|Medication|Substance|Group) ]; #
    1..1 Reference to the group member
```

```
    fhir:Group.member.period [ Period ]; # 0..1 Period member belonged to the group
    fhir:Group.member.inactive [ boolean ]; # 0..1 If member is no longer in group
  ], ...;
]
```

Changes since Release 3

Group
Group.active | Default Value "true" removed
Group.type | Change value set from http://hl7.org/fhir/ValueSet/group-type to http://hl7.org/fhir/ValueSet/group-type|4.0.1
Group.managingEntity | Added Element
Group.characteristic.value[x] | Add Type Reference
Group.characteristic.exclude | No longer marked as Modifier
Group.member.entity | Type Reference: Added Target Types PractitionerRole, Group
Group.member.inactive | Default Value "false" removed

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.3.3.1 Terminology Bindings

Path | Definition | Type | Reference
Group.type | Types of resources that are part of group. | Required | GroupType
Group.code | Kind of particular resource; e.g. cow, syringe, lake, etc. | Unknown | No details provided yet
Group.characteristic.code | List of characteristics used to describe group members; e.g. gender, age, owner, location, etc. | Unknown | No details provided yet
Group.characteristic.value[x] | Value of descriptive member characteristic; e.g. red, male, pneumonia, Caucasian, etc. | Unknown | No details provided yet

8.3.3.2 Constraints

id | Level | Location | Description | Expression
grp-1 | Rule | (base) | Can only have members if group is "actual" | member.empty() or (actual = true)

8.3.3.3 Mixing Characteristics and Members

If both Group.characteristic and Group.member are present, then the members are the individuals who were found who met the characteristic. It's possible that there might be other candidate members who meet the characteristic and aren't (yet) in the list. All members SHALL have the listed characteristics.

8.3.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actual | token | Descriptive or actual | Group.actual
characteristic | token | Kind of characteristic | Group.characteristic.code
characteristic-value | composite | A composite of both characteristic and value | On Group.characteristic: characteristic: code value: value
code | token | The kind of resources contained | Group.code
exclude | token | Group includes or excludes | Group.characteristic.exclude
identifier | token | Unique id | Group.identifier
managing-entity | reference | Entity that is the custodian of the Group's definition | Group.managingEntity (Practitioner, Organization, PractitionerRole, RelatedPerson)
member | reference | Reference to the group member | Group.member.entity (Practitioner, Group, Device, Medication, Patient, Substance, PractitionerRole)
type | token | The type of resources the group contains | Group.type
value | token | Value held by characteristic | (Group.characteristic.value as CodeableConcept) | (Group.characteristic.value as boolean)

8.3.5 Resource Group - Examples

URL: https://hl7.org/fhir/R4/group-examples.html

8.3.5 Resource Group - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner

Example Name | id | Format
General Group Example, Characteristic Based | 101 | XML | JSON | Turtle
General Group Example, Member Based | 102 | XML | JSON | Turtle
Herd of 2500 breeding sows | herd1 | XML | JSON | Turtle

Patients primarily attributed to Practitioner 123 | example-patientlist | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.3.6 Resource Group - Detailed Descriptions

URL: https://hl7.org/fhir/R4/group-definitions.html

8.3.6 Resource Group - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Business | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Group resource.

Group
Element Id | Group
Definition | Represents a defined collection of entities that may be discussed or acted upon collectively but which are not expected to act collectively, and are not formally or legally recognized; i.e. a collection of entities that isn't an Organization.
Cardinality | 0..*
Type | DomainResource
Comments | If both Group.characteristic and Group.member are present, then the members are the individuals who were found who met the characteristic. It's possible that there might be other candidate members who meet the characteristic and aren't (yet) in the list. All members SHALL have the listed characteristics.
Invariants | Defined on this element grp-1 Rule Can only have members if group is "actual" member.empty() or (actual = true)
Defined on this element
grp-1 | Rule | Can only have members if group is "actual" | member.empty() or (actual = true)
Group.identifier
Element Id | Group.identifier
Definition | A unique business identifier for this group.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows the group to be referenced from external specifications.
Summary | true
Group.active
Element Id | Group.active
Definition | Indicates whether the record for the group is available for use or is merely being retained for historical purposes.
Cardinality | 0..1
Type | boolean
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a group record as not to be used because it was created in error or is otherwise no longer available (e.g. a herd that no longer exists).
Summary | true
Group.type
Element Id | Group.type
Definition | Identifies the broad classification of the kind of resources the group includes.
Cardinality | 1..1
Terminology Binding | GroupType (Required)
Type | code
Requirements | Identifies what type of resources the group is made up of.
Summary | true
Comments | Group members SHALL be of the appropriate resource type (Patient for person or animal; or Practitioner, Device, Medication or Substance for the other types.).
Group.actual
Element Id | Group.actual
Definition | If true, indicates that the resource refers to a specific group of real individuals. If false, the group defines a set of intended individuals.
Cardinality | 1..1
Type | boolean
Requirements | There are use-cases for groups that define specific collections of individuals, and other groups that define "types" of intended individuals. The requirements for both kinds of groups are similar, so we use a single resource, distinguished by this flag.
Summary | true
Invariants | Affect this element grp-1 Rule Can only have members if group is "actual" member.empty() or (actual = true)
Affect this element
grp-1 | Rule | Can only have members if group is "actual" | member.empty() or (actual = true)
Group.code
Element Id | Group.code
Definition | Provides a specific type of resource the group includes; e.g. "cow", "syringe", etc.
Cardinality | 0..1
Terminology Binding | GroupKind:
Type | CodeableConcept
Summary | true

Comments | This would generally be omitted for Person resources.
Group.name
Element Id | Group.name
Definition | A label assigned to the group for human identification and communication.
Cardinality | 0..1
Type | string
Requirements | Used to identify the group in human communication.
Summary | true
Group.quantity
Element Id | Group.quantity
Definition | A count of the number of resource instances that are part of the group.
Cardinality | 0..1
Type | unsignedInt
Requirements | Group size is a common defining characteristic.
Summary | true
Comments | Note that the quantity may be less than the number of members if some of the members are not active.
Group.managingEntity
Element Id | Group.managingEntity
Definition | Entity responsible for defining and maintaining Group characteristics and/or registered members.
Cardinality | 0..1
Type | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole)
Summary | true
Comments | This does not strictly align with ownership of a herd or flock, but may suffice to represent that relationship in simple cases. More complex cases will require an extension.
Group.characteristic
Element Id | Group.characteristic
Definition | Identifies traits whose presence r absence is shared by members of the group.
Cardinality | 0..*
Requirements | Needs to be a generic mechanism for identifying what individuals can be part of a group.
Comments | All the identified characteristics must be true for an entity to a member of the group.
Group.characteristic.code
Element Id | Group.characteristic.code
Definition | A code that identifies the kind of trait being asserted.
Cardinality | 1..1
Terminology Binding | GroupCharacteristicKind:
Type | CodeableConcept
Requirements | Need a formal way of identifying the characteristic being described.
Group.characteristic.value[x]
Element Id | Group.characteristic.value[x]
Definition | The value of the trait that holds (or does not hold - see 'exclude') for members of the group.
Cardinality | 1..1
Terminology Binding | GroupCharacteristicValue:
Type | CodeableConcept|boolean|Quantity|Range|Reference
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The value of the characteristic is what determines group membership.
Comments | For Range, it means members of the group have a value that falls somewhere within the specified range.
Group.characteristic.exclude
Element Id | Group.characteristic.exclude
Definition | If true, indicates the characteristic is one that is NOT held by members of the group.
Cardinality | 1..1
Type | boolean
Requirements | Sometimes group membership is determined by characteristics not possessed.
Comments | This is labeled as "Is Modifier" because applications cannot wrongly include excluded members as included or vice versa.
Group.characteristic.period
Element Id | Group.characteristic.period
Definition | The period over which the characteristic is tested; e.g. the patient had an operation during the month of June.
Cardinality | 0..1
Type | Period
Group.member
Element Id | Group.member
Definition | Identifies the resource instances that are members of the group.
Cardinality | 0..*
Requirements | Often the only thing of interest about a group is "who's in it".
Invariants | Affect this element grp-1 Rule Can only have members if group is "actual" member.empty() or (actual = true)
Affect this element
grp-1 | Rule | Can only have members if group is "actual" | member.empty() or (actual = true)
Group.member.entity
Element Id | Group.member.entity
Definition | A reference to the entity that is a member of the group. Must be consistent with Group.type. If the entity is another group, then the type must be the same.
Cardinality | 1..1

Type | Reference(Patient | Practitioner | PractitionerRole | Device | Medication | Substance | Group)
Group.member.period
Element Id | Group.member.period
Definition | The period that the member was in the group, if known.
Cardinality | 0..1
Type | Period
Meaning if Missing | The member is in the group at this time
Requirements | Need to track who was in a group at a particular time.
Group.member.inactive
Element Id | Group.member.inactive
Definition | A flag to indicate that the member is no longer in the group, but previously may have been a member.
Cardinality | 0..1
Type | boolean
Meaning if Missing | Members are considered active unless explicitly specified otherwise
Requirements | Sometimes you don't know when someone stopped being in a group, but not when.

8.3.7 Resource Group - Mappings

URL: https://hl7.org/fhir/R4/group-mappings.html

8.3.7 Resource Group - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the group resource (see Mappings to Other Standards for further information & status).

8.3.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Group
identifier | FiveWs.identifier
active | FiveWs.status
type | FiveWs.class
actual | FiveWs.class
code | FiveWs.what[x]

8.3.7.2 RIM Mapping (http://hl7.org/v3)

Group | Entity[determinerCode="GRP" or determinerCode="GRP_KIND"]
identifier | unique(./id[displayable="true" and scope="OBJ"])
active | ./statusCode
type | person: ./classCode="PSN" animal: ./classCode="ANM" device: ./classCode="DEV" medication: ./classCode="MMAT" and ./playedRole[isNormalRole() and classCode="THER"] food: ./classCode="FOOD"
actual | true: ./determinerCode="GRP" false: ./determinerCode="GRP_KIND"
code | ./code
name | ./name[type="ST"]
quantity | ./quantity
characteristic | ./playedRole[isNormalRole()]/participation[isNormalParticipation() and typeCode="SBJ"]/act[isNormalAct and classCode="OBS" and moodCode="EVN"]
code | ./code
value[x] | ./value
exclude | ./valueNegationInd
member | ./scopedRole[isNormalRole() and classCode="MBR"]/player

8.3.8 Resource Group - Extensions & Profiles

URL: https://hl7.org/fhir/R4/group-profiles.html

8.3.8 Resource Group - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the Group resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.3.8.1 Profiles

Profile | Description | Context
ActualGroup | Enforces an actual group, rather than a definitional group | Actual Group
GroupDefinition | Enforces a descriptive group that can be used in definitional resources | Group Definition

8.3.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.3.8.3 Search Extensions

No Search Extensions defined for this resource

8.3.9 Resource Group - Operations

URL: https://hl7.org/fhir/R4/group-operations.html

8.3.9 Resource Group - Operations

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Business | Compartments: Device, Patient, Practitioner

This resource has 1 operation operations associated with it:

$everything | Fetch a group of Patient Records

For more information about operations, including how they are invoked, see Operations.

8.3.9.1 Operation $everything on Group

URL: https://hl7.org/fhir/R4/group-operation-everything.html

8.3.9.1 Operation $everything on Group

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: N/A

This operation is used to return all the information related to one or more patients that are part of the group on which this operation is invoked. The response is a bundle of type "searchset". At a minimum, the patient resource(s) itself is returned, along with any other resources that the server has that are related to the patient(s), and that are available for the given user. The server also returns whatever resources are needed to support the records - e.g. linked practitioners, medications, locations, organizations etc. The intended use for this operation is for a provider or other user to perform a bulk data download. The server SHOULD return at least all resources that it has that are in the patient compartment for the identified patient(s), and any resource referenced from those, including binaries and attachments. In the US Realm, at a mimimum, the resources returned SHALL include all the data covered by the meaningful use common data elements as defined in US-Core . Other applicable implementation guides may make additional rules about how much information that is returned.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Group-everything
```

Formal Definition (as a OperationDefinition).

URL: [base]/Group/[id]/$everything

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
start | 0..1 | date | | | The date range relates to care dates, not record currency dates - e.g. all records relating to care provided in a certain date range. If no start date is provided, all records prior to the end date are in scope.
end | 0..1 | date | | | The date range relates to care dates, not record currency dates - e.g. all records relating to care provided in a certain date range. If no end date is provided, all records subsequent to the start date are in scope.
_since | 0..1 | instant | | | Resources updated after this period will be included in the response. The intent of this parameter is to allow a client to request only records that have changed since the last request, based on either the return header time, or or (for asynchronous use), the transaction time
_type | 0..* | code | | | One or more parameters, each containing one or more comma-delimited FHIR resource types to include in the return resources. In the absense of any specified types, the server returns all resource types
_count | 0..1 | integer | | | See discussion below on the utility of paging through the results of the $everything operation
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The bundle type is "searchset" Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this operation and simply searching the group's patients compartment are:

• unless the client requests otherwise, the server returns the entire result set in a single bundle (rather than using paging)

• the server is responsible for determining what resources to return as included resources (rather than the client specifying which ones).

This frees the client from needing to determine what it could or should ask for, particularly with regard to included resources.

It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a particular group, or determine whether the context has the rights to the nominated group, if there is one, or can determine an appropriate list of groups to provide data for from the context of the request. If there is no nominated group (GET /Group/$everything) and the context is not associated with a single group record, the actual list of groups is all groups that the user associated with the request has access to. In such cases, the server may choose to return an error rather than all the records (and is likely to do so, but not required to). Specifying the relationship between the context, a user and groups is outside the scope of this specification (though see The Smart App Launch Implementation Guide .

The return bundle from this operation is usually rather a lot of data; servers typically choose to require that such requests are made asynchronously, and associated with bulk data formats. Alternatively, clients may choose to page through the result set (or servers may require this). Paging through the results is done the same as for Searching, using the _count parameter, and Bundle links. Implementers should note that paging will be slower than simply returning all the results at once (more network traffic, multiple latency delays) but may be required in order not to exhaust available memory reading or writing the whole response in a single package. Unlike searching, there is no inherent user-display order for the $everything operation. Servers might consider sorting the returned resources in descending order of last record update, but are not required to do so.

The _since parameter is provided to support periodic queries to get additional information that has changed about the group since the last query. This means that the _since parameter is based on record time. The value of the _since parameter should be set to the time from the server. If using direct response, this is the timestamp in the response header. If using the async interface, this is the transaction timestamp in the json response. Servers should ensure that the timestamps a managed such that the client does not miss any changes. Clients should be able to handle getting the same response more than once in the case that the transaction falls on a time boundary. Clients should ensure that the other query parameters are constant to ensure a coherent set of records when doing periodic queries.

8.3.9.1.1 Examples

Request: Get all resources linked to a single patient

```
GET /open/Group/example/$everything
[some headers]
```

Response: Everything linked to the nominated patient (in this case, built automatically by the publication tooling)

```
HTTP/1.1 200 OK
[other headers]

<Bundle xml="http://hl7.org/fhir">
  <id value="1a969b36-120a-4306-a8bb-6e8fb0c2031f"/>
  <type value="searchset"/>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Patient/f001"/>
    <resource>
     <Patient xmlns="http://hl7.org/fhir">
       <id value="f001"/>
       <identifier>
         <use value="usual"/>
         <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!--  BSN identification system  -->
         <value value="738472983"/>
       </identifier>
       <identifier>
         <use value="usual"/>
         <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!--  BSN identification system  -->
       </identifier>
       <active value="true"/>
       <name>
         <use value="usual"/>
         <family value="van de Heuvel"/>
         <given value="Pieter"/>
         <suffix value="MSc"/>
       </name>
       <telecom>
         <system value="phone"/>
```

```
        <value value="0648352638"/>
        <use value="mobile"/>
      </telecom>
      <telecom>
        <system value="email"/>
        <value value="p.heuvel@gmail.com"/>
        <use value="home"/>
      </telecom>
      <gender value="male"/>
      <birthDate value="1944-11-17"/>
      <deceasedBoolean value="false"/>
      <address>
        <use value="home"/>
        <line value="Van Egmondkade 23"/>
        <city value="Amsterdam"/>
        <postalCode value="1024 RJ"/>
        <country value="NLD"/>
      <!--  ISO 3166 Codes (Countries)  -->
      </address>
      <maritalStatus>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-MaritalStatus"/>
          <code value="M"/>
          <display value="Married"/>
        </coding>
        <text value="Getrouwd"/>
      </maritalStatus>
      <multipleBirthBoolean value="true"/>
      <contact>
        <relationship>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v2-0131"/>
            <code value="C"/>
          </coding>
        </relationship>
        <name>
          <use value="usual"/>
          <family value="Abels"/>
          <given value="Sarah"/>
        </name>
        <telecom>
          <system value="phone"/>
          <value value="0690383372"/>
          <use value="mobile"/>
        </telecom>
      </contact>
      <communication>
        <language>
          <coding>
            <system value="urn:ietf:bcp:47"/>
          <!--  IETF language tag  -->
            <code value="nl"/>
            <display value="Dutch"/>
          </coding>
          <text value="Nederlands"/>
        </language>
        <preferred value="true"/>
      </communication>
      <managingOrganization>
        <reference value="Organization/f001"/>
        <display value="Burgers University Medical Centre"/>
      </managingOrganization>
    </Patient>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f003"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f003"/><!--   urn:oid:2.16.840.1.113883.4.642.1.7   --><!--   2.16.840.1.113883.4.642.1.118
  --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6325"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="11557-6"/>
        <display value="Carbon dioxide in blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
```

```
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="6.2"/>
      <unit value="kPa"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="kPa"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="4.8"/>
        <unit value="kPa"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kPa"/>
      </low>
      <high>
        <value value="6.0"/>
        <unit value="kPa"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kPa"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f003"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="172960003"/>
               <display value="Incision of retropharyngeal abscess"/>
           </coding>
       </code>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
    <encounter>
           <reference value="Encounter/f003"/>
       </encounter>
    <performedPeriod>
           <start value="2013-03-24T09:30:10+01:00"/>
           <end value="2013-03-24T10:30:10+01:00"/>
       </performedPeriod>
    <performer>
           <function>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
           </function>
      <actor>
               <reference value="Practitioner/f001"/>
               <display value="E.M.J.M. van den broek"/>
           </actor>
       </performer>
       <reasonCode>
         <text value="abcess in retropharyngeal area"/>
       </reasonCode>
       <bodySite>

               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="83030008"/>
                   <display value="Retropharyngeal area"/>
               </coding>

       </bodySite>
       <outcome>
         <text value="removal of the retropharyngeal abscess"/>
```

```
      </outcome>
      <report>
          <reference value="DiagnosticReport/f001"/>
          <display value="Lab results blood test"/>
      </report>
      <followUp>
        <text value="described in care plan"/>
      </followUp>
   </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notTime"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notTime"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 6. Withhold or withdraw consent for disclosure of records related published during a
  timeframe
    </p><p>
       Patient "P. van de Heuvel" wishes to withhold disclosure of all data from a timeframe to any provider.
      </p>
     </div>
    </text>
    <status value="active"/>
    <scope>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
        <code value="patient-privacy"/>
      </coding>
      </scope>
        <category>
      <coding>
        <system value="http://loinc.org"/>
        <code value="59284-0"/>
      </coding>
    </category>
    <patient>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </patient>
    <dateTime value="2015-11-18"/>
    <!-- not bound by a timeframe - Consent.period -->
    <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
    <organization>
      <reference value="Organization/f001"/>
    </organization>
    <!-- the text terms of the consent in lawyer speak -->
    <sourceAttachment>
      <title value="The terms of the consent in lawyer speak."/>
      <!-- likely use url pointer to common text -->
    </sourceAttachment>
    <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
    <policyRule>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
        <code value="OPTIN"/>
      </coding>
    </policyRule>
    <provision>
      <period>
        <start value="2015-01-01"/>
        <end value="2015-02-01"/>
      </period>
    </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notOrg"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notOrg"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
```

```
        or withholding consent for that purpose. In other jurisdictions, where an express consent
        model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
        "withhold" or "withdraw" consent for. <p> specific to use-case 3) Withhold or withdraw consent
          for disclosure to a specific provider organization </p><p> Patient "P. van de Heuvel"
          ex-spouse is a Physician that is employed by Good Health Clinic. P. decides that he would
          like to withdraw/withhold consent to disclose any instance of his health information to Good
          Health Clinic. </p>
      </div>
    </text>

    <issued value="2015-11-18"/>
    <!-- not bound by a timeframe - Contract.applies -->

    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>

    <authority>
      <reference value="Organization/3"/>
      <display value="Michigan Health"/>
    </authority>

    <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
    <domain>
      <reference value="Location/ukp"/>
      <display value="UK Pharmacies"/>
    </domain>

    <type>
      <coding>
        <system value="http://loinc.org"/>
        <code value="57016-8"/>
      </coding>
    </type>

    <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
    <subType>
      <coding>
        <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
        <code value="Opt-In"/>
        <display value="Default Authorization with exceptions."/>
      </coding>
    </subType>

    <term>
      <type>
        <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-from"/>
          <display value="Withhold all data from specified actor entity."/>
        </coding>
      </type>
      <offer>
        <topic>
          <reference value="Organization/2.16.840.1.113883.19.5"/>
          <display value="Good Health Clinic"/>
        </topic>
        <!-- by not specifying term.actor I assume this means everyone? -->
        <text value="Withhold this order and any results or related objects from any provider."/>
      </offer>
    </term>

    <!-- the terms of the consent in friendly consumer speak -->
    <friendly>
      <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
    </friendly>

    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f003"/>
  <resource>
```

```
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f003"/>
     <clinicalStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
         <code value="active"/>
       </coding>
     </clinicalStatus>

     <verificationStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
         <code value="confirmed"/>
       </coding>
     </verificationStatus>

     <category>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="439401001"/>
         <display value="diagnosis"/>
       </coding>
     </category>
     <severity>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="371923003"/>
         <display value="Mild to moderate"/>
       </coding>
     </severity>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="18099001"/>
         <display value="Retropharyngeal abscess"/>
       </coding>
     </code>
     <bodySite>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="280193007"/>
         <display value="Entire retropharyngeal area"/>
       </coding>
     </bodySite>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <encounter>
       <reference value="Encounter/f003"/>
     </encounter>
     <onsetDateTime value="2012-02-27"/>
     <recordedDate value="2012-02-20"/>
     <asserter>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </asserter>
     <evidence>
       <code>
         <coding>
           <system value="http://snomed.info/sct"/>
           <code value="169068008"/>
           <display value="CT of neck"/>
         </coding>
       </code>
     </evidence>
   </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-Emergency"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-Emergency"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <p>
    Withhold Authorization for Treatment rule for Emegency Treatment
    </p>
         <p>
      Patient "P. van de Heuvel" wishes to have no data at the Good Health Psychiatric Hospital
      available rule for Emergency treatment use.
      An overall Deny consent Directive, with an exception "Permit" of purposeOfUse "ETREAT"
      at "Good Health" hospital, and exception "Deny" all other purposeOfUse from good health hospital.
      </p>
       </div>
     </text>
     <status value="active"/>
```

```
      <scope>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
          <code value="patient-privacy"/>
        </coding>
            </scope>
     <category>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
          <code value="EMRGONLY"/>
        </coding>
      </category>
      <patient>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </patient>
      <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->
      <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
    Consent.authority element -->
      <organization>
        <reference value="Organization/f001"/>
      </organization>
      <!-- the text terms of the consent in lawyer speak -->
      <sourceAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
      </sourceAttachment>
      <!--
        this is opt-out - e.g. nothing approved unless otherwise stated.
      -->
      <policyRule>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
          <code value="OPTOUT"/>
        </coding>
      </policyRule>
      <provision>
        <actor>
          <role>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
              <code value="CST"/>
            </coding>
          </role>
          <reference>
            <reference value="Organization/f001"/>
          </reference>
        </actor>
        <purpose>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ActReason"/>
          <code value="ETREAT"/>
        </purpose>
        <!-- this rule overrides the other one -->
        <provision>
          <type value="deny"/>
          <actor>
            <role>
              <coding>
                <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
                <code value="CST"/>
              </coding>
            </role>
            <reference>
              <reference value="Organization/f001"/>
            </reference>
          </actor>
        </provision>
      </provision>
    </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f001"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
      <id value="f001"/>
        <status value="completed"/>
        <code>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="34068001"/>
                <display value="Heart valve replacement"/>
            </coding>
        </code>
     <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
```

```
    <encounter>
           <reference value="Encounter/f001"/>
       </encounter>
       <performedPeriod>
           <start value="2011-06-26"/>
           <end value="2011-06-27"/>
       </performedPeriod>
    <performer>
           <function>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
           </function>
      <actor>
                <reference value="Practitioner/f002"/>
                <display value="P. Voigt"/>
           </actor>
       </performer>
    <reasonCode>
         <text value="Heart valve disorder"/>
       </reasonCode>
       <bodySite>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="17401000"/>
                   <display value="Heart valve structure"/>
               </coding>
       </bodySite>
       <outcome>
         <text value="improved blood circulation"/>
       </outcome>
       <report>
           <reference value="DiagnosticReport/f001"/>
           <display value="Lab results blood test"/>
       </report>
       <followUp>
         <text value="described in care plan"/>
       </followUp>
   </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notThem"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notThem"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 4) Withhold or withdraw consent for disclosure to a specific provider agent (an individual
  within an organization)
    </p><p>
       Patient "P. van de Heuvel" ex-spouse, Bill T Lookafter is a Nurse that is employed by Good Health Hospital.
      P. realizes that she may receive treatment at GHH, however she does not want her ex-spouse to have any
      access to her IIHI.  She indicates that she would like to withdraw/withhold consent to disclose any instance
      of her health information to her ex-spouse as a result of his employment at Good Health Clinic.
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
```

```
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <!-- no read or write access for this individual -->
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="PRCP"/>
             <!-- PRCP: recipient of the information -->
           </coding>
         </role>
         <reference>
           <reference value="Practitioner/f204"/>
           <display value="Fictive Nurse"/>
         </reference>
       </actor>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="access"/>
         </coding>
       </action>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="correct"/>
         </coding>
       </action>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notAuthor"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notAuthor"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
         or withholding consent for that purpose. In other jurisdictions, where an express consent
         model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
         "withhold" or "withdraw" consent for. <p> specific to use-case 5) Withhold or withdraw consent
           for disclosure of records that were authored by a specific organization (or service delivery
           location). </p><p> Patient "P. van de Heuvel" wishes to have all of the PHI collected at the
           Good Health Psychiatric Hospital restricted from disclosure to every provider. </p>
       </div>
     </text>

     <issued value="2015-11-18"/>
     <!-- not bound by a timeframe - Contract.applies -->

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <authority>
       <reference value="Organization/3"/>
       <display value="Michigan Health"/>
     </authority>

     <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
     <domain>
       <reference value="Location/ukp"/>
       <display value="UK Pharmacies"/>
     </domain>

     <type>
       <coding>
         <system value="http://loinc.org"/>
```

```
        <code value="57016-8"/>
      </coding>
    </type>

    <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
    <subType>
      <coding>
        <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
        <code value="Opt-In"/>
        <display value="Default Authorization with exceptions."/>
      </coding>
    </subType>

    <term>
      <type>
        <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-authored-by"/>
          <display value="Withhold all data authored by specified actor entity."/>
        </coding>
      </type>
      <offer>
        <topic>
          <reference value="Organization/2.16.840.1.113883.19.5"/>
          <display value="Good Health Clinic"/>
        </topic>

        <text value="Withhold all data authored by Good Health provider."/>
      </offer>
    </term>

    <!-- the terms of the consent in friendly consumer speak -->
    <friendly>
      <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
    </friendly>

    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-grantor"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
    <id value="consent-example-grantor"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on a question asked on John Moehrke's
   blog in the
    <a href="http://healthcaresecprivacy.blogspot.com/2016/04/consent-given-to-authorized.html">Consent given to
  authorized representative</a>
      <p>
   This is a case where a Patient is giving a limited (read-only) access to an individual.
    </p><p>
      Patient "P. van de Heuvel" wishes to provide XXX with read-only access to the data at Good Health Psychiatric
  Hospital
     available for patient requested access use.
     </p>
     </div>
    </text>
    <status value="active"/>
    <scope>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
        <code value="patient-privacy"/>
      </coding>
         </scope>
   <category>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
        <code value="INFAO"/>
      </coding>
    </category>
    <patient>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
```

```
      </patient>
      <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->
      <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
    Consent.authority element -->
      <organization>
        <reference value="Organization/f001"/>
      </organization>
      <!-- the text terms of the consent in lawyer speak -->
      <sourceAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
      </sourceAttachment>
      <!-- this is opt-out - e.g. nothing approved unless otherwise stated -->
      <policyRule>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
          <code value="OPTOUT"/>
        </coding>
      </policyRule>
      <provision>
        <actor>
          <role>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
              <code value="CST"/>
            </coding>
          </role>
          <reference>
            <reference value="Organization/f001"/>
          </reference>
        </actor>
        <actor>
          <role>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
              <code value="PRCP"/>
            </coding>
          </role>
          <reference>
            <reference value="Patient/example"/>
            <display value="Good Health Clinic"/>
          </reference>
        </actor>
        <action>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
            <code value="access"/>
          </coding>
        </action>
      </provision>
    </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/unsat"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
     <id value="unsat"/>
   <!--  urn:oid:2.16.840.1.113883.4.642.1.7  -->
   <!--  2.16.840.1.113883.4.642.1.118  -->
     <identifier>
       <use value="official"/>
       <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
       <value value="6323"/>
     </identifier>
     <status value="cancelled"/>
     <code>
       <coding>
         <system value="http://loinc.org"/>
         <code value="15074-8"/>
         <display value="Glucose [Moles/volume] in Blood"/>
       </coding>
     </code>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <effectivePeriod>
       <start value="2013-04-02T09:30:10+01:00"/>
       <end value="2013-04-05T09:30:10+01:00"/>
     </effectivePeriod>
     <issued value="2013-04-03T15:30:10+01:00"/>
     <performer>
       <reference value="Practitioner/f005"/>
       <display value="A. Langeveld"/>
     </performer>
     <dataAbsentReason>
       <coding>
```

```
          <system value="http://snomed.info/sct"/>
          <code value="125154007"/>
          <display value="Specimen unsatisfactory for evaluation"/>
        </coding>
      </dataAbsentReason>
      <note><text value="Tube broken in transit and sample leaked"/></note>
      <referenceRange>
        <low>
          <value value="3.1"/>
          <unit value="mmol/l"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mmol/L"/>
        </low>
        <high>
          <value value="6.2"/>
          <unit value="mmol/l"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mmol/L"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-basic"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
    <id value="consent-example-basic"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">
        <p>
    Authorize Normal access for Treatment
        </p>
        <p>
       Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
      available for normal treatment use.
        </p>
      </div>
    </text>
    <status value="active"/>
    <scope>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
        <code value="patient-privacy"/>
      </coding>
    </scope>
    <category>
      <coding>
        <system value="http://loinc.org"/>
        <code value="59284-0"/>
      </coding>
    </category>
    <patient>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </patient>
    <dateTime value="2016-05-11"/>
    <organization>
      <reference value="Organization/f001"/>
    </organization>
    <!-- the terms of the consent as signed by the patient in ink -->
    <sourceAttachment>
      <title value="The terms of the consent in lawyer speak."/>
      <!-- likely use url pointer to common text -->
    </sourceAttachment>
    <!-- this is plain out opt-in with no exceptions -->
    <policyRule>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
        <code value="OPTIN"/>
      </coding>
    </policyRule>
    <provision>
      <!-- bound by a timeframe between 1964 and 2016 - consent.applies -->
      <period>
        <start value="1964-01-01"/>
        <end value="2016-01-01"/>
      </period>
    </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-Out"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-Out"/>
     <text>
```

```
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
      <p>
    Specific to use-case 0) Withhold Authorization of sharing data for Treatment
    </p><p>
       Patient "P. van de Heuvel" wishes to have no data at the Good Health Psychiatric Hospital
      available for normal treatment use.
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-out - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTOUT"/>
       </coding>
     </policyRule>
     <provision>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
           <reference value="Organization/f001"/>
         </reference>
       </actor>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f003"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f001"/>
            <display value="E.M. van den broek"/>
            </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <lifecycleStatus value="completed"/>
      <achievementStatus>
```

```
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/goal-achievement"/>
        <code value="achieved"/>
        <display value="Achieved"/>
      </coding>
      <text value="Achieved"/>
      </achievementStatus>
         <description>
           <text value="Retropharyngeal abscess removal"/>
         </description>
         <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished without complications"/>
         </note>
       </Goal>
     </contained>
     <identifier>
       <use value="official"/>
       <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
       <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
       <value value="CP3953"/>
     </identifier>
     <status value="completed"/>
     <intent value="plan"/>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <period>
       <start value="2013-03-08T09:00:10+01:00"/>
       <end value="2013-03-08T09:30:10+01:00"/>
     </period>
     <careTeam>
       <reference value="#careteam"/>
     </careTeam>
     <addresses>
       <reference value="Condition/f201"/>
       <!--TODO Correcte referentie-->
       <display value="?????"/>
     </addresses>
     <goal>
       <reference value="#goal"/>
     </goal>
     <activity>
       <detail>
         <kind value="ServiceRequest"/>
         <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
         <code>
           <coding>
             <system value="http://snomed.info/sct"/>
             <code value="172960003"/>
             <display value="Incision of retropharyngeal abscess"/>
           </coding>
         </code>
         <status value="completed"/>
         <doNotPerform value="true"/>
         <scheduledString value="2011-06-27T09:30:10+01:00"/>
         <performer>
           <reference value="Practitioner/f001"/>
           <display value="E.M. van den broek"/>
         </performer>
       </detail>
     </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notAuthor"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notAuthor"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 5) Withhold or withdraw consent for disclosure of records that were authored by a specific
  organization (or service delivery location).
    </p><p>
       Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
      restricted from disclosure to every provider.
```

```
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
       <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
           <reference value="Organization/f001"/>
         </reference>
       </actor>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f002"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
     <id value="f002"/>
       <identifier>
           <use value="official"/>
           <system value="http://www.bmc.nl/zorgportal/identifiers/encounters"/>
           <value value="v3251"/>
       </identifier>
       <status value="finished"/>
       <class>
           <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
           <code value="AMB"/> <!-- outpatient -->
           <display value="ambulatory"/>
       </class>
       <type>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="270427003"/>
               <display value="Patient-initiated encounter"/>
           </coding>
       </type>
       <priority>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="103391001"/>
               <display value="Urgent"/>
           </coding>
       </priority>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
       <participant>
           <individual>
```

```
                <reference value="Practitioner/f003"/>
                <display value="M.I.M Versteegh"/>
            </individual>
        </participant>
        <length>
            <value value="140"/>
            <unit value="min"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="min"/>
        </length>
        <reasonCode>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="34068001"/>
                <display value="Partial lobectomy of lung"/>
            </coding>
        </reasonCode>
        <hospitalization>
            <preAdmissionIdentifier>
                <use value="official"/>
                <system value="http://www.bmc.nl/zorgportal/identifiers/pre-admissions"/>
                <value value="98682"/>
            </preAdmissionIdentifier>
        <!--    <preAdmissionTest>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="399208008"/>
                    <display value="Chest X-ray"/>
                </coding>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="396550006"/>
                    <display value="Blood test"/>
                </coding>
            </preAdmissionTest> -->
            <admitSource>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="305997006"/>
                    <display value="Referral by radiologist"/>
                </coding>
            </admitSource>

            <dischargeDisposition>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="306689006"/>
                    <display value="Discharge to home"/>
                </coding>
            </dischargeDisposition>
        </hospitalization>
        <serviceProvider>
            <reference value="Organization/f001"/>
            <display value="BMC"/>
        </serviceProvider>
   </Encounter>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f002"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f002"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="359615001"/>
               <display value="Partial lobectomy of lung"/>
           </coding>
       </code>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
    <encounter>
           <reference value="Encounter/f002"/>
       </encounter>
       <performedPeriod>
           <start value="2013-03-08T09:00:10+01:00"/>
           <end value="2013-03-08T09:30:10+01:00"/>
       </performedPeriod>
    <performer>
           <function>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
```

```
            </function>
        <actor>
                <reference value="Practitioner/f003"/>
                <display value="M.I.M. Versteegh"/>
            </actor>
        </performer>
      <reasonCode>
          <text value="Malignant tumor of lung"/>
        </reasonCode>
        <bodySite>

                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="39607008"/>
                    <display value="Lung structure"/>
                </coding>

        </bodySite>
        <outcome>
          <text value="improved blood circulation"/>
        </outcome>
        <report>
            <reference value="DiagnosticReport/f001"/>
            <display value="Lab results blood test"/>
        </report>
        <followUp>
          <text value="described in care plan"/>
        </followUp>
    </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f004"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f004"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="48387007"/>
               <display value="Tracheotomy"/>
           </coding>
       </code>
     <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
     <encounter>
           <reference value="Encounter/f003"/>
       </encounter>
       <performedPeriod>
           <start value="2013-03-22T09:30:10+01:00"/>
           <end value="2013-03-22T10:30:10+01:00"/>
       </performedPeriod>
       <performer>
           <function>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
           </function>
       <actor>
                <reference value="Practitioner/f005"/>
                <display value="A. Langeveld"/>
            </actor>
       </performer>
       <reasonCode>
         <text value="ensure breathing during surgery"/>
       </reasonCode>
       <bodySite>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="83030008"/>
               <display value="Retropharyngeal area"/>
           </coding>
       </bodySite>
       <outcome>
         <text value="removal of the retropharyngeal abscess"/>
       </outcome>
       <report>
           <reference value="DiagnosticReport/f001"/>
           <display value="???????????"/>
       </report>
       <followUp>
         <text value="described in care plan"/>
       </followUp>
```

```
   </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/List/long"/>
  <resource>
   <List xmlns="http://hl7.org/fhir">
     <id value="long"/>
     <status value="current"/>
     <mode value="changes"/>
     <date value="2018-02-21T12:17:00+11:00"/>
     <entry>
       <item>
         <reference value="Patient/example"/>
       </item>
     </entry>
     <entry>
       <item>
         <reference value="Patient/pat1"/>
       </item>
     </entry>
     <entry>
       <item>
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```

```
      <entry>
        <item>
          <reference value="Patient/f001"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/f201"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/glossy"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/proband"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/genetics-example1"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/ch-example"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/newborn"/>
        </item>
      </entry>
      <entry>
        <item>
          <reference value="Patient/mom"/>
        </item>
      </entry>
    </List>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f001"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f001"/>
     <clinicalStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
         <code value="active"/>
       </coding>
     </clinicalStatus>

     <verificationStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
         <code value="confirmed"/>
       </coding>
     </verificationStatus>
     <category>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="439401001"/>
         <display value="diagnosis"/>
       </coding>
     </category>
     <severity>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="6736007"/>
         <display value="Moderate"/>
       </coding>
     </severity>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="368009"/>
         <display value="Heart valve disorder"/>
       </coding>
     </code>
     <bodySite>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="40768004"/>
         <display value="Left thorax"/>
       </coding>
       <text value="heart structure"/>
```

```
      </bodySite>
      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>
      <encounter>
        <reference value="Encounter/f001"/>
      </encounter>
      <onsetDateTime value="2011-08-05"/>
      <recordedDate value="2011-10-05"/>
      <asserter>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </asserter>
      <evidence>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="426396005"/>
            <display value="Cardiac chest pain"/>
          </coding>
        </code>
      </evidence>
    </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f001"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f001"/><!--  urn:oid:2.16.840.1.113883.4.642.1.7  --><!--  2.16.840.1.113883.4.642.1.118
  --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6323"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="15074-8"/>
        <display value="Glucose [Moles/volume] in Blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T09:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="6.3"/>
      <unit value="mmol/l"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mmol/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="3.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="6.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notOrg"/>
  <resource>
```

```
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notOrg"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
    are realized in existing systems in Canada.
     The default policy is one of implied consent for the provision of care, so these scenarios all deal with
   withdrawal or withholding consent for that purpose.
     In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
   phrase "consent to" rather
     than "withhold" or "withdraw" consent for.
        <p>
     specific to use-case 3) Withhold or withdraw consent for disclosure to a specific provider organization
     </p><p>
        Patient "P. van de Heuvel" ex-spouse is a Physician that is employed by Good Health Clinic.  P. decides
       that he would like to withdraw/withhold consent to disclose any instance of his health information
       to Good Health Clinic.
       </p>
       </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
   Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <!-- no read or write access for this org -->
       <type value="deny"/>
       <actor>
         <role>
           <coding>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
             <code value="PRCP"/>
             <!-- PRCP: recipient of the information -->
           </coding>
         </role>
         <reference>
           <reference value="Organization/f001"/>
         </reference>
       </actor>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="access"/>
         </coding>
       </action>
       <action>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/consentaction"/>
           <code value="correct"/>
         </coding>
       </action>
     </provision>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RelatedPerson/f001"/>
```

```
  <resource>
   <RelatedPerson xmlns="http://hl7.org/fhir">
    <id value="f001"/>
     <text>

       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
        Sarah Abels
       </div>
     </text>

     <identifier>
       <use value="official"/>
       <type><text value="BSN"/></type>
       <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!-- Dutch BSN identification system -->
     </identifier>

     <patient>
       <reference value="Patient/f001"/>
     </patient>

     <relationship>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
         <code value="SIGOTHR"/>
       </coding>
     </relationship>

     <name>
       <use value="usual"/>
       <family value="Abels"/>
       <given value="Sarah"/>
     </name>
     <telecom>
       <system value="phone"/>
       <value value="0690383372"/>
       <use value="mobile"/>
     </telecom>
     <telecom>
       <system value="email"/>
       <value value="s.abels@kpn.nl"/>
       <use value="home"/>
     </telecom>
     <gender value="female"/>

   </RelatedPerson>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f002"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f002"/>
     <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f003"/>
            <display value="M.I.M. Versteegh"/>
            </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <lifecycleStatus value="completed"/>
      <achievementStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/goal-achievement"/>
        <code value="achieved"/>
        <display value="Achieved"/>
      </coding>
      <text value="Achieved"/>
      </achievementStatus>
         <description>
           <text value="succesful surgery on lung of patient"/>
         </description>
         <subject>
    <reference value="Patient/f001"/>
    <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished with minor complications"/>
         </note>
       </Goal>
     </contained>
```

```
      <identifier>
        <use value="official"/>
        <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
        <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
        <value value="CP2934"/>
      </identifier>
      <status value="completed"/>
      <intent value="plan"/>
      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>
      <period>
        <start value="2011-07-06"/>
        <end value="2013-07-07"/>
      </period>
      <careTeam>
        <reference value="#careteam"/>
      </careTeam>
      <addresses>
        <reference value="Condition/f201"/>
        <!--TODO Correcte referentie-->
        <display value="?????"/>
      </addresses>
      <goal>
        <reference value="#goal"/>
      </goal>
      <activity>
        <detail>
          <kind value="ServiceRequest"/>
          <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
          <code>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="359615001"/>
              <display value="Partial lobectomy of lung"/>
            </coding>
          </code>
          <status value="completed"/>
          <doNotPerform value="true"/>
          <scheduledString value="2011-07-07T09:30:10+01:00"/>
          <performer>
            <reference value="Practitioner/f003"/>
            <display value="M.I.M. Versteegh"/>
          </performer>
        </detail>
      </activity>
    </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example3"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
     <id value="example3"/>

     <status value="draft"/>
     <intent value="order"/>
     <code>
       <text value="Refill Request"/>
     </code>
     <focus>
       <reference value="MedicationRequest/medrx002"/>
     </focus>
     <for>
       <reference value="Patient/f001"/>
     </for>
     <authoredOn value="2016-03-10T22:39:32-04:00"/>
     <lastModified value="2016-03-10T22:39:32-04:00"/>
     <requester>
         <reference value="Patient/example"/>
     </requester>
     <owner>
       <reference value="Practitioner/example"/>
     </owner>
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f001"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f001"/>
     <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f002"/>
```

```
            <display value="P. Voigt"/>
               </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <lifecycleStatus value="completed"/>
      <achievementStatus>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/goal-achievement"/>
        <code value="achieved"/>
        <display value="Achieved"/>
      </coding>
      <text value="Achieved"/>
      </achievementStatus>
         <description>
           <text value="recovery surgery on heart of patient"/>
         </description>
         <subject>
    <reference value="Patient/f001"/>
    <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished without complications"/>
         </note>
       </Goal>
     </contained>
     <identifier>
       <use value="official"/>
       <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
       <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
       <value value="CP2903"/>
     </identifier>
     <status value="completed"/>
     <intent value="plan"/>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <period>
       <start value="2011-06-26"/>
       <end value="2011-06-27"/>
     </period>
     <careTeam>
       <reference value="#careteam"/>
     </careTeam>
     <addresses>
       <reference value="Condition/f201"/>
       <!--TODO Correcte referentie-->
       <display value="?????"/>
     </addresses>
     <goal>
       <reference value="#goal"/>
     </goal>
     <activity>
       <detail>
         <kind value="ServiceRequest"/>
         <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
         <code>
           <coding>
             <system value="http://snomed.info/sct"/>
             <code value="64915003"/>
             <display value="Operation on heart"/>
           </coding>
         </code>
         <status value="completed"/>
         <doNotPerform value="true"/>
         <scheduledString value="2011-06-27T09:30:10+01:00"/>
         <performer>
           <reference value="Practitioner/f002"/>
           <display value="P. Voigt"/>
         </performer>
       </detail>
     </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f002"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f002"/><!--  urn:oid:2.16.840.1.113883.4.642.1.7  --><!--  2.16.840.1.113883.4.642.1.118
  --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6324"/>
    </identifier>
```

```
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="11555-0"/>
        <display value="Base excess in Blood by calculation"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="12.6"/>
      <unit value="mmol/l"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mmol/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="7.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="11.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notThis"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notThis"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
         or withholding consent for that purpose. In other jurisdictions, where an express consent
         model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
         "withhold" or "withdraw" consent for. <p> specific to use-case 2. Withhold or withdraw consent
           for disclosure of a specific record (e.g. Lab Order/Result) </p><p> Patient "P. van de
           Heuvel" Primary Care Provider, Dr. Philip Primary, has ordered a set of lab test which Adam
           wishes to keep as private as possible. At the sample collection facility, he indicates that
           he would like withhold consent to disclose the order and all results associated with that
           specific order from all other providers </p>
       </div>
     </text>

     <issued value="2015-11-18"/>
     <!-- not bound by a timeframe - Contract.applies -->

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <authority>
       <reference value="Organization/3"/>
       <display value="Michigan Health"/>
     </authority>

     <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
     <domain>
```

```
      <reference value="Location/ukp"/>
      <display value="UK Pharmacies"/>
    </domain>

    <type>
      <coding>
        <system value="http://loinc.org"/>
        <code value="57016-8"/>
      </coding>
    </type>

    <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
    <subType>
      <coding>
        <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
        <code value="Opt-In"/>
        <display value="Default Authorization with exceptions."/>
      </coding>
    </subType>

    <!-- I can easily say to exclude a type of FHIR Resource, but lab-orders is not a direct Resource type. Might we
  put in FHIR based query parameters? -->
    <term>
      <identifier>
        <system value="http://example.org/fhir/term-items"/>
        <value value="3347689"/>
      </identifier>
      <issued value="2015-11-01"/>
      <applies>
        <start value="2015-11-18"/>
      </applies>
      <type>
        <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-identified-object-and-related"/>
          <display value="Withhold the identified object and any other resources that are related to this object."/>
        </coding>
      </type>
      <offer>
        <topic>
          <reference value="ServiceRequest/lipid"/>
        </topic>
        <!-- by not specifying term.actor I assume this means everyone? -->
        <text value="Withhold this order and any results or related objects from any provider."/>
      </offer>
    </term>

    <!-- the terms of the consent in friendly consumer speak -->
    <friendly>
      <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
    </friendly>

    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notLabs"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
    <id value="pcd-example-notLabs"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
        jurisdictional policy and are realized in existing systems in Canada. The default policy is
        one of implied consent for the provision of care, so these scenarios all deal with withdrawal
        or withholding consent for that purpose. In other jurisdictions, where an express consent
        model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
        "withhold" or "withdraw" consent for. <p> specific to use-case 1. Withhold or withdraw consent
          for disclosure of records related to specific domain (e.g. DI, LAB, etc.) </p><p> Patient
          "P. van de Heuvel" wishes to withhold disclosure of all of her lab orders and results to any
          provider. </p>
      </div>
    </text>
```

```
   <issued value="2014-08-17"/>
   <!-- not bound by a timeframe - Contract.applies -->

   <subject>
     <reference value="Patient/f001"/>
     <display value="P. van de Heuvel"/>
   </subject>

   <authority>
     <reference value="Organization/3"/>
     <display value="Michigan Health"/>
   </authority>

   <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
Juristiction) -->
   <domain>
     <reference value="Location/ukp"/>
     <display value="UK Pharmacies"/>
   </domain>

   <type>
     <coding>
       <system value="http://loinc.org"/>
       <code value="57016-8"/>
     </coding>
   </type>

   <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
-->
   <subType>
     <coding>
       <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
       <code value="Opt-In"/>
       <display value="Default Authorization with exceptions."/>
     </coding>
   </subType>

   <!-- I can easily say to exclude a type of FHIR Resource, but lab-orders is not a direct Resource type. Might we
put in FHIR based query parameters? -->
   <term>
     <offer>
       <text value="sample"/>
     </offer>
     <group>
       <type>
         <coding>
           <!-- made up code system -->
           <system value="http://example.org/fhir/consent-term-type-codes"/>
           <code value="withhold-object-type"/>
         </coding>
       </type>
       <subType>
         <coding>
           <system value="http://hl7.org/fhir/resource-types"/>
           <code value="ServiceRequest"/>
         </coding>
       </subType>
       <!-- by not specifying term.actor I assume this means everyone? -->
       <offer>
         <text value="Withhold orders from any provider."/>
       </offer>
     </group>
     <group>
       <type>
         <coding>
           <!-- made up code system -->
           <system value="http://example.org/fhir/consent-term-type-codes"/>
           <code value="withhold-object-type"/>
         </coding>
       </type>
       <subType>
         <coding>
           <system value="http://hl7.org/fhir/resource-types"/>
           <code value="DiagnosticReport"/>
         </coding>
       </subType>
       <!-- by not specifying term.actor I assume this means everyone? -->
       <offer>
         <text value="Withhold order results from any provider."/>
       </offer>
     </group>
   </term>

   <!-- the terms of the consent in friendly consumer speak -->
   <friendly>
     <contentAttachment>
       <title value="The terms of the consent in friendly consumer speak."/>
       <!-- likely use url pointer to common text -->
     </contentAttachment>
```

```
     </friendly>

     <!-- the legal terms of the consent in lawyer speak -->
     <legal>
       <contentAttachment>
         <title value="The terms of the consent in lawyer speak."/>
         <!-- likely use url pointer to common text -->
         <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
       </contentAttachment>
     </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f003"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
    <id value="f003"/>
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/encounters"/>
      <value value="v6751"/>
    </identifier>
    <status value="finished"/>
    <class>
      <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
      <code value="AMB"/>
      <!-- outpatient -->
      <display value="ambulatory"/>
    </class>
    <type>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="270427003"/>
        <display value="Patient-initiated encounter"/>
      </coding>
    </type>
    <priority>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="103391001"/>
        <display value="Non-urgent ear, nose and throat admission"/>
      </coding>
    </priority>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <participant>
      <individual>
        <reference value="Practitioner/f001"/>
        <display value="E.M. van den Broek"/>
      </individual>
    </participant>
    <length>
      <value value="90"/>
      <unit value="min"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="min"/>
    </length>
    <reasonCode>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="18099001"/>
        <display value="Retropharyngeal abscess"/>
      </coding>
    </reasonCode>
    <hospitalization>
      <preAdmissionIdentifier>
        <use value="official"/>
        <system value="http://www.bmc.nl/zorgportal/identifiers/pre-admissions"/>
        <value value="93042"/>
      </preAdmissionIdentifier>
      <!--     <preAdmissionTest>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="168719007"/>
                   <display value="Neck soft tissue X-ray"/>
               </coding>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="396550006"/>
                   <display value="Blood test"/>
               </coding>
          </preAdmissionTest> -->
      <admitSource>
        <coding>
          <system value="http://snomed.info/sct"/>
```

```
          <code value="305956004"/>
          <display value="Referral by physician"/>
        </coding>
      </admitSource>
      <dischargeDisposition>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="306689006"/>
          <display value="Discharge to home"/>
        </coding>
      </dischargeDisposition>
    </hospitalization>
    <serviceProvider>
      <reference value="Organization/f001"/>
    </serviceProvider>
   </Encounter>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notThis"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notThis"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 2. Withhold or withdraw consent for disclosure of a specific record (e.g. Lab Order/Result)
    </p><p>
       Patient "P. van de Heuvel" Primary Care Provider, Dr. Philip Primary, has ordered a set of lab test which Adam
      wishes to keep as private as possible.  At the sample collection facility, he indicates that he would
      like withhold consent to disclose the order and all results associated with that specific order from
      all other providers
      </p>
      </div>
     </text>
     <status value="active"/>
     <scope>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/consentscope"/>
         <code value="patient-privacy"/>
       </coding>
       </scope>
         <category>
       <coding>
         <system value="http://loinc.org"/>
         <code value="59284-0"/>
       </coding>
     </category>
     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>
     <dateTime value="2015-11-18"/>
     <!-- not bound by a timeframe - Consent.period -->
     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/f001"/>
     </organization>
     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
     </sourceAttachment>
     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
         <code value="OPTIN"/>
       </coding>
     </policyRule>
     <provision>
       <data>
         <meaning value="related"/>
         <reference>
           <reference value="Task/example3"/>
         </reference>
       </data>
     </provision>
   </Consent>
  </resource>
</entry>
```

```
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notThem"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notThem"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
         jurisdictional policy and are realized in existing systems in Canada. The default policy is
         one of implied consent for the provision of care, so these scenarios all deal with withdrawal
         or withholding consent for that purpose. In other jurisdictions, where an express consent
         model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
         "withhold" or "withdraw" consent for. <p> specific to use-case 4) Withhold or withdraw consent
           for disclosure to a specific provider agent (an individual within an organization) </p><p>
           Patient "P. van de Heuvel" ex-spouse, Bill T Lookafter is a Nurse that is employed by Good
           Health Hospital. P. realizes that she may receive treatment at GHH, however she does not
           want her ex-spouse to have any access to her IIHI. She indicates that she would like to
           withdraw/withhold consent to disclose any instance of her health information to her
           ex-spouse as a result of his employment at Good Health Clinic. </p>
       </div>
     </text>

     <issued value="2015-11-18"/>

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <authority>
       <reference value="Organization/3"/>
       <display value="Michigan Health"/>
     </authority>

     <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
   Juristiction) -->
     <domain>
       <reference value="Location/ukp"/>
       <display value="UK Pharmacies"/>
     </domain>

     <type>
       <coding>
         <system value="http://loinc.org"/>
         <code value="57016-8"/>
       </coding>
     </type>

     <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
   -->
     <subType>
       <coding>
         <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
         <code value="Opt-In"/>
         <display value="Default Authorization with exceptions."/>
       </coding>
     </subType>

     <term>
       <type>
         <coding>
           <!-- made up code system -->
           <system value="http://example.org/fhir/consent-term-type-codes"/>
           <code value="withhold-from"/>
           <display value="Withhold all data from specified actor entity."/>
         </coding>
       </type>
       <offer>
         <topic>
           <reference value="Practitioner/f204"/>
           <display value="Fictive Nurse"/>
         </topic>

         <text value="Withhold this order and any results or related objects from specified nurse provider."/>
       </offer>
     </term>

     <signer>
       <type>
         <system value="http://terminology.hl7.org/CodeSystem/contractsignertypecodes"/>
         <code value="COVPTY"/>
       </type>
       <party>
         <reference value="Patient/f001"/>
       </party>
       <signature>
         <type>
           <system value="urn:iso-astm:E1762-95:2013"/>
```

```
          <code value="1.2.840.10065.1.12.1.1"/>
        </type>
        <when value="2013-06-08T10:57:34-07:00"/>
        <who>
          <reference value="Patient/f001"/>
        </who>
      </signature>
    </signer>

    <!-- the terms of the consent in friendly consumer speak -->
    <friendly>
      <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
    </friendly>

    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
    <!-- not bound by a timeframe - Contract.applies -->
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f004"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f004"/><!--   urn:oid:2.16.840.1.113883.4.642.1.7   --><identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6326"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="789-8"/>
        <display value="Erythrocytes [#/volume] in Blood by Automated count"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="4.12"/><!--  Actually, 4.12E+12  --><unit value="10^12/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*12/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
    </interpretation><!-- this example demonstrates using the reference range text element for stuff that is not
  easily structured !--><referenceRange>
      <text value=" 12-14 y Male: 4.4 - 5.2  x  10^12/L ; 12-14 y Female: 4.2 - 4.8  x  10^12/L ; 15-17 y Male: 4.6 -
  5.4  x  10^12/L ; 15-17 y Female: 4.2 - 4.8  x  10^12/L ; 18-64 y Male: 4.6 - 5.4  x  10^12/L ; 18-64 y Female: 4.0
  - 4.8  x  10^12/L ; 65-74 y Male: 4.3 - 5.3  x  10^12/L ; 65-74 y Female: 4.1 - 4.9  x  10^12/L       "/>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f002"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f002"/>
     <clinicalStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
         <code value="active"/>
```

```
      </coding>
     </clinicalStatus>

     <verificationStatus>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
         <code value="confirmed"/>
       </coding>
     </verificationStatus>
     <category>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="439401001"/>
         <display value="diagnosis"/>
       </coding>
     </category>
     <severity>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="24484000"/>
         <display value="Severe"/>
       </coding>
     </severity>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="254637007"/>
         <display value="NSCLC - Non-small cell lung cancer"/>
       </coding>
     </code>
     <bodySite>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="51185008"/>
         <display value="Thorax"/>
       </coding>
     </bodySite>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <encounter>
       <reference value="Encounter/f002"/>
     </encounter>
     <onsetDateTime value="2011-05-05"/>
     <recordedDate value="2012-06-03"/>
     <asserter>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </asserter>
     <stage>
       <summary>
         <coding>
           <system value="http://snomed.info/sct"/>
           <code value="258219007"/>
           <display value="stage II"/>
         </coding>
       </summary>
     <type>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="260998006"/>
            <display value="Clinical staging (qualifier value)"/>
          </coding>
       </type>
     </stage>
     <evidence>
       <code>
         <coding>
           <system value="http://snomed.info/sct"/>
           <code value="169069000"/>
           <display value="CT of thorax"/>
         </coding>
       </code>
     </evidence>
    </Condition>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/Observation/f005"/>
   <resource>
    <Observation xmlns="http://hl7.org/fhir">
     <id value="f005"/><!--  urn:oid:2.16.840.1.113883.4.642.1.7  --><identifier>
       <use value="official"/>
       <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
       <value value="6327"/>
     </identifier>
     <status value="final"/>
     <code>
```

```
      <coding>
        <system value="http://loinc.org"/>
        <code value="718-7"/>
        <display value="Hemoglobin [Mass/volume] in Blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-05T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-05T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="7.2"/>
      <unit value="g/dl"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="g/dL"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="7.5"/>
        <unit value="g/dl"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/dL"/>
      </low>
      <high>
        <value value="10"/>
        <unit value="g/dl"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/dL"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/ekg"/>
  <resource>
   <!-- Sampled ECG potentials (Leads I, II, II). There are 2 chunks (numbered as 0 and 1) for each lead. The voltage
  of each lead is near zero. --><Observation xmlns="http://hl7.org/fhir">
    <id value="ekg"/>
    <status value="final"/>
    <category>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
        <code value="procedure"/>
        <display value="Procedure"/>
      </coding>
    </category>
    <code>
      <coding>
        <system value="urn:oid:2.16.840.1.113883.6.24"/>
        <code value="131328"/>
        <display value="MDC_ECG_ELEC_POTL"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectiveDateTime value="2015-02-19T09:30:35+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <device>
      <display value="12 lead EKG Device Metric"/>
    </device>
    <!-- Lead I data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131329"/>
          <display value="MDC_ECG_ELEC_POTL_I"/>
```

```
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
  2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
  2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
  2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
  2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
  2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
    <!-- Lead II data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131330"/>
          <display value="MDC_ECG_ELEC_POTL_II"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
  2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
  2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
  2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
  2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
  2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
    <!-- Lead III data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131389"/>
          <display value="MDC_ECG_ELEC_POTL_III"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
  2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
  2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
  2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
  2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
  2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
  </Observation>
 </resource>
</entry>
```

```
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f001"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
     <id value="f001"/>
       <identifier>
           <use value="official"/>
           <system value="http://www.amc.nl/zorgportal/identifiers/visits"/>
           <value value="v1451"/>
       </identifier>
       <status value="finished"/>
       <class>
           <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
           <code value="AMB"/> <!-- outpatient -->
           <display value="ambulatory"/>
       </class>
       <type>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="270427003"/>
               <display value="Patient-initiated encounter"/>
           </coding>
       </type>
       <priority>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="310361003"/>
               <display value="Non-urgent cardiological admission"/>
           </coding>
       </priority>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
       <participant>
           <individual>
               <reference value="Practitioner/f002"/>
               <display value="P. Voigt"/>
           </individual>
       </participant>
       <length>
           <value value="140"/>
           <unit value="min"/>
           <system value="http://unitsofmeasure.org"/>
           <code value="min"/>
       </length>
       <reasonCode>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="34068001"/>
               <display value="Heart valve replacement"/>
           </coding>
       </reasonCode>
       <hospitalization>
           <preAdmissionIdentifier>
               <use value="official"/>
               <system value="http://www.amc.nl/zorgportal/identifiers/pre-admissions"/>
               <value value="93042"/>
           </preAdmissionIdentifier>
    <!--       <preAdmissionTest>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="164847006"/>
                   <display value="Standard ECG"/>
               </coding>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="396550006"/>
                   <display value="Blood test"/>
               </coding>
           </preAdmissionTest> -->
           <admitSource>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="305956004"/>
                   <display value="Referral by physician"/>
               </coding>
           </admitSource>
           <dischargeDisposition>
               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="306689006"/>
                   <display value="Discharge to home"/>
               </coding>
           </dischargeDisposition>
       </hospitalization>
       <serviceProvider>
           <reference value="Organization/f001"/>
           <display value="Burgers University Medical Center"/>
```

```
        </serviceProvider>
     </Encounter>
    </resource>
  </entry>
</Bundle>
```

For more information about operations, including how they are invoked, see Operations.

8.3.10 Resource Group - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/group-version-maps.html

No validation errors - all conversions are clean

8.3.10 Resource Group - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for Group.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.3.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Group3to4" = "R3 to R4 Conversions for Group"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Group" alias GroupR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Group" alias group as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group group(source src : GroupR3, target tgt : group) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.type -> tgt.type;
  src.actual -> tgt.actual;
  src.code -> tgt.code;
  src.name -> tgt.name;
  src.quantity -> tgt.quantity;
  src.characteristic as vs0 -> tgt.characteristic as vt0 then characteristic(vs0, vt0);
  src.member as vs0 -> tgt.member as vt0 then member(vs0, vt0);
}

group characteristic(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value : CodeableConcept as vs -> tgt.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.value : boolean as vs -> tgt.value = create('boolean') as vt then boolean(vs, vt);
  src.value : Quantity as vs -> tgt.value = create('Quantity') as vt then Quantity(vs, vt);
  src.value : Range as vs -> tgt.value = create('Range') as vt then Range(vs, vt);
  src.exclude -> tgt.exclude;
  src.period -> tgt.period;
}

group member(source src, target tgt) extends BackboneElement {
  src.period -> tgt.period;
  src.entity -> tgt.entity;
  src.inactive -> tgt.inactive;
}
```

8.3.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Group4to3" = "R4 to R3 Conversion for Group"

uses "http://hl7.org/fhir/StructureDefinition/Group" alias Group as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Group" alias GroupR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Group(source src : Group, target tgt : GroupR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.type -> tgt.type;
  src.actual -> tgt.actual;
  src.code -> tgt.code;
  src.name -> tgt.name;
  src.quantity -> tgt.quantity;
  src.characteristic as vs0 -> tgt.characteristic as vt0 then characteristic(vs0, vt0);
  src.member as vs0 -> tgt.member as vt0 then member(vs0, vt0);
}

group characteristic(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value : CodeableConcept as vs -> tgt.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.value : boolean as vs -> tgt.value = create('boolean') as vt then boolean(vs, vt);
  src.value : Quantity as vs -> tgt.value = create('Quantity') as vt then Quantity(vs, vt);
  src.value : Range as vs -> tgt.value = create('Range') as vt then Range(vs, vt);
  src.exclude -> tgt.exclude;
  src.period -> tgt.period;
```

```
}

group member(source src, target tgt) extends BackboneElement {
  src.period -> tgt.period;
  src.entity -> tgt.entity;
  src.inactive -> tgt.inactive;
}
```

8.3.10.3 R4 Validation Errors

8.3.11 Group Definition

URL: https://hl7.org/fhir/R4/group-definition.html

Enforces a descriptive group that can be used in definitional resources

8.3.11 Group Definition

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

8.3.11.1 Content

Profiles:
GroupDefinition | Enforces a descriptive group that can be used in definitional resources

8.3.11.2 StructureDefinition: GroupDefinition

URL: https://hl7.org/fhir/R4/groupdefinition.html

8.3.11.2 StructureDefinition: GroupDefinition

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/groupdefinition`

Enforces a descriptive group that can be used in definitional resources

This profile was published on [no date] as a draft by HL7.

8.3.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from Group.

Summary

Mandatory: 1 element Fixed Value: 1 element Prohibited: 1 element

This structure is derived from Group.

Name | Flags | Card. | Type | Description & Constraints
Group | | 1..1
actual | | 1..1 | boolean | Fixed Value: false
member | | 0..0

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Group | I | 1..1 | | Group of multiple entities
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Unique id
active | ΣI | 0..1 | boolean | Whether this group's record is in active use
type | ΣI | 1..1 | code | person | animal | practitioner | device | medication | substance Binding: GroupType (required)
actual | ΣI | 1..1 | boolean | Descriptive or actual Fixed Value: false
code | ΣI | 0..1 | CodeableConcept | Kind of Group members Binding: (unbound) (example)
name | ΣI | 0..1 | string | Label for Group

quantity | ΣI | 0..1 | unsignedInt | Number of members
managingEntity | ΣI | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole) | Entity that is the custodian of the Group's definition
characteristic | I | 0..* | BackboneElement | Include / Exclude group members by Trait
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | CodeableConcept | Kind of characteristic Binding: (unbound) (example)
value[x] | I | 1..1 | | Value held by characteristic Binding: (unbound) (example)
valueCodeableConcept | | | CodeableConcept
valueBoolean | | | boolean
valueQuantity | | | Quantity
valueRange | | | Range
valueReference | | | Reference()
exclude | I | 1..1 | boolean | Group includes or excludes
period | I | 0..1 | Period | Period over which characteristic is tested
Documentation for this format

```
<!-- Group Definition -->

<Group xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!--  0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!--  1..1 person | animal | practitioner | device | medication | substance   -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!--  0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!--  0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!--  0..1 Number of members -->
 <managingEntity><!--  0..1 Reference(Organization)
     Entity that is the custodian of the Group's definition --></managingEntity>
 <characteristic>  0..* BackboneElement  <!--  0..* Include / Exclude group members by Trait -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept
      Kind of characteristic --></code>
  <value[x]><!--  1..1 CodeableConcept|boolean|Quantity|
    Range|Reference Value held by characteristic --></value[x]>
  <exclude value="[boolean]"/><!--  1..1 Group includes or excludes -->
  <period><!--  0..1 Period
      Period over which characteristic is tested --></period>
 </characteristic>
 <member>  0..0 BackboneElement  <!--  0..0 Who or what is in group -->
 </member>
</Group>
```

```
{ // Group Definition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Unique id
    "active" : <boolean>, // C?Whether this group's record is in active use
    "type" : "<code>", // C? R! person | animal | practitioner | device | medication | substance
    "actual" : false, // C? R!
    "code" : { CodeableConcept }, // C?Kind of Group members
    "name" : "<string>", // C?Label for Group
    "quantity" : "<unsignedInt>", // C?Number of members
    "managingEntity" : { Reference(Organization) }, // C?Entity that is the custodian of the Group's definition
    "characteristic" : [{ BackboneElement }] // C?Include / Exclude group members by Trait
  }
```

This structure is derived from Group.

Summary

Mandatory: 1 element Fixed Value: 1 element Prohibited: 1 element

Differential View

This structure is derived from Group.

Name | Flags | Card. | Type | Description & Constraints
Group | | 1..1
actual | | 1..1 | boolean | Fixed Value: false
member | | 0..0

Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Group | I | 1..1 | | Group of multiple entities
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Unique id
active | ΣI | 0..1 | boolean | Whether this group's record is in active use
type | ΣI | 1..1 | code | person | animal | practitioner | device | medication | substance Binding: GroupType (required)
actual | ΣI | 1..1 | boolean | Descriptive or actual Fixed Value: false
code | ΣI | 0..1 | CodeableConcept | Kind of Group members Binding: (unbound) (example)
name | ΣI | 0..1 | string | Label for Group
quantity | ΣI | 0..1 | unsignedInt | Number of members
managingEntity | ΣI | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole) | Entity that is the custodian of the Group's definition
characteristic | I | 0..* | BackboneElement | Include / Exclude group members by Trait
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | CodeableConcept | Kind of characteristic Binding: (unbound) (example)
value[x] | I | 1..1 | | Value held by characteristic Binding: (unbound) (example)
valueCodeableConcept | | | CodeableConcept
valueBoolean | | | boolean
valueQuantity | | | Quantity
valueRange | | | Range
valueReference | | | Reference()
exclude | I | 1..1 | boolean | Group includes or excludes
period | I | 0..1 | Period | Period over which characteristic is tested
Documentation for this format

XML Template

```
<!-- Group Definition -->

<Group xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!--  0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!--  1..1 person | animal | practitioner | device | medication | substance   -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!--  0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!--  0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!--  0..1 Number of members -->
 <managingEntity><!--  0..1 Reference(Organization)
     Entity that is the custodian of the Group's definition --></managingEntity>
 <characteristic>  0..* BackboneElement  <!--  0..* Include / Exclude group members by Trait -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept
      Kind of characteristic --></code>
  <value[x]><!--  1..1 CodeableConcept|boolean|Quantity|
    Range|Reference Value held by characteristic --></value[x]>
  <exclude value="[boolean]"/><!--  1..1 Group includes or excludes -->
  <period><!--  0..1 Period
      Period over which characteristic is tested --></period>
 </characteristic>
 <member>  0..0 BackboneElement  <!--  0..0 Who or what is in group -->
 </member>
</Group>
```

JSON Template

```
{ // Group Definition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
```

```
  "contained" : [{ Resource }], //Contained, inline Resources
  "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
  "identifier" : [{ Identifier }], // C?Unique id
  "active" : <boolean>, // C?Whether this group's record is in active use
  "type" : "<code>", // C? R! person | animal | practitioner | device | medication | substance
  "actual" : false, // C? R!
  "code" : { CodeableConcept }, // C?Kind of Group members
  "name" : "<string>", // C?Label for Group
  "quantity" : "<unsignedInt>", // C?Number of members
  "managingEntity" : { Reference(Organization) }, // C?Entity that is the custodian of the Group's definition
  "characteristic" : [{ BackboneElement }] // C?Include / Exclude group members by Trait
}
```

Alternate definitions: Master Definition (XML, JSON), Schematron

8.3.11.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Group.language | Common Languages | preferred | Common Languages
Group.type | GroupType | required | GroupType
Group.code | ?ext | example
Group.characteristic.code | ?ext | example
Group.characteristic.value[x] | ?ext | example

8.3.12 Actual Group

URL: https://hl7.org/fhir/R4/group-actual.html

Enforces an actual group, rather than a definitional group

8.3.12 Actual Group

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

8.3.12.1 Content

Profiles:
ActualGroup | Enforces an actual group, rather than a definitional group

8.3.12.2 StructureDefinition: ActualGroup

URL: https://hl7.org/fhir/R4/actualgroup.html

8.3.12.2 StructureDefinition: ActualGroup

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/actualgroup`

Enforces an actual group, rather than a definitional group

This profile was published on [no date] as a draft by HL7.

8.3.12.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Group.

Summary

Mandatory: 1 element Fixed Value: 1 element Prohibited: 1 element

This structure is derived from Group.

Name | Flags | Card. | Type | Description & Constraints
Group | | 1..1
actual | | 1..1 | boolean | Fixed Value: true
characteristic | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Group | I | 1..1 | | Group of multiple entities
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource

implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Unique id
active | ΣI | 0..1 | boolean | Whether this group's record is in active use
type | ΣI | 1..1 | code | person | animal | practitioner | device | medication | substance Binding: GroupType (required)
actual | ΣI | 1..1 | boolean | Descriptive or actual Fixed Value: true
code | ΣI | 0..1 | CodeableConcept | Kind of Group members Binding: (unbound) (example)
name | ΣI | 0..1 | string | Label for Group
quantity | ΣI | 0..1 | unsignedInt | Number of members
managingEntity | ΣI | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole) | Entity that is the custodian of the Group's definition
member | I | 0..* | BackboneElement | Who or what is in group
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
entity | I | 1..1 | Reference(Patient | Practitioner | PractitionerRole | Device | Medication | Substance | Group) | Reference to the group member
period | I | 0..1 | Period | Period member belonged to the group
inactive | I | 0..1 | boolean | If member is no longer in group
Documentation for this format

```
<!-- Actual Group -->

<Group xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!--  0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!--  1..1 person | animal | practitioner | device | medication | substance   -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!--  0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!--  0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!--  0..1 Number of members -->
 <managingEntity><!--  0..1 Reference(Organization)
     Entity that is the custodian of the Group's definition --></managingEntity>
 <characteristic>  0..0 BackboneElement  <!--  0..0 Include / Exclude group members by Trait -->
 </characteristic>
 <member>  0..* BackboneElement  <!--  0..* Who or what is in group -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <entity><!--  1..1 Reference(Patient) Reference to the group member --></entity>
  <period><!--  0..1 Period Period member belonged to the group --></period>
  <inactive value="[boolean]"/><!--  0..1 If member is no longer in group -->
 </member>
</Group>
```

```
{ // Actual Group
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Unique id
    "active" : <boolean>, // C?Whether this group's record is in active use
    "type" : "<code>", // C? R! person | animal | practitioner | device | medication | substance
    "actual" : true, // C? R!
    "code" : { CodeableConcept }, // C?Kind of Group members
    "name" : "<string>", // C?Label for Group
    "quantity" : "<unsignedInt>", // C?Number of members
    "managingEntity" : { Reference(Organization) }, // C?Entity that is the custodian of the Group's definition
    "member" : [{ BackboneElement }] // C?Who or what is in group
  }
```

This structure is derived from Group.

Summary

Mandatory: 1 element Fixed Value: 1 element Prohibited: 1 element

Differential View

This structure is derived from Group.

Name | Flags | Card. | Type | Description & Constraints

Group | | 1..1
actual | | 1..1 | boolean | Fixed Value: true
characteristic | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Group | I | 1..1 | | Group of multiple entities
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Unique id
active | ΣI | 0..1 | boolean | Whether this group's record is in active use
type | ΣI | 1..1 | code | person | animal | practitioner | device | medication | substance Binding: GroupType (required)
actual | ΣI | 1..1 | boolean | Descriptive or actual Fixed Value: true
code | ΣI | 0..1 | CodeableConcept | Kind of Group members Binding: (unbound) (example)
name | ΣI | 0..1 | string | Label for Group
quantity | ΣI | 0..1 | unsignedInt | Number of members
managingEntity | ΣI | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole) | Entity that is the custodian of the Group's definition
member | I | 0..* | BackboneElement | Who or what is in group
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
entity | I | 1..1 | Reference(Patient | Practitioner | PractitionerRole | Device | Medication | Substance | Group) | Reference to the group member
period | I | 0..1 | Period | Period member belonged to the group
inactive | I | 0..1 | boolean | If member is no longer in group
Documentation for this format

XML Template

```
<!-- Actual Group -->

<Group xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!--  0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!--  1..1 person | animal | practitioner | device | medication | substance   -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!--  0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!--  0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!--  0..1 Number of members -->
 <managingEntity><!--  0..1 Reference(Organization)
     Entity that is the custodian of the Group's definition --></managingEntity>
 <characteristic>  0..0 BackboneElement  <!--  0..0 Include / Exclude group members by Trait -->
 </characteristic>
 <member>  0..* BackboneElement  <!--  0..* Who or what is in group -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <entity><!--  1..1 Reference(Patient) Reference to the group member --></entity>
  <period><!--  0..1 Period Period member belonged to the group --></period>
  <inactive value="[boolean]"/><!--  0..1 If member is no longer in group -->
 </member>
</Group>
```

JSON Template

```
{ // Actual Group
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Unique id
    "active" : <boolean>, // C?Whether this group's record is in active use
    "type" : "<code>", // C? R! person | animal | practitioner | device | medication | substance
    "actual" : true, // C? R!
    "code" : { CodeableConcept }, // C?Kind of Group members
```

```
    "name" : "<string>", // C?Label for Group
    "quantity" : "<unsignedInt>", // C?Number of members
    "managingEntity" : { Reference(Organization) }, // C?Entity that is the custodian of the Group's definition
    "member" : [{ BackboneElement }] // C?Who or what is in group
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

8.3.12.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Group.language	Common Languages	preferred	Common Languages
Group.type	GroupType	required	GroupType
Group.code	?ext	example	
Group.characteristic.code	?ext	example	
Group.characteristic.value[x]	?ext	example	

8.3.13 JSON Schema for Group

URL: https://hl7.org/fhir/R4/group.schema.json.html

8.3.13 JSON Schema for Group

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for Group, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Group",
  "$ref": "#/definitions/Group",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Group": {
      "description": "Represents a defined collection of entities that may be discussed or acted upon collectively but
    which are not expected to act collectively, and are not formally or legally recognized; i.e. a collection of
    entities that isn\u0027t an Organization.",
      "properties": {
        "resourceType": {
          "description": "This is a Group resource",
          "const": "Group"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A unique business identifier for this group.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Indicates whether the record for the group is available for use or is merely being retained for historical purposes.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Identifies the broad classification of the kind of resources the group includes.",
      "enum": [
        "person",
        "animal",
        "practitioner",
        "device",
        "medication",
        "substance"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "actual": {
      "description": "If true, indicates that the resource refers to a specific group of real individuals.  If false, the group defines a set of intended individuals.",
      "$ref": "#/definitions/boolean"
    },
    "_actual": {
      "description": "Extensions for actual",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "Provides a specific type of resource the group includes; e.g. \"cow\", \"syringe\", etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "name": {
      "description": "A label assigned to the group for human identification and communication.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "quantity": {
      "description": "A count of the number of resource instances that are part of the group.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_quantity": {
      "description": "Extensions for quantity",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "managingEntity": {
      "description": "Entity responsible for defining and maintaining Group characteristics and/or registered members.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "characteristic": {
      "description": "Identifies traits whose presence r absence is shared by members of the group.",
      "items": {
```

```
        "$ref": "#/definitions/Group_Characteristic"
      },
      "type": "array"
    },
    "member": {
      "description": "Identifies the resource instances that are members of the group.",
      "items": {
        "$ref": "#/definitions/Group_Member"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Group_Characteristic": {
  "description": "Represents a defined collection of entities that may be discussed or acted upon collectively but
which are not expected to act collectively, and are not formally or legally recognized; i.e. a collection of
entities that isn\u0027t an Organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A code that identifies the kind of trait being asserted.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueCodeableConcept": {
      "description": "The value of the trait that holds (or does not hold - see \u0027exclude\u0027) for members of
the group.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueBoolean": {
      "description": "The value of the trait that holds (or does not hold - see \u0027exclude\u0027) for members of
the group.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueQuantity": {
      "description": "The value of the trait that holds (or does not hold - see \u0027exclude\u0027) for members of
the group.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "valueRange": {
      "description": "The value of the trait that holds (or does not hold - see \u0027exclude\u0027) for members of
the group.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "valueReference": {
      "description": "The value of the trait that holds (or does not hold - see \u0027exclude\u0027) for members of
the group.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "exclude": {
      "description": "If true, indicates the characteristic is one that is NOT held by members of the group.",
      "$ref": "#/definitions/boolean"
    },
```

```
        "_exclude": {
          "description": "Extensions for exclude",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "period": {
          "description": "The period over which the characteristic is tested; e.g. the patient had an operation during
    the month of June.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false,
      "required": [
        "code"
      ]
    },
    "Group_Member": {
      "description": "Represents a defined collection of entities that may be discussed or acted upon collectively but
    which are not expected to act collectively, and are not formally or legally recognized; i.e. a collection of
    entities that isn\u0027t an Organization.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "entity": {
          "description": "A reference to the entity that is a member of the group. Must be consistent with Group.type.
    If the entity is another group, then the type must be the same.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "period": {
          "description": "The period that the member was in the group, if known.",
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "inactive": {
          "description": "A flag to indicate that the member is no longer in the group, but previously may have been a
    member.",
          "$ref": "#/definitions/boolean"
        },
        "_inactive": {
          "description": "Extensions for inactive",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "entity"
      ]
    }
  }
}
```


8.3.14 Resource Group - Maturity Dependency View

URL: https://hl7.org/fhir/R4/group-dependencies.html

8.3.14 Resource Group - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for Group FMM level 1

Name | Card. | Type | Dependency Analysis
Group | | DomainResource

identifier | 0..* | Identifier
active | 0..1 | boolean
type | 1..1 | code
actual | 1..1 | boolean
code | 0..1 | CodeableConcept
name | 0..1 | string
quantity | 0..1 | unsignedInt
managingEntity | 0..1 | Reference(Organization | RelatedPerson | Practitioner | PractitionerRole)
characteristic | 0..* | BackboneElement
code | 1..1 | CodeableConcept
value[x] | 1..1
valueCodeableConcept | | CodeableConcept
valueBoolean | | boolean
valueQuantity | | Quantity
valueRange | | Range
valueReference | | Reference(
exclude | 1..1 | boolean
period | 0..1 | Period
member | 0..* | BackboneElement
entity | 1..1 | Reference(Patient | Practitioner | PractitionerRole | Device | Medication | Substance | Group)
period | 0..1 | Period
inactive | 0..1 | boolean

8.4 Resource Practitioner - Content

URL: https://hl7.org/fhir/R4/practitioner.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Individual | Compartments: Practitioner

A person who is directly or indirectly involved in the provisioning of healthcare.

8.4.1 Scope and Usage

Practitioner covers all individuals who are engaged in the healthcare process and healthcare-related services as part of their formal responsibilities and this Resource is used for attribution of activities and responsibilities to these individuals. Practitioners include (but are not limited to):

• physicians, dentists, pharmacists
• physician assistants, nurses, scribes
• midwives, dietitians, therapists, optometrists, paramedics
• medical technicians, laboratory scientists, prosthetic technicians, radiographers
• social workers, professional homecare providers, official volunteers
• receptionists handling patient registration
• IT personnel merging or unmerging patient records
• Service animal (e.g., ward assigned dog capable of detecting cancer in patients)

8.4.2 Boundaries and Relationships

The Resource SHALL NOT be used for persons involved without a formal responsibility like individuals taking care for friends, relatives or neighbors. These can be registered as a Patient's Contact. If performing some action or being referenced by another resource, use the RelatedPerson resource.

The primary distinction between a Practitioner and a RelatedPerson is based on whether:

• The person/animal operates on behalf of the care delivery organization over multiple patients (Practitioner) or,

• Where the person/animal is not associated with the organization, and instead is allocated tasks specifically for the RelatedPerson's Patient (RelatedPerson).

A standard extension animalSpecies can be used to indicate the species of a service animal.

The PractitionerRole resource provides the details of roles that the practitioner is approved to perform for which organizations (and at which locations, and optionally what services too).

Practitioners are also often grouped into CareTeams independently of roles, where the CareTeam defines what specific role that they are fulfilling within the team, and might or might not have actual practitioner role resources created for the practitioner (and in the care team context, the organization the practitioner is representing)

8.4.3 Background and Context

Practitioner performs different roles within the same or even different organizations. Depending on jurisdiction and custom, it may be necessary to maintain a specific Practitioner Resource for each such role or have a single Practitioner with multiple roles. The role can be limited to a specific period, after which

authorization for this role ends. Note that the represented organization need not necessarily be the (direct) employer of a Practitioner.

This resource is referenced by Annotation, Signature, Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BiologicallyDerivedProduct, CarePlan, CareTeam, CatalogEntry, ChargeItem, Claim, ClaimResponse, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, CoverageEligibilityRequest, CoverageEligibilityResponse, DetectedIssue, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, Encounter, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, ExplanationOfBenefit, Flag, Goal, Group, ImagingStudy, Immunization, Invoice, Linkage, List, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MessageHeader, NutritionOrder, Observation, Patient, PaymentNotice, PaymentReconciliation, Person, PractitionerRole, Procedure, Provenance, QuestionnaireResponse, RequestGroup, ResearchStudy, RiskAssessment, Schedule, ServiceRequest, Specimen, SupplyDelivery, SupplyRequest, Task, VerificationResult and VisionPrescription

8.4.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Practitioner | TU | | DomainResource | A person with a formal responsibility in the provisioning of healthcare or related services
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | An identifier for the person as this agent
active | Σ | 0..1 | boolean | Whether this practitioner's record is in active use
name | Σ | 0..* | HumanName | The name(s) associated with the practitioner
telecom | Σ | 0..* | ContactPoint | A contact detail for the practitioner (that apply to all roles)
address | Σ | 0..* | Address | Address(es) of the practitioner that are not role specific (typically home address)
gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | Σ | 0..1 | date | The date on which the practitioner was born
photo | | 0..* | Attachment | Image of the person
qualification | | 0..* | BackboneElement | Certification, licenses, or training pertaining to the provision of care
identifier | | 0..* | Identifier | An identifier for this qualification for the practitioner
code | | 1..1 | CodeableConcept | Coded representation of the qualification v2 table 0360, Version 2.7 (Example)
period | | 0..1 | Period | Period during which the qualification is valid
issuer | | 0..1 | Reference(Organization) | Organization that regulates and issues the qualification
communication | | 0..* | CodeableConcept | A language the practitioner can use in patient communication Common Languages (Preferred but limited to AllLanguages)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Practitioner xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for the person as this agent --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner's record is in active use -->
 <name><!-- 0..* HumanName The name(s) associated with the practitioner --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the practitioner (that apply to all roles) --></telecom>
 <address><!-- 0..* Address Address(es) of the practitioner that are not role specific (typically home address)
    --></address>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date  on which the practitioner was born -->
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <qualification>  <!-- 0..* Certification, licenses, or training pertaining to the provision of care -->
  <identifier><!-- 0..* Identifier An identifier for this qualification for the practitioner --></identifier>
  <code><!-- 1..1 CodeableConcept Coded representation of the qualification --></code>
  <period><!-- 0..1 Period Period during which the qualification is valid --></period>
  <issuer><!-- 0..1 Reference(Organization) Organization that regulates and issues the qualification --></issuer>
 </qualification>
 <communication><!-- 0..* CodeableConcept A language the practitioner can use in patient communication
    --></communication>
</Practitioner>
```

JSON Template

```
{
  "resourceType" : "Practitioner",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for the person as this agent
  "active" : <boolean>, // Whether this practitioner's record is in active use
  "name" : [{ HumanName }], // The name(s) associated with the practitioner
  "telecom" : [{ ContactPoint }], // A contact detail for the practitioner (that apply to all roles)
  "address" : [{ Address }], // Address(es) of the practitioner that are not role specific (typically home address)
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date  on which the practitioner was born
  "photo" : [{ Attachment }], // Image of the person
  "qualification" : [{ // Certification, licenses, or training pertaining to the provision of care
    "identifier" : [{ Identifier }], // An identifier for this qualification for the practitioner
    "code" : { CodeableConcept }, // R!  Coded representation of the qualification
    "period" : { Period }, // Period during which the qualification is valid
    "issuer" : { Reference(Organization) } // Organization that regulates and issues the qualification
  }],
  "communication" : [{ CodeableConcept }] // A language the practitioner can use in patient communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Practitioner;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Practitioner.identifier [ Identifier ], ... ; # 0..* An identifier for the person as this agent
  fhir:Practitioner.active [ boolean ]; # 0..1 Whether this practitioner's record is in active use
  fhir:Practitioner.name [ HumanName ], ... ; # 0..* The name(s) associated with the practitioner
  fhir:Practitioner.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the practitioner (that apply to all
    roles)
  fhir:Practitioner.address [ Address ], ... ; # 0..* Address(es) of the practitioner that are not role specific
    (typically home address)
  fhir:Practitioner.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Practitioner.birthDate [ date ]; # 0..1 The date  on which the practitioner was born
  fhir:Practitioner.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:Practitioner.qualification [ # 0..* Certification, licenses, or training pertaining to the provision of care
    fhir:Practitioner.qualification.identifier [ Identifier ], ... ; # 0..* An identifier for this qualification for the
     practitioner
    fhir:Practitioner.qualification.code [ CodeableConcept ]; # 1..1 Coded representation of the qualification
    fhir:Practitioner.qualification.period [ Period ]; # 0..1 Period during which the qualification is valid
    fhir:Practitioner.qualification.issuer [ Reference(Organization) ]; # 0..1 Organization that regulates and issues
    the qualification
  ], ...;
  fhir:Practitioner.communication [ CodeableConcept ], ... ; # 0..* A language the practitioner can use in patient
    communication
]
```

Changes since R3

Practitioner

Practitioner.active | Default Value "true" removed

Practitioner.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Practitioner.communication | Change binding strength from extensible to preferred

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Practitioner | TU | | DomainResource | A person with a formal responsibility in the provisioning of healthcare or related services

Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | An identifier for the person as this agent

active | Σ | 0..1 | boolean | Whether this practitioner's record is in active use

name | Σ | 0..* | HumanName | The name(s) associated with the practitioner

telecom | Σ | 0..* | ContactPoint | A contact detail for the practitioner (that apply to all roles)

address | Σ | 0..* | Address | Address(es) of the practitioner that are not role specific (typically home address)

gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

birthDate | Σ | 0..1 | date | The date on which the practitioner was born

photo | | 0..* | Attachment | Image of the person

qualification | | 0..* | BackboneElement | Certification, licenses, or training pertaining to the provision of care

identifier | | 0..* | Identifier | An identifier for this qualification for the practitioner

code | | 1..1 | CodeableConcept | Coded representation of the qualification v2 table 0360, Version 2.7 (Example)

period | | 0..1 | Period | Period during which the qualification is valid

issuer | | 0..1 | Reference(Organization) | Organization that regulates and issues the qualification

communication | | 0..* | CodeableConcept | A language the practitioner can use in patient communication Common Languages (Preferred but limited to AllLanguages)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Practitioner xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for the person as this agent --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner's record is in active use -->
 <name><!-- 0..* HumanName The name(s) associated with the practitioner --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the practitioner (that apply to all roles) --></telecom>
 <address><!-- 0..* Address Address(es) of the practitioner that are not role specific (typically home address)
    --></address>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date  on which the practitioner was born -->
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <qualification>  <!-- 0..* Certification, licenses, or training pertaining to the provision of care -->
  <identifier><!-- 0..* Identifier An identifier for this qualification for the practitioner --></identifier>
  <code><!-- 1..1 CodeableConcept Coded representation of the qualification --></code>
  <period><!-- 0..1 Period Period during which the qualification is valid --></period>
  <issuer><!-- 0..1 Reference(Organization) Organization that regulates and issues the qualification --></issuer>
 </qualification>
 <communication><!-- 0..* CodeableConcept A language the practitioner can use in patient communication
    --></communication>
</Practitioner>
```

JSON Template

```
{
  "resourceType" : "Practitioner",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for the person as this agent
  "active" : <boolean>, // Whether this practitioner's record is in active use
  "name" : [{ HumanName }], // The name(s) associated with the practitioner
  "telecom" : [{ ContactPoint }], // A contact detail for the practitioner (that apply to all roles)
  "address" : [{ Address }], // Address(es) of the practitioner that are not role specific (typically home address)
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date  on which the practitioner was born
  "photo" : [{ Attachment }], // Image of the person
  "qualification" : [{ // Certification, licenses, or training pertaining to the provision of care
    "identifier" : [{ Identifier }], // An identifier for this qualification for the practitioner
    "code" : { CodeableConcept }, // R!  Coded representation of the qualification
    "period" : { Period }, // Period during which the qualification is valid
    "issuer" : { Reference(Organization) } // Organization that regulates and issues the qualification
  }],
  "communication" : [{ CodeableConcept }] // A language the practitioner can use in patient communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Practitioner;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Practitioner.identifier [ Identifier ], ... ; # 0..* An identifier for the person as this agent
  fhir:Practitioner.active [ boolean ]; # 0..1 Whether this practitioner's record is in active use
  fhir:Practitioner.name [ HumanName ], ... ; # 0..* The name(s) associated with the practitioner
  fhir:Practitioner.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the practitioner (that apply to all
    roles)
  fhir:Practitioner.address [ Address ], ... ; # 0..* Address(es) of the practitioner that are not role specific
    (typically home address)
  fhir:Practitioner.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Practitioner.birthDate [ date ]; # 0..1 The date  on which the practitioner was born
  fhir:Practitioner.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:Practitioner.qualification [ # 0..* Certification, licenses, or training pertaining to the provision of care
    fhir:Practitioner.qualification.identifier [ Identifier ], ... ; # 0..* An identifier for this qualification for the
     practitioner
    fhir:Practitioner.qualification.code [ CodeableConcept ]; # 1..1 Coded representation of the qualification
    fhir:Practitioner.qualification.period [ Period ]; # 0..1 Period during which the qualification is valid
    fhir:Practitioner.qualification.issuer [ Reference(Organization) ]; # 0..1 Organization that regulates and issues
    the qualification
  ], ...;
  fhir:Practitioner.communication [ CodeableConcept ], ... ; # 0..* A language the practitioner can use in patient
    communication
]
```

Changes since Release 3

Practitioner

Practitioner.active | Default Value "true" removed

Practitioner.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Practitioner.communication | Change binding strength from extensible to preferred

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.4.4.1 Terminology Bindings

Path | Definition | Type | Reference
Practitioner.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender
Practitioner.qualification.code | Specific qualification the practitioner has to provide a service. | Example | v2.0360.2.7
Practitioner.communication | A human language. | Preferred, but limited to AllLanguages | CommonLanguages

8.4.5 Notes:

• The practitioner's Qualifications are acquired by the practitioner independent of any organization or role, and do not imply that they are allowed/authorized to perform roles relevant to the qualification at any specific Organization/Location.

8.4.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Whether the practitioner record is active | Practitioner.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Practitioner.address | 3 Resources
address-city | string | A city specified in an address | Practitioner.address.city | 3 Resources
address-country | string | A country specified in an address | Practitioner.address.country | 3 Resources
address-postalcode | string | A postalCode specified in an address | Practitioner.address.postalCode | 3 Resources
address-state | string | A state specified in an address | Practitioner.address.state | 3 Resources
address-use | token | A use code specified in an address | Practitioner.address.use | 3 Resources
communication | token | One of the languages that the practitioner can communicate with | Practitioner.communication
email | token | A value in an email contact | Practitioner.telecom.where(system='email') | 4 Resources
family | string | A portion of the family name | Practitioner.name.family | 1 Resources
gender | token | Gender of the practitioner | Practitioner.gender | 3 Resources
given | string | A portion of the given name | Practitioner.name.given | 1 Resources
identifier | token | A practitioner's Identifier | Practitioner.identifier
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Practitioner.name
phone | token | A value in a phone contact | Practitioner.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of either family or given name using some kind of phonetic matching algorithm | Practitioner.name | 3 Resources
telecom | token | The value in any kind of contact | Practitioner.telecom | 4 Resources

8.4.7 Resource Practitioner - Examples

URL: https://hl7.org/fhir/R4/practitioner-examples.html

8.4.7 Resource Practitioner - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

Example Name | id | Format
General Person Example | example | XML | JSON | Turtle
CDA Example Author | xcda-author | XML | JSON | Turtle
HL7 Defined Practitioner Examples | 3ad0687e-f477-468c-afd5-fcc2bf897809 | XML | JSON | Turtle
Fictive KNO-physician | f001 | XML | JSON | Turtle
Fictive Cardiothoracal surgeon | f002 | XML | JSON | Turtle
Fictive Cardiothoracal surgeon | f003 | XML | JSON | Turtle
Fictive KNO-physician | f004 | XML | JSON | Turtle
Fictive KNO-physician | f005 | XML | JSON | Turtle
Fictive Oncologist/Pulmonologist | f201 | XML | JSON | Turtle
Fictive Lab worker | f202 | XML | JSON | Turtle
Fictive Physiotherapist | f203 | XML | JSON | Turtle
Fictive Nurse | f204 | XML | JSON | Turtle
Fictive Pharmacist | f006 | XML | JSON | Turtle
Fictive physician | f007 | XML | JSON | Turtle
2nd CDA Example Author | xcda1 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.4.8 Resource Practitioner - Detailed Descriptions

URL: https://hl7.org/fhir/R4/practitioner-definitions.html

8.4.8 Resource Practitioner - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Individual | Compartments: Practitioner

Detailed Descriptions for the elements in the Practitioner resource.

Practitioner
Element Id | Practitioner
Definition | A person who is directly or indirectly involved in the provisioning of healthcare.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to track doctors, staff, locums etc. for both healthcare practitioners, funders, etc.
Practitioner.identifier
Element Id | Practitioner.identifier
Definition | An identifier that applies to this person in this role.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the agent.
Summary | true
Practitioner.active
Element Id | Practitioner.active
Definition | Whether this practitioner's record is in active use.
Cardinality | 0..1
Type | boolean
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a practitioner record as not to be used because it was created in error.
Summary | true
Comments | If the practitioner is not in use by one organization, then it should mark the period on the PractitonerRole with an end date (even if they are active) as they may be active in another role.
Practitioner.name
Element Id | Practitioner.name
Definition | The name(s) associated with the practitioner.
Cardinality | 0..*
Type | HumanName
Requirements | The name(s) that a Practitioner is known by. Where there are multiple, the name that the practitioner is usually known as should be used in the display.
Summary | true
Comments | The selection of the use property should ensure that there is a single usual name specified, and others use the nickname (alias), old, or other values as appropriate. In general, select the value to be used in the ResourceReference.display based on this: There is more than 1 name Use = usual Period is current to the date of the usage Use = official Other order as decided by internal business rules.
Practitioner.telecom
Element Id | Practitioner.telecom
Definition | A contact detail for the practitioner, e.g. a telephone number or an email address.
Cardinality | 0..*
Type | ContactPoint
Requirements | Need to know how to reach a practitioner independent to any roles the practitioner may have.
Summary | true
Comments | Person may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently and to help with identification. These typically will have home numbers, or mobile numbers that are not role specific.
Practitioner.address
Element Id | Practitioner.address
Definition | Address(es) of the practitioner that are not role specific (typically home address). Work addresses are not typically entered in this property as they are usually role dependent.
Cardinality | 0..*
Type | Address
Requirements | The home/mailing address of the practitioner is often required for employee administration purposes, and also for some rostering services where the start point (practitioners home) can be used in calculations.
Summary | true
Comments | The PractitionerRole does not have an address value on it, as it is expected that the location property be used for this purpose (which has an address).
Practitioner.gender
Element Id | Practitioner.gender
Definition | Administrative Gender - the gender that the person is considered to have for administration and record keeping purposes.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed to address the person correctly.
Summary | true
Practitioner.birthDate

Element Id | Practitioner.birthDate
Definition | The date of birth for the practitioner.
Cardinality | 0..1
Type | date
Requirements | Needed for identification.
Summary | true
Practitioner.photo
Element Id | Practitioner.photo
Definition | Image of the person.
Cardinality | 0..*
Type | Attachment
Requirements | Many EHR systems have the capability to capture an image of patients and personnel. Fits with newer social media usage too.
Practitioner.qualification
Element Id | Practitioner.qualification
Definition | The official certifications, training, and licenses that authorize or otherwise pertain to the provision of care by the practitioner. For example, a medical license issued by a medical board authorizing the practitioner to practice medicine within a certian locality.
Cardinality | 0..*
Practitioner.qualification.identifier
Element Id | Practitioner.qualification.identifier
Definition | An identifier that applies to this person's qualification in this role.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the qualification.
Practitioner.qualification.code
Element Id | Practitioner.qualification.code
Definition | Coded representation of the qualification.
Cardinality | 1..1
Terminology Binding | v2 table 0360, Version 2.7 (Example)
Type | CodeableConcept
Practitioner.qualification.period
Element Id | Practitioner.qualification.period
Definition | Period during which the qualification is valid.
Cardinality | 0..1
Type | Period
Requirements | Qualifications are often for a limited period of time, and can be revoked.
Practitioner.qualification.issuer
Element Id | Practitioner.qualification.issuer
Definition | Organization that regulates and issues the qualification.
Cardinality | 0..1
Type | Reference(Organization)
Practitioner.communication
Element Id | Practitioner.communication
Definition | A language the practitioner can use in patient communication.
Cardinality | 0..*
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | CodeableConcept
Requirements | Knowing which language a practitioner speaks can help in facilitating communication with patients.
Comments | The structure aa-BB with this exact casing is one the most widely used notations for locale. However not all systems code this but instead have it as free text. Hence CodeableConcept instead of code as the data type.

8.4.9 Resource Practitioner - Mappings

URL: https://hl7.org/fhir/R4/practitioner-mappings.html

8.4.9 Resource Practitioner - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

ServD

Mappings for the practitioner resource (see Mappings to Other Standards for further information & status).

8.4.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Practitioner

identifier | FiveWs.identifier
active | FiveWs.status

8.4.9.2 HL7 v2 Mapping (http://hl7.org/v2)

Practitioner | PRD (as one example)
identifier | PRD-7 (or XCN.1)
active
name | XCN Components
telecom | PRT-15, STF-10, ROL-12
address | ORC-24, STF-11, ROL-11, PRT-14
gender | STF-5
birthDate | STF-6
photo
qualification | CER?
identifier
code
period
issuer
communication | PID-15, NK1-20, LAN-2

8.4.9.3 RIM Mapping (http://hl7.org/v3)

Practitioner | Role
identifier | ./id
active | ./statusCode
name | ./name
telecom | ./telecom
address | ./addr
gender | ./administrativeGender
birthDate | ./birthTime
photo | ./subjectOf/ObservationEvent[code="photo"]/value
qualification | .playingEntity.playingRole[classCode=QUAL].code
identifier | .playingEntity.playingRole[classCode=QUAL].id
code | .playingEntity.playingRole[classCode=QUAL].code
period | .playingEntity.playingRole[classCode=QUAL].effectiveTime
issuer | .playingEntity.playingRole[classCode=QUAL].scoper
communication | ./languageCommunication

8.4.9.4 ServD (http://www.omg.org/spec/ServD/1.0/)

Practitioner | Provider
identifier | ./Identifiers
active
name | ./PreferredName (GivenNames, FamilyName, TitleCode)
telecom | ./ContactPoints
address | ./Addresses
gender | ./GenderCode
birthDate | (not represented in ServD)
photo | ./ImageURI (only supports the URI reference)
qualification | ./Qualifications
identifier
code | ./Qualifications.Value
period | ./Qualifications.StartDate and ./Qualifications.EndDate
issuer
communication | ./Languages.LanguageSpokenCode

8.4.10 Resource Practitioner - Extensions & Profiles

URL: https://hl7.org/fhir/R4/practitioner-profiles.html

8.4.10 Resource Practitioner - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

This table lists profiles and extensions for the Practitioner resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.4.10.1 Profiles

No Profiles defined for this resource

8.4.10.2 Extensions

practitioner-animalSpecies | Practitioner | animalSpecies | for Practitioner HL7 Extensions

Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

consent-Transcriber | Transcriber | for Consent HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
cqf-initiatingPerson | initiatingPerson | for Clinical Reasoning Extensions
cqf-receivingPerson | receivingPerson | for Clinical Reasoning Extensions
goal-acceptance | acceptance | for FHIR Core Goal Profile
procedure-directedBy | directedBy | for Procedure HL7 Extensions
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
questionnaireresponse-reviewer | reviewer | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.4.10.3 Search Extensions

No Search Extensions defined for this resource

8.4.11 Resource Practitioner - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/practitioner-version-maps.html

No validation errors - all conversions are clean

8.4.11 Resource Practitioner - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

R3 : R4 Conversion maps for Practitioner.

Functional status for this map: 14 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.4.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Practitioner3to4" = "R3 to R4 Conversions for Practitioner"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Practitioner" alias PractitionerR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Practitioner" alias Practitioner as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Practitioner(source src : PractitionerR3, target tgt : Practitioner) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.photo -> tgt.photo;
  src.qualification as vs0 -> tgt.qualification as vt0 then {
    vs0.identifier -> vt0.identifier;
    vs0.code -> vt0.code;
    vs0.period -> vt0.period;
    vs0.issuer -> vt0.issuer;
  };
  src.communication -> tgt.communication;
}
```

8.4.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Practitioner4to3" = "R4 to R3 Conversion for Practitioner"

uses "http://hl7.org/fhir/StructureDefinition/Practitioner" alias Practitioner as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Practitioner" alias PractitionerR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Practitioner(source src : Practitioner, target tgt : PractitionerR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
```

```
  src.address -> tgt.address;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.photo -> tgt.photo;
  src.qualification as vs0 -> tgt.qualification as vt0 then {
    vs0.identifier -> vt0.identifier;
    vs0.code -> vt0.code;
    vs0.period -> vt0.period;
    vs0.issuer -> vt0.issuer;
  };
  src.communication -> tgt.communication;
}
```

8.4.11.3 R4 Validation Errors

8.4.12 Practitioner HL7 Extensions

URL: https://hl7.org/fhir/R4/practitioner-extensions.html

Defines common extensions used with or related to the Practitioner resource

8.4.12 Practitioner HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.4.12.1 Content

Extensions:

practitioner-animalSpecies | animalSpecies : This extension should be used to specifiy that a practioner or RelatedPerson resource is a service animal.

8.4.13 JSON Schema for Practitioner

URL: https://hl7.org/fhir/R4/practitioner.schema.json.html

8.4.13 JSON Schema for Practitioner

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Practitioner

This is a representation of the json schema for Practitioner, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Practitioner",
  "$ref": "#/definitions/Practitioner",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Practitioner": {
      "description": "A person who is directly or indirectly involved in the provisioning of healthcare.",
      "properties": {
        "resourceType": {
          "description": "This is a Practitioner resource",
          "const": "Practitioner"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
```

```
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "An identifier that applies to this person in this role.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Whether this practitioner\u0027s record is in active use.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "The name(s) associated with the practitioner.",
      "items": {
        "$ref": "HumanName.schema.json#/definitions/HumanName"
      },
      "type": "array"
    },
    "telecom": {
      "description": "A contact detail for the practitioner, e.g. a telephone number or an email address.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "address": {
      "description": "Address(es) of the practitioner that are not role specific (typically home address). \rWork
addresses are not typically entered in this property as they are usually role dependent.",
      "items": {
        "$ref": "Address.schema.json#/definitions/Address"
      },
      "type": "array"
    },
    "gender": {
      "description": "Administrative Gender - the gender that the person is considered to have for administration
and record keeping purposes.",
      "enum": [
        "male",
        "female",
        "other",
        "unknown"
      ]
    },
    "_gender": {
      "description": "Extensions for gender",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "birthDate": {
      "description": "The date of birth for the practitioner.",
      "$ref": "#/definitions/date"
    },
```

```
    "_birthDate": {
      "description": "Extensions for birthDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "photo": {
      "description": "Image of the person.",
      "items": {
        "$ref": "Attachment.schema.json#/definitions/Attachment"
      },
      "type": "array"
    },
    "qualification": {
      "description": "The official certifications, training, and licenses that authorize or otherwise pertain to the
 provision of care by the practitioner.  For example, a medical license issued by a medical board authorizing the
practitioner to practice medicine within a certian locality.",
      "items": {
        "$ref": "#/definitions/Practitioner_Qualification"
      },
      "type": "array"
    },
    "communication": {
      "description": "A language the practitioner can use in patient communication.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Practitioner_Qualification": {
  "description": "A person who is directly or indirectly involved in the provisioning of healthcare.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "An identifier that applies to this person\u0027s qualification in this role.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "code": {
      "description": "Coded representation of the qualification.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "period": {
      "description": "Period during which the qualification is valid.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "issuer": {
      "description": "Organization that regulates and issues the qualification.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "code"
  ]
```

```
    }
  }
}
```

8.4.14 Resource Practitioner - Maturity Dependency View

URL: https://hl7.org/fhir/R4/practitioner-dependencies.html

8.4.14 Resource Practitioner - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Practitioner

Dependency Graph for Practitioner FMM level 3

Name | Card. | Type | Dependency Analysis
Practitioner | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
name | 0..* | HumanName
telecom | 0..* | ContactPoint
address | 0..* | Address
gender | 0..1 | code
birthDate | 0..1 | date
photo | 0..* | Attachment
qualification | 0..* | BackboneElement
identifier | 0..* | Identifier
code | 1..1 | CodeableConcept
period | 0..1 | Period
issuer | 0..1 | Reference(Organization)
communication | 0..* | CodeableConcept

8.5 Resource PractitionerRole - Content

URL: https://hl7.org/fhir/R4/practitionerrole.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Individual | Compartments: Practitioner

A specific set of Roles/Locations/specialties/services that a practitioner may perform at an organization for a period of time.

8.5.1 Scope and Usage

PractitionerRole covers the recording of the location and types of services that Practitioners are able to provide for an organization.

The role, specialty, Location telecom and HealthcareService properties can be repeated if required in other instances of the PractitionerRole. Some systems record a collection of service values for a single location, others record the single service and the list of locations it is available. Both are acceptable options for representing this data. Where availability, telecom, or other details are not the same across all healthcareservices, or locations a seperate PractitionerRole instance should be created.

8.5.2 Boundaries and Relationships

Qualifications (from the Practitioner resource) do not imply a Role, but might be considered when an Organization allocates practitioners to roles within their organization, and could provide useful information (such as expiry information) which could need to be tracked in some situations to ensure they continue to be eligible for a specific role.

The CareTeam resource is also often used to provide details of a role that a practitioner is allocated to perform, but is usually scoped to a much finer granularity of care, and often within the specific context of a Patient, or functional role (e.g. Crisis planning team). In contrast the PractitionerRole is used in a more general sense to cover all the places that the practitioner is allocated to work (and specific details relevant to that role - such as a specific contact number, or electronic services endpoint).

8.5.3 Background and Context

Practitioner performs different roles within the same or even different organizations. Depending on jurisdiction and custom, it may be necessary to maintain a specific Practitioner Resource for each such role or have a single Practitioner with multiple roles. The role can be limited to a specific period, after which authorization for this role ends. Note that the represented organization need not necessarily be the (direct) employer of a Practitioner.

This resource is referenced by Signature, Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BiologicallyDerivedProduct, CarePlan, CareTeam, CatalogEntry, ChargeItem, Claim, ClaimResponse, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, CoverageEligibilityRequest, CoverageEligibilityResponse, DetectedIssue, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest,

DocumentReference, Encounter, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, ExplanationOfBenefit, Flag, Goal, Group, ImagingStudy, Immunization, Invoice, Linkage, List, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MedicinalProduct, MessageHeader, NutritionOrder, Observation, Patient, PaymentNotice, PaymentReconciliation, Procedure, Provenance, QuestionnaireResponse, RequestGroup, ResearchStudy, RiskAssessment, Schedule, ServiceRequest, Specimen, SupplyDelivery, SupplyRequest, Task, VerificationResult and VisionPrescription

8.5.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
PractitionerRole | TU | | DomainResource | Roles/organizations the practitioner is associated with Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifiers that are specific to a role/location
active | Σ | 0..1 | boolean | Whether this practitioner role record is in active use
period | Σ | 0..1 | Period | The period during which the practitioner is authorized to perform in these role(s)
practitioner | Σ | 0..1 | Reference(Practitioner) | Practitioner that is able to provide the defined services for the organization
organization | Σ | 0..1 | Reference(Organization) | Organization where the roles are available
code | Σ | 0..* | CodeableConcept | Roles which this practitioner may perform Practitioner role (Example)
specialty | Σ | 0..* | CodeableConcept | Specific specialty of the practitioner Practice Setting Code Value Set (Preferred)
location | Σ | 0..* | Reference(Location) | The location(s) at which this practitioner provides care
healthcareService | | 0..* | Reference(HealthcareService) | The list of healthcare services that this worker provides for this role's Organization/Location(s)
telecom | Σ | 0..* | ContactPoint | Contact details that are specific to the role/location/service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not available from this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the practitioner with this role
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PractitionerRole xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifiers that are specific to a role/location --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner role record is in active use -->
 <period><!-- 0..1 Period The period during which the practitioner is authorized to perform in these role(s)
    --></period>
 <practitioner><!-- 0..1 Reference(Practitioner) Practitioner that is able to provide the defined services for the
    organization --></practitioner>
 <organization><!-- 0..1 Reference(Organization) Organization where the roles are available --></organization>
 <code><!-- 0..* CodeableConcept Roles which this practitioner may perform --></code>
 <specialty><!-- 0..* CodeableConcept Specific specialty of the practitioner --></specialty>
 <location><!-- 0..* Reference(Location) The location(s) at which this practitioner provides care --></location>
 <healthcareService><!-- 0..* Reference(HealthcareService) The list of healthcare services that this worker provides for
     this role's Organization/Location(s) --></healthcareService>
 <telecom><!-- 0..* ContactPoint Contact details that are specific to the role/location/service --></telecom>
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not available from this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
```

```
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the practitioner
    with this role --></endpoint>
</PractitionerRole>
```

JSON Template

```
{
  "resourceType" : "PractitionerRole",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifiers that are specific to a role/location
  "active" : <boolean>, // Whether this practitioner role record is in active use
  "period" : { Period }, // The period during which the practitioner is authorized to perform in these role(s)
  "practitioner" : { Reference(Practitioner) }, // Practitioner that is able to provide the defined services for the
    organization
  "organization" : { Reference(Organization) }, // Organization where the roles are available
  "code" : [{ CodeableConcept }], // Roles which this practitioner may perform
  "specialty" : [{ CodeableConcept }], // Specific specialty of the practitioner
  "location" : [{ Reference(Location) }], // The location(s) at which this practitioner provides care
  "healthcareService" : [{ Reference(HealthcareService) }], // The list of healthcare services that this worker provides
     for this role's Organization/Location(s)
  "telecom" : [{ ContactPoint }], // Contact details that are specific to the role/location/service
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not available from this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    practitioner with this role
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PractitionerRole;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PractitionerRole.identifier [ Identifier ], ... ; # 0..* Business Identifiers that are specific to a
    role/location
  fhir:PractitionerRole.active [ boolean ]; # 0..1 Whether this practitioner role record is in active use
  fhir:PractitionerRole.period [ Period ]; # 0..1 The period during which the practitioner is authorized to perform in
    these role(s)
  fhir:PractitionerRole.practitioner [ Reference(Practitioner) ]; # 0..1 Practitioner that is able to provide the
    defined services for the organization
  fhir:PractitionerRole.organization [ Reference(Organization) ]; # 0..1 Organization where the roles are available
  fhir:PractitionerRole.code [ CodeableConcept ], ... ; # 0..* Roles which this practitioner may perform
  fhir:PractitionerRole.specialty [ CodeableConcept ], ... ; # 0..* Specific specialty of the practitioner
  fhir:PractitionerRole.location [ Reference(Location) ], ... ; # 0..* The location(s) at which this practitioner
    provides care
  fhir:PractitionerRole.healthcareService [ Reference(HealthcareService) ], ... ; # 0..* The list of healthcare services
     that this worker provides for this role's Organization/Location(s)
  fhir:PractitionerRole.telecom [ ContactPoint ], ... ; # 0..* Contact details that are specific to the
    role/location/service
  fhir:PractitionerRole.availableTime [ # 0..* Times the Service Site is available
    fhir:PractitionerRole.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:PractitionerRole.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:PractitionerRole.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:PractitionerRole.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay = true)
  ], ...;
  fhir:PractitionerRole.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:PractitionerRole.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:PractitionerRole.notAvailable.during [ Period ]; # 0..1 Service not available from this date
  ], ...;
  fhir:PractitionerRole.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:PractitionerRole.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the practitioner with this role
]
```

Changes since R3

PractitionerRole

PractitionerRole.active | Default Value "true" removed

PractitionerRole.availableTime.daysOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
PractitionerRole | TU | | DomainResource | Roles/organizations the practitioner is associated with Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifiers that are specific to a role/location
active | Σ | 0..1 | boolean | Whether this practitioner role record is in active use
period | Σ | 0..1 | Period | The period during which the practitioner is authorized to perform in these role(s)
practitioner | Σ | 0..1 | Reference(Practitioner) | Practitioner that is able to provide the defined services for the organization
organization | Σ | 0..1 | Reference(Organization) | Organization where the roles are available
code | Σ | 0..* | CodeableConcept | Roles which this practitioner may perform Practitioner role (Example)
specialty | Σ | 0..* | CodeableConcept | Specific specialty of the practitioner Practice Setting Code Value Set (Preferred)
location | Σ | 0..* | Reference(Location) | The location(s) at which this practitioner provides care
healthcareService | | 0..* | Reference(HealthcareService) | The list of healthcare services that this worker provides for this role's Organization/Location(s)
telecom | Σ | 0..* | ContactPoint | Contact details that are specific to the role/location/service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not available from this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the practitioner with this role
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PractitionerRole xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifiers that are specific to a role/location --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner role record is in active use -->
 <period><!-- 0..1 Period The period during which the practitioner is authorized to perform in these role(s)
    --></period>
 <practitioner><!-- 0..1 Reference(Practitioner) Practitioner that is able to provide the defined services for the
    organization --></practitioner>
 <organization><!-- 0..1 Reference(Organization) Organization where the roles are available --></organization>
 <code><!-- 0..* CodeableConcept Roles which this practitioner may perform --></code>
 <specialty><!-- 0..* CodeableConcept Specific specialty of the practitioner --></specialty>
 <location><!-- 0..* Reference(Location) The location(s) at which this practitioner provides care --></location>
 <healthcareService><!-- 0..* Reference(HealthcareService) The list of healthcare services that this worker provides for
     this role's Organization/Location(s) --></healthcareService>
 <telecom><!-- 0..* ContactPoint Contact details that are specific to the role/location/service --></telecom>
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not available from this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the practitioner
    with this role --></endpoint>
</PractitionerRole>
```

JSON Template

```
{
  "resourceType" : "PractitionerRole",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifiers that are specific to a role/location
  "active" : <boolean>, // Whether this practitioner role record is in active use
  "period" : { Period }, // The period during which the practitioner is authorized to perform in these role(s)
  "practitioner" : { Reference(Practitioner) }, // Practitioner that is able to provide the defined services for the
    organization
  "organization" : { Reference(Organization) }, // Organization where the roles are available
  "code" : [{ CodeableConcept }], // Roles which this practitioner may perform
  "specialty" : [{ CodeableConcept }], // Specific specialty of the practitioner
  "location" : [{ Reference(Location) }], // The location(s) at which this practitioner provides care
  "healthcareService" : [{ Reference(HealthcareService) }], // The list of healthcare services that this worker provides
     for this role's Organization/Location(s)
  "telecom" : [{ ContactPoint }], // Contact details that are specific to the role/location/service
  "availableTime" : [{ // Times the Service Site is available
```

```
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not available from this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    practitioner with this role
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PractitionerRole;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PractitionerRole.identifier [ Identifier ], ... ; # 0..* Business Identifiers that are specific to a
    role/location
  fhir:PractitionerRole.active [ boolean ]; # 0..1 Whether this practitioner role record is in active use
  fhir:PractitionerRole.period [ Period ]; # 0..1 The period during which the practitioner is authorized to perform in
    these role(s)
  fhir:PractitionerRole.practitioner [ Reference(Practitioner) ]; # 0..1 Practitioner that is able to provide the
    defined services for the organization
  fhir:PractitionerRole.organization [ Reference(Organization) ]; # 0..1 Organization where the roles are available
  fhir:PractitionerRole.code [ CodeableConcept ], ... ; # 0..* Roles which this practitioner may perform
  fhir:PractitionerRole.specialty [ CodeableConcept ], ... ; # 0..* Specific specialty of the practitioner
  fhir:PractitionerRole.location [ Reference(Location) ], ... ; # 0..* The location(s) at which this practitioner
    provides care
  fhir:PractitionerRole.healthcareService [ Reference(HealthcareService) ], ... ; # 0..* The list of healthcare services
     that this worker provides for this role's Organization/Location(s)
  fhir:PractitionerRole.telecom [ ContactPoint ], ... ; # 0..* Contact details that are specific to the
    role/location/service
  fhir:PractitionerRole.availableTime [ # 0..* Times the Service Site is available
    fhir:PractitionerRole.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:PractitionerRole.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:PractitionerRole.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:PractitionerRole.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay = true)
  ], ...;
  fhir:PractitionerRole.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:PractitionerRole.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:PractitionerRole.notAvailable.during [ Period ]; # 0..1 Service not available from this date
  ], ...;
  fhir:PractitionerRole.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:PractitionerRole.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the practitioner with this role
]
```

Changes since Release 3

PractitionerRole

PractitionerRole.active | Default Value "true" removed

PractitionerRole.availableTime.daysOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.5.4.1 Terminology Bindings

Path | Definition | Type | Reference

PractitionerRole.code | The role a person plays representing an organization. | Example | PractitionerRole

PractitionerRole.specialty | Specific specialty associated with the agency. | Preferred | PracticeSettingCodeValueSet

PractitionerRole.availableTime.daysOfWeek | The days of the week. | Required | DaysOfWeek

8.5.5 Notes

• There is no address on the PractitionerRole as the location that is defined here contains the address. This prevents having to duplicate the address values across multiple resources.

8.5.6 Multiple Locations in a PractitionerRole

As briefly noted in the boundaries and relationships section the PractitionerRole resource can be used to represent multiple locations in the one resource instance, however this should be done with care. When representing multiple locations in an instance, and/or multiple specialties, all details apply to all locations. Using that approach you cannot differentiate a different contact number, different availabilities, or different services at each location. Hence maintaining these records needs to ensure that when changing a value, such as availability, it applies to all locations. If different values are required for the different locations, then a new instance will need to be created, and then split as appropriate. Existing resources referencing the original PractitionerRole instance will need to be updated to refer to the appropriate record. For this reason we expect that it will be common to profile the location element down to a single location, simplifying overall usage, and the availabilities are clear that they apply to this location, and any referenced healthcare services.

8.5.7 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Whether this practitioner role record is in active use | PractitionerRole.active
date | date | The period during which the practitioner is authorized to perform in these role(s) | PractitionerRole.period
email | token | A value in an email contact | PractitionerRole.telecom.where(system='email') | 4 Resources
endpoint | reference | Technical endpoints providing access to services operated for the practitioner with this role | PractitionerRole.endpoint (Endpoint)
identifier | token | A practitioner's Identifier | PractitionerRole.identifier
location | reference | One of the locations at which this practitioner provides care | PractitionerRole.location (Location)
organization | reference | The identity of the organization the practitioner represents / acts on behalf of | PractitionerRole.organization (Organization)
phone | token | A value in a phone contact | PractitionerRole.telecom.where(system='phone') | 4 Resources
practitioner | reference | Practitioner that is able to provide the defined services for the organization | PractitionerRole.practitioner (Practitioner)
role | token | The practitioner can perform this role at for the organization | PractitionerRole.code
service | reference | The list of healthcare services that this worker provides for this role's Organization/Location(s) | PractitionerRole.healthcareService (HealthcareService)
specialty | token | The practitioner has this specialty at an organization | PractitionerRole.specialty
telecom | token | The value in any kind of contact | PractitionerRole.telecom | 4 Resources

8.5.8 Resource PractitionerRole - Examples

URL: https://hl7.org/fhir/R4/practitionerrole-examples.html

8.5.8 Resource PractitionerRole - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

Example Name | id | Format
Dr Adam Careful's role at the Acme Hospital | example | XML | JSON | Turtle
HL7 Defined Practitioner Role examples | 3ad0687e-f477-468c-afd5-fcc2bf897808 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.5.9 Resource PractitionerRole - Detailed Descriptions

URL: https://hl7.org/fhir/R4/practitionerrole-definitions.html

8.5.9 Resource PractitionerRole - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Individual | Compartments: Practitioner

Detailed Descriptions for the elements in the PractitionerRole resource.

PractitionerRole
Element Id | PractitionerRole
Definition | A specific set of Roles/Locations/specialties/services that a practitioner may perform at an organization for a period of time.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to track services that a healthcare provider is able to provide at an organization's location, and the services that they can perform there.
PractitionerRole.identifier
Element Id | PractitionerRole.identifier
Definition | Business Identifiers that are specific to a role/location.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the agent.

Summary | true
PractitionerRole.active
Element Id | PractitionerRole.active
Definition | Whether this practitioner role record is in active use.
Cardinality | 0..1
Type | boolean
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a practitioner role record as not to be used because it was created in error, or otherwise no longer in active use.
Summary | true
Comments | If this value is false, you may refer to the period to see when the role was in active use. If there is no period specified, no inference can be made about when it was active.
PractitionerRole.period
Element Id | PractitionerRole.period
Definition | The period during which the person is authorized to act as a practitioner in these role(s) for the organization.
Cardinality | 0..1
Type | Period
Requirements | Even after the agencies is revoked, the fact that it existed must still be recorded.
Summary | true
PractitionerRole.practitioner
Element Id | PractitionerRole.practitioner
Definition | Practitioner that is able to provide the defined services for the organization.
Cardinality | 0..1
Type | Reference(Practitioner)
Summary | true
PractitionerRole.organization
Element Id | PractitionerRole.organization
Definition | The organization where the Practitioner performs the roles associated.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
PractitionerRole.code
Element Id | PractitionerRole.code
Definition | Roles which this practitioner is authorized to perform for the organization.
Cardinality | 0..*
Terminology Binding | Practitioner role (Example)
Type | CodeableConcept
Requirements | Need to know what authority the practitioner has - what can they do?
Summary | true
Comments | A person may have more than one role.
PractitionerRole.specialty
Element Id | PractitionerRole.specialty
Definition | Specific specialty of the practitioner.
Cardinality | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
PractitionerRole.location
Element Id | PractitionerRole.location
Definition | The location(s) at which this practitioner provides care.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
PractitionerRole.healthcareService
Element Id | PractitionerRole.healthcareService
Definition | The list of healthcare services that this worker provides for this role's Organization/Location(s).
Cardinality | 0..*
Type | Reference(HealthcareService)
PractitionerRole.telecom
Element Id | PractitionerRole.telecom
Definition | Contact details that are specific to the role/location/service.
Cardinality | 0..*
Type | ContactPoint
Requirements | Often practitioners have a dedicated line for each location (or service) that they work at, and need to be able to define separate contact details for each of these.
Summary | true
PractitionerRole.availableTime
Element Id | PractitionerRole.availableTime
Definition | A collection of times the practitioner is available or performing this role at the location and/or healthcareservice.
Cardinality | 0..*
Comments | More detailed availability information may be provided in associated Schedule/Slot resources.
PractitionerRole.availableTime.daysOfWeek

Element Id | PractitionerRole.availableTime.daysOfWeek
Definition | Indicates which days of the week are available between the start and end Times.
Cardinality | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
PractitionerRole.availableTime.allDay
Element Id | PractitionerRole.availableTime.allDay
Definition | Is this always available? (hence times are irrelevant) e.g. 24 hour service.
Cardinality | 0..1
Type | boolean
PractitionerRole.availableTime.availableStartTime
Element Id | PractitionerRole.availableTime.availableStartTime
Definition | The opening time of day. Note: If the AllDay flag is set, then this time is ignored.
Cardinality | 0..1
Type | time
Comments | The timezone is expected to be for where this HealthcareService is provided at.
PractitionerRole.availableTime.availableEndTime
Element Id | PractitionerRole.availableTime.availableEndTime
Definition | The closing time of day. Note: If the AllDay flag is set, then this time is ignored.
Cardinality | 0..1
Type | time
Comments | The timezone is expected to be for where this HealthcareService is provided at.
PractitionerRole.notAvailable
Element Id | PractitionerRole.notAvailable
Definition | The practitioner is not available or performing this role during this period of time due to the provided reason.
Cardinality | 0..*
PractitionerRole.notAvailable.description
Element Id | PractitionerRole.notAvailable.description
Definition | The reason that can be presented to the user as to why this time is not available.
Cardinality | 1..1
Type | string
PractitionerRole.notAvailable.during
Element Id | PractitionerRole.notAvailable.during
Definition | Service is not available (seasonally or for a public holiday) from this date.
Cardinality | 0..1
Type | Period
PractitionerRole.availabilityExceptions
Element Id | PractitionerRole.availabilityExceptions
Definition | A description of site availability exceptions, e.g. public holiday availability. Succinctly describing all possible exceptions to normal site availability as details in the available Times and not available Times.
Cardinality | 0..1
Type | string
PractitionerRole.endpoint
Element Id | PractitionerRole.endpoint
Definition | Technical endpoints providing access to services operated for the practitioner with this role.
Cardinality | 0..*
Type | Reference(Endpoint)
Requirements | Organizations have multiple systems that provide various services and ,ay also be different for practitioners too. So the endpoint satisfies the need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.5.10 Resource PractitionerRole - Mappings

URL: https://hl7.org/fhir/R4/practitionerrole-mappings.html

8.5.10 Resource PractitionerRole - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

ServD

Mappings for the practitionerrole resource (see Mappings to Other Standards for further information & status).

8.5.10.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

PractitionerRole
identifier | FiveWs.identifier
active | FiveWs.status

period | FiveWs.done[x]
location | FiveWs.where[x]

8.5.10.2 HL7 v2 Mapping (http://hl7.org/v2)

PractitionerRole | PRD (as one example)
identifier | PRD-7 (or XCN.1)
active | STF-7
period | PRD-8/9 / PRA-5.4
practitioner
organization
code | PRD-1 / STF-18 / PRA-3 / PRT-4 / ROL-3 / ORC-12 / OBR-16 / PV1-7 / PV1-8 / PV1-9 / PV1-17
specialty | PRA-5
location
healthcareService | EDU-2 / AFF-3
telecom
availableTime
daysOfWeek
allDay
availableStartTime
availableEndTime
notAvailable
description
during
availabilityExceptions
endpoint

8.5.10.3 RIM Mapping (http://hl7.org/v3)

PractitionerRole | Role
identifier | .id
active | .statusCode
period | .performance[@typeCode <= 'PPRF'].ActDefinitionOrEvent.effectiveTime
practitioner | .player
organization | .scoper
code | .code
specialty | .player.HealthCareProvider[@classCode = 'PROV'].code
location | .performance.ActDefinitionOrEvent.ServiceDeliveryLocation[@classCode = 'SDLOC']
healthcareService | .player.QualifiedEntity[@classCode = 'QUAL'].code
telecom | .telecom
availableTime | .effectiveTime
daysOfWeek | .effectiveTime
allDay | .effectiveTime
availableStartTime | .effectiveTime
availableEndTime | .effectiveTime
notAvailable | .effectiveTime
description | n/a
during | .effectiveTime
availabilityExceptions | .effectiveTime
endpoint | n/a

8.5.10.4 ServD (http://www.omg.org/spec/ServD/1.0/)

PractitionerRole | ServiceSiteProvider
identifier | ./Identifiers
active
period | (ServD maps Practitioners and Organizations via another entity, so this concept is not available)
practitioner
organization
code | (ServD maps Practitioners and Organizations via another entity, so this concept is not available)
specialty | ./Specialty
location | (ServD maps Practitioners and Organizations via another entity, so this concept is not available)
 However these are accessed via the Site.ServiceSite.ServiceSiteProvider record. (The Site has the location)
healthcareService
telecom
availableTime
daysOfWeek
allDay
availableStartTime
availableEndTime
notAvailable
description
during
availabilityExceptions

endpoint

8.5.11 Resource PractitionerRole - Extensions & Profiles

URL: https://hl7.org/fhir/R4/practitionerrole-profiles.html

8.5.11 Resource PractitionerRole - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

This table lists profiles and extensions for the PractitionerRole resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.5.11.1 Profiles

No Profiles defined for this resource

8.5.11.2 Extensions

practitionerrole-primaryInd | PractitionerRole.specialty | primaryInd | for PractitionerRole HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

procedure-directedBy | directedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.5.11.3 Search Extensions

No Search Extensions defined for this resource

8.5.12 Resource PractitionerRole - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/practitionerrole-version-maps.html

No validation errors - all conversions are clean

8.5.12 Resource PractitionerRole - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Individual | Compartments: Practitioner

R3 : R4 Conversion maps for PractitionerRole.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.5.12.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/PractitionerRole3to4" = "R3 to R4 Conversions for PractitionerRole"

uses "http://hl7.org/fhir/3.0/StructureDefinition/PractitionerRole" alias PractitionerRoleR3 as source
uses "http://hl7.org/fhir/StructureDefinition/PractitionerRole" alias PractitionerRole as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group PractitionerRole(source src : PractitionerRoleR3, target tgt : PractitionerRole) extends DomainResource <<type+>>
    {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.period -> tgt.period;
  src.practitioner -> tgt.practitioner;
  src.organization -> tgt.organization;
  src.code -> tgt.code;
  src.specialty -> tgt.specialty;
  src.location -> tgt.location;
  src.healthcareService -> tgt.healthcareService;
  src.telecom -> tgt.telecom;
  src.availableTime as vs0 -> tgt.availableTime as vt0 then {
    vs0.daysOfWeek -> vt0.daysOfWeek;
    vs0.allDay -> vt0.allDay;
    vs0.availableStartTime -> vt0.availableStartTime;
    vs0.availableEndTime -> vt0.availableEndTime;
  };
```

```
  src.notAvailable as vs0 -> tgt.notAvailable as vt0 then {
    vs0.description -> vt0.description;
    vs0.during -> vt0.during;
  };
  src.availabilityExceptions -> tgt.availabilityExceptions;
  src.endpoint -> tgt.endpoint;
}
```

8.5.12.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/PractitionerRole4to3" = "R4 to R3 Conversion for PractitionerRole"

uses "http://hl7.org/fhir/StructureDefinition/PractitionerRole" alias PractitionerRole as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/PractitionerRole" alias PractitionerRoleR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group PractitionerRole(source src : PractitionerRole, target tgt : PractitionerRoleR3) extends DomainResource <<type+>>
    {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.period -> tgt.period;
  src.practitioner -> tgt.practitioner;
  src.organization -> tgt.organization;
  src.code -> tgt.code;
  src.specialty -> tgt.specialty;
  src.location -> tgt.location;
  src.healthcareService -> tgt.healthcareService;
  src.telecom -> tgt.telecom;
  src.availableTime as vs0 -> tgt.availableTime as vt0 then {
    vs0.daysOfWeek -> vt0.daysOfWeek;
    vs0.allDay -> vt0.allDay;
    vs0.availableStartTime -> vt0.availableStartTime;
    vs0.availableEndTime -> vt0.availableEndTime;
  };
  src.notAvailable as vs0 -> tgt.notAvailable as vt0 then {
    vs0.description -> vt0.description;
    vs0.during -> vt0.during;
  };
  src.availabilityExceptions -> tgt.availabilityExceptions;
  src.endpoint -> tgt.endpoint;
}
```

8.5.12.3 R4 Validation Errors

8.5.13 PractitionerRole HL7 Extensions

URL: https://hl7.org/fhir/R4/practitionerrole-extensions.html

Defines common extensions used with or related to the PractitionerRole resource

8.5.13 PractitionerRole HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.5.13.1 Content

Extensions:

practitionerrole-primaryInd | primaryInd : Flag indicating if the specialty is the primary specialty of the provider. Normally, a practitioner will have one primary specialty, but in some cases more than one can be primary.

8.5.14 JSON Schema for PractitionerRole

URL: https://hl7.org/fhir/R4/practitionerrole.schema.json.html

8.5.14 JSON Schema for PractitionerRole

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Practitioner

This is a representation of the json schema for PractitionerRole, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/PractitionerRole",
  "$ref": "#/definitions/PractitionerRole",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "PractitionerRole": {
      "description": "A specific set of Roles/Locations/specialties/services that a practitioner may perform at an
    organization for a period of time.",
      "properties": {
        "resourceType": {
          "description": "This is a PractitionerRole resource",
          "const": "PractitionerRole"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
```

```
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business Identifiers that are specific to a role/location.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Whether this practitioner role record is in active use.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The period during which the person is authorized to act as a practitioner in these role(s) for
 the organization.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "practitioner": {
      "description": "Practitioner that is able to provide the defined services for the organization.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
```

```
    "organization": {
      "description": "The organization where the Practitioner performs the roles associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "code": {
      "description": "Roles which this practitioner is authorized to perform for the organization.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialty": {
      "description": "Specific specialty of the practitioner.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "location": {
      "description": "The location(s) at which this practitioner provides care.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "healthcareService": {
      "description": "The list of healthcare services that this worker provides for this role\u0027s
Organization/Location(s).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "telecom": {
      "description": "Contact details that are specific to the role/location/service.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "availableTime": {
      "description": "A collection of times the practitioner is available or performing this role at the location
and/or healthcareservice.",
      "items": {
        "$ref": "#/definitions/PractitionerRole_AvailableTime"
      },
      "type": "array"
    },
    "notAvailable": {
      "description": "The practitioner is not available or performing this role during this period of time due to
the provided reason.",
      "items": {
        "$ref": "#/definitions/PractitionerRole_NotAvailable"
      },
      "type": "array"
    },
    "availabilityExceptions": {
      "description": "A description of site availability exceptions, e.g. public holiday availability. Succinctly
describing all possible exceptions to normal site availability as details in the available Times and not available
Times.",
      "$ref": "#/definitions/string"
    },
    "_availabilityExceptions": {
      "description": "Extensions for availabilityExceptions",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "endpoint": {
      "description": "Technical endpoints providing access to services operated for the practitioner with this
role.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"PractitionerRole_AvailableTime": {
  "description": "A specific set of Roles/Locations/specialties/services that a practitioner may perform at an
organization for a period of time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
```

```
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "daysOfWeek": {
      "description": "Indicates which days of the week are available between the start and end Times.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_daysOfWeek": {
      "description": "Extensions for daysOfWeek",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "allDay": {
      "description": "Is this always available? (hence times are irrelevant) e.g. 24 hour service.",
      "$ref": "#/definitions/boolean"
    },
    "_allDay": {
      "description": "Extensions for allDay",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "availableStartTime": {
      "description": "The opening time of day. Note: If the AllDay flag is set, then this time is ignored.",
      "$ref": "#/definitions/time"
    },
    "_availableStartTime": {
      "description": "Extensions for availableStartTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "availableEndTime": {
      "description": "The closing time of day. Note: If the AllDay flag is set, then this time is ignored.",
      "$ref": "#/definitions/time"
    },
    "_availableEndTime": {
      "description": "Extensions for availableEndTime",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"PractitionerRole_NotAvailable": {
  "description": "A specific set of Roles/Locations/specialties/services that a practitioner may perform at an
organization for a period of time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
```

```
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "description": {
          "description": "The reason that can be presented to the user as to why this time is not available.",
          "$ref": "#/definitions/string"
        },
        "_description": {
          "description": "Extensions for description",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "during": {
          "description": "Service is not available (seasonally or for a public holiday) from this date.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false
    }
  }
}
```

8.5.15 Resource PractitionerRole - Maturity Dependency View

URL: https://hl7.org/fhir/R4/practitionerrole-dependencies.html

8.5.15 Resource PractitionerRole - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

Dependency Graph for PractitionerRole FMM level 2

Name | Card. | Type | Dependency Analysis
PractitionerRole | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
period | 0..1 | Period
practitioner | 0..1 | Reference(Practitioner)
organization | 0..1 | Reference(Organization)
code | 0..* | CodeableConcept
specialty | 0..* | CodeableConcept
location | 0..* | Reference(Location)
healthcareService | 0..* | Reference(HealthcareService)
telecom | 0..* | ContactPoint
availableTime | 0..* | BackboneElement
daysOfWeek | 0..* | code
allDay | 0..1 | boolean
availableStartTime | 0..1 | time
availableEndTime | 0..1 | time
notAvailable | 0..* | BackboneElement
description | 1..1 | string
during | 0..1 | Period
availabilityExceptions | 0..1 | string
endpoint | 0..* | Reference(Endpoint)

8.6 Resource Organization - Content

URL: https://hl7.org/fhir/R4/organization.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

A formally or informally recognized grouping of people or organizations formed for the purpose of achieving some form of collective action. Includes companies, institutions, corporations, departments, community groups, healthcare practice groups, payer/insurer, etc.

8.6.1 Scope and Usage

This resource may be used in a shared registry of contact and other information for various organizations or it can be used merely as a support for other resources that need to reference organizations, perhaps as a document, message or as a contained resource. If using a registry approach, it's entirely possible for multiple registries to exist, each dealing with different types or levels of organization.

8.6.2 Boundaries and Relationships

The Organization resource is used for collections of people that have come together to achieve an objective. The Group resource is used to identify a collection of people (or animals, devices, etc.) that are gathered for the purpose of analysis or acting upon, but are not expected to act themselves.

The Organization resource often exists as a hierarchy of organization resources, using the part-of property to provide the association of the child to its parent organization. This organizational hierarchy helps communicate the conceptual structure, whereas the Location resource provides the physical representation of the hierarchy. The linkage between Organization and Location is from each point in the location hierarchy to the appropriate level in the Organization hierarchy. These links don't all have to be to the top level Organization. When populating the organization and location hierarchies there is often not a clear distinction between these 2, however to assist in making the decision, Locations are always used for recording where a service occurs, and hence where encounters and observations are associated. The Organization property on these resources might not be the location where the service took place.

This resource is referenced by Annotation, Identifier, Signature, UsageContext, Account, AuditEvent, Basic, BiologicallyDerivedProduct, CapabilityStatement, CarePlan, CareTeam, CatalogEntry, ChargeItem, Claim, ClaimResponse, Communication, CommunicationRequest, Composition, Consent, Contract, Coverage, CoverageEligibilityRequest, CoverageEligibilityResponse, Device, DeviceDefinition, DeviceRequest, DiagnosticReport, DocumentManifest, DocumentReference, Encounter, Endpoint, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, ExplanationOfBenefit, Flag, Goal, Group, HealthcareService, ImagingStudy, Immunization, ImmunizationEvaluation, ImmunizationRecommendation, InsurancePlan, Invoice, Linkage, Location, MeasureReport, Media, Medication, MedicationDispense, MedicationKnowledge, MedicationRequest, MedicationStatement, MedicinalProduct, MedicinalProductAuthorization, MedicinalProductIngredient, MedicinalProductManufactured, MedicinalProductPackaged, MessageHeader, MolecularSequence, Observation, itself, OrganizationAffiliation, Patient, PaymentNotice, PaymentReconciliation, Person, Practitioner, PractitionerRole, Procedure, Provenance, QuestionnaireResponse, ResearchStudy, ServiceRequest, SupplyDelivery, SupplyRequest, Task and VerificationResult

8.6.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Organization | ITU | | DomainResource | A grouping of people or organizations with a common purpose + Rule: The organization SHALL at least have a name or an identifier, and possibly more than one Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ΣI | 0..* | Identifier | Identifies this organization across multiple systems
active | ?!Σ | 0..1 | boolean | Whether the organization's record is still in active use
type | Σ | 0..* | CodeableConcept | Kind of organization Organization type (Example)
name | ΣI | 0..1 | string | Name used for the organization
alias | | 0..* | string | A list of alternate names that the organization is known as, or was known as in the past
telecom | I | 0..* | ContactPoint | A contact detail for the organization + Rule: The telecom of an organization can never be of use 'home'
address | I | 0..* | Address | An address for the organization + Rule: An address of an organization can never be of use 'home'
partOf | Σ | 0..1 | Reference(Organization) | The organization of which this organization forms a part
contact | | 0..* | BackboneElement | Contact for the organization for a certain purpose
purpose | | 0..1 | CodeableConcept | The type of contact Contact entity type (Extensible)
name | | 0..1 | HumanName | A name associated with the contact
telecom | | 0..* | ContactPoint | Contact details (telephone, email, etc.) for a contact
address | | 0..1 | Address | Visiting or postal addresses for the contact
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Organization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!--  0..* Identifier Identifies this organization  across multiple systems --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether the organization's record is still in active use -->
 <type><!-- 0..* CodeableConcept Kind of organization --></type>
 <name value="[string]"/><!--  0..1 Name used for the organization -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the organization is known as, or was known as in the
```

```
    past -->
 <telecom><!--  0..* ContactPoint A contact detail for the organization --></telecom>
 <address><!--  0..* Address An address for the organization --></address>
 <partOf><!-- 0..1 Reference(Organization) The organization of which this organization forms a part --></partOf>
 <contact>  <!-- 0..* Contact for the organization for a certain purpose -->
  <purpose><!-- 0..1 CodeableConcept The type of contact --></purpose>
  <name><!-- 0..1 HumanName A name associated with the contact --></name>
  <telecom><!-- 0..* ContactPoint Contact details (telephone, email, etc.)  for a contact --></telecom>
  <address><!-- 0..1 Address Visiting or postal addresses for the contact --></address>
 </contact>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the organization
    --></endpoint>
</Organization>
```

JSON Template

```
{
  "resourceType" : "Organization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // C? Identifies this organization  across multiple systems
  "active" : <boolean>, // Whether the organization's record is still in active use
  "type" : [{ CodeableConcept }], // Kind of organization
  "name" : "<string>", // C? Name used for the organization
  "alias" : ["<string>"], // A list of alternate names that the organization is known as, or was known as in the past
  "telecom" : [{ ContactPoint }], // C? A contact detail for the organization
  "address" : [{ Address }], // C? An address for the organization
  "partOf" : { Reference(Organization) }, // The organization of which this organization forms a part
  "contact" : [{ // Contact for the organization for a certain purpose
    "purpose" : { CodeableConcept }, // The type of contact
    "name" : { HumanName }, // A name associated with the contact
    "telecom" : [{ ContactPoint }], // Contact details (telephone, email, etc.)  for a contact
    "address" : { Address } // Visiting or postal addresses for the contact
  }],
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Organization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Organization.identifier [ Identifier ], ... ; # 0..* Identifies this organization  across multiple systems
  fhir:Organization.active [ boolean ]; # 0..1 Whether the organization's record is still in active use
  fhir:Organization.type [ CodeableConcept ], ... ; # 0..* Kind of organization
  fhir:Organization.name [ string ]; # 0..1 Name used for the organization
  fhir:Organization.alias [ string ], ... ; # 0..* A list of alternate names that the organization is known as, or was
    known as in the past
  fhir:Organization.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the organization
  fhir:Organization.address [ Address ], ... ; # 0..* An address for the organization
  fhir:Organization.partOf [ Reference(Organization) ]; # 0..1 The organization of which this organization forms a part
  fhir:Organization.contact [ # 0..* Contact for the organization for a certain purpose
    fhir:Organization.contact.purpose [ CodeableConcept ]; # 0..1 The type of contact
    fhir:Organization.contact.name [ HumanName ]; # 0..1 A name associated with the contact
    fhir:Organization.contact.telecom [ ContactPoint ], ... ; # 0..* Contact details (telephone, email, etc.)  for a
    contact
    fhir:Organization.contact.address [ Address ]; # 0..1 Visiting or postal addresses for the contact
  ], ...;
  fhir:Organization.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the organization
]
```

Changes since R3

Organization

Organization.active | Default Value "true" removed

Organization.contact.purpose | Change code system for extensibly bound codes from "http://hl7.org/fhir/contactentity-type" to "http://terminology.hl7.org/CodeSystem/contactentity-type"

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Organization | ITU | | DomainResource | A grouping of people or organizations with a common purpose + Rule: The organization SHALL at least have a name or an identifier, and possibly more than one Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ΣI | 0..* | Identifier | Identifies this organization across multiple systems

active | ?!Σ | 0..1 | boolean | Whether the organization's record is still in active use

type | Σ | 0..* | CodeableConcept | Kind of organization Organization type (Example)

name | ΣI | 0..1 | string | Name used for the organization
alias | | 0..* | string | A list of alternate names that the organization is known as, or was known as in the past
telecom | I | 0..* | ContactPoint | A contact detail for the organization + Rule: The telecom of an organization can never be of use 'home'
address | I | 0..* | Address | An address for the organization + Rule: An address of an organization can never be of use 'home'
partOf | Σ | 0..1 | Reference(Organization) | The organization of which this organization forms a part
contact | | 0..* | BackboneElement | Contact for the organization for a certain purpose
purpose | | 0..1 | CodeableConcept | The type of contact Contact entity type (Extensible)
name | | 0..1 | HumanName | A name associated with the contact
telecom | | 0..* | ContactPoint | Contact details (telephone, email, etc.) for a contact
address | | 0..1 | Address | Visiting or postal addresses for the contact
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Organization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!--  0..* Identifier Identifies this organization  across multiple systems --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether the organization's record is still in active use -->
 <type><!-- 0..* CodeableConcept Kind of organization --></type>
 <name value="[string]"/><!--  0..1 Name used for the organization -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the organization is known as, or was known as in the
    past -->
 <telecom><!--  0..* ContactPoint A contact detail for the organization --></telecom>
 <address><!--  0..* Address An address for the organization --></address>
 <partOf><!-- 0..1 Reference(Organization) The organization of which this organization forms a part --></partOf>
 <contact>  <!-- 0..* Contact for the organization for a certain purpose -->
  <purpose><!-- 0..1 CodeableConcept The type of contact --></purpose>
  <name><!-- 0..1 HumanName A name associated with the contact --></name>
  <telecom><!-- 0..* ContactPoint Contact details (telephone, email, etc.)  for a contact --></telecom>
  <address><!-- 0..1 Address Visiting or postal addresses for the contact --></address>
 </contact>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the organization
    --></endpoint>
</Organization>
```

JSON Template

```
{
  "resourceType" : "Organization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // C? Identifies this organization  across multiple systems
  "active" : <boolean>, // Whether the organization's record is still in active use
  "type" : [{ CodeableConcept }], // Kind of organization
  "name" : "<string>", // C? Name used for the organization
  "alias" : ["<string>"], // A list of alternate names that the organization is known as, or was known as in the past
  "telecom" : [{ ContactPoint }], // C? A contact detail for the organization
  "address" : [{ Address }], // C? An address for the organization
  "partOf" : { Reference(Organization) }, // The organization of which this organization forms a part
  "contact" : [{ // Contact for the organization for a certain purpose
    "purpose" : { CodeableConcept }, // The type of contact
    "name" : { HumanName }, // A name associated with the contact
    "telecom" : [{ ContactPoint }], // Contact details (telephone, email, etc.)  for a contact
    "address" : { Address } // Visiting or postal addresses for the contact
  }],
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Organization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Organization.identifier [ Identifier ], ... ; # 0..* Identifies this organization  across multiple systems
  fhir:Organization.active [ boolean ]; # 0..1 Whether the organization's record is still in active use
  fhir:Organization.type [ CodeableConcept ], ... ; # 0..* Kind of organization
  fhir:Organization.name [ string ]; # 0..1 Name used for the organization
  fhir:Organization.alias [ string ], ... ; # 0..* A list of alternate names that the organization is known as, or was
    known as in the past
  fhir:Organization.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the organization
  fhir:Organization.address [ Address ], ... ; # 0..* An address for the organization
  fhir:Organization.partOf [ Reference(Organization) ]; # 0..1 The organization of which this organization forms a part
  fhir:Organization.contact [ # 0..* Contact for the organization for a certain purpose
    fhir:Organization.contact.purpose [ CodeableConcept ]; # 0..1 The type of contact
    fhir:Organization.contact.name [ HumanName ]; # 0..1 A name associated with the contact
    fhir:Organization.contact.telecom [ ContactPoint ], ... ; # 0..* Contact details (telephone, email, etc.)  for a
    contact
    fhir:Organization.contact.address [ Address ]; # 0..1 Visiting or postal addresses for the contact
```

```
  ], ...;
  fhir:Organization.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the organization
]
```

Changes since Release 3

Organization

Organization.active | Default Value "true" removed

Organization.contact.purpose | Change code system for extensibly bound codes from "http://hl7.org/fhir/contactentity-type" to "http://terminology.hl7.org/CodeSystem/contactentity-type"

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.6.3.1 Terminology Bindings

Path | Definition | Type | Reference

Organization.type | Used to categorize the organization. | Example | OrganizationType

Organization.contact.purpose | The purpose for which you would contact a contact party. | Extensible | ContactEntityType

8.6.3.2 Constraints

id | Level | Location | Description | Expression

org-1 | Rule | (base) | The organization SHALL at least have a name or an identifier, and possibly more than one | (identifier.count() + name.count()) > 0

org-2 | Rule | Organization.address | An address of an organization can never be of use 'home' | where(use = 'home').empty()

org-3 | Rule | Organization.telecom | The telecom of an organization can never be of use 'home' | where(use = 'home').empty()

8.6.4 Notes:

• There are two places for contact information: one on Organization itself and zero or more using the ContactEntity construct. The first one is to be used for the generic, public organization point of contact. The ContactEntity is to be used for reaching a person or party that has been designated by the organization to be contacted for a specific purpose or goal.

8.6.5 Example Organization Hierarchy:

An example organization hierarchy should help give some guidance as to one example of how a location hierarchy could look within a fictitious Medical Organization. (The nesting here would be the "part-of" structure of the Organization resource)

```
Burgers University Medical Center
    Eastern Services (prov)
        Emergency Dept
        Oncology Dept
            Nuclear Medicine Research Trials (edu)
        Maternity Ward
        Childrens Ward
        Day Procedures Unit
    Mobile Services (Ambulance)
    Research Center (edu)
        Nuclear Medicine  (edu)
    Burgers University (edu)
        Nuclear Medicine Faculty (edu)
        Undergraduate Medicine (edu)
        ...
```

Note that physical structures of this hierarchy are not present - these are defined by a Location hierarchy.

8.6.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

active | token | Is the Organization record active | Organization.active

address | string | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Organization.address

address-city | string | A city specified in an address | Organization.address.city

address-country | string | A country specified in an address | Organization.address.country

address-postalcode | string | A postal code specified in an address | Organization.address.postalCode

address-state | string | A state specified in an address | Organization.address.state

address-use | token | A use code specified in an address | Organization.address.use

endpoint | reference | Technical endpoints providing access to services operated for the organization | Organization.endpoint (Endpoint)

identifier | token | Any identifier for the organization (not the accreditation issuer's identifier) | Organization.identifier

name | string | A portion of the organization's name or alias | Organization.name | Organization.alias
partof | reference | An organization of which this organization forms a part | Organization.partOf (Organization)
phonetic | string | A portion of the organization's name using some kind of phonetic matching algorithm | Organization.name
type | token | A code for the type of organization | Organization.type

8.6.7 Resource Organization - Examples

URL: https://hl7.org/fhir/R4/organization-examples.html

8.6.7 Resource Organization - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
HL7 itself | hl7 | XML | JSON | Turtle
Gastroenterology team at ACME Healthcare | 1 | XML | JSON | Turtle
Clinical Laboratory at ACME Healthcare | 1832473e-2fe0-452d-abe9-3cdb9879522f | XML | JSON | Turtle
XYZ Insurance | 2 | XML | JSON | Turtle
Good Health Clinic | 2.16.840.1.113883.19.5 | XML | JSON | Turtle
Real-world organization example (Burgers MC) | f001 | XML | JSON | Turtle
Real-world organization example (Burgers MC Cardiology) | f002 | XML | JSON | Turtle
Real-world organization example (Artis UMC) | f201 | XML | JSON | Turtle
Real-world organization example (Blijdorp MC) | f203 | XML | JSON | Turtle
Real-world organization example (ENT) | f003 | XML | JSON | Turtle
Medication Manufacturer | mmanu | XML | JSON | Turtle
Government Department of Health | 3 | XML | JSON | Turtle
HL7 Payer Network | hl7pay | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.6.8 Resource Organization - Detailed Descriptions

URL: https://hl7.org/fhir/R4/organization-definitions.html

8.6.8 Resource Organization - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Organization resource.

Organization
Element Id | Organization
Definition | A formally or informally recognized grouping of people or organizations formed for the purpose of achieving some form of collective action. Includes companies, institutions, corporations, departments, community groups, healthcare practice groups, payer/insurer, etc.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element org-1 Rule The organization SHALL at least have a name or an identifier, and possibly more than one (identifier.count() + name.count()) > 0
Defined on this element
org-1 | Rule | The organization SHALL at least have a name or an identifier, and possibly more than one | (identifier.count() + name.count()) > 0
Organization.identifier
Element Id | Organization.identifier
Definition | Identifier for the organization that is used to identify the organization across multiple disparate systems.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Organizations are known by a variety of ids. Some institutions maintain several, and most collect identifiers for exchange with other organizations concerning the organization.
Summary | true
Invariants | Affect this element org-1 Rule The organization SHALL at least have a name or an identifier, and possibly more than one (identifier.count() + name.count()) > 0
Affect this element
org-1 | Rule | The organization SHALL at least have a name or an identifier, and possibly more than one | (identifier.count() + name.count()) > 0
Organization.active
Element Id | Organization.active
Definition | Whether the organization's record is still in active use.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)

Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need a flag to indicate a record is no longer to be used and should generally be hidden for the user in the UI.
Summary | true
Comments | This active flag is not intended to be used to mark an organization as temporarily closed or under construction. Instead the Location(s) within the Organization should have the suspended status. If further details of the reason for the suspension are required, then an extension on this element should be used. This element is labeled as a modifier because it may be used to mark that the resource was created in error.
Organization.type
Element Id | Organization.type
Definition | The kind(s) of organization that this is.
Cardinality | 0..*
Terminology Binding | Organization type (Example)
Type | CodeableConcept
Requirements | Need to be able to track the kind of organization that this is - different organization types have different uses.
Summary | true
Comments | Organizations can be corporations, wards, sections, clinical teams, government departments, etc. Note that code is generally a classifier of the type of organization; in many applications, codes are used to identity a particular organization (say, ward) as opposed to another of the same type - these are identifiers, not codes When considering if multiple types are appropriate, you should evaluate if child organizations would be a more appropriate use of the concept, as different types likely are in different sub-areas of the organization. This is most likely to be used where type values have orthogonal values, such as a religious, academic and medical center. We expect that some jurisdictions will profile this optionality to be a single cardinality.
Organization.name
Element Id | Organization.name
Definition | A name associated with the organization.
Cardinality | 0..1
Type | string
Requirements | Need to use the name as the label of the organization.
Summary | true
Comments | If the name of an organization changes, consider putting the old name in the alias column so that it can still be located through searches.
Invariants | Affect this element org-1 Rule The organization SHALL at least have a name or an identifier, and possibly more than one (identifier.count() + name.count()) > 0
Affect this element
org-1 | Rule | The organization SHALL at least have a name or an identifier, and possibly more than one | (identifier.count() + name.count()) > 0
Organization.alias
Element Id | Organization.alias
Definition | A list of alternate names that the organization is known as, or was known as in the past.
Cardinality | 0..*
Type | string
Requirements | Over time locations and organizations go through many changes and can be known by different names. For searching knowing previous names that the organization was known by can be very useful.
Comments | There are no dates associated with the alias/historic names, as this is not intended to track when names were used, but to assist in searching so that older names can still result in identifying the organization.
Organization.telecom
Element Id | Organization.telecom
Definition | A contact detail for the organization.
Cardinality | 0..*
Type | ContactPoint
Requirements | Human contact for the organization.
Comments | The use code 'home' is not to be used. Note that these contacts are not the contact details of people who are employed by or represent the organization, but official contacts for the organization itself.
Invariants | Defined on this element org-3 Rule The telecom of an organization can never be of use 'home' where(use = 'home').empty() Affect this element
Defined on this element
org-3 | Rule | The telecom of an organization can never be of use 'home' | where(use = 'home').empty()
Affect this element
Organization.address
Element Id | Organization.address
Definition | An address for the organization.
Cardinality | 0..*
Type | Address
Requirements | May need to keep track of the organization's addresses for contacting, billing or reporting requirements.
Comments | Organization may have multiple addresses with different uses or applicable periods. The use code 'home' is not to be used.
Invariants | Defined on this element org-2 Rule An address of an organization can never be of use 'home' where(use = 'home').empty() Affect this element
Defined on this element
org-2 | Rule | An address of an organization can never be of use 'home' | where(use = 'home').empty()
Affect this element
Organization.partOf
Element Id | Organization.partOf
Definition | The organization of which this organization forms a part.
Cardinality | 0..1

Type | Reference(Organization)
Hierarchy | This reference is part of a strict Hierarchy
Requirements | Need to be able to track the hierarchy of organizations within an organization.
Summary | true
Organization.contact
Element Id | Organization.contact
Definition | Contact for the organization for a certain purpose.
Cardinality | 0..*
Requirements | Need to keep track of assigned contact points within bigger organization.
Comments | Where multiple contacts for the same purpose are provided there is a standard extension that can be used to determine which one is the preferred contact to use.
Organization.contact.purpose
Element Id | Organization.contact.purpose
Definition | Indicates a purpose for which the contact can be reached.
Cardinality | 0..1
Terminology Binding | Contact entity type (Extensible)
Type | CodeableConcept
Requirements | Need to distinguish between multiple contact persons.
Organization.contact.name
Element Id | Organization.contact.name
Definition | A name associated with the contact.
Cardinality | 0..1
Type | HumanName
Requirements | Need to be able to track the person by name.
Organization.contact.telecom
Element Id | Organization.contact.telecom
Definition | A contact detail (e.g. a telephone number or an email address) by which the party may be contacted.
Cardinality | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Organization.contact.address
Element Id | Organization.contact.address
Definition | Visiting or postal addresses for the contact.
Cardinality | 0..1
Type | Address
Requirements | May need to keep track of a contact party's address for contacting, billing or reporting requirements.
Organization.endpoint
Element Id | Organization.endpoint
Definition | Technical endpoints providing access to services operated for the organization.
Cardinality | 0..*
Type | Reference(Endpoint)
Requirements | Organizations have multiple systems that provide various services and need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.6.9 Resource Organization - Mappings

URL: https://hl7.org/fhir/R4/organization-mappings.html

8.6.9 Resource Organization - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

ServD

Mappings for the organization resource (see Mappings to Other Standards for further information & status).

8.6.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Organization
identifier | FiveWs.identifier
active | FiveWs.status
type | FiveWs.class

8.6.9.2 HL7 v2 Mapping (http://hl7.org/v2)

Organization | (also see master files messages)
identifier | XON.10 / XON.3
active | No equivalent in HL7 v2
type | No equivalent in v2

name | XON.1
alias
telecom | ORC-22?
address | ORC-23?
partOf | No equivalent in HL7 v2
contact
purpose
name | PID-5, PID-9
telecom | PID-13, PID-14
address | PID-11
endpoint

8.6.9.3 RIM Mapping (http://hl7.org/v3)

Organization | Organization(classCode=ORG, determinerCode=INST)
identifier | .scopes[Role](classCode=IDENT)
active | .status
type | .code
name | .name
alias | .name
telecom | .telecom
address | .address
partOf | .playedBy[classCode=Part].scoper
contact | .contactParty
purpose | ./type
name | ./name
telecom | ./telecom
address | ./addr
endpoint | n/a

8.6.9.4 ServD (http://www.omg.org/spec/ServD/1.0/)

Organization | Organization
identifier | ./Identifiers
active | ./Status (however this concept in ServD more covers why the organization is active or not, could be delisted, deregistered, not operational yet) this could alternatively be derived from ./StartDate and ./EndDate and given a context date.
type | n/a
name | .PreferredName/Name
alias
telecom | ./ContactPoints
address | ./PrimaryAddress and ./OtherAddresses
partOf | n/a
contact
purpose
name
telecom
address
endpoint

8.6.10 Resource Organization - Extensions & Profiles

URL: https://hl7.org/fhir/R4/organization-profiles.html

8.6.10 Resource Organization - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Organization resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.6.10.1 Profiles

No Profiles defined for this resource

8.6.10.2 Extensions

organization-period | Organization | period | for Organization HL7 Extensions
organization-preferredContact | Organization.contact | preferredContact | for Organization HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

consent-Transcriber | Transcriber | for Consent HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions

cqf-initiatingOrganization | initiatingOrganization | for Clinical Reasoning Extensions
cqf-receivingOrganization | receivingOrganization | for Clinical Reasoning Extensions
task-candidateList | candidateList | for Task HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.6.10.3 Search Extensions

No Search Extensions defined for this resource

8.6.11 Resource Organization - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/organization-version-maps.html

No validation errors - all conversions are clean

8.6.11 Resource Organization - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Organization.

Functional status for this map: 11 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

8.6.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Organization3to4" = "R3 to R4 Conversions for Organization"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Organization" alias OrganizationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Organization" alias Organization as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Organization(source src : OrganizationR3, target tgt : Organization) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.type -> tgt.type;
  src.name -> tgt.name;
  src.alias -> tgt.alias;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
  src.partOf -> tgt.partOf;
  src.contact as vs0 -> tgt.contact as vt0 then contact(vs0, vt0);
  src.endpoint -> tgt.endpoint;
}

group contact(source src, target tgt) extends BackboneElement {
  src.purpose -> tgt.purpose;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
}
```

8.6.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Organization4to3" = "R4 to R3 Conversion for Organization"

uses "http://hl7.org/fhir/StructureDefinition/Organization" alias Organization as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Organization" alias OrganizationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Organization(source src : Organization, target tgt : OrganizationR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.type -> tgt.type;
  src.name -> tgt.name;
  src.alias -> tgt.alias;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
  src.partOf -> tgt.partOf;
  src.contact as vs0 -> tgt.contact as vt0 then contact(vs0, vt0);
  src.endpoint -> tgt.endpoint;
}

group contact(source src, target tgt) extends BackboneElement {
  src.purpose -> tgt.purpose;
```

```
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
}
```

8.6.11.3 R4 Validation Errors

8.6.12 Organization HL7 Extensions

URL: https://hl7.org/fhir/R4/organization-extensions.html

Defines common extensions used with or related to the Organization resource

8.6.12 Organization HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.6.12.1 Content

Extensions:

organization-period | period : The date range that this organization should be considered available.

organization-preferredContact | preferredContact : This Contact is the preferred contact at this organization for the purpose of the contact There can be multiple contacts on an Organizations record with this value set to true, but these should all have different purpose values.

8.6.13 JSON Schema for Organization

URL: https://hl7.org/fhir/R4/organization.schema.json.html

8.6.13 JSON Schema for Organization

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Organization, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Organization",
  "$ref": "#/definitions/Organization",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Organization": {
      "description": "A formally or informally recognized grouping of people or organizations formed for the purpose of
    achieving some form of collective action.  Includes companies, institutions, corporations, departments, community
    groups, healthcare practice groups, payer/insurer, etc.",
      "properties": {
        "resourceType": {
          "description": "This is a Organization resource",
          "const": "Organization"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
```

```
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for the organization that is used to identify the organization across multiple
disparate systems.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Whether the organization\u0027s record is still in active use.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The kind(s) of organization that this is.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "name": {
      "description": "A name associated with the organization.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "alias": {
      "description": "A list of alternate names that the organization is known as, or was known as in the past.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_alias": {
      "description": "Extensions for alias",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "telecom": {
      "description": "A contact detail for the organization.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "address": {
      "description": "An address for the organization.",
      "items": {
        "$ref": "Address.schema.json#/definitions/Address"
      },
      "type": "array"
```

```
        },
        "partOf": {
          "description": "The organization of which this organization forms a part.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "contact": {
          "description": "Contact for the organization for a certain purpose.",
          "items": {
            "$ref": "#/definitions/Organization_Contact"
          },
          "type": "array"
        },
        "endpoint": {
          "description": "Technical endpoints providing access to services operated for the organization.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    },
    "Organization_Contact": {
      "description": "A formally or informally recognized grouping of people or organizations formed for the purpose of
    achieving some form of collective action.  Includes companies, institutions, corporations, departments, community
    groups, healthcare practice groups, payer/insurer, etc.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "purpose": {
          "description": "Indicates a purpose for which the contact can be reached.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "name": {
          "description": "A name associated with the contact.",
          "$ref": "HumanName.schema.json#/definitions/HumanName"
        },
        "telecom": {
          "description": "A contact detail (e.g. a telephone number or an email address) by which the party may be
    contacted.",
          "items": {
            "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
          },
          "type": "array"
        },
        "address": {
          "description": "Visiting or postal addresses for the contact.",
          "$ref": "Address.schema.json#/definitions/Address"
        }
      },
      "additionalProperties": false
    }
  }
}
```


8.6.14 Resource Organization - Maturity Dependency View

URL: https://hl7.org/fhir/R4/organization-dependencies.html

8.6.14 Resource Organization - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Organization FMM level 3

Name | Card. | Type | Dependency Analysis
Organization | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
type | 0..* | CodeableConcept
name | 0..1 | string
alias | 0..* | string
telecom | 0..* | ContactPoint
address | 0..* | Address
partOf | 0..1 | Reference(Organization)
contact | 0..* | BackboneElement
purpose | 0..1 | CodeableConcept
name | 0..1 | HumanName
telecom | 0..* | ContactPoint
address | 0..1 | Address
endpoint | 0..* | Reference(Endpoint) | Type Warning: (Endpoint = FMM2-Trial Use vs. Element = FMM3-Trial Use)

8.7 Resource Location - Content

URL: https://hl7.org/fhir/R4/location.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Details and position information for a physical place where services are provided and resources and participants may be stored, found, contained, or accommodated.

8.7.1 Scope and Usage

A Location includes both incidental locations (a place which is used for healthcare without prior designation or authorization) and dedicated, formally appointed locations. Locations may be private, public, mobile or fixed and scale from small freezers to full hospital buildings or parking garages.

Examples of Locations are:

- Building, ward, corridor, room or bed
- Mobile Clinic
- Freezer, incubator
- Vehicle or lift
- Home, shed, or a garage
- Road, parking place, a park
- Ambulance (generic)
- Ambulance (specific)
- Patient's Home (generic)
- Jurisdiction

These locations are not intended to cover locations on a patient where something occurred (i.e. a patient's broken leg), but can happily cover the location where the patient broke the leg (the playground)

8.7.2 Boundaries and Relationships

Locations and Organizations are very closely related resources and can often be mixed/matched/confused. The Location is intended to describe the more physical structures managed/operated by an organization, whereas the Organization is intended to represent the more conceptual hierarchies, such as a ward.

A Location is valid without an address in cases where it could be purely described by a geo-coded location in remote areas, or when recorded by a device. Locations with a mode = "kind" would also likely not have an address, as they are just a type of location, but could also have an address where they can be found at the address.

Another use of location could be for describing a Jurisdiction. This jurisdiction may be considered a classified boundary which could be a combination of a physical boundary, and some other discriminator(s):

- Nation - Country-wide community or Federal Government (Ministry of Health)
- Province or State (community or Government)
- Business (throughout an enterprise)
- Business scope (CDC/FDA)
- Business segment (UK Pharmacy)

This resource is referenced by UsageContext, Account, ActivityDefinition, AdverseEvent, Appointment, AppointmentResponse, AuditEvent, CarePlan, Claim, ClaimResponse, Contract, CoverageEligibilityRequest, Device, DeviceRequest, DiagnosticReport, Encounter, ExplanationOfBenefit, Flag, HealthcareService, ImagingStudy, Immunization, InsurancePlan, List, itself, MeasureReport, Media, MedicationDispense, Observation, OrganizationAffiliation, PractitionerRole, Procedure, Provenance, ResearchStudy, Schedule, ServiceRequest, Specimen, SupplyDelivery, SupplyRequest and Task

8.7.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Location | TU | | DomainResource | Details and position information for a physical place Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique code or number identifying the location to its users
status | ?!Σ | 0..1 | code | active | suspended | inactive LocationStatus (Required)
operationalStatus | Σ | 0..1 | Coding | The operational status of the location (typically only for a bed/room) v2 BED STATUS (Preferred)
name | Σ | 0..1 | string | Name of the location as used by humans
alias | | 0..* | string | A list of alternate names that the location is known as, or was known as, in the past
description | Σ | 0..1 | string | Additional details about the location that could be displayed as further information to identify the location beyond its name
mode | Σ | 0..1 | code | instance | kind LocationMode (Required)
type | Σ | 0..* | CodeableConcept | Type of function performed V3 Value SetServiceDeliveryLocationRoleType (Extensible)
telecom | | 0..* | ContactPoint | Contact details of the location
address | | 0..1 | Address | Physical location
physicalType | Σ | 0..1 | CodeableConcept | Physical form of the location Location type (Example)
position | | 0..1 | BackboneElement | The absolute geographic location
longitude | | 1..1 | decimal | Longitude with WGS84 datum
latitude | | 1..1 | decimal | Latitude with WGS84 datum
altitude | | 0..1 | decimal | Altitude with WGS84 datum
managingOrganization | Σ | 0..1 | Reference(Organization) | Organization responsible for provisioning and upkeep
partOf | | 0..1 | Reference(Location) | Another Location this one is physically a part of
hoursOfOperation | | 0..* | BackboneElement | What days/times during a week is this location usually open
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | The Location is open all day
openingTime | | 0..1 | time | Time that the Location opens
closingTime | | 0..1 | time | Time that the Location closes
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the location
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Location xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique code or number identifying the location to its users --></identifier>
 <status value="[code]"/><!-- 0..1 active | suspended | inactive -->
 <operationalStatus><!-- 0..1 Coding The operational status of the location (typically only for a bed/room)
    --></operationalStatus>
 <name value="[string]"/><!-- 0..1 Name of the location as used by humans -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the location is known as, or was known as, in the
    past -->
 <description value="[string]"/><!-- 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name -->
 <mode value="[code]"/><!-- 0..1 instance | kind -->
 <type><!-- 0..* CodeableConcept Type of function performed --></type>
 <telecom><!-- 0..* ContactPoint Contact details of the location --></telecom>
 <address><!-- 0..1 Address Physical location --></address>
 <physicalType><!-- 0..1 CodeableConcept Physical form of the location --></physicalType>
 <position>  <!-- 0..1 The absolute geographic location -->
  <longitude value="[decimal]"/><!-- 1..1 Longitude with WGS84 datum -->
  <latitude value="[decimal]"/><!-- 1..1 Latitude with WGS84 datum -->
  <altitude value="[decimal]"/><!-- 0..1 Altitude with WGS84 datum -->
 </position>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization responsible for provisioning and upkeep
    --></managingOrganization>
```

```
 <partOf><!-- 0..1 Reference(Location) Another Location this one is physically a part of --></partOf>
 <hoursOfOperation>  <!-- 0..* What days/times during a week is this location usually open -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 The Location is open all day -->
  <openingTime value="[time]"/><!-- 0..1 Time that the Location opens -->
  <closingTime value="[time]"/><!-- 0..1 Time that the Location closes -->
 </hoursOfOperation>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the location
    --></endpoint>
</Location>
```

JSON Template

```
{
  "resourceType" : "Location",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique code or number identifying the location to its users
  "status" : "<code>", // active | suspended | inactive
  "operationalStatus" : { Coding }, // The operational status of the location (typically only for a bed/room)
  "name" : "<string>", // Name of the location as used by humans
  "alias" : ["<string>"], // A list of alternate names that the location is known as, or was known as, in the past
  "description" : "<string>", // Additional details about the location that could be displayed as further information to
     identify the location beyond its name
  "mode" : "<code>", // instance | kind
  "type" : [{ CodeableConcept }], // Type of function performed
  "telecom" : [{ ContactPoint }], // Contact details of the location
  "address" : { Address }, // Physical location
  "physicalType" : { CodeableConcept }, // Physical form of the location
  "position" : { // The absolute geographic location
    "longitude" : <decimal>, // R!  Longitude with WGS84 datum
    "latitude" : <decimal>, // R!  Latitude with WGS84 datum
    "altitude" : <decimal> // Altitude with WGS84 datum
  },
  "managingOrganization" : { Reference(Organization) }, // Organization responsible for provisioning and upkeep
  "partOf" : { Reference(Location) }, // Another Location this one is physically a part of
  "hoursOfOperation" : [{ // What days/times during a week is this location usually open
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // The Location is open all day
    "openingTime" : "<time>", // Time that the Location opens
    "closingTime" : "<time>" // Time that the Location closes
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the location
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Location;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Location.identifier [ Identifier ], ... ; # 0..* Unique code or number identifying the location to its users
  fhir:Location.status [ code ]; # 0..1 active | suspended | inactive
  fhir:Location.operationalStatus [ Coding ]; # 0..1 The operational status of the location (typically only for a
    bed/room)
  fhir:Location.name [ string ]; # 0..1 Name of the location as used by humans
  fhir:Location.alias [ string ], ... ; # 0..* A list of alternate names that the location is known as, or was known as,
     in the past
  fhir:Location.description [ string ]; # 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name
  fhir:Location.mode [ code ]; # 0..1 instance | kind
  fhir:Location.type [ CodeableConcept ], ... ; # 0..* Type of function performed
  fhir:Location.telecom [ ContactPoint ], ... ; # 0..* Contact details of the location
  fhir:Location.address [ Address ]; # 0..1 Physical location
  fhir:Location.physicalType [ CodeableConcept ]; # 0..1 Physical form of the location
  fhir:Location.position [ # 0..1 The absolute geographic location
    fhir:Location.position.longitude [ decimal ]; # 1..1 Longitude with WGS84 datum
    fhir:Location.position.latitude [ decimal ]; # 1..1 Latitude with WGS84 datum
    fhir:Location.position.altitude [ decimal ]; # 0..1 Altitude with WGS84 datum
  ];
  fhir:Location.managingOrganization [ Reference(Organization) ]; # 0..1 Organization responsible for provisioning and
    upkeep
  fhir:Location.partOf [ Reference(Location) ]; # 0..1 Another Location this one is physically a part of
  fhir:Location.hoursOfOperation [ # 0..* What days/times during a week is this location usually open
    fhir:Location.hoursOfOperation.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:Location.hoursOfOperation.allDay [ boolean ]; # 0..1 The Location is open all day
    fhir:Location.hoursOfOperation.openingTime [ time ]; # 0..1 Time that the Location opens
    fhir:Location.hoursOfOperation.closingTime [ time ]; # 0..1 Time that the Location closes
  ], ...;
  fhir:Location.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:Location.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services operated
     for the location
]
```

Changes since R3

Location
Location.status | Change value set from http://hl7.org/fhir/ValueSet/location-status to http://hl7.org/fhir/ValueSet/location-status|4.0.1
Location.mode | Change value set from http://hl7.org/fhir/ValueSet/location-mode to http://hl7.org/fhir/ValueSet/location-mode|4.0.1 No longer marked as Modifier
Location.type | Max Cardinality changed from 1 to *
Location.hoursOfOperation | Added Element
Location.hoursOfOperation.daysOfWeek | Added Element
Location.hoursOfOperation.allDay | Added Element
Location.hoursOfOperation.openingTime | Added Element
Location.hoursOfOperation.closingTime | Added Element
Location.availabilityExceptions | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Location | TU | | DomainResource | Details and position information for a physical place Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique code or number identifying the location to its users
status | ?!Σ | 0..1 | code | active | suspended | inactive LocationStatus (Required)
operationalStatus | Σ | 0..1 | Coding | The operational status of the location (typically only for a bed/room) v2 BED STATUS (Preferred)
name | Σ | 0..1 | string | Name of the location as used by humans
alias | | 0..* | string | A list of alternate names that the location is known as, or was known as, in the past
description | Σ | 0..1 | string | Additional details about the location that could be displayed as further information to identify the location beyond its name
mode | Σ | 0..1 | code | instance | kind LocationMode (Required)
type | Σ | 0..* | CodeableConcept | Type of function performed V3 Value SetServiceDeliveryLocationRoleType (Extensible)
telecom | | 0..* | ContactPoint | Contact details of the location
address | | 0..1 | Address | Physical location
physicalType | Σ | 0..1 | CodeableConcept | Physical form of the location Location type (Example)
position | | 0..1 | BackboneElement | The absolute geographic location
longitude | | 1..1 | decimal | Longitude with WGS84 datum
latitude | | 1..1 | decimal | Latitude with WGS84 datum
altitude | | 0..1 | decimal | Altitude with WGS84 datum
managingOrganization | Σ | 0..1 | Reference(Organization) | Organization responsible for provisioning and upkeep
partOf | | 0..1 | Reference(Location) | Another Location this one is physically a part of
hoursOfOperation | | 0..* | BackboneElement | What days/times during a week is this location usually open
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | The Location is open all day
openingTime | | 0..1 | time | Time that the Location opens
closingTime | | 0..1 | time | Time that the Location closes
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the location
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Location xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique code or number identifying the location to its users --></identifier>
 <status value="[code]"/><!-- 0..1 active | suspended | inactive -->
 <operationalStatus><!-- 0..1 Coding The operational status of the location (typically only for a bed/room)
    --></operationalStatus>
 <name value="[string]"/><!-- 0..1 Name of the location as used by humans -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the location is known as, or was known as, in the
    past -->
 <description value="[string]"/><!-- 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name -->
 <mode value="[code]"/><!-- 0..1 instance | kind -->
 <type><!-- 0..* CodeableConcept Type of function performed --></type>
 <telecom><!-- 0..* ContactPoint Contact details of the location --></telecom>
 <address><!-- 0..1 Address Physical location --></address>
 <physicalType><!-- 0..1 CodeableConcept Physical form of the location --></physicalType>
 <position>  <!-- 0..1 The absolute geographic location -->
  <longitude value="[decimal]"/><!-- 1..1 Longitude with WGS84 datum -->
  <latitude value="[decimal]"/><!-- 1..1 Latitude with WGS84 datum -->
  <altitude value="[decimal]"/><!-- 0..1 Altitude with WGS84 datum -->
 </position>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization responsible for provisioning and upkeep
    --></managingOrganization>
 <partOf><!-- 0..1 Reference(Location) Another Location this one is physically a part of --></partOf>
 <hoursOfOperation>  <!-- 0..* What days/times during a week is this location usually open -->
```

```
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 The Location is open all day -->
  <openingTime value="[time]"/><!-- 0..1 Time that the Location opens -->
  <closingTime value="[time]"/><!-- 0..1 Time that the Location closes -->
 </hoursOfOperation>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the location
    --></endpoint>
</Location>
```

JSON Template

```
{
  "resourceType" : "Location",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique code or number identifying the location to its users
  "status" : "<code>", // active | suspended | inactive
  "operationalStatus" : { Coding }, // The operational status of the location (typically only for a bed/room)
  "name" : "<string>", // Name of the location as used by humans
  "alias" : ["<string>"], // A list of alternate names that the location is known as, or was known as, in the past
  "description" : "<string>", // Additional details about the location that could be displayed as further information to
     identify the location beyond its name
  "mode" : "<code>", // instance | kind
  "type" : [{ CodeableConcept }], // Type of function performed
  "telecom" : [{ ContactPoint }], // Contact details of the location
  "address" : { Address }, // Physical location
  "physicalType" : { CodeableConcept }, // Physical form of the location
  "position" : { // The absolute geographic location
    "longitude" : <decimal>, // R!  Longitude with WGS84 datum
    "latitude" : <decimal>, // R!  Latitude with WGS84 datum
    "altitude" : <decimal> // Altitude with WGS84 datum
  },
  "managingOrganization" : { Reference(Organization) }, // Organization responsible for provisioning and upkeep
  "partOf" : { Reference(Location) }, // Another Location this one is physically a part of
  "hoursOfOperation" : [{ // What days/times during a week is this location usually open
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // The Location is open all day
    "openingTime" : "<time>", // Time that the Location opens
    "closingTime" : "<time>" // Time that the Location closes
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the location
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Location;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Location.identifier [ Identifier ], ... ; # 0..* Unique code or number identifying the location to its users
  fhir:Location.status [ code ]; # 0..1 active | suspended | inactive
  fhir:Location.operationalStatus [ Coding ]; # 0..1 The operational status of the location (typically only for a
    bed/room)
  fhir:Location.name [ string ]; # 0..1 Name of the location as used by humans
  fhir:Location.alias [ string ], ... ; # 0..* A list of alternate names that the location is known as, or was known as,
     in the past
  fhir:Location.description [ string ]; # 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name
  fhir:Location.mode [ code ]; # 0..1 instance | kind
  fhir:Location.type [ CodeableConcept ], ... ; # 0..* Type of function performed
  fhir:Location.telecom [ ContactPoint ], ... ; # 0..* Contact details of the location
  fhir:Location.address [ Address ]; # 0..1 Physical location
  fhir:Location.physicalType [ CodeableConcept ]; # 0..1 Physical form of the location
  fhir:Location.position [ # 0..1 The absolute geographic location
    fhir:Location.position.longitude [ decimal ]; # 1..1 Longitude with WGS84 datum
    fhir:Location.position.latitude [ decimal ]; # 1..1 Latitude with WGS84 datum
    fhir:Location.position.altitude [ decimal ]; # 0..1 Altitude with WGS84 datum
  ];
  fhir:Location.managingOrganization [ Reference(Organization) ]; # 0..1 Organization responsible for provisioning and
    upkeep
  fhir:Location.partOf [ Reference(Location) ]; # 0..1 Another Location this one is physically a part of
  fhir:Location.hoursOfOperation [ # 0..* What days/times during a week is this location usually open
    fhir:Location.hoursOfOperation.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:Location.hoursOfOperation.allDay [ boolean ]; # 0..1 The Location is open all day
    fhir:Location.hoursOfOperation.openingTime [ time ]; # 0..1 Time that the Location opens
    fhir:Location.hoursOfOperation.closingTime [ time ]; # 0..1 Time that the Location closes
  ], ...;
  fhir:Location.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:Location.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services operated
     for the location
]
```

Changes since Release 3

Location

Location.status | Change value set from http://hl7.org/fhir/ValueSet/location-status to http://hl7.org/fhir/ValueSet/location-status|4.0.1
Location.mode | Change value set from http://hl7.org/fhir/ValueSet/location-mode to http://hl7.org/fhir/ValueSet/location-mode|4.0.1 No longer marked as Modifier
Location.type | Max Cardinality changed from 1 to *
Location.hoursOfOperation | Added Element
Location.hoursOfOperation.daysOfWeek | Added Element
Location.hoursOfOperation.allDay | Added Element
Location.hoursOfOperation.openingTime | Added Element
Location.hoursOfOperation.closingTime | Added Element
Location.availabilityExceptions | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.7.3.1 Terminology Bindings

Path | Definition | Type | Reference
Location.status | Indicates whether the location is still in use. | Required | LocationStatus
Location.operationalStatus | The operational status if the location (where typically a bed/room). | Preferred | v2.0116
Location.mode | Indicates whether a resource instance represents a specific location or a class of locations. | Required | LocationMode
Location.type | Indicates the type of function performed at the location. | Extensible | v3.ServiceDeliveryLocationRoleType
Location.physicalType | Physical form of the location. | Example | LocationType
Location.hoursOfOperation.daysOfWeek | The days of the week. | Required | DaysOfWeek

8.7.4 Notes

• Multiple Organizations or Practitioners may provide services at a Location. These references are not kept in Location, but can be found in the models for Organization and Practitioner instead.

• Locations may range from whole buildings to cabinets; it is possible to relate smaller Locations to their containing bigger Location using the Location.partOf element.

• Location.position is expressed using the same syntax, datum and reference system as used in Google Earth's KML files, see Google/OGS's KML .

8.7.4.1 Location Mode

The Location.mode element can be used to indicate whether a Location resource represents a specific (potentially identifiable) Location ('instance'), or a class of Locations ('kind'). Especially Resources capturing orders, resource scheduling, plans and definitions may refer to Locations in 'kind' mode. For these domains, it is often not necessary to refer to a specific Location, but rather to a class of Locations. An example of this is found in planning, where we need to allocate an "isolation room" for a patient, or need to dispatch "an ambulance" at a certain time. In these cases it is not important to identify exactly which isolation room or ambulance is allocated, it is sufficient to just indicate a 'kind' of Location.

Note that 'kind' should not be used to represent Locations where an actual instance of a Location was involved, but by identifying missing information. E.g. when a patient arrived 'by ambulance', but it is not known by which ambulance, this should be represented using a Location in 'instance' mode with a missing identifier, not a Location of 'kind' ambulance.

Some of Location's data elements are only relevant when mode is 'instance' and should not be used when mode is 'kind': (however this information could still be included if was relevant, such as when it is a generic item, but not globally generic, e.g. a Burgers Medical Centre ambulance)

• Location.identifier
• Location.telecom
• Location.address
• Location.position
• Location.status
• Location.managingOrganization

8.7.5 Example Location Hierarchy

An example location hierarchy should help give some guidance as to one example of how a location hierarchy could look within a fictitious Hospital. (The nesting here would be the "part-of" structure of the location)

```
Hospital A Building C (instance)
    East Wing (instance)
        Level 1 (instance)
```

```
            Reception (instance)
            Nurses Station EM-ns1 (instance)
                Medication Cupboard A (instance)
            Room 1 (instance)
                Room 1a (instance) - space in room separatable via a curtain
                    Bed 1a (instance) - always in this room
                Room 1b (instance)
                    Trolley 43 (instance) - moves about
                Room 1d (instance)
                    Trolley 19 (instance) - moves about
                Room 2 (instance)
                    ...
            Theatre EM-TA (instance)
            Coridor (generic)
        Level 2 (instance)
            Reception (instance)
                ...
            Nurses Station EM-ns1 (instance)
                Medication Cupboard A (instance)
            Coridor (generic)
Mobile Services (kind)
    Ambulance (kind)
        Ambulance AMB1 (instance)
        Ambulance AMB2 (instance)
```

Note: Wards/departments are not part of this structure - these would form part of the Organizational Hierarchy.

8.7.5.1 Positional Searching

Searching for locations often require that a facility is within a specified distance of a specified point. For example, to locate healthcare facilities within 11.2 kms of a client's home, or the current geo-coded position of a practitioner travelling between patients (read from a mobile phone or device).

```
GET [base]/Location?near=-83.694810|42.256500|11.20|km...
```

The distance and distance unit parameter components are optional, if the units are missing, kms are to be assumed. If the distance parameter component is missing, then the server may choose its own interpretation of what near enough is to be included in the search results.

Note: The STU3 version of this functionality did not support the multiple separator , or chaining. The update to this format now supports both of these use cases. (And the near-distance was deprecated as a result of this change too)

The distance between the location and the provided point is often used as one of the determining factors for selection of the location. So this value is included in the results. However the value cannot be inside the Location resource as it is different depending on the point of reference in the search. So the distance between is included in the search section of the bundle entry. Where multiple near positions are included, the distance to the closest point provided may be included.

```
<entry>
    <resource>
        <Location>
            <!-- location details -->
        </Location>
    </resource>
    <search>
        <extension url="http://hl7.org/fhir/StructureDefinition/location-distance">
            <valueDistance >
                <!-- The distance that this location resource is from the provided point in the query -->
                <value value="10.5"/>
                <unit value="km"/>
            </valueDistance>
        </extension>
    </search>
</entry>
```

8.7.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
address | string | A (part of the) address of the location | Location.address
address-city | string | A city specified in an address | Location.address.city
address-country | string | A country specified in an address | Location.address.country
address-postalcode | string | A postal code specified in an address | Location.address.postalCode
address-state | string | A state specified in an address | Location.address.state
address-use | token | A use code specified in an address | Location.address.use
endpoint | reference | Technical endpoints providing access to services operated for the location | Location.endpoint (Endpoint)
identifier | token | An identifier for the location | Location.identifier
name | string | A portion of the location's name or alias | Location.name | Location.alias
near | special | Search for locations where the location.position is near to, or within a specified distance of, the provided coordinates expressed as [latitude]|[longitude]|[distance]|[units] (using the WGS84 datum, see notes). If the units are omitted, then kms should be assumed. If the distance is omitted, then the server can use its own discretion as to what distances should be

considered near (and units are irrelevant) Servers may search using various techniques that might have differing accuracies, depending on implementation efficiency. Requires the near-distance parameter to be provided also | Location.position
operational-status | token | Searches for locations (typically bed/room) that have an operational status (e.g. contaminated, housekeeping) | Location.operationalStatus
organization | reference | Searches for locations that are managed by the provided organization | Location.managingOrganization (Organization)
partof | reference | A location of which this location is a part | Location.partOf (Location)
status | token | Searches for locations with a specific kind of status | Location.status
type | token | A code for the type of location | Location.type

8.7.7 Resource Location - Examples

URL: https://hl7.org/fhir/R4/location-examples.html

8.7.7 Resource Location - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Wing within a hospital | 1 | XML | JSON | Turtle
Operation room within hospital | 2 | XML | JSON | Turtle
Ambulance provided by Burgers Healthcare | amb | XML | JSON | Turtle
Patient's Home which can be used for remote encounters | ph | XML | JSON | Turtle
UK Pharmacy Jurisdiction | ukp | XML | JSON | Turtle
HL7 HQ Location | hl7 | XML | JSON | Turtle
General location examples | 3ad0687e-f477-468c-afd5-fcc2bf897819 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.7.8 Resource Location - Detailed Descriptions

URL: https://hl7.org/fhir/R4/location-definitions.html

8.7.8 Resource Location - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Location resource.

Location
Element Id | Location
Definition | Details and position information for a physical place where services are provided and resources and participants may be stored, found, contained, or accommodated.
Cardinality | 0..*
Type | DomainResource
Location.identifier
Element Id | Location.identifier
Definition | Unique code or number identifying the location to its users.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Organization label locations in registries, need to keep track of those.
Summary | true
Location.status
Element Id | Location.status
Definition | The status property covers the general availability of the resource, not the current value which may be covered by the operationStatus, or by a schedule/slots if they are configured for the location.
Cardinality | 0..1
Terminology Binding | LocationStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid)
Summary | true
Location.operationalStatus
Element Id | Location.operationalStatus
Definition | The operational status covers operation values most relevant to beds (but can also apply to rooms/units/chairs/etc. such as an isolation unit/dialysis chair). This typically covers concepts such as contamination, housekeeping, and other activities like maintenance.
Cardinality | 0..1
Terminology Binding | v2 BED STATUS (Preferred)
Type | Coding
Summary | true
Location.name

Element Id | Location.name
Definition | Name of the location as used by humans. Does not need to be unique.
Cardinality | 0..1
Type | string
Summary | true
Comments | If the name of a location changes, consider putting the old name in the alias column so that it can still be located through searches.
Location.alias
Element Id | Location.alias
Definition | A list of alternate names that the location is known as, or was known as, in the past.
Cardinality | 0..*
Type | string
Requirements | Over time locations and organizations go through many changes and can be known by different names. For searching knowing previous names that the location was known by can be very useful.
Comments | There are no dates associated with the alias/historic names, as this is not intended to track when names were used, but to assist in searching so that older names can still result in identifying the location.
Location.description
Element Id | Location.description
Definition | Description of the Location, which helps in finding or referencing the place.
Cardinality | 0..1
Type | string
Requirements | Humans need additional information to verify a correct location has been identified.
Summary | true
Location.mode
Element Id | Location.mode
Definition | Indicates whether a resource instance represents a specific location or a class of locations.
Cardinality | 0..1
Terminology Binding | LocationMode (Required)
Type | code
Requirements | When using a Location resource for scheduling or orders, we need to be able to refer to a class of Locations instead of a specific Location.
Summary | true
Comments | This is labeled as a modifier because whether or not the location is a class of locations changes how it can be used and understood.
Location.type
Element Id | Location.type
Definition | Indicates the type of function performed at the location.
Cardinality | 0..*
Terminology Binding | V3 Value SetServiceDeliveryLocationRoleType (Extensible)
Type | CodeableConcept
Summary | true
Location.telecom
Element Id | Location.telecom
Definition | The contact details of communication devices available at the location. This can include phone numbers, fax numbers, mobile numbers, email addresses and web sites.
Cardinality | 0..*
Type | ContactPoint
Location.address
Element Id | Location.address
Definition | Physical location.
Cardinality | 0..1
Type | Address
Requirements | If locations can be visited, we need to keep track of their address.
Comments | Additional addresses should be recorded using another instance of the Location resource, or via the Organization.
Location.physicalType
Element Id | Location.physicalType
Definition | Physical form of the location, e.g. building, room, vehicle, road.
Cardinality | 0..1
Terminology Binding | Location type (Example)
Type | CodeableConcept
Requirements | For purposes of showing relevant locations in queries, we need to categorize locations.
Summary | true
Location.position
Element Id | Location.position
Definition | The absolute geographic location of the Location, expressed using the WGS84 datum (This is the same co-ordinate system used in KML).
Cardinality | 0..1
Requirements | For mobile applications and automated route-finding knowing the exact location of the Location is required.
Location.position.longitude
Element Id | Location.position.longitude
Definition | Longitude. The value domain and the interpretation are the same as for the text of the longitude element in KML (see notes below).
Cardinality | 1..1

Type | decimal
Location.position.latitude
Element Id | Location.position.latitude
Definition | Latitude. The value domain and the interpretation are the same as for the text of the latitude element in KML (see notes below).
Cardinality | 1..1
Type | decimal
Location.position.altitude
Element Id | Location.position.altitude
Definition | Altitude. The value domain and the interpretation are the same as for the text of the altitude element in KML (see notes below).
Cardinality | 0..1
Type | decimal
Location.managingOrganization
Element Id | Location.managingOrganization
Definition | The organization responsible for the provisioning and upkeep of the location.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | Need to know who manages the location.
Summary | true
Comments | This can also be used as the part of the organization hierarchy where this location provides services. These services can be defined through the HealthcareService resource.
Location.partOf
Element Id | Location.partOf
Definition | Another Location of which this Location is physically a part of.
Cardinality | 0..1
Type | Reference(Location)
Hierarchy | This reference is part of a strict Hierarchy
Requirements | For purposes of location, display and identification, knowing which locations are located within other locations is important.
Location.hoursOfOperation
Element Id | Location.hoursOfOperation
Definition | What days/times during a week is this location usually open.
Cardinality | 0..*
Comments | This type of information is commonly found published in directories and on websites informing customers when the facility is available. Specific services within the location may have their own hours which could be shorter (or longer) than the locations hours.
Location.hoursOfOperation.daysOfWeek
Element Id | Location.hoursOfOperation.daysOfWeek
Definition | Indicates which days of the week are available between the start and end Times.
Cardinality | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
Location.hoursOfOperation.allDay
Element Id | Location.hoursOfOperation.allDay
Definition | The Location is open all day.
Cardinality | 0..1
Type | boolean
Location.hoursOfOperation.openingTime
Element Id | Location.hoursOfOperation.openingTime
Definition | Time that the Location opens.
Cardinality | 0..1
Type | time
Location.hoursOfOperation.closingTime
Element Id | Location.hoursOfOperation.closingTime
Definition | Time that the Location closes.
Cardinality | 0..1
Type | time
Location.availabilityExceptions
Element Id | Location.availabilityExceptions
Definition | A description of when the locations opening ours are different to normal, e.g. public holiday availability. Succinctly describing all possible exceptions to normal site availability as detailed in the opening hours Times.
Cardinality | 0..1
Type | string
Location.endpoint
Element Id | Location.endpoint
Definition | Technical endpoints providing access to services operated for the location.
Cardinality | 0..*
Type | Reference(Endpoint)
Requirements | Organizations may have different systems at different locations that provide various services and need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.7.9 Resource Location - Mappings

URL: https://hl7.org/fhir/R4/location-mappings.html

8.7.9 Resource Location - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the location resource (see Mappings to Other Standards for further information & status).

8.7.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Location
identifier | FiveWs.identifier
status | FiveWs.status
operationalStatus | FiveWs.status
mode | FiveWs.class
type | FiveWs.class
physicalType | FiveWs.class

8.7.9.2 RIM Mapping (http://hl7.org/v3)

Location | .Role[classCode=SDLC]
identifier | .id
status | .statusCode
operationalStatus | n/a
name | .name
alias | .name
description | .playingEntity[classCode=PLC determinerCode=INSTANCE].desc
mode | .playingEntity[classCode=PLC].determinerCode
type | .code
telecom | .telecom
address | .addr
physicalType | .playingEntity [classCode=PLC].code
position | .playingEntity [classCode=PLC determinerCode=INSTANCE].positionText
longitude | (RIM Opted not to map the sub-elements of GPS location, is now an OBS)
latitude | (RIM Opted not to map the sub-elements of GPS location, is now an OBS)
altitude | (RIM Opted not to map the sub-elements of GPS location, is now an OBS)
managingOrganization | .scopingEntity[classCode=ORG determinerKind=INSTANCE]
partOf | .inboundLink[typeCode=PART].source[classCode=SDLC]
hoursOfOperation | .effectiveTime
daysOfWeek | .effectiveTime
allDay | .effectiveTime
openingTime | .effectiveTime
closingTime | .effectiveTime
availabilityExceptions | n/a
endpoint | n/a

8.7.10 Resource Location - Extensions & Profiles

URL: https://hl7.org/fhir/R4/location-profiles.html

8.7.10 Resource Location - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Location resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.7.10.1 Profiles

No Profiles defined for this resource

8.7.10.2 Extensions

location-boundary-geojson | Location | boundary-geojson | for Location HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

communicationrequest-initiatingLocation | initiatingLocation | for CommunicationRequest HL7 Extensions

consent-location | location | for Consent HL7 Extensions
diagnosticReport-locationPerformed | locationPerformed | for DiagnosticReport HL7 Extensions
event-location | location | for Event Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.7.10.3 Search Extensions

No Search Extensions defined for this resource

8.7.11 Resource Location - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/location-version-maps.html

No validation errors - all conversions are clean

8.7.11 Resource Location - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Location.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

8.7.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Location3to4" = "R3 to R4 Conversions for Location"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Location" alias LocationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Location" alias Location as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Location(source src : LocationR3, target tgt : Location) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.operationalStatus -> tgt.operationalStatus;
  src.name -> tgt.name;
  src.alias -> tgt.alias;
  src.description -> tgt.description;
  src.mode -> tgt.mode;
  src.type -> tgt.type;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
  src.physicalType -> tgt.physicalType;
  src.position as vs0 -> tgt.position as vt0 then position(vs0, vt0);
  src.managingOrganization -> tgt.managingOrganization;
  src.partOf -> tgt.partOf;
  src.endpoint -> tgt.endpoint;
}

group position(source src, target tgt) extends BackboneElement {
  src.longitude -> tgt.longitude;
  src.latitude -> tgt.latitude;
  src.altitude -> tgt.altitude;
}
```

8.7.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Location4to3" = "R4 to R3 Conversion for Location"

uses "http://hl7.org/fhir/StructureDefinition/Location" alias Location as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Location" alias LocationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Location(source src : Location, target tgt : LocationR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.operationalStatus -> tgt.operationalStatus;
  src.name -> tgt.name;
  src.alias -> tgt.alias;
  src.description -> tgt.description;
  src.mode -> tgt.mode;
  src.type -> tgt.type;
  src.telecom -> tgt.telecom;
  src.address -> tgt.address;
```

```
  src.physicalType -> tgt.physicalType;
  src.position as vs0 -> tgt.position as vt0 then position(vs0, vt0);
  src.managingOrganization -> tgt.managingOrganization;
  src.partOf -> tgt.partOf;
  src.endpoint -> tgt.endpoint;
}

group position(source src, target tgt) extends BackboneElement {
  src.longitude -> tgt.longitude;
  src.latitude -> tgt.latitude;
  src.altitude -> tgt.altitude;
}
```

8.7.11.3 R4 Validation Errors

8.7.12 Location HL7 Extensions

URL: https://hl7.org/fhir/R4/location-extensions.html

Defines common extensions used with or related to the Location resource

8.7.12 Location HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.7.12.1 Content

Extensions:

location-boundary-geojson | boundary-geojson : A boundary shape that represents the outside edge of the location (in GeoJSON format) This shape may have holes, and disconnected shapes.

8.7.13 JSON Schema for Location

URL: https://hl7.org/fhir/R4/location.schema.json.html

8.7.13 JSON Schema for Location

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Location, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Location",
  "$ref": "#/definitions/Location",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Location": {
      "description": "Details and position information for a physical place where services are provided and resources
    and participants may be stored, found, contained, or accommodated.",
      "properties": {
        "resourceType": {
          "description": "This is a Location resource",
          "const": "Location"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
```

```
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique code or number identifying the location to its users.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The status property covers the general availability of the resource, not the current value
which may be covered by the operationStatus, or by a schedule/slots if they are configured for the location.",
      "enum": [
        "active",
        "suspended",
        "inactive"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "operationalStatus": {
      "description": "The operational status covers operation values most relevant to beds (but can also apply to
rooms/units/chairs/etc. such as an isolation unit/dialysis chair). This typically covers concepts such as
contamination, housekeeping, and other activities like maintenance.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "name": {
      "description": "Name of the location as used by humans. Does not need to be unique.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "alias": {
      "description": "A list of alternate names that the location is known as, or was known as, in the past.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_alias": {
      "description": "Extensions for alias",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "description": {
      "description": "Description of the Location, which helps in finding or referencing the place.",
      "$ref": "#/definitions/string"
    },
    "_description": {
```

```
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "mode": {
      "description": "Indicates whether a resource instance represents a specific location or a class of
locations.",
      "enum": [
        "instance",
        "kind"
      ]
    },
    "_mode": {
      "description": "Extensions for mode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Indicates the type of function performed at the location.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "telecom": {
      "description": "The contact details of communication devices available at the location. This can include phone
 numbers, fax numbers, mobile numbers, email addresses and web sites.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "address": {
      "description": "Physical location.",
      "$ref": "Address.schema.json#/definitions/Address"
    },
    "physicalType": {
      "description": "Physical form of the location, e.g. building, room, vehicle, road.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "position": {
      "description": "The absolute geographic location of the Location, expressed using the WGS84 datum (This is the
 same co-ordinate system used in KML).",
      "$ref": "#/definitions/Location_Position"
    },
    "managingOrganization": {
      "description": "The organization responsible for the provisioning and upkeep of the location.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "partOf": {
      "description": "Another Location of which this Location is physically a part of.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "hoursOfOperation": {
      "description": "What days/times during a week is this location usually open.",
      "items": {
        "$ref": "#/definitions/Location_HoursOfOperation"
      },
      "type": "array"
    },
    "availabilityExceptions": {
      "description": "A description of when the locations opening ours are different to normal, e.g. public holiday
availability. Succinctly describing all possible exceptions to normal site availability as detailed in the opening
hours Times.",
      "$ref": "#/definitions/string"
    },
    "_availabilityExceptions": {
      "description": "Extensions for availabilityExceptions",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "endpoint": {
      "description": "Technical endpoints providing access to services operated for the location.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Location_Position": {
  "description": "Details and position information for a physical place where services are provided and resources
and participants may be stored, found, contained, or accommodated.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
```

```
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "longitude": {
      "description": "Longitude. The value domain and the interpretation are the same as for the text of the
longitude element in KML (see notes below).",
      "$ref": "#/definitions/decimal"
    },
    "_longitude": {
      "description": "Extensions for longitude",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "latitude": {
      "description": "Latitude. The value domain and the interpretation are the same as for the text of the latitude
 element in KML (see notes below).",
      "$ref": "#/definitions/decimal"
    },
    "_latitude": {
      "description": "Extensions for latitude",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "altitude": {
      "description": "Altitude. The value domain and the interpretation are the same as for the text of the altitude
 element in KML (see notes below).",
      "$ref": "#/definitions/decimal"
    },
    "_altitude": {
      "description": "Extensions for altitude",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Location_HoursOfOperation": {
  "description": "Details and position information for a physical place where services are provided and resources
and participants may be stored, found, contained, or accommodated.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "daysOfWeek": {
```

```
          "description": "Indicates which days of the week are available between the start and end Times.",
          "items": {
            "$ref": "#/definitions/code"
          },
          "type": "array"
        },
        "_daysOfWeek": {
          "description": "Extensions for daysOfWeek",
          "items": {
            "$ref": "Element.schema.json#/definitions/Element"
          },
          "type": "array"
        },
        "allDay": {
          "description": "The Location is open all day.",
          "$ref": "#/definitions/boolean"
        },
        "_allDay": {
          "description": "Extensions for allDay",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "openingTime": {
          "description": "Time that the Location opens.",
          "$ref": "#/definitions/time"
        },
        "_openingTime": {
          "description": "Extensions for openingTime",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "closingTime": {
          "description": "Time that the Location closes.",
          "$ref": "#/definitions/time"
        },
        "_closingTime": {
          "description": "Extensions for closingTime",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


8.7.14 Resource Location - Maturity Dependency View

URL: https://hl7.org/fhir/R4/location-dependencies.html

8.7.14 Resource Location - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Location FMM level 3

Name | Card. | Type | Dependency Analysis
Location | | DomainResource
identifier | 0..* | Identifier
status | 0..1 | code
operationalStatus | 0..1 | Coding
name | 0..1 | string
alias | 0..* | string
description | 0..1 | string
mode | 0..1 | code
type | 0..* | CodeableConcept
telecom | 0..* | ContactPoint
address | 0..1 | Address
physicalType | 0..1 | CodeableConcept
position | 0..1 | BackboneElement
longitude | 1..1 | decimal
latitude | 1..1 | decimal
altitude | 0..1 | decimal
managingOrganization | 0..1 | Reference(Organization)
partOf | 0..1 | Reference(Location)
hoursOfOperation | 0..* | BackboneElement
daysOfWeek | 0..* | code
allDay | 0..1 | boolean
openingTime | 0..1 | time
closingTime | 0..1 | time
availabilityExceptions | 0..1 | string
endpoint | 0..* | Reference(Endpoint) | Type Warning: (Endpoint = FMM2-Trial Use vs. Element = FMM3-Trial Use)

8.7.15 Value Set Sc-valueset-location-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-location-status.html

8.7.15 Value Set Sc-valueset-location-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.7.16 LocationStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-location-status)

Mapping from http://hl7.org/fhir/ValueSet/location-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Indicates whether the location is still in use."

Source Code | Equivalence | Destination Code
active | equivalent | active
suspended | equivalent | suspended
inactive | equivalent | inactive

8.8 Resource HealthcareService - Content

URL: https://hl7.org/fhir/R4/healthcareservice.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The details of a healthcare service available at a location.

8.8.1 Scope and Usage

The HealthcareService resource is used to describe a single healthcare service or category of services that are provided by an organization at a location. The location of the services could be virtual, as with telemedicine services.

Common examples of HealthcareServices resources are:

• Allied Health
• Clinical Neuropsychologist
• Podiatry Service
• Smallville Hospital Emergency Services
• Respite care provided at a nursing home or hostel
• 24hr crisis telephone counseling service
• Information, advice and/or referral services; Disability, Telecommunications
• Rural TeleHealth Services
• Hospital in the home
• Yellow Cabs
• Pharmacy
• Active Rehab
• Social Support
• Drug and/or alcohol counseling
• Day Programs, Adult Training & Support Services
• Consulting psychologists and/or psychology services
• Group Hydrotherapy
• Little River Home Maintenance

HealthcareService resources do not represent Computer related Services (not SOA)

Example uses of HealthcareService resources are:

• National Services Directory - Consumer Focus
• National Services Directory - Practitioner Referrals Searching
• Organization's Client Portal - to locate services / book appointments
• Address book of services for Referrals including references to Questionnaires for assessments that are required as part of the referral

• Health Network internal directory Used for tracking available services offered internally, and also those offered by business partners. This information may also include costing information.

8.8.2 Boundaries and Relationships

The HealthcareService resource can be used with the Schedule resource to define actual availability of the service. This would be done by using the Schedule's Actor property.

When creating an Appointment, the HealthcareService is to be assigned to one of the participants. It is up to the scheduling system to determine if the service is available and can be accepted.

The HealthcareService resource is used with the following resources:

• Organization: The organization provides the services, the healthcareservice describes the services

• ServiceRequest: This is a subject specific request for a specific service, it may be to a specific healthcare service, and/or simply a coded service value. The granularity included in the request codes may be finer than defined in the healthcare service.

• OrganizationAffiliation: Might constrain a list of healthcare services that are available between 2 organizations

• Location: Specifies the place where the service(s) are offered/available within.

This resource is referenced by UsageContext, Account, Appointment, AppointmentResponse, CarePlan, CatalogEntry, Communication, CommunicationRequest, DeviceRequest, OrganizationAffiliation, PractitionerRole, Schedule, ServiceRequest, SupplyRequest and Task

8.8.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
HealthcareService | TU | | DomainResource | The details of a healthcare service available at a location Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External identifiers for this item
active | ?!Σ | 0..1 | boolean | Whether this HealthcareService record is in active use
providedBy | Σ | 0..1 | Reference(Organization) | Organization that provides this service
category | Σ | 0..* | CodeableConcept | Broad category of service being performed or delivered Service category (Example)
type | Σ | 0..* | CodeableConcept | Type of service that may be delivered or performed Service type (Example)
specialty | Σ | 0..* | CodeableConcept | Specialties handled by the HealthcareService Practice Setting Code Value Set (Preferred)
location | Σ | 0..* | Reference(Location) | Location(s) where service may be provided
name | Σ | 0..1 | string | Description of service as presented to a consumer while searching
comment | Σ | 0..1 | string | Additional description and/or any specific issues not covered elsewhere
extraDetails | | 0..1 | markdown | Extra details about the service that can't be placed in the other fields
photo | Σ | 0..1 | Attachment | Facilitates quick identification of the service
telecom | | 0..* | ContactPoint | Contacts related to the healthcare service
coverageArea | | 0..* | Reference(Location) | Location(s) service is intended for/available to
serviceProvisionCode | | 0..* | CodeableConcept | Conditions under which service is available/offered ServiceProvisionConditions (Example)
eligibility | | 0..* | BackboneElement | Specific eligibility requirements required to use the service
code | | 0..1 | CodeableConcept | Coded value for the eligibility
comment | | 0..1 | markdown | Describes the eligibility conditions for the service
program | | 0..* | CodeableConcept | Programs that this service is applicable to Program (Example)
characteristic | | 0..* | CodeableConcept | Collection of characteristics (attributes)
communication | | 0..* | CodeableConcept | The language that this service is offered in Common Languages (Preferred but limited to AllLanguages)
referralMethod | | 0..* | CodeableConcept | Ways that the service accepts referrals ReferralMethod (Example)
appointmentRequired | | 0..1 | boolean | If an appointment is required for access to this service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not available from this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions

endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to electronic services operated for the healthcare service

Documentation for this format

UML Diagram (Legend)

XML Template

```
<HealthcareService xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifiers for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this HealthcareService record is in active use -->
 <providedBy><!-- 0..1 Reference(Organization) Organization that provides this service --></providedBy>
 <category><!-- 0..* CodeableConcept Broad category of service being performed or delivered --></category>
 <type><!-- 0..* CodeableConcept Type of service that may be delivered or performed --></type>
 <specialty><!-- 0..* CodeableConcept Specialties handled by the HealthcareService --></specialty>
 <location><!-- 0..* Reference(Location) Location(s) where service may be provided --></location>
 <name value="[string]"/><!-- 0..1 Description of service as presented to a consumer while searching -->
 <comment value="[string]"/><!-- 0..1 Additional description and/or any specific issues not covered elsewhere -->
 <extraDetails value="[markdown]"/><!-- 0..1 Extra details about the service that can't be placed in the other fields
    -->
 <photo><!-- 0..1 Attachment Facilitates quick identification of the service --></photo>
 <telecom><!-- 0..* ContactPoint Contacts related to the healthcare service --></telecom>
 <coverageArea><!-- 0..* Reference(Location) Location(s) service is intended for/available to --></coverageArea>
 <serviceProvisionCode><!-- 0..* CodeableConcept Conditions under which service is available/offered
    --></serviceProvisionCode>
 <eligibility>  <!-- 0..* Specific eligibility requirements required to use the service -->
  <code><!-- 0..1 CodeableConcept Coded value for the eligibility --></code>
  <comment value="[markdown]"/><!-- 0..1 Describes the eligibility conditions for the service -->
 </eligibility>
 <program><!-- 0..* CodeableConcept Programs that this service is applicable to --></program>
 <characteristic><!-- 0..* CodeableConcept Collection of characteristics (attributes) --></characteristic>
 <communication><!-- 0..* CodeableConcept The language that this service is offered in --></communication>
 <referralMethod><!-- 0..* CodeableConcept Ways that the service accepts referrals --></referralMethod>
 <appointmentRequired value="[boolean]"/><!-- 0..1 If an appointment is required for access to this service -->
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not available from this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to electronic services operated for the
    healthcare service --></endpoint>
</HealthcareService>
```

JSON Template

```
{
  "resourceType" : "HealthcareService",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifiers for this item
  "active" : <boolean>, // Whether this HealthcareService record is in active use
  "providedBy" : { Reference(Organization) }, // Organization that provides this service
  "category" : [{ CodeableConcept }], // Broad category of service being performed or delivered
  "type" : [{ CodeableConcept }], // Type of service that may be delivered or performed
  "specialty" : [{ CodeableConcept }], // Specialties handled by the HealthcareService
  "location" : [{ Reference(Location) }], // Location(s) where service may be provided
  "name" : "<string>", // Description of service as presented to a consumer while searching
  "comment" : "<string>", // Additional description and/or any specific issues not covered elsewhere
  "extraDetails" : "<markdown>", // Extra details about the service that can't be placed in the other fields
  "photo" : { Attachment }, // Facilitates quick identification of the service
  "telecom" : [{ ContactPoint }], // Contacts related to the healthcare service
  "coverageArea" : [{ Reference(Location) }], // Location(s) service is intended for/available to
  "serviceProvisionCode" : [{ CodeableConcept }], // Conditions under which service is available/offered
  "eligibility" : [{ // Specific eligibility requirements required to use the service
    "code" : { CodeableConcept }, // Coded value for the eligibility
    "comment" : "<markdown>" // Describes the eligibility conditions for the service
  }],
  "program" : [{ CodeableConcept }], // Programs that this service is applicable to
  "characteristic" : [{ CodeableConcept }], // Collection of characteristics (attributes)
  "communication" : [{ CodeableConcept }], // The language that this service is offered in
  "referralMethod" : [{ CodeableConcept }], // Ways that the service accepts referrals
  "appointmentRequired" : <boolean>, // If an appointment is required for access to this service
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not available from this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
```

```
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to electronic services operated for the
     healthcare service
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:HealthcareService;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:HealthcareService.identifier [ Identifier ], ... ; # 0..* External identifiers for this item
  fhir:HealthcareService.active [ boolean ]; # 0..1 Whether this HealthcareService record is in active use
  fhir:HealthcareService.providedBy [ Reference(Organization) ]; # 0..1 Organization that provides this service
  fhir:HealthcareService.category [ CodeableConcept ], ... ; # 0..* Broad category of service being performed or
    delivered
  fhir:HealthcareService.type [ CodeableConcept ], ... ; # 0..* Type of service that may be delivered or performed
  fhir:HealthcareService.specialty [ CodeableConcept ], ... ; # 0..* Specialties handled by the HealthcareService
  fhir:HealthcareService.location [ Reference(Location) ], ... ; # 0..* Location(s) where service may be provided
  fhir:HealthcareService.name [ string ]; # 0..1 Description of service as presented to a consumer while searching
  fhir:HealthcareService.comment [ string ]; # 0..1 Additional description and/or any specific issues not covered
    elsewhere
  fhir:HealthcareService.extraDetails [ markdown ]; # 0..1 Extra details about the service that can't be placed in the
    other fields
  fhir:HealthcareService.photo [ Attachment ]; # 0..1 Facilitates quick identification of the service
  fhir:HealthcareService.telecom [ ContactPoint ], ... ; # 0..* Contacts related to the healthcare service
  fhir:HealthcareService.coverageArea [ Reference(Location) ], ... ; # 0..* Location(s) service is intended
    for/available to
  fhir:HealthcareService.serviceProvisionCode [ CodeableConcept ], ... ; # 0..* Conditions under which service is
    available/offered
  fhir:HealthcareService.eligibility [ # 0..* Specific eligibility requirements required to use the service
    fhir:HealthcareService.eligibility.code [ CodeableConcept ]; # 0..1 Coded value for the eligibility
    fhir:HealthcareService.eligibility.comment [ markdown ]; # 0..1 Describes the eligibility conditions for the service
  ], ...;
  fhir:HealthcareService.program [ CodeableConcept ], ... ; # 0..* Programs that this service is applicable to
  fhir:HealthcareService.characteristic [ CodeableConcept ], ... ; # 0..* Collection of characteristics (attributes)
  fhir:HealthcareService.communication [ CodeableConcept ], ... ; # 0..* The language that this service is offered in
  fhir:HealthcareService.referralMethod [ CodeableConcept ], ... ; # 0..* Ways that the service accepts referrals
  fhir:HealthcareService.appointmentRequired [ boolean ]; # 0..1 If an appointment is required for access to this
    service
  fhir:HealthcareService.availableTime [ # 0..* Times the Service Site is available
    fhir:HealthcareService.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:HealthcareService.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:HealthcareService.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:HealthcareService.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay =
    true)
  ], ...;
  fhir:HealthcareService.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:HealthcareService.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:HealthcareService.notAvailable.during [ Period ]; # 0..1 Service not available from this date
  ], ...;
  fhir:HealthcareService.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:HealthcareService.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to
    electronic services operated for the healthcare service
]
```

Changes since R3

HealthcareService
HealthcareService.active | Default Value "true" removed
HealthcareService.category | Max Cardinality changed from 1 to *
HealthcareService.extraDetails | Type changed from string to markdown
HealthcareService.eligibility | Max Cardinality changed from 1 to * Type changed from CodeableConcept to BackboneElement
HealthcareService.eligibility.code | Added Element
HealthcareService.eligibility.comment | Added Element
HealthcareService.program | Added Element
HealthcareService.communication | Added Element
HealthcareService.availableTime.daysOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
HealthcareService.eligibilityNote | deleted
HealthcareService.programName | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
HealthcareService | TU | | DomainResource | The details of a healthcare service available at a location Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | External identifiers for this item
active | ?!Σ | 0..1 | boolean | Whether this HealthcareService record is in active use
providedBy | Σ | 0..1 | Reference(Organization) | Organization that provides this service
category | Σ | 0..* | CodeableConcept | Broad category of service being performed or delivered Service category (Example)
type | Σ | 0..* | CodeableConcept | Type of service that may be delivered or performed Service type (Example)
specialty | Σ | 0..* | CodeableConcept | Specialties handled by the HealthcareService Practice Setting Code Value Set (Preferred)
location | Σ | 0..* | Reference(Location) | Location(s) where service may be provided
name | Σ | 0..1 | string | Description of service as presented to a consumer while searching
comment | Σ | 0..1 | string | Additional description and/or any specific issues not covered elsewhere
extraDetails | | 0..1 | markdown | Extra details about the service that can't be placed in the other fields
photo | Σ | 0..1 | Attachment | Facilitates quick identification of the service
telecom | | 0..* | ContactPoint | Contacts related to the healthcare service
coverageArea | | 0..* | Reference(Location) | Location(s) service is intended for/available to
serviceProvisionCode | | 0..* | CodeableConcept | Conditions under which service is available/offered ServiceProvisionConditions (Example)
eligibility | | 0..* | BackboneElement | Specific eligibility requirements required to use the service
code | | 0..1 | CodeableConcept | Coded value for the eligibility
comment | | 0..1 | markdown | Describes the eligibility conditions for the service
program | | 0..* | CodeableConcept | Programs that this service is applicable to Program (Example)
characteristic | | 0..* | CodeableConcept | Collection of characteristics (attributes)
communication | | 0..* | CodeableConcept | The language that this service is offered in Common Languages (Preferred but limited to AllLanguages)
referralMethod | | 0..* | CodeableConcept | Ways that the service accepts referrals ReferralMethod (Example)
appointmentRequired | | 0..1 | boolean | If an appointment is required for access to this service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not available from this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to electronic services operated for the healthcare service
Documentation for this format

UML Diagram (Legend)

XML Template

```
<HealthcareService xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifiers for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this HealthcareService record is in active use -->
 <providedBy><!-- 0..1 Reference(Organization) Organization that provides this service --></providedBy>
 <category><!-- 0..* CodeableConcept Broad category of service being performed or delivered --></category>
 <type><!-- 0..* CodeableConcept Type of service that may be delivered or performed --></type>
 <specialty><!-- 0..* CodeableConcept Specialties handled by the HealthcareService --></specialty>
 <location><!-- 0..* Reference(Location) Location(s) where service may be provided --></location>
 <name value="[string]"/><!-- 0..1 Description of service as presented to a consumer while searching -->
 <comment value="[string]"/><!-- 0..1 Additional description and/or any specific issues not covered elsewhere -->
 <extraDetails value="[markdown]"/><!-- 0..1 Extra details about the service that can't be placed in the other fields
    -->
 <photo><!-- 0..1 Attachment Facilitates quick identification of the service --></photo>
 <telecom><!-- 0..* ContactPoint Contacts related to the healthcare service --></telecom>
 <coverageArea><!-- 0..* Reference(Location) Location(s) service is intended for/available to --></coverageArea>
 <serviceProvisionCode><!-- 0..* CodeableConcept Conditions under which service is available/offered
    --></serviceProvisionCode>
 <eligibility>  <!-- 0..* Specific eligibility requirements required to use the service -->
  <code><!-- 0..1 CodeableConcept Coded value for the eligibility --></code>
  <comment value="[markdown]"/><!-- 0..1 Describes the eligibility conditions for the service -->
 </eligibility>
 <program><!-- 0..* CodeableConcept Programs that this service is applicable to --></program>
 <characteristic><!-- 0..* CodeableConcept Collection of characteristics (attributes) --></characteristic>
 <communication><!-- 0..* CodeableConcept The language that this service is offered in --></communication>
 <referralMethod><!-- 0..* CodeableConcept Ways that the service accepts referrals --></referralMethod>
 <appointmentRequired value="[boolean]"/><!-- 0..1 If an appointment is required for access to this service -->
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not available from this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to electronic services operated for the
    healthcare service --></endpoint>
```

```
</HealthcareService>
```

JSON Template

```
{
  "resourceType" : "HealthcareService",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifiers for this item
  "active" : <boolean>, // Whether this HealthcareService record is in active use
  "providedBy" : { Reference(Organization) }, // Organization that provides this service
  "category" : [{ CodeableConcept }], // Broad category of service being performed or delivered
  "type" : [{ CodeableConcept }], // Type of service that may be delivered or performed
  "specialty" : [{ CodeableConcept }], // Specialties handled by the HealthcareService
  "location" : [{ Reference(Location) }], // Location(s) where service may be provided
  "name" : "<string>", // Description of service as presented to a consumer while searching
  "comment" : "<string>", // Additional description and/or any specific issues not covered elsewhere
  "extraDetails" : "<markdown>", // Extra details about the service that can't be placed in the other fields
  "photo" : { Attachment }, // Facilitates quick identification of the service
  "telecom" : [{ ContactPoint }], // Contacts related to the healthcare service
  "coverageArea" : [{ Reference(Location) }], // Location(s) service is intended for/available to
  "serviceProvisionCode" : [{ CodeableConcept }], // Conditions under which service is available/offered
  "eligibility" : [{ // Specific eligibility requirements required to use the service
    "code" : { CodeableConcept }, // Coded value for the eligibility
    "comment" : "<markdown>" // Describes the eligibility conditions for the service
  }],
  "program" : [{ CodeableConcept }], // Programs that this service is applicable to
  "characteristic" : [{ CodeableConcept }], // Collection of characteristics (attributes)
  "communication" : [{ CodeableConcept }], // The language that this service is offered in
  "referralMethod" : [{ CodeableConcept }], // Ways that the service accepts referrals
  "appointmentRequired" : <boolean>, // If an appointment is required for access to this service
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not available from this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to electronic services operated for the
     healthcare service
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:HealthcareService;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:HealthcareService.identifier [ Identifier ], ... ; # 0..* External identifiers for this item
  fhir:HealthcareService.active [ boolean ]; # 0..1 Whether this HealthcareService record is in active use
  fhir:HealthcareService.providedBy [ Reference(Organization) ]; # 0..1 Organization that provides this service
  fhir:HealthcareService.category [ CodeableConcept ], ... ; # 0..* Broad category of service being performed or
    delivered
  fhir:HealthcareService.type [ CodeableConcept ], ... ; # 0..* Type of service that may be delivered or performed
  fhir:HealthcareService.specialty [ CodeableConcept ], ... ; # 0..* Specialties handled by the HealthcareService
  fhir:HealthcareService.location [ Reference(Location) ], ... ; # 0..* Location(s) where service may be provided
  fhir:HealthcareService.name [ string ]; # 0..1 Description of service as presented to a consumer while searching
  fhir:HealthcareService.comment [ string ]; # 0..1 Additional description and/or any specific issues not covered
    elsewhere
  fhir:HealthcareService.extraDetails [ markdown ]; # 0..1 Extra details about the service that can't be placed in the
    other fields
  fhir:HealthcareService.photo [ Attachment ]; # 0..1 Facilitates quick identification of the service
  fhir:HealthcareService.telecom [ ContactPoint ], ... ; # 0..* Contacts related to the healthcare service
  fhir:HealthcareService.coverageArea [ Reference(Location) ], ... ; # 0..* Location(s) service is intended
    for/available to
  fhir:HealthcareService.serviceProvisionCode [ CodeableConcept ], ... ; # 0..* Conditions under which service is
    available/offered
  fhir:HealthcareService.eligibility [ # 0..* Specific eligibility requirements required to use the service
    fhir:HealthcareService.eligibility.code [ CodeableConcept ]; # 0..1 Coded value for the eligibility
    fhir:HealthcareService.eligibility.comment [ markdown ]; # 0..1 Describes the eligibility conditions for the service
  ], ...;
  fhir:HealthcareService.program [ CodeableConcept ], ... ; # 0..* Programs that this service is applicable to
  fhir:HealthcareService.characteristic [ CodeableConcept ], ... ; # 0..* Collection of characteristics (attributes)
  fhir:HealthcareService.communication [ CodeableConcept ], ... ; # 0..* The language that this service is offered in
  fhir:HealthcareService.referralMethod [ CodeableConcept ], ... ; # 0..* Ways that the service accepts referrals
  fhir:HealthcareService.appointmentRequired [ boolean ]; # 0..1 If an appointment is required for access to this
    service
  fhir:HealthcareService.availableTime [ # 0..* Times the Service Site is available
    fhir:HealthcareService.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:HealthcareService.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:HealthcareService.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:HealthcareService.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay =
```

```
    true)
  ], ...;
  fhir:HealthcareService.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:HealthcareService.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:HealthcareService.notAvailable.during [ Period ]; # 0..1 Service not available from this date
  ], ...;
  fhir:HealthcareService.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:HealthcareService.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to
    electronic services operated for the healthcare service
]
```

Changes since Release 3

HealthcareService
HealthcareService.active | Default Value "true" removed
HealthcareService.category | Max Cardinality changed from 1 to *
HealthcareService.extraDetails | Type changed from string to markdown
HealthcareService.eligibility | Max Cardinality changed from 1 to * Type changed from CodeableConcept to BackboneElement
HealthcareService.eligibility.code | Added Element
HealthcareService.eligibility.comment | Added Element
HealthcareService.program | Added Element
HealthcareService.communication | Added Element
HealthcareService.availableTime.daysOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
HealthcareService.eligibilityNote | deleted
HealthcareService.programName | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.8.3.1 Terminology Bindings

Path | Definition | Type | Reference
HealthcareService.category | A category of the service(s) that could be provided. | Example | ServiceCategory
HealthcareService.type | Additional details about where the content was created (e.g. clinical specialty). | Example | ServiceType
HealthcareService.specialty | A specialty that a healthcare service may provide. | Preferred | PracticeSettingCodeValueSet
HealthcareService.serviceProvisionCode | The code(s) that detail the conditions under which the healthcare service is available/offered. | Example | ServiceProvisionConditions
HealthcareService.eligibility.code | Coded values underwhich a specific service is made available. | Unknown | No details provided yet
HealthcareService.program | Government or local programs that this service applies to. | Example | Program
HealthcareService.characteristic | A custom attribute that could be provided at a service (e.g. Wheelchair accessibiliy). | Unknown | No details provided yet
HealthcareService.communication | A human language. | Preferred, but limited to AllLanguages | CommonLanguages
HealthcareService.referralMethod | The methods of referral can be used when referring to a specific HealthCareService resource. | Example | ReferralMethod
HealthcareService.availableTime.daysOfWeek | The days of the week. | Required | DaysOfWeek

8.8.4 Notes:

• The HealthcareService could be mapped to components of the IHE Care Services Directory, and/or the OMG ServD standards

8.8.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | The Healthcare Service is currently marked as active | HealthcareService.active
characteristic | token | One of the HealthcareService's characteristics | HealthcareService.characteristic
coverage-area | reference | Location(s) service is intended for/available to | HealthcareService.coverageArea (Location)
endpoint | reference | Technical endpoints providing access to electronic services operated for the healthcare service | HealthcareService.endpoint (Endpoint)
identifier | token | External identifiers for this item | HealthcareService.identifier
location | reference | The location of the Healthcare Service | HealthcareService.location (Location)
name | string | A portion of the Healthcare service name | HealthcareService.name
organization | reference | The organization that provides this Healthcare Service | HealthcareService.providedBy (Organization)
program | token | One of the Programs supported by this HealthcareService | HealthcareService.program
service-category | token | Service Category of the Healthcare Service | HealthcareService.category
service-type | token | The type of service provided by this healthcare service | HealthcareService.type
specialty | token | The specialty of the service provided by this healthcare service | HealthcareService.specialty

8.8.6 Resource HealthcareService - Examples

URL: https://hl7.org/fhir/R4/healthcareservice-examples.html

8.8.6 Resource HealthcareService - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of healthcareservice | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.8.7 Resource HealthcareService - Detailed Descriptions

URL: https://hl7.org/fhir/R4/healthcareservice-definitions.html

8.8.7 Resource HealthcareService - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the HealthcareService resource.

HealthcareService
Element Id | HealthcareService
Definition | The details of a healthcare service available at a location.
Cardinality | 0..*
Type | DomainResource
HealthcareService.identifier
Element Id | HealthcareService.identifier
Definition | External identifiers for this item.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
HealthcareService.active
Element Id | HealthcareService.active
Definition | This flag is used to mark the record to not be used. This is not used when a center is closed for maintenance, or for holidays, the notAvailable period is to be used for this.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
HealthcareService.providedBy
Element Id | HealthcareService.providedBy
Definition | The organization that provides this healthcare service.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
Comments | This property is recommended to be the same as the Location's managingOrganization, and if not provided should be interpreted as such. If the Location does not have a managing Organization, then this property should be populated.
HealthcareService.category
Element Id | HealthcareService.category
Definition | Identifies the broad category of service being performed or delivered.
Cardinality | 0..*
Terminology Binding | Service category (Example)
Type | CodeableConcept
Alternate Names | service category
Summary | true
Comments | Selecting a Service Category then determines the list of relevant service types that can be selected in the primary service type.
HealthcareService.type
Element Id | HealthcareService.type
Definition | The specific type of service that may be delivered or performed.
Cardinality | 0..*
Terminology Binding | Service type (Example)
Type | CodeableConcept
Alternate Names | service type
Summary | true
HealthcareService.specialty

Element Id | HealthcareService.specialty
Definition | Collection of specialties handled by the service site. This is more of a medical term.
Cardinality | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
HealthcareService.location
Element Id | HealthcareService.location
Definition | The location(s) where this healthcare service may be provided.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
HealthcareService.name
Element Id | HealthcareService.name
Definition | Further description of the service as it would be presented to a consumer while searching.
Cardinality | 0..1
Type | string
Summary | true
HealthcareService.comment
Element Id | HealthcareService.comment
Definition | Any additional description of the service and/or any specific issues not covered by the other attributes, which can be displayed as further detail under the serviceName.
Cardinality | 0..1
Type | string
Summary | true
Comments | Would expect that a user would not see this information on a search results, and it would only be available when viewing the complete details of the service.
HealthcareService.extraDetails
Element Id | HealthcareService.extraDetails
Definition | Extra details about the service that can't be placed in the other fields.
Cardinality | 0..1
Type | markdown
HealthcareService.photo
Element Id | HealthcareService.photo
Definition | If there is a photo/symbol associated with this HealthcareService, it may be included here to facilitate quick identification of the service in a list.
Cardinality | 0..1
Type | Attachment
Summary | true
HealthcareService.telecom
Element Id | HealthcareService.telecom
Definition | List of contacts related to this specific healthcare service.
Cardinality | 0..*
Type | ContactPoint
Comments | If this is empty, then refer to the location's contacts.
HealthcareService.coverageArea
Element Id | HealthcareService.coverageArea
Definition | The location(s) that this service is available to (not where the service is provided).
Cardinality | 0..*
Type | Reference(Location)
Comments | The locations referenced by the coverage area can include both specific locations, including areas, and also conceptual domains too (mode = kind), such as a physical area (tri-state area) and some other attribute (covered by Example Care Organization). These types of Locations are often not managed by any specific organization. This could also include generic locations such as "in-home".
HealthcareService.serviceProvisionCode
Element Id | HealthcareService.serviceProvisionCode
Definition | The code(s) that detail the conditions under which the healthcare service is available/offered.
Cardinality | 0..*
Terminology Binding | ServiceProvisionConditions (Example)
Type | CodeableConcept
Comments | The provision means being commissioned by, contractually obliged or financially sourced. Types of costings that may apply to this healthcare service, such if the service may be available for free, some discounts available, or fees apply.
HealthcareService.eligibility
Element Id | HealthcareService.eligibility
Definition | Does this service have specific eligibility requirements that need to be met in order to use the service?
Cardinality | 0..*
HealthcareService.eligibility.code
Element Id | HealthcareService.eligibility.code
Definition | Coded value for the eligibility.
Cardinality | 0..1
Terminology Binding | ServiceEligibility:
Type | CodeableConcept
HealthcareService.eligibility.comment

Element Id | HealthcareService.eligibility.comment
Definition | Describes the eligibility conditions for the service.
Cardinality | 0..1
Type | markdown
Comments | The description of service eligibility should, in general, not exceed one or two paragraphs. It should be sufficient for a prospective consumer to determine if they are likely to be eligible or not. Where eligibility requirements and conditions are complex, it may simply be noted that an eligibility assessment is required. Where eligibility is determined by an outside source, such as an Act of Parliament, this should be noted, preferably with a reference to a commonly available copy of the source document such as a web page.
HealthcareService.program
Element Id | HealthcareService.program
Definition | Programs that this service is applicable to.
Cardinality | 0..*
Terminology Binding | Program (Example)
Type | CodeableConcept
Comments | Programs are often defined externally to an Organization, commonly by governments; e.g. Home and Community Care Programs, Homeless Program, ….
HealthcareService.characteristic
Element Id | HealthcareService.characteristic
Definition | Collection of characteristics (attributes).
Cardinality | 0..*
Terminology Binding | ServiceCharacteristic:
Type | CodeableConcept
Comments | These could be such things as is wheelchair accessible.
HealthcareService.communication
Element Id | HealthcareService.communication
Definition | Some services are specifically made available in multiple languages, this property permits a directory to declare the languages this is offered in. Typically this is only provided where a service operates in communities with mixed languages used.
Cardinality | 0..*
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | CodeableConcept
Comments | When using this property it indicates that the service is available with this language, it is not derived from the practitioners, and not all are required to use this language, just that this language is available while scheduling.
HealthcareService.referralMethod
Element Id | HealthcareService.referralMethod
Definition | Ways that the service accepts referrals, if this is not provided then it is implied that no referral is required.
Cardinality | 0..*
Terminology Binding | ReferralMethod (Example)
Type | CodeableConcept
HealthcareService.appointmentRequired
Element Id | HealthcareService.appointmentRequired
Definition | Indicates whether or not a prospective consumer will require an appointment for a particular service at a site to be provided by the Organization. Indicates if an appointment is required for access to this service.
Cardinality | 0..1
Type | boolean
HealthcareService.availableTime
Element Id | HealthcareService.availableTime
Definition | A collection of times that the Service Site is available.
Cardinality | 0..*
Comments | More detailed availability information may be provided in associated Schedule/Slot resources.
HealthcareService.availableTime.daysOfWeek
Element Id | HealthcareService.availableTime.daysOfWeek
Definition | Indicates which days of the week are available between the start and end Times.
Cardinality | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
HealthcareService.availableTime.allDay
Element Id | HealthcareService.availableTime.allDay
Definition | Is this always available? (hence times are irrelevant) e.g. 24 hour service.
Cardinality | 0..1
Type | boolean
HealthcareService.availableTime.availableStartTime
Element Id | HealthcareService.availableTime.availableStartTime
Definition | The opening time of day. Note: If the AllDay flag is set, then this time is ignored.
Cardinality | 0..1
Type | time
Comments | The time zone is expected to be for where this HealthcareService is provided at.
HealthcareService.availableTime.availableEndTime
Element Id | HealthcareService.availableTime.availableEndTime
Definition | The closing time of day. Note: If the AllDay flag is set, then this time is ignored.
Cardinality | 0..1
Type | time
Comments | The time zone is expected to be for where this HealthcareService is provided at.

HealthcareService.notAvailable
Element Id | HealthcareService.notAvailable
Definition | The HealthcareService is not available during this period of time due to the provided reason.
Cardinality | 0..*
HealthcareService.notAvailable.description
Element Id | HealthcareService.notAvailable.description
Definition | The reason that can be presented to the user as to why this time is not available.
Cardinality | 1..1
Type | string
HealthcareService.notAvailable.during
Element Id | HealthcareService.notAvailable.during
Definition | Service is not available (seasonally or for a public holiday) from this date.
Cardinality | 0..1
Type | Period
HealthcareService.availabilityExceptions
Element Id | HealthcareService.availabilityExceptions
Definition | A description of site availability exceptions, e.g. public holiday availability. Succinctly describing all possible exceptions to normal site availability as details in the available Times and not available Times.
Cardinality | 0..1
Type | string
HealthcareService.endpoint
Element Id | HealthcareService.endpoint
Definition | Technical endpoints providing access to services operated for the specific healthcare services defined at this resource.
Cardinality | 0..*
Type | Reference(Endpoint)

8.8.8 Resource HealthcareService - Mappings

URL: https://hl7.org/fhir/R4/healthcareservice-mappings.html

8.8.8 Resource HealthcareService - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the healthcareservice resource (see Mappings to Other Standards for further information & status).

8.8.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

HealthcareService
identifier | FiveWs.identifier
active | FiveWs.status
category | FiveWs.class
location | FiveWs.where[x]

8.8.8.2 RIM Mapping (http://hl7.org/v3)

HealthcareService | act[classCode=ACT][moodCode=DEF]
identifier | .id
active | .statusCode
providedBy | .scopingRole.Organization
category | .code
type | .actrelationship[typeCode=COMP.act[classCode=ACT][moodCode=DEF].code
specialty | .actrelationship[typeCode=COMP.act[classCode=ACT][moodCode=DEF].code
location | .location.role[classCode=SDLOC]
name | .name
comment | .location.role[classCode=SDLOC].desc
extraDetails | .actrelationship[typeCode=COMP.act[classCode=ACT][moodCode=DEF].text
photo | .actrelationship[typeCode=SBJ].observation.value
telecom | .telecom
coverageArea | .location.role[classCode=SDLOC].subjectOf.A_SpatialCoordinate
serviceProvisionCode | .actrelationship[typeCode=PRCN].observation[moodCode=EVN.CRT]
code | .actrelationship[typeCode=PRCN].observation[moodCode=EVN.CRT]
comment | .actrelationship[typeCode=PRCN].observation[moodCode=EVN.CRT].text
program | .actrelationship[typeCode=PERT].observation
characteristic | .actrelationship[typeCode=PERT].observation
referralMethod | .actrelationship[typeCode=PERT].observation
appointmentRequired | .actrelationship[typeCode=PERT].observation
availableTime | .effectiveTime
daysOfWeek | .effectiveTime

allDay | .effectiveTime
availableStartTime | .effectiveTime
availableEndTime | .effectiveTime
notAvailable | .effectiveTime
description | n/a
during | .effectiveTime
availabilityExceptions | .effectiveTime
endpoint | n/a

8.8.9 Resource HealthcareService - Extensions & Profiles

URL: https://hl7.org/fhir/R4/healthcareservice-profiles.html

8.8.9 Resource HealthcareService - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the HealthcareService resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.8.9.1 Profiles

No Profiles defined for this resource

8.8.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.8.9.3 Search Extensions

No Search Extensions defined for this resource

8.8.10 Resource HealthcareService - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/healthcareservice-version-maps.html

No validation errors - all conversions are clean

8.8.10 Resource HealthcareService - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for HealthcareService.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

8.8.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/HealthcareService3to4" = "R3 to R4 Conversions for HealthcareService"

uses "http://hl7.org/fhir/3.0/StructureDefinition/HealthcareService" alias HealthcareServiceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/HealthcareService" alias HealthcareService as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group HealthcareService(source src : HealthcareServiceR3, target tgt : HealthcareService) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.providedBy -> tgt.providedBy;
  src.category -> tgt.category;
  src.type -> tgt.type;
  src.specialty -> tgt.specialty;
  src.location -> tgt.location;
```

```
  src.name -> tgt.name;
  src.comment -> tgt.comment;
  src.extraDetails -> tgt.extraDetails;
  src.photo -> tgt.photo;
  src.telecom -> tgt.telecom;
  src.coverageArea -> tgt.coverageArea;
  src.serviceProvisionCode -> tgt.serviceProvisionCode;
  src.eligibility as vs -> tgt.eligibility as vt then eligibility(vs, vt, src);
  src.programName as v ->  tgt.program as cc,  cc.text = v;
  src.characteristic -> tgt.characteristic;
  src.referralMethod -> tgt.referralMethod;
  src.appointmentRequired -> tgt.appointmentRequired;
  src.availableTime as vs -> tgt.availableTime as vt then availableTime(vs, vt);
  src.notAvailable as vs -> tgt.notAvailable as vt then notAvailable(vs, vt);
  src.availabilityExceptions -> tgt.availabilityExceptions;
  src.endpoint -> tgt.endpoint;
}

group eligibility(source src, target tgt, source res) {
  src as vs -> tgt.code as vt then CodeableConcept(vs, vt) "code";
  res.eligibilityNote -> tgt.comment;
}

group availableTime(source src, target tgt) extends BackboneElement {
  src.daysOfWeek -> tgt.daysOfWeek;
  src.allDay -> tgt.allDay;
  src.availableStartTime -> tgt.availableStartTime;
  src.availableEndTime -> tgt.availableEndTime;
}

group notAvailable(source src, target tgt) extends BackboneElement {
  src.description -> tgt.description;
  src.during -> tgt.during;
}
```

8.8.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/HealthcareService4to3" = "R4 to R3 Conversion for HealthcareService"

uses "http://hl7.org/fhir/StructureDefinition/HealthcareService" alias HealthcareService as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/HealthcareService" alias HealthcareServiceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group HealthcareService(source src : HealthcareService, target tgt : HealthcareServiceR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.providedBy -> tgt.providedBy;
  src.category -> tgt.category;
  src.type -> tgt.type;
  src.specialty -> tgt.specialty;
  src.location -> tgt.location;
  src.name -> tgt.name;
  src.comment -> tgt.comment;
  src.extraDetails -> tgt.extraDetails;
  src.photo -> tgt.photo;
  src.telecom -> tgt.telecom;
  src.coverageArea -> tgt.coverageArea;
  src.serviceProvisionCode -> tgt.serviceProvisionCode;
  src.eligibility as vs -> tgt.eligibility as vt then eligibility(vs, vt, tgt);
  src.program as cc then {
    cc.text -> tgt.programName;
  };
  src.characteristic -> tgt.characteristic;
  src.referralMethod -> tgt.referralMethod;
  src.appointmentRequired -> tgt.appointmentRequired;
  src.availableTime as vs -> tgt.availableTime as vt then availableTime(vs, vt);
  src.notAvailable as vs -> tgt.notAvailable as vt then notAvailable(vs, vt);
  src.availabilityExceptions -> tgt.availabilityExceptions;
  src.endpoint -> tgt.endpoint;
}

group eligibility(source src, target tgt, target res) {
  src.code as vs then CodeableConcept(vs, tgt);
  src.comment -> res.eligibilityNote;
}

group availableTime(source src, target tgt) extends BackboneElement {
  src.daysOfWeek -> tgt.daysOfWeek;
  src.allDay -> tgt.allDay;
  src.availableStartTime -> tgt.availableStartTime;
  src.availableEndTime -> tgt.availableEndTime;
}

group notAvailable(source src, target tgt) extends BackboneElement {
  src.description -> tgt.description;
  src.during -> tgt.during;
}
```

8.8.10.3 R4 Validation Errors

8.8.11 JSON Schema for HealthcareService

URL: https://hl7.org/fhir/R4/healthcareservice.schema.json.html

8.8.11 JSON Schema for HealthcareService

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for HealthcareService, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/HealthcareService",
  "$ref": "#/definitions/HealthcareService",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "HealthcareService": {
      "description": "The details of a healthcare service available at a location.",
      "properties": {
        "resourceType": {
          "description": "This is a HealthcareService resource",
          "const": "HealthcareService"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
```

```
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "External identifiers for this item.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "This flag is used to mark the record to not be used. This is not used when a center is closed
for maintenance, or for holidays, the notAvailable period is to be used for this.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "providedBy": {
      "description": "The organization that provides this healthcare service.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "category": {
      "description": "Identifies the broad category of service being performed or delivered.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "type": {
      "description": "The specific type of service that may be delivered or performed.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialty": {
      "description": "Collection of specialties handled by the service site. This is more of a medical term.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "location": {
      "description": "The location(s) where this healthcare service may be provided.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "name": {
      "description": "Further description of the service as it would be presented to a consumer while searching.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "Any additional description of the service and/or any specific issues not covered by the other
attributes, which can be displayed as further detail under the serviceName.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "extraDetails": {
      "description": "Extra details about the service that can\u0027t be placed in the other fields.",
      "$ref": "#/definitions/markdown"
    },
    "_extraDetails": {
      "description": "Extensions for extraDetails",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "photo": {
      "description": "If there is a photo/symbol associated with this HealthcareService, it may be included here to
facilitate quick identification of the service in a list.",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "telecom": {
      "description": "List of contacts related to this specific healthcare service.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
```

```
    },
    "coverageArea": {
      "description": "The location(s) that this service is available to (not where the service is provided).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "serviceProvisionCode": {
      "description": "The code(s) that detail the conditions under which the healthcare service is
available/offered.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "eligibility": {
      "description": "Does this service have specific eligibility requirements that need to be met in order to use
the service?",
      "items": {
        "$ref": "#/definitions/HealthcareService_Eligibility"
      },
      "type": "array"
    },
    "program": {
      "description": "Programs that this service is applicable to.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "characteristic": {
      "description": "Collection of characteristics (attributes).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "communication": {
      "description": "Some services are specifically made available in multiple languages, this property permits a
directory to declare the languages this is offered in. Typically this is only provided where a service operates in
communities with mixed languages used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "referralMethod": {
      "description": "Ways that the service accepts referrals, if this is not provided then it is implied that no
referral is required.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "appointmentRequired": {
      "description": "Indicates whether or not a prospective consumer will require an appointment for a particular
service at a site to be provided by the Organization. Indicates if an appointment is required for access to this
service.",
      "$ref": "#/definitions/boolean"
    },
    "_appointmentRequired": {
      "description": "Extensions for appointmentRequired",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "availableTime": {
      "description": "A collection of times that the Service Site is available.",
      "items": {
        "$ref": "#/definitions/HealthcareService_AvailableTime"
      },
      "type": "array"
    },
    "notAvailable": {
      "description": "The HealthcareService is not available during this period of time due to the provided
reason.",
      "items": {
        "$ref": "#/definitions/HealthcareService_NotAvailable"
      },
      "type": "array"
    },
    "availabilityExceptions": {
      "description": "A description of site availability exceptions, e.g. public holiday availability. Succinctly
describing all possible exceptions to normal site availability as details in the available Times and not available
Times.",
      "$ref": "#/definitions/string"
    },
    "_availabilityExceptions": {
      "description": "Extensions for availabilityExceptions",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "endpoint": {
      "description": "Technical endpoints providing access to services operated for the specific healthcare services
 defined at this resource.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"HealthcareService_Eligibility": {
  "description": "The details of a healthcare service available at a location.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Coded value for the eligibility.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "comment": {
      "description": "Describes the eligibility conditions for the service.",
      "$ref": "#/definitions/markdown"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"HealthcareService_AvailableTime": {
  "description": "The details of a healthcare service available at a location.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
```

```
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "daysOfWeek": {
      "description": "Indicates which days of the week are available between the start and end Times.",
      "items": {
        "enum": [
          "mon",
          "tue",
          "wed",
          "thu",
          "fri",
          "sat",
          "sun"
        ]
      },
      "type": "array"
    },
    "_daysOfWeek": {
      "description": "Extensions for daysOfWeek",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "allDay": {
      "description": "Is this always available? (hence times are irrelevant) e.g. 24 hour service.",
      "$ref": "#/definitions/boolean"
    },
    "_allDay": {
      "description": "Extensions for allDay",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "availableStartTime": {
      "description": "The opening time of day. Note: If the AllDay flag is set, then this time is ignored.",
      "$ref": "#/definitions/time"
    },
    "_availableStartTime": {
      "description": "Extensions for availableStartTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "availableEndTime": {
      "description": "The closing time of day. Note: If the AllDay flag is set, then this time is ignored.",
      "$ref": "#/definitions/time"
    },
    "_availableEndTime": {
      "description": "Extensions for availableEndTime",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"HealthcareService_NotAvailable": {
  "description": "The details of a healthcare service available at a location.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "The reason that can be presented to the user as to why this time is not available.",
      "$ref": "#/definitions/string"
```

```
        },
        "_description": {
          "description": "Extensions for description",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "during": {
          "description": "Service is not available (seasonally or for a public holiday) from this date.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false
    }
  }
}
```


8.8.12 Resource HealthcareService - Maturity Dependency View

URL: https://hl7.org/fhir/R4/healthcareservice-dependencies.html

8.8.12 Resource HealthcareService - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for HealthcareService FMM level 2

Name | Card. | Type | Dependency Analysis
HealthcareService | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
providedBy | 0..1 | Reference(Organization)
category | 0..* | CodeableConcept
type | 0..* | CodeableConcept
specialty | 0..* | CodeableConcept
location | 0..* | Reference(Location)
name | 0..1 | string
comment | 0..1 | string
extraDetails | 0..1 | markdown
photo | 0..1 | Attachment
telecom | 0..* | ContactPoint
coverageArea | 0..* | Reference(Location)
serviceProvisionCode | 0..* | CodeableConcept
eligibility | 0..* | BackboneElement
code | 0..1 | CodeableConcept
comment | 0..1 | markdown
program | 0..* | CodeableConcept
characteristic | 0..* | CodeableConcept
communication | 0..* | CodeableConcept
referralMethod | 0..* | CodeableConcept
appointmentRequired | 0..1 | boolean
availableTime | 0..* | BackboneElement
daysOfWeek | 0..* | code
allDay | 0..1 | boolean
availableStartTime | 0..1 | time
availableEndTime | 0..1 | time
notAvailable | 0..* | BackboneElement
description | 1..1 | string
during | 0..1 | Period
availabilityExceptions | 0..1 | string
endpoint | 0..* | Reference(Endpoint)

8.9 Resource Endpoint - Content

URL: https://hl7.org/fhir/R4/endpoint.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The technical details of an endpoint that can be used for electronic services, such as for web services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context information.

8.9.1 Scope and Usage

An endpoint describes the technical details of a location that can be connected to for the delivery/retrieval of information. Sufficient information is required to ensure that a connection can be made securely, and appropriate data transmitted as defined by the endpoint owner. This is not a description of details of the

current system, as found in CapabilityStatement, but of another (potentially external) system. These may be locally hosted services, regional services, or national service.

These resources are typically used to identify where to locate endpoint details for:

• Questionnaires: Where to send information (currently an SDC extension with just the address)

• ValueSet: Where related Terminology Services can be found (where not local)

• Subscription: The destination to send the subscribed data (or to pull)

• Messaging: (currently defined in the Message Header, but only as the address)

• Referrals: Where to send referral requests (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

• Referrals - Templates: Where to locate referral templates (Questionnaires) (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

• CarePlans: Where a shared CarePlan can be found

• Scheduling: Where to lookup to discover schedules/availability information (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

• Scheduling: Where to lookup to send appointment requests (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

• Patient/Person: Location of Master Patient/Person Indexes

• Service Provider Directories: Location of related directories (parent/child/federated)

• DICOM/imaging: Location of where to query for imaging content and metadata (QIDO-RS, WADO-RS, or WADO-URI)

8.9.2 Boundaries and Relationships

8.9.2.1 CapabilityStatement

The endpoint is distinct from a capability statement in that the CapabilityStatement statement describes the entire capability of a server (and in the metadata case, just this server) Where the endpoint resource describes the technical details for how to connect, and for what purposes (which could be a small sub-set of the server's capabilities, and might not be a FHIR endpoint).

8.9.2.2 ContactPoint

Endpoints are used to facilitate system to system communication, providing the details of the addressing of the system, purpose of use, protocols required/provided, and any other details required to communicate between the systems. (such as configuration parameters, require headers ...) The address value in the endpoint can only be used in the context of the provided details.

E.g. If the endpoint has a URL for an XDS service, using that in a browser by a user is meaningless.

E.g.2 If the endpoint has a US Direct address, using this to send a normal email message will not work, as they are not true email addresses (even though they appear to be)

A ContactPoint contains contact information, such as a phone number, web address that a user can directly interact with, no additional information would be required to use the value. The value is generally used to communicate with an entity directly, not a system.

8.9.3 Background and Context

8.9.3.1 Expected Implementations

• Any solution where there are distributed FHIR servers deployed and need discovery/configuration

• (Refer to the scope and usage section for common uses)

This resource is referenced by HealthcareService, ImagingStudy, InsurancePlan, Location, Organization, OrganizationAffiliation and PractitionerRole

8.9.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints

Endpoint | TU | | DomainResource | The technical details of an endpoint that can be used for electronic services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Identifies this endpoint across multiple systems

status | ?!Σ | 1..1 | code | active | suspended | error | off | entered-in-error | test EndpointStatus (Required)

connectionType | Σ | 1..1 | Coding | Protocol/Profile/Standard to be used with this endpoint connection Endpoint Connection Type (Extensible)

name | Σ | 0..1 | string | A name that this endpoint can be identified by

managingOrganization | Σ | 0..1 | Reference(Organization) | Organization that manages this endpoint (might not be the organization that exposes the endpoint)

contact | | 0..* | ContactPoint | Contact details for source (e.g. troubleshooting)

period | Σ | 0..1 | Period | Interval the endpoint is expected to be operational

payloadType | Σ | 1..* | CodeableConcept | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) Endpoint Payload Type (Example)

payloadMimeType | Σ | 0..* | code | Mimetype to send. If not specified, the content could be anything (including no payload, if the connectionType defined this) MimeType (Required)

address | Σ | 1..1 | url | The technical base address for connecting to this endpoint

header | | 0..* | string | Usage depends on the channel type

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Endpoint xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifies this endpoint across multiple systems --></identifier>
 <status value="[code]"/><!-- 1..1 active | suspended | error | off | entered-in-error | test -->
 <connectionType><!-- 1..1 Coding Protocol/Profile/Standard to be used with this endpoint connection
    --></connectionType>
 <name value="[string]"/><!-- 0..1 A name that this endpoint can be identified by -->
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that manages this endpoint (might not be the
    organization that exposes the endpoint) --></managingOrganization>
 <contact><!-- 0..* ContactPoint Contact details for source (e.g. troubleshooting) --></contact>
 <period><!-- 0..1 Period Interval the endpoint is expected to be operational --></period>
 <payloadType><!-- 1..* CodeableConcept The type of content that may be used at this endpoint (e.g. XDS Discharge
    summaries) --></payloadType>
 <payloadMimeType value="[code]"/><!-- 0..* Mimetype to send. If not specified, the content could be anything (including
     no payload, if the connectionType defined this) -->
 <address value="[url]"/><!-- 1..1 The technical base address for connecting to this endpoint -->
 <header value="[string]"/><!-- 0..* Usage depends on the channel type -->
</Endpoint>
```

JSON Template

```
{
  "resourceType" : "Endpoint",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifies this endpoint across multiple systems
  "status" : "<code>", // R!  active | suspended | error | off | entered-in-error | test
  "connectionType" : { Coding }, // R!  Protocol/Profile/Standard to be used with this endpoint connection
  "name" : "<string>", // A name that this endpoint can be identified by
  "managingOrganization" : { Reference(Organization) }, // Organization that manages this endpoint (might not be the
    organization that exposes the endpoint)
  "contact" : [{ ContactPoint }], // Contact details for source (e.g. troubleshooting)
  "period" : { Period }, // Interval the endpoint is expected to be operational
  "payloadType" : [{ CodeableConcept }], // R!  The type of content that may be used at this endpoint (e.g. XDS
    Discharge summaries)
  "payloadMimeType" : ["<code>"], // Mimetype to send. If not specified, the content could be anything (including no
    payload, if the connectionType defined this)
  "address" : "<url>", // R!  The technical base address for connecting to this endpoint
  "header" : ["<string>"] // Usage depends on the channel type
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Endpoint;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Endpoint.identifier [ Identifier ], ... ; # 0..* Identifies this endpoint across multiple systems
  fhir:Endpoint.status [ code ]; # 1..1 active | suspended | error | off | entered-in-error | test
  fhir:Endpoint.connectionType [ Coding ]; # 1..1 Protocol/Profile/Standard to be used with this endpoint connection
  fhir:Endpoint.name [ string ]; # 0..1 A name that this endpoint can be identified by
  fhir:Endpoint.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that manages this endpoint (might
    not be the organization that exposes the endpoint)
  fhir:Endpoint.contact [ ContactPoint ], ... ; # 0..* Contact details for source (e.g. troubleshooting)
  fhir:Endpoint.period [ Period ]; # 0..1 Interval the endpoint is expected to be operational
  fhir:Endpoint.payloadType [ CodeableConcept ], ... ; # 1..* The type of content that may be used at this endpoint
    (e.g. XDS Discharge summaries)
  fhir:Endpoint.payloadMimeType [ code ], ... ; # 0..* Mimetype to send. If not specified, the content could be anything
     (including no payload, if the connectionType defined this)
  fhir:Endpoint.address [ url ]; # 1..1 The technical base address for connecting to this endpoint
```

```
  fhir:Endpoint.header [ string ], ... ; # 0..* Usage depends on the channel type
]
```

Changes since R3

Endpoint

Endpoint.status | Change value set from http://hl7.org/fhir/ValueSet/endpoint-status to http://hl7.org/fhir/ValueSet/endpoint-status|4.0.1

Endpoint.connectionType | Change code system for extensibly bound codes from "http://hl7.org/fhir/endpoint-connection-type" to "http://terminology.hl7.org/CodeSystem/endpoint-connection-type"

Endpoint.payloadMimeType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

Endpoint.address | Type changed from uri to url

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Endpoint | TU | | DomainResource | The technical details of an endpoint that can be used for electronic services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Identifies this endpoint across multiple systems

status | ?!Σ | 1..1 | code | active | suspended | error | off | entered-in-error | test EndpointStatus (Required)

connectionType | Σ | 1..1 | Coding | Protocol/Profile/Standard to be used with this endpoint connection Endpoint Connection Type (Extensible)

name | Σ | 0..1 | string | A name that this endpoint can be identified by

managingOrganization | Σ | 0..1 | Reference(Organization) | Organization that manages this endpoint (might not be the organization that exposes the endpoint)

contact | | 0..* | ContactPoint | Contact details for source (e.g. troubleshooting)

period | Σ | 0..1 | Period | Interval the endpoint is expected to be operational

payloadType | Σ | 1..* | CodeableConcept | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) Endpoint Payload Type (Example)

payloadMimeType | Σ | 0..* | code | Mimetype to send. If not specified, the content could be anything (including no payload, if the connectionType defined this) MimeType (Required)

address | Σ | 1..1 | url | The technical base address for connecting to this endpoint

header | | 0..* | string | Usage depends on the channel type

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Endpoint xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifies this endpoint across multiple systems --></identifier>
 <status value="[code]"/><!-- 1..1 active | suspended | error | off | entered-in-error | test -->
 <connectionType><!-- 1..1 Coding Protocol/Profile/Standard to be used with this endpoint connection
    --></connectionType>
 <name value="[string]"/><!-- 0..1 A name that this endpoint can be identified by -->
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that manages this endpoint (might not be the
    organization that exposes the endpoint) --></managingOrganization>
 <contact><!-- 0..* ContactPoint Contact details for source (e.g. troubleshooting) --></contact>
 <period><!-- 0..1 Period Interval the endpoint is expected to be operational --></period>
 <payloadType><!-- 1..* CodeableConcept The type of content that may be used at this endpoint (e.g. XDS Discharge
    summaries) --></payloadType>
 <payloadMimeType value="[code]"/><!-- 0..* Mimetype to send. If not specified, the content could be anything (including
     no payload, if the connectionType defined this) -->
 <address value="[url]"/><!-- 1..1 The technical base address for connecting to this endpoint -->
 <header value="[string]"/><!-- 0..* Usage depends on the channel type -->
</Endpoint>
```

JSON Template

```
{
  "resourceType" : "Endpoint",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifies this endpoint across multiple systems
  "status" : "<code>", // R!  active | suspended | error | off | entered-in-error | test
  "connectionType" : { Coding }, // R!  Protocol/Profile/Standard to be used with this endpoint connection
  "name" : "<string>", // A name that this endpoint can be identified by
  "managingOrganization" : { Reference(Organization) }, // Organization that manages this endpoint (might not be the
    organization that exposes the endpoint)
  "contact" : [{ ContactPoint }], // Contact details for source (e.g. troubleshooting)
  "period" : { Period }, // Interval the endpoint is expected to be operational
  "payloadType" : [{ CodeableConcept }], // R!  The type of content that may be used at this endpoint (e.g. XDS
    Discharge summaries)
  "payloadMimeType" : ["<code>"], // Mimetype to send. If not specified, the content could be anything (including no
    payload, if the connectionType defined this)
  "address" : "<url>", // R!  The technical base address for connecting to this endpoint
  "header" : ["<string>"] // Usage depends on the channel type
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Endpoint;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Endpoint.identifier [ Identifier ], ... ; # 0..* Identifies this endpoint across multiple systems
  fhir:Endpoint.status [ code ]; # 1..1 active | suspended | error | off | entered-in-error | test
  fhir:Endpoint.connectionType [ Coding ]; # 1..1 Protocol/Profile/Standard to be used with this endpoint connection
  fhir:Endpoint.name [ string ]; # 0..1 A name that this endpoint can be identified by
  fhir:Endpoint.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that manages this endpoint (might
    not be the organization that exposes the endpoint)
  fhir:Endpoint.contact [ ContactPoint ], ... ; # 0..* Contact details for source (e.g. troubleshooting)
  fhir:Endpoint.period [ Period ]; # 0..1 Interval the endpoint is expected to be operational
  fhir:Endpoint.payloadType [ CodeableConcept ], ... ; # 1..* The type of content that may be used at this endpoint
    (e.g. XDS Discharge summaries)
  fhir:Endpoint.payloadMimeType [ code ], ... ; # 0..* Mimetype to send. If not specified, the content could be anything
     (including no payload, if the connectionType defined this)
  fhir:Endpoint.address [ url ]; # 1..1 The technical base address for connecting to this endpoint
  fhir:Endpoint.header [ string ], ... ; # 0..* Usage depends on the channel type
]
```

Changes since Release 3

Endpoint

Endpoint.status | Change value set from http://hl7.org/fhir/ValueSet/endpoint-status to http://hl7.org/fhir/ValueSet/endpoint-status|4.0.1

Endpoint.connectionType | Change code system for extensibly bound codes from "http://hl7.org/fhir/endpoint-connection-type" to "http://terminology.hl7.org/CodeSystem/endpoint-connection-type"

Endpoint.payloadMimeType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

Endpoint.address | Type changed from uri to url

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.9.4.1 Terminology Bindings

Path | Definition | Type | Reference

Endpoint.status | The status of the endpoint. | Required | EndpointStatus

Endpoint.connectionType | | Extensible | EndpointConnectionType

Endpoint.payloadType | | Example | EndpointPayloadType

Endpoint.payloadMimeType | The mime type of an attachment. Any valid mime type is allowed. | Required | Mime Types

8.9.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

connection-type | token | Protocol/Profile/Standard to be used with this endpoint connection | Endpoint.connectionType

identifier | token | Identifies this endpoint across multiple systems | Endpoint.identifier

name | string | A name that this endpoint can be identified by | Endpoint.name

organization | reference | The organization that is managing the endpoint | Endpoint.managingOrganization (Organization)

payload-type | token | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) | Endpoint.payloadType

status | token | The current status of the Endpoint (usually expected to be active) | Endpoint.status

8.9.6 Resource Endpoint - Examples

URL: https://hl7.org/fhir/R4/endpoint-examples.html

8.9.6 Resource Endpoint - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example Endpoint | example | XML | JSON | Turtle

Collection of endpoint examples | b0a5e4277-83c4-4adb-87e2-e3efe3369b6f | XML | JSON | Turtle

Example of IHE-IID endpoint | example-iid | XML | JSON | Turtle

Example of DICOM WADO-RS endpoint | example-wadors | XML | JSON | Turtle

Example of a DIRECT endpoint | direct-endpoint | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.9.7 Resource Endpoint - Detailed Descriptions

URL: https://hl7.org/fhir/R4/endpoint-definitions.html

8.9.7 Resource Endpoint - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Endpoint resource.

Endpoint
Element Id | Endpoint
Definition | The technical details of an endpoint that can be used for electronic services, such as for web services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context information.
Cardinality | 0..*
Type | DomainResource
Endpoint.identifier
Element Id | Endpoint.identifier
Definition | Identifier for the organization that is used to identify the endpoint across multiple disparate systems.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Endpoint.status
Element Id | Endpoint.status
Definition | active | suspended | error | off | test.
Cardinality | 1..1
Terminology Binding | EndpointStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the endpoint as not currently valid.
Endpoint.connectionType
Element Id | Endpoint.connectionType
Definition | A coded value that represents the technical details of the usage of this endpoint, such as what WSDLs should be used in what way. (e.g. XDS.b/DICOM/cds-hook).
Cardinality | 1..1
Terminology Binding | Endpoint Connection Type (Extensible)
Type | Coding
Summary | true
Comments | For additional connectivity details for the protocol, extensions will be used at this point, as in the XDS example.
Endpoint.name
Element Id | Endpoint.name
Definition | A friendly name that this endpoint can be referred to with.
Cardinality | 0..1
Type | string
Summary | true
Endpoint.managingOrganization
Element Id | Endpoint.managingOrganization
Definition | The organization that manages this endpoint (even if technically another organization is hosting this in the cloud, it is the organization associated with the data).
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
Comments | This property is not typically used when searching for Endpoint resources for usage. The typical usage is via the reference from an applicable Organization/Location/Practitioner resource, which is where the context is provided. Multiple Locations may reference a single endpoint, and don't have to be within the same organization resource, but most likely within the same organizational hierarchy.
Endpoint.contact
Element Id | Endpoint.contact
Definition | Contact details for a human to contact about the subscription. The primary use of this for system administrator troubleshooting.
Cardinality | 0..*
Type | ContactPoint
Endpoint.period
Element Id | Endpoint.period
Definition | The interval during which the endpoint is expected to be operational.
Cardinality | 0..1
Type | Period

Summary | true
Endpoint.payloadType
Element Id | Endpoint.payloadType
Definition | The payload type describes the acceptable content that can be communicated on the endpoint.
Cardinality | 1..*
Terminology Binding | Endpoint Payload Type (Example)
Type | CodeableConcept
Summary | true
Comments | The payloadFormat describes the serialization format of the data, where the payloadType indicates the specific document/schema that is being transferred; e.g. DischargeSummary or CarePlan.
Endpoint.payloadMimeType
Element Id | Endpoint.payloadMimeType
Definition | The mime type to send the payload in - e.g. application/fhir+xml, application/fhir+json. If the mime type is not specified, then the sender could send any content (including no content depending on the connectionType).
Cardinality | 0..*
Terminology Binding | MimeType (Required)
Type | code
Summary | true
Comments | Sending the payload has obvious security consequences. The server is responsible for ensuring that the content is appropriately secured.
Endpoint.address
Element Id | Endpoint.address
Definition | The uri that describes the actual end-point to connect to.
Cardinality | 1..1
Type | url
Summary | true
Comments | For rest-hook, and websocket, the end-point must be an http: or https: URL; for email, a mailto: url, for sms, a tel: url, and for message the endpoint can be in any form of url the server understands (usually, http: or mllp:). The URI is allowed to be relative; in which case, it is relative to the server end-point (since there may be more than one, clients should avoid using relative URIs) This address will be to the service base, without any parameters, or sub-services or resources tacked on. E.g. for a WADO-RS endpoint, the url should be "https://pacs.hospital.org/wado-rs" and not "https://pacs.hospital.org/wado-rs/studies/1.2.250.1.59.40211.12345678.678910/series/1.2.250.1.59.40211.789001276.14556172.67789/instances/...".
Endpoint.header
Element Id | Endpoint.header
Definition | Additional headers / information to send as part of the notification.
Cardinality | 0..*
Type | string
Summary | false
Comments | Exactly what these mean depends on the channel type. The can convey additional information to the recipient and/or meet security requirements.

8.9.8 Resource Endpoint - Mappings

URL: https://hl7.org/fhir/R4/endpoint-mappings.html

8.9.8 Resource Endpoint - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the endpoint resource (see Mappings to Other Standards for further information & status).

8.9.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Endpoint
identifier | FiveWs.identifier
status | FiveWs.status
connectionType | FiveWs.class
name | FiveWs.what[x]
period | FiveWs.done[x]

8.9.8.2 RIM Mapping (http://hl7.org/v3)

Endpoint | n/a

8.9.9 Resource Endpoint - Extensions & Profiles

URL: https://hl7.org/fhir/R4/endpoint-profiles.html

8.9.9 Resource Endpoint - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Endpoint resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.9.9.1 Profiles

No Profiles defined for this resource

8.9.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.9.9.3 Search Extensions

No Search Extensions defined for this resource

8.9.10 Resource Endpoint - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/endpoint-version-maps.html

No validation errors - all conversions are clean

8.9.10 Resource Endpoint - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Endpoint.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.9.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Endpoint3to4" = "R3 to R4 Conversions for Endpoint"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Endpoint" alias EndpointR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Endpoint" alias Endpoint as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Endpoint(source src : EndpointR3, target tgt : Endpoint) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.connectionType -> tgt.connectionType;
  src.name -> tgt.name;
  src.managingOrganization -> tgt.managingOrganization;
  src.contact -> tgt.contact;
  src.period -> tgt.period;
  src.payloadType -> tgt.payloadType;
  src.payloadMimeType -> tgt.payloadMimeType;
  src.address -> tgt.address;
  src.header -> tgt.header;
}
```

8.9.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Endpoint4to3" = "R4 to R3 Conversion for Endpoint"

uses "http://hl7.org/fhir/StructureDefinition/Endpoint" alias Endpoint as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Endpoint" alias EndpointR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Endpoint(source src : Endpoint, target tgt : EndpointR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.connectionType -> tgt.connectionType;
  src.name -> tgt.name;
```

```
  src.managingOrganization -> tgt.managingOrganization;
  src.contact -> tgt.contact;
  src.period -> tgt.period;
  src.payloadType -> tgt.payloadType;
  src.payloadMimeType -> tgt.payloadMimeType;
  src.address -> tgt.address;
  src.header -> tgt.header;
}
```

8.9.10.3 R4 Validation Errors

8.9.11 JSON Schema for Endpoint

URL: https://hl7.org/fhir/R4/endpoint.schema.json.html

8.9.11 JSON Schema for Endpoint

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Endpoint, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Endpoint",
  "$ref": "#/definitions/Endpoint",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Endpoint": {
      "description": "The technical details of an endpoint that can be used for electronic services, such as for web
    services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context
    information.",
      "properties": {
        "resourceType": {
          "description": "This is a Endpoint resource",
          "const": "Endpoint"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
```

```
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for the organization that is used to identify the endpoint across multiple
disparate systems.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "active | suspended | error | off | test.",
      "enum": [
        "active",
        "suspended",
        "error",
        "off",
        "entered-in-error",
        "test"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "connectionType": {
      "description": "A coded value that represents the technical details of the usage of this endpoint, such as
what WSDLs should be used in what way. (e.g. XDS.b/DICOM/cds-hook).",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "name": {
      "description": "A friendly name that this endpoint can be referred to with.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "managingOrganization": {
      "description": "The organization that manages this endpoint (even if technically another organization is
hosting this in the cloud, it is the organization associated with the data).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "contact": {
      "description": "Contact details for a human to contact about the subscription. The primary use of this for
system administrator troubleshooting.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "period": {
      "description": "The interval during which the endpoint is expected to be operational.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "payloadType": {
      "description": "The payload type describes the acceptable content that can be communicated on the endpoint.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "payloadMimeType": {
      "description": "The mime type to send the payload in - e.g. application/fhir+xml, application/fhir+json. If
the mime type is not specified, then the sender could send any content (including no content depending on the
connectionType).",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_payloadMimeType": {
      "description": "Extensions for payloadMimeType",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
```

```
          },
          "type": "array"
        },
        "address": {
          "description": "The uri that describes the actual end-point to connect to.",
          "$ref": "#/definitions/url"
        },
        "_address": {
          "description": "Extensions for address",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "header": {
          "description": "Additional headers / information to send as part of the notification.",
          "items": {
            "$ref": "#/definitions/string"
          },
          "type": "array"
        },
        "_header": {
          "description": "Extensions for header",
          "items": {
            "$ref": "Element.schema.json#/definitions/Element"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "payloadType",
        "connectionType",
        "resourceType"
      ]
    }
  }
}
```

8.9.12 Resource Endpoint - Maturity Dependency View

URL: https://hl7.org/fhir/R4/endpoint-dependencies.html

8.9.12 Resource Endpoint - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Endpoint FMM level 2

Name | Card. | Type | Dependency Analysis
Endpoint | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
connectionType | 1..1 | Coding
name | 0..1 | string
managingOrganization | 0..1 | Reference(Organization)
contact | 0..* | ContactPoint
period | 0..1 | Period
payloadType | 1..* | CodeableConcept
payloadMimeType | 0..* | code
address | 1..1 | url
header | 0..* | string

8.9.13 Value Set Sc-valueset-endpoint-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-endpoint-status.html

8.9.13 Value Set Sc-valueset-endpoint-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.9.14 EndpointStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-endpoint-status)

Mapping from http://hl7.org/fhir/ValueSet/endpoint-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the endpoint."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error

test | equivalent | draft
active | equivalent | active
suspended | equivalent | suspended
error | equivalent | failed
off | equivalent | inactive

8.10 Resource EpisodeOfCare - Content

URL: https://hl7.org/fhir/R4/episodeofcare.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

An association between a patient and an organization / healthcare provider(s) during which time encounters may occur. The managing organization assumes a level of responsibility for the patient during this time.

8.10.1 Scope and Usage

The EpisodeOfCare Resource contains information about an association of a Patient with a Healthcare Provider for a period of time under which related healthcare activities may occur.

In many cases, this represents a period of time where the Healthcare Provider has some level of responsibility for the care of the patient regarding a specific condition or problem, even if not currently participating in an encounter.

These resources are typically known in existing systems as:

• EpisodeOfCare: Case, Program, Problem, Episode

• Encounter: Visit, Contact

8.10.1.1 Multiple Organizations and Transfer of Care

Many organizations can be involved in an EpisodeOfCare; however each organization will have its own EpisodeOfCare resource instance that tracks its responsibility with the patient.

When an Organization completes their involvement with the patient and transfers care to another Organization. This is often in the form of a referral to another Organization (or Organizations).

When an incoming referral is received a new EpisodeOfCare may be created for this organization. The initial step(s) in the intake workflow for the referral often involve some form of assessment(s), eligibility, capacity, care levels, which could take some time. Once the intake process is completed and the patient is accepted, a CarePlan is often created.

8.10.2 Boundaries and Relationships

The primary difference between the EpisodeOfCare and the Encounter is that the Encounter records the details of an activity directly relating to the patient, while the EpisodeOfCare is the container that can link a series of Encounters together for problems/issues. The Example scenarios below give some good examples as to when you might want to be using an EpisodeOfCare.

This difference is a similar difference between the EpisodeOfCare and a CarePlan. The EpisodeOfCare is a tracking resource, rather than a planning resource. The EpisodeOfCare usually exists before the CarePlan. You don't need a CarePlan to use an EpisodeOfCare.

8.10.3 Background and Context

Systems collect a coherent group of activities (such as encounters) related to a patient's health condition or problem often referred to as a Care Episode. Information about an episode is often shared across systems, and in some cases organizational and disciplinary boundaries. An EpisodeOfCare contains details about the purpose of the care and can exist without any activities. The minimal information that would be required in an episode of care would be a patient, organization and a reason for the ongoing association. Other reasons for creating an EpisodeOfCare could be for tracking the details required for government reporting or billing.

8.10.3.1 Expected Implementations

• Chronic Disease Management Systems

• Community Care Systems Tracking progress of a specific condition Tracking government funding

• Problem based General Practice systems

• Disability Support Systems

• Aged Care Systems (Community and Residential)

This resource is referenced by ChargeItem, Contract, DocumentReference, Encounter, MedicationAdministration, MedicationDispense and MedicationStatement

8.10.4 Resource Content

• Structure

• UML

- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
EpisodeOfCare | TU | | DomainResource | An association of a Patient with an Organization and Healthcare Provider(s) for a period of time that the Organization assumes some level of responsibility Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier(s) relevant for this EpisodeOfCare
status | ?!Σ | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)
statusHistory | | 0..* | BackboneElement | Past list of status codes (the current status may be included to cover the start date of the status)
status | | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)
period | | 1..1 | Period | Duration the EpisodeOfCare was in the specified status
type | Σ | 0..* | CodeableConcept | Type/class - e.g. specialist referral, disease management Episode of care type (Example)
diagnosis | Σ | 0..* | BackboneElement | The list of diagnosis relevant to this episode of care
condition | Σ | 1..1 | Reference(Condition) | Conditions/problems/diagnoses this episode of care is for
role | Σ | 0..1 | CodeableConcept | Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge ...) DiagnosisRole (Preferred)
rank | Σ | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
patient | Σ | 1..1 | Reference(Patient) | The patient who is the focus of this episode of care
managingOrganization | Σ | 0..1 | Reference(Organization) | Organization that assumes care
period | Σ | 0..1 | Period | Interval during responsibility is assumed
referralRequest | | 0..* | Reference(ServiceRequest) | Originating Referral Request(s)
careManager | | 0..1 | Reference(Practitioner | PractitionerRole) | Care manager/care coordinator for the patient
team | | 0..* | Reference(CareTeam) | Other practitioners facilitating this episode of care
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this EpisodeOfCare
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EpisodeOfCare xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier(s) relevant for this EpisodeOfCare --></identifier>
 <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
 <statusHistory>  <!-- 0..* Past list of status codes (the current status may be included to cover the start date of the
     status) -->
  <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
  <period><!-- 1..1 Period Duration the EpisodeOfCare was in the specified status --></period>
 </statusHistory>
 <type><!-- 0..* CodeableConcept Type/class  - e.g. specialist referral, disease management --></type>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this episode of care -->
  <condition><!-- 1..1 Reference(Condition) Conditions/problems/diagnoses this episode of care is for --></condition>
  <role><!-- 0..1 CodeableConcept Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge â€¦) --></role>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <patient><!-- 1..1 Reference(Patient) The patient who is the focus of this episode of care --></patient>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that assumes care --></managingOrganization>
 <period><!-- 0..1 Period Interval during responsibility is assumed --></period>
 <referralRequest><!-- 0..* Reference(ServiceRequest) Originating Referral Request(s) --></referralRequest>
 <careManager><!-- 0..1 Reference(Practitioner|PractitionerRole) Care manager/care coordinator for the patient
    --></careManager>
 <team><!-- 0..* Reference(CareTeam) Other practitioners facilitating this episode of care --></team>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this EpisodeOfCare
    --></account>
</EpisodeOfCare>
```

JSON Template

```
{
  "resourceType" : "EpisodeOfCare",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier(s) relevant for this EpisodeOfCare
  "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
  "statusHistory" : [{ // Past list of status codes (the current status may be included to cover the start date of the
    status)
    "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
    "period" : { Period } // R!  Duration the EpisodeOfCare was in the specified status
  }],
  "type" : [{ CodeableConcept }], // Type/class  - e.g. specialist referral, disease management
  "diagnosis" : [{ // The list of diagnosis relevant to this episode of care
    "condition" : { Reference(Condition) }, // R!  Conditions/problems/diagnoses this episode of care is for
    "role" : { CodeableConcept }, // Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge â€¦)
```

```
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "patient" : { Reference(Patient) }, // R!  The patient who is the focus of this episode of care
  "managingOrganization" : { Reference(Organization) }, // Organization that assumes care
  "period" : { Period }, // Interval during responsibility is assumed
  "referralRequest" : [{ Reference(ServiceRequest) }], // Originating Referral Request(s)
  "careManager" : { Reference(Practitioner|PractitionerRole) }, // Care manager/care coordinator for the patient
  "team" : [{ Reference(CareTeam) }], // Other practitioners facilitating this episode of care
  "account" : [{ Reference(Account) }] // The set of accounts that may be used for billing for this EpisodeOfCare
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EpisodeOfCare;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EpisodeOfCare.identifier [ Identifier ], ... ; # 0..* Business Identifier(s) relevant for this EpisodeOfCare
  fhir:EpisodeOfCare.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-
    error
  fhir:EpisodeOfCare.statusHistory [ # 0..* Past list of status codes (the current status may be included to cover the
    start date of the status)
    fhir:EpisodeOfCare.statusHistory.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled
     | entered-in-error
    fhir:EpisodeOfCare.statusHistory.period [ Period ]; # 1..1 Duration the EpisodeOfCare was in the specified status
  ], ...;
  fhir:EpisodeOfCare.type [ CodeableConcept ], ... ; # 0..* Type/class  - e.g. specialist referral, disease management
  fhir:EpisodeOfCare.diagnosis [ # 0..* The list of diagnosis relevant to this episode of care
    fhir:EpisodeOfCare.diagnosis.condition [ Reference(Condition) ]; # 1..1 Conditions/problems/diagnoses this episode
    of care is for
    fhir:EpisodeOfCare.diagnosis.role [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the episode of
    care (e.g. admission, billing, discharge â€¦)
    fhir:EpisodeOfCare.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:EpisodeOfCare.patient [ Reference(Patient) ]; # 1..1 The patient who is the focus of this episode of care
  fhir:EpisodeOfCare.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that assumes care
  fhir:EpisodeOfCare.period [ Period ]; # 0..1 Interval during responsibility is assumed
  fhir:EpisodeOfCare.referralRequest [ Reference(ServiceRequest) ], ... ; # 0..* Originating Referral Request(s)
  fhir:EpisodeOfCare.careManager [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Care manager/care coordinator for
    the patient
  fhir:EpisodeOfCare.team [ Reference(CareTeam) ], ... ; # 0..* Other practitioners facilitating this episode of care
  fhir:EpisodeOfCare.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for
    this EpisodeOfCare
]
```

Changes since R3

EpisodeOfCare

EpisodeOfCare.status | Change value set from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/episode-of-care-status|4.0.1

EpisodeOfCare.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/episode-of-care-status|4.0.1

EpisodeOfCare.referralRequest | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest

EpisodeOfCare.careManager | Type Reference: Added Target Type PractitionerRole

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

EpisodeOfCare | TU | | DomainResource | An association of a Patient with an Organization and Healthcare Provider(s) for a period of time that the Organization assumes some level of responsibility Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Business Identifier(s) relevant for this EpisodeOfCare

status | ?!Σ | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)

statusHistory | | 0..* | BackboneElement | Past list of status codes (the current status may be included to cover the start date of the status)

status | | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)

period | | 1..1 | Period | Duration the EpisodeOfCare was in the specified status

type | Σ | 0..* | CodeableConcept | Type/class - e.g. specialist referral, disease management Episode of care type (Example)

diagnosis | Σ | 0..* | BackboneElement | The list of diagnosis relevant to this episode of care

condition | Σ | 1..1 | Reference(Condition) | Conditions/problems/diagnoses this episode of care is for

role | Σ | 0..1 | CodeableConcept | Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)

rank | Σ | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)

patient | Σ | 1..1 | Reference(Patient) | The patient who is the focus of this episode of care

managingOrganization | Σ | 0..1 | Reference(Organization) | Organization that assumes care
period | Σ | 0..1 | Period | Interval during responsibility is assumed
referralRequest | | 0..* | Reference(ServiceRequest) | Originating Referral Request(s)
careManager | | 0..1 | Reference(Practitioner | PractitionerRole) | Care manager/care coordinator for the patient
team | | 0..* | Reference(CareTeam) | Other practitioners facilitating this episode of care
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this EpisodeOfCare
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EpisodeOfCare xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier(s) relevant for this EpisodeOfCare --></identifier>
 <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
 <statusHistory>  <!-- 0..* Past list of status codes (the current status may be included to cover the start date of the
     status) -->
  <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
  <period><!-- 1..1 Period Duration the EpisodeOfCare was in the specified status --></period>
 </statusHistory>
 <type><!-- 0..* CodeableConcept Type/class  - e.g. specialist referral, disease management --></type>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this episode of care -->
  <condition><!-- 1..1 Reference(Condition) Conditions/problems/diagnoses this episode of care is for --></condition>
  <role><!-- 0..1 CodeableConcept Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge â€¦) --></role>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <patient><!-- 1..1 Reference(Patient) The patient who is the focus of this episode of care --></patient>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that assumes care --></managingOrganization>
 <period><!-- 0..1 Period Interval during responsibility is assumed --></period>
 <referralRequest><!-- 0..* Reference(ServiceRequest) Originating Referral Request(s) --></referralRequest>
 <careManager><!-- 0..1 Reference(Practitioner|PractitionerRole) Care manager/care coordinator for the patient
    --></careManager>
 <team><!-- 0..* Reference(CareTeam) Other practitioners facilitating this episode of care --></team>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this EpisodeOfCare
    --></account>
</EpisodeOfCare>
```

JSON Template

```
{
  "resourceType" : "EpisodeOfCare",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier(s) relevant for this EpisodeOfCare
  "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
  "statusHistory" : [{ // Past list of status codes (the current status may be included to cover the start date of the
    status)
    "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
    "period" : { Period } // R!  Duration the EpisodeOfCare was in the specified status
  }],
  "type" : [{ CodeableConcept }], // Type/class  - e.g. specialist referral, disease management
  "diagnosis" : [{ // The list of diagnosis relevant to this episode of care
    "condition" : { Reference(Condition) }, // R!  Conditions/problems/diagnoses this episode of care is for
    "role" : { CodeableConcept }, // Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge â€¦)
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "patient" : { Reference(Patient) }, // R!  The patient who is the focus of this episode of care
  "managingOrganization" : { Reference(Organization) }, // Organization that assumes care
  "period" : { Period }, // Interval during responsibility is assumed
  "referralRequest" : [{ Reference(ServiceRequest) }], // Originating Referral Request(s)
  "careManager" : { Reference(Practitioner|PractitionerRole) }, // Care manager/care coordinator for the patient
  "team" : [{ Reference(CareTeam) }], // Other practitioners facilitating this episode of care
  "account" : [{ Reference(Account) }] // The set of accounts that may be used for billing for this EpisodeOfCare
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EpisodeOfCare;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EpisodeOfCare.identifier [ Identifier ], ... ; # 0..* Business Identifier(s) relevant for this EpisodeOfCare
  fhir:EpisodeOfCare.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-
    error
  fhir:EpisodeOfCare.statusHistory [ # 0..* Past list of status codes (the current status may be included to cover the
    start date of the status)
    fhir:EpisodeOfCare.statusHistory.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled
     | entered-in-error
    fhir:EpisodeOfCare.statusHistory.period [ Period ]; # 1..1 Duration the EpisodeOfCare was in the specified status
  ], ...;
  fhir:EpisodeOfCare.type [ CodeableConcept ], ... ; # 0..* Type/class  - e.g. specialist referral, disease management
  fhir:EpisodeOfCare.diagnosis [ # 0..* The list of diagnosis relevant to this episode of care
```

```
    fhir:EpisodeOfCare.diagnosis.condition [ Reference(Condition) ]; # 1..1 Conditions/problems/diagnoses this episode
    of care is for
    fhir:EpisodeOfCare.diagnosis.role [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the episode of
    care (e.g. admission, billing, discharge â€¦)
    fhir:EpisodeOfCare.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:EpisodeOfCare.patient [ Reference(Patient) ]; # 1..1 The patient who is the focus of this episode of care
  fhir:EpisodeOfCare.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that assumes care
  fhir:EpisodeOfCare.period [ Period ]; # 0..1 Interval during responsibility is assumed
  fhir:EpisodeOfCare.referralRequest [ Reference(ServiceRequest) ], ... ; # 0..* Originating Referral Request(s)
  fhir:EpisodeOfCare.careManager [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Care manager/care coordinator for
    the patient
  fhir:EpisodeOfCare.team [ Reference(CareTeam) ], ... ; # 0..* Other practitioners facilitating this episode of care
  fhir:EpisodeOfCare.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for
    this EpisodeOfCare
]
```

Changes since Release 3

EpisodeOfCare

EpisodeOfCare.status | Change value set from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/episode-of-care-status|4.0.1

EpisodeOfCare.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/episode-of-care-status|4.0.1

EpisodeOfCare.referralRequest | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest

EpisodeOfCare.careManager | Type Reference: Added Target Type PractitionerRole

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.10.4.1 Terminology Bindings

Path | Definition | Type | Reference

EpisodeOfCare.status EpisodeOfCare.statusHistory.status | The status of the episode of care. | Required | EpisodeOfCareStatus

EpisodeOfCare.type | The type of the episode of care. | Example | EpisodeOfCareType

EpisodeOfCare.diagnosis.role | The type of diagnosis this condition represents. | Preferred | DiagnosisRole

8.10.5 Status Management

8.10.5.1 History and Period

When an organization assumes responsibility for a patient, then the EpisodeOfCare is created and a start date entered to show when it has begun. As the organization's responsibility changes, so does the status of the EpisodeOfCare. This is described via an example below for an intake workflow.

8.10.5.2 Leave Handling

With long term care there is often a concept of the provision of care being suspended for various reasons. Many systems have extensive Leave Management/Tracking solutions which consider the complexities of this space, however this EpisodeOfCare resource is NOT intended to provide this level of tracking. Extension(s) may be used on the status/status history to track the on-hold reason, which can facilitate the processing.

A more complete Leave Management solution may have to deal with:

• Leave Types

• Leave Entitlements

• Billing/Funding implications while on different types of leave

8.10.5.3 Example Intake Workflow

This example sequence demonstrates some status transitions and how other resources interact. The context could be in a Community/Aged Care/Disability/Mental Health setting.

• ServiceRequest received

• intake clerk processes referral and decides that the first level eligibility has been met (e.g. Have capacity in the facility for the patient, the patient is covered by VA)

• EpisodeOfCare created with status of planned which is allocated as fulfilling the ServiceRequest

• Further assessment of needs is scheduled to be taken, a care manager is probably allocated at this point

• Assessment Practitioner sees the Patient and completes a series of relevant Questionnaires to rank the patient

• The assessments are reviewed and a formal CarePlan is created

• The EpisodeOfCare is updated to be marked as active, and the CareTeam is likely filled in

• The provision of care is then managed through the care plan, with all activities will also being linked to the EpisodeOfCare

• The patient is admitted to hospital for some procedures, and the EpisodeOfCare is marked as on hold Some of the services on the CarePlan (or scheduled appointments) would be reviewed to determine if they can be performed without the patient (e.g., home maintenance), or if they should be suspended while the patient is on hold.

• The patient returns from the hospital and the EpisodeOfCare is marked as active again (and services reviewed again)

• Patient wished to move to another area to be closer to family

• Organization creates an outgoing ServiceRequest to a new Organization to continue the care

• The EpisodeOfCare is closed

In some jurisdictions an Organization may be funded by a government body for the days that a patient is under their care. These are known as "active days". This does not mean that they are actively receiving a service (an encounter), but that the organization is responsible for managing their care. This monthly reporting value can be easily extracted from the status history as described above. The actual provision of services may also be funded separately, and this would be via the Encounters.

8.10.5.4 EpisodeOfCare Outcomes Review

An Organization may perform analytics on their EpisodeOfCare resources to have an understanding of how their business is performing. Observing that there was a 60/40 split of episodes being finished/cancelled is not very informative. The organization would prefer to know the reason why the episodes are completing so that they can plan their business effectively. They would be more interested in knowing whether it was due to services hitting their mandatory end date, client passing away, client transitioning to a higher level of services provided by them or to another provider etc.

Currently there are no attributes on this resource to provide this information. This would be very specific to each implementation and usage, so it would be recommended to use extensions to achieve this functionality.

8.10.6 Example Scenarios

A General Practitioner wants to review how well his patient is managing his diabetes over time from information within his clinic and also the regional community care organization's system(s).

The EpisodeOfCare enables the practitioner to easily separate the diabetes activities from the mental health problem's activities.

A Community Care organization wants to track all activities that occur with a patient relating to their disability to simplify the reporting to the government to receive funding to care for the patient

8.10.7 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
care-manager | reference | Care manager/care coordinator for the patient | EpisodeOfCare.careManager.where(resolve() is Practitioner) (Practitioner)
condition | reference | Conditions/problems/diagnoses this episode of care is for | EpisodeOfCare.diagnosis.condition (Condition)
date | date | The provided date search value falls within the episode of care's period | EpisodeOfCare.period | 17 Resources
identifier | token | Business Identifier(s) relevant for this EpisodeOfCare | EpisodeOfCare.identifier | 30 Resources
incoming-referral | reference | Incoming Referral Request | EpisodeOfCare.referralRequest (ServiceRequest)
organization | reference | The organization that has assumed the specific responsibilities of this EpisodeOfCare | EpisodeOfCare.managingOrganization (Organization)
patient | reference | The patient who is the focus of this episode of care | EpisodeOfCare.patient (Patient) | 33 Resources
status | token | The current status of the Episode of Care as provided (does not check the status history collection) | EpisodeOfCare.status
type | token | Type/class - e.g. specialist referral, disease management | EpisodeOfCare.type | 5 Resources

8.10.8 Resource EpisodeOfCare - Examples

URL: https://hl7.org/fhir/R4/episodeofcare-examples.html

8.10.8 Resource EpisodeOfCare - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Example Name | id | Format
Episode of Care example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.10.9 Resource EpisodeOfCare - Detailed Descriptions

URL: https://hl7.org/fhir/R4/episodeofcare-definitions.html

8.10.9 Resource EpisodeOfCare - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the EpisodeOfCare resource.

EpisodeOfCare
Element Id | EpisodeOfCare
Definition | An association between a patient and an organization / healthcare provider(s) during which time encounters may occur. The managing organization assumes a level of responsibility for the patient during this time.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Case Program Problem
EpisodeOfCare.identifier
Element Id | EpisodeOfCare.identifier
Definition | The EpisodeOfCare may be known by different identifiers for different contexts of use, such as when an external agency is tracking the Episode for funding purposes.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
EpisodeOfCare.status
Element Id | EpisodeOfCare.status
Definition | planned | waitlist | active | onhold | finished | cancelled.
Cardinality | 1..1
Terminology Binding | EpisodeOfCareStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the episode as not currently valid.
EpisodeOfCare.statusHistory
Element Id | EpisodeOfCare.statusHistory
Definition | The history of statuses that the EpisodeOfCare has been through (without requiring processing the history of the resource).
Cardinality | 0..*
EpisodeOfCare.statusHistory.status
Element Id | EpisodeOfCare.statusHistory.status
Definition | planned | waitlist | active | onhold | finished | cancelled.
Cardinality | 1..1
Terminology Binding | EpisodeOfCareStatus (Required)
Type | code
EpisodeOfCare.statusHistory.period
Element Id | EpisodeOfCare.statusHistory.period
Definition | The period during this EpisodeOfCare that the specific status applied.
Cardinality | 1..1
Type | Period
EpisodeOfCare.type
Element Id | EpisodeOfCare.type
Definition | A classification of the type of episode of care; e.g. specialist referral, disease management, type of funded care.
Cardinality | 0..*
Terminology Binding | Episode of care type (Example)
Type | CodeableConcept
Summary | true
Comments | The type can be very important in processing as this could be used in determining if the EpisodeOfCare is relevant to specific government reporting, or other types of classifications.
EpisodeOfCare.diagnosis
Element Id | EpisodeOfCare.diagnosis
Definition | The list of diagnosis relevant to this episode of care.
Cardinality | 0..*
Summary | true
EpisodeOfCare.diagnosis.condition
Element Id | EpisodeOfCare.diagnosis.condition
Definition | A list of conditions/problems/diagnoses that this episode of care is intended to be providing care for.
Cardinality | 1..1
Type | Reference(Condition)
Summary | true
EpisodeOfCare.diagnosis.role
Element Id | EpisodeOfCare.diagnosis.role
Definition | Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge …).
Cardinality | 0..1

Terminology Binding | DiagnosisRole (Preferred)
Type | CodeableConcept
Summary | true
EpisodeOfCare.diagnosis.rank
Element Id | EpisodeOfCare.diagnosis.rank
Definition | Ranking of the diagnosis (for each role type).
Cardinality | 0..1
Type | positiveInt
Summary | true
EpisodeOfCare.patient
Element Id | EpisodeOfCare.patient
Definition | The patient who is the focus of this episode of care.
Cardinality | 1..1
Type | Reference(Patient)
Summary | true
EpisodeOfCare.managingOrganization
Element Id | EpisodeOfCare.managingOrganization
Definition | The organization that has assumed the specific responsibilities for the specified duration.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
EpisodeOfCare.period
Element Id | EpisodeOfCare.period
Definition | The interval during which the managing organization assumes the defined responsibility.
Cardinality | 0..1
Type | Period
Summary | true
EpisodeOfCare.referralRequest
Element Id | EpisodeOfCare.referralRequest
Definition | Referral Request(s) that are fulfilled by this EpisodeOfCare, incoming referrals.
Cardinality | 0..*
Type | Reference(ServiceRequest)
EpisodeOfCare.careManager
Element Id | EpisodeOfCare.careManager
Definition | The practitioner that is the care manager/care coordinator for this patient.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
EpisodeOfCare.team
Element Id | EpisodeOfCare.team
Definition | The list of practitioners that may be facilitating this episode of care for specific purposes.
Cardinality | 0..*
Type | Reference(CareTeam)
Alternate Names | CareTeam
EpisodeOfCare.account
Element Id | EpisodeOfCare.account
Definition | The set of accounts that may be used for billing for this EpisodeOfCare.
Cardinality | 0..*
Type | Reference(Account)
Comments | The billing system may choose to allocate billable items associated with the EpisodeOfCare to different referenced Accounts based on internal business rules.

8.10.10 Resource EpisodeOfCare - Mappings

URL: https://hl7.org/fhir/R4/episodeofcare-mappings.html

8.10.10 Resource EpisodeOfCare - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the episodeofcare resource (see Mappings to Other Standards for further information & status).

8.10.10.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

EpisodeOfCare
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class
condition | FiveWs.what[x]

patient | FiveWs.subject[x]
period | FiveWs.init

8.10.10.2 Workflow Pattern (http://hl7.org/fhir/workflow)

EpisodeOfCare | Event
identifier | Event.identifier
status | Event.status
type | Event.code
condition | Event.reasonReference
patient | Event.subject
managingOrganization | Event.performer.actor
period | Event.occurrence[x]
referralRequest | Event.basedOn
careManager | Event.performer.actor
team | Event.performer.actor

8.10.10.3 RIM Mapping (http://hl7.org/v3)

EpisodeOfCare | n/a

8.10.11 Resource EpisodeOfCare - Extensions & Profiles

URL: https://hl7.org/fhir/R4/episodeofcare-profiles.html

8.10.11 Resource EpisodeOfCare - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

This table lists profiles and extensions for the EpisodeOfCare resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.10.11.1 Profiles

No Profiles defined for this resource

8.10.11.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-episodeOfCare | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.10.11.3 Search Extensions

No Search Extensions defined for this resource

8.10.12 Resource EpisodeOfCare - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/episodeofcare-version-maps.html

No validation errors - all conversions are clean

8.10.12 Resource EpisodeOfCare - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

R3 : R4 Conversion maps for EpisodeOfCare.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.10.12.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/EpisodeOfCare3to4" = "R3 to R4 Conversions for EpisodeOfCare"

uses "http://hl7.org/fhir/3.0/StructureDefinition/EpisodeOfCare" alias EpisodeOfCareR3 as source
```

```
uses "http://hl7.org/fhir/StructureDefinition/EpisodeOfCare" alias EpisodeOfCare as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group EpisodeOfCare(source src : EpisodeOfCareR3, target tgt : EpisodeOfCare) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.statusHistory as vs0 -> tgt.statusHistory as vt0 then statusHistory(vs0, vt0);
  src.type -> tgt.type;
  src.diagnosis as vs0 -> tgt.diagnosis as vt0 then diagnosis(vs0, vt0);
  src.patient -> tgt.patient;
  src.managingOrganization -> tgt.managingOrganization;
  src.period -> tgt.period;
  src.referralRequest -> tgt.referralRequest;
  src.careManager -> tgt.careManager;
  src.team -> tgt.team;
  src.account -> tgt.account;
}

group statusHistory(source src, target tgt) extends BackboneElement {
  src.status -> tgt.status;
  src.period -> tgt.period;
}

group diagnosis(source src, target tgt) extends BackboneElement {
  src.condition -> tgt.condition;
  src.role -> tgt.role;
  src.rank -> tgt.rank;
}
```

8.10.12.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/EpisodeOfCare4to3" = "R4 to R3 Conversion for EpisodeOfCare"

uses "http://hl7.org/fhir/StructureDefinition/EpisodeOfCare" alias EpisodeOfCare as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/EpisodeOfCare" alias EpisodeOfCareR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group EpisodeOfCare(source src : EpisodeOfCare, target tgt : EpisodeOfCareR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.statusHistory as vs0 -> tgt.statusHistory as vt0 then statusHistory(vs0, vt0);
  src.type -> tgt.type;
  src.diagnosis as vs0 -> tgt.diagnosis as vt0 then diagnosis(vs0, vt0);
  src.patient -> tgt.patient;
  src.managingOrganization -> tgt.managingOrganization;
  src.period -> tgt.period;
  src.referralRequest -> tgt.referralRequest;
  src.careManager -> tgt.careManager;
  src.team -> tgt.team;
  src.account -> tgt.account;
}

group statusHistory(source src, target tgt) extends BackboneElement {
  src.status -> tgt.status;
  src.period -> tgt.period;
}

group diagnosis(source src, target tgt) extends BackboneElement {
  src.condition -> tgt.condition;
  src.role -> tgt.role;
  src.rank -> tgt.rank;
}
```

8.10.12.3 R4 Validation Errors

8.10.13 JSON Schema for EpisodeOfCare

URL: https://hl7.org/fhir/R4/episodeofcare.schema.json.html

8.10.13 JSON Schema for EpisodeOfCare

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is a representation of the json schema for EpisodeOfCare, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/EpisodeOfCare",
  "$ref": "#/definitions/EpisodeOfCare",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "EpisodeOfCare": {
      "description": "An association between a patient and an organization / healthcare provider(s) during which time
    encounters may occur. The managing organization assumes a level of responsibility for the patient during this
    time.",
      "properties": {
        "resourceType": {
```

```
      "description": "This is a EpisodeOfCare resource",
      "const": "EpisodeOfCare"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "The EpisodeOfCare may be known by different identifiers for different contexts of use, such as
 when an external agency is tracking the Episode for funding purposes.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "planned | waitlist | active | onhold | finished | cancelled.",
      "enum": [
        "planned",
        "waitlist",
        "active",
        "onhold",
        "finished",
        "cancelled",
        "entered-in-error"
      ]
```

```
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusHistory": {
      "description": "The history of statuses that the EpisodeOfCare has been through (without requiring processing
the history of the resource).",
      "items": {
        "$ref": "#/definitions/EpisodeOfCare_StatusHistory"
      },
      "type": "array"
    },
    "type": {
      "description": "A classification of the type of episode of care; e.g. specialist referral, disease management,
 type of funded care.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "diagnosis": {
      "description": "The list of diagnosis relevant to this episode of care.",
      "items": {
        "$ref": "#/definitions/EpisodeOfCare_Diagnosis"
      },
      "type": "array"
    },
    "patient": {
      "description": "The patient who is the focus of this episode of care.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "managingOrganization": {
      "description": "The organization that has assumed the specific responsibilities for the specified duration.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "period": {
      "description": "The interval during which the managing organization assumes the defined responsibility.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "referralRequest": {
      "description": "Referral Request(s) that are fulfilled by this EpisodeOfCare, incoming referrals.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "careManager": {
      "description": "The practitioner that is the care manager/care coordinator for this patient.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "team": {
      "description": "The list of practitioners that may be facilitating this episode of care for specific
purposes.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "account": {
      "description": "The set of accounts that may be used for billing for this EpisodeOfCare.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "patient",
    "resourceType"
  ]
},
"EpisodeOfCare_StatusHistory": {
  "description": "An association between a patient and an organization / healthcare provider(s) during which time
encounters may occur. The managing organization assumes a level of responsibility for the patient during this
time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "status": {
      "description": "planned | waitlist | active | onhold | finished | cancelled.",
      "enum": [
        "planned",
        "waitlist",
        "active",
        "onhold",
        "finished",
        "cancelled",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The period during this EpisodeOfCare that the specific status applied.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false,
  "required": [
    "period"
  ]
},
"EpisodeOfCare_Diagnosis": {
  "description": "An association between a patient and an organization / healthcare provider(s) during which time
encounters may occur. The managing organization assumes a level of responsibility for the patient during this
time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "condition": {
      "description": "A list of conditions/problems/diagnoses that this episode of care is intended to be providing
care for.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "role": {
      "description": "Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge
…).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "rank": {
```

```
          "description": "Ranking of the diagnosis (for each role type).",
          "$ref": "#/definitions/positiveInt"
        },
        "_rank": {
          "description": "Extensions for rank",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "condition"
      ]
    }
  }
}
```


8.10.14 Resource EpisodeOfCare - Maturity Dependency View

URL: https://hl7.org/fhir/R4/episodeofcare-dependencies.html

8.10.14 Resource EpisodeOfCare - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner

Dependency Graph for EpisodeOfCare FMM level 2

Name | Card. | Type | Dependency Analysis
EpisodeOfCare | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
statusHistory | 0..* | BackboneElement
status | 1..1 | code
period | 1..1 | Period
type | 0..* | CodeableConcept
diagnosis | 0..* | BackboneElement
condition | 1..1 | Reference(Condition)
role | 0..1 | CodeableConcept
rank | 0..1 | positiveInt
patient | 1..1 | Reference(Patient)
managingOrganization | 0..1 | Reference(Organization)
period | 0..1 | Period
referralRequest | 0..* | Reference(ServiceRequest)
careManager | 0..1 | Reference(Practitioner | PractitionerRole)
team | 0..* | Reference(CareTeam)
account | 0..* | Reference(Account)

8.10.15 Value Set Sc-valueset-episode-of-care-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-episode-of-care-status.html

8.10.15 Value Set Sc-valueset-episode-of-care-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.10.16 EpisodeOfCareStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-episode-of-care-status)

Mapping from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the episode of care."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
planned | equivalent | planned
waitlist | equivalent | draft
active | equivalent | active
onhold | equivalent | suspended
finished | equivalent | complete
cancelled | equivalent | abandoned

8.11 Resource Encounter - Content

URL: https://hl7.org/fhir/R4/encounter.html

An interaction between a patient and healthcare provider(s) for the purpose of providing healthcare service(s) or assessing the health status of a patient.

8.11.1 Scope and Usage

A patient encounter is further characterized by the setting in which it takes place. Amongst them are ambulatory, emergency, home health, inpatient and virtual encounters. An Encounter encompasses the lifecycle from pre-admission, the actual encounter (for ambulatory encounters), and admission, stay and discharge (for inpatient encounters). During the encounter the patient may move from practitioner to practitioner and location to location.

Because of the broad scope of Encounter, not all elements will be relevant in all settings. For this reason, admission/discharge related information is kept in a separate Hospitalization component within Encounter. The class element is used to distinguish between these settings, which will guide further validation and application of business rules.

There is also substantial variance from organization to organization (and between jurisdictions and countries) on which business events translate to the start of a new Encounter, or what level of aggregation is used for Encounter. For example, each single visit of a practitioner during a hospitalization may lead to a new instance of Encounter, but depending on local practice and the systems involved, it may well be that this is aggregated to a single instance for a whole hospitalization. Even more aggregation may occur where jurisdictions introduce groups of Encounters for financial or other reasons. Encounters can be aggregated or grouped under other Encounters using the partOf element. See below for examples.

Encounter instances may exist before the actual encounter takes place to convey pre-admission information, including using Encounters elements to reflect the planned start date or planned encounter locations. In this case the status element is set to 'planned'.

The Hospitalization component is intended to store the extended information relating to a hospitalization event. It is always expected to be the same period as the encounter itself. Where the period is different, another encounter instance should be used to capture this information as a partOf this encounter instance.

The Procedure and encounter have references to each other, and these should be to different procedures; one for the procedure that was performed during the encounter (stored in Procedure.encounter), and another for cases where an encounter is a result of another procedure (stored in Encounter.indication) such as a follow-up encounter to resolve complications from an earlier procedure.

8.11.1.1 Status Management

During the life-cycle of an encounter it will pass through many statuses. Typically these are in order or the organization's workflow: planned, in-progress, finished/cancelled. This status information is often used for other things, and often an analysis of the status history is required. This could be done by scanning through all the versions of the encounter, checking the period of each, and then doing some form of post processing. To ease the burden of this (or where a system doesn't support resource histories) a status history component is included.

There is no direct indication purely by the status field as to whether an encounter is considered "admitted". The context of the encounter and business practices/policies/workflows/types can influence this definition. (e.g., acute care facility, aged care center, outpatient clinic, emergency department, community-based clinic). Statuses of "arrived", "triaged" or "in progress" could be considered the start of the admission, and also have the presence of the hospitalization sub-component entered.

The "on leave" status might or might not be a part of the admission, for example if the patient was permitted to go home for a weekend or some other form of external event. The location is also likely to be filled in with a location status of "present". For other examples such as an outpatient visit (day procedure - colonoscopy), the patient could also be considered to be admitted, hence the encounter doesn't have a fixed definition of admitted. At a minimum, we do believe that a patient IS admitted when the status is in-progress.

8.11.2 Boundaries and Relationships

The Encounter resource is not to be used to store appointment information, the Appointment resource is intended to be used for that. Note that in many systems outpatient encounters (which are in scope for Encounter) and Appointment are used concurrently. In FHIR, Appointment is used for establishing a date for the encounter, while Encounter is applicable to information about the actual Encounter, i.e., the patient showing up. As such, an encounter in the "planned" status is not identical to the appointment that scheduled it, but it is the encounter prior to its actual occurrence, with the expectation that encounter will be updated as it progresses to completion. Patient arrival at a location does not necessarily mean the start of the encounter (e.g. a patient arrives an hour earlier than he is actually seen by a practitioner).

An appointment is normally used for the planning stage of an appointment, searching, locating an available time, then making the appointment. Once this process is completed and the appointment is about to start, then the appointment will be marked as fulfilled, and linked to the newly created encounter. This new

encounter may start in an "arrived" status when they are admitted at a location of the facility, and then will move to the ward where another part-of encounter may begin.

Communication resources are used for a simultaneous interaction between a practitioner and a patient where there is no direct contact. Examples include a phone message, or transmission of some correspondence documentation. There is no duration recorded for a communication resource, but it could contain sent and received times.

Standard Extension: Associated Encounter This extension should be used to reference an encounter where there is no property that already defines this association on the resource.

This resource is referenced by AdverseEvent, AllergyIntolerance, CarePlan, CareTeam, ChargeItem, Claim, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, Contract, DeviceRequest, DiagnosticReport, DocumentReference, itself, ExplanationOfBenefit, Flag, GuidanceResponse, ImagingStudy, Immunization, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, NutritionOrder, Observation, Procedure, QuestionnaireResponse, RequestGroup, RiskAssessment, ServiceRequest, Task and VisionPrescription

8.11.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Encounter | TU | | DomainResource | An interaction during which services are provided to the patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifier(s) by which this encounter is known
status | ?!Σ | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
statusHistory | | 0..* | BackboneElement | List of past encounter statuses
status | | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
period | | 1..1 | Period | The time that the episode was in the specified status
class | Σ | 1..1 | Coding | Classification of patient encounter V3 Value SetActEncounterCode (Extensible)
classHistory | | 0..* | BackboneElement | List of past encounter classes
class | | 1..1 | Coding | inpatient | outpatient | ambulatory | emergency + V3 Value SetActEncounterCode (Extensible)
period | | 1..1 | Period | The time that the episode was in the specified class
type | Σ | 0..* | CodeableConcept | Specific type of encounter Encounter type (Example)
serviceType | Σ | 0..1 | CodeableConcept | Specific type of service Service type (Example)
priority | | 0..1 | CodeableConcept | Indicates the urgency of the encounter v3 Code System ActPriority (Example)
subject | Σ | 0..1 | Reference(Patient | Group) | The patient or group present at the encounter
episodeOfCare | Σ | 0..* | Reference(EpisodeOfCare) | Episode(s) of care that this encounter should be recorded against
basedOn | | 0..* | Reference(ServiceRequest) | The ServiceRequest that initiated this encounter
participant | Σ | 0..* | BackboneElement | List of participants involved in the encounter
type | Σ | 0..* | CodeableConcept | Role of participant in encounter Participant type (Extensible)
period | | 0..1 | Period | Period of time during the encounter that the participant participated
individual | Σ | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson) | Persons involved in the encounter other than the patient
appointment | Σ | 0..* | Reference(Appointment) | The appointment that scheduled this encounter
period | | 0..1 | Period | The start and end time of the encounter
length | | 0..1 | Duration | Quantity of time the encounter lasted (less time absent)
reasonCode | Σ | 0..* | CodeableConcept | Coded reason the encounter takes place Encounter Reason Codes (Preferred)
reasonReference | Σ | 0..* | Reference(Condition | Procedure | Observation | ImmunizationRecommendation) | Reason the encounter takes place (reference)
diagnosis | Σ | 0..* | BackboneElement | The list of diagnosis relevant to this encounter
condition | Σ | 1..1 | Reference(Condition | Procedure) | The diagnosis or procedure relevant to the encounter
use | | 0..1 | CodeableConcept | Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)
rank | | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this Encounter
hospitalization | | 0..1 | BackboneElement | Details about the admission to a healthcare service
preAdmissionIdentifier | | 0..1 | Identifier | Pre-admission identifier
origin | | 0..1 | Reference(Location | Organization) | The location/organization from which the patient came before admission
admitSource | | 0..1 | CodeableConcept | From where patient was admitted (physician referral, transfer) Admit source (Preferred)
reAdmission | | 0..1 | CodeableConcept | The type of hospital re-admission that has occurred (if any). If the value is absent, then this is not identified as a readmission v2 RE-ADMISSION INDICATOR (Example)
dietPreference | | 0..* | CodeableConcept | Diet preferences reported by the patient Diet (Example)
specialCourtesy | | 0..* | CodeableConcept | Special courtesies (VIP, board member) Special courtesy (Preferred)

specialArrangement | | 0..* | CodeableConcept | Wheelchair, translator, stretcher, etc. Special arrangements (Preferred)
destination | | 0..1 | Reference(Location | Organization) | Location/organization to which the patient is discharged
dischargeDisposition | | 0..1 | CodeableConcept | Category or kind of location after discharge Discharge disposition (Example)
location | | 0..* | BackboneElement | List of locations where the patient has been
location | | 1..1 | Reference(Location) | Location the encounter takes place
status | | 0..1 | code | planned | active | reserved | completed EncounterLocationStatus (Required)
physicalType | | 0..1 | CodeableConcept | The physical type of the location (usually the level in the location hierachy - bed room ward etc.) Location type (Example)
period | | 0..1 | Period | Time period during which the patient was present at the location
serviceProvider | | 0..1 | Reference(Organization) | The organization (facility) responsible for this encounter
partOf | | 0..1 | Reference(Encounter) | Another Encounter this encounter is part of
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Encounter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier(s) by which this encounter is known --></identifier>
 <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
 <statusHistory>  <!-- 0..* List of past encounter statuses -->
  <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
  <period><!-- 1..1 Period The time that the episode was in the specified status --></period>
 </statusHistory>
 <class><!-- 1..1 Coding Classification of patient encounter --></class>
 <classHistory>  <!-- 0..* List of past encounter classes -->
  <class><!-- 1..1 Coding inpatient | outpatient | ambulatory | emergency + --></class>
  <period><!-- 1..1 Period The time that the episode was in the specified class --></period>
 </classHistory>
 <type><!-- 0..* CodeableConcept Specific type of encounter --></type>
 <serviceType><!-- 0..1 CodeableConcept Specific type of service --></serviceType>
 <priority><!-- 0..1 CodeableConcept Indicates the urgency of the encounter --></priority>
 <subject><!-- 0..1 Reference(Patient|Group) The patient or group present at the encounter --></subject>
 <episodeOfCare><!-- 0..* Reference(EpisodeOfCare) Episode(s) of care that this encounter should be recorded against
    --></episodeOfCare>
 <basedOn><!-- 0..* Reference(ServiceRequest) The ServiceRequest that initiated this encounter --></basedOn>
 <participant>  <!-- 0..* List of participants involved in the encounter -->
  <type><!-- 0..* CodeableConcept Role of participant in encounter --></type>
  <period><!-- 0..1 Period Period of time during the encounter that the participant participated --></period>
  <individual><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson) Persons involved in the encounter other
    than the patient --></individual>
 </participant>
 <appointment><!-- 0..* Reference(Appointment) The appointment that scheduled this encounter --></appointment>
 <period><!-- 0..1 Period The start and end time of the encounter --></period>
 <length><!-- 0..1 Duration Quantity of time the encounter lasted (less time absent) --></length>
 <reasonCode><!-- 0..* CodeableConcept Coded reason the encounter takes place --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) Reason the encounter takes place (reference) --></reasonReference>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this encounter -->
  <condition><!-- 1..1 Reference(Condition|Procedure) The diagnosis or procedure relevant to the encounter
    --></condition>
  <use><!-- 0..1 CodeableConcept Role that this diagnosis has within the encounter (e.g. admission, billing, discharge
    â€¦) --></use>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this Encounter --></account>
 <hospitalization>  <!-- 0..1 Details about the admission to a healthcare service -->
  <preAdmissionIdentifier><!-- 0..1 Identifier Pre-admission identifier --></preAdmissionIdentifier>
  <origin><!-- 0..1 Reference(Location|Organization) The location/organization from which the patient came before
    admission --></origin>
  <admitSource><!-- 0..1 CodeableConcept From where patient was admitted (physician referral, transfer)
    --></admitSource>
  <reAdmission><!-- 0..1 CodeableConcept The type of hospital re-admission that has occurred (if any). If the value is
    absent, then this is not identified as a readmission --></reAdmission>
  <dietPreference><!-- 0..* CodeableConcept Diet preferences reported by the patient --></dietPreference>
  <specialCourtesy><!-- 0..* CodeableConcept Special courtesies (VIP, board member) --></specialCourtesy>
  <specialArrangement><!-- 0..* CodeableConcept Wheelchair, translator, stretcher, etc. --></specialArrangement>
  <destination><!-- 0..1 Reference(Location|Organization) Location/organization to which the patient is discharged
    --></destination>
  <dischargeDisposition><!-- 0..1 CodeableConcept Category or kind of location after discharge
    --></dischargeDisposition>
 </hospitalization>
 <location>  <!-- 0..* List of locations where the patient has been -->
  <location><!-- 1..1 Reference(Location) Location the encounter takes place --></location>
  <status value="[code]"/><!-- 0..1 planned | active | reserved | completed -->
  <physicalType><!-- 0..1 CodeableConcept The physical type of the location (usually the level in the location hierachy
    - bed room ward etc.) --></physicalType>
  <period><!-- 0..1 Period Time period during which the patient was present at the location --></period>
 </location>
 <serviceProvider><!-- 0..1 Reference(Organization) The organization (facility) responsible for this encounter
    --></serviceProvider>
 <partOf><!-- 0..1 Reference(Encounter) Another Encounter this encounter is part of --></partOf>
</Encounter>
```

JSON Template

```
{
  "resourceType" : "Encounter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier(s) by which this encounter is known
  "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  "statusHistory" : [{ // List of past encounter statuses
    "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
    "period" : { Period } // R!  The time that the episode was in the specified status
  }],
  "class" : { Coding }, // R!  Classification of patient encounter
  "classHistory" : [{ // List of past encounter classes
    "class" : { Coding }, // R!  inpatient | outpatient | ambulatory | emergency +
    "period" : { Period } // R!  The time that the episode was in the specified class
  }],
  "type" : [{ CodeableConcept }], // Specific type of encounter
  "serviceType" : { CodeableConcept }, // Specific type of service
  "priority" : { CodeableConcept }, // Indicates the urgency of the encounter
  "subject" : { Reference(Patient|Group) }, // The patient or group present at the encounter
  "episodeOfCare" : [{ Reference(EpisodeOfCare) }], // Episode(s) of care that this encounter should be recorded against
  "basedOn" : [{ Reference(ServiceRequest) }], // The ServiceRequest that initiated this encounter
  "participant" : [{ // List of participants involved in the encounter
    "type" : [{ CodeableConcept }], // Role of participant in encounter
    "period" : { Period }, // Period of time during the encounter that the participant participated
    "individual" : { Reference(Practitioner|PractitionerRole|RelatedPerson) } // Persons involved in the encounter other
     than the patient
  }],
  "appointment" : [{ Reference(Appointment) }], // The appointment that scheduled this encounter
  "period" : { Period }, // The start and end time of the encounter
  "length" : { Duration }, // Quantity of time the encounter lasted (less time absent)
  "reasonCode" : [{ CodeableConcept }], // Coded reason the encounter takes place
  "reasonReference" : [{ Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) }], // Reason the encounter takes place (reference)
  "diagnosis" : [{ // The list of diagnosis relevant to this encounter
    "condition" : { Reference(Condition|Procedure) }, // R!  The diagnosis or procedure relevant to the encounter
    "use" : { CodeableConcept }, // Role that this diagnosis has within the encounter (e.g. admission, billing,
    discharge â€¦)
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "account" : [{ Reference(Account) }], // The set of accounts that may be used for billing for this Encounter
  "hospitalization" : { // Details about the admission to a healthcare service
    "preAdmissionIdentifier" : { Identifier }, // Pre-admission identifier
    "origin" : { Reference(Location|Organization) }, // The location/organization from which the patient came before
    admission
    "admitSource" : { CodeableConcept }, // From where patient was admitted (physician referral, transfer)
    "reAdmission" : { CodeableConcept }, // The type of hospital re-admission that has occurred (if any). If the value
    is absent, then this is not identified as a readmission
    "dietPreference" : [{ CodeableConcept }], // Diet preferences reported by the patient
    "specialCourtesy" : [{ CodeableConcept }], // Special courtesies (VIP, board member)
    "specialArrangement" : [{ CodeableConcept }], // Wheelchair, translator, stretcher, etc.
    "destination" : { Reference(Location|Organization) }, // Location/organization to which the patient is discharged
    "dischargeDisposition" : { CodeableConcept } // Category or kind of location after discharge
  },
  "location" : [{ // List of locations where the patient has been
    "location" : { Reference(Location) }, // R!  Location the encounter takes place
    "status" : "<code>", // planned | active | reserved | completed
    "physicalType" : { CodeableConcept }, // The physical type of the location (usually the level in the location
    hierachy - bed room ward etc.)
    "period" : { Period } // Time period during which the patient was present at the location
  }],
  "serviceProvider" : { Reference(Organization) }, // The organization (facility) responsible for this encounter
  "partOf" : { Reference(Encounter) } // Another Encounter this encounter is part of
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Encounter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Encounter.identifier [ Identifier ], ... ; # 0..* Identifier(s) by which this encounter is known
  fhir:Encounter.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  fhir:Encounter.statusHistory [ # 0..* List of past encounter statuses
    fhir:Encounter.statusHistory.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished
    | cancelled +
    fhir:Encounter.statusHistory.period [ Period ]; # 1..1 The time that the episode was in the specified status
  ], ...;
  fhir:Encounter.class [ Coding ]; # 1..1 Classification of patient encounter
  fhir:Encounter.classHistory [ # 0..* List of past encounter classes
    fhir:Encounter.classHistory.class [ Coding ]; # 1..1 inpatient | outpatient | ambulatory | emergency +
    fhir:Encounter.classHistory.period [ Period ]; # 1..1 The time that the episode was in the specified class
  ], ...;
  fhir:Encounter.type [ CodeableConcept ], ... ; # 0..* Specific type of encounter
  fhir:Encounter.serviceType [ CodeableConcept ]; # 0..1 Specific type of service
  fhir:Encounter.priority [ CodeableConcept ]; # 0..1 Indicates the urgency of the encounter
  fhir:Encounter.subject [ Reference(Patient|Group) ]; # 0..1 The patient or group present at the encounter
```

```
  fhir:Encounter.episodeOfCare [ Reference(EpisodeOfCare) ], ... ; # 0..* Episode(s) of care that this encounter should
    be recorded against
  fhir:Encounter.basedOn [ Reference(ServiceRequest) ], ... ; # 0..* The ServiceRequest that initiated this encounter
  fhir:Encounter.participant [ # 0..* List of participants involved in the encounter
    fhir:Encounter.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in encounter
    fhir:Encounter.participant.period [ Period ]; # 0..1 Period of time during the encounter that the participant
    participated
    fhir:Encounter.participant.individual [ Reference(Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Persons
    involved in the encounter other than the patient
  ], ...;
  fhir:Encounter.appointment [ Reference(Appointment) ], ... ; # 0..* The appointment that scheduled this encounter
  fhir:Encounter.period [ Period ]; # 0..1 The start and end time of the encounter
  fhir:Encounter.length [ Duration ]; # 0..1 Quantity of time the encounter lasted (less time absent)
  fhir:Encounter.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason the encounter takes place
  fhir:Encounter.reasonReference [ Reference(Condition|Procedure|Observation|ImmunizationRecommendation) ], ... ; # 0..*
     Reason the encounter takes place (reference)
  fhir:Encounter.diagnosis [ # 0..* The list of diagnosis relevant to this encounter
    fhir:Encounter.diagnosis.condition [ Reference(Condition|Procedure) ]; # 1..1 The diagnosis or procedure relevant to
     the encounter
    fhir:Encounter.diagnosis.use [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the encounter (e.g.
    admission, billing, discharge â€¦)
    fhir:Encounter.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:Encounter.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for this
    Encounter
  fhir:Encounter.hospitalization [ # 0..1 Details about the admission to a healthcare service
    fhir:Encounter.hospitalization.preAdmissionIdentifier [ Identifier ]; # 0..1 Pre-admission identifier
    fhir:Encounter.hospitalization.origin [ Reference(Location|Organization) ]; # 0..1 The location/organization from
    which the patient came before admission
    fhir:Encounter.hospitalization.admitSource [ CodeableConcept ]; # 0..1 From where patient was admitted (physician
    referral, transfer)
    fhir:Encounter.hospitalization.reAdmission [ CodeableConcept ]; # 0..1 The type of hospital re-admission that has
    occurred (if any). If the value is absent, then this is not identified as a readmission
    fhir:Encounter.hospitalization.dietPreference [ CodeableConcept ], ... ; # 0..* Diet preferences reported by the
    patient
    fhir:Encounter.hospitalization.specialCourtesy [ CodeableConcept ], ... ; # 0..* Special courtesies (VIP, board
    member)
    fhir:Encounter.hospitalization.specialArrangement [ CodeableConcept ], ... ; # 0..* Wheelchair, translator,
    stretcher, etc.
    fhir:Encounter.hospitalization.destination [ Reference(Location|Organization) ]; # 0..1 Location/organization to
    which the patient is discharged
    fhir:Encounter.hospitalization.dischargeDisposition [ CodeableConcept ]; # 0..1 Category or kind of location after
    discharge
  ];
  fhir:Encounter.location [ # 0..* List of locations where the patient has been
    fhir:Encounter.location.location [ Reference(Location) ]; # 1..1 Location the encounter takes place
    fhir:Encounter.location.status [ code ]; # 0..1 planned | active | reserved | completed
    fhir:Encounter.location.physicalType [ CodeableConcept ]; # 0..1 The physical type of the location (usually the
    level in the location hierachy - bed room ward etc.)
    fhir:Encounter.location.period [ Period ]; # 0..1 Time period during which the patient was present at the location
  ], ...;
  fhir:Encounter.serviceProvider [ Reference(Organization) ]; # 0..1 The organization (facility) responsible for this
    encounter
  fhir:Encounter.partOf [ Reference(Encounter) ]; # 0..1 Another Encounter this encounter is part of
]
```

Changes since R3

Encounter

Encounter.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/encounter-status|4.0.1

Encounter.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/encounter-status|4.0.1

Encounter.class | Min Cardinality changed from 0 to 1

Encounter.serviceType | Added Element

Encounter.basedOn | Renamed from incomingReferral to basedOn Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest

Encounter.participant.individual | Type Reference: Added Target Type PractitionerRole

Encounter.appointment | Max Cardinality changed from 1 to *

Encounter.reasonCode | Added Element

Encounter.reasonReference | Added Element

Encounter.diagnosis.use | Added Element

Encounter.hospitalization.origin | Type Reference: Added Target Type Organization

Encounter.hospitalization.destination | Type Reference: Added Target Type Organization

Encounter.location.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-location-status to http://hl7.org/fhir/ValueSet/encounter-location-status|4.0.1

Encounter.location.physicalType | Added Element

Encounter.reason | deleted

Encounter.diagnosis.role | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 10 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Encounter | TU | | DomainResource | An interaction during which services are provided to the patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifier(s) by which this encounter is known
status | ?!Σ | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
statusHistory | | 0..* | BackboneElement | List of past encounter statuses
status | | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
period | | 1..1 | Period | The time that the episode was in the specified status
class | Σ | 1..1 | Coding | Classification of patient encounter V3 Value SetActEncounterCode (Extensible)
classHistory | | 0..* | BackboneElement | List of past encounter classes
class | | 1..1 | Coding | inpatient | outpatient | ambulatory | emergency + V3 Value SetActEncounterCode (Extensible)
period | | 1..1 | Period | The time that the episode was in the specified class
type | Σ | 0..* | CodeableConcept | Specific type of encounter Encounter type (Example)
serviceType | Σ | 0..1 | CodeableConcept | Specific type of service Service type (Example)
priority | | 0..1 | CodeableConcept | Indicates the urgency of the encounter v3 Code System ActPriority (Example)
subject | Σ | 0..1 | Reference(Patient | Group) | The patient or group present at the encounter
episodeOfCare | Σ | 0..* | Reference(EpisodeOfCare) | Episode(s) of care that this encounter should be recorded against
basedOn | | 0..* | Reference(ServiceRequest) | The ServiceRequest that initiated this encounter
participant | Σ | 0..* | BackboneElement | List of participants involved in the encounter
type | Σ | 0..* | CodeableConcept | Role of participant in encounter Participant type (Extensible)
period | | 0..1 | Period | Period of time during the encounter that the participant participated
individual | Σ | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson) | Persons involved in the encounter other than the patient
appointment | Σ | 0..* | Reference(Appointment) | The appointment that scheduled this encounter
period | | 0..1 | Period | The start and end time of the encounter
length | | 0..1 | Duration | Quantity of time the encounter lasted (less time absent)
reasonCode | Σ | 0..* | CodeableConcept | Coded reason the encounter takes place Encounter Reason Codes (Preferred)
reasonReference | Σ | 0..* | Reference(Condition | Procedure | Observation | ImmunizationRecommendation) | Reason the encounter takes place (reference)
diagnosis | Σ | 0..* | BackboneElement | The list of diagnosis relevant to this encounter
condition | Σ | 1..1 | Reference(Condition | Procedure) | The diagnosis or procedure relevant to the encounter
use | | 0..1 | CodeableConcept | Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)
rank | | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this Encounter
hospitalization | | 0..1 | BackboneElement | Details about the admission to a healthcare service
preAdmissionIdentifier | | 0..1 | Identifier | Pre-admission identifier
origin | | 0..1 | Reference(Location | Organization) | The location/organization from which the patient came before admission
admitSource | | 0..1 | CodeableConcept | From where patient was admitted (physician referral, transfer) Admit source (Preferred)
reAdmission | | 0..1 | CodeableConcept | The type of hospital re-admission that has occurred (if any). If the value is absent, then this is not identified as a readmission v2 RE-ADMISSION INDICATOR (Example)
dietPreference | | 0..* | CodeableConcept | Diet preferences reported by the patient Diet (Example)
specialCourtesy | | 0..* | CodeableConcept | Special courtesies (VIP, board member) Special courtesy (Preferred)
specialArrangement | | 0..* | CodeableConcept | Wheelchair, translator, stretcher, etc. Special arrangements (Preferred)
destination | | 0..1 | Reference(Location | Organization) | Location/organization to which the patient is discharged
dischargeDisposition | | 0..1 | CodeableConcept | Category or kind of location after discharge Discharge disposition (Example)
location | | 0..* | BackboneElement | List of locations where the patient has been
location | | 1..1 | Reference(Location) | Location the encounter takes place
status | | 0..1 | code | planned | active | reserved | completed EncounterLocationStatus (Required)
physicalType | | 0..1 | CodeableConcept | The physical type of the location (usually the level in the location hierachy - bed room ward etc.) Location type (Example)
period | | 0..1 | Period | Time period during which the patient was present at the location
serviceProvider | | 0..1 | Reference(Organization) | The organization (facility) responsible for this encounter
partOf | | 0..1 | Reference(Encounter) | Another Encounter this encounter is part of
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Encounter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier(s) by which this encounter is known --></identifier>
 <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
 <statusHistory>  <!-- 0..* List of past encounter statuses -->
  <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
  <period><!-- 1..1 Period The time that the episode was in the specified status --></period>
 </statusHistory>
 <class><!-- 1..1 Coding Classification of patient encounter --></class>
 <classHistory>  <!-- 0..* List of past encounter classes -->
  <class><!-- 1..1 Coding inpatient | outpatient | ambulatory | emergency + --></class>
  <period><!-- 1..1 Period The time that the episode was in the specified class --></period>
 </classHistory>
 <type><!-- 0..* CodeableConcept Specific type of encounter --></type>
 <serviceType><!-- 0..1 CodeableConcept Specific type of service --></serviceType>
```

```
 <priority><!-- 0..1 CodeableConcept Indicates the urgency of the encounter --></priority>
 <subject><!-- 0..1 Reference(Patient|Group) The patient or group present at the encounter --></subject>
 <episodeOfCare><!-- 0..* Reference(EpisodeOfCare) Episode(s) of care that this encounter should be recorded against
    --></episodeOfCare>
 <basedOn><!-- 0..* Reference(ServiceRequest) The ServiceRequest that initiated this encounter --></basedOn>
 <participant>  <!-- 0..* List of participants involved in the encounter -->
  <type><!-- 0..* CodeableConcept Role of participant in encounter --></type>
  <period><!-- 0..1 Period Period of time during the encounter that the participant participated --></period>
  <individual><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson) Persons involved in the encounter other
    than the patient --></individual>
 </participant>
 <appointment><!-- 0..* Reference(Appointment) The appointment that scheduled this encounter --></appointment>
 <period><!-- 0..1 Period The start and end time of the encounter --></period>
 <length><!-- 0..1 Duration Quantity of time the encounter lasted (less time absent) --></length>
 <reasonCode><!-- 0..* CodeableConcept Coded reason the encounter takes place --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) Reason the encounter takes place (reference) --></reasonReference>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this encounter -->
  <condition><!-- 1..1 Reference(Condition|Procedure) The diagnosis or procedure relevant to the encounter
    --></condition>
  <use><!-- 0..1 CodeableConcept Role that this diagnosis has within the encounter (e.g. admission, billing, discharge
    â€¦) --></use>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this Encounter --></account>
 <hospitalization>  <!-- 0..1 Details about the admission to a healthcare service -->
  <preAdmissionIdentifier><!-- 0..1 Identifier Pre-admission identifier --></preAdmissionIdentifier>
  <origin><!-- 0..1 Reference(Location|Organization) The location/organization from which the patient came before
    admission --></origin>
  <admitSource><!-- 0..1 CodeableConcept From where patient was admitted (physician referral, transfer)
    --></admitSource>
  <reAdmission><!-- 0..1 CodeableConcept The type of hospital re-admission that has occurred (if any). If the value is
    absent, then this is not identified as a readmission --></reAdmission>
  <dietPreference><!-- 0..* CodeableConcept Diet preferences reported by the patient --></dietPreference>
  <specialCourtesy><!-- 0..* CodeableConcept Special courtesies (VIP, board member) --></specialCourtesy>
  <specialArrangement><!-- 0..* CodeableConcept Wheelchair, translator, stretcher, etc. --></specialArrangement>
  <destination><!-- 0..1 Reference(Location|Organization) Location/organization to which the patient is discharged
    --></destination>
  <dischargeDisposition><!-- 0..1 CodeableConcept Category or kind of location after discharge
    --></dischargeDisposition>
 </hospitalization>
 <location>  <!-- 0..* List of locations where the patient has been -->
  <location><!-- 1..1 Reference(Location) Location the encounter takes place --></location>
  <status value="[code]"/><!-- 0..1 planned | active | reserved | completed -->
  <physicalType><!-- 0..1 CodeableConcept The physical type of the location (usually the level in the location hierachy
    - bed room ward etc.) --></physicalType>
  <period><!-- 0..1 Period Time period during which the patient was present at the location --></period>
 </location>
 <serviceProvider><!-- 0..1 Reference(Organization) The organization (facility) responsible for this encounter
    --></serviceProvider>
 <partOf><!-- 0..1 Reference(Encounter) Another Encounter this encounter is part of --></partOf>
</Encounter>
```

JSON Template

```
{
  "resourceType" : "Encounter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier(s) by which this encounter is known
  "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  "statusHistory" : [{ // List of past encounter statuses
    "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
    "period" : { Period } // R!  The time that the episode was in the specified status
  }],
  "class" : { Coding }, // R!  Classification of patient encounter
  "classHistory" : [{ // List of past encounter classes
    "class" : { Coding }, // R!  inpatient | outpatient | ambulatory | emergency +
    "period" : { Period } // R!  The time that the episode was in the specified class
  }],
  "type" : [{ CodeableConcept }], // Specific type of encounter
  "serviceType" : { CodeableConcept }, // Specific type of service
  "priority" : { CodeableConcept }, // Indicates the urgency of the encounter
  "subject" : { Reference(Patient|Group) }, // The patient or group present at the encounter
  "episodeOfCare" : [{ Reference(EpisodeOfCare) }], // Episode(s) of care that this encounter should be recorded against
  "basedOn" : [{ Reference(ServiceRequest) }], // The ServiceRequest that initiated this encounter
  "participant" : [{ // List of participants involved in the encounter
    "type" : [{ CodeableConcept }], // Role of participant in encounter
    "period" : { Period }, // Period of time during the encounter that the participant participated
    "individual" : { Reference(Practitioner|PractitionerRole|RelatedPerson) } // Persons involved in the encounter other
     than the patient
  }],
  "appointment" : [{ Reference(Appointment) }], // The appointment that scheduled this encounter
  "period" : { Period }, // The start and end time of the encounter
  "length" : { Duration }, // Quantity of time the encounter lasted (less time absent)
  "reasonCode" : [{ CodeableConcept }], // Coded reason the encounter takes place
  "reasonReference" : [{ Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) }], // Reason the encounter takes place (reference)
  "diagnosis" : [{ // The list of diagnosis relevant to this encounter
    "condition" : { Reference(Condition|Procedure) }, // R!  The diagnosis or procedure relevant to the encounter
```

```
    "use" : { CodeableConcept }, // Role that this diagnosis has within the encounter (e.g. admission, billing,
    discharge â€¦)
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "account" : [{ Reference(Account) }], // The set of accounts that may be used for billing for this Encounter
  "hospitalization" : { // Details about the admission to a healthcare service
    "preAdmissionIdentifier" : { Identifier }, // Pre-admission identifier
    "origin" : { Reference(Location|Organization) }, // The location/organization from which the patient came before
    admission
    "admitSource" : { CodeableConcept }, // From where patient was admitted (physician referral, transfer)
    "reAdmission" : { CodeableConcept }, // The type of hospital re-admission that has occurred (if any). If the value
    is absent, then this is not identified as a readmission
    "dietPreference" : [{ CodeableConcept }], // Diet preferences reported by the patient
    "specialCourtesy" : [{ CodeableConcept }], // Special courtesies (VIP, board member)
    "specialArrangement" : [{ CodeableConcept }], // Wheelchair, translator, stretcher, etc.
    "destination" : { Reference(Location|Organization) }, // Location/organization to which the patient is discharged
    "dischargeDisposition" : { CodeableConcept } // Category or kind of location after discharge
  },
  "location" : [{ // List of locations where the patient has been
    "location" : { Reference(Location) }, // R!  Location the encounter takes place
    "status" : "<code>", // planned | active | reserved | completed
    "physicalType" : { CodeableConcept }, // The physical type of the location (usually the level in the location
    hierachy - bed room ward etc.)
    "period" : { Period } // Time period during which the patient was present at the location
  }],
  "serviceProvider" : { Reference(Organization) }, // The organization (facility) responsible for this encounter
  "partOf" : { Reference(Encounter) } // Another Encounter this encounter is part of
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Encounter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Encounter.identifier [ Identifier ], ... ; # 0..* Identifier(s) by which this encounter is known
  fhir:Encounter.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  fhir:Encounter.statusHistory [ # 0..* List of past encounter statuses
    fhir:Encounter.statusHistory.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished
    | cancelled +
    fhir:Encounter.statusHistory.period [ Period ]; # 1..1 The time that the episode was in the specified status
  ], ...;
  fhir:Encounter.class [ Coding ]; # 1..1 Classification of patient encounter
  fhir:Encounter.classHistory [ # 0..* List of past encounter classes
    fhir:Encounter.classHistory.class [ Coding ]; # 1..1 inpatient | outpatient | ambulatory | emergency +
    fhir:Encounter.classHistory.period [ Period ]; # 1..1 The time that the episode was in the specified class
  ], ...;
  fhir:Encounter.type [ CodeableConcept ], ... ; # 0..* Specific type of encounter
  fhir:Encounter.serviceType [ CodeableConcept ]; # 0..1 Specific type of service
  fhir:Encounter.priority [ CodeableConcept ]; # 0..1 Indicates the urgency of the encounter
  fhir:Encounter.subject [ Reference(Patient|Group) ]; # 0..1 The patient or group present at the encounter
  fhir:Encounter.episodeOfCare [ Reference(EpisodeOfCare) ], ... ; # 0..* Episode(s) of care that this encounter should
    be recorded against
  fhir:Encounter.basedOn [ Reference(ServiceRequest) ], ... ; # 0..* The ServiceRequest that initiated this encounter
  fhir:Encounter.participant [ # 0..* List of participants involved in the encounter
    fhir:Encounter.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in encounter
    fhir:Encounter.participant.period [ Period ]; # 0..1 Period of time during the encounter that the participant
    participated
    fhir:Encounter.participant.individual [ Reference(Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Persons
    involved in the encounter other than the patient
  ], ...;
  fhir:Encounter.appointment [ Reference(Appointment) ], ... ; # 0..* The appointment that scheduled this encounter
  fhir:Encounter.period [ Period ]; # 0..1 The start and end time of the encounter
  fhir:Encounter.length [ Duration ]; # 0..1 Quantity of time the encounter lasted (less time absent)
  fhir:Encounter.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason the encounter takes place
  fhir:Encounter.reasonReference [ Reference(Condition|Procedure|Observation|ImmunizationRecommendation) ], ... ; # 0..*
     Reason the encounter takes place (reference)
  fhir:Encounter.diagnosis [ # 0..* The list of diagnosis relevant to this encounter
    fhir:Encounter.diagnosis.condition [ Reference(Condition|Procedure) ]; # 1..1 The diagnosis or procedure relevant to
     the encounter
    fhir:Encounter.diagnosis.use [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the encounter (e.g.
    admission, billing, discharge â€¦)
    fhir:Encounter.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:Encounter.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for this
    Encounter
  fhir:Encounter.hospitalization [ # 0..1 Details about the admission to a healthcare service
    fhir:Encounter.hospitalization.preAdmissionIdentifier [ Identifier ]; # 0..1 Pre-admission identifier
    fhir:Encounter.hospitalization.origin [ Reference(Location|Organization) ]; # 0..1 The location/organization from
    which the patient came before admission
    fhir:Encounter.hospitalization.admitSource [ CodeableConcept ]; # 0..1 From where patient was admitted (physician
    referral, transfer)
    fhir:Encounter.hospitalization.reAdmission [ CodeableConcept ]; # 0..1 The type of hospital re-admission that has
    occurred (if any). If the value is absent, then this is not identified as a readmission
    fhir:Encounter.hospitalization.dietPreference [ CodeableConcept ], ... ; # 0..* Diet preferences reported by the
    patient
```

```
    fhir:Encounter.hospitalization.specialCourtesy [ CodeableConcept ], ... ; # 0..* Special courtesies (VIP, board
    member)
    fhir:Encounter.hospitalization.specialArrangement [ CodeableConcept ], ... ; # 0..* Wheelchair, translator,
    stretcher, etc.
    fhir:Encounter.hospitalization.destination [ Reference(Location|Organization) ]; # 0..1 Location/organization to
    which the patient is discharged
    fhir:Encounter.hospitalization.dischargeDisposition [ CodeableConcept ]; # 0..1 Category or kind of location after
    discharge
  ];
  fhir:Encounter.location [ # 0..* List of locations where the patient has been
    fhir:Encounter.location.location [ Reference(Location) ]; # 1..1 Location the encounter takes place
    fhir:Encounter.location.status [ code ]; # 0..1 planned | active | reserved | completed
    fhir:Encounter.location.physicalType [ CodeableConcept ]; # 0..1 The physical type of the location (usually the
    level in the location hierachy - bed room ward etc.)
    fhir:Encounter.location.period [ Period ]; # 0..1 Time period during which the patient was present at the location
  ], ...;
  fhir:Encounter.serviceProvider [ Reference(Organization) ]; # 0..1 The organization (facility) responsible for this
    encounter
  fhir:Encounter.partOf [ Reference(Encounter) ]; # 0..1 Another Encounter this encounter is part of
]
```

Changes since Release 3

Encounter
Encounter.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/encounter-status|4.0.1
Encounter.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/encounter-status|4.0.1
Encounter.class | Min Cardinality changed from 0 to 1
Encounter.serviceType | Added Element
Encounter.basedOn | Renamed from incomingReferral to basedOn Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest
Encounter.participant.individual | Type Reference: Added Target Type PractitionerRole
Encounter.appointment | Max Cardinality changed from 1 to *
Encounter.reasonCode | Added Element
Encounter.reasonReference | Added Element
Encounter.diagnosis.use | Added Element
Encounter.hospitalization.origin | Type Reference: Added Target Type Organization
Encounter.hospitalization.destination | Type Reference: Added Target Type Organization
Encounter.location.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-location-status to http://hl7.org/fhir/ValueSet/encounter-location-status|4.0.1
Encounter.location.physicalType | Added Element
Encounter.reason | deleted
Encounter.diagnosis.role | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 10 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.11.3.1 Terminology Bindings

Path | Definition | Type | Reference
Encounter.status Encounter.statusHistory.status | Current state of the encounter. | Required | EncounterStatus
Encounter.class Encounter.classHistory.class | Classification of the encounter. | Extensible | v3.ActEncounterCode
Encounter.type | The type of encounter. | Example | EncounterType
Encounter.serviceType | Broad categorization of the service that is to be provided. | Example | ServiceType
Encounter.priority | Indicates the urgency of the encounter. | Example | v3.ActPriority
Encounter.participant.type | Role of participant in encounter. | Extensible | ParticipantType
Encounter.reasonCode | Reason why the encounter takes place. | Preferred | EncounterReasonCodes
Encounter.diagnosis.use | The type of diagnosis this condition represents. | Preferred | DiagnosisRole
Encounter.hospitalization.admitSource | From where the patient was admitted. | Preferred | AdmitSource
Encounter.hospitalization.reAdmission | The reason for re-admission of this hospitalization encounter. | Example | v2.0092
Encounter.hospitalization.dietPreference | Medical, cultural or ethical food preferences to help with catering requirements. | Example | Diet
Encounter.hospitalization.specialCourtesy | Special courtesies. | Preferred | SpecialCourtesy
Encounter.hospitalization.specialArrangement | Special arrangements. | Preferred | SpecialArrangements
Encounter.hospitalization.dischargeDisposition | Discharge Disposition. | Example | DischargeDisposition
Encounter.location.status | The status of the location. | Required | EncounterLocationStatus
Encounter.location.physicalType | Physical form of the location. | Example | LocationType

8.11.4 Notes

• The class element describes the setting (in/outpatient etc.) in which the Encounter took place. Since this is important for interpreting the context of the encounter, choosing the appropriate business rules to enforce and for the management of the process, this element is required.

• In future versions of FHIR, some kind of charge posting vehicle (e.g. Account) will be added.

8.11.5 Example usage

As stated, Encounter allows a flexible nesting of Encounters using the partOf element. For example:

• A patient is admitted for two weeks - This could be modeled using a single Encounter instance, in which the start and length are given for the duration of the whole stay. The admitting doctor and the responsible doctor during the stay are specified using the Participant component.

• During the encounter, the patient moves from the admitting department to the Intensive Care unit and back - Three more detailed additional Encounters can be created, one for each location in which the patient stayed. Each of these Encounters has a single location (twice the admitting department and once the Intensive Care unit) and one or more participants at that location. These Encounters may use the partOf relationship to indicate these movements occurred during the longer overarching Encounter.

• During the last part of the stay, the patient is visited by the members of the multi-disciplinary team that treated him for final evaluation - If relevant, for each of these short visits, an Encounter may be created with a single participant. Since these took place during the last part of the stay, the partOf element can be used to associate these short visits with either the third patient movement or the bigger overall encounter.

Exactly how the Encounter is used depends on information available in the source system, the relevance of exchange of each level of Encounter and demands specific to the communicating partners. The expectation is that for each domain of exchange, profiles are used to limit the flexibility of Encounter to meet the demands of the use case.

8.11.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
account | reference | The set of accounts that may be used for billing for this Encounter | Encounter.account (Account)
appointment | reference | The appointment that scheduled this encounter | Encounter.appointment (Appointment)
based-on | reference | The ServiceRequest that initiated this encounter | Encounter.basedOn (ServiceRequest)
class | token | Classification of patient encounter | Encounter.class
date | date | A date within the period the Encounter lasted | Encounter.period | 17 Resources
diagnosis | reference | The diagnosis or procedure relevant to the encounter | Encounter.diagnosis.condition (Condition, Procedure)
episode-of-care | reference | Episode(s) of care that this encounter should be recorded against | Encounter.episodeOfCare (EpisodeOfCare)
identifier | token | Identifier(s) by which this encounter is known | Encounter.identifier | 30 Resources
length | quantity | Length of encounter in days | Encounter.length
location | reference | Location the encounter takes place | Encounter.location.location (Location)
location-period | date | Time period during which the patient was present at the location | Encounter.location.period
part-of | reference | Another Encounter this encounter is part of | Encounter.partOf (Encounter)
participant | reference | Persons involved in the encounter other than the patient | Encounter.participant.individual (Practitioner, PractitionerRole, RelatedPerson)
participant-type | token | Role of participant in encounter | Encounter.participant.type
patient | reference | The patient or group present at the encounter | Encounter.subject.where(resolve() is Patient) (Patient) | 33 Resources
practitioner | reference | Persons involved in the encounter other than the patient | Encounter.participant.individual.where(resolve() is Practitioner) (Practitioner)
reason-code | token | Coded reason the encounter takes place | Encounter.reasonCode
reason-reference | reference | Reason the encounter takes place (reference) | Encounter.reasonReference (Condition, Observation, Procedure, ImmunizationRecommendation)
service-provider | reference | The organization (facility) responsible for this encounter | Encounter.serviceProvider (Organization)
special-arrangement | token | Wheelchair, translator, stretcher, etc. | Encounter.hospitalization.specialArrangement
status | token | planned | arrived | triaged | in-progress | onleave | finished | cancelled + | Encounter.status
subject | reference | The patient or group present at the encounter | Encounter.subject (Group, Patient)
type | token | Specific type of encounter | Encounter.type | 5 Resources

8.11.7 Resource Encounter - Examples

URL: https://hl7.org/fhir/R4/encounter-examples.html

8.11.7 Resource Encounter - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Encounter example | example | XML | JSON | Turtle
Encounter example - virtual encounter | home | XML | JSON | Turtle
Real-world encounter example | f201 | XML | JSON | Turtle
Real-world encounter example (with primaryDiagnosis extension added) | f202 | XML | JSON | Turtle

Real-world encounter example | f203 | XML | JSON | Turtle
Real-world encounter example | f001 | XML | JSON | Turtle
Real-world encounter example | f002 | XML | JSON | Turtle
Real-world encounter example | f003 | XML | JSON | Turtle
for Clinical Document example patient | xcda | XML | JSON | Turtle
Emergency transitioning into inpatient example | emerg | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.11.8 Resource Encounter - Detailed Descriptions

URL: https://hl7.org/fhir/R4/encounter-definitions.html

8.11.8 Resource Encounter - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Encounter resource.

Encounter
Element Id | Encounter
Definition | An interaction between a patient and healthcare provider(s) for the purpose of providing healthcare service(s) or assessing the health status of a patient.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Visit
Encounter.identifier
Element Id | Encounter.identifier
Definition | Identifier(s) by which this encounter is known.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Encounter.status
Element Id | Encounter.status
Definition | planned | arrived | triaged | in-progress | onleave | finished | cancelled +.
Cardinality | 1..1
Terminology Binding | EncounterStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | Note that internal business rules will determine the appropriate transitions that may occur between statuses (and also classes).
Encounter.statusHistory
Element Id | Encounter.statusHistory
Definition | The status history permits the encounter resource to contain the status history without needing to read through the historical versions of the resource, or even have the server store them.
Cardinality | 0..*
Comments | The current status is always found in the current version of the resource, not the status history.
Encounter.statusHistory.status
Element Id | Encounter.statusHistory.status
Definition | planned | arrived | triaged | in-progress | onleave | finished | cancelled +.
Cardinality | 1..1
Terminology Binding | EncounterStatus (Required)
Type | code
Encounter.statusHistory.period
Element Id | Encounter.statusHistory.period
Definition | The time that the episode was in the specified status.
Cardinality | 1..1
Type | Period
Encounter.class
Element Id | Encounter.class
Definition | Concepts representing classification of patient encounter such as ambulatory (outpatient), inpatient, emergency, home health or others due to local variations.
Cardinality | 1..1
Terminology Binding | V3 Value SetActEncounterCode (Extensible)
Type | Coding
Summary | true
Encounter.classHistory
Element Id | Encounter.classHistory
Definition | The class history permits the tracking of the encounters transitions without needing to go through the resource history. This would be used for a case where an admission starts of as an emergency encounter, then transitions into an inpatient

scenario. Doing this and not restarting a new encounter ensures that any lab/diagnostic results can more easily follow the patient and not require re-processing and not get lost or cancelled during a kind of discharge from emergency to inpatient.
Cardinality | 0..*
Encounter.classHistory.class
Element Id | Encounter.classHistory.class
Definition | inpatient | outpatient | ambulatory | emergency +.
Cardinality | 1..1
Terminology Binding | V3 Value SetActEncounterCode (Extensible)
Type | Coding
Encounter.classHistory.period
Element Id | Encounter.classHistory.period
Definition | The time that the episode was in the specified class.
Cardinality | 1..1
Type | Period
Encounter.type
Element Id | Encounter.type
Definition | Specific type of encounter (e.g. e-mail consultation, surgical day-care, skilled nursing, rehabilitation).
Cardinality | 0..*
Terminology Binding | Encounter type (Example)
Type | CodeableConcept
Summary | true
Comments | Since there are many ways to further classify encounters, this element is 0..*.
Encounter.serviceType
Element Id | Encounter.serviceType
Definition | Broad categorization of the service that is to be provided (e.g. cardiology).
Cardinality | 0..1
Terminology Binding | Service type (Example)
Type | CodeableConcept
Summary | true
Encounter.priority
Element Id | Encounter.priority
Definition | Indicates the urgency of the encounter.
Cardinality | 0..1
Terminology Binding | v3 Code System ActPriority (Example)
Type | CodeableConcept
Encounter.subject
Element Id | Encounter.subject
Definition | The patient or group present at the encounter.
Cardinality | 0..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
Comments | While the encounter is always about the patient, the patient might not actually be known in all contexts of use, and there may be a group of patients that could be anonymous (such as in a group therapy for Alcoholics Anonymous - where the recording of the encounter could be used for billing on the number of people/staff and not important to the context of the specific patients) or alternately in veterinary care a herd of sheep receiving treatment (where the animals are not individually tracked).
Encounter.episodeOfCare
Element Id | Encounter.episodeOfCare
Definition | Where a specific encounter should be classified as a part of a specific episode(s) of care this field should be used. This association can facilitate grouping of related encounters together for a specific purpose, such as government reporting, issue tracking, association via a common problem. The association is recorded on the encounter as these are typically created after the episode of care and grouped on entry rather than editing the episode of care to append another encounter to it (the episode of care could span years).
Cardinality | 0..*
Type | Reference(EpisodeOfCare)
Summary | true
Encounter.basedOn
Element Id | Encounter.basedOn
Definition | The request this encounter satisfies (e.g. incoming referral or procedure request).
Cardinality | 0..*
Type | Reference(ServiceRequest)
Alternate Names | incomingReferral
Encounter.participant
Element Id | Encounter.participant
Definition | The list of people responsible for providing the service.
Cardinality | 0..*
Summary | true
Encounter.participant.type
Element Id | Encounter.participant.type
Definition | Role of participant in encounter.
Cardinality | 0..*
Terminology Binding | Participant type (Extensible)
Type | CodeableConcept

Summary | true
Comments | The participant type indicates how an individual participates in an encounter. It includes non-practitioner participants, and for practitioners this is to describe the action type in the context of this encounter (e.g. Admitting Dr, Attending Dr, Translator, Consulting Dr). This is different to the practitioner roles which are functional roles, derived from terms of employment, education, licensing, etc.
Encounter.participant.period
Element Id | Encounter.participant.period
Definition | The period of time that the specified participant participated in the encounter. These can overlap or be sub-sets of the overall encounter's period.
Cardinality | 0..1
Type | Period
Encounter.participant.individual
Element Id | Encounter.participant.individual
Definition | Persons involved in the encounter other than the patient.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | RelatedPerson)
Summary | true
Encounter.appointment
Element Id | Encounter.appointment
Definition | The appointment that scheduled this encounter.
Cardinality | 0..*
Type | Reference(Appointment)
Summary | true
Encounter.period
Element Id | Encounter.period
Definition | The start and end time of the encounter.
Cardinality | 0..1
Type | Period
Comments | If not (yet) known, the end of the Period may be omitted.
Encounter.length
Element Id | Encounter.length
Definition | Quantity of time the encounter lasted. This excludes the time during leaves of absence.
Cardinality | 0..1
Type | Duration
Comments | May differ from the time the Encounter.period lasted because of leave of absence.
Encounter.reasonCode
Element Id | Encounter.reasonCode
Definition | Reason the encounter takes place, expressed as a code. For admissions, this can be used for a coded admission diagnosis.
Cardinality | 0..*
Terminology Binding | Encounter Reason Codes (Preferred)
Type | CodeableConcept
Alternate Names | Indication; Admission diagnosis
Summary | true
Comments | For systems that need to know which was the primary diagnosis, these will be marked with the standard extension primaryDiagnosis (which is a sequence value rather than a flag, 1 = primary diagnosis).
Encounter.reasonReference
Element Id | Encounter.reasonReference
Definition | Reason the encounter takes place, expressed as a code. For admissions, this can be used for a coded admission diagnosis.
Cardinality | 0..*
Type | Reference(Condition | Procedure | Observation | ImmunizationRecommendation)
Alternate Names | Indication; Admission diagnosis
Summary | true
Comments | For systems that need to know which was the primary diagnosis, these will be marked with the standard extension primaryDiagnosis (which is a sequence value rather than a flag, 1 = primary diagnosis).
Encounter.diagnosis
Element Id | Encounter.diagnosis
Definition | The list of diagnosis relevant to this encounter.
Cardinality | 0..*
Summary | true
Encounter.diagnosis.condition
Element Id | Encounter.diagnosis.condition
Definition | Reason the encounter takes place, as specified using information from another resource. For admissions, this is the admission diagnosis. The indication will typically be a Condition (with other resources referenced in the evidence.detail), or a Procedure.
Cardinality | 1..1
Type | Reference(Condition | Procedure)
Alternate Names | Admission diagnosis; discharge diagnosis; indication
Summary | true
Comments | For systems that need to know which was the primary diagnosis, these will be marked with the standard extension primaryDiagnosis (which is a sequence value rather than a flag, 1 = primary diagnosis).
Encounter.diagnosis.use

Element Id | Encounter.diagnosis.use
Definition | Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …).
Cardinality | 0..1
Terminology Binding | DiagnosisRole (Preferred)
Type | CodeableConcept
Encounter.diagnosis.rank
Element Id | Encounter.diagnosis.rank
Definition | Ranking of the diagnosis (for each role type).
Cardinality | 0..1
Type | positiveInt
Encounter.account
Element Id | Encounter.account
Definition | The set of accounts that may be used for billing for this Encounter.
Cardinality | 0..*
Type | Reference(Account)
Comments | The billing system may choose to allocate billable items associated with the Encounter to different referenced Accounts based on internal business rules.
Encounter.hospitalization
Element Id | Encounter.hospitalization
Definition | Details about the admission to a healthcare service.
Cardinality | 0..1
Comments | An Encounter may cover more than just the inpatient stay. Contexts such as outpatients, community clinics, and aged care facilities are also included. The duration recorded in the period of this encounter covers the entire scope of this hospitalization record.
Encounter.hospitalization.preAdmissionIdentifier
Element Id | Encounter.hospitalization.preAdmissionIdentifier
Definition | Pre-admission identifier.
Cardinality | 0..1
Type | Identifier
Encounter.hospitalization.origin
Element Id | Encounter.hospitalization.origin
Definition | The location/organization from which the patient came before admission.
Cardinality | 0..1
Type | Reference(Location | Organization)
Encounter.hospitalization.admitSource
Element Id | Encounter.hospitalization.admitSource
Definition | From where patient was admitted (physician referral, transfer).
Cardinality | 0..1
Terminology Binding | Admit source (Preferred)
Type | CodeableConcept
Encounter.hospitalization.reAdmission
Element Id | Encounter.hospitalization.reAdmission
Definition | Whether this hospitalization is a readmission and why if known.
Cardinality | 0..1
Terminology Binding | v2 RE-ADMISSION INDICATOR (Example)
Type | CodeableConcept
Encounter.hospitalization.dietPreference
Element Id | Encounter.hospitalization.dietPreference
Definition | Diet preferences reported by the patient.
Cardinality | 0..*
Terminology Binding | Diet (Example)
Type | CodeableConcept
Requirements | Used to track patient's diet restrictions and/or preference. For a complete description of the nutrition needs of a patient during their stay, one should use the nutritionOrder resource which links to Encounter.
Comments | For example, a patient may request both a dairy-free and nut-free diet preference (not mutually exclusive).
Encounter.hospitalization.specialCourtesy
Element Id | Encounter.hospitalization.specialCourtesy
Definition | Special courtesies (VIP, board member).
Cardinality | 0..*
Terminology Binding | Special courtesy (Preferred)
Type | CodeableConcept
Encounter.hospitalization.specialArrangement
Element Id | Encounter.hospitalization.specialArrangement
Definition | Any special requests that have been made for this hospitalization encounter, such as the provision of specific equipment or other things.
Cardinality | 0..*
Terminology Binding | Special arrangements (Preferred)
Type | CodeableConcept
Encounter.hospitalization.destination
Element Id | Encounter.hospitalization.destination
Definition | Location/organization to which the patient is discharged.
Cardinality | 0..1
Type | Reference(Location | Organization)

Encounter.hospitalization.dischargeDisposition
Element Id | Encounter.hospitalization.dischargeDisposition
Definition | Category or kind of location after discharge.
Cardinality | 0..1
Terminology Binding | Discharge disposition (Example)
Type | CodeableConcept
Encounter.location
Element Id | Encounter.location
Definition | List of locations where the patient has been during this encounter.
Cardinality | 0..*
Comments | Virtual encounters can be recorded in the Encounter by specifying a location reference to a location of type "kind" such as "client's home" and an encounter.class = "virtual".
Encounter.location.location
Element Id | Encounter.location.location
Definition | The location where the encounter takes place.
Cardinality | 1..1
Type | Reference(Location)
Encounter.location.status
Element Id | Encounter.location.status
Definition | The status of the participants' presence at the specified location during the period specified. If the participant is no longer at the location, then the period will have an end date/time.
Cardinality | 0..1
Terminology Binding | EncounterLocationStatus (Required)
Type | code
Comments | When the patient is no longer active at a location, then the period end date is entered, and the status may be changed to completed.
Encounter.location.physicalType
Element Id | Encounter.location.physicalType
Definition | This will be used to specify the required levels (bed/ward/room/etc.) desired to be recorded to simplify either messaging or query.
Cardinality | 0..1
Terminology Binding | Location type (Example)
Type | CodeableConcept
Comments | This information is de-normalized from the Location resource to support the easier understanding of the encounter resource and processing in messaging or query. There may be many levels in the hierachy, and this may only pic specific levels that are required for a specific usage scenario.
Encounter.location.period
Element Id | Encounter.location.period
Definition | Time period during which the patient was present at the location.
Cardinality | 0..1
Type | Period
Encounter.serviceProvider
Element Id | Encounter.serviceProvider
Definition | The organization that is primarily responsible for this Encounter's services. This MAY be the same as the organization on the Patient record, however it could be different, such as if the actor performing the services was from an external organization (which may be billed seperately) for an external consultation. Refer to the example bundle showing an abbreviated set of Encounters for a colonoscopy.
Cardinality | 0..1
Type | Reference(Organization)
Encounter.partOf
Element Id | Encounter.partOf
Definition | Another Encounter of which this encounter is a part of (administratively or in time).
Cardinality | 0..1
Type | Reference(Encounter)
Hierarchy | This reference is part of a strict Hierarchy
Comments | This is also used for associating a child's encounter back to the mother's encounter. Refer to the Notes section in the Patient resource for further details.

8.11.9 Resource Encounter - Mappings

URL: https://hl7.org/fhir/R4/encounter-mappings.html

8.11.9 Resource Encounter - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the encounter resource (see Mappings to Other Standards for further information & status).

8.11.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Encounter
identifier | FiveWs.identifier
status | FiveWs.status
class | FiveWs.class
type | FiveWs.class
priority | FiveWs.grade
subject | FiveWs.subject[x]
episodeOfCare | FiveWs.context
individual | FiveWs.who
period | FiveWs.done[x]
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]
condition | FiveWs.why[x]
location | FiveWs.where[x]

8.11.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Encounter | Event
identifier | Event.identifier
status | Event.status
type | Event.code
serviceType | Event.code
subject | Event.subject
episodeOfCare | Event.context
basedOn | Event.basedOn
participant | Event.performer
type | Event.performer.function
individual | Event.performer.actor
appointment | Event.basedOn
period | Event.occurrence[x]
length | Event.occurrence[x]
reasonCode | Event.reasonCode
reasonReference | Event.reasonCode
condition | Event.reasonReference
location | Event.location
serviceProvider | Event.performer.actor
partOf | Event.partOf

8.11.9.3 HL7 v2 Mapping (http://hl7.org/v2)

Encounter
identifier | PV1-19
status | No clear equivalent in HL7 v2; active/finished could be inferred from PV1-44, PV1-45, PV2-24; inactive could be inferred from PV2-16
statusHistory
status
period
class | PV1-2
classHistory
class
period
type | PV1-4 / PV1-18
serviceType | PV1-10
priority | PV2-25
subject | PID-3
episodeOfCare | PV1-54, PV1-53
basedOn
participant | ROL
type | ROL-3 (or maybe PRT-4)
period | ROL-5, ROL-6 (or maybe PRT-5)
individual | ROL-4
appointment | SCH-1 / SCH-2
period | PV1-44, PV1-45
length | (PV1-45 less PV1-44) iff ((PV1-44 not empty) and (PV1-45 not empty)); units in minutes
reasonCode | EVN-4 / PV2-3 (note: PV2-3 is nominally constrained to inpatient admissions; HL7 v2 makes no vocabulary suggestions for PV2-3; would not expect PV2 segment or PV2-3 to be in use in all implementations)
reasonReference | EVN-4 / PV2-3 (note: PV2-3 is nominally constrained to inpatient admissions; HL7 v2 makes no vocabulary suggestions for PV2-3; would not expect PV2 segment or PV2-3 to be in use in all implementations)
diagnosis
condition | Resources that would commonly referenced at Encounter.indication would be Condition and/or Procedure. These most closely align with DG1/PRB and PR1 respectively.

use
rank
account
hospitalization
preAdmissionIdentifier | PV1-5
origin
admitSource | PV1-14
reAdmission | PV1-13
dietPreference | PV1-38
specialCourtesy | PV1-16
specialArrangement | PV1-15 / OBR-30 / OBR-43
destination | PV1-37
dischargeDisposition | PV1-36
location
location | PV1-3 / PV1-6 / PV1-11 / PV1-42 / PV1-43
status
physicalType
period
serviceProvider | PL.6 & PL.1
partOf

8.11.9.4 RIM Mapping (http://hl7.org/v3)

Encounter | Encounter[@moodCode='EVN']
identifier | .id
status | .statusCode
statusHistory | n/a
status | n/a
period | n/a
class | .inboundRelationship[typeCode=SUBJ].source[classCode=LIST].code
classHistory | n/a
class | n/a
period | n/a
type | .code
serviceType | n/a
priority | .priorityCode
subject | .participation[typeCode=SBJ]/role[classCode=PAT]
episodeOfCare | n/a
basedOn | .reason.ClinicalDocument
participant | .participation[typeCode=PFM]
type | .functionCode
period | .time
individual | .role
appointment | .outboundRelationship[typeCode=FLFS].target[classCode=ENC, moodCode=APT]
period | .effectiveTime (low & high)
length | .lengthOfStayQuantity
reasonCode | .reasonCode
reasonReference | .reasonCode
diagnosis | .outboundRelationship[typeCode=RSON]
condition | .outboundRelationship[typeCode=RSON].target
use | n/a
rank | .outboundRelationship[typeCode=RSON].priority
account | .pertains.A_Account
hospitalization | .outboundRelationship[typeCode=COMP].target[classCode=ENC, moodCode=EVN]
preAdmissionIdentifier | .id
origin | .participation[typeCode=ORG].role
admitSource | .admissionReferralSourceCode
reAdmission | n/a
dietPreference | .outboundRelationship[typeCode=COMP].target[classCode=SBADM, moodCode=EVN, code="diet"]
specialCourtesy | .specialCourtesiesCode
specialArrangement | .specialArrangementCode
destination | .participation[typeCode=DST]
dischargeDisposition | .dischargeDispositionCode
location | .participation[typeCode=LOC]
location | .role
status | .role.statusCode
period | .time
serviceProvider | .particiaption[typeCode=PFM].role
partOf | .inboundRelationship[typeCode=COMP].source[classCode=COMP, moodCode=EVN]

8.11.10 Resource Encounter - Extensions & Profiles

URL: https://hl7.org/fhir/R4/encounter-profiles.html

8.11.10 Resource Encounter - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Encounter resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.11.10.1 Profiles

No Profiles defined for this resource

8.11.10.2 Extensions

encounter-associatedEncounter | Encounter | associatedEncounter | for Encounter HL7 Extensions
encounter-modeOfArrival | Encounter | modeOfArrival | for Encounter HL7 Extensions
encounter-reasonCancelled | Encounter | reasonCancelled | for Encounter HL7 Extensions
workflow-researchStudy | Encounter | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

encounter-associatedEncounter | associatedEncounter | for Encounter HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.11.10.3 Search Extensions

No Search Extensions defined for this resource

8.11.11 Resource Encounter - Operations

URL: https://hl7.org/fhir/R4/encounter-operations.html

8.11.11 Resource Encounter - Operations

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This resource has 1 operation operations associated with it:

$everything | Fetch Encounter Record

For more information about operations, including how they are invoked, see Operations.

8.11.11.1 Operation $everything on Encounter

URL: https://hl7.org/fhir/R4/encounter-operation-everything.html

8.11.11.1 Operation $everything on Encounter

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

This operation is used to return all the information related to an encounter described in the resource on which this operation is invoked. The response is a bundle of type "searchset". At a minimum, the encounter resource itself is returned, along with any other resources that the server has available for the given encounter for the user. The server also returns whatever resources are needed to support the records - e.g. linked practitioners, locations, organizations etc. The principle intended use for this operation is to provide a patient with access to their record, or to allow a client to retrieve everything for an encounter for efficient display).

The server SHOULD return all resources it has that:

• are included in the encounter compartment for the identified encounter (have a reference to the encounter)

• are referenced by the standard extenstion for associating an encounter (where no reference element exists) http://hl7.org/fhir/StructureDefinition/encounter-associatedEncounter

• the server believes are relevant to the context of the encounter for any other reason (internally defined/decided)

• any resource referenced by the above, including binaries and attachments (to make a more complete package)

In the US Realm, at a mimimum, the resources returned SHALL include all the data covered by the meaningful use common data elements (see DAF for further guidance). Other applicable implementation guides may make additional rules about the information that is returned. Note that for many resources, the exact nature of the link to encounter can be ambiguous (e.g. for a DiagnosticReport, is it the encounter when it was initiated, or when it was reported?).

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Encounter-everything
```

Formal Definition (as a OperationDefinition).

URL: [base]/Encounter/[id]/$everything

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

_since | 0..1 | instant | | | Resources updated after this period will be included in the response. The intent of this parameter is to allow a client to request only records that have changed since the last request, based on either the return header time, or or (for asynchronous use), the transaction time

_type | 0..* | code | | | One or more parameters, each containing one or more comma-delimited FHIR resource types to include in the return resources. In the absense of any specified types, the server returns all resource types

_count | 0..1 | integer | | | See discussion below on the utility of paging through the results of the $everything operation

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The bundle type is "searchset" Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this operation and simply searching the encounter compartment are:

• unless the client requests otherwise, the server returns the entire result set in a single bundle (rather than using paging)

• the server is responsible for determining what resources to return as included resources (rather than the client specifying which ones)

This frees the client from needing to determine what it could or should ask for, particularly with regard to included resources. Servers should consider returning appropriate Provenance and AuditTrail on the returned resources, even though these are not directly part of the patient compartment.

It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single encounter, or determine whether the context has the rights to the nominated encounter, if there is one, or can determine an appropriate list of encouners to provide data for from the context of the request. If there is no nominated encounter (GET /Encounter/$everything) and the context is not associated with a single encounter record, the actual list of encounters is all encounters that the user associated with the request has access to. In such cases, the server may choose to return an error rather than all the records. Specifying the relationship between the context, a user and encounter records is outside the scope of this specification (though see The Smart App Launch Implementation Guide .

When this operation is used to access multiple encounter records at once, the return bundle could be rather a lot of data; servers may choose to require that such requests are made asynchronously, and associated with bulk data formats. Alternatively, clients may choose to page through the result set (or servers may require this). Paging through the results is done the same as for Searching, using the _count parameter, and Bundle links. Implementers should note that paging will be slower than simply returning all the results at once (more network traffic, multiple latency delays) but may be required in order not to exhaust available memory reading or writing the whole response in a single package. Unlike searching, there is no inherent user-display order for the $everything operation. Servers might consider sorting the returned resources in descending order of last record update, but are not required to do so. Servers should consider returning appropriate Provenance and AuditTrail on the returned resources, even though these are not directly part of the patient compartment.

The _since parameter is provided to support periodic queries to get additional information that has changed about the encounter since the last query. This means that the _since parameter is based on record time. The value of the _since parameter should be set to the time from the server. If using direct response, this is the timestamp in the response header. If using the async interface, this is the transaction timestamp in the json response. Servers should ensure that the timestamps a managed such that the client does not miss any changes. Clients should be able to handle getting the same response more than once in the case that the transaction falls on a time boundary. Clients should ensure that the other query parameters are constant to ensure a coherent set of records when doing periodic queries.

8.11.11.1.1 Examples

Request: Get all resources linked to a single patient

```
GET /open/Encounter/example/$everything
[some headers]
```

Response: Everything linked to the nominated patient (in this case, built automatically by the publication tooling)

```
HTTP/1.1 200 OK
[other headers]

<Bundle xml="http://hl7.org/fhir">
  <id value="3785cf96-fa4c-4443-94b2-4f80cc7354c0"/>
  <type value="searchset"/>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Encounter/example"/>
    <resource>
     <Encounter xmlns="http://hl7.org/fhir">
       <id value="example"/>
       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">Encounter with patient @example</div>
       </text>
       <status value="in-progress"/>
         <class>
             <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
             <code value="IMP"/>
             <display value="inpatient encounter"/>
         </class>
      <subject>
         <reference value="Patient/example"/>
       </subject>
     </Encounter>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/CareTeam/example"/>
    <resource>
     <CareTeam xmlns="http://hl7.org/fhir">
       <id value="example"/>

       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">Care Team</div>
       </text>

       <contained>
         <Practitioner>
           <id value="pr1"/>
           <name>
             <family value="Dietician"/>
             <given value="Dorothy"/>
           </name>
         </Practitioner>
       </contained>
       <identifier>
            <value value="12345"/>
       </identifier>
       <status value="active"/>
       <category>
          <coding>
            <system value="http://loinc.org"/>
            <code value="LA27976-2"/>
         <display value="Encounter-focused care team"/>
          </coding>
       </category>
       <name value="Peter James Charlmers Care Plan for Inpatient Encounter"/>
       <subject>
         <reference value="Patient/example"/>
         <display value="Peter James Chalmers"/>
       </subject>
       <encounter>
      <reference value="Encounter/example"/>
       </encounter>
       <period>
         <end value="2013-01-01"/>
       </period>

       <participant>
         <role>
           <text value="responsiblePerson"/>
         </role>
         <member>
           <reference value="Patient/example"/>
           <display value="Peter James Chalmers"/>
         </member>
      </participant>

      <participant>
        <role>
          <text value="adviser"/>
        </role>
        <member>
          <reference value="#pr1"/>
          <display value="Dorothy Dietition"/>
        </member>
        <onBehalfOf>
```

```
        <reference value="Organization/f001"/>
      </onBehalfOf>
      <period>
        <end value="2013-01-01"/>
      </period>
    </participant>
    <managingOrganization>
           <reference value="Organization/f001"/>
       </managingOrganization>
   </CareTeam>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/example"/>
  <resource>
   <!--
    This is an example of a simple weight measurement.
    --><Observation xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <!--  the mandatory quality flags:  -->
    <status value="final"/>
    <!-- category code is A code that classifies the general type of observation being made. This is used for
  searching, sorting and display purposes.-->
    <category>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
        <code value="vital-signs"/>
        <display value="Vital Signs"/>
      </coding>
    </category>
    <!--
       Observations are often coded in multiple code systems.
         - LOINC provides codes of varying granularity (though not usefully more specific in this particular case) and
   more generic LOINCs  can be mapped to more specific codes as shown here
         - snomed provides a clinically relevant code that is usually less granular than LOINC
         - the source system provides its own code, which may be less or more granular than LOINC
      -->
    <code>
      <!--  LOINC - always recommended to have a LOINC code  -->
      <coding>
        <system value="http://loinc.org"/>
        <code value="29463-7"/> <!--more generic methodless LOINC-->
        <display value="Body Weight"/>
      </coding>
      <coding>
        <system value="http://loinc.org"/>
        <code value="3141-9"/><!--translation is more specific method = measured LOINC-->
        <display value="Body weight Measured"/>
      </coding>
      <!--  SNOMED CT Codes - becoming more common  -->
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="27113001"/>
        <display value="Body weight"/>
      </coding>
      <!--  Also, a local code specific to the source system  -->
      <coding>
        <system value="http://acme.org/devices/clinical-codes"/>
        <code value="body-weight"/>
        <display value="Body Weight"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <effectiveDateTime value="2016-03-28"/>
    <!--  In FHIR, units may be represented twice. Once in the
       agreed human representation, and once in a coded form.
       Both is best, since it's not always possible to infer
       one from the other in code.

       When a computable unit is provided, UCUM (http://unitsofmeasure.org)
       is always preferred, but it doesn't provide notional units (such as
       "tablet"), etc. For these, something else is required (e.g. SNOMED CT)
       -->
    <valueQuantity>
      <value value="185"/>
      <unit value="lbs"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="[lb_av]"/>
    </valueQuantity>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/gpvisit"/>
```

```
<resource>
 <CarePlan xmlns="http://hl7.org/fhir">
   <id value="gpvisit"/>
   <text>
     <status value="additional"/>
     <div xmlns="http://www.w3.org/1999/xhtml">
       <p>  Represents the flow of a patient within a practice. The plan is created when
         they arrive and represents the 'care' of the patient over the course of that encounter.
         They first see the nurse for basic observations (BP, pulse, temp) then the doctor for
         the consultation and finally the nurse again for a tetanus immunization. As the plan is
         updated (e.g. a new activity added), different versions of the plan exist, and workflow timings
         for reporting can be gained by examining the plan history. This example is the version after
         seeing the doctor, and waiting for the nurse.The plan can either be created 'ad hoc' and modified as
         the parient progresses, or start with a standard template (which can, of course, be altered to suit the
patient.</p>
     </div>
   </text>
   <contained>
     <!-- This is the reason for the encounter. It is referenced by the concern -->
     <Condition>
       <id value="p1"/>
       <clinicalStatus>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
      <code value="active"/>
    </coding>
    </clinicalStatus>

    <verificationStatus>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
      <code value="confirmed"/>
    </coding>
    </verificationStatus>
       <code>
         <!-- Could coded if we wanted to...-->
         <text value="Overseas encounter"/>
       </code>
       <subject>
         <reference value="Patient/100"/>
         <display value="Peter James Chalmers"/>
       </subject>
     </Condition>
   </contained>

   <contained>
     <CareTeam>
         <id value="careteam"/>
         <participant id="part1">

      <role>

        <coding>

        <system value="http://example.org/local"/>

        <code value="nur"/>

        </coding>

    <text value="nurse"/>
      </role>

      <member>

    <reference value="Practitioner/13"/>

    <display value="Nurse Nancy"/>

      </member>

  </participant>

  <participant id="part2">

      <role>

    <coding>

        <system value="http://example.org/local"/>

        <code value="doc"/>

          </coding>

    <text value="doctor"/>

       </role>
```

```
      <member>

    <reference value="Practitioner/14"/>

    <display value="Doctor Dave"/>

      </member>

  </participant>

    </CareTeam>
  </contained>


 <contained>
    <Goal>
      <id value="goal"/>
      <lifecycleStatus value="planned"/>
    <description>
        <text value="Complete consultation"/>
      </description>
    <subject>
        <reference value="Patient/100"/>
        <display value="Peter James Chalmers"/>
      </subject>
    </Goal>
  </contained>
  <status value="active"/>
  <intent value="plan"/>
  <subject>
    <reference value="Patient/100"/>
    <display value="Peter James Chalmers"/>
  </subject>
  <period>
    <!-- This is the time the plan started - i.e. when they arrived-->
    <start value="2013-01-01T10:30:00+00:00"/>
    <!-- No end yet as the encounter is still in progress.-->
  </period>
  <careTeam>
    <reference value="#careteam"/>
  </careTeam>
  <addresses>
    <reference value="#p1"/>
    <display value="obesity"/>
  </addresses>
  <goal>
    <reference value="#goal"/>
  </goal>
  <activity>
    <outcomeReference>
   <!-- This is a link to the nurse encounter. The assumption is that all contacts with practitioners
   are modelled as separate encounters. Ideally, there will be  a 'master/parent' encounter that ties them
together.
    If there is a single encounter, then all participants will be linked to that encounter.-->
    <reference value="Encounter/example"/>
    </outcomeReference>
    <detail>
      <!-- This activity is for the initial nurse encounter where vitals are taken. It has been completed.-->
      <kind value="Appointment"/>
      <code>
        <coding>
          <system value="http://example.org/local"/>
          <code value="nursecon"/>
        </coding>
        <text value="Nurse Consultation"/>
      </code>
      <status value="completed"/>
      <doNotPerform value="false"/>
      <scheduledPeriod>
        <!-- the nurse saw the patient between 10:38 and 10:50 -->
        <start value="2013-01-01T10:38:00+00:00"/>
        <end value="2013-01-01T10:50:00+00:00"/>
      </scheduledPeriod>
      <performer>
        <!-- refer to the participant (the nurse) in this resource-->
        <reference value="Practitioner/13"/>
        <display value="Nurse Nancy"/>
      </performer>
    </detail>
  </activity>
  <activity>
    <detail>
      <!-- This activity is for the encounter with the doctor. It is scheduled, but not yet started,
             so there is no timing[x]  or actionTaken element yet -->
      <kind value="Appointment"/>
      <code>
        <coding>
          <system value="http://example.org/local"/>
          <code value="doccon"/>
```

```
          </coding>
          <text value="Doctor Consultation"/>
         </code>
         <status value="scheduled"/>
         <doNotPerform value="false"/>
         <!-- The status is 'scheduled' so the doctor knows the patient is waiting.  -->
         <performer>
           <!-- refer to the participant (the nurse) in this resource-->
           <reference value="Practitioner/14"/>
           <display value="Doctor Dave"/>
         </performer>
       </detail>
     </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Immunization/protocol"/>
  <resource>
   <!--Authored by Joginder Madra--><Immunization xmlns="http://hl7.org/fhir">
    <id value="protocol"/>
    <identifier>
      <system value="urn:ietf:rfc:3986"/>
      <value value="urn:oid:1.3.6.1.4.1.21367.2005.3.7.1234"/>
    </identifier>
    <status value="completed"/>
    <vaccineCode>
      <coding>
        <system value="http://hl7.org/fhir/sid/cvx"/>
        <code value="104"/>
      </coding>
      <text value="Twinrix (HepA/HepB)"/>
    </vaccineCode>
    <patient>
      <reference value="Patient/example"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <occurrenceDateTime value="2018-06-18"/>
    <primarySource value="true"/>
    <location>
      <reference value="Location/1"/>
    </location>
    <manufacturer>
      <reference value="Organization/hl7"/>
    </manufacturer>
    <lotNumber value="PT123F"/>
    <expirationDate value="2018-12-15"/>
    <site>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActSite"/>
        <code value="LA"/>
        <display value="left arm"/>
      </coding>
    </site>
    <route>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration"/>
        <code value="IM"/>
        <display value="Injection, intramuscular"/>
      </coding>
    </route>
    <doseQuantity>
      <value value="5"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mg"/>
    </doseQuantity>
    <performer>
      <function>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0443"/>
          <code value="OP"/>
        </coding>
      </function>
        <actor>
            <reference value="Practitioner/example"/>
        </actor>
    </performer>
    <performer>
      <function>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0443"/>
          <code value="AP"/>
        </coding>
      </function>
        <actor>
            <reference value="Practitioner/example"/>
        </actor>
```

```
    </performer>
    <isSubpotent value="false"/>
    <programEligibility>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-program-eligibility"/>
        <code value="ineligible"/>
      </coding>
    </programEligibility>
    <fundingSource>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-funding-source"/>
        <code value="private"/>
      </coding>
    </fundingSource>
    <protocolApplied>
      <series value="2-dose"/>
      <targetDisease>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="40468003"/>
        </coding>
      </targetDisease>
      <doseNumberPositiveInt value="1"/>
    </protocolApplied>
    <protocolApplied>
      <series value="3-dose"/>
      <targetDisease>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="66071002"/>
        </coding>
      </targetDisease>
      <doseNumberPositiveInt value="2"/>
    </protocolApplied>
   </Immunization>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Communication/example"/>
  <resource>
   <Communication xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Patient has very high serum potassium</div>
     </text>
     <identifier>
       <type>
         <text value="Paging System"/>
       </type>
       <system value="urn:oid:1.3.4.5.6.7"/>
       <value value="2345678901"/>
     </identifier>
     <instantiatesUri value="http://example.org/hyperkalemia"/>
     <partOf>
    <display value="Serum Potassium Observation"/>
     </partOf>
     <status value="completed"/>
     <category>
       <coding>
         <system value="http://acme.org/messagetypes"/>
         <code value="Alert"/>
       </coding>
       <text value="Alert"/>
     </category>
     <medium>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationMode"/>
         <code value="WRITTEN"/>
         <display value="written"/>
       </coding>
       <text value="written"/>
     </medium>
     <subject>
       <reference value="Patient/example"/>
     </subject>

     <encounter>
       <reference value="Encounter/example"/>
     </encounter>
     <sent value="2014-12-12T18:01:10-08:00"/>
     <received value="2014-12-12T18:01:11-08:00"/>
     <recipient>
       <reference value="Practitioner/example"/>
     </recipient>
     <sender>
       <reference value="Device/f001"/>
     </sender>
```

```
      <payload>
        <contentString value="Patient 1 has a very high serum potassium value (7.2 mmol/L on 2014-Dec-12 at 5:55 pm)"/>
      </payload>
      <payload>
        <!--Reference to the lab result -->
        <contentReference>
          <display value="Serum Potassium Observation"/>
        </contentReference>
      </payload>
    </Communication>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Flag/example-encounter"/>
  <resource>
   <Flag xmlns="http://hl7.org/fhir">
     <id value="example-encounter"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Follow Infection Control Level 3 Protocol</div>
     </text>
     <status value="active"/>
     <category>
       <coding>
         <system value="http://example.org/local"/>
         <code value="infection"/>
         <display value="Infection Control Level"/>
       </coding>
     </category>
     <code>
       <coding>
         <system value="http://example.org/local/if1"/>
         <code value="l3"/>
         <display value="Follow Level 3 Protocol"/>
       </coding>
     </code>

     <subject>
       <reference value="Patient/example"/>
       <display value="Peter Patient"/>
     </subject>

     <encounter>
       <reference value="Encounter/example"/>
     </encounter>


   </Flag>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/infantenteral"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><id value="infantenteral"/>
    <identifier>
      <system value="http://www.acme.org/nutritionorders"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <enteralFormula>
      <baseFormulaType>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="412414007"/>
          <display value="infant formula + iron"/>
        </coding>
      </baseFormulaType>
      <baseFormulaProductName value="Acme Infant Formula + Iron"/>

      <additiveType>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/entformula-additive"/>
          <code value="carbohydrate"/>
          <display value="Carbohydrate"/>
```

```
        </coding>
       </additiveType>
       <additiveProductName value="Acme High Carbohydrate Additive"/>
       <caloricDensity>
         <value value="20"/>
         <unit value="calories per ounce"/>
         <system value="http://unitsofmeasure.org"/>
         <code value="cal/[foz_us]"/>
       </caloricDensity>
       <routeofAdministration>
         <coding>
           <system value="http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration"/>
           <code value="PO"/>
           <display value="Swallow, oral"/>
           <userSelected value="true"/>
         </coding>
       </routeofAdministration>
       <administration>
       <!-- scheduled 4 oz every 3hrs not to exceed 32 oz/day-->
       <schedule>
         <repeat>
         <boundsPeriod>
           <start value="2014-09-17"/>
         </boundsPeriod>
           <frequency value="1"/>
           <period value="3"/>
           <periodUnit value="h"/>
         </repeat>
       </schedule>
       <quantity>
         <value value="4"/>
         <unit value="ounces"/>
         <system value="http://unitsofmeasure.org"/>
         <code value="[foz_us]"/>
       </quantity>
       </administration>
       <maxVolumeToDeliver>
         <value value="32"/>
         <unit value="ounces"/>
         <system value="http://unitsofmeasure.org"/>
         <code value="[foz_us]"/>
       </maxVolumeToDeliver>
       <!-- EH infant feeding instruction and additional instructions -->
       <administrationInstruction value="Add high calorie high carbohydrate additive to increase cal/oz from 24 cal/oz
   to 27 cal/oz."/>

     </enteralFormula>
    </NutritionOrder>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/RiskAssessment/riskexample"/>
   <resource>
    <RiskAssessment xmlns="http://hl7.org/fhir">
      <id value="riskexample"/>
      <text>
        <status value="generated"/>
        <div xmlns="http://www.w3.org/1999/xhtml">Risk assessment for reference only</div>
      </text>
      <identifier>
        <use value="official"/>
        <system value="http://example.org"/>
        <value value="risk-assessment-breastcancer1"/>
      </identifier>
      <basedOn>
        <reference value="ImmunizationRecommendation/example"/>
      </basedOn>
      <parent>
        <reference value="DiagnosticReport/example"/>
      </parent>
      <status value="final"/>
       <code>
        <coding>
          <system value="http://browser.ihtsdotools.org/"/>
          <code value="709510001"/>
          <display value="Assessment of risk for disease (procedure)"/>
        </coding>
      </code>
      <subject>
        <reference value="Patient/example"/>
      </subject>
      <encounter>
        <reference value="Encounter/example"/>
        <display value="Encounter with patient @example"/>
      </encounter>
      <occurrenceDateTime value="2017-10-10"/>
      <condition>
        <reference value="Condition/example"/>
      </condition>
```

```
      <performer>
        <reference value="Practitioner/example"/>
      </performer>
      <basis>
        <reference value="DiagnosticReport/example"/>
      </basis>
      <basis>
        <reference value="Observation/example"/>
      </basis>
      <prediction>
        <outcome>
          <text value="Breast Cancer"/>
        </outcome>
        <probabilityDecimal value="0.000368"/>
        <whenRange>
          <low>
            <value value="54"/>
            <unit value="years"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="a"/>
          </low>
          <high>
            <value value="57"/>
            <unit value="years"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="a"/>
          </high>
        </whenRange>
      </prediction>
      <note>
        <text value="This risk assessment is for reference only"/>
      </note>

    </RiskAssessment>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/GuidanceResponse/example"/>
  <resource>
    <GuidanceResponse xmlns="http://hl7.org/fhir">
      <id value="example"/>
      <!-- This example demonstrates the response to an $evaluate operation using the GuidanceResponse resource. -->
      <contained>
        <Parameters>
          <id value="outputParameters1"/>
          <parameter>
            <name value="score"/>
            <valueDecimal value="7"/>
          </parameter>
          <parameter>
            <name value="item-assessed"/>
       <valueCodeableConcept>
         <coding>
         <system value="http://www.ama-assn.org/go/cpt"/>
         <code value="70450"/>
         <display value="CT, head, wo iv contrast"/>
         </coding>
       </valueCodeableConcept>
          </parameter>
          <parameter>
            <name value="device"/>
            <valueReference>
              <reference value="Device/software"/>
            </valueReference>
          </parameter>
          <parameter>
            <name value="guideline-followed"/>
            <valueUri value="http://someguidelineprovider.org/radiology-appropriateness-guidelines.html"/>
          </parameter>
        </Parameters>
      </contained>
      <requestIdentifier>
        <system value="http://example.org"/>
        <value value="guidanceRequest1"/>
      </requestIdentifier>
      <identifier>
        <system value="http://example.org"/>
        <value value="guidanceResponse1"/>
      </identifier>
      <moduleUri value="http://someguidelineprovider.org/radiology-appropriateness-guidelines.html"/>
      <status value="success"/>
      <subject>
        <reference value="Patient/example"/>
      </subject>
      <encounter>
        <reference value="Encounter/example"/>
      </encounter>
      <occurrenceDateTime value="2017-03-10T16:02:00Z"/>
      <performer>
```

```
        <reference value="Device/software"/>
       </performer>
       <reasonCode>
         <text value="Guideline Appropriate Ordering Assessment"/>
       </reasonCode>
       <outputParameters>
         <reference value="#outputParameters1"/>
       </outputParameters>
     </GuidanceResponse>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/Observation/abdo-tender"/>
   <resource>
    <Observation xmlns="http://hl7.org/fhir">
     <id value="abdo-tender"/>


     <status value="final"/>
     <category>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
         <code value="exam"/>
         <display value="Exam"/>
       </coding>
     </category>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="43478001"/>
         <display value="Abdominal tenderness (finding)"/>
       </coding>
       <text value="Abdominal tenderness"/>
     </code>
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <encounter>
       <reference value="Encounter/example"/>
     </encounter>
     <effectivePeriod>
       <start value="2018-04-02T10:30:10+01:00"/>
     </effectivePeriod>
     <issued value="2018-04-03T15:30:10+01:00"/>
     <valueBoolean value="true"/>
     <interpretation>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
         <code value="A"/>
         <display value="Abnormal"/>
       </coding>
       <text value="Abnormal"/>
     </interpretation>
    </Observation>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/NutritionOrder/energysupplement"/>
   <resource>
    <NutritionOrder xmlns="http://hl7.org/fhir">
    edited by Eric Haas Health eData Inc
    --><!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use supplement
   element --><id value="energysupplement"/>
     <identifier>
       <system value="http://goodhealthhospital.org/nutrition-requests"/>
       <value value="123"/>
     </identifier>
     <status value="active"/><intent value="order"/>
     <patient>
       <reference value="Patient/example"/>
       <display value="Peter Chalmers"/>
     </patient>
     <encounter>
       <reference value="Encounter/example"/>
       <display value="Inpatient"/>
     </encounter>
     <dateTime value="2014-09-17"/>
     <orderer>
       <reference value="Practitioner/example"/>
       <display value="Dr Adam Careful"/>
     </orderer>
     <allergyIntolerance>
       <reference value="AllergyIntolerance/example"/>
       <display value="Cashew Nuts"/>
     </allergyIntolerance>
     <foodPreferenceModifier>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/diet"/>
         <code value="kosher"/>
```

```
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <supplement>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="442971000124100"/>
          <display value="Adult high energy formula"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/supplement-type-codes"/>
          <code value="1010"/>
          <display value="Adult high energy drink"/>
        </coding>
        <text value="Adult high energy drink"/>
      </type>
      <productName value="Ensure"/>
      <!--EH:  1 8 oz can tid using coded data -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <frequency value="3"/>
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <quantity>
        <value value="1"/>
        <unit value="8 oz can"/>
      </quantity>
    </supplement>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RequestGroup/kdn5-example"/>
  <resource>
   <RequestGroup xmlns="http://hl7.org/fhir">
     <id value="kdn5-example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Administer gemcitabine and carboplatin.</div>
     </text>
     <contained>
       <MedicationRequest>
         <id value="1111"/>
         <status value="unknown"/>
         <intent value="proposal"/>
         <medicationCodeableConcept>
           <coding>
             <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
             <code value="12574"/>
             <display value="gemcitabine"/>
           </coding>
           <text value="gemcitabine"/>
         </medicationCodeableConcept>
         <subject>
           <reference value="Patient/example"/>
         </subject>
         <dosageInstruction>
           <text value="1250 mg/m² IV over 30 minutes"/>
           <timing>
             <repeat>
               <duration value="30"/>
               <durationUnit value="min"/>
             </repeat>
           </timing>
           <route>
             <text value="IV"/>
           </route>
           <doseAndRate>
             <type>
               <coding>
                 <system value="http://terminology.hl7.org/CodeSystem/dose-rate-type"/>
                 <code value="ordered"/>
                 <display value="Ordered"/>
               </coding>
             </type>
             <doseQuantity>
```

```
        <value value="1250"/>
        <unit value="mg/m²"/>
        </doseQuantity>
      </doseAndRate>
    </dosageInstruction>
  </MedicationRequest>
</contained>
<contained>
  <MedicationRequest>
    <id value="2222"/>
    <status value="unknown"/>
    <intent value="proposal"/>
    <medicationCodeableConcept>
      <coding>
        <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
        <code value="40048"/>
        <display value="Carboplatin"/>
      </coding>
      <text value="CARBOplatin"/>
    </medicationCodeableConcept>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <dosageInstruction>
      <text value="AUC 5 IV over 30 minutes"/>
      <timing>
        <repeat>
          <duration value="30"/>
          <durationUnit value="min"/>
        </repeat>
      </timing>
      <route>
        <text value="IV"/>
      </route>
      <doseAndRate>
        <type>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/dose-rate-type"/>
            <code value="ordered"/>
            <display value="Ordered"/>
          </coding>
        </type>
        <doseQuantity>
          <!-- AUC as dose unit or as an extension? -->
          <extension url="http://example.org/fhir/AUC-dose">
            <valueInteger value="5"/>
          </extension>
          <!-- <value value="5" />
        <unit value="AUC" />
        <system value="http://snomed.info/sct" />
        <code value="413573002" /> -->
        </doseQuantity>
      </doseAndRate>

    </dosageInstruction>
  </MedicationRequest>
</contained>
<identifier>
  <value value="requestgroup-kdn5"/>
</identifier>
<instantiatesCanonical value="PlanDefinition/KDN5"/>
<status value="draft"/>
<intent value="plan"/>
<priority value="routine"/>
<subject>
  <reference value="Patient/example"/>
</subject>
<encounter>
  <reference value="Encounter/example"/>
</encounter>
<authoredOn value="2017-03-06T17:31:00Z"/>
<author>
  <reference value="Practitioner/1"/>
</author>

<action>
  <!-- list of regimen options. in this case only one -->
  <selectionBehavior value="exactly-one"/>
  <action>
    <!-- list of regimen parts. in this case only one -->
    <selectionBehavior value="all"/>
    <action>
      <!-- list of cycle definitions. in this case only one -->
      <groupingBehavior value="sentence-group"/>
      <selectionBehavior value="exactly-one"/>
      <action id="cycle-definition-1">
        <textEquivalent value="21-day cycle for 6 cycles"/>

        <timingTiming>
```

```
              <repeat>
                <count value="6"/>
                <duration value="21"/>
                <durationUnit value="d"/>
              </repeat>
            </timingTiming>

            <!-- list of regimen entry references for this regimen part -->
            <!-- references would duplicated in each cycle definition option
            (see http://build.fhir.org/codesystem-action-grouping-behavior.html#sentence-group)-->
            <action id="action-1">
              <extension url="http://hl7.org/fhir/StructureDefinition/timing-daysOfCycle">
                <extension url="day">
                  <valueInteger value="1"/>
                </extension>
                <extension url="day">
                  <valueInteger value="8"/>
                </extension>
              </extension>
              <textEquivalent value="Gemcitabine 1250 mg/m² IV over 30 minutes on days 1 and 8"/>
              <resource>
                <reference value="#1111"/>
              </resource>
            </action>
            <action id="action-2">
              <extension url="http://hl7.org/fhir/StructureDefinition/timing-daysOfCycle">
                <extension url="day">
                  <valueInteger value="1"/>
                </extension>
              </extension>
              <textEquivalent value="CARBOplatin AUC 5 IV over 30 minutes on Day 1"/>
              <relatedAction>
                <actionId value="action-1"/>
                <relationship value="concurrent-with-start"/>
       </relatedAction>
              <resource>
                <reference value="#2222"/>
              </resource>
            </action> <!-- end of regimen entries -->
          </action> <!-- end of cycle definitions -->
        </action> <!-- end of regimen parts -->
      </action> <!-- end of regimen options -->
    </action>
  </RequestGroup>
 </resource>
</entry>
<entry>
 <fullUrl value="http://hl7.org/fhir/Immunization/subpotent"/>
 <resource>
  <!--Authored by Joginder Madra--><Immunization xmlns="http://hl7.org/fhir">
   <id value="subpotent"/>
   <identifier>
     <system value="urn:ietf:rfc:3986"/>
     <value value="urn:oid:1.3.6.1.4.1.21367.2005.3.7.1234"/>
   </identifier>
   <status value="completed"/>
   <vaccineCode>
     <coding>
       <system value="urn:oid:1.2.36.1.2001.1005.17"/>
       <code value="GNHEP"/>
     </coding>
     <text value="Hepatitis B"/>
   </vaccineCode>
   <patient>
     <reference value="Patient/example"/>
   </patient>
   <encounter>
     <reference value="Encounter/example"/>
   </encounter>
   <occurrenceDateTime value="2015-01-15"/>
   <primarySource value="true"/>
   <location>
     <reference value="Location/1"/>
   </location>
   <manufacturer>
     <reference value="Organization/hl7"/>
   </manufacturer>
   <lotNumber value="AAJN11K"/>
   <expirationDate value="2015-02-28"/>
   <site>
     <coding>
       <system value="http://terminology.hl7.org/CodeSystem/v3-ActSite"/>
       <code value="LT"/>
       <display value="left thigh"/>
     </coding>
   </site>
   <route>
     <coding>
       <system value="http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration"/>
```

```
        <code value="IM"/>
        <display value="Injection, intramuscular"/>
      </coding>
    </route>
    <doseQuantity>
      <value value="0.5"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="ml"/>
    </doseQuantity>
    <performer>
      <function>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0443"/>
          <code value="OP"/>
        </coding>
      </function>
        <actor>
            <reference value="Practitioner/example"/>
        </actor>
    </performer>
    <performer>
      <function>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0443"/>
          <code value="AP"/>
        </coding>
      </function>
        <actor>
            <reference value="Practitioner/example"/>
        </actor>
    </performer>
    <note>
      <text value="Notes on adminstration of vaccine"/>
    </note>
    <isSubpotent value="false"/>
    <subpotentReason>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-subpotent-reason"/>
        <code value="partial"/>
      </coding>
    </subpotentReason>
    <education>
      <documentType value="2530886983000103111120702"/>
      <publicationDate value="2012-07-02"/>
      <presentationDate value="2013-01-10"/>
    </education>
    <programEligibility>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-program-eligibility"/>
        <code value="ineligible"/>
      </coding>
    </programEligibility>
    <fundingSource>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-funding-source"/>
        <code value="private"/>
      </coding>
    </fundingSource>
   </Immunization>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example5"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a servicerequest and to illustrate how to populate
  many of the task elements this is the first in a series of related Task examples:  example1 action lipid panel -in
  progress =>. example2 specimen collection subtask -accepted => example4  specimen collection subtask - completed
  with output Specimen=>example 5 action lipid panel -in progress specimen collected with output Specimen => example 6
   action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example5"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-001"/>
    </identifier>
    <basedOn>
      <display value="General Wellness Careplan"/>
    </basedOn>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <status value="in-progress"/>
    <!--fictive business status populated for this example -->
    <businessStatus>
      <text value="specimen received, test in progress"/>
    </businessStatus>
```

```
    <intent value="order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Lipid Panel"/>
    </code>
    <description value="Create order for getting specimen, Set up inhouse testing,  generate order for any sendouts
  and submit with specimen"/>
    <!--  Use the lipid servicerequest example -->
    <focus>
      <reference value="ServiceRequest/lipid"/>
      <display value="Lipid Panel Request"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </encounter>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:25:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:25:05+10:00"/>
    <!--  updated last modified date to track task activity-->
    <lastModified value="2016-10-31T16:45:05+10:00"/>
    <requester>
        <reference value="Practitioner/example"/>
        <display value="Dr Adam Careful"/>
    </requester>
    <performerType>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <!--  populate the reason just to demonstrate it use in this example
   -->
    <reasonCode>
      <text value="The Task.reason should only be included if there is no Task.focus or if it differs from the reason
  indicated on the focus"/>
    </reasonCode>
    <note>
      <text value="This is an example to demonstrate using task for actioning a servicerequest and to illustrate how
  to populate many of the task elements - this is the parent task that will be broken into subtask to grab the
  specimen and a sendout lab test "/>
    </note>
    <!--  add the link to the signature on the task to show how this element is usedemonstrating in this example the
  relevant history element using a contained resource for the purpose of this example only typically would reference
  an external resource

   -->
    <!--  complete the request within 48 hrs
   -->
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-02T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a DiagnosticRequest and will generated once the task is near completion
  -->
      <output>
      <type>
        <text value="collected specimen"/>
      </type>
      <valueReference>
        <reference value="Specimen/101"/>
      </valueReference>
    </output>

   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/enteralcontinuous"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   editor Eric Haas Health eData Inc
```

```
--><!-- Tube feedings given as continuous rate --><id value="enteralcontinuous"/>
 <identifier>
   <system value="http://www.acme.org/nutritionorders"/>
   <value value="123"/>
 </identifier>
 <status value="active"/><intent value="order"/>
 <patient>
   <reference value="Patient/example"/>
   <display value="Peter Chalmers"/>
 </patient>
 <encounter>
   <reference value="Encounter/example"/>
   <display value="Inpatient"/>
 </encounter>
 <dateTime value="2014-09-17"/>
 <orderer>
   <reference value="Practitioner/example"/>
   <display value="Dr Adam Careful"/>
 </orderer>
 <enteralFormula>
   <baseFormulaType>
     <coding>
       <system value="http://snomed.info/sct"/>
       <code value="6547210000124112"/>
       <display value="Diabetic specialty enteral formula"/>
     </coding>
   </baseFormulaType>
   <baseFormulaProductName value=" Acme Diabetes Formula"/>

   <caloricDensity>
     <value value="1"/>
     <unit value="calories per milliliter"/>
     <system value="http://unitsofmeasure.org"/>
     <code value="cal/mL"/>
   </caloricDensity>
   <routeofAdministration>
     <coding>
       <system value="http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration"/>
       <code value="NGT"/>
       <display value="Instillation, nasogastric tube"/>
     </coding>
   <!--step up rate of tube feeding by 20 ml/hr every 4 hours -->
   </routeofAdministration>
   <administration>

     <schedule>
     <repeat>
       <boundsPeriod>
         <start value="2014-09-17T07:00:00Z"/>
       </boundsPeriod>
     </repeat>
     </schedule>
     <rateQuantity>
       <value value="60"/>
       <unit value="ml/hr"/>
       <system value="http://unitsofmeasure.org"/>
       <code value="mL/h"/>
     </rateQuantity>
   </administration>
   <administration>
     <schedule>
     <repeat>
       <boundsPeriod>
         <start value="2014-09-17T11:00:00Z"/>
       </boundsPeriod>
     </repeat>
     </schedule>
     <rateQuantity>
       <value value="80"/>
       <unit value="ml/hr"/>
       <system value="http://unitsofmeasure.org"/>
       <code value="mL/h"/>
     </rateQuantity>
   </administration>
   <administration>
     <schedule>
     <repeat>
       <boundsPeriod>
         <start value="2014-09-17T15:00:00Z"/>
       </boundsPeriod>
     </repeat>
     </schedule>
     <rateQuantity>
       <value value="100"/>
       <unit value="ml/hr"/>
       <system value="http://unitsofmeasure.org"/>
       <code value="mL/h"/>
     </rateQuantity>
   </administration>
```

```
      <maxVolumeToDeliver>
        <value value="880"/>
        <unit value="milliliter/day"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mL/d"/>
      </maxVolumeToDeliver>
      <administrationInstruction value="Hold feedings from 7 pm to 7 am. Add MCT oil to increase calories from 1.0
  cal/mL to 1.5 cal/mL"/>

    </enteralFormula>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Media/xray"/>
  <resource>
   <Media xmlns="http://hl7.org/fhir">
     <id value="xray"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Xray of left hand for Patient Henry Levin (MRN 12345)
  2016-03-15</div>
     </text>
     <basedOn>
       <identifier>
         <system value="http://someclinic.org/fhir/NamingSystem/imaging-orders"/>
         <value value="111222"/>
         <assigner>
           <display value="XYZ Medical Clinic"/>
         </assigner>
       </identifier>
     </basedOn>

   <status value="completed"/>


     <modality>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="39714003"/>
         <display value="Skeletal X-ray of wrist and hand"/>
       </coding>
     </modality>

     <subject>
       <reference value="Patient/example"/>
     </subject>
     <encounter>
       <reference value="Encounter/example"/>
     </encounter>
     <createdDateTime value="2016-03-15"/>
     <bodySite>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="85151006"/>
         <display value="Structure of left hand (body structure)"/>
       </coding>
     </bodySite>

     <height value="432"/>
     <width value="640"/>

     <content id="a1">
       <contentType value="image/jpeg"/>
       <url value="http://someimagingcenter.org/fhir/Binary/A12345"/>
       <!-- the date this image was created -->
       <creation value="2016-03-15"/>
     </content>
   </Media>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ImagingStudy/example-xr"/>
  <resource>
   <ImagingStudy xmlns="http://hl7.org/fhir">
    <id value="example-xr"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">XR Wrist 3+ Views. John Smith (MRN: 09236). Accession: W12342398.
  Performed: 2017-01-01. 1 series, 2 images.</div>
    </text>
    <identifier>
      <use value="official"/>
      <system value="urn:dicom:uid"/>
      <value value="urn:oid:2.16.124.113543.6003.1154777499.30246.19789.3503430046"/>
    </identifier>
    <identifier>
      <use value="usual"/>
```

```
  <type>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
      <code value="ACSN"/>
    </coding>
  </type>
  <value value="W12342398"/>
  <assigner>
    <reference value="Organization/dicom-organization"/>
  </assigner>
</identifier>
<identifier>
  <use value="secondary"/>
  <value value="55551234"/>
  <assigner>
    <reference value="Organization/dicom-organization"/>
  </assigner>
</identifier>
<status value="available"/>
<modality>
  <system value="http://dicom.nema.org/resources/ontology/DCM"/>
  <code value="DX"/>
</modality>
<subject>
  <reference value="Patient/dicom"/>
</subject>
<encounter>
  <reference value="Encounter/example"/>
</encounter>
<started value="2017-01-01T11:01:20+03:00"/>
<basedOn>
  <reference value="ServiceRequest/example"/>
</basedOn>
<referrer>
  <reference value="Practitioner/example"/>
</referrer>
<interpreter>
  <reference value="Practitioner/example"/>
</interpreter>
<endpoint>
  <reference value="Endpoint/example-wadors"/>
</endpoint>
<numberOfSeries value="1"/>
<numberOfInstances value="2"/>
<procedureReference>
  <reference value="Procedure/example"/>
</procedureReference>
<procedureCode>
  <coding>
    <system value="http://www.radlex.org"/>
    <code value="RPID2589"/>
    <display value="XR Wrist 3+ Views"/>
  </coding>
  <text value="XR Wrist 3+ Views"/>
</procedureCode>
<reasonCode>
  <coding>
    <system value="http://snomed.info/sct"/>
    <code value="357009"/>
    <display value="Closed fracture of trapezoidal bone of wrist"/>
  </coding>
</reasonCode>
<note>
    <text value="XR Wrist 3+ Views"/>
</note>
<series>
  <uid value="2.16.124.113543.6003.1154777499.30246.19789.3503430045.1"/>
  <number value="3"/>
  <modality>
    <system value="http://dicom.nema.org/resources/ontology/DCM"/>
    <code value="DX"/>
  </modality>
  <description value="XR Wrist 3+ Views"/>
  <numberOfInstances value="2"/>
  <endpoint>
    <reference value="Endpoint/example-wadors"/>
  </endpoint>
  <bodySite>
    <!--   Added body site to support duplicate checking example   -->
    <system value="http://snomed.info/sct"/>
    <code value="T-15460"/>
    <display value="Wrist Joint"/>
  </bodySite>
  <laterality>
    <system value="http://snomed.info/sct"/>
    <code value="419161000"/>
    <display value="Unilateral left"/>
  </laterality>
  <started value="2011-01-01T11:01:20+03:00"/>
```

```
      <performer>
        <function>
        <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
        <code value="PRF"/>
        </coding>
        </function>
        <actor>
        <reference value="Practitioner/example"/>
        </actor>
      </performer>
      <instance>
        <uid value="2.16.124.113543.6003.1154777499.30246.19789.3503430045.1.1"/>
        <sopClass>
          <system value="urn:ietf:rfc:3986"/>
          <code value="urn:oid:1.2.840.10008.5.1.4.1.1.2"/>
        </sopClass>
        <number value="1"/>
        <title value="PA VIEW"/>
      </instance>
      <instance>
        <uid value="2.16.124.113543.6003.1154777499.30246.19789.3503430045.1.2"/>
        <sopClass>
          <system value="urn:ietf:rfc:3986"/>
          <code value="urn:oid:1.2.840.10008.5.1.4.1.1.2"/>
        </sopClass>
        <number value="2"/>
        <title value="LL VIEW"/>
      </instance>
    </series>
   </ImagingStudy>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ChargeItem/example"/>
  <resource>
   <?xml-model href="../../publish/chargeitem.sch" type="application/xml"
  schematypens="http://purl.oclc.org/dsdl/schematron"?><ChargeItem xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">Example of ChargeItem Usage in Context of the German EBM Billing code
  system</div>
    </text>
    <identifier>
      <system value="http://myHospital.org/ChargeItems"/>
      <value value="654321"/>
    </identifier>
    <definitionUri value="http://www.kbv.de/tools/ebm/html/01520_2904360860826220813632.html"/>
    <status value="billable"/>

    <code>
      <coding>
        <code value="01510"/>
        <display value="Zusatzpauschale für Beobachtung nach diagnostischer Koronarangiografie"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <occurrencePeriod>
      <start value="2017-01-25T08:00:00+01:00"/>
      <end value="2017-01-25T12:35:00+01:00"/>
    </occurrencePeriod>
    <performer>
      <function>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="17561000"/>
          <display value="Cardiologist"/>
        </coding>
      </function>
      <actor>

        <reference value="Practitioner/example"/>
      </actor>
    </performer>
    <performer>
      <function>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="224542009"/>
          <display value="Coronary Care Nurse"/>
        </coding>
      </function>
      <actor>
        <reference value="Practitioner/example"/>
```

```
      </actor>
    </performer>
    <performingOrganization>
      <identifier>
        <system value="http://myhospital/NamingSystem/departments"/>
        <value value="CARD_INTERMEDIATE_CARE"/>
      </identifier>
    </performingOrganization>
    <requestingOrganization>
      <identifier>
        <system value="http://myhospital/NamingSystem/departments"/>
        <value value="CARD_U1"/>
      </identifier>
    </requestingOrganization>
    <quantity>
      <value value="1"/>
    </quantity>
    <factorOverride value="0.8"/>
    <priceOverride>
        <value value="40"/>
           <currency value="EUR"/>
    </priceOverride>
    <overrideReason value="Patient is Cardiologist's golf buddy, so he gets a 20% discount!"/>
    <enterer>
      <reference value="Practitioner/example"/>
    </enterer>
    <enteredDate value="2017-01-25T23:55:04+01:00"/>
    <reason>
        <coding>
          <system value="http://hl7.org/fhir/sid/icd-10"/>
          <code value="123456"/>
          <display value="DIAG-1"/>
        </coding>
    </reason>
    <service>
      <reference value="Procedure/example"/>
    </service>
    <account>
      <reference value="Account/example"/>
    </account>
    <note>
      <authorReference>
        <reference value="Practitioner/example"/>
      </authorReference>
      <time value="2017-01-25T23:55:04+01:00"/>
      <text value="The code is only applicable for periods longer than 4h"/>
    </note>

   </ChargeItem>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/cardiacdiet"/>
  <resource>
   <?xml-stylesheet type="text/xsl" href="file://Mac/Home/Documents/FHIRBuild/reorder-nodes-
  template.xslt"?><NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   -->
   <!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
  -->
   <id value="cardiacdiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
```

```
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <oralDiet>
      <!-- EH:  list type of diet showing standard code and local codes -->
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="386619000"/>
          <display value="Low sodium diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1040"/>
          <display value="Low Sodium Diet"/>
        </coding>
        <text value="Low sodium diet"/>
      </type>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="226208002"/>
          <display value="Fluid restricted diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1040"/>
          <display value="Fluid restricted diet"/>
        </coding>
        <text value="Fluid restricted diet"/>
      </type>
      <!-- list nutrients and amounts here -->
      <nutrient>
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="39972003"/>
            <display value="Sodium"/>
          </coding>
        </modifier>
        <amount>
          <value value="2"/>
          <unit value="grams"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="g"/>
        </amount>
      </nutrient>
      <nutrient>
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="33463005"/>
            <display value="Fluid"/>
          </coding>
        </modifier>
        <amount>
          <value value="1500"/>
          <unit value="milliliter"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL"/>
        </amount>
      </nutrient>
      <!--instructions as free text -->
      <instruction value="Starting on 2/10 breakfast, maximum 400 ml fluids per meal"/>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/pureeddiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
   --><id value="pureeddiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
```

```
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <oralDiet>
      <!-- EH:  list type of diet showing standard code and local codes -->
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="226211001"/>
          <display value="Pureed diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1010"/>
          <display value="Pureed diet"/>
        </coding>
        <text value="Pureed diet"/>
      </type>
      <!-- EH example of three times a day starting on 2/10  -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <frequency value="3"/>
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <!--EH: this example showing how to code for texture modification -->
      <texture>
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="228055009"/>
            <display value="Liquidized food"/>
          </coding>
          <text value="Pureed"/>
        </modifier>
      </texture>
      <fluidConsistencyType>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="439021000124105"/>
          <display value="Dietary liquid consistency - nectar thick liquid"/>
        </coding>
        <text value="Nectar thick liquids"/>
      </fluidConsistencyType>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/renaldiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!-- EH: use oral Diet element --><id value="renaldiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
```

```
  <display value="Peter Chalmers"/>
</patient>
<encounter>
  <reference value="Encounter/example"/>
  <display value="Inpatient"/>
</encounter>
<dateTime value="2014-09-17"/>
<orderer>
  <reference value="Practitioner/example"/>
  <display value="Dr Adam Careful"/>
</orderer>
<allergyIntolerance>
  <!--EH  Show all the elements in this examples for preference, and intolerance -->
  <reference value="AllergyIntolerance/example"/>
  <display value="Cashew Nuts"/>
</allergyIntolerance>
<foodPreferenceModifier>
  <coding>
    <system value="http://terminology.hl7.org/CodeSystem/diet"/>
    <code value="dairy-free"/>
  </coding>
</foodPreferenceModifier>
<excludeFoodModifier>
  <coding>
    <system value="http://snomed.info/sct"/>
    <version value="20140730"/>
    <code value="227493005"/>
    <display value="Cashew Nut"/>
  </coding>
</excludeFoodModifier>
<oralDiet>
  <type>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="33489005"/>
      <display value="Renal diet"/>
    </coding>
    <coding>
      <system value="http://goodhealthhospital.org/diet-type-codes"/>
      <code value="1050"/>
      <display value="Renal"/>
    </coding>
    <text value="Renal Diet"/>
  </type>
  <!-- EH example of three times a day starting on 2/10 -->
  <schedule>
    <repeat>
       <boundsPeriod>
         <start value="2015-02-10"/>
       </boundsPeriod>
      <frequency value="3"/>
      <period value="1"/>
      <periodUnit value="d"/>
    </repeat>
  </schedule>
  <nutrient>
    <!-- list nutrients and amounts here -->
    <modifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="39972003"/>
        <display value="Sodium"/>
      </coding>
    </modifier>
    <amount>
      <value value="2"/>
      <unit value="grams"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="g"/>
    </amount>
  </nutrient>
  <nutrient>
    <modifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="30820000"/>
        <display value="Phosphorus"/>
      </coding>
    </modifier>
    <amount>
      <value value="1000"/>
      <unit value="milligrams"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mg"/>
    </amount>
  </nutrient>
  <nutrient>
    <modifier>
      <coding>
```

```
            <system value="http://snomed.info/sct"/>
            <code value="88480006"/>
            <display value="Potassium"/>
          </coding>
        </modifier>
        <amount>
          <value value="2"/>
          <unit value="grams"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="g"/>
        </amount>
      </nutrient>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/proteinsupplement"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!-- EH: This example has no food exclusions or preferences --><!-- EH: use supplement element --><id
  value="proteinsupplement"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <supplement>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="442991000124104"/>
          <display value="Adult high protein formula"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/supplement-type-codes"/>
          <code value="1000"/>
          <display value="High Protein Powder"/>
        </coding>
        <text value="High Protein Powder"/>
      </type>
      <productName value="Beneprotein"/>
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <when value="C"/>
        </repeat>
      </schedule>
      <quantity>
        <value value="1"/>
        <unit value="scoop"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="{scoop}"/>
      </quantity>
      <!-- EH:  use the .instructions element to list diet  supplement order-->
      <instruction value="Beneprotein 1 scoop TID with meal.starting on  2015-02-10"/>
    </supplement>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/QuestionnaireResponse/3141"/>
  <resource>
   <QuestionnaireResponse xmlns="http://hl7.org/fhir">
     <id value="3141"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <pre>
               Comorbidity? YES
                 Cardial Comorbidity? YES
                   Angina? YES
                   MI? NO
```

```
            Vascular Comorbidity?
              (no answers)
            ...
          Histopathology
            Abdominal
              pT category: 1a
            ...
        </pre>
  </div>
</text>
<contained>
  <Patient>
    <id value="patsub"/>
    <identifier>
      <system value="http://cancer.questionnaire.org/systems/id/patientnr"/>
      <value value="A34442332"/>
    </identifier>
    <identifier>
      <type>
        <text value="Dutch BSN"/>
      </type>
      <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
      <value value="188912345"/>
    </identifier>
    <gender value="male"/>
    <birthDate value="1972-11-30"/>
  </Patient>
</contained>
<contained>
  <ServiceRequest>
    <id value="order"/>
    <status value="unknown"/>
    <intent value="order"/>
    <subject>
      <reference value="#patsub"/>
    </subject>
    <requester>
        <reference value="Practitioner/example"/>
    </requester>
  </ServiceRequest>
</contained>
<contained>
  <Practitioner>
    <id value="questauth"/>
    <identifier>
      <type>
        <text value="AUMC, Den Helder"/>
      </type>
      <system value="http://cancer.questionnaire.org/systems/id/org"/>
      <value value="AUMC"/>
    </identifier>
  </Practitioner>
</contained>
<!--   TODO: decide how to do this
<contained>

  <Observation>
    <id value="obs.pt-category"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="53786006"/>
        <display value="pT1 category"/>
      </coding>
    </code>
    <subject>
      <reference value="#patsub"/>
    </subject>
    <performer>
      <reference value="#questauth"/>
    </performer>
    <valueCodeableConcept>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="443357004"/>
        <display value="pT1a category"/>
      </coding>
    </valueCodeableConcept>
  </Observation>
</contained> -->
<identifier>
  <system value="http://example.org/fhir/NamingSystem/questionnaire-ids"/>
  <value value="Q12349876"/>
</identifier>
<basedOn>
  <reference value="#order"/>
</basedOn>
<partOf>
```

```
      <reference value="Procedure/f201"/>
    </partOf>
    <status value="completed"/>
    <subject>
      <reference value="#patsub"/>
    </subject>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <authored value="2013-02-19T14:15:00-05:00"/>
    <author>
      <reference value="#questauth"/>
    </author>
    <item>
      <!--   COMORBIDITY   -->
      <!--   First main section of the form, questions about comorbidity   -->
      <!--   section contains one question: whether there is comorbidity   -->
      <linkId value="1"/>
      <item>
        <linkId value="1.1"/>
        <answer>
          <valueCoding>
            <system value="http://cancer.questionnaire.org/system/code/yesno"/>
            <code value="1"/>
            <display value="Yes"/>
          </valueCoding>
          <item>
            <!--   COMORBIDITY/CARDIAL   -->
            <!--   Subsection about specific comorbidity: cardial   -->
            <linkId value="1.1.1"/>
            <item>
              <linkId value="1.1.1.1"/>
              <answer>
                <valueCoding>
                  <system value="http://cancer.questionnaire.org/system/code/yesno"/>
                  <code value="1"/>
                </valueCoding>
              </answer>
            </item>
            <item>
              <!--   This answer carries both the questionnaire-specific name and an equivalent SNOMED CT code   -->
              <linkId value="1.1.1.2"/>
              <answer>
                <valueCoding>
                  <system value="http://cancer.questionnaire.org/system/code/yesno"/>
                  <code value="1"/>
                </valueCoding>
              </answer>
            </item>
            <item>
              <linkId value="1.1.1.3"/>
              <answer>
                <valueCoding>

                  <system value="http://cancer.questionnaire.org/system/code/yesno"/>
                  <code value="0"/>
                </valueCoding>
              </answer>
            </item>
          </item>
        </answer>
      </item>
    </item>
    <!--   HISTOPATHOLOGY   -->
    <!--   TODO: decide how to do this
    <item>

      <item>
        <item>
         <dataResource>
            <reference value="#obs.pt-category"/>
          </dataResource>
        </item>
      </item>
    </item>   -->
   </QuestionnaireResponse>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example1"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a servicerequest and to illustrate how to populate
  many of the task elements - this is the parent task that will be broken into subtask to grab the specimen
   example1 action lipid panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen
  collection subtask - completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected
  => example 6 action lipid panel - completed with output DiagnosticReport -->
    <id value="example1"/>
    <contained>
      <!-- using contained resource for the purpose of this example only typically this would be a separate
```

```
independent resource -->
    <Provenance>
      <id value="signature"/>
      <target>
        <reference value="ServiceRequest/physiotherapy/_history/1"/>
      </target>
      <recorded value="2016-10-31T08:25:05+10:00"/>
      <agent>
        <role>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
            <code value="AUT"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/f202"/>
          <display value="Luigi Maas"/>
        </who>
      </agent>
      <signature>
        <type>
          <system value="urn:iso-astm:E1762-95:2013"/>
          <code value="1.2.840.10065.1.12.1.1"/>
          <display value="Author's Signature"/>
        </type>
        <when value="2016-10-31T08:25:05+10:00"/>
        <who>
          <reference value="Practitioner/example"/>
          <display value="Dr Adam Careful"/>
        </who>
        <targetFormat value="application/fhir+xml"/>
        <sigFormat value="application/signature+xml"/>
        <data value="dGhpcyBibG9iIGlzIHNuaXBwZWQ="/>
      </signature>
    </Provenance>
  </contained>
  <identifier>
    <use value="official"/>
    <system value="http:/goodhealth.org/identifiers"/>
    <value value="20170201-001"/>
  </identifier>
  <basedOn>
    <display value="General Wellness Careplan"/>
  </basedOn>
  <groupIdentifier>
    <use value="official"/>
    <system value="http:/goodhealth.org/accession/identifiers"/>
    <value value="G20170201-001"/>
  </groupIdentifier>
  <status value="in-progress"/>
  <!--fictive business status populated for this example -->
  <businessStatus>
    <text value="waiting for specimen"/>
  </businessStatus>
  <intent value="order"/>
  <priority value="routine"/>
  <!--fictive code and description elements populated for this example -->
  <code>
    <text value="Lipid Panel"/>
  </code>
  <description value="Create order for getting specimen, Set up inhouse testing,  generate order for any sendouts
and submit with specimen"/>
  <!--  Use the lipid servicerequest example -->
  <focus>
    <reference value="ServiceRequest/lipid"/>
    <display value="Lipid Panel Request"/>
  </focus>
  <!--  who the task is for
 -->
  <for>
    <reference value="Patient/example"/>
    <display value="Peter James Chalmers"/>
  </for>
  <encounter>
    <reference value="Encounter/example"/>
    <display value="Example In-Patient Encounter"/>
  </encounter>
  <!--  Lots of dates -->
  <executionPeriod>
    <start value="2016-10-31T08:25:05+10:00"/>
  </executionPeriod>
  <authoredOn value="2016-10-31T08:25:05+10:00"/>
  <lastModified value="2016-10-31T09:45:05+10:00"/>
  <requester>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
  </requester>
  <performerType>
    <coding>
```

```
        <system value="http://terminology.hl7.org/CodeSystem/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <!--  populate the reason just to demonstrate it use in this example
   -->
    <reasonCode>
      <text value="The Task.reason should only be included if there is no Task.focus or if it differs from the reason
  indicated on the focus"/>
    </reasonCode>
    <note>
      <text value="This is an example to demonstrate using task for actioning a servicerequest and to illustrate how
  to populate many of the task elements - this is the parent task that will be broken into subtask to grab the
  specimen and a sendout lab test "/>
    </note>
    <!--  add the link to the signature on the task to show how this element is usedemonstrating in this example the
  relevant history element using a contained resource for the purpose of this example only typically would reference
  an external resource

   -->
    <relevantHistory>
      <reference value="#signature"/>
      <display value="Author's Signature"/>
    </relevantHistory>
    <!--  complete the request within 48 hrs
   -->
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-02T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a DiagnosticRequest and will generated once the task is near completion
  -->
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ClinicalImpression/example"/>
  <resource>
   <ClinicalImpression xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <identifier>
          <value value="12345"/>
     </identifier>
     <status value="completed"/>
     <description value="This 26 yo male patient is brought into ER by ambulance after being involved in a motor
  vehicle accident"/>
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <encounter>
       <reference value="Encounter/example"/>
     </encounter>
     <effectivePeriod>
         <start value="2014-12-06T20:00:00+11:00"/>
         <end value="2014-12-06T22:33:00+11:00"/>
     </effectivePeriod>
     <date value="2014-12-06T22:33:00+11:00"/>
     <assessor>
       <reference value="Practitioner/example"/>
     </assessor>
     <problem>
       <display value="MVA"/> <!-- todo: reference to condition with snomed code 418399005 : Motor vehicle accident
  (event) -->
     </problem>
     <investigation>
       <code>
         <text value="Initial Examination"/>
       </code>
       <!-- todo: turn all of these into observations -->
       <item>
         <display value="deep laceration of the scalp (left temporo-occipital)"/>
       </item>
       <item>
         <display value="decreased level of consciousness"/>
       </item>
       <item>
         <display value="disoriented to time and place"/>
       </item>
       <item>
         <display value="restless"/>
       </item>
```

```
     </investigation>
     <summary value="provisional diagnoses of laceration of head and traumatic brain injury (TBI)"/>
     <finding>
       <itemCodeableConcept>
         <coding>
           <system value="http://hl7.org/fhir/sid/icd-9"/>
           <code value="850.0"/>
         </coding>
       </itemCodeableConcept>
     </finding>
   </ClinicalImpression>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example2"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a filler generated servicerequest and to illustrate
   how to populate many of the task elements - this is the child task from task example1 to grab the specimen
   example1 action lipid panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen
  collection subtask - completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected
  with output Specimen => example 6 action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example2"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-002"/>
    </identifier>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <!--  This Task is a child task of the submitted lipid panel requet
   -->
    <partOf>
      <reference value="Task/example1"/>
      <display value="Lipid Panel"/>
    </partOf>
    <status value="accepted"/>
    <!--fictive business status populated for this example-->
    <businessStatus>
      <text value="waiting for patient"/>
    </businessStatus>
    <intent value="filler-order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Specimen Collection"/>
    </code>
    <!--  Filler generated order using a fictive  servicerequest to complete the loop-->
    <focus>
      <display value="BloodDraw ServiceRequest"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </encounter>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:45:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:45:05+10:00"/>
    <lastModified value="2016-10-31T09:45:05+10:00"/>
    <!--requester is filler-->
    <requester>
        <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
        <display value="Clinical Laboratory @ Acme Hospital"/>
    </requester>
    <performerType>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <!--assume at draw station-->
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <restriction>
```

```
      <repetitions value="1"/>
      <period>
        <end value="2016-11-01T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a Specimen and will generated once the task is completed -->
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/diabeticdiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc-->
   <!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
  --><id value="diabeticdiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <oralDiet>
      <!-- EH:  list type of diet showing standard code and local codes -->
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="160670007"/>
          <display value="Diabetic diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1030"/>
          <display value="DD - Diabetic diet"/>
        </coding>
        <text value="DD - Diabetic diet"/>
      </type>
      <!-- EH example of three times a day starting on 2/10 using schedule -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <frequency value="3"/>
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <!-- list nutrients and amounts here -->
      <nutrient>
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="2331003"/>
            <display value="Carbohydrate"/>
          </coding>
        </modifier>
        <amount>
          <value value="75"/>
```

```
          <unit value="grams"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="g"/>
        </amount>
      </nutrient>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/enteralbolus"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   Tube feedings are given in boluses several times a day   edited by Eric Haas Health eData Inc
   --><!--EH  Show all the elements in this examples for preference, and intolerance --><id value="enteralbolus"/>
    <identifier>
      <system value="http://www.acme.org/nutritionorders"/>
      <value value="123"/>
    </identifier>
    <status value="active"/><intent value="order"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <enteralFormula>

      <baseFormulaType>
        <coding>
          <!--EH:  bolus tube feeding instruction in text-->
          <system value="http://snomed.info/sct"/>
          <code value="442991000124104"/>
          <display value="Adult high protein formula"/>
        </coding>
      </baseFormulaType>
      <baseFormulaProductName value="Acme High Protein Formula"/>

      <additiveType>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/entformula-additive"/>
          <code value="lipid"/>
          <display value="Lipid"/>
        </coding>
      </additiveType>
      <additiveProductName value="Acme Lipid Additive"/>
      <caloricDensity>
        <value value="1.5"/>
        <unit value="calories per milliliter"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="cal/mL"/>
      </caloricDensity>
      <routeofAdministration>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration"/>
          <code value="GT"/>
          <display value="Instillation, gastrostomy tube"/>
        </coding>
      </routeofAdministration>
      <administration>
        <schedule>
          <repeat>
          <boundsPeriod>
            <start value="2014-09-17T16:00:00Z"/>
```

```
          </boundsPeriod>
            <frequency value="1"/>
            <period value="4"/>
            <periodUnit value="h"/>
          </repeat>
        </schedule>
        <quantity>
          <value value="240"/>
          <unit value="milliliters"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL"/>
        </quantity>
      </administration>
      <maxVolumeToDeliver>
        <value value="1440"/>
        <unit value="milliliter/day"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mL/d"/>
      </maxVolumeToDeliver>
      <administrationInstruction value="240 mls every 4hrs "/>
    </enteralFormula>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CommunicationRequest/example"/>
  <resource>
   <CommunicationRequest xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">To be filled out at a later time</div>
     </text>
     <status value="active"/>
     <!-- insert contents here -->
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <encounter>
    <reference value="Encounter/example"/>
     </encounter>
   </CommunicationRequest>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RequestGroup/example"/>
  <resource>
   <RequestGroup xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Example RequestGroup illustrating related actions to administer
  medications in sequence with time delay.</div>
     </text>
     <contained>
       <MedicationRequest>
         <id value="medicationrequest-1"/>
         <status value="unknown"/>
         <intent value="proposal"/>
         <medicationCodeableConcept>
           <text value="Medication 1"/>
         </medicationCodeableConcept>
         <subject>
           <reference value="Patient/example"/>
         </subject>
       </MedicationRequest>
     </contained>
     <contained>
       <MedicationRequest>
         <id value="medicationrequest-2"/>
         <status value="unknown"/>
         <intent value="proposal"/>
         <medicationCodeableConcept>
           <text value="Medication 2"/>
         </medicationCodeableConcept>
         <subject>
           <reference value="Patient/example"/>
         </subject>
       </MedicationRequest>
     </contained>
     <identifier>
       <value value="requestgroup-1"/>
     </identifier>
     <groupIdentifier>
       <system value="http://example.org/treatment-group"/>
       <value value="00001"/>
     </groupIdentifier>
     <status value="draft"/>
     <intent value="plan"/>
```

```
      <priority value="routine"/>
      <subject>
        <reference value="Patient/example"/>
      </subject>
      <encounter>
        <reference value="Encounter/example"/>
      </encounter>
      <authoredOn value="2017-03-06T17:31:00Z"/>
      <author>
        <reference value="Practitioner/1"/>
      </author>
      <reasonCode>
        <text value="Treatment"/>
      </reasonCode>
      <note>
        <text value="Additional notes about the request group"/>
      </note>
      <action>
        <prefix value="1"/>
        <title value="Administer Medications"/>
        <description value="Administer medications at the appropriate time"/>
        <textEquivalent value="Administer medication 1, followed an hour later by medication 2"/>
        <timingDateTime value="2017-03-06T19:00:00Z"/>
        <participant>
          <reference value="Practitioner/1"/>
        </participant>
        <groupingBehavior value="logical-group"/>
        <selectionBehavior value="all"/>
        <requiredBehavior value="must"/>
        <precheckBehavior value="yes"/>
        <cardinalityBehavior value="single"/>
        <action id="medication-action-1">
          <description value="Administer medication 1"/>
          <type>
         <coding><code value="create"/></coding>
          </type>
          <resource>
            <reference value="#medicationrequest-1"/>
          </resource>
        </action>
        <action id="medication-action-2">
          <description value="Administer medication 2"/>
          <relatedAction>
            <actionId value="medication-action-1"/>
            <relationship value="after-end"/>
            <offsetDuration>
              <value value="1"/>
              <unit value="h"/>
            </offsetDuration>
          </relatedAction>
          <type>
         <coding><code value="create"/></coding>
          </type>
          <resource>
            <reference value="#medicationrequest-2"/>
          </resource>
        </action>
      </action>
    </RequestGroup>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/List/example"/>
  <resource>
   <List xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <table>
           <thead>
             <tr>
               <th>Condition</th>
               <th>Severity</th>
               <th>Date</th>
               <th>Location</th>
               <th>Status</th>
             </tr>
           </thead>
           <tbody>
             <tr>
               <td>Burnt Ear</td>
               <td>Severe</td>
               <td>24-May 2012</td>
               <td>Left Ear</td>
               <td>deleted</td>
             </tr>
             <tr>
               <td>Asthma</td>
```

```
              <td>Mild</td>
              <td>21-Nov 2012</td>
              <td>--</td>
              <td>added</td>
            </tr>
          </tbody>
        </table>
      </div>
    </text>
    <identifier>
      <system value="urn:uuid:a9fcea7c-fcdf-4d17-a5e0-f26dda030b59"/>
      <value value="23974652"/>
    </identifier>
    <status value="current"/>
    <mode value="changes"/>
    <!-- This list doesn't have a code. In actual fact,
      it's a Condition list produced at the end of an encounter
      to a regular primary care practitioner. But the only way
      to know this is to hunt down the place it is used
      and find out -->
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <date value="2012-11-25T22:17:00+11:00"/>
    <source>
      <reference value="Patient/example"/>
    </source>
    <entry>
      <flag>
        <text value="Deleted due to error"/>
      </flag>
      <deleted value="true"/>
      <item>
        <reference value="Condition/example"/>
      </item>
    </entry>
    <entry>
      <flag>
        <text value="Added"/>
      </flag>
      <item>
        <reference value="Condition/example2"/>
      </item>
    </entry>
  </List>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example6"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a servicerequest and to illustrate how to populate
  many of the task elements this is the first in a series of related Task examples:  example1 action lipid panel -in
  progress =>. example2 specimen collection subtask -accepted => example4  specimen collection subtask - completed
  with output Specimen=>example 5 action lipid panel -in progress specimen collected with output Specimen => example 6
   action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example6"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-001"/>
    </identifier>
    <basedOn>
      <display value="General Wellness Careplan"/>
    </basedOn>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <status value="completed"/>
    <!--fictive business status populated for this example -->
    <businessStatus>
      <text value="test completed and posted"/>
    </businessStatus>
    <intent value="order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Lipid Panel"/>
    </code>
    <description value="Create order for getting specimen, Set up inhouse testing,  generate order for any sendouts
  and submit with specimen"/>
    <!--  Use the lipid servicerequest example -->
    <focus>
      <reference value="ServiceRequest/lipid"/>
```

```
      <display value="Lipid Panel Request"/>
    </focus>
    <!--  who the task is for
    -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </encounter>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:25:05+10:00"/>
      <!--added completion date-->
      <end value="2016-10-31T18:45:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:25:05+10:00"/>
    <!--  updated last modified date to track task activity-->
    <lastModified value="2016-10-31T18:45:05+10:00"/>
    <requester>
        <reference value="Practitioner/example"/>
        <display value="Dr Adam Careful"/>
    </requester>
    <performerType>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <!--  populate the reason just to demonstrate it use in this example
    -->
    <reasonCode>
      <text value="The Task.reason should only be included if there is no Task.focus or if it differs from the reason
  indicated on the focus"/>
    </reasonCode>
    <note>
      <text value="This is an example to demonstrate using task for actioning a servicerequest and to illustrate how
  to populate many of the task elements - this is the parent task that will be broken into subtask to grab the
  specimen and a sendout lab test "/>
    </note>
    <!--  add the link to the signature on the task to show how this element is usedemonstrating in this example the
  relevant history element using a contained resource for the purpose of this example only typically would reference
  an external resource

    -->
    <!--  complete the request within 48 hrs
    -->
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-02T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a DiagnosticReport and Specimen-->
    <output>
      <type>
        <text value="DiagnosticReport generated"/>
      </type>
      <valueReference>
        <reference value="DiagnosticReport/lipids"/>
      </valueReference>
    </output>
      <output>
      <type>
        <text value="collected specimen"/>
      </type>
      <valueReference>
        <reference value="Specimen/101"/>
      </valueReference>
    </output>

   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CommunicationRequest/fm-solicit"/>
  <resource>
   <CommunicationRequest xmlns="http://hl7.org/fhir">
     <id value="fm-solicit"/>
     <text>
       <status value="generated"/>
```

```
      <div xmlns="http://www.w3.org/1999/xhtml">Request for Accident Report</div>
    </text>
    <!-- insert contents here -->

      <contained>
      <Organization>
        <id value="provider"/>
        <identifier>
          <system value="http://www.jurisdiction.com/provideroffices"/>
          <value value="3456"/>
        </identifier>
      </Organization>
    </contained>

    <contained>
      <Organization>
        <id value="payor"/>
        <identifier>
          <system value="http://www.jurisdiction.com/insurer"/>
          <value value="123456"/>
        </identifier>
      </Organization>
    </contained>

    <contained>
      <Practitioner>
        <id value="requester"/>
        <identifier>
           <value value="6789"/>
        </identifier>
      </Practitioner>
    </contained>
    <!-- body of the resource -->

    <identifier>
      <system value="http://www.jurisdiction.com/insurer/123456"/>
      <value value="ABC123"/> <!-- this is the value to which the response will refer -->
    </identifier>
     <basedOn>
      <display value="EligibilityRequest"/>
    </basedOn>
     <replaces>
      <display value="prior CommunicationRequest"/>
    </replaces>
    <groupIdentifier>
         <value value="12345"/>
    </groupIdentifier>
    <status value="active"/>

    <category>
       <coding>
         <system value="http://acme.org/messagetypes"/>
         <code value="SolicitedAttachmentRequest"/>
       </coding>
    </category>
    <priority value="routine"/>
    <medium>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationMode"/>
        <code value="WRITTEN"/>
        <display value="written"/>
      </coding>
      <text value="written"/>
    </medium>

    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <payload>
      <contentString value="Please provide the accident report and any associated pictures to support your Claim#
  DEF5647."/>
    </payload>
    <occurrenceDateTime value="2016-06-10T11:01:10-08:00"/>
    <authoredOn value="2016-06-10T11:01:10-08:00"/>
    <requester>
     <reference value="#requester"/>
     </requester>
     <recipient>
        <reference value="#provider"/>
    </recipient>
     <sender>
       <reference value="#payor"/>
     </sender>


   </CommunicationRequest>
  </resource>
</entry>
<entry>
```

```
<fullUrl value="http://hl7.org/fhir/RiskAssessment/cardiac"/>
<resource>
 <RiskAssessment xmlns="http://hl7.org/fhir">
  <id value="cardiac"/>
   <text>
     <status value="additional"/>
     <div xmlns="http://www.w3.org/1999/xhtml">
       <table>
         <tr>
           <td colspan="2">
             <h1>Information about your risk score:</h1>
           </td>
         </tr>
         <tr>
           <td>
             <strong>Age:</strong>
           </td>
           <td>39</td>
         </tr>
         <tr>
           <td>
             <strong>Gender:</strong>
           </td>
           <td>male</td>
         </tr>
         <tr>
           <td>
             <strong>Total Cholesterol:</strong>
           </td>
           <td>244 mg/dL</td>
         </tr>
         <tr>
           <td>
             <strong>HDL Cholesterol:</strong>
           </td>
           <td>50 mg/dL
           </td>
         </tr>
         <tr>
           <td>
             <strong>Smoker:</strong>
           </td>
           <td>No</td>
         </tr>
         <tr>
           <td>
             <strong>Systolic Blood Pressure:</strong>
           </td>
           <td>107 mm/Hg</td>
         </tr>
         <tr>
           <td>
             <strong>On medication for HBP:</strong>
           </td>
           <td>No</td>
         </tr>
         <tr>
           <td valign="top">
             <b>Risk Score*</b>
           </td>
           <td>2%<br/>Means 2 of 100 people with this level of risk will have a heart attack in the next 10
years.</td>
         </tr>
         <tr>
           <td colspan="2">* Your risk score was calculated using an equation.  Other NCEP products, such as printed
 ATP III materials, use a point system to determine a risk score that is close to the equation score.</td>
         </tr>
         <tr>
           <td colspan="2">To find out what your risk score means and how to lower your risk for a heart attack, go
to<br/>
             <a href="http://www.nhlbi.nih.gov/health/public/heart/chol/hbc_what.htm">"High Blood Cholesterol—What
You Need to Know"</a>.<br/>
             <a href="http://www.nhlbi.nih.gov/health/public/heart/chol/chol_tlc.htm">Your Guide to Lowering
Cholesterol with Therapeutic Lifestyle Changes (TLC)</a><br/>
             <a href="http://www.nhlbi.nih.gov/health/public/heart/other/hhw/index.htm">The Healthy Heart Handbook
for Women</a>
           </td>
         </tr>
       </table>
     </div>
   </text>
   <identifier>
     <use value="official"/>
     <system value="http://example.org"/>
     <value value="risk-assessment-cardiac"/>
   </identifier>
   <status value="final"/>
   <subject>
     <reference value="Patient/pat2"/>
```

```
      </subject>
      <encounter>
        <reference value="Encounter/example"/>
      </encounter>
      <occurrenceDateTime value="2014-07-19T16:04:00Z"/>
      <performer>
        <display value="http://cvdrisk.nhlbi.nih.gov/#cholesterol"/>
      </performer>
      <basis>
        <reference value="Patient/pat2"/>
      </basis>
      <basis>
        <reference value="DiagnosticReport/lipids"/>
      </basis>
      <basis>
        <reference value="Observation/blood-pressure"/>
      </basis>
      <prediction>
        <outcome>
          <text value="Heart Attack"/>
        </outcome>
        <probabilityDecimal value="0.02"/>
        <whenRange>
          <low>
            <value value="39"/>
            <unit value="years"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="a"/>
          </low>
          <high>
            <value value="49"/>
            <unit value="years"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="a"/>
          </high>
        </whenRange>
      </prediction>
    </RiskAssessment>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/map-sitting"/>
  <resource>
    <Observation xmlns="http://hl7.org/fhir">
      <id value="map-sitting"/>

      <!--  To be used only when the body position in not precoordinated in the observation code.-->
      <extension url="http://hl7.org/fhir/StructureDefinition/observation-bodyPosition">
        <valueCodeableConcept>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="33586001"/>
            <display value="Sitting position (finding)"/>
          </coding>
        </valueCodeableConcept>
      </extension>
        <!--  The qualitative change in the value relative to the previous measurement. Usually only recorded if the
  change is clinically significant.-->
      <extension url="http://hl7.org/fhir/StructureDefinition/observation-delta">
        <valueCodeableConcept>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="1250004"/>
            <display value="Decreased (qualifier value)"/>
          </coding>
        </valueCodeableConcept>
      </extension>
      <status value="final"/>
      <category>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
          <code value="laboratory"/>
          <display value="Laboratory"/>
        </coding>
      </category>
      <code>
        <coding>
          <system value="http://loinc.org"/>
          <code value="8478-0"/>
          <display value="Mean blood pressure"/>
        </coding>
      </code>
      <subject>
        <reference value="Patient/example"/>
      </subject>
      <encounter>
        <reference value="Encounter/example"/>
      </encounter>
      <effectivePeriod>
```

```
      <start value="2018-04-02T10:30:10+01:00"/>
      <end value="2018-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2018-04-03T15:30:10+01:00"/>
    <valueQuantity>
      <value value="60"/>
      <unit value="mm Hg"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mm[Hg]"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v2-0078"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
      <text value="Below low normal"/>
    </interpretation>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ExplanationOfBenefit/EB3500"/>
  <resource>
   <ExplanationOfBenefit xmlns="http://hl7.org/fhir">
    <id value="EB3500"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">A human-readable rendering of the ExplanationOfBenefit</div>
    </text>

    <identifier>
      <system value="http://www.BenefitsInc.com/fhir/explanationofbenefit"/>
      <value value="987654321"/>
    </identifier>

    <status value="active"/>

    <type>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/claim-type"/>
        <code value="oral"/>
      </coding>
    </type>

    <use value="claim"/>

    <patient>
      <reference value="Patient/pat1"/>
    </patient>

    <created value="2014-08-16"/>

    <enterer>
      <reference value="Practitioner/1"/>
    </enterer>

    <insurer>
      <reference value="Organization/3"/>
    </insurer>

    <provider>
      <reference value="Practitioner/1"/>
    </provider>

    <payee>
      <type>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/payeetype"/>
          <code value="provider"/>
        </coding>
      </type>
      <party>
        <reference value="Organization/2"/>
      </party>
    </payee>

    <facility>
      <reference value="Location/1"/>
    </facility>

    <claim>
      <reference value="Claim/100150"/>
    </claim>

    <claimResponse>
      <reference value="ClaimResponse/R3500"/>
    </claimResponse>
```

```
<outcome value="complete"/>

<disposition value="Claim settled as per contract."/>

<careTeam>
  <sequence value="1"/>
  <provider>
    <reference value="Practitioner/example"/>
  </provider>
</careTeam>

<insurance>
  <focal value="true"/>
  <coverage>
    <reference value="Coverage/9876B1"/>
  </coverage>
</insurance>

<item>
  <sequence value="1"/>
  <careTeamSequence value="1"/>
  <productOrService>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/ex-USCLS"/>
      <code value="1205"/>
    </coding>
  </productOrService>
  <servicedDate value="2014-08-16"/>
  <unitPrice>
    <value value="135.57"/>
    <currency value="USD"/>
  </unitPrice>
  <net>
    <value value="135.57"/>
    <currency value="USD"/>
  </net>
  <udi>
    <reference value="Device/example"/>
  </udi>
  <encounter>
    <reference value="Encounter/example"/>
  </encounter>
  <adjudication>
    <category>
      <coding>
        <code value="eligible"/>
      </coding>
    </category>
    <amount>
      <value value="120.00"/>
      <currency value="USD"/>
    </amount>
  </adjudication>
  <adjudication>
    <category>
      <coding>
        <code value="eligpercent"/>
      </coding>
    </category>
    <value value="0.80"/>
  </adjudication>
  <adjudication>
    <category>
      <coding>
        <code value="benefit"/>
      </coding>
    </category>
    <amount>
      <value value="96.00"/>
      <currency value="USD"/>
    </amount>
  </adjudication>
</item>

<item>
  <sequence value="2"/>
  <careTeamSequence value="1"/>
  <productOrService>
    <coding>
      <code value="group"/>
    </coding>
  </productOrService>
  <servicedDate value="2014-08-16"/>
  <net>
    <value value="200.00"/>
    <currency value="USD"/>
  </net>
  <adjudication>
    <category>
```

```
      <coding>
        <code value="benefit"/>
      </coding>
    </category>
    <amount>
      <value value="180.00"/>
      <currency value="USD"/>
    </amount>
  </adjudication>
  <detail>
    <sequence value="1"/>
    <productOrService>
      <coding>
        <code value="group"/>
      </coding>
    </productOrService>
    <net>
      <value value="200.00"/>
      <currency value="USD"/>
    </net>
    <udi>
      <reference value="Device/example"/>
    </udi>
    <adjudication>
      <category>
        <coding>
          <code value="benefit"/>
        </coding>
      </category>
      <amount>
        <value value="180.00"/>
        <currency value="USD"/>
      </amount>
    </adjudication>
    <subDetail>
      <sequence value="1"/>
      <productOrService>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/ex-USCLS"/>
          <code value="1205"/>
        </coding>
      </productOrService>
      <unitPrice>
        <value value="200.00"/>
        <currency value="USD"/>
      </unitPrice>
      <net>
        <value value="200.00"/>
        <currency value="USD"/>
      </net>
      <udi>
        <reference value="Device/example"/>
      </udi>
      <adjudication>
        <category>
          <coding>
            <code value="eligible"/>
          </coding>
        </category>
        <amount>
          <value value="200.00"/>
          <currency value="USD"/>
        </amount>
      </adjudication>
      <adjudication>
        <category>
          <coding>
            <code value="eligpercent"/>
          </coding>
        </category>
        <value value="0.90"/>
      </adjudication>
      <adjudication>
        <category>
          <coding>
            <code value="benefit"/>
          </coding>
        </category>
        <amount>
          <value value="180.00"/>
          <currency value="USD"/>
        </amount>
      </adjudication>
    </subDetail>
  </detail>
</item>

<total>
  <category>
```

```
        <coding>
          <code value="submitted"/>
        </coding>
      </category>
      <amount>
        <value value="135.57"/>
        <currency value="USD"/>
      </amount>
    </total>

    <total>
      <category>
        <coding>
          <code value="benefit"/>
        </coding>
      </category>
      <amount>
        <value value="96.00"/>
        <currency value="USD"/>
      </amount>
    </total>

   </ExplanationOfBenefit>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Claim/960150"/>
  <resource>
   <Claim xmlns="http://hl7.org/fhir">
    <id value="960150"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">A human-readable rendering of the Claim</div>
    </text>

    <identifier>
      <system value="http://happyhospital.com/claim"/>
      <value value="9612345"/>
    </identifier>

    <status value="active"/>

    <type>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/claim-type"/>
        <code value="institutional"/>
      </coding>
    </type>

    <subType>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/ex-claimsubtype"/>
        <code value="emergency"/>
      </coding>
    </subType>

    <use value="claim"/>

    <patient>
      <reference value="Patient/1"/>
    </patient>

    <created value="2014-08-16"/>

    <enterer>
      <identifier>
        <system value="http://jurisdiction.org/facilities/HOSP1234/users"/>
        <value value="UC1234"/>
      </identifier>
    </enterer>

    <insurer>
      <reference value="Organization/2"/>
    </insurer>

    <provider>
      <reference value="Organization/1"/>
    </provider>

    <priority>
      <coding>
        <code value="normal"/>
      </coding>
    </priority>

    <payee>
      <type>
        <coding>
```

```
          <code value="provider"/>
        </coding>
      </type>
    </payee>

    <facility>
      <identifier>
        <system value="http://jurisdiction.org/facilities"/>
        <value value="HOSP1234"/>
      </identifier>
    </facility>

    <careTeam>
      <sequence value="1"/>
      <provider>
        <reference value="Practitioner/example"/>
      </provider>
    </careTeam>

    <diagnosis>
      <sequence value="1"/>
      <diagnosisCodeableConcept>
        <coding>
          <code value="654456"/>
        </coding>
      </diagnosisCodeableConcept>
    </diagnosis>

    <procedure>
      <sequence value="1"/>
      <type>
        <coding>
          <code value="primary"/>
        </coding>
      </type>
      <date value="2014-08-16"/>
      <procedureCodeableConcept>
        <coding>
          <code value="SDI9901"/>
        </coding>
        <text value="Subcutaneous diagnostic implant"/>
      </procedureCodeableConcept>
      <udi>
        <reference value="Device/example"/>
      </udi>
    </procedure>

    <insurance>
      <sequence value="1"/>
      <focal value="true"/>
      <coverage>
        <reference value="Coverage/9876B1"/>
      </coverage>
    </insurance>

    <item>
      <sequence value="1"/>
      <careTeamSequence value="1"/>
      <productOrService>
        <coding>
          <system value="http://hl7.org/fhir/ex-serviceproduct"/>
          <code value="exam"/>
        </coding>
      </productOrService>
      <servicedDate value="2014-08-16"/>
      <unitPrice>
        <value value="125.00"/>
        <currency value="USD"/>
      </unitPrice>
      <net>
        <value value="125.00"/>
        <currency value="USD"/>
      </net>
      <encounter>
        <reference value="Encounter/example"/>
      </encounter>
    </item>

    <total>
      <value value="125.00"/>
      <currency value="USD"/>
    </total>
   </Claim>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/diabeticsupplement"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
```

```
    edited by Eric Haas Health eData Inc

    --><!-- EH: use supplement element --><id value="diabeticsupplement"/>
     <identifier>
       <system value="http://goodhealthhospital.org/nutrition-requests"/>
       <value value="123"/>
     </identifier>
     <status value="active"/><intent value="order"/>
     <patient>
       <reference value="Patient/example"/>
       <display value="Peter Chalmers"/>
     </patient>
     <encounter>
       <reference value="Encounter/example"/>
       <display value="Inpatient"/>
     </encounter>
     <dateTime value="2014-09-17"/>
     <orderer>
       <reference value="Practitioner/example"/>
       <display value="Dr Adam Careful"/>
     </orderer>
     <allergyIntolerance>
       <reference value="AllergyIntolerance/example"/>
       <display value="Cashew Nuts"/>
     </allergyIntolerance>
     <foodPreferenceModifier>
       <coding>
         <system value="http://terminology.hl7.org/CodeSystem/diet"/>
         <code value="kosher"/>
       </coding>
     </foodPreferenceModifier>
     <excludeFoodModifier>
       <coding>
         <system value="http://snomed.info/sct"/>
         <version value="20140730"/>
         <code value="227493005"/>
         <display value="Cashew Nut"/>
       </coding>
     </excludeFoodModifier>
     <supplement>
       <type>
         <coding>
           <system value="http://snomed.info/sct"/>
           <code value="443051000124104"/>
           <display value="Adult diabetes specialty formula"/>
         </coding>
         <coding>
           <system value="http://goodhealthhospital.org/supplement-type-codes"/>
           <code value="1010"/>
           <display value="Adult diabetic formula"/>
         </coding>
         <text value="Adult diabetic formula"/>
       </type>
       <productName value="Glucerna"/>
       <!--EH:Glucerna Shake at 3pm and HS-->
       <schedule>
         <repeat>
           <boundsPeriod>
             <start value="2015-02-10T15:00:00Z"/>
           </boundsPeriod>
           <frequency value="1"/>
           <period value="24"/>
           <periodUnit value="h"/>
         </repeat>
       </schedule>
       <schedule>
         <repeat>
           <duration value="1"/>
           <durationUnit value="h"/>
           <when value="HS"/>
         </repeat>
       </schedule>
       <quantity>
         <value value="1"/>
         <unit value="8 oz bottle"/>
       </quantity>
       <!-- EH:  use the .instructions element to list diet  supplement order
       <instruction value="Glucerna Shake at 3pm and HS"/>
    -->
     </supplement></NutritionOrder>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/Task/example4"/>
   <resource>
    <Task xmlns="http://hl7.org/fhir">
     <!--This example show an updated status and output - a generated task to collect a specimen
    example1 action lipid panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen
   collection subtask - completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected
```

```
  with output Specimen => example 6 action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example4"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-002"/>
    </identifier>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <!--  This Task is a child task of the submitted lipid panel requet
   -->
    <partOf>
      <reference value="Task/example1"/>
      <display value="Lipid Panel"/>
    </partOf>
    <status value="completed"/>
    <!--fictive business status populated for this example-->
    <intent value="filler-order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Specimen Collection"/>
    </code>
    <!--  Filler generated order using a fictive  servicerequest to complete the loop-->
    <focus>
      <display value="BloodDraw ServiceRequest"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </encounter>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:45:05+10:00"/>
      <end value="2016-10-31T14:45:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:45:05+10:00"/>
    <lastModified value="2016-10-31T09:45:05+10:00"/>
    <!--requester is filler-->
    <requester>
        <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
        <display value="Clinical Laboratory @ Acme Hospital"/>
    </requester>
    <performerType>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <!--assume at draw station-->
    <owner>
      <reference value="Practitioner/f202"/>
      <display value="Luigi Maas"/>
    </owner>
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-01T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a Specimen and will generated once the task is completed -->
    <output>
      <type>
        <text value="collected specimen"/>
      </type>
      <valueReference>
        <reference value="Specimen/101"/>
      </valueReference>
    </output>
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/clinical-gender"/>
  <resource>
   <!-- Clinical Gender Observation example see http://build.fhir.org/patient.html#gender --><Observation
  xmlns="http://hl7.org/fhir">
     <id value="clinical-gender"/>
```

```
      <status value="final"/>
      <category>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
          <code value="social-history"/>
          <display value="Social History"/>
        </coding>
      </category>
      <code>
        <coding>
          <system value="http://loinc.org"/>
          <!-- 0..1 Identity of the terminology system -->
          <code value="76691-5"/>
          <!-- 0..1 Symbol in syntax defined by the system -->
          <display value="Gender identity"/>
          <!-- 0..1 Representation defined by the system -->
        </coding>
        <text value="Gender identity"/>
        <!-- 0..1 Plain text representation of the concept -->
      </code>
      <subject>
        <reference value="Patient/example"/>
      </subject>
      <encounter>
        <reference value="Encounter/example"/>
      </encounter>
      <effectiveDateTime value="2018-02-01"/>
      <performer>
        <!-- 0..* Reference(Practitioner|PractitionerRole|Organization|     CareTeam|Patient|RelatedPerson)Who is
    responsible for the observation -->
        <reference value="Encounter/example"/>
      </performer>
      <valueCodeableConcept>
        <coding>
          <system value="http://snomed.info/sct"/>
          <!-- 0..1 Identity of the terminology system -->
          <code value="191788006"/>
          <!-- 0..1 Symbol in syntax defined by the system -->
          <display value="Feminism in boy (finding)"/>
          <!-- 0..1 Representation defined by the system -->
        </coding>
        <text value="Feminism in boy"/>
        <!-- 0..1 Plain text representation of the concept -->
      </valueCodeableConcept>
    </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ServiceRequest/lipid"/>
  <resource>
   <ServiceRequest xmlns="http://hl7.org/fhir" xmlns:xhtml="http://www.w3.org/1999/xhtml">
    <id value="lipid"/>
    <contained>
      <Observation>
        <id value="fasting"/>
        <!--    the mandatory quality flag:    -->
        <status value="final"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="49541-6"/>
            <display value="Fasting status - Reported"/>
          </coding>
        </code>
        <subject>
          <reference value="Patient/example"/>
        </subject>
        <valueCodeableConcept>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v2-0136"/>
            <code value="Y"/>
            <display value="Yes"/>
          </coding>
        </valueCodeableConcept>
      </Observation>
    </contained>
    <contained>
      <Specimen>
        <id value="serum"/>
        <identifier>
          <system value="http://acme.org/specimens"/>
          <value value="20150107-0012"/>
        </identifier>
        <type>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="119364003"/>
            <display value="Serum sample"/>
          </coding>
```

```
        </type>
        <subject>
          <reference value="Patient/example"/>
        </subject>
        <collection>
          <collectedDateTime value="2015-08-16T06:40:17Z"/>
        </collection>
      </Specimen>
    </contained>
    <identifier>
      <type>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
          <code value="PLAC"/>
        </coding>
        <text value="Placer"/>
      </type>
      <system value="urn:oid:1.3.4.5.6.7"/>
      <value value="2345234234234"/>
    </identifier>
    <status value="active"/>
    <intent value="original-order"/>
    <code>
      <coding>
        <system value="http://acme.org/tests"/>
        <code value="LIPID"/>
      </coding>
      <text value="Lipid Panel"/>
    </code>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <occurrenceDateTime value="2013-05-02T16:16:00-07:00"/>
    <requester>
        <reference value="Practitioner/example"/>
    </requester>
    <performer>
      <reference value="Practitioner/f202"/>
    </performer>
    <reasonCode>
      <coding>
        <system value="http://hl7.org/fhir/sid/icd-9"/>
        <code value="V173"/>
        <display value="Fam hx-ischem heart dis"/>
      </coding>
    </reasonCode>
    <supportingInfo>
      <reference value="#fasting"/>
      <display value="Fasting status"/>
    </supportingInfo>
    <specimen>
      <reference value="#serum"/>
      <display value="Serum specimen"/>
    </specimen>
    <note>
      <text value="patient is afraid of needles"/>
    </note>
   </ServiceRequest>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Immunization/example"/>
  <resource>
   <!--Authored by Joginder Madra--><Immunization xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <identifier>
      <system value="urn:ietf:rfc:3986"/>
      <value value="urn:oid:1.3.6.1.4.1.21367.2005.3.7.1234"/>
    </identifier>
    <status value="completed"/>
    <vaccineCode>
      <coding>
        <system value="urn:oid:1.2.36.1.2001.1005.17"/>
        <code value="FLUVAX"/>
      </coding>
      <text value="Fluvax (Influenza)"/>
    </vaccineCode>
    <patient>
      <reference value="Patient/example"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <occurrenceDateTime value="2013-01-10"/>
    <primarySource value="true"/>
    <location>
```

```
      <reference value="Location/1"/>
    </location>
    <manufacturer>
      <reference value="Organization/hl7"/>
    </manufacturer>
    <lotNumber value="AAJN11K"/>
    <expirationDate value="2015-02-15"/>
    <site>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-ActSite"/>
        <code value="LA"/>
        <display value="left arm"/>
      </coding>
    </site>
    <route>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration"/>
        <code value="IM"/>
        <display value="Injection, intramuscular"/>
      </coding>
    </route>
    <doseQuantity>
      <value value="5"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mg"/>
    </doseQuantity>
    <performer>
      <function>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0443"/>
          <code value="OP"/>
        </coding>
      </function>
        <actor>
            <reference value="Practitioner/example"/>
        </actor>
    </performer>
    <performer>
      <function>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v2-0443"/>
          <code value="AP"/>
        </coding>
      </function>
        <actor>
            <reference value="Practitioner/example"/>
        </actor>
    </performer>
    <note>
      <text value="Notes on adminstration of vaccine"/>
    </note>
    <reasonCode>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="429060002"/>
      </coding>
    </reasonCode>
    <isSubpotent value="true"/>
    <education>
      <documentType value="25308869830001031112070 2"/>
      <publicationDate value="2012-07-02"/>
      <presentationDate value="2013-01-10"/>
    </education>
    <programEligibility>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-program-eligibility"/>
        <code value="ineligible"/>
      </coding>
    </programEligibility>
    <fundingSource>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/immunization-funding-source"/>
        <code value="private"/>
      </coding>
    </fundingSource>
   </Immunization>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/fiberrestricteddiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
   --><id value="fiberrestricteddiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
```

```
 <status value="active"/><intent value="order"/>
 <patient>
   <reference value="Patient/example"/>
   <display value="Peter Chalmers"/>
 </patient>
 <encounter>
   <reference value="Encounter/example"/>
   <display value="Inpatient"/>
 </encounter>
 <dateTime value="2014-09-17"/>
 <orderer>
   <reference value="Practitioner/example"/>
   <display value="Dr Adam Careful"/>
 </orderer>
 <allergyIntolerance>
   <reference value="AllergyIntolerance/example"/>
   <display value="Cashew Nuts"/>
 </allergyIntolerance>
 <foodPreferenceModifier>
   <coding>
     <system value="http://terminology.hl7.org/CodeSystem/diet"/>
     <code value="dairy-free"/>
   </coding>
 </foodPreferenceModifier>
 <excludeFoodModifier>
   <coding>
     <system value="http://snomed.info/sct"/>
     <version value="20140730"/>
     <code value="227493005"/>
     <display value="Cashew Nut"/>
   </coding>
 </excludeFoodModifier>
 <oralDiet>
   <!-- EH:  list type of diet showing standard code and local codes -->
   <type>
     <coding>
       <system value="http://snomed.info/sct"/>
       <code value="15108003"/>
       <display value="Restricted fiber diet"/>
     </coding>
     <coding>
       <system value="http://goodhealthhospital.org/diet-type-codes"/>
       <code value="1000"/>
       <display value="Fiber restricted"/>
     </coding>
     <text value="Fiber restricted diet"/>
   </type>
   <type>
     <coding>
       <system value="http://snomed.info/sct"/>
       <code value="16208003"/>
       <display value="Low fat diet"/>
     </coding>
     <coding>
       <system value="http://goodhealthhospital.org/diet-type-codes"/>
       <code value="1100"/>
       <display value="Low Fat"/>
     </coding>
     <text value="Low fat diet"/>
   </type>
   <!-- EH example of three times a day starting on 2/10  -->
   <schedule>
     <repeat>
        <boundsPeriod>
          <start value="2015-02-10"/>
        </boundsPeriod>
       <frequency value="3"/>
       <period value="1"/>
       <periodUnit value="d"/>
     </repeat>
   </schedule>
   <nutrient>
     <!-- list nutrients and amounts here -->
     <modifier>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="256674009"/>
         <display value="Fat"/>
       </coding>
     </modifier>
     <amount>
       <value value="50"/>
       <unit value="grams"/>
       <system value="http://unitsofmeasure.org"/>
       <code value="g"/>
     </amount>
   </nutrient>
 </oralDiet>
</NutritionOrder>
```

```
    </resource>
  </entry>
</Bundle>
```

For more information about operations, including how they are invoked, see Operations.

8.11.12 Resource Encounter - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/encounter-version-maps.html

No validation errors - all conversions are clean

8.11.12 Resource Encounter - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Encounter.

Functional status for this map: 10 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

8.11.12.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Encounter3to4" = "R3 to R4 Conversions for Encounter"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Encounter" alias EncounterR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Encounter" alias Encounter as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Encounter(source src : EncounterR3, target tgt : Encounter) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.statusHistory as vs0 -> tgt.statusHistory as vt0 then statusHistory(vs0, vt0);
  src.class -> tgt.class;
  src.classHistory as vs0 -> tgt.classHistory as vt0 then classHistory(vs0, vt0);
  src.type -> tgt.type;
  src.priority -> tgt.priority;
  src.subject -> tgt.subject;
  src.episodeOfCare -> tgt.episodeOfCare;
  src.incomingReferral -> tgt.basedOn;
  src.participant as vs0 -> tgt.participant as vt0 then participant(vs0, vt0);
  src.appointment -> tgt.appointment;
  src.period -> tgt.period;
  src.length -> tgt.length;
  src.reason -> tgt.reasonCode;
  src.diagnosis as vs0 -> tgt.diagnosis as vt0 then diagnosis(vs0, vt0);
  src.account -> tgt.account;
  src.hospitalization as vs0 -> tgt.hospitalization as vt0 then hospitalization(vs0, vt0);
  src.location as vs0 -> tgt.location as vt0 then location(vs0, vt0);
  src.serviceProvider -> tgt.serviceProvider;
  src.partOf -> tgt.partOf;
}

group statusHistory(source src, target tgt) extends BackboneElement {
  src.status -> tgt.status;
  src.period -> tgt.period;
}

group classHistory(source src, target tgt) extends BackboneElement {
  src.class -> tgt.class;
  src.period -> tgt.period;
}

group participant(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.period -> tgt.period;
  src.individual -> tgt.individual;
}

group diagnosis(source src, target tgt) extends BackboneElement {
  src.condition -> tgt.condition;
  src.role -> tgt.use;
  src.rank -> tgt.rank;
}

group hospitalization(source src, target tgt) extends BackboneElement {
  src.preAdmissionIdentifier -> tgt.preAdmissionIdentifier;
  src.origin -> tgt.origin;
  src.admitSource -> tgt.admitSource;
  src.reAdmission -> tgt.reAdmission;
  src.dietPreference -> tgt.dietPreference;
  src.specialCourtesy -> tgt.specialCourtesy;
  src.specialArrangement -> tgt.specialArrangement;
  src.destination -> tgt.destination;
  src.dischargeDisposition -> tgt.dischargeDisposition;
}

group location(source src, target tgt) extends BackboneElement {
```

```
  src.location -> tgt.location;
  src.status -> tgt.status;
  src.period -> tgt.period;
}
```

8.11.12.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Encounter4to3" = "R4 to R3 Conversion for Encounter"

uses "http://hl7.org/fhir/StructureDefinition/Encounter" alias Encounter as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Encounter" alias EncounterR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Encounter(source src : Encounter, target tgt : EncounterR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.statusHistory as vs0 -> tgt.statusHistory as vt0 then statusHistory(vs0, vt0);
  src.class -> tgt.class;
  src.classHistory as vs0 -> tgt.classHistory as vt0 then classHistory(vs0, vt0);
  src.type -> tgt.type;
  src.priority -> tgt.priority;
  src.subject -> tgt.subject;
  src.episodeOfCare -> tgt.episodeOfCare;
  src.basedOn -> tgt.incomingReferral;
  src.participant as vs0 -> tgt.participant as vt0 then participant(vs0, vt0);
  src.appointment -> tgt.appointment;
  src.period -> tgt.period;
  src.length -> tgt.length;
  src.reasonCode -> tgt.reason;
  src.diagnosis as vs0 -> tgt.diagnosis as vt0 then diagnosis(vs0, vt0);
  src.account -> tgt.account;
  src.hospitalization as vs0 -> tgt.hospitalization as vt0 then hospitalization(vs0, vt0);
  src.location as vs0 -> tgt.location as vt0 then location(vs0, vt0);
  src.serviceProvider -> tgt.serviceProvider;
  src.partOf -> tgt.partOf;
}

group statusHistory(source src, target tgt) extends BackboneElement {
  src.status -> tgt.status;
  src.period -> tgt.period;
}

group classHistory(source src, target tgt) extends BackboneElement {
  src.class -> tgt.class;
  src.period -> tgt.period;
}

group participant(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.period -> tgt.period;
  src.individual -> tgt.individual;
}

group diagnosis(source src, target tgt) extends BackboneElement {
  src.condition -> tgt.condition;
  src.use -> tgt.role;
  src.rank -> tgt.rank;
}

group hospitalization(source src, target tgt) extends BackboneElement {
  src.preAdmissionIdentifier -> tgt.preAdmissionIdentifier;
  src.origin -> tgt.origin;
  src.admitSource -> tgt.admitSource;
  src.reAdmission -> tgt.reAdmission;
  src.dietPreference -> tgt.dietPreference;
  src.specialCourtesy -> tgt.specialCourtesy;
  src.specialArrangement -> tgt.specialArrangement;
  src.destination -> tgt.destination;
  src.dischargeDisposition -> tgt.dischargeDisposition;
}

group location(source src, target tgt) extends BackboneElement {
  src.location -> tgt.location;
  src.status -> tgt.status;
  src.period -> tgt.period;
}
```

8.11.12.3 R4 Validation Errors

8.11.13 Encounter HL7 Extensions

URL: https://hl7.org/fhir/R4/encounter-extensions.html

Defines common extensions used with or related to the Encounter resource

8.11.13 Encounter HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.11.13.1 Content

Extensions:

encounter-reasonCancelled | reasonCancelled : If the encountered was cancelled after it was planned, why? Applies only if the status is cancelled.

encounter-associatedEncounter | associatedEncounter : This encounter occurs within the scope of the referenced encounter.

encounter-modeOfArrival | modeOfArrival : Identifies whether a patient arrives at the reporting facility via ambulance and the type of ambulance that was used.

8.11.14 JSON Schema for Encounter

URL: https://hl7.org/fhir/R4/encounter.schema.json.html

8.11.14 JSON Schema for Encounter

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Encounter, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Encounter",
  "$ref": "#/definitions/Encounter",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Encounter": {
      "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
    healthcare service(s) or assessing the health status of a patient.",
      "properties": {
        "resourceType": {
          "description": "This is a Encounter resource",
          "const": "Encounter"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
```

```
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier(s) by which this encounter is known.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "planned | arrived | triaged | in-progress | onleave | finished | cancelled +.",
      "enum": [
        "planned",
        "arrived",
        "triaged",
        "in-progress",
        "onleave",
        "finished",
        "cancelled",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusHistory": {
      "description": "The status history permits the encounter resource to contain the status history without
needing to read through the historical versions of the resource, or even have the server store them.",
      "items": {
        "$ref": "#/definitions/Encounter_StatusHistory"
      },
      "type": "array"
    },
    "class": {
      "description": "Concepts representing classification of patient encounter such as ambulatory (outpatient),
inpatient, emergency, home health or others due to local variations.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "classHistory": {
      "description": "The class history permits the tracking of the encounters transitions without needing to go
through the resource history.  This would be used for a case where an admission starts of as an emergency encounter,
 then transitions into an inpatient scenario. Doing this and not restarting a new encounter ensures that any
lab/diagnostic results can more easily follow the patient and not require re-processing and not get lost or
cancelled during a kind of discharge from emergency to inpatient.",
      "items": {
        "$ref": "#/definitions/Encounter_ClassHistory"
      },
      "type": "array"
    },
    "type": {
      "description": "Specific type of encounter (e.g. e-mail consultation, surgical day-care, skilled nursing,
rehabilitation).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "serviceType": {
      "description": "Broad categorization of the service that is to be provided (e.g. cardiology).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "priority": {
      "description": "Indicates the urgency of the encounter.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The patient or group present at the encounter.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "episodeOfCare": {
      "description": "Where a specific encounter should be classified as a part of a specific episode(s) of care
this field should be used. This association can facilitate grouping of related encounters together for a specific
purpose, such as government reporting, issue tracking, association via a common problem.  The association is
recorded on the encounter as these are typically created after the episode of care and grouped on entry rather than
```

```
editing the episode of care to append another encounter to it (the episode of care could span years).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "The request this encounter satisfies (e.g. incoming referral or procedure request).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "participant": {
      "description": "The list of people responsible for providing the service.",
      "items": {
        "$ref": "#/definitions/Encounter_Participant"
      },
      "type": "array"
    },
    "appointment": {
      "description": "The appointment that scheduled this encounter.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "period": {
      "description": "The start and end time of the encounter.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "length": {
      "description": "Quantity of time the encounter lasted. This excludes the time during leaves of absence.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "reasonCode": {
      "description": "Reason the encounter takes place, expressed as a code. For admissions, this can be used for a
coded admission diagnosis.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Reason the encounter takes place, expressed as a code. For admissions, this can be used for a
coded admission diagnosis.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "diagnosis": {
      "description": "The list of diagnosis relevant to this encounter.",
      "items": {
        "$ref": "#/definitions/Encounter_Diagnosis"
      },
      "type": "array"
    },
    "account": {
      "description": "The set of accounts that may be used for billing for this Encounter.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "hospitalization": {
      "description": "Details about the admission to a healthcare service.",
      "$ref": "#/definitions/Encounter_Hospitalization"
    },
    "location": {
      "description": "List of locations where  the patient has been during this encounter.",
      "items": {
        "$ref": "#/definitions/Encounter_Location"
      },
      "type": "array"
    },
    "serviceProvider": {
      "description": "The organization that is primarily responsible for this Encounter\u0027s services. This MAY be
 the same as the organization on the Patient record, however it could be different, such as if the actor performing
the services was from an external organization (which may be billed seperately) for an external consultation.  Refer
 to the example bundle showing an abbreviated set of Encounters for a colonoscopy.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "partOf": {
      "description": "Another Encounter of which this encounter is a part of (administratively or in time).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
```

```
  "required": [
    "class",
    "resourceType"
  ]
},
"Encounter_StatusHistory": {
  "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
healthcare service(s) or assessing the health status of a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "status": {
      "description": "planned | arrived | triaged | in-progress | onleave | finished | cancelled +.",
      "enum": [
        "planned",
        "arrived",
        "triaged",
        "in-progress",
        "onleave",
        "finished",
        "cancelled",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The time that the episode was in the specified status.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false,
  "required": [
    "period"
  ]
},
"Encounter_ClassHistory": {
  "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
healthcare service(s) or assessing the health status of a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
```

```
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "class": {
      "description": "inpatient | outpatient | ambulatory | emergency +.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "period": {
      "description": "The time that the episode was in the specified class.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false,
  "required": [
    "period",
    "class"
  ]
},
"Encounter_Participant": {
  "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
healthcare service(s) or assessing the health status of a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Role of participant in encounter.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "period": {
      "description": "The period of time that the specified participant participated in the encounter. These can
overlap or be sub-sets of the overall encounter\u0027s period.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "individual": {
      "description": "Persons involved in the encounter other than the patient.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"Encounter_Diagnosis": {
  "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
healthcare service(s) or assessing the health status of a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "condition": {
      "description": "Reason the encounter takes place, as specified using information from another resource. For
admissions, this is the admission diagnosis. The indication will typically be a Condition (with other resources
referenced in the evidence.detail), or a Procedure.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "use": {
      "description": "Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "rank": {
      "description": "Ranking of the diagnosis (for each role type).",
      "$ref": "#/definitions/positiveInt"
    },
    "_rank": {
      "description": "Extensions for rank",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "condition"
  ]
},
"Encounter_Hospitalization": {
  "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
healthcare service(s) or assessing the health status of a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "preAdmissionIdentifier": {
      "description": "Pre-admission identifier.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "origin": {
      "description": "The location/organization from which the patient came before admission.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "admitSource": {
      "description": "From where patient was admitted (physician referral, transfer).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
    },
    "reAdmission": {
      "description": "Whether this hospitalization is a readmission and why if known.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "dietPreference": {
      "description": "Diet preferences reported by the patient.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialCourtesy": {
      "description": "Special courtesies (VIP, board member).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialArrangement": {
      "description": "Any special requests that have been made for this hospitalization encounter, such as the
provision of specific equipment or other things.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "destination": {
      "description": "Location/organization to which the patient is discharged.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "dischargeDisposition": {
      "description": "Category or kind of location after discharge.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"Encounter_Location": {
  "description": "An interaction between a patient and healthcare provider(s) for the purpose of providing
healthcare service(s) or assessing the health status of a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "location": {
      "description": "The location where the encounter takes place.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "status": {
      "description": "The status of the participants\u0027 presence at the specified location during the period
specified. If the participant is no longer at the location, then the period will have an end date/time.",
      "enum": [
        "planned",
        "active",
        "reserved",
        "completed"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
        "physicalType": {
          "description": "This will be used to specify the required levels (bed/ward/room/etc.) desired to be recorded
    to simplify either messaging or query.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "period": {
          "description": "Time period during which the patient was present at the location.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false,
      "required": [
        "location"
      ]
    }
  }
}
```


8.11.15 Resource Encounter - Maturity Dependency View

URL: https://hl7.org/fhir/R4/encounter-dependencies.html

8.11.15 Resource Encounter - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for Encounter FMM level 2

Name | Card. | Type | Dependency Analysis
Encounter | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
statusHistory | 0..* | BackboneElement
status | 1..1 | code
period | 1..1 | Period
class | 1..1 | Coding
classHistory | 0..* | BackboneElement
class | 1..1 | Coding
period | 1..1 | Period
type | 0..* | CodeableConcept
serviceType | 0..1 | CodeableConcept
priority | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
episodeOfCare | 0..* | Reference(EpisodeOfCare)
basedOn | 0..* | Reference(ServiceRequest)
participant | 0..* | BackboneElement
type | 0..* | CodeableConcept
period | 0..1 | Period
individual | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson)
appointment | 0..* | Reference(Appointment)
period | 0..1 | Period
length | 0..1 | Duration
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Procedure | Observation | ImmunizationRecommendation) | Type Warning: (ImmunizationRecommendation = FMM1-Trial Use vs. Element = FMM2-Trial Use)
diagnosis | 0..* | BackboneElement
condition | 1..1 | Reference(Condition | Procedure)
use | 0..1 | CodeableConcept
rank | 0..1 | positiveInt
account | 0..* | Reference(Account)
hospitalization | 0..1 | BackboneElement
preAdmissionIdentifier | 0..1 | Identifier
origin | 0..1 | Reference(Location | Organization)
admitSource | 0..1 | CodeableConcept
reAdmission | 0..1 | CodeableConcept
dietPreference | 0..* | CodeableConcept
specialCourtesy | 0..* | CodeableConcept
specialArrangement | 0..* | CodeableConcept
destination | 0..1 | Reference(Location | Organization)
dischargeDisposition | 0..1 | CodeableConcept
location | 0..* | BackboneElement
location | 1..1 | Reference(Location)
status | 0..1 | code
physicalType | 0..1 | CodeableConcept
period | 0..1 | Period
serviceProvider | 0..1 | Reference(Organization)
partOf | 0..1 | Reference(Encounter)

8.11.16 Value Set Sc-valueset-encounter-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-encounter-status.html

8.11.16 Value Set Sc-valueset-encounter-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.11.17 EncounterStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-encounter-status)

Mapping from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Current state of the encounter."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
planned | equivalent | planned
triaged | equivalent | accepted
arrived | equivalent | arrived
in-progress | equivalent | active
onleave | equivalent | suspended
finished | equivalent | complete
cancelled | equivalent | abandoned
unknown | equivalent | unknown

8.11.18 Value Set Sc-valueset-encounter-location-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-encounter-location-status.html

8.11.18 Value Set Sc-valueset-encounter-location-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.11.19 EncounterLocationStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-encounter-location-status)

Mapping from http://hl7.org/fhir/ValueSet/encounter-location-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the location."

Source Code | Equivalence | Destination Code
planned | equivalent | planned
reserved | equivalent | accepted
active | equivalent | active
completed | equivalent | complete

8.12 Resource Account - Content

URL: https://hl7.org/fhir/R4/account.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

A financial tool for tracking value accrued for a particular purpose. In the healthcare field, used to track charges for a patient, cost centers, etc.

8.12.1 Scope and Usage

The Account resource acts as a central record against which charges, payments, and adjustments are applied. It contains information about which parties are responsible for payment of the account.

While the Account does conceptually have a balance, expressing that balance directly as a resource property is challenging due to the complexity of pricing contracts. An operation to retrieve the current balance of an

account is in consideration as future work.

8.12.2 Boundaries and Relationships

The Account itself does not include information about the charges, payments or adjustments, but rather those resources, such as ChargeItem point to the account to which they apply. Payment and adjustment resources have not yet been developed.

8.12.3 Background and Context

The Account resource can be considered a "bucket" to which ChargeItem resources are linked. These charges are processed by a billing system, which determines the responsible parties for the balance of the account. The billing system then submits claims or sends statements to the appropriate parties. Once payment is received, an adjustment is applied to the Account. The internal calculation of balances and allocation of responsibility is expected to be internal to the billing systems. Only the inputs and outputs of the billing process is communicated in the relevant financial FHIR resources.

This resource is referenced by itself, ChargeItem, Encounter, EpisodeOfCare and Invoice

8.12.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Account | TU | | DomainResource | Tracks balance, charges, for patient or cost center Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Account number
status | ?!Σ | 1..1 | code | active | inactive | entered-in-error | on-hold | unknown AccountStatus (Required)
type | Σ | 0..1 | CodeableConcept | E.g. patient, expense, depreciation Account Types (Example)
name | Σ | 0..1 | string | Human-readable label
subject | Σ | 0..* | Reference(Patient | Device | Practitioner | PractitionerRole | Location | HealthcareService | Organization) | The entity that caused the expenses
servicePeriod | Σ | 0..1 | Period | Transaction window
coverage | Σ | 0..* | BackboneElement | The party(s) that are responsible for covering the payment of this account, and what order should they be applied to the account
coverage | Σ | 1..1 | Reference(Coverage) | The party(s), such as insurances, that may contribute to the payment of this account
priority | Σ | 0..1 | positiveInt | The priority of the coverage in the context of this account
owner | Σ | 0..1 | Reference(Organization) | Entity managing the Account
description | Σ | 0..1 | string | Explanation of purpose/use
guarantor | | 0..* | BackboneElement | The parties ultimately responsible for balancing the Account
party | | 1..1 | Reference(Patient | RelatedPerson | Organization) | Responsible entity
onHold | | 0..1 | boolean | Credit or other hold applied
period | | 0..1 | Period | Guarantee account during
partOf | | 0..1 | Reference(Account) | Reference to a parent Account
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Account xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Account number --></identifier>
 <status value="[code]"/><!-- 1..1 active | inactive | entered-in-error | on-hold | unknown -->
 <type><!-- 0..1 CodeableConcept E.g. patient, expense, depreciation --></type>
 <name value="[string]"/><!-- 0..1 Human-readable label -->
 <subject><!-- 0..* Reference(Patient|Device|Practitioner|PractitionerRole|
   Location|HealthcareService|Organization) The entity that caused the expenses --></subject>
 <servicePeriod><!-- 0..1 Period Transaction window --></servicePeriod>
 <coverage>  <!-- 0..* The party(s) that are responsible for covering the payment of this account, and what order should
     they be applied to the account -->
  <coverage><!-- 1..1 Reference(Coverage) The party(s), such as insurances, that may contribute to the payment of this
    account --></coverage>
  <priority value="[positiveInt]"/><!-- 0..1 The priority of the coverage in the context of this account -->
 </coverage>
 <owner><!-- 0..1 Reference(Organization) Entity managing the Account --></owner>
 <description value="[string]"/><!-- 0..1 Explanation of purpose/use -->
 <guarantor>  <!-- 0..* The parties ultimately responsible for balancing the Account -->
  <party><!-- 1..1 Reference(Patient|RelatedPerson|Organization) Responsible entity --></party>
```

```
  <onHold value="[boolean]"/><!-- 0..1 Credit or other hold applied -->
  <period><!-- 0..1 Period Guarantee account during --></period>
 </guarantor>
 <partOf><!-- 0..1 Reference(Account) Reference to a parent Account --></partOf>
</Account>
```

JSON Template

```
{
  "resourceType" : "Account",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Account number
  "status" : "<code>", // R!  active | inactive | entered-in-error | on-hold | unknown
  "type" : { CodeableConcept }, // E.g. patient, expense, depreciation
  "name" : "<string>", // Human-readable label
  "subject" : [{ Reference(Patient|Device|Practitioner|PractitionerRole|
   Location|HealthcareService|Organization) }], // The entity that caused the expenses
  "servicePeriod" : { Period }, // Transaction window
  "coverage" : [{ // The party(s) that are responsible for covering the payment of this account, and what order should
    they be applied to the account
    "coverage" : { Reference(Coverage) }, // R!  The party(s), such as insurances, that may contribute to the payment of
     this account
    "priority" : "<positiveInt>" // The priority of the coverage in the context of this account
  }],
  "owner" : { Reference(Organization) }, // Entity managing the Account
  "description" : "<string>", // Explanation of purpose/use
  "guarantor" : [{ // The parties ultimately responsible for balancing the Account
    "party" : { Reference(Patient|RelatedPerson|Organization) }, // R!  Responsible entity
    "onHold" : <boolean>, // Credit or other hold applied
    "period" : { Period } // Guarantee account during
  }],
  "partOf" : { Reference(Account) } // Reference to a parent Account
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Account;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Account.identifier [ Identifier ], ... ; # 0..* Account number
  fhir:Account.status [ code ]; # 1..1 active | inactive | entered-in-error | on-hold | unknown
  fhir:Account.type [ CodeableConcept ]; # 0..1 E.g. patient, expense, depreciation
  fhir:Account.name [ string ]; # 0..1 Human-readable label
  fhir:Account.subject [ Reference(Patient|Device|Practitioner|PractitionerRole|Location|HealthcareService|
  Organization) ], ... ; # 0..* The entity that caused the expenses
  fhir:Account.servicePeriod [ Period ]; # 0..1 Transaction window
  fhir:Account.coverage [ # 0..* The party(s) that are responsible for covering the payment of this account, and what
    order should they be applied to the account
    fhir:Account.coverage.coverage [ Reference(Coverage) ]; # 1..1 The party(s), such as insurances, that may contribute
     to the payment of this account
    fhir:Account.coverage.priority [ positiveInt ]; # 0..1 The priority of the coverage in the context of this account
  ], ...;
  fhir:Account.owner [ Reference(Organization) ]; # 0..1 Entity managing the Account
  fhir:Account.description [ string ]; # 0..1 Explanation of purpose/use
  fhir:Account.guarantor [ # 0..* The parties ultimately responsible for balancing the Account
    fhir:Account.guarantor.party [ Reference(Patient|RelatedPerson|Organization) ]; # 1..1 Responsible entity
    fhir:Account.guarantor.onHold [ boolean ]; # 0..1 Credit or other hold applied
    fhir:Account.guarantor.period [ Period ]; # 0..1 Guarantee account during
  ], ...;
  fhir:Account.partOf [ Reference(Account) ]; # 0..1 Reference to a parent Account
]
```

Changes since R3

Account

Account.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/account-status to http://hl7.org/fhir/ValueSet/account-status|4.0.1

Account.subject | Max Cardinality changed from 1 to * Type Reference: Added Target Type PractitionerRole

Account.servicePeriod | Renamed from period to servicePeriod

Account.partOf | Added Element

Account.active | deleted

Account.balance | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Account | TU | | DomainResource | Tracks balance, charges, for patient or cost center Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Account number
status | ?!Σ | 1..1 | code | active | inactive | entered-in-error | on-hold | unknown AccountStatus (Required)
type | Σ | 0..1 | CodeableConcept | E.g. patient, expense, depreciation Account Types (Example)
name | Σ | 0..1 | string | Human-readable label
subject | Σ | 0..* | Reference(Patient | Device | Practitioner | PractitionerRole | Location | HealthcareService | Organization) | The entity that caused the expenses
servicePeriod | Σ | 0..1 | Period | Transaction window
coverage | Σ | 0..* | BackboneElement | The party(s) that are responsible for covering the payment of this account, and what order should they be applied to the account
coverage | Σ | 1..1 | Reference(Coverage) | The party(s), such as insurances, that may contribute to the payment of this account
priority | Σ | 0..1 | positiveInt | The priority of the coverage in the context of this account
owner | Σ | 0..1 | Reference(Organization) | Entity managing the Account
description | Σ | 0..1 | string | Explanation of purpose/use
guarantor | | 0..* | BackboneElement | The parties ultimately responsible for balancing the Account
party | | 1..1 | Reference(Patient | RelatedPerson | Organization) | Responsible entity
onHold | | 0..1 | boolean | Credit or other hold applied
period | | 0..1 | Period | Guarantee account during
partOf | | 0..1 | Reference(Account) | Reference to a parent Account
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Account xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Account number --></identifier>
 <status value="[code]"/><!-- 1..1 active | inactive | entered-in-error | on-hold | unknown -->
 <type><!-- 0..1 CodeableConcept E.g. patient, expense, depreciation --></type>
 <name value="[string]"/><!-- 0..1 Human-readable label -->
 <subject><!-- 0..* Reference(Patient|Device|Practitioner|PractitionerRole|
   Location|HealthcareService|Organization) The entity that caused the expenses --></subject>
 <servicePeriod><!-- 0..1 Period Transaction window --></servicePeriod>
 <coverage>  <!-- 0..* The party(s) that are responsible for covering the payment of this account, and what order should
     they be applied to the account -->
  <coverage><!-- 1..1 Reference(Coverage) The party(s), such as insurances, that may contribute to the payment of this
    account --></coverage>
  <priority value="[positiveInt]"/><!-- 0..1 The priority of the coverage in the context of this account -->
 </coverage>
 <owner><!-- 0..1 Reference(Organization) Entity managing the Account --></owner>
 <description value="[string]"/><!-- 0..1 Explanation of purpose/use -->
 <guarantor>  <!-- 0..* The parties ultimately responsible for balancing the Account -->
  <party><!-- 1..1 Reference(Patient|RelatedPerson|Organization) Responsible entity --></party>
  <onHold value="[boolean]"/><!-- 0..1 Credit or other hold applied -->
  <period><!-- 0..1 Period Guarantee account during --></period>
 </guarantor>
 <partOf><!-- 0..1 Reference(Account) Reference to a parent Account --></partOf>
</Account>
```

JSON Template

```
{
  "resourceType" : "Account",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Account number
  "status" : "<code>", // R!  active | inactive | entered-in-error | on-hold | unknown
  "type" : { CodeableConcept }, // E.g. patient, expense, depreciation
  "name" : "<string>", // Human-readable label
  "subject" : [{ Reference(Patient|Device|Practitioner|PractitionerRole|
   Location|HealthcareService|Organization) }], // The entity that caused the expenses
  "servicePeriod" : { Period }, // Transaction window
  "coverage" : [{ // The party(s) that are responsible for covering the payment of this account, and what order should
    they be applied to the account
    "coverage" : { Reference(Coverage) }, // R!  The party(s), such as insurances, that may contribute to the payment of
     this account
    "priority" : "<positiveInt>" // The priority of the coverage in the context of this account
  }],
  "owner" : { Reference(Organization) }, // Entity managing the Account
  "description" : "<string>", // Explanation of purpose/use
  "guarantor" : [{ // The parties ultimately responsible for balancing the Account
    "party" : { Reference(Patient|RelatedPerson|Organization) }, // R!  Responsible entity
    "onHold" : <boolean>, // Credit or other hold applied
    "period" : { Period } // Guarantee account during
  }],
  "partOf" : { Reference(Account) } // Reference to a parent Account
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Account;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
```

```
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Account.identifier [ Identifier ], ... ; # 0..* Account number
  fhir:Account.status [ code ]; # 1..1 active | inactive | entered-in-error | on-hold | unknown
  fhir:Account.type [ CodeableConcept ]; # 0..1 E.g. patient, expense, depreciation
  fhir:Account.name [ string ]; # 0..1 Human-readable label
  fhir:Account.subject [ Reference(Patient|Device|Practitioner|PractitionerRole|Location|HealthcareService|
  Organization) ], ... ; # 0..* The entity that caused the expenses
  fhir:Account.servicePeriod [ Period ]; # 0..1 Transaction window
  fhir:Account.coverage [ # 0..* The party(s) that are responsible for covering the payment of this account, and what
    order should they be applied to the account
    fhir:Account.coverage.coverage [ Reference(Coverage) ]; # 1..1 The party(s), such as insurances, that may contribute
     to the payment of this account
    fhir:Account.coverage.priority [ positiveInt ]; # 0..1 The priority of the coverage in the context of this account
  ], ...;
  fhir:Account.owner [ Reference(Organization) ]; # 0..1 Entity managing the Account
  fhir:Account.description [ string ]; # 0..1 Explanation of purpose/use
  fhir:Account.guarantor [ # 0..* The parties ultimately responsible for balancing the Account
    fhir:Account.guarantor.party [ Reference(Patient|RelatedPerson|Organization) ]; # 1..1 Responsible entity
    fhir:Account.guarantor.onHold [ boolean ]; # 0..1 Credit or other hold applied
    fhir:Account.guarantor.period [ Period ]; # 0..1 Guarantee account during
  ], ...;
  fhir:Account.partOf [ Reference(Account) ]; # 0..1 Reference to a parent Account
]
```

Changes since Release 3

Account
Account.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/account-status to http://hl7.org/fhir/ValueSet/account-status|4.0.1
Account.subject | Max Cardinality changed from 1 to * Type Reference: Added Target Type PractitionerRole
Account.servicePeriod | Renamed from period to servicePeriod
Account.partOf | Added Element
Account.active | deleted
Account.balance | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.12.4.1 Terminology Bindings

Path | Definition | Type | Reference
Account.status | Indicates whether the account is available to be used. | Required | AccountStatus
Account.type | The usage type of this account, permits categorization of accounts. | Example | AccountTypes

8.12.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | Account number | Account.identifier
name | string | Human-readable label | Account.name
owner | reference | Entity managing the Account | Account.owner (Organization)
patient | reference | The entity that caused the expenses | Account.subject.where(resolve() is Patient) (Patient)
period | date | Transaction window | Account.servicePeriod
status | token | active | inactive | entered-in-error | on-hold | unknown | Account.status
subject | reference | The entity that caused the expenses | Account.subject (Practitioner, Organization, Device, Patient, HealthcareService, PractitionerRole, Location)
type | token | E.g. patient, expense, depreciation | Account.type

8.12.6 Resource Account - Examples

URL: https://hl7.org/fhir/R4/account-examples.html

8.12.6 Resource Account - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Example Name | id | Format
General Account example | example | XML | JSON | Turtle
Example with a Guarantor | ewg | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.12.7 Resource Account - Detailed Descriptions

URL: https://hl7.org/fhir/R4/account-definitions.html

8.12.7 Resource Account - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Account resource.

Account
Element Id | Account
Definition | A financial tool for tracking value accrued for a particular purpose. In the healthcare field, used to track charges for a patient, cost centers, etc.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Cost center; Record
Account.identifier
Element Id | Account.identifier
Definition | Unique identifier used to reference the account. Might or might not be intended for human use (e.g. credit card number).
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Account.status
Element Id | Account.status
Definition | Indicates whether the account is presently used/usable or not.
Cardinality | 1..1
Terminology Binding | AccountStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes inactive and entered-in-error that mark the Account as not currently valid.
Account.type
Element Id | Account.type
Definition | Categorizes the account for reporting and searching purposes.
Cardinality | 0..1
Terminology Binding | Account Types (Example)
Type | CodeableConcept
Summary | true
Account.name
Element Id | Account.name
Definition | Name used for the account when displaying it to humans in reports, etc.
Cardinality | 0..1
Type | string
Alternate Names | Title
Summary | true
Account.subject
Element Id | Account.subject
Definition | Identifies the entity which incurs the expenses. While the immediate recipients of services or goods might be entities related to the subject, the expenses were ultimately incurred by the subject of the Account.
Cardinality | 0..*
Type | Reference(Patient | Device | Practitioner | PractitionerRole | Location | HealthcareService | Organization)
Alternate Names | target
Summary | true
Comments | Accounts can be applied to non-patients for tracking other non-patient related activities, such as group services (patients not tracked, and costs charged to another body), or might not be allocated.
Account.servicePeriod
Element Id | Account.servicePeriod
Definition | The date range of services associated with this account.
Cardinality | 0..1
Type | Period
Summary | true
Comments | It is possible for transactions to be posted outside the service period, as long as the service was provided within the defined service period.
Account.coverage
Element Id | Account.coverage
Definition | The party(s) that are responsible for covering the payment of this account, and what order should they be applied to the account.
Cardinality | 0..*
Summary | true

Comments | Typically. this may be some form of insurance, internal charges, or self-pay. Local or jurisdictional business rules may determine which coverage covers which types of billable items charged to the account, and in which order. Where the order is important, a local/jurisdictional extension may be defined to specify the order for the type of charge.
Account.coverage.coverage
Element Id | Account.coverage.coverage
Definition | The party(s) that contribute to payment (or part of) of the charges applied to this account (including self-pay). A coverage may only be responsible for specific types of charges, and the sequence of the coverages in the account could be important when processing billing.
Cardinality | 1..1
Type | Reference(Coverage)
Summary | true
Account.coverage.priority
Element Id | Account.coverage.priority
Definition | The priority of the coverage in the context of this account.
Cardinality | 0..1
Type | positiveInt
Alternate Names | coordination of benefit priority
Summary | true
Comments | It is common in some jurisdictions for there to be multiple coverages allocated to an account, and a sequence is required to order the settling of the account (often with insurance claiming).
Account.owner
Element Id | Account.owner
Definition | Indicates the service area, hospital, department, etc. with responsibility for managing the Account.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
Account.description
Element Id | Account.description
Definition | Provides additional information about what the account tracks and how it is used.
Cardinality | 0..1
Type | string
Summary | true
Account.guarantor
Element Id | Account.guarantor
Definition | The parties responsible for balancing the account if other payment options fall short.
Cardinality | 0..*
Account.guarantor.party
Element Id | Account.guarantor.party
Definition | The entity who is responsible.
Cardinality | 1..1
Type | Reference(Patient | RelatedPerson | Organization)
Account.guarantor.onHold
Element Id | Account.guarantor.onHold
Definition | A guarantor may be placed on credit hold or otherwise have their role temporarily suspended.
Cardinality | 0..1
Type | boolean
Account.guarantor.period
Element Id | Account.guarantor.period
Definition | The timeframe during which the guarantor accepts responsibility for the account.
Cardinality | 0..1
Type | Period
Account.partOf
Element Id | Account.partOf
Definition | Reference to a parent Account.
Cardinality | 0..1
Type | Reference(Account)

8.12.8 Resource Account - Mappings

URL: https://hl7.org/fhir/R4/account-mappings.html

8.12.8 Resource Account - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the account resource (see Mappings to Other Standards for further information & status).

8.12.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Account
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class
name | FiveWs.what[x]
subject | FiveWs.subject[x]
servicePeriod | FiveWs.done[x]

8.12.8.2 RIM Mapping (http://hl7.org/v3)

Account | Account
identifier | .id
status | .statusCode
type | .code
name | .title
subject | .subject
servicePeriod | .coverage.insurancePolicy.effectiveTime
coverage | .coverage
coverage | .coverage.insurancePolicy.author
priority | .coverage.sequenceNumber
owner | .holder
description | .subjectOf.observationEvent[code="annotation"].value
guarantor | .holder.guarantorRole
party | .holder.guarantorRole[guarantorPerson|guarantorOrganization]
onHold | .holder.guarantorRole.statusCode
period | .holder.guarantorRole.effectiveTime

8.12.9 Resource Account - Extensions & Profiles

URL: https://hl7.org/fhir/R4/account-profiles.html

8.12.9 Resource Account - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the Account resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.12.9.1 Profiles

No Profiles defined for this resource

8.12.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.12.9.3 Search Extensions

No Search Extensions defined for this resource

8.12.10 Resource Account - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/account-version-maps.html

No validation errors - all conversions are clean

8.12.10 Resource Account - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for Account.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid. (see documentation)

8.12.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Account3to4" = "R3 to R4 Conversions for Account"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Account" alias AccountR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Account" alias Account as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Account(source src : AccountR3, target tgt : Account) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.name -> tgt.name;
  src.subject -> tgt.subject;
  src.period -> tgt.servicePeriod;
  src.coverage as s -> tgt.coverage as t then AccountCoverage(s, t);
  src.owner -> tgt.owner;
  src.description -> tgt.description;
  src.guarantor as s -> tgt.guarantor as t then AccountGuarantor(s, t);
  src.partOf -> tgt.partOf;
}

group AccountCoverage(source src, target tgt) extends BackboneElement {
  src.coverage -> tgt.coverage;
  src.priority -> tgt.priority;
}

group AccountGuarantor(source src, target tgt) extends BackboneElement {
  src.party -> tgt.party;
  src.onHold -> tgt.onHold;
  src.period -> tgt.period;
}
```

8.12.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Account4to3" = "R4 to R3 Conversion for Account"

uses "http://hl7.org/fhir/StructureDefinition/Account" alias Account as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Account" alias AccountR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Account(source src : AccountR3, target tgt : Account) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.name -> tgt.name;
  src.subject -> tgt.subject;
  src.servicePeriod -> tgt.period;
  src.coverage as s -> tgt.coverage as t then AccountCoverage(s, t);
  src.owner -> tgt.owner;
  src.description -> tgt.description;
  src.guarantor as s -> tgt.guarantor as t then AccountGuarantor(s, t);
  src.partOf -> tgt.partOf;
}

group AccountCoverage(source src, target tgt) extends BackboneElement {
  src.coverage -> tgt.coverage;
  src.priority -> tgt.priority;
}

group AccountGuarantor(source src, target tgt) extends BackboneElement {
  src.party -> tgt.party;
  src.onHold -> tgt.onHold;
  src.period -> tgt.period;
}
```

8.12.10.3 R4 Validation Errors

8.12.11 JSON Schema for Account

URL: https://hl7.org/fhir/R4/account.schema.json.html

8.12.11 JSON Schema for Account

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for Account, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Account",
  "$ref": "#/definitions/Account",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
```

```
"definitions": {
  "Account": {
    "description": "A financial tool for tracking value accrued for a particular purpose.  In the healthcare field,
  used to track charges for a patient, cost centers, etc.",
    "properties": {
      "resourceType": {
        "description": "This is a Account resource",
        "const": "Account"
      },
      "id": {
        "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
   never changes.",
        "$ref": "id.schema.json#/definitions/id"
      },
      "meta": {
        "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
  Changes to the content might not always be associated with version changes to the resource.",
        "$ref": "Meta.schema.json#/definitions/Meta"
      },
      "implicitRules": {
        "description": "A reference to a set of rules that were followed when the resource was constructed, and which
  must be understood when processing the content. Often, this is a reference to an implementation guide that defines
  the special rules along with other profiles etc.",
        "$ref": "#/definitions/uri"
      },
      "_implicitRules": {
        "description": "Extensions for implicitRules",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "language": {
        "description": "The base language in which the resource is written.",
        "$ref": "#/definitions/code"
      },
      "_language": {
        "description": "Extensions for language",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "text": {
        "description": "A human-readable narrative that contains a summary of the resource and can be used to
  represent the content of the resource to a human. The narrative need not encode all the structured data, but is
  required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
  Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
        "$ref": "Narrative.schema.json#/definitions/Narrative"
      },
      "contained": {
        "description": "These resources do not have an independent existence apart from the resource that contains
  them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
        "items": {
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
   definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource and that modifies the understanding of the element that contains it and/or the understanding of the
  containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
  of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
  met as part of the definition of the extension. Applications processing a resource are required to check for
  modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
  DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "identifier": {
        "description": "Unique identifier used to reference the account.  Might or might not be intended for human use
   (e.g. credit card number).",
        "items": {
          "$ref": "Identifier.schema.json#/definitions/Identifier"
        },
        "type": "array"
      },
      "status": {
        "description": "Indicates whether the account is presently used/usable or not.",
        "enum": [
          "active",
          "inactive",
```

```
        "entered-in-error",
        "on-hold",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Categorizes the account for reporting and searching purposes.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "name": {
      "description": "Name used for the account when displaying it to humans in reports, etc.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subject": {
      "description": "Identifies the entity which incurs the expenses. While the immediate recipients of services or
 goods might be entities related to the subject, the expenses were ultimately incurred by the subject of the
Account.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "servicePeriod": {
      "description": "The date range of services associated with this account.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "coverage": {
      "description": "The party(s) that are responsible for covering the payment of this account, and what order
should they be applied to the account.",
      "items": {
        "$ref": "#/definitions/Account_Coverage"
      },
      "type": "array"
    },
    "owner": {
      "description": "Indicates the service area, hospital, department, etc. with responsibility for managing the
Account.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "description": {
      "description": "Provides additional information about what the account tracks and how it is used.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "guarantor": {
      "description": "The parties responsible for balancing the account if other payment options fall short.",
      "items": {
        "$ref": "#/definitions/Account_Guarantor"
      },
      "type": "array"
    },
    "partOf": {
      "description": "Reference to a parent Account.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Account_Coverage": {
  "description": "A financial tool for tracking value accrued for a particular purpose.  In the healthcare field,
used to track charges for a patient, cost centers, etc.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
```

```
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "coverage": {
      "description": "The party(s) that contribute to payment (or part of) of the charges applied to this account
(including self-pay).\n\nA coverage may only be responsible for specific types of charges, and the sequence of the
coverages in the account could be important when processing billing.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "priority": {
      "description": "The priority of the coverage in the context of this account.",
      "$ref": "#/definitions/positiveInt"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "coverage"
  ]
},
"Account_Guarantor": {
  "description": "A financial tool for tracking value accrued for a particular purpose.  In the healthcare field,
used to track charges for a patient, cost centers, etc.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "party": {
      "description": "The entity who is responsible.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onHold": {
      "description": "A guarantor may be placed on credit hold or otherwise have their role temporarily suspended.",
      "$ref": "#/definitions/boolean"
    },
    "_onHold": {
      "description": "Extensions for onHold",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The timeframe during which the guarantor accepts responsibility for the account.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false,
  "required": [
    "party"
```

```
      ]
    }
  }
}
```

8.12.12 Resource Account - Maturity Dependency View

URL: https://hl7.org/fhir/R4/account-dependencies.html

8.12.12 Resource Account - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for Account FMM level 2

Name | Card. | Type | Dependency Analysis
Account | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
type | 0..1 | CodeableConcept
name | 0..1 | string
subject | 0..* | Reference(Patient | Device | Practitioner | PractitionerRole | Location | HealthcareService | Organization)
servicePeriod | 0..1 | Period
coverage | 0..* | BackboneElement
coverage | 1..1 | Reference(Coverage)
priority | 0..1 | positiveInt
owner | 0..1 | Reference(Organization)
description | 0..1 | string
guarantor | 0..* | BackboneElement
party | 1..1 | Reference(Patient | RelatedPerson | Organization)
onHold | 0..1 | boolean
period | 0..1 | Period
partOf | 0..1 | Reference(Account)

8.12.13 Value Set Sc-valueset-account-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-account-status.html

8.12.13 Value Set Sc-valueset-account-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.12.14 AccountStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-account-status)

Mapping from http://hl7.org/fhir/ValueSet/account-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Indicates whether the account is available to be used."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
active | equivalent | active
on-hold | equivalent | suspended
inactive | equivalent | inactive
unknown | equivalent | unknown

8.13 Resource Flag - Content

URL: https://hl7.org/fhir/R4/flag.html
Patient Care Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Prospective warnings of potential issues when providing care to the patient.

8.13.1 Scope and Usage

A flag is a warning or notification of some sort presented to the user - who may be a clinician or some other person involve in patient care. It usually represents something of sufficient significance to warrant a special display of some sort - rather than just a note in the resource. A flag has a subject representing the resource that will trigger its display. This subject can be of different types, as described in the examples below:

• A note that a patient has an overdue account, which the provider may wish to discuss with them - in case of hardship for example (subject = Patient)

• An outbreak of Ebola in a particular region (subject=Location) so that all patients from that region have a higher risk of having that condition

• A particular provider is unavailable for referrals over a given period (subject = Practitioner)

• A patient who is enrolled in a clinical trial (subject=Group)

• Special guidance or caveats to be aware of when following a protocol (subject=PlanDefinition)

• Warnings about using a drug in a formulary requires special approval (subject=Medication)

• etc.

A flag is typically presented as a label in a prominent location in the record to notify the clinician of the potential issues, though it may also appear in other contexts; e.g. notes applicable to a radiology technician, or to a clinician performing a home visit. For patients, the information in the flag will often be derived from the record, and therefore, for a thorough and careful clinician, who has the time to review the notes will be redundant. However, given the volume of information frequently found in patients' records and the potentially serious consequences of losing sight of some facts, this redundancy is deemed appropriate. As well, some flags may reflect information not captured by any other resource in the record. (E.g. "Patient has large dog at home")

In line with its purpose, a flag is concise, highlighting a small set of high-priority issues among the much larger set of data in the chart. Readers who want more detail should consult the chart or other source of information. Caution should be exercised in creating Flag instances. If entries are created for information that could be gleaned in a sufficiently timely fashion by reviewing the patient record, the flag list will itself become overwhelming and will cease to serve its intended purpose.

Flags are expected to persist in a record for some period of time and are, at most, targeted to particular types of practitioners or to practitioners in particular system.

Examples of Patient related issues that might appear in flags:

• Risks to the patient (functional risk of falls, spousal restraining order, latex allergy)

• Patient's needs for special accommodations (hard of hearing, need for easy-open caps)

• Risks to providers (dog in house, patient may bite, infection control precautions)

• Administrative concerns (incomplete information, pre-payment required due to credit risk)

Examples of issues that should not appear only in flags:

• Potential allergy or drug interaction to planned therapy (use DetectedIssue)

• Known adverse reaction to a substance (use AllergyIntolerance)

Note that we include "latex allergy" in the "in scope" list, and "allergy" in the "not in scope" list. The Flag resource is not designed to replace the normal order checking process, and one should not expect to see all allergies in Flags. However, if there is an activity that might occur prior to careful evaluation of the record (e.g. donning of latex gloves) and that activity might pose a risk to the patient, that is the sort of eventuality the Flag is intended to support.

Specific guidelines about what type of information is appropriate to expose using Flag, as well as what categories of individuals should see particular flags, will vary by interoperability community.

8.13.2 Boundaries and Relationships

Flags may highlight a highly condensed view of information found in the AllergyIntolerance, Condition, Observation, Procedure and possibly other resources. A common extension allows the linkage of a Flag to the supporting detail resource. The purpose of these other resources is to provide detailed clinical information. The purpose of a Flag is to alert practitioners to information that is important to influence their interaction with a Patient prior to detailed review of the record.

Flags are not used to convey information to a specific individual or organization (e.g. an abnormal lab result reported to the ordering clinician, reporting of an adverse reaction to a regulatory authority). These are handled using the CommunicationRequest and the Communication resources.

Flags are not raised as a result of a reported or proposed action (e.g. drug-drug interactions, duplicate therapy warnings). These would be handled using DetectedIssue.

8.13.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Flag | TU | | DomainResource | Key information to flag to healthcare providers Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | active | inactive | entered-in-error FlagStatus (Required)
category | Σ | 0..* | CodeableConcept | Clinical, administrative, etc. Flag Category (Example)
code | Σ | 1..1 | CodeableConcept | Coded or textual message to display to user Flag Code (Example)
subject | Σ | 1..1 | Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure) | Who/What is flag about?
period | Σ | 0..1 | Period | Time period when flag is active
encounter | Σ | 0..1 | Reference(Encounter) | Alert relevant during encounter
author | Σ | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole) | Flag creator
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Flag xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 active | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Clinical, administrative, etc. --></category>
 <code><!-- 1..1 CodeableConcept Coded or textual message to display to user --></code>
 <subject><!-- 1..1 Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) Who/What is flag about? --></subject>
 <period><!-- 0..1 Period Time period when flag is active --></period>
 <encounter><!-- 0..1 Reference(Encounter) Alert relevant during encounter --></encounter>
 <author><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole) Flag creator --></author>
</Flag>
```

JSON Template

```
{
  "resourceType" : "Flag",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  active | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Clinical, administrative, etc.
  "code" : { CodeableConcept }, // R!  Coded or textual message to display to user
  "subject" : { Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) }, // R!  Who/What is flag about?
  "period" : { Period }, // Time period when flag is active
  "encounter" : { Reference(Encounter) }, // Alert relevant during encounter
  "author" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole) } // Flag creator
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Flag;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Flag.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Flag.status [ code ]; # 1..1 active | inactive | entered-in-error
  fhir:Flag.category [ CodeableConcept ], ... ; # 0..* Clinical, administrative, etc.
  fhir:Flag.code [ CodeableConcept ]; # 1..1 Coded or textual message to display to user
  fhir:Flag.subject [ Reference(Patient|Location|Group|Organization|Practitioner|PlanDefinition|Medication|
  Procedure) ]; # 1..1 Who/What is flag about?
  fhir:Flag.period [ Period ]; # 0..1 Time period when flag is active
  fhir:Flag.encounter [ Reference(Encounter) ]; # 0..1 Alert relevant during encounter
  fhir:Flag.author [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole) ]; # 0..1 Flag creator
]
```

Changes since R3

Flag
Flag.status | Change value set from http://hl7.org/fhir/ValueSet/flag-status to http://hl7.org/fhir/ValueSet/flag-status|4.0.1
Flag.category | Max Cardinality changed from 1 to *
Flag.author | Type Reference: Added Target Type PractitionerRole

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Flag | TU | | DomainResource | Key information to flag to healthcare providers Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | active | inactive | entered-in-error FlagStatus (Required)
category | Σ | 0..* | CodeableConcept | Clinical, administrative, etc. Flag Category (Example)
code | Σ | 1..1 | CodeableConcept | Coded or textual message to display to user Flag Code (Example)
subject | Σ | 1..1 | Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure) | Who/What is flag about?
period | Σ | 0..1 | Period | Time period when flag is active
encounter | Σ | 0..1 | Reference(Encounter) | Alert relevant during encounter
author | Σ | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole) | Flag creator
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Flag xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 active | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Clinical, administrative, etc. --></category>
 <code><!-- 1..1 CodeableConcept Coded or textual message to display to user --></code>
 <subject><!-- 1..1 Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) Who/What is flag about? --></subject>
 <period><!-- 0..1 Period Time period when flag is active --></period>
 <encounter><!-- 0..1 Reference(Encounter) Alert relevant during encounter --></encounter>
 <author><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole) Flag creator --></author>
</Flag>
```

JSON Template

```
{
  "resourceType" : "Flag",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  active | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Clinical, administrative, etc.
  "code" : { CodeableConcept }, // R!  Coded or textual message to display to user
  "subject" : { Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) }, // R!  Who/What is flag about?
  "period" : { Period }, // Time period when flag is active
  "encounter" : { Reference(Encounter) }, // Alert relevant during encounter
  "author" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole) } // Flag creator
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Flag;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Flag.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Flag.status [ code ]; # 1..1 active | inactive | entered-in-error
  fhir:Flag.category [ CodeableConcept ], ... ; # 0..* Clinical, administrative, etc.
  fhir:Flag.code [ CodeableConcept ]; # 1..1 Coded or textual message to display to user
  fhir:Flag.subject [ Reference(Patient|Location|Group|Organization|Practitioner|PlanDefinition|Medication|
  Procedure) ]; # 1..1 Who/What is flag about?
  fhir:Flag.period [ Period ]; # 0..1 Time period when flag is active
  fhir:Flag.encounter [ Reference(Encounter) ]; # 0..1 Alert relevant during encounter
  fhir:Flag.author [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole) ]; # 0..1 Flag creator
]
```

Changes since Release 3

Flag
Flag.status | Change value set from http://hl7.org/fhir/ValueSet/flag-status to http://hl7.org/fhir/ValueSet/flag-status|4.0.1
Flag.category | Max Cardinality changed from 1 to *
Flag.author | Type Reference: Added Target Type PractitionerRole

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.13.3.1 Terminology Bindings

Path | Definition | Type | Reference
Flag.status | Indicates whether this flag is active and needs to be displayed to a user, or whether it is no longer needed or was entered in error. | Required | FlagStatus
Flag.category | A general category for flags for filtering/display purposes. | Example | FlagCategory
Flag.code | Detail codes identifying specific flagged issues. | Example | FlagCode

8.13.4 Notes

The Flag resource is sometimes used as "patient notes" and MAY be used to warn of issues such as:

• Issues that impact on the patient's ability to receive/respond to care the care provision process itself (e.g., poor language comprehension, low compliance expected)

• Issues that impact on the ability to provide care (e.g., patient has a big dog at home)

• Financial matters (e.g., patient is a bad debtor)

8.13.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | Flag creator | Flag.author (Practitioner, Organization, Device, Patient, PractitionerRole)
date | date | Time period when flag is active | Flag.period | 17 Resources
encounter | reference | Alert relevant during encounter | Flag.encounter (Encounter) | 12 Resources
identifier | token | Business identifier | Flag.identifier
patient | reference | The identity of a subject to list flags for | Flag.subject.where(resolve() is Patient) (Patient) | 33 Resources
subject | reference | The identity of a subject to list flags for | Flag.subject (Practitioner, Group, Organization, Medication, Patient, PlanDefinition, Procedure, Location)

8.13.6 Resource Flag - Examples

URL: https://hl7.org/fhir/R4/flag-examples.html

8.13.6 Resource Flag - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Example Name | id | Format
Example of flag | example | XML | JSON | Turtle
Encounter Specific Alert | example-encounter | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.13.7 Resource Flag - Detailed Descriptions

URL: https://hl7.org/fhir/R4/flag-definitions.html

8.13.7 Resource Flag - Detailed Descriptions

Patient Care Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Flag resource.

Flag
Element Id | Flag
Definition | Prospective warnings of potential issues when providing care to the patient.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Barriers to Care; Alert; Warning
Flag.identifier
Element Id | Flag.identifier
Definition | Business identifiers assigned to this flag by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the flag as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
Flag.status
Element Id | Flag.status

Definition | Supports basic workflow.
Cardinality | 1..1
Terminology Binding | FlagStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Flag.category
Element Id | Flag.category
Definition | Allows a flag to be divided into different categories like clinical, administrative etc. Intended to be used as a means of filtering which flags are displayed to particular user or in a given context.
Cardinality | 0..*
Terminology Binding | Flag Category (Example)
Type | CodeableConcept
Summary | true
Comments | The value set will often need to be adjusted based on local business rules and usage context.
Flag.code
Element Id | Flag.code
Definition | The coded value or textual component of the flag to display to the user.
Cardinality | 1..1
Terminology Binding | Flag Code (Example)
Type | CodeableConcept
Summary | true
Comments | If non-coded, use CodeableConcept.text. This element should always be included in the narrative.
Flag.subject
Element Id | Flag.subject
Definition | The patient, location, group, organization, or practitioner etc. this is about record this flag is associated with.
Cardinality | 1..1
Type | Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure)
Summary | true
Flag.period
Element Id | Flag.period
Definition | The period of time from the activation of the flag to inactivation of the flag. If the flag is active, the end of the period should be unspecified.
Cardinality | 0..1
Type | Period
Summary | true
Flag.encounter
Element Id | Flag.encounter
Definition | This alert is only relevant during the encounter.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | If both Flag.encounter and Flag.period are valued, then Flag.period.start shall not be before Encounter.period.start and Flag.period.end shall not be after Encounter.period.end.
Flag.author
Element Id | Flag.author
Definition | The person, organization or device that created the flag.
Cardinality | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | PractitionerRole)
Summary | true

8.13.8 Resource Flag - Mappings

URL: https://hl7.org/fhir/R4/flag-mappings.html

8.13.8 Resource Flag - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the flag resource (see Mappings to Other Standards for further information & status).

8.13.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Flag
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class

code | FiveWs.what[x]
subject | FiveWs.subject[x]
period | FiveWs.context
encounter | FiveWs.context
author | FiveWs.author

8.13.8.2 RIM Mapping (http://hl7.org/v3)

Flag | Observation[classCode=ISSUE, moodCode=EVN]
identifier | .id
status | .status
category | .code
code | .value
subject | .participation[typeCode=SBJ].role[classCode=PAT | LOCE | ASSIGNED]
author | .participation[typeCode=AUT].role

8.13.9 Resource Flag - Extensions & Profiles

URL: https://hl7.org/fhir/R4/flag-profiles.html

8.13.9 Resource Flag - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the Flag resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.13.9.1 Profiles

No Profiles defined for this resource

8.13.9.2 Extensions

flag-detail | Flag | detail | for Flag HL7 Extensions
flag-priority | Flag | priority | for Flag HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.13.9.3 Search Extensions

No Search Extensions defined for this resource

8.13.10 Resource Flag - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/flag-version-maps.html
No validation errors - all conversions are clean

8.13.10 Resource Flag - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for Flag.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.13.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Flag3to4" = "R3 to R4 Conversions for Flag"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Flag" alias FlagR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Flag" alias Flag as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group Flag(source src : FlagR3, target tgt : Flag) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.period -> tgt.period;
  src.encounter -> tgt.encounter;
  src.author -> tgt.author;
}
```

8.13.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Flag4to3" = "R4 to R3 Conversion for Flag"

uses "http://hl7.org/fhir/StructureDefinition/Flag" alias Flag as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Flag" alias FlagR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Flag(source src : Flag, target tgt : FlagR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.period -> tgt.period;
  src.encounter -> tgt.encounter;
  src.author -> tgt.author;
}
```

8.13.10.3 R4 Validation Errors

8.13.11 Flag HL7 Extensions

URL: https://hl7.org/fhir/R4/flag-extensions.html

Defines common extensions used with or related to the Flag resource

8.13.11 Flag HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

8.13.11.1 Content

Extensions:

flag-detail | detail : Points to the Observation, AllergyIntolerance or other record that provides additional supporting information about this flag.

flag-priority | priority : A code that identifies the priority of the alert, for example the Alert Priority flags column in IHE PCD TF 2 Table B.8-4.

8.13.12 JSON Schema for Flag

URL: https://hl7.org/fhir/R4/flag.schema.json.html

8.13.12 JSON Schema for Flag

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for Flag, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Flag",
  "$ref": "#/definitions/Flag",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Flag": {
      "description": "Prospective warnings of potential issues when providing care to the patient.",
      "properties": {
        "resourceType": {
          "description": "This is a Flag resource",
          "const": "Flag"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
```

```
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this flag by the performer or other systems which remain
constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Supports basic workflow.",
      "enum": [
        "active",
        "inactive",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Allows a flag to be divided into different categories like clinical, administrative etc.
Intended to be used as a means of filtering which flags are displayed to particular user or in a given context.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "code": {
      "description": "The coded value or textual component of the flag to display to the user.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The patient, location, group, organization, or practitioner etc. this is about record this
flag is associated with.",
```

```
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "period": {
          "description": "The period of time from the activation of the flag to inactivation of the flag. If the flag is
    active, the end of the period should be unspecified.",
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "encounter": {
          "description": "This alert is only relevant during the encounter.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "author": {
          "description": "The person, organization or device that created the flag.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "code",
        "subject",
        "resourceType"
      ]
    }
  }
}
```


8.13.13 Resource Flag - Maturity Dependency View

URL: https://hl7.org/fhir/R4/flag-dependencies.html

8.13.13 Resource Flag - Maturity Dependency View

Patient Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for Flag FMM level 1

Name	Card.	Type	Dependency Analysis
Flag			DomainResource
identifier	0..*	Identifier	
status	1..1	code	
category	0..*	CodeableConcept	
code	1..1	CodeableConcept	
subject	1..1	Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure)	
period	0..1	Period	
encounter	0..1	Reference(Encounter)	
author	0..1	Reference(Device | Organization | Patient | Practitioner | PractitionerRole)	

8.13.14 Value Set Sc-valueset-flag-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-flag-status.html

8.13.14 Value Set Sc-valueset-flag-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.13.15 FlagStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-flag-status)

Mapping from http://hl7.org/fhir/ValueSet/flag-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Indicates whether this flag is active and needs to be displayed to a user, or whether it is no longer needed or was entered in error."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
active	equivalent	active
inactive	equivalent	inactive

8.14 Resource Device - Content

URL: https://hl7.org/fhir/R4/device.html

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

A type of a manufactured item that is used in the provision of healthcare without being substantially changed through that activity. The device may be a medical or non-medical device.

8.14.1 Scope and Usage

This resource is an administrative resource that tracks individual instances of a device and their location. It is referenced by other resources for recording which device performed an action such as a procedure or an observation, referenced when prescribing and dispensing devices for patient use or for ordering supplies, and used to record and transmit Unique Device Identifier (UDI) information about a device such as a patient's implant.

8.14.2 Boundaries and Relationships

These are the device related resources

• Device (this resource)

• DeviceDefinition - Describes a "kind" of device - not a physical instance, cut a "catalog entry" where a device is defined by the manufacturer, reseller, or regulator.

• DeviceMetric - Describes a measurement, calculation or setting capability of a medical device.

In FHIR, the "Device" is the "administrative" resource for the device (it does not change much and has manufacturer information etc.), whereas the DeviceComponent and DeviceMetric (which is really a kind of DeviceComponent) model the physical part, including operation status and is much more volatile. The physical composition of a Device is represented by the Devices pointing to their "parent".

Devices differ from medications because they are not "used up" - they remain active in a patient in an ongoing fashion. However, the specific boundary between medications and devices is defined at the implementation level and this standard does not enforce a boundary with the exception of devices that are implanted in a patient. The Medication resource should not be used to represent implanted devices.

This resource is referenced by Signature, Account, AdverseEvent, Appointment, AppointmentResponse, AuditEvent, CarePlan, CatalogEntry, ChargeItem, ChargeItemDefinition, Claim, Communication, CommunicationRequest, Composition, Consent, Contract, DetectedIssue, itself, DeviceMetric, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, ExplanationOfBenefit, Flag, Group, GuidanceResponse, ImagingStudy, Invoice, List, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MessageHeader, MolecularSequence, Observation, Procedure, Provenance, QuestionnaireResponse, RequestGroup, RiskAssessment, Schedule, ServiceRequest, Specimen, SupplyDelivery, SupplyRequest and Task

8.14.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Device | TU | | DomainResource | Item used in healthcare Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Instance identifier
definition | | 0..1 | Reference(DeviceDefinition) | The reference to the definition for the device
udiCarrier | Σ | 0..* | BackboneElement | Unique Device Identifier (UDI) Barcode string
deviceIdentifier | Σ | 0..1 | string | Mandatory fixed portion of UDI
issuer | | 0..1 | uri | UDI Issuing Organization
jurisdiction | | 0..1 | uri | Regional UDI authority
carrierAIDC | Σ | 0..1 | base64Binary | UDI Machine Readable Barcode String
carrierHRF | Σ | 0..1 | string | UDI Human Readable Barcode String
entryType | | 0..1 | code | barcode | rfid | manual + UDIEntryType (Required)
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error | unknown FHIRDeviceStatus (Required)
statusReason | | 0..* | CodeableConcept | online | paused | standby | offline | not-ready | transduc-discon | hw-discon | off FHIRDeviceStatusReason (Extensible)
distinctIdentifier | | 0..1 | string | The distinct identification string
manufacturer | | 0..1 | string | Name of device manufacturer
manufactureDate | | 0..1 | dateTime | Date when the device was made
expirationDate | | 0..1 | dateTime | Date and time of expiry of this device (if applicable)
lotNumber | | 0..1 | string | Lot number of manufacture
serialNumber | | 0..1 | string | Serial number assigned by the manufacturer
deviceName | | 0..* | BackboneElement | The name of the device as given by the manufacturer
name | | 1..1 | string | The name of the device
type | | 1..1 | code | udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-name | other DeviceNameType (Required)

modelNumber | | 0..1 | string | The model number for the device
partNumber | | 0..1 | string | The part number of the device
type | | 0..1 | CodeableConcept | The kind or type of device Device Type (Example)
specialization | | 0..* | BackboneElement | The capabilities supported on a device, the standards to which the device conforms for a particular purpose, and used for the communication
systemType | | 1..1 | CodeableConcept | The standard that is used to operate and communicate
version | | 0..1 | string | The version of the standard that is used to operate and communicate
version | | 0..* | BackboneElement | The actual design of the device or software version running on the device
type | | 0..1 | CodeableConcept | The type of the device version
component | | 0..1 | Identifier | A single component of the device version
value | | 1..1 | string | The version text
property | | 0..* | BackboneElement | The actual configuration settings of a device as it actually operates, e.g., regulation status, time properties
type | | 1..1 | CodeableConcept | Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
valueQuantity | | 0..* | Quantity | Property value as a quantity
valueCode | | 0..* | CodeableConcept | Property value as a code, e.g., NTP4 (synced to NTP)
patient | | 0..1 | Reference(Patient) | Patient to whom Device is affixed
owner | | 0..1 | Reference(Organization) | Organization responsible for device
contact | | 0..* | ContactPoint | Details for human/organization for support
location | | 0..1 | Reference(Location) | Where the device is found
url | | 0..1 | uri | Network address to contact device
note | | 0..* | Annotation | Device notes and comments
safety | Σ | 0..* | CodeableConcept | Safety Characteristics of Device
parent | | 0..1 | Reference(Device) | The parent device
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Device xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <definition><!-- 0..1 Reference(DeviceDefinition) The reference to the definition for the device --></definition>
 <udiCarrier>  <!-- 0..* Unique Device Identifier (UDI) Barcode string -->
  <deviceIdentifier value="[string]"/><!-- 0..1 Mandatory fixed portion of UDI -->
  <issuer value="[uri]"/><!-- 0..1 UDI Issuing Organization -->
  <jurisdiction value="[uri]"/><!-- 0..1 Regional UDI authority -->
  <carrierAIDC value="[base64Binary]"/><!-- 0..1 UDI Machine Readable Barcode String -->
  <carrierHRF value="[string]"/><!-- 0..1 UDI Human Readable Barcode String -->
  <entryType value="[code]"/><!-- 0..1 barcode | rfid | manual + -->
 </udiCarrier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error | unknown -->
 <statusReason><!-- 0..* CodeableConcept online | paused | standby | offline | not-ready | transduc-discon | hw-discon |
     off --></statusReason>
 <distinctIdentifier value="[string]"/><!-- 0..1 The distinct identification string -->
 <manufacturer value="[string]"/><!-- 0..1 Name of device manufacturer -->
 <manufactureDate value="[dateTime]"/><!-- 0..1 Date when the device was made -->
 <expirationDate value="[dateTime]"/><!-- 0..1 Date and time of expiry of this device (if applicable) -->
 <lotNumber value="[string]"/><!-- 0..1 Lot number of manufacture -->
 <serialNumber value="[string]"/><!-- 0..1 Serial number assigned by the manufacturer -->
 <deviceName>  <!-- 0..* The name of the device as given by the manufacturer -->
  <name value="[string]"/><!-- 1..1 The name of the device -->
  <type value="[code]"/><!-- 1..1 udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name |
     model-name | other -->
 </deviceName>
 <modelNumber value="[string]"/><!-- 0..1 The model number for the device -->
 <partNumber value="[string]"/><!-- 0..1 The part number of the device -->
 <type><!-- 0..1 CodeableConcept The kind or type of device --></type>
 <specialization>  <!-- 0..* The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication -->
  <systemType><!-- 1..1 CodeableConcept The standard that is used to operate and communicate --></systemType>
  <version value="[string]"/><!-- 0..1 The version of the standard that is used to operate and communicate -->
 </specialization>
 <version>  <!-- 0..* The actual design of the device or software version running on the device -->
  <type><!-- 0..1 CodeableConcept The type of the device version --></type>
  <component><!-- 0..1 Identifier A single component of the device version --></component>
  <value value="[string]"/><!-- 1..1 The version text -->
 </version>
 <property>  <!-- 0..* The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties -->
  <type><!-- 1..1 CodeableConcept Code that specifies the property DeviceDefinitionPropetyCode (Extensible) --></type>
  <valueQuantity><!-- 0..* Quantity Property value as a quantity --></valueQuantity>
  <valueCode><!-- 0..* CodeableConcept Property value as a code, e.g., NTP4 (synced to NTP) --></valueCode>
 </property>
 <patient><!-- 0..1 Reference(Patient) Patient to whom Device is affixed --></patient>
 <owner><!-- 0..1 Reference(Organization) Organization responsible for device --></owner>
 <contact><!-- 0..* ContactPoint Details for human/organization for support --></contact>
 <location><!-- 0..1 Reference(Location) Where the device is found --></location>
 <url value="[uri]"/><!-- 0..1 Network address to contact device -->
 <note><!-- 0..* Annotation Device notes and comments --></note>
 <safety><!-- 0..* CodeableConcept Safety Characteristics of Device --></safety>
 <parent><!-- 0..1 Reference(Device) The parent device --></parent>
</Device>
```

JSON Template

```
{
  "resourceType" : "Device",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "definition" : { Reference(DeviceDefinition) }, // The reference to the definition for the device
  "udiCarrier" : [{ // Unique Device Identifier (UDI) Barcode string
    "deviceIdentifier" : "<string>", // Mandatory fixed portion of UDI
    "issuer" : "<uri>", // UDI Issuing Organization
    "jurisdiction" : "<uri>", // Regional UDI authority
    "carrierAIDC" : "<base64Binary>", // UDI Machine Readable Barcode String
    "carrierHRF" : "<string>", // UDI Human Readable Barcode String
    "entryType" : "<code>" // barcode | rfid | manual +
  }],
  "status" : "<code>", // active | inactive | entered-in-error | unknown
  "statusReason" : [{ CodeableConcept }], // online | paused | standby | offline | not-ready | transduc-discon | hw-
    discon | off
  "distinctIdentifier" : "<string>", // The distinct identification string
  "manufacturer" : "<string>", // Name of device manufacturer
  "manufactureDate" : "<dateTime>", // Date when the device was made
  "expirationDate" : "<dateTime>", // Date and time of expiry of this device (if applicable)
  "lotNumber" : "<string>", // Lot number of manufacture
  "serialNumber" : "<string>", // Serial number assigned by the manufacturer
  "deviceName" : [{ // The name of the device as given by the manufacturer
    "name" : "<string>", // R!  The name of the device
    "type" : "<code>" // R!  udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-
    name | other
  }],
  "modelNumber" : "<string>", // The model number for the device
  "partNumber" : "<string>", // The part number of the device
  "type" : { CodeableConcept }, // The kind or type of device
  "specialization" : [{ // The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication
    "systemType" : { CodeableConcept }, // R!  The standard that is used to operate and communicate
    "version" : "<string>" // The version of the standard that is used to operate and communicate
  }],
  "version" : [{ // The actual design of the device or software version running on the device
    "type" : { CodeableConcept }, // The type of the device version
    "component" : { Identifier }, // A single component of the device version
    "value" : "<string>" // R!  The version text
  }],
  "property" : [{ // The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties
    "type" : { CodeableConcept }, // R!  Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
    "valueQuantity" : [{ Quantity }], // Property value as a quantity
    "valueCode" : [{ CodeableConcept }] // Property value as a code, e.g., NTP4 (synced to NTP)
  }],
  "patient" : { Reference(Patient) }, // Patient to whom Device is affixed
  "owner" : { Reference(Organization) }, // Organization responsible for device
  "contact" : [{ ContactPoint }], // Details for human/organization for support
  "location" : { Reference(Location) }, // Where the device is found
  "url" : "<uri>", // Network address to contact device
  "note" : [{ Annotation }], // Device notes and comments
  "safety" : [{ CodeableConcept }], // Safety Characteristics of Device
  "parent" : { Reference(Device) } // The parent device
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Device;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Device.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:Device.definition [ Reference(DeviceDefinition) ]; # 0..1 The reference to the definition for the device
  fhir:Device.udiCarrier [ # 0..* Unique Device Identifier (UDI) Barcode string
    fhir:Device.udiCarrier.deviceIdentifier [ string ]; # 0..1 Mandatory fixed portion of UDI
    fhir:Device.udiCarrier.issuer [ uri ]; # 0..1 UDI Issuing Organization
    fhir:Device.udiCarrier.jurisdiction [ uri ]; # 0..1 Regional UDI authority
    fhir:Device.udiCarrier.carrierAIDC [ base64Binary ]; # 0..1 UDI Machine Readable Barcode String
    fhir:Device.udiCarrier.carrierHRF [ string ]; # 0..1 UDI Human Readable Barcode String
    fhir:Device.udiCarrier.entryType [ code ]; # 0..1 barcode | rfid | manual +
  ], ...;
  fhir:Device.status [ code ]; # 0..1 active | inactive | entered-in-error | unknown
  fhir:Device.statusReason [ CodeableConcept ], ... ; # 0..* online | paused | standby | offline | not-ready | transduc-
    discon | hw-discon | off
  fhir:Device.distinctIdentifier [ string ]; # 0..1 The distinct identification string
  fhir:Device.manufacturer [ string ]; # 0..1 Name of device manufacturer
  fhir:Device.manufactureDate [ dateTime ]; # 0..1 Date when the device was made
  fhir:Device.expirationDate [ dateTime ]; # 0..1 Date and time of expiry of this device (if applicable)
  fhir:Device.lotNumber [ string ]; # 0..1 Lot number of manufacture
  fhir:Device.serialNumber [ string ]; # 0..1 Serial number assigned by the manufacturer
  fhir:Device.deviceName [ # 0..* The name of the device as given by the manufacturer
    fhir:Device.deviceName.name [ string ]; # 1..1 The name of the device
```

```
    fhir:Device.deviceName.type [ code ]; # 1..1 udi-label-name | user-friendly-name | patient-reported-name |
    manufacturer-name | model-name | other
  ], ...;
  fhir:Device.modelNumber [ string ]; # 0..1 The model number for the device
  fhir:Device.partNumber [ string ]; # 0..1 The part number of the device
  fhir:Device.type [ CodeableConcept ]; # 0..1 The kind or type of device
  fhir:Device.specialization [ # 0..* The capabilities supported on a  device, the standards to which the device
    conforms for a particular purpose, and used for the communication
    fhir:Device.specialization.systemType [ CodeableConcept ]; # 1..1 The standard that is used to operate and
    communicate
    fhir:Device.specialization.version [ string ]; # 0..1 The version of the standard that is used to operate and
    communicate
  ], ...;
  fhir:Device.version [ # 0..* The actual design of the device or software version running on the device
    fhir:Device.version.type [ CodeableConcept ]; # 0..1 The type of the device version
    fhir:Device.version.component [ Identifier ]; # 0..1 A single component of the device version
    fhir:Device.version.value [ string ]; # 1..1 The version text
  ], ...;
  fhir:Device.property [ # 0..* The actual configuration settings of a device as it actually operates, e.g., regulation
    status, time properties
    fhir:Device.property.type [ CodeableConcept ]; # 1..1 Code that specifies the property DeviceDefinitionPropetyCode
    (Extensible)
    fhir:Device.property.valueQuantity [ Quantity ], ... ; # 0..* Property value as a quantity
    fhir:Device.property.valueCode [ CodeableConcept ], ... ; # 0..* Property value as a code, e.g., NTP4 (synced to
    NTP)
  ], ...;
  fhir:Device.patient [ Reference(Patient) ]; # 0..1 Patient to whom Device is affixed
  fhir:Device.owner [ Reference(Organization) ]; # 0..1 Organization responsible for device
  fhir:Device.contact [ ContactPoint ], ... ; # 0..* Details for human/organization for support
  fhir:Device.location [ Reference(Location) ]; # 0..1 Where the device is found
  fhir:Device.url [ uri ]; # 0..1 Network address to contact device
  fhir:Device.note [ Annotation ], ... ; # 0..* Device notes and comments
  fhir:Device.safety [ CodeableConcept ], ... ; # 0..* Safety Characteristics of Device
  fhir:Device.parent [ Reference(Device) ]; # 0..1 The parent device
]
```

Changes since R3

Device
Device.definition | Added Element
Device.udiCarrier | Added Element
Device.udiCarrier.deviceIdentifier | Added Element
Device.udiCarrier.issuer | Added Element
Device.udiCarrier.jurisdiction | Added Element
Device.udiCarrier.carrierAIDC | Added Element
Device.udiCarrier.carrierHRF | Added Element
Device.udiCarrier.entryType | Added Element
Device.status | Change value set from http://hl7.org/fhir/ValueSet/device-status to http://hl7.org/fhir/ValueSet/device-status|4.0.1
Device.statusReason | Added Element
Device.distinctIdentifier | Added Element
Device.serialNumber | Added Element
Device.deviceName | Added Element
Device.deviceName.name | Added Mandatory Element
Device.deviceName.type | Added Mandatory Element
Device.modelNumber | Added Element
Device.partNumber | Added Element
Device.specialization | Added Element
Device.specialization.systemType | Added Mandatory Element
Device.specialization.version | Added Element
Device.version | Max Cardinality changed from 1 to * Type changed from string to BackboneElement
Device.version.type | Added Element
Device.version.component | Added Element
Device.version.value | Added Mandatory Element
Device.property | Added Element
Device.property.type | Added Mandatory Element
Device.property.valueQuantity | Added Element
Device.property.valueCode | Added Element
Device.parent | Added Element
Device.udi | deleted
Device.model | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Device | TU | | DomainResource | Item used in healthcare Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Instance identifier
definition | | 0..1 | Reference(DeviceDefinition) | The reference to the definition for the device
udiCarrier | Σ | 0..* | BackboneElement | Unique Device Identifier (UDI) Barcode string
deviceIdentifier | Σ | 0..1 | string | Mandatory fixed portion of UDI
issuer | | 0..1 | uri | UDI Issuing Organization
jurisdiction | | 0..1 | uri | Regional UDI authority
carrierAIDC | Σ | 0..1 | base64Binary | UDI Machine Readable Barcode String
carrierHRF | Σ | 0..1 | string | UDI Human Readable Barcode String
entryType | | 0..1 | code | barcode | rfid | manual + UDIEntryType (Required)
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error | unknown FHIRDeviceStatus (Required)
statusReason | | 0..* | CodeableConcept | online | paused | standby | offline | not-ready | transduc-discon | hw-discon | off FHIRDeviceStatusReason (Extensible)
distinctIdentifier | | 0..1 | string | The distinct identification string
manufacturer | | 0..1 | string | Name of device manufacturer
manufactureDate | | 0..1 | dateTime | Date when the device was made
expirationDate | | 0..1 | dateTime | Date and time of expiry of this device (if applicable)
lotNumber | | 0..1 | string | Lot number of manufacture
serialNumber | | 0..1 | string | Serial number assigned by the manufacturer
deviceName | | 0..* | BackboneElement | The name of the device as given by the manufacturer
name | | 1..1 | string | The name of the device
type | | 1..1 | code | udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-name | other DeviceNameType (Required)
modelNumber | | 0..1 | string | The model number for the device
partNumber | | 0..1 | string | The part number of the device
type | | 0..1 | CodeableConcept | The kind or type of device Device Type (Example)
specialization | | 0..* | BackboneElement | The capabilities supported on a device, the standards to which the device conforms for a particular purpose, and used for the communication
systemType | | 1..1 | CodeableConcept | The standard that is used to operate and communicate
version | | 0..1 | string | The version of the standard that is used to operate and communicate
version | | 0..* | BackboneElement | The actual design of the device or software version running on the device
type | | 0..1 | CodeableConcept | The type of the device version
component | | 0..1 | Identifier | A single component of the device version
value | | 1..1 | string | The version text
property | | 0..* | BackboneElement | The actual configuration settings of a device as it actually operates, e.g., regulation status, time properties
type | | 1..1 | CodeableConcept | Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
valueQuantity | | 0..* | Quantity | Property value as a quantity
valueCode | | 0..* | CodeableConcept | Property value as a code, e.g., NTP4 (synced to NTP)
patient | | 0..1 | Reference(Patient) | Patient to whom Device is affixed
owner | | 0..1 | Reference(Organization) | Organization responsible for device
contact | | 0..* | ContactPoint | Details for human/organization for support
location | | 0..1 | Reference(Location) | Where the device is found
url | | 0..1 | uri | Network address to contact device
note | | 0..* | Annotation | Device notes and comments
safety | Σ | 0..* | CodeableConcept | Safety Characteristics of Device
parent | | 0..1 | Reference(Device) | The parent device
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Device xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <definition><!-- 0..1 Reference(DeviceDefinition) The reference to the definition for the device --></definition>
 <udiCarrier>  <!-- 0..* Unique Device Identifier (UDI) Barcode string -->
  <deviceIdentifier value="[string]"/><!-- 0..1 Mandatory fixed portion of UDI -->
  <issuer value="[uri]"/><!-- 0..1 UDI Issuing Organization -->
  <jurisdiction value="[uri]"/><!-- 0..1 Regional UDI authority -->
  <carrierAIDC value="[base64Binary]"/><!-- 0..1 UDI Machine Readable Barcode String -->
  <carrierHRF value="[string]"/><!-- 0..1 UDI Human Readable Barcode String -->
  <entryType value="[code]"/><!-- 0..1 barcode | rfid | manual + -->
 </udiCarrier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error | unknown -->
 <statusReason><!-- 0..* CodeableConcept online | paused | standby | offline | not-ready | transduc-discon | hw-discon |
     off --></statusReason>
 <distinctIdentifier value="[string]"/><!-- 0..1 The distinct identification string -->
 <manufacturer value="[string]"/><!-- 0..1 Name of device manufacturer -->
 <manufactureDate value="[dateTime]"/><!-- 0..1 Date when the device was made -->
 <expirationDate value="[dateTime]"/><!-- 0..1 Date and time of expiry of this device (if applicable) -->
 <lotNumber value="[string]"/><!-- 0..1 Lot number of manufacture -->
 <serialNumber value="[string]"/><!-- 0..1 Serial number assigned by the manufacturer -->
 <deviceName>  <!-- 0..* The name of the device as given by the manufacturer -->
  <name value="[string]"/><!-- 1..1 The name of the device -->
  <type value="[code]"/><!-- 1..1 udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name |
    model-name | other -->
 </deviceName>
 <modelNumber value="[string]"/><!-- 0..1 The model number for the device -->
```

```
 <partNumber value="[string]"/><!-- 0..1 The part number of the device -->
 <type><!-- 0..1 CodeableConcept The kind or type of device --></type>
 <specialization>  <!-- 0..* The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication -->
  <systemType><!-- 1..1 CodeableConcept The standard that is used to operate and communicate --></systemType>
  <version value="[string]"/><!-- 0..1 The version of the standard that is used to operate and communicate -->
 </specialization>
 <version>  <!-- 0..* The actual design of the device or software version running on the device -->
  <type><!-- 0..1 CodeableConcept The type of the device version --></type>
  <component><!-- 0..1 Identifier A single component of the device version --></component>
  <value value="[string]"/><!-- 1..1 The version text -->
 </version>
 <property>  <!-- 0..* The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties -->
  <type><!-- 1..1 CodeableConcept Code that specifies the property DeviceDefinitionPropetyCode (Extensible) --></type>
  <valueQuantity><!-- 0..* Quantity Property value as a quantity --></valueQuantity>
  <valueCode><!-- 0..* CodeableConcept Property value as a code, e.g., NTP4 (synced to NTP) --></valueCode>
 </property>
 <patient><!-- 0..1 Reference(Patient) Patient to whom Device is affixed --></patient>
 <owner><!-- 0..1 Reference(Organization) Organization responsible for device --></owner>
 <contact><!-- 0..* ContactPoint Details for human/organization for support --></contact>
 <location><!-- 0..1 Reference(Location) Where the device is found --></location>
 <url value="[uri]"/><!-- 0..1 Network address to contact device -->
 <note><!-- 0..* Annotation Device notes and comments --></note>
 <safety><!-- 0..* CodeableConcept Safety Characteristics of Device --></safety>
 <parent><!-- 0..1 Reference(Device) The parent device --></parent>
</Device>
```

JSON Template

```
{
  "resourceType" : "Device",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "definition" : { Reference(DeviceDefinition) }, // The reference to the definition for the device
  "udiCarrier" : [{ // Unique Device Identifier (UDI) Barcode string
    "deviceIdentifier" : "<string>", // Mandatory fixed portion of UDI
    "issuer" : "<uri>", // UDI Issuing Organization
    "jurisdiction" : "<uri>", // Regional UDI authority
    "carrierAIDC" : "<base64Binary>", // UDI Machine Readable Barcode String
    "carrierHRF" : "<string>", // UDI Human Readable Barcode String
    "entryType" : "<code>" // barcode | rfid | manual +
  }],
  "status" : "<code>", // active | inactive | entered-in-error | unknown
  "statusReason" : [{ CodeableConcept }], // online | paused | standby | offline | not-ready | transduc-discon | hw-
    discon | off
  "distinctIdentifier" : "<string>", // The distinct identification string
  "manufacturer" : "<string>", // Name of device manufacturer
  "manufactureDate" : "<dateTime>", // Date when the device was made
  "expirationDate" : "<dateTime>", // Date and time of expiry of this device (if applicable)
  "lotNumber" : "<string>", // Lot number of manufacture
  "serialNumber" : "<string>", // Serial number assigned by the manufacturer
  "deviceName" : [{ // The name of the device as given by the manufacturer
    "name" : "<string>", // R!  The name of the device
    "type" : "<code>" // R!  udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-
    name | other
  }],
  "modelNumber" : "<string>", // The model number for the device
  "partNumber" : "<string>", // The part number of the device
  "type" : { CodeableConcept }, // The kind or type of device
  "specialization" : [{ // The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication
    "systemType" : { CodeableConcept }, // R!  The standard that is used to operate and communicate
    "version" : "<string>" // The version of the standard that is used to operate and communicate
  }],
  "version" : [{ // The actual design of the device or software version running on the device
    "type" : { CodeableConcept }, // The type of the device version
    "component" : { Identifier }, // A single component of the device version
    "value" : "<string>" // R!  The version text
  }],
  "property" : [{ // The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties
    "type" : { CodeableConcept }, // R!  Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
    "valueQuantity" : [{ Quantity }], // Property value as a quantity
    "valueCode" : [{ CodeableConcept }] // Property value as a code, e.g., NTP4 (synced to NTP)
  }],
  "patient" : { Reference(Patient) }, // Patient to whom Device is affixed
  "owner" : { Reference(Organization) }, // Organization responsible for device
  "contact" : [{ ContactPoint }], // Details for human/organization for support
  "location" : { Reference(Location) }, // Where the device is found
  "url" : "<uri>", // Network address to contact device
  "note" : [{ Annotation }], // Device notes and comments
  "safety" : [{ CodeableConcept }], // Safety Characteristics of Device
  "parent" : { Reference(Device) } // The parent device
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:Device;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Device.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:Device.definition [ Reference(DeviceDefinition) ]; # 0..1 The reference to the definition for the device
  fhir:Device.udiCarrier [ # 0..* Unique Device Identifier (UDI) Barcode string
    fhir:Device.udiCarrier.deviceIdentifier [ string ]; # 0..1 Mandatory fixed portion of UDI
    fhir:Device.udiCarrier.issuer [ uri ]; # 0..1 UDI Issuing Organization
    fhir:Device.udiCarrier.jurisdiction [ uri ]; # 0..1 Regional UDI authority
    fhir:Device.udiCarrier.carrierAIDC [ base64Binary ]; # 0..1 UDI Machine Readable Barcode String
    fhir:Device.udiCarrier.carrierHRF [ string ]; # 0..1 UDI Human Readable Barcode String
    fhir:Device.udiCarrier.entryType [ code ]; # 0..1 barcode | rfid | manual +
  ], ...;
  fhir:Device.status [ code ]; # 0..1 active | inactive | entered-in-error | unknown
  fhir:Device.statusReason [ CodeableConcept ], ... ; # 0..* online | paused | standby | offline | not-ready | transduc-
    discon | hw-discon | off
  fhir:Device.distinctIdentifier [ string ]; # 0..1 The distinct identification string
  fhir:Device.manufacturer [ string ]; # 0..1 Name of device manufacturer
  fhir:Device.manufactureDate [ dateTime ]; # 0..1 Date when the device was made
  fhir:Device.expirationDate [ dateTime ]; # 0..1 Date and time of expiry of this device (if applicable)
  fhir:Device.lotNumber [ string ]; # 0..1 Lot number of manufacture
  fhir:Device.serialNumber [ string ]; # 0..1 Serial number assigned by the manufacturer
  fhir:Device.deviceName [ # 0..* The name of the device as given by the manufacturer
    fhir:Device.deviceName.name [ string ]; # 1..1 The name of the device
    fhir:Device.deviceName.type [ code ]; # 1..1 udi-label-name | user-friendly-name | patient-reported-name |
    manufacturer-name | model-name | other
  ], ...;
  fhir:Device.modelNumber [ string ]; # 0..1 The model number for the device
  fhir:Device.partNumber [ string ]; # 0..1 The part number of the device
  fhir:Device.type [ CodeableConcept ]; # 0..1 The kind or type of device
  fhir:Device.specialization [ # 0..* The capabilities supported on a  device, the standards to which the device
    conforms for a particular purpose, and used for the communication
    fhir:Device.specialization.systemType [ CodeableConcept ]; # 1..1 The standard that is used to operate and
    communicate
    fhir:Device.specialization.version [ string ]; # 0..1 The version of the standard that is used to operate and
    communicate
  ], ...;
  fhir:Device.version [ # 0..* The actual design of the device or software version running on the device
    fhir:Device.version.type [ CodeableConcept ]; # 0..1 The type of the device version
    fhir:Device.version.component [ Identifier ]; # 0..1 A single component of the device version
    fhir:Device.version.value [ string ]; # 1..1 The version text
  ], ...;
  fhir:Device.property [ # 0..* The actual configuration settings of a device as it actually operates, e.g., regulation
    status, time properties
    fhir:Device.property.type [ CodeableConcept ]; # 1..1 Code that specifies the property DeviceDefinitionPropetyCode
    (Extensible)
    fhir:Device.property.valueQuantity [ Quantity ], ... ; # 0..* Property value as a quantity
    fhir:Device.property.valueCode [ CodeableConcept ], ... ; # 0..* Property value as a code, e.g., NTP4 (synced to
    NTP)
  ], ...;
  fhir:Device.patient [ Reference(Patient) ]; # 0..1 Patient to whom Device is affixed
  fhir:Device.owner [ Reference(Organization) ]; # 0..1 Organization responsible for device
  fhir:Device.contact [ ContactPoint ], ... ; # 0..* Details for human/organization for support
  fhir:Device.location [ Reference(Location) ]; # 0..1 Where the device is found
  fhir:Device.url [ uri ]; # 0..1 Network address to contact device
  fhir:Device.note [ Annotation ], ... ; # 0..* Device notes and comments
  fhir:Device.safety [ CodeableConcept ], ... ; # 0..* Safety Characteristics of Device
  fhir:Device.parent [ Reference(Device) ]; # 0..1 The parent device
]
```

Changes since Release 3

Device
Device.definition | Added Element
Device.udiCarrier | Added Element
Device.udiCarrier.deviceIdentifier | Added Element
Device.udiCarrier.issuer | Added Element
Device.udiCarrier.jurisdiction | Added Element
Device.udiCarrier.carrierAIDC | Added Element
Device.udiCarrier.carrierHRF | Added Element
Device.udiCarrier.entryType | Added Element
Device.status | Change value set from http://hl7.org/fhir/ValueSet/device-status to http://hl7.org/fhir/ValueSet/device-status|4.0.1
Device.statusReason | Added Element
Device.distinctIdentifier | Added Element
Device.serialNumber | Added Element
Device.deviceName | Added Element
Device.deviceName.name | Added Mandatory Element
Device.deviceName.type | Added Mandatory Element
Device.modelNumber | Added Element
Device.partNumber | Added Element
Device.specialization | Added Element
Device.specialization.systemType | Added Mandatory Element
Device.specialization.version | Added Element

Device.version | Max Cardinality changed from 1 to * Type changed from string to BackboneElement
Device.version.type | Added Element
Device.version.component | Added Element
Device.version.value | Added Mandatory Element
Device.property | Added Element
Device.property.type | Added Mandatory Element
Device.property.valueQuantity | Added Element
Device.property.valueCode | Added Element
Device.parent | Added Element
Device.udi | deleted
Device.model | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.14.3.1 Terminology Bindings

Path | Definition | Type | Reference
Device.udiCarrier.entryType | Codes to identify how UDI data was entered. | Required | UDIEntryType
Device.status | The availability status of the device. | Required | FHIRDeviceStatus
Device.statusReason | The availability status reason of the device. | Extensible | FHIRDeviceStatusReason
Device.deviceName.type | The type of name the device is referred by. | Required | DeviceNameType
Device.type | Codes to identify medical devices. | Example | DeviceType

8.14.3.2 Notes

8.14.3.2.1 Device Identifier and Device Type

Nearly all devices are assigned a string of characters to represent one or more identifiers or codes, which are usually printed or affixed to the device using either barcodes or RFIDs. The identifier or code can come from the manufacturer (for example, a 'serial number', 'reference number', or 'catalog number'), various institution and registries. Any of these identifiers or codes assigned to the device can and should be recorded in the device resource. However, there can there can be confusion where to represent them in the resource because codes and identifiers are represented in FHIR as semantically distinct elements and because organizations may conflate the term 'code' for an identifier or 'identifier' for a code in their names.

The identifier element is only intended for use when it's an actual identifier for a specific instance of a device. That would mean that each device would have a separate serial number and would be represented using this element - devices without serial numbers (for example, a box of syringes) would not. Concepts such as a reference number or catalog number or GTIN describe a code which represents a kind of device and are conveyed using the type element. Some sources of standard codes for devices and translations within type are listed below:

• SNOMED CT - the example binding used here

• Global Medical Device Nomenclature (GMDN®)

• Rosetta Terminology Mapping (RTM)

8.14.3.2.2 Unique Device Identifier (UDI)

The International Medical Device Regulators Forum IMDRF UDI Working Group published UDI System for Medical Devices (Version 2.0) , the base specification for Unique Device Identifiers (UDI). The United States Food and Drug Administration has produced an implementation guide for Unique Device Identifiers (UDI) which implements the IMDRF specification and other jurisdictions may produce similar IMDRF implementation guides as well. The full UDI string that represents the barcode as printed on the packaging of the device or Automatic Identification and Data Capture (AIDC) representation is called the "UDI carrier". The UDI has 2 components*:

• Device identifier (DI)**, which is the actual identification component

• Production identifier(s)(PI) which provide the means to track a device through its manufacture, distribution and use.

*non-UDI elements may also appear within the UDI carrier. **a "GTIN" (sometimes also called an EAN number) is a code developed by GS1 for the kind of device not an identifier for the device. A GTIN may appear on its own or it may appear in a UDI string as the DI component.

The DI of the UDI may be stored in a jurisdictional repository and used as the primary key to access other device information. For example, in the United States, the DI of the UDI is submitted in a device record to the Global Unique Device Identification Database (GUDID) . The UDI may identify an instance of a device uniquely (when the PI includes a serial number), or it may just identify the type of the device. The UDI is parsed into its constituent parts (DI, PI and other elements) by parsing rules developed by each Issuing Agency standard.

Where the device has an assigned UDI, the other details carried in the resource (e.g., lot, expiration date, etc.) SHALL be consistent with the information encoded in the UDI string or registered in the local repository.

Best practice guidelines for transmitting UDI data using the Device resource dictate transmitting both the UDI Carrier and all components found within the UDI as described in Device UDI Mapping. Several examples are provided for further guidance.

8.14.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
device-name | string | A server defined search that may match any of the string fields in Device.deviceName or Device.type. | Device.deviceName.name | Device.type.coding.display | Device.type.text
identifier | token | Instance id from manufacturer, owner, and others | Device.identifier
location | reference | A location, where the resource is found | Device.location (Location)
manufacturer | string | The manufacturer of the device | Device.manufacturer
model | string | The model of the device | Device.modelNumber
organization | reference | The organization responsible for the device | Device.owner (Organization)
patient | reference | Patient information, if the resource is affixed to a person | Device.patient (Patient)
status | token | active | inactive | entered-in-error | unknown | Device.status
type | token | The type of the device | Device.type
udi-carrier | string | UDI Barcode (RFID or other technology) string in *HRF* format. | Device.udiCarrier.carrierHRF
udi-di | string | The udi Device Identifier (DI) | Device.udiCarrier.deviceIdentifier
url | uri | Network address to contact device | Device.url

8.14.5 Resource Device - Examples

URL: https://hl7.org/fhir/R4/device-examples.html

8.14.5 Resource Device - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
General Device Example | example | XML | JSON | Turtle
Feeding Tube | f001 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.14.6 Resource Device - Detailed Descriptions

URL: https://hl7.org/fhir/R4/device-definitions.html

8.14.6 Resource Device - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Device resource.

Device
Element Id | Device
Definition | A type of a manufactured item that is used in the provision of healthcare without being substantially changed through that activity. The device may be a medical or non-medical device.
Cardinality | 0..*
Type | DomainResource
Requirements | Allows institutions to track their devices.
Device.identifier
Element Id | Device.identifier
Definition | Unique instance identifiers assigned to a device by manufacturers other organizations or owners.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Comments | The barcode string from a barcode present on a device label or package may identify the instance, include names given to the device in local usage, or may identify the type of device. If the identifier identifies the type of device, Device.type element should be used.
Device.definition
Element Id | Device.definition
Definition | The reference to the definition for the device.
Cardinality | 0..1
Type | Reference(DeviceDefinition)
Device.udiCarrier
Element Id | Device.udiCarrier

Definition | Unique device identifier (UDI) assigned to device label or package. Note that the Device may include multiple udiCarriers as it either may include just the udiCarrier for the jurisdiction it is sold, or for multiple jurisdictions it could have been sold.
Cardinality | 0..*
Summary | true
Comments | UDI may identify an unique instance of a device, or it may only identify the type of the device. See UDI mappings for a complete mapping of UDI parts to Device.
Device.udiCarrier.deviceIdentifier
Element Id | Device.udiCarrier.deviceIdentifier
Definition | The device identifier (DI) is a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device.
Cardinality | 0..1
Type | string
Alternate Names | DI
Summary | true
Device.udiCarrier.issuer
Element Id | Device.udiCarrier.issuer
Definition | Organization that is charged with issuing UDIs for devices. For example, the US FDA issuers include : GS1: http://hl7.org/fhir/NamingSystem/gs1-di, HIBCC: http://hl7.org/fhir/NamingSystem/hibcc-dI, ICCBBA for blood containers: http://hl7.org/fhir/NamingSystem/iccbba-blood-di, ICCBA for other devices: http://hl7.org/fhir/NamingSystem/iccbba-other-di.
Cardinality | 0..1
Type | uri
Alternate Names | Barcode System
Device.udiCarrier.jurisdiction
Element Id | Device.udiCarrier.jurisdiction
Definition | The identity of the authoritative source for UDI generation within a jurisdiction. All UDIs are globally unique within a single namespace with the appropriate repository uri as the system. For example, UDIs of devices managed in the U.S. by the FDA, the value is http://hl7.org/fhir/NamingSystem/fda-udi.
Cardinality | 0..1
Type | uri
Requirements | Allows a recipient of a UDI to know which database will contain the UDI-associated metadata.
Device.udiCarrier.carrierAIDC
Element Id | Device.udiCarrier.carrierAIDC
Definition | The full UDI carrier of the Automatic Identification and Data Capture (AIDC) technology representation of the barcode string as printed on the packaging of the device - e.g., a barcode or RFID. Because of limitations on character sets in XML and the need to round-trip JSON data through XML, AIDC Formats SHALL be base64 encoded.
Cardinality | 0..1
Type | base64Binary
Alternate Names | Automatic Identification and Data Capture
Summary | true
Comments | The AIDC form of UDIs should be scanned or otherwise used for the identification of the device whenever possible to minimize errors in records resulting from manual transcriptions. If separate barcodes for DI and PI are present, concatenate the string with DI first and in order of human readable expression on label.
Device.udiCarrier.carrierHRF
Element Id | Device.udiCarrier.carrierHRF
Definition | The full UDI carrier as the human readable form (HRF) representation of the barcode string as printed on the packaging of the device.
Cardinality | 0..1
Type | string
Alternate Names | Human Readable Form
Summary | true
Comments | If separate barcodes for DI and PI are present, concatenate the string with DI first and in order of human readable expression on label.
Device.udiCarrier.entryType
Element Id | Device.udiCarrier.entryType
Definition | A coded entry to indicate how the data was entered.
Cardinality | 0..1
Terminology Binding | UDIEntryType (Required)
Type | code
Requirements | Supports a way to distinguish hand entered from machine read data.
Device.status
Element Id | Device.status
Definition | Status of the Device availability.
Cardinality | 0..1
Terminology Binding | FHIRDeviceStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes inactive and entered-in-error that mark the device (record)as not currently valid.
Device.statusReason
Element Id | Device.statusReason
Definition | Reason for the dtatus of the Device availability.

Cardinality | 0..*
Terminology Binding | FHIRDeviceStatusReason (Extensible)
Type | CodeableConcept
Device.distinctIdentifier
Element Id | Device.distinctIdentifier
Definition | The distinct identification string as required by regulation for a human cell, tissue, or cellular and tissue-based product.
Cardinality | 0..1
Type | string
Alternate Names | Distinct Identification Code (DIC)
Comments | For example, this applies to devices in the United States regulated under Code of Federal Regulation 21CFR§1271.290(c).
Device.manufacturer
Element Id | Device.manufacturer
Definition | A name of the manufacturer.
Cardinality | 0..1
Type | string
Device.manufactureDate
Element Id | Device.manufactureDate
Definition | The date and time when the device was manufactured.
Cardinality | 0..1
Type | dateTime
Device.expirationDate
Element Id | Device.expirationDate
Definition | The date and time beyond which this device is no longer valid or should not be used (if applicable).
Cardinality | 0..1
Type | dateTime
Device.lotNumber
Element Id | Device.lotNumber
Definition | Lot number assigned by the manufacturer.
Cardinality | 0..1
Type | string
Device.serialNumber
Element Id | Device.serialNumber
Definition | The serial number assigned by the organization when the device was manufactured.
Cardinality | 0..1
Type | string
Comments | Alphanumeric Maximum 20.
Device.deviceName
Element Id | Device.deviceName
Definition | This represents the manufacturer's name of the device as provided by the device, from a UDI label, or by a person describing the Device. This typically would be used when a person provides the name(s) or when the device represents one of the names available from DeviceDefinition.
Cardinality | 0..*
Device.deviceName.name
Element Id | Device.deviceName.name
Definition | The name of the device.
Cardinality | 1..1
Type | string
Alternate Names | Σ
Device.deviceName.type
Element Id | Device.deviceName.type
Definition | The type of deviceName. UDILabelName | UserFriendlyName | PatientReportedName | ManufactureDeviceName | ModelName.
Cardinality | 1..1
Terminology Binding | DeviceNameType (Required)
Type | code
Device.modelNumber
Element Id | Device.modelNumber
Definition | The model number for the device.
Cardinality | 0..1
Type | string
Device.partNumber
Element Id | Device.partNumber
Definition | The part number of the device.
Cardinality | 0..1
Type | string
Comments | Alphanumeric Maximum 20.
Device.type
Element Id | Device.type
Definition | The kind or type of device.
Cardinality | 0..1
Terminology Binding | Device Type (Example)
Type | CodeableConcept

Device.specialization
Element Id | Device.specialization
Definition | The capabilities supported on a device, the standards to which the device conforms for a particular purpose, and used for the communication.
Cardinality | 0..*
Device.specialization.systemType
Element Id | Device.specialization.systemType
Definition | The standard that is used to operate and communicate.
Cardinality | 1..1
Type | CodeableConcept
Alternate Names | Σ
Device.specialization.version
Element Id | Device.specialization.version
Definition | The version of the standard that is used to operate and communicate.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Device.version
Element Id | Device.version
Definition | The actual design of the device or software version running on the device.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..*
Device.version.type
Element Id | Device.version.type
Definition | The type of the device version.
Cardinality | 0..1
Type | CodeableConcept
Alternate Names | Σ
Device.version.component
Element Id | Device.version.component
Definition | A single component of the device version.
Cardinality | 0..1
Type | Identifier
Device.version.value
Element Id | Device.version.value
Definition | The version text.
Cardinality | 1..1
Type | string
Device.property
Element Id | Device.property
Definition | The actual configuration settings of a device as it actually operates, e.g., regulation status, time properties.
Cardinality | 0..*
Device.property.type
Element Id | Device.property.type
Definition | Code that specifies the property DeviceDefinitionPropetyCode (Extensible).
Cardinality | 1..1
Type | CodeableConcept
Device.property.valueQuantity
Element Id | Device.property.valueQuantity
Definition | Property value as a quantity.
Cardinality | 0..*
Type | Quantity
Device.property.valueCode
Element Id | Device.property.valueCode
Definition | Property value as a code, e.g., NTP4 (synced to NTP).
Cardinality | 0..*
Type | CodeableConcept
Device.patient
Element Id | Device.patient
Definition | Patient information, If the device is affixed to a person.
Cardinality | 0..1
Type | Reference(Patient)
Requirements | If the device is implanted in a patient, then need to associate the device to the patient.
Device.owner
Element Id | Device.owner
Definition | An organization that is responsible for the provision and ongoing maintenance of the device.
Cardinality | 0..1
Type | Reference(Organization)
Device.contact
Element Id | Device.contact
Definition | Contact details for an organization or a particular human that is responsible for the device.
Cardinality | 0..*

Type | ContactPoint
Comments | used for troubleshooting etc.
Device.location
Element Id | Device.location
Definition | The place where the device can be found.
Cardinality | 0..1
Type | Reference(Location)
Requirements | Device.location can be used to track device location.
Device.url
Element Id | Device.url
Definition | A network address on which the device may be contacted directly.
Cardinality | 0..1
Type | uri
Comments | If the device is running a FHIR server, the network address should be the Base URL from which a conformance statement may be retrieved.
Device.note
Element Id | Device.note
Definition | Descriptive information, usage information or implantation information that is not captured in an existing element.
Cardinality | 0..*
Type | Annotation
Device.safety
Element Id | Device.safety
Definition | Provides additional safety characteristics about a medical device. For example devices containing latex.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
Device.parent
Element Id | Device.parent
Definition | The parent device.
Cardinality | 0..1
Type | Reference(Device)

8.14.7 Resource Device - Mappings

URL: https://hl7.org/fhir/R4/device-mappings.html

8.14.7 Resource Device - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

UDI Mapping

Mappings for the device resource (see Mappings to Other Standards for further information & status).

8.14.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Device
identifier | FiveWs.identifier
udiCarrier | FiveWs.identifier
deviceIdentifier | FiveWs.what[x]
status | FiveWs.status
statusReason | FiveWs.status
distinctIdentifier | FiveWs.what[x]
manufacturer | FiveWs.what[x]
manufactureDate | FiveWs.what[x]
expirationDate | FiveWs.what[x]
lotNumber | FiveWs.what[x]
serialNumber | FiveWs.what[x]
type | FiveWs.what[x]
modelNumber | FiveWs.what[x]
partNumber | FiveWs.what[x]
version | FiveWs.what[x]
component | FiveWs.what[x]
patient | FiveWs.subject[x]
owner | FiveWs.source
contact | FiveWs.source
location | FiveWs.where[x]
url | FiveWs.where[x]

8.14.7.2 RIM Mapping (http://hl7.org/v3)

Device | Device
identifier | .id
udiCarrier | .id and .code
deviceIdentifier | Role.id.extension
issuer | Role.id.root
jurisdiction | Role.scoper
carrierAIDC | Role.id.extension
carrierHRF | Role.id.extension
entryType | NA
status | .statusCode
distinctIdentifier | .lotNumberText
manufacturer | .playedRole[typeCode=MANU].scoper.name
manufactureDate | .existenceTime.low
expirationDate | .expirationTime
lotNumber | .lotNumberText
serialNumber | .playedRole[typeCode=MANU].id
type | .playedRole[typeCode=MANU].code
modelNumber | .softwareName (included as part)
partNumber | .playedRole[typeCode=MANU].id
patient | .playedRole[typeCode=USED].scoper.playedRole[typeCode=PAT]
owner | .playedRole[typeCode=OWN].scoper
contact | .scopedRole[typeCode=CON].player
location | .playedRole[typeCode=LOCE].scoper
url | .telecom
note | .text
safety | NA

8.14.7.3 UDI Mapping (http://fda.gov/UDI)

UDI is a unique numeric or alphanumeric code that consists of two parts: (1) a device identifier (DI), a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device, and (2) a production identifier (PI), a conditional, variable portion of a UDI that identifies one or more of the following when included on the label of a device: (2a) the lot or batch number within which a device was manufactured; (2b) the serial number of a specific device; (2c) the expiration date of a specific device; (2d) the date a specific device was manufactured; (2e) the distinct identification code required by §1271.290(c) for a human cell, tissue, or cellular and tissue-based product (HCT/P) regulated as a device.

Device
identifier | The serial number which is a component of the production identifier (PI), a conditional, variable portion of a UDI. The identifier.type code should be set to “SNO”(Serial Number) and the system left empty.
definition
udiCarrier
deviceIdentifier | The device identifier (DI), a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device.
issuer | All UDIs are to be issued under a system operated by an Jurisdiction-accredited issuing agency. GS1 DIs: http://hl7.org/fhir/NamingSystem/gs1 HIBCC DIs: http://hl7.org/fhir/NamingSystem/hibcc ICCBBA DIs for blood containers: http://hl7.org/fhir/NamingSystem/iccbba-blood ICCBA DIs for other devices: http://hl7.org/fhir/NamingSystem/iccbba-other
jurisdiction
carrierAIDC | A unique device identifier (UDI) on a device label a form that uses automatic identification and data capture (AIDC) technology.
carrierHRF | A unique device identifier (UDI) on a device label in plain text
entryType
status
statusReason
distinctIdentifier | The lot or batch number within which a device was manufactured - which is a component of the production identifier (PI), a conditional, variable portion of a UDI.
manufacturer | N/A
manufactureDate | The date a specific device was manufactured - which is a component of the production identifier (PI), a conditional, variable portion of a UDI. For FHIR, The datetime syntax must converted to YYYY-MM-DD[THH:MM:SS]. If hour is present, the minutes and seconds should both be set to “00”.
expirationDate | the expiration date of a specific device - which is a component of the production identifier (PI), a conditional, variable portion of a UDI. For FHIR, The datetime syntax must converted to YYYY-MM-DD[THH:MM:SS]. If hour is present, the minutes and seconds should both be set to “00”.
lotNumber | The lot or batch number within which a device was manufactured - which is a component of the production identifier (PI), a conditional, variable portion of a UDI.
serialNumber
deviceName
name
type
modelNumber
partNumber
type

specialization
systemType
version
version
type
component
value
property
type
valueQuantity
valueCode
patient
owner
contact
location
url
note
safety
parent

8.14.8 Resource Device - Extensions & Profiles

URL: https://hl7.org/fhir/R4/device-profiles.html

8.14.8 Resource Device - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Device resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.14.8.1 Profiles

No Profiles defined for this resource

8.14.8.2 Extensions

device-implantStatus | Device | implantStatus | for Device HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

observation-gatewayDevice | gatewayDevice | for Observation HL7 Extensions
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.14.8.3 Search Extensions

din | token | The donation identification number (DIN) Expression: Device.extension('http://hl7.org/fhir/SearchParameter/device-extensions-Device-din')

8.14.9 Resource Device - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/device-version-maps.html

No validation errors - all conversions are clean

8.14.9 Resource Device - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Device.

Functional status for this map: 9 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

8.14.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Device3to4" = "R3 to R4 Conversions for Device"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Device" alias DeviceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Device" alias Device as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Device(source src : DeviceR3, target tgt : Device) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.udi as vs0 -> tgt.udiCarrier as vt0 then {
    vs0.deviceIdentifier -> vt0.deviceIdentifier;
    vs0.name as vs ->  tgt.deviceName as vt,  vt.name = vs,  vt.type = 'udi-label-name';
    vs0.jurisdiction -> vt0.jurisdiction;
    vs0.carrierHRF -> vt0.carrierHRF;
    vs0.carrierAIDC -> vt0.carrierAIDC;
    vs0.issuer -> vt0.issuer;
    vs0.entryType -> vt0.entryType;
  };
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.lotNumber -> tgt.lotNumber;
  src.manufacturer -> tgt.manufacturer;
  src.manufactureDate -> tgt.manufactureDate;
  src.expirationDate -> tgt.expirationDate;
  src.model -> tgt.modelNumber;
  src.version as vs ->  tgt.version as vt,  vt.value = vs;
  src.patient -> tgt.patient;
  src.owner -> tgt.owner;
  src.contact -> tgt.contact;
  src.location -> tgt.location;
  src.url -> tgt.url;
  src.note -> tgt.note;
  src.safety -> tgt.safety;
}
```

8.14.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Device4to3" = "R4 to R3 Conversion for Device"

uses "http://hl7.org/fhir/StructureDefinition/Device" alias Device as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Device" alias DeviceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Device(source src : Device, target tgt : DeviceR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.udiCarrier as vs0 -> tgt.udi as vt0 then {
    vs0.deviceIdentifier -> vt0.deviceIdentifier;
    vs0.jurisdiction -> vt0.jurisdiction;
    vs0.carrierHRF -> vt0.carrierHRF;
    vs0.carrierAIDC -> vt0.carrierAIDC;
    vs0.issuer -> vt0.issuer;
    vs0.entryType -> vt0.entryType;
  };
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.lotNumber -> tgt.lotNumber;
  src.manufacturer -> tgt.manufacturer;
  src.manufactureDate -> tgt.manufactureDate;
  src.expirationDate -> tgt.expirationDate;
  src.modelNumber -> tgt.model;
  src.version as vs -> tgt as vt then {
    vs.value -> tgt.version;
  };
  src.deviceName as vs -> tgt.udi as vt then {
    vs.name -> vt.name;
  };
  src.patient -> tgt.patient;
  src.owner -> tgt.owner;
  src.contact -> tgt.contact;
  src.location -> tgt.location;
  src.url -> tgt.url;
  src.note -> tgt.note;
  src.safety -> tgt.safety;
}
```

8.14.9.3 R4 Validation Errors

8.14.10 Device HL7 Extensions

URL: https://hl7.org/fhir/R4/device-extensions.html

Defines common extensions used with or related to the Device resource

8.14.10 Device HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

8.14.10.1 Content

Extensions:
device-implantStatus | implantStatus : Codes to represent the functional status of a device implanted in a patient. Both overall device status and an implant status need to be considered. The implant status should only be used when the device status is active.

8.14.10.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
din | token | The donation identification number (DIN) | | XML / JSON

8.14.10.3 Device-extensions-Device-din

URL: https://hl7.org/fhir/R4/device-extensions-Device-din.html

8.14.10.3 Device-extensions-Device-din

Search Parameter din (type: token) defined as part of the device-extensions Profile.

Orders and Observations | Maturity Level: 0 | Informative

The donation identification number (DIN)

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/device-extensions-Device-din

• Paths :

• Targets :

8.14.11 JSON Schema for Device

URL: https://hl7.org/fhir/R4/device.schema.json.html

8.14.11 JSON Schema for Device

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Device, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Device",
  "$ref": "#/definitions/Device",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Device": {
      "description": "A type of a manufactured item that is used in the provision of healthcare without being
    substantially changed through that activity. The device may be a medical or non-medical device.",
      "properties": {
        "resourceType": {
          "description": "This is a Device resource",
          "const": "Device"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique instance identifiers assigned to a device by manufacturers other organizations or
owners.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "definition": {
      "description": "The reference to the definition for the device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "udiCarrier": {
      "description": "Unique device identifier (UDI) assigned to device label or package.  Note that the Device may
include multiple udiCarriers as it either may include just the udiCarrier for the jurisdiction it is sold, or for
multiple jurisdictions it could have been sold.",
      "items": {
        "$ref": "#/definitions/Device_UdiCarrier"
      },
      "type": "array"
    },
    "status": {
      "description": "Status of the Device availability.",
      "enum": [
        "active",
        "inactive",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Reason for the dtatus of the Device availability.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "distinctIdentifier": {
      "description": "The distinct identification string as required by regulation for a human cell, tissue, or
cellular and tissue-based product.",
      "$ref": "#/definitions/string"
    },
    "_distinctIdentifier": {
      "description": "Extensions for distinctIdentifier",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "manufacturer": {
      "description": "A name of the manufacturer.",
      "$ref": "#/definitions/string"
    },
    "_manufacturer": {
      "description": "Extensions for manufacturer",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "manufactureDate": {
      "description": "The date and time when the device was manufactured.",
      "$ref": "#/definitions/dateTime"
    },
    "_manufactureDate": {
      "description": "Extensions for manufactureDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expirationDate": {
      "description": "The date and time beyond which this device is no longer valid or should not be used (if
applicable).",
      "$ref": "#/definitions/dateTime"
    },
    "_expirationDate": {
      "description": "Extensions for expirationDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lotNumber": {
      "description": "Lot number assigned by the manufacturer.",
      "$ref": "#/definitions/string"
    },
    "_lotNumber": {
      "description": "Extensions for lotNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "serialNumber": {
      "description": "The serial number assigned by the organization when the device was manufactured.",
      "$ref": "#/definitions/string"
    },
    "_serialNumber": {
      "description": "Extensions for serialNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "deviceName": {
      "description": "This represents the manufacturer\u0027s name of the device as provided by the device, from a
UDI label, or by a person describing the Device.  This typically would be used when a person provides the name(s) or
 when the device represents one of the names available from DeviceDefinition.",
      "items": {
        "$ref": "#/definitions/Device_DeviceName"
      },
      "type": "array"
    },
    "modelNumber": {
      "description": "The model number for the device.",
      "$ref": "#/definitions/string"
    },
    "_modelNumber": {
      "description": "Extensions for modelNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "partNumber": {
      "description": "The part number of the device.",
      "$ref": "#/definitions/string"
    },
    "_partNumber": {
      "description": "Extensions for partNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The kind or type of device.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "specialization": {
      "description": "The capabilities supported on a  device, the standards to which the device conforms for a
particular purpose, and used for the communication.",
      "items": {
        "$ref": "#/definitions/Device_Specialization"
      },
      "type": "array"
    },
    "version": {
      "description": "The actual design of the device or software version running on the device.",
      "items": {
        "$ref": "#/definitions/Device_Version"
      },
      "type": "array"
    },
    "property": {
      "description": "The actual configuration settings of a device as it actually operates, e.g., regulation
status, time properties.",
```

```
      "items": {
        "$ref": "#/definitions/Device_Property"
      },
      "type": "array"
    },
    "patient": {
      "description": "Patient information, If the device is affixed to a person.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "owner": {
      "description": "An organization that is responsible for the provision and ongoing maintenance of the device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "contact": {
      "description": "Contact details for an organization or a particular human that is responsible for the
device.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "location": {
      "description": "The place where the device can be found.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "url": {
      "description": "A network address on which the device may be contacted directly.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "Descriptive information, usage information or implantation information that is not captured in
 an existing element.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "safety": {
      "description": "Provides additional safety characteristics about a medical device.  For example devices
containing latex.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "parent": {
      "description": "The parent device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Device_UdiCarrier": {
  "description": "A type of a manufactured item that is used in the provision of healthcare without being
substantially changed through that activity. The device may be a medical or non-medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "deviceIdentifier": {
      "description": "The device identifier (DI) is a mandatory, fixed portion of a UDI that identifies the labeler
and the specific version or model of a device.",
      "$ref": "#/definitions/string"
    },
    "_deviceIdentifier": {
      "description": "Extensions for deviceIdentifier",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "issuer": {
      "description": "Organization that is charged with issuing UDIs for devices.  For example, the US FDA issuers
include :\n1) GS1: \nhttp://hl7.org/fhir/NamingSystem/gs1-di, \n2) HIBCC:\nhttp://hl7.org/fhir/NamingSystem/hibcc-
dI, \n3) ICCBBA for blood containers:\nhttp://hl7.org/fhir/NamingSystem/iccbba-blood-di, \n4) ICCBA for other
devices:\nhttp://hl7.org/fhir/NamingSystem/iccbba-other-di.",
      "$ref": "#/definitions/uri"
    },
    "_issuer": {
      "description": "Extensions for issuer",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "jurisdiction": {
      "description": "The identity of the authoritative source for UDI generation within a  jurisdiction.  All UDIs
are globally unique within a single namespace with the appropriate repository uri as the system.  For example,  UDIs
 of devices managed in the U.S. by the FDA, the value is  http://hl7.org/fhir/NamingSystem/fda-udi.",
      "$ref": "#/definitions/uri"
    },
    "_jurisdiction": {
      "description": "Extensions for jurisdiction",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "carrierAIDC": {
      "description": "The full UDI carrier of the Automatic Identification and Data Capture (AIDC) technology
representation of the barcode string as printed on the packaging of the device - e.g., a barcode or RFID.   Because
of limitations on character sets in XML and the need to round-trip JSON data through XML, AIDC Formats *SHALL* be
base64 encoded.",
      "$ref": "#/definitions/base64Binary"
    },
    "_carrierAIDC": {
      "description": "Extensions for carrierAIDC",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "carrierHRF": {
      "description": "The full UDI carrier as the human readable form (HRF) representation of the barcode string as
printed on the packaging of the device.",
      "$ref": "#/definitions/string"
    },
    "_carrierHRF": {
      "description": "Extensions for carrierHRF",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "entryType": {
      "description": "A coded entry to indicate how the data was entered.",
      "enum": [
        "barcode",
        "rfid",
        "manual",
        "card",
        "self-reported",
        "unknown"
      ]
    },
    "_entryType": {
      "description": "Extensions for entryType",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Device_DeviceName": {
  "description": "A type of a manufactured item that is used in the provision of healthcare without being
substantially changed through that activity. The device may be a medical or non-medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The name of the device.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of deviceName.\nUDILabelName | UserFriendlyName | PatientReportedName |
ManufactureDeviceName | ModelName.",
      "enum": [
        "udi-label-name",
        "user-friendly-name",
        "patient-reported-name",
        "manufacturer-name",
        "model-name",
        "other"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Device_Specialization": {
  "description": "A type of a manufactured item that is used in the provision of healthcare without being
substantially changed through that activity. The device may be a medical or non-medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "systemType": {
      "description": "The standard that is used to operate and communicate.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "version": {
      "description": "The version of the standard that is used to operate and communicate.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
```

```
  "required": [
    "systemType"
  ]
},
"Device_Version": {
  "description": "A type of a manufactured item that is used in the provision of healthcare without being
substantially changed through that activity. The device may be a medical or non-medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of the device version.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "component": {
      "description": "A single component of the device version.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "value": {
      "description": "The version text.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Device_Property": {
  "description": "A type of a manufactured item that is used in the provision of healthcare without being
substantially changed through that activity. The device may be a medical or non-medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
        "type": {
          "description": "Code that specifies the property DeviceDefinitionPropetyCode (Extensible).",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "valueQuantity": {
          "description": "Property value as a quantity.",
          "items": {
            "$ref": "Quantity.schema.json#/definitions/Quantity"
          },
          "type": "array"
        },
        "valueCode": {
          "description": "Property value as a code, e.g., NTP4 (synced to NTP).",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "type"
      ]
    }
  }
}
```


8.14.12 Resource Device - Maturity Dependency View

URL: https://hl7.org/fhir/R4/device-dependencies.html

8.14.12 Resource Device - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Device FMM level 2

Name | Card. | Type | Dependency Analysis
Device | | DomainResource
identifier | 0..* | Identifier
definition | 0..1 | Reference(DeviceDefinition) | Type Warning: (DeviceDefinition = FMM0-Trial Use vs. Element = FMM2-Trial Use)
udiCarrier | 0..* | BackboneElement
deviceIdentifier | 0..1 | string
issuer | 0..1 | uri
jurisdiction | 0..1 | uri
carrierAIDC | 0..1 | base64Binary
carrierHRF | 0..1 | string
entryType | 0..1 | code
status | 0..1 | code
statusReason | 0..* | CodeableConcept
distinctIdentifier | 0..1 | string
manufacturer | 0..1 | string
manufactureDate | 0..1 | dateTime
expirationDate | 0..1 | dateTime
lotNumber | 0..1 | string
serialNumber | 0..1 | string
deviceName | 0..* | BackboneElement
name | 1..1 | string
type | 1..1 | code
modelNumber | 0..1 | string
partNumber | 0..1 | string
type | 0..1 | CodeableConcept
specialization | 0..* | BackboneElement
systemType | 1..1 | CodeableConcept
version | 0..1 | string
version | 0..* | BackboneElement
type | 0..1 | CodeableConcept
component | 0..1 | Identifier
value | 1..1 | string
property | 0..* | BackboneElement
type | 1..1 | CodeableConcept
valueQuantity | 0..* | Quantity
valueCode | 0..* | CodeableConcept
patient | 0..1 | Reference(Patient)
owner | 0..1 | Reference(Organization)
contact | 0..* | ContactPoint
location | 0..1 | Reference(Location)
url | 0..1 | uri
note | 0..* | Annotation

safety | 0..* | CodeableConcept
parent | 0..1 | Reference(Device)

8.14.13 Value Set Sc-valueset-device-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-device-status.html

8.14.13 Value Set Sc-valueset-device-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.14.14 FHIRDeviceStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-device-status)

Mapping from http://hl7.org/fhir/ValueSet/device-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The availability status of the device."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
active | equivalent | active
inactive | equivalent | inactive
unknown | equivalent | unknown

8.15 Resource DeviceDefinition - Content

URL: https://hl7.org/fhir/R4/devicedefinition.html
Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The characteristics, operational status and capabilities of a medical-related component of a medical device.

8.15.1 Scope and Usage

The DeviceDefinition resource is used to describe the characteristics and capabilities of a medical device.

8.15.2 Boundaries and Relationships

The DeviceDefinition resource contains the "catalog" definition of a device - whether that definition is authored by the regulatory entities, or it is a local definition that includes assembled device configurations. The DeviceDefinition allows defining hierarchical device configurations (devices as part of other devices).

Device vs deviceDefinition: The Device resource is meant to refer to a physical instance of a device - hence having attributes like lot number, patient, location, which the DeviceDefinition resource does not have.

8.15.3 Background and Context

8.15.3.1

This resource is referenced by Device, itself, MedicinalProductPackaged and MedicinalProductPharmaceutical

8.15.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceDefinition | TU | | DomainResource | An instance of a medical-related component of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Instance identifier
udiDeviceIdentifier | | 0..* | BackboneElement | Unique Device Identifier (UDI) Barcode string
deviceIdentifier | | 1..1 | string | The identifier that is to be associated with every Device that references this DeviceDefintiion for the issuer and jurisdication porvided in the DeviceDefinition.udiDeviceIdentifier

issuer | | 1..1 | uri | The organization that assigns the identifier algorithm
jurisdiction | | 1..1 | uri | The jurisdiction to which the deviceIdentifier applies
manufacturer[x] | | 0..1 | | Name of device manufacturer
manufacturerString | | | string
manufacturerReference | | | Reference(Organization)
deviceName | | 0..* | BackboneElement | A name given to the device to identify it
name | | 1..1 | string | The name of the device
type | | 1..1 | code | udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-name | other DeviceNameType (Required)
modelNumber | | 0..1 | string | The model number for the device
type | | 0..1 | CodeableConcept | What kind of device or device system this is FHIR Device Types (Example)
specialization | | 0..* | BackboneElement | The capabilities supported on a device, the standards to which the device conforms for a particular purpose, and used for the communication
systemType | | 1..1 | string | The standard that is used to operate and communicate
version | | 0..1 | string | The version of the standard that is used to operate and communicate
version | | 0..* | string | Available versions
safety | Σ | 0..* | CodeableConcept | Safety characteristics of the device Device safety (Example)
shelfLifeStorage | | 0..* | ProductShelfLife | Shelf Life and storage information
physicalCharacteristics | | 0..1 | ProdCharacteristic | Dimensions, color etc.
languageCode | | 0..* | CodeableConcept | Language code for the human-readable text strings produced by the device (all supported)
capability | | 0..* | BackboneElement | Device capabilities
type | | 1..1 | CodeableConcept | Type of capability
description | | 0..* | CodeableConcept | Description of capability
property | | 0..* | BackboneElement | The actual configuration settings of a device as it actually operates, e.g., regulation status, time properties
type | | 1..1 | CodeableConcept | Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
valueQuantity | | 0..* | Quantity | Property value as a quantity
valueCode | | 0..* | CodeableConcept | Property value as a code, e.g., NTP4 (synced to NTP)
owner | | 0..1 | Reference(Organization) | Organization responsible for device
contact | | 0..* | ContactPoint | Details for human/organization for support
url | | 0..1 | uri | Network address to contact device
onlineInformation | | 0..1 | uri | Access to on-line information
note | | 0..* | Annotation | Device notes and comments
quantity | | 0..1 | Quantity | The quantity of the device present in the packaging (e.g. the number of devices present in a pack, or the number of devices in the same package of the medicinal product)
parentDevice | Σ | 0..1 | Reference(DeviceDefinition) | The parent device it can be part of
material | | 0..* | BackboneElement | A substance used to create the material(s) of which the device is made
substance | | 1..1 | CodeableConcept | The substance
alternate | | 0..1 | boolean | Indicates an alternative material of the device
allergenicIndicator | | 0..1 | boolean | Whether the substance is a known or suspected allergen
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <udiDeviceIdentifier>  <!-- 0..* Unique Device Identifier (UDI) Barcode string -->
  <deviceIdentifier value="[string]"/><!-- 1..1 The identifier that is to be associated with every Device that
     references this DeviceDefintiion for the issuer and jurisdication porvided in the
     DeviceDefinition.udiDeviceIdentifier -->
  <issuer value="[uri]"/><!-- 1..1 The organization that assigns the identifier algorithm -->
  <jurisdiction value="[uri]"/><!-- 1..1 The jurisdiction to which the deviceIdentifier applies -->
 </udiDeviceIdentifier>
 <manufacturer[x]><!-- 0..1 string|Reference(Organization) Name of device manufacturer --></manufacturer[x]>
 <deviceName>  <!-- 0..* A name given to the device to identify it -->
  <name value="[string]"/><!-- 1..1 The name of the device -->
  <type value="[code]"/><!-- 1..1 udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name |
     model-name | other -->
 </deviceName>
 <modelNumber value="[string]"/><!-- 0..1 The model number for the device -->
 <type><!-- 0..1 CodeableConcept What kind of device or device system this is --></type>
 <specialization>  <!-- 0..* The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication -->
  <systemType value="[string]"/><!-- 1..1 The standard that is used to operate and communicate -->
  <version value="[string]"/><!-- 0..1 The version of the standard that is used to operate and communicate -->
 </specialization>
 <version value="[string]"/><!-- 0..* Available versions -->
 <safety><!-- 0..* CodeableConcept Safety characteristics of the device --></safety>
 <shelfLifeStorage><!-- 0..* ProductShelfLife Shelf Life and storage information --></shelfLifeStorage>
 <physicalCharacteristics><!-- 0..1 ProdCharacteristic Dimensions, color etc. --></physicalCharacteristics>
 <languageCode><!-- 0..* CodeableConcept Language code for the human-readable text strings produced by the device (all
    supported) --></languageCode>
 <capability>  <!-- 0..* Device capabilities -->
  <type><!-- 1..1 CodeableConcept Type of capability --></type>
  <description><!-- 0..* CodeableConcept Description of capability --></description>
 </capability>
```

```
 <property>  <!-- 0..* The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties -->
  <type><!-- 1..1 CodeableConcept Code that specifies the property DeviceDefinitionPropetyCode (Extensible) --></type>
  <valueQuantity><!-- 0..* Quantity Property value as a quantity --></valueQuantity>
  <valueCode><!-- 0..* CodeableConcept Property value as a code, e.g., NTP4 (synced to NTP) --></valueCode>
 </property>
 <owner><!-- 0..1 Reference(Organization) Organization responsible for device --></owner>
 <contact><!-- 0..* ContactPoint Details for human/organization for support --></contact>
 <url value="[uri]"/><!-- 0..1 Network address to contact device -->
 <onlineInformation value="[uri]"/><!-- 0..1 Access to on-line information -->
 <note><!-- 0..* Annotation Device notes and comments --></note>
 <quantity><!-- 0..1 Quantity The quantity of the device present in the packaging (e.g. the number of devices present in
     a pack, or the number of devices in the same package of the medicinal product) --></quantity>
 <parentDevice><!-- 0..1 Reference(DeviceDefinition) The parent device it can be part of --></parentDevice>
 <material>  <!-- 0..* A substance used to create the material(s) of which the device is made -->
  <substance><!-- 1..1 CodeableConcept The substance --></substance>
  <alternate value="[boolean]"/><!-- 0..1 Indicates an alternative material of the device -->
  <allergenicIndicator value="[boolean]"/><!-- 0..1 Whether the substance is a known or suspected allergen -->
 </material>
</DeviceDefinition>
```

JSON Template

```
{
  "resourceType" : "DeviceDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "udiDeviceIdentifier" : [{ // Unique Device Identifier (UDI) Barcode string
    "deviceIdentifier" : "<string>", // R!  The identifier that is to be associated with every Device that references
    this DeviceDefintiion for the issuer and jurisdication porvided in the DeviceDefinition.udiDeviceIdentifier
    "issuer" : "<uri>", // R!  The organization that assigns the identifier algorithm
    "jurisdiction" : "<uri>" // R!  The jurisdiction to which the deviceIdentifier applies
  }],
  // manufacturer[x]: Name of device manufacturer. One of these 2:
  "manufacturerString" : "<string>",
  "manufacturerReference" : { Reference(Organization) },
  "deviceName" : [{ // A name given to the device to identify it
    "name" : "<string>", // R!  The name of the device
    "type" : "<code>" // R!  udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-
    name | other
  }],
  "modelNumber" : "<string>", // The model number for the device
  "type" : { CodeableConcept }, // What kind of device or device system this is
  "specialization" : [{ // The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication
    "systemType" : "<string>", // R!  The standard that is used to operate and communicate
    "version" : "<string>" // The version of the standard that is used to operate and communicate
  }],
  "version" : ["<string>"], // Available versions
  "safety" : [{ CodeableConcept }], // Safety characteristics of the device
  "shelfLifeStorage" : [{ ProductShelfLife }], // Shelf Life and storage information
  "physicalCharacteristics" : { ProdCharacteristic }, // Dimensions, color etc.
  "languageCode" : [{ CodeableConcept }], // Language code for the human-readable text strings produced by the device
    (all supported)
  "capability" : [{ // Device capabilities
    "type" : { CodeableConcept }, // R!  Type of capability
    "description" : [{ CodeableConcept }] // Description of capability
  }],
  "property" : [{ // The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties
    "type" : { CodeableConcept }, // R!  Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
    "valueQuantity" : [{ Quantity }], // Property value as a quantity
    "valueCode" : [{ CodeableConcept }] // Property value as a code, e.g., NTP4 (synced to NTP)
  }],
  "owner" : { Reference(Organization) }, // Organization responsible for device
  "contact" : [{ ContactPoint }], // Details for human/organization for support
  "url" : "<uri>", // Network address to contact device
  "onlineInformation" : "<uri>", // Access to on-line information
  "note" : [{ Annotation }], // Device notes and comments
  "quantity" : { Quantity }, // The quantity of the device present in the packaging (e.g. the number of devices present
    in a pack, or the number of devices in the same package of the medicinal product)
  "parentDevice" : { Reference(DeviceDefinition) }, // The parent device it can be part of
  "material" : [{ // A substance used to create the material(s) of which the device is made
    "substance" : { CodeableConcept }, // R!  The substance
    "alternate" : <boolean>, // Indicates an alternative material of the device
    "allergenicIndicator" : <boolean> // Whether the substance is a known or suspected allergen
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceDefinition.identifier [ Identifier ], ... ; # 0..* Instance identifier
```

```
  fhir:DeviceDefinition.udiDeviceIdentifier [ # 0..* Unique Device Identifier (UDI) Barcode string
    fhir:DeviceDefinition.udiDeviceIdentifier.deviceIdentifier [ string ]; # 1..1 The identifier that is to be
    associated with every Device that references this DeviceDefintiion for the issuer and jurisdication porvided in the
    DeviceDefinition.udiDeviceIdentifier
    fhir:DeviceDefinition.udiDeviceIdentifier.issuer [ uri ]; # 1..1 The organization that assigns the identifier
    algorithm
    fhir:DeviceDefinition.udiDeviceIdentifier.jurisdiction [ uri ]; # 1..1 The jurisdiction to which the
    deviceIdentifier applies
  ], ...;
  # DeviceDefinition.manufacturer[x] : 0..1 Name of device manufacturer. One of these 2
    fhir:DeviceDefinition.manufacturerString [ string ]
    fhir:DeviceDefinition.manufacturerReference [ Reference(Organization) ]
  fhir:DeviceDefinition.deviceName [ # 0..* A name given to the device to identify it
    fhir:DeviceDefinition.deviceName.name [ string ]; # 1..1 The name of the device
    fhir:DeviceDefinition.deviceName.type [ code ]; # 1..1 udi-label-name | user-friendly-name | patient-reported-name |
     manufacturer-name | model-name | other
  ], ...;
  fhir:DeviceDefinition.modelNumber [ string ]; # 0..1 The model number for the device
  fhir:DeviceDefinition.type [ CodeableConcept ]; # 0..1 What kind of device or device system this is
  fhir:DeviceDefinition.specialization [ # 0..* The capabilities supported on a  device, the standards to which the
    device conforms for a particular purpose, and used for the communication
    fhir:DeviceDefinition.specialization.systemType [ string ]; # 1..1 The standard that is used to operate and
    communicate
    fhir:DeviceDefinition.specialization.version [ string ]; # 0..1 The version of the standard that is used to operate
    and communicate
  ], ...;
  fhir:DeviceDefinition.version [ string ], ... ; # 0..* Available versions
  fhir:DeviceDefinition.safety [ CodeableConcept ], ... ; # 0..* Safety characteristics of the device
  fhir:DeviceDefinition.shelfLifeStorage [ ProductShelfLife ], ... ; # 0..* Shelf Life and storage information
  fhir:DeviceDefinition.physicalCharacteristics [ ProdCharacteristic ]; # 0..1 Dimensions, color etc.
  fhir:DeviceDefinition.languageCode [ CodeableConcept ], ... ; # 0..* Language code for the human-readable text strings
     produced by the device (all supported)
  fhir:DeviceDefinition.capability [ # 0..* Device capabilities
    fhir:DeviceDefinition.capability.type [ CodeableConcept ]; # 1..1 Type of capability
    fhir:DeviceDefinition.capability.description [ CodeableConcept ], ... ; # 0..* Description of capability
  ], ...;
  fhir:DeviceDefinition.property [ # 0..* The actual configuration settings of a device as it actually operates, e.g.,
    regulation status, time properties
    fhir:DeviceDefinition.property.type [ CodeableConcept ]; # 1..1 Code that specifies the property
    DeviceDefinitionPropetyCode (Extensible)
    fhir:DeviceDefinition.property.valueQuantity [ Quantity ], ... ; # 0..* Property value as a quantity
    fhir:DeviceDefinition.property.valueCode [ CodeableConcept ], ... ; # 0..* Property value as a code, e.g., NTP4
    (synced to NTP)
  ], ...;
  fhir:DeviceDefinition.owner [ Reference(Organization) ]; # 0..1 Organization responsible for device
  fhir:DeviceDefinition.contact [ ContactPoint ], ... ; # 0..* Details for human/organization for support
  fhir:DeviceDefinition.url [ uri ]; # 0..1 Network address to contact device
  fhir:DeviceDefinition.onlineInformation [ uri ]; # 0..1 Access to on-line information
  fhir:DeviceDefinition.note [ Annotation ], ... ; # 0..* Device notes and comments
  fhir:DeviceDefinition.quantity [ Quantity ]; # 0..1 The quantity of the device present in the packaging (e.g. the
    number of devices present in a pack, or the number of devices in the same package of the medicinal product)
  fhir:DeviceDefinition.parentDevice [ Reference(DeviceDefinition) ]; # 0..1 The parent device it can be part of
  fhir:DeviceDefinition.material [ # 0..* A substance used to create the material(s) of which the device is made
    fhir:DeviceDefinition.material.substance [ CodeableConcept ]; # 1..1 The substance
    fhir:DeviceDefinition.material.alternate [ boolean ]; # 0..1 Indicates an alternative material of the device
    fhir:DeviceDefinition.material.allergenicIndicator [ boolean ]; # 0..1 Whether the substance is a known or suspected
     allergen
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

DeviceDefinition | TU | | DomainResource | An instance of a medical-related component of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Instance identifier

udiDeviceIdentifier | | 0..* | BackboneElement | Unique Device Identifier (UDI) Barcode string

deviceIdentifier | | 1..1 | string | The identifier that is to be associated with every Device that references this DeviceDefintiion for the issuer and jurisdication porvided in the DeviceDefinition.udiDeviceIdentifier

issuer | | 1..1 | uri | The organization that assigns the identifier algorithm

jurisdiction | | 1..1 | uri | The jurisdiction to which the deviceIdentifier applies

manufacturer[x] | | 0..1 | | Name of device manufacturer

manufacturerString | | | string

manufacturerReference | | | Reference(Organization)

deviceName | | 0..* | BackboneElement | A name given to the device to identify it

name | | 1..1 | string | The name of the device

type | | 1..1 | code | udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-name | other DeviceNameType (Required)

modelNumber | | 0..1 | string | The model number for the device

type | | 0..1 | CodeableConcept | What kind of device or device system this is FHIR Device Types (Example)

specialization | | 0..* | BackboneElement | The capabilities supported on a device, the standards to which the device conforms for a particular purpose, and used for the communication
systemType | | 1..1 | string | The standard that is used to operate and communicate
version | | 0..1 | string | The version of the standard that is used to operate and communicate
version | | 0..* | string | Available versions
safety | Σ | 0..* | CodeableConcept | Safety characteristics of the device Device safety (Example)
shelfLifeStorage | | 0..* | ProductShelfLife | Shelf Life and storage information
physicalCharacteristics | | 0..1 | ProdCharacteristic | Dimensions, color etc.
languageCode | | 0..* | CodeableConcept | Language code for the human-readable text strings produced by the device (all supported)
capability | | 0..* | BackboneElement | Device capabilities
type | | 1..1 | CodeableConcept | Type of capability
description | | 0..* | CodeableConcept | Description of capability
property | | 0..* | BackboneElement | The actual configuration settings of a device as it actually operates, e.g., regulation status, time properties
type | | 1..1 | CodeableConcept | Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
valueQuantity | | 0..* | Quantity | Property value as a quantity
valueCode | | 0..* | CodeableConcept | Property value as a code, e.g., NTP4 (synced to NTP)
owner | | 0..1 | Reference(Organization) | Organization responsible for device
contact | | 0..* | ContactPoint | Details for human/organization for support
url | | 0..1 | uri | Network address to contact device
onlineInformation | | 0..1 | uri | Access to on-line information
note | | 0..* | Annotation | Device notes and comments
quantity | | 0..1 | Quantity | The quantity of the device present in the packaging (e.g. the number of devices present in a pack, or the number of devices in the same package of the medicinal product)
parentDevice | Σ | 0..1 | Reference(DeviceDefinition) | The parent device it can be part of
material | | 0..* | BackboneElement | A substance used to create the material(s) of which the device is made
substance | | 1..1 | CodeableConcept | The substance
alternate | | 0..1 | boolean | Indicates an alternative material of the device
allergenicIndicator | | 0..1 | boolean | Whether the substance is a known or suspected allergen
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <udiDeviceIdentifier>  <!-- 0..* Unique Device Identifier (UDI) Barcode string -->
  <deviceIdentifier value="[string]"/><!-- 1..1 The identifier that is to be associated with every Device that
     references this DeviceDefintiion for the issuer and jurisdication porvided in the
     DeviceDefinition.udiDeviceIdentifier -->
  <issuer value="[uri]"/><!-- 1..1 The organization that assigns the identifier algorithm -->
  <jurisdiction value="[uri]"/><!-- 1..1 The jurisdiction to which the deviceIdentifier applies -->
 </udiDeviceIdentifier>
 <manufacturer[x]><!-- 0..1 string|Reference(Organization) Name of device manufacturer --></manufacturer[x]>
 <deviceName>  <!-- 0..* A name given to the device to identify it -->
  <name value="[string]"/><!-- 1..1 The name of the device -->
  <type value="[code]"/><!-- 1..1 udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name |
     model-name | other -->
 </deviceName>
 <modelNumber value="[string]"/><!-- 0..1 The model number for the device -->
 <type><!-- 0..1 CodeableConcept What kind of device or device system this is --></type>
 <specialization>  <!-- 0..* The capabilities supported on a  device, the standards to which the device conforms for a
     particular purpose, and used for the communication -->
  <systemType value="[string]"/><!-- 1..1 The standard that is used to operate and communicate -->
  <version value="[string]"/><!-- 0..1 The version of the standard that is used to operate and communicate -->
 </specialization>
 <version value="[string]"/><!-- 0..* Available versions -->
 <safety><!-- 0..* CodeableConcept Safety characteristics of the device --></safety>
 <shelfLifeStorage><!-- 0..* ProductShelfLife Shelf Life and storage information --></shelfLifeStorage>
 <physicalCharacteristics><!-- 0..1 ProdCharacteristic Dimensions, color etc. --></physicalCharacteristics>
 <languageCode><!-- 0..* CodeableConcept Language code for the human-readable text strings produced by the device (all
     supported) --></languageCode>
 <capability>  <!-- 0..* Device capabilities -->
  <type><!-- 1..1 CodeableConcept Type of capability --></type>
  <description><!-- 0..* CodeableConcept Description of capability --></description>
 </capability>
 <property>  <!-- 0..* The actual configuration settings of a device as it actually operates, e.g., regulation status,
     time properties -->
  <type><!-- 1..1 CodeableConcept Code that specifies the property DeviceDefinitionPropetyCode (Extensible) --></type>
  <valueQuantity><!-- 0..* Quantity Property value as a quantity --></valueQuantity>
  <valueCode><!-- 0..* CodeableConcept Property value as a code, e.g., NTP4 (synced to NTP) --></valueCode>
 </property>
 <owner><!-- 0..1 Reference(Organization) Organization responsible for device --></owner>
 <contact><!-- 0..* ContactPoint Details for human/organization for support --></contact>
 <url value="[uri]"/><!-- 0..1 Network address to contact device -->
 <onlineInformation value="[uri]"/><!-- 0..1 Access to on-line information -->
 <note><!-- 0..* Annotation Device notes and comments --></note>
 <quantity><!-- 0..1 Quantity The quantity of the device present in the packaging (e.g. the number of devices present in
     a pack, or the number of devices in the same package of the medicinal product) --></quantity>
 <parentDevice><!-- 0..1 Reference(DeviceDefinition) The parent device it can be part of --></parentDevice>
```

```
 <material>  <!-- 0..* A substance used to create the material(s) of which the device is made -->
  <substance><!-- 1..1 CodeableConcept The substance --></substance>
  <alternate value="[boolean]"/><!-- 0..1 Indicates an alternative material of the device -->
  <allergenicIndicator value="[boolean]"/><!-- 0..1 Whether the substance is a known or suspected allergen -->
 </material>
</DeviceDefinition>
```

JSON Template

```
{
  "resourceType" : "DeviceDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "udiDeviceIdentifier" : [{ // Unique Device Identifier (UDI) Barcode string
    "deviceIdentifier" : "<string>", // R!  The identifier that is to be associated with every Device that references
    this DeviceDefintiion for the issuer and jurisdication porvided in the DeviceDefinition.udiDeviceIdentifier
    "issuer" : "<uri>", // R!  The organization that assigns the identifier algorithm
    "jurisdiction" : "<uri>" // R!  The jurisdiction to which the deviceIdentifier applies
  }],
  // manufacturer[x]: Name of device manufacturer. One of these 2:
  "manufacturerString" : "<string>",
  "manufacturerReference" : { Reference(Organization) },
  "deviceName" : [{ // A name given to the device to identify it
    "name" : "<string>", // R!  The name of the device
    "type" : "<code>" // R!  udi-label-name | user-friendly-name | patient-reported-name | manufacturer-name | model-
    name | other
  }],
  "modelNumber" : "<string>", // The model number for the device
  "type" : { CodeableConcept }, // What kind of device or device system this is
  "specialization" : [{ // The capabilities supported on a  device, the standards to which the device conforms for a
    particular purpose, and used for the communication
    "systemType" : "<string>", // R!  The standard that is used to operate and communicate
    "version" : "<string>" // The version of the standard that is used to operate and communicate
  }],
  "version" : ["<string>"], // Available versions
  "safety" : [{ CodeableConcept }], // Safety characteristics of the device
  "shelfLifeStorage" : [{ ProductShelfLife }], // Shelf Life and storage information
  "physicalCharacteristics" : { ProdCharacteristic }, // Dimensions, color etc.
  "languageCode" : [{ CodeableConcept }], // Language code for the human-readable text strings produced by the device
    (all supported)
  "capability" : [{ // Device capabilities
    "type" : { CodeableConcept }, // R!  Type of capability
    "description" : [{ CodeableConcept }] // Description of capability
  }],
  "property" : [{ // The actual configuration settings of a device as it actually operates, e.g., regulation status,
    time properties
    "type" : { CodeableConcept }, // R!  Code that specifies the property DeviceDefinitionPropetyCode (Extensible)
    "valueQuantity" : [{ Quantity }], // Property value as a quantity
    "valueCode" : [{ CodeableConcept }] // Property value as a code, e.g., NTP4 (synced to NTP)
  }],
  "owner" : { Reference(Organization) }, // Organization responsible for device
  "contact" : [{ ContactPoint }], // Details for human/organization for support
  "url" : "<uri>", // Network address to contact device
  "onlineInformation" : "<uri>", // Access to on-line information
  "note" : [{ Annotation }], // Device notes and comments
  "quantity" : { Quantity }, // The quantity of the device present in the packaging (e.g. the number of devices present
    in a pack, or the number of devices in the same package of the medicinal product)
  "parentDevice" : { Reference(DeviceDefinition) }, // The parent device it can be part of
  "material" : [{ // A substance used to create the material(s) of which the device is made
    "substance" : { CodeableConcept }, // R!  The substance
    "alternate" : <boolean>, // Indicates an alternative material of the device
    "allergenicIndicator" : <boolean> // Whether the substance is a known or suspected allergen
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceDefinition.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:DeviceDefinition.udiDeviceIdentifier [ # 0..* Unique Device Identifier (UDI) Barcode string
    fhir:DeviceDefinition.udiDeviceIdentifier.deviceIdentifier [ string ]; # 1..1 The identifier that is to be
    associated with every Device that references this DeviceDefintiion for the issuer and jurisdication porvided in the
    DeviceDefinition.udiDeviceIdentifier
    fhir:DeviceDefinition.udiDeviceIdentifier.issuer [ uri ]; # 1..1 The organization that assigns the identifier
    algorithm
    fhir:DeviceDefinition.udiDeviceIdentifier.jurisdiction [ uri ]; # 1..1 The jurisdiction to which the
    deviceIdentifier applies
  ], ...;
  # DeviceDefinition.manufacturer[x] : 0..1 Name of device manufacturer. One of these 2
    fhir:DeviceDefinition.manufacturerString [ string ]
    fhir:DeviceDefinition.manufacturerReference [ Reference(Organization) ]
  fhir:DeviceDefinition.deviceName [ # 0..* A name given to the device to identify it
    fhir:DeviceDefinition.deviceName.name [ string ]; # 1..1 The name of the device
```

```
    fhir:DeviceDefinition.deviceName.type [ code ]; # 1..1 udi-label-name | user-friendly-name | patient-reported-name |
     manufacturer-name | model-name | other
  ], ...;
  fhir:DeviceDefinition.modelNumber [ string ]; # 0..1 The model number for the device
  fhir:DeviceDefinition.type [ CodeableConcept ]; # 0..1 What kind of device or device system this is
  fhir:DeviceDefinition.specialization [ # 0..* The capabilities supported on a  device, the standards to which the
    device conforms for a particular purpose, and used for the communication
    fhir:DeviceDefinition.specialization.systemType [ string ]; # 1..1 The standard that is used to operate and
    communicate
    fhir:DeviceDefinition.specialization.version [ string ]; # 0..1 The version of the standard that is used to operate
    and communicate
  ], ...;
  fhir:DeviceDefinition.version [ string ], ... ; # 0..* Available versions
  fhir:DeviceDefinition.safety [ CodeableConcept ], ... ; # 0..* Safety characteristics of the device
  fhir:DeviceDefinition.shelfLifeStorage [ ProductShelfLife ], ... ; # 0..* Shelf Life and storage information
  fhir:DeviceDefinition.physicalCharacteristics [ ProdCharacteristic ]; # 0..1 Dimensions, color etc.
  fhir:DeviceDefinition.languageCode [ CodeableConcept ], ... ; # 0..* Language code for the human-readable text strings
     produced by the device (all supported)
  fhir:DeviceDefinition.capability [ # 0..* Device capabilities
    fhir:DeviceDefinition.capability.type [ CodeableConcept ]; # 1..1 Type of capability
    fhir:DeviceDefinition.capability.description [ CodeableConcept ], ... ; # 0..* Description of capability
  ], ...;
  fhir:DeviceDefinition.property [ # 0..* The actual configuration settings of a device as it actually operates, e.g.,
    regulation status, time properties
    fhir:DeviceDefinition.property.type [ CodeableConcept ]; # 1..1 Code that specifies the property
    DeviceDefinitionPropetyCode (Extensible)
    fhir:DeviceDefinition.property.valueQuantity [ Quantity ], ... ; # 0..* Property value as a quantity
    fhir:DeviceDefinition.property.valueCode [ CodeableConcept ], ... ; # 0..* Property value as a code, e.g., NTP4
    (synced to NTP)
  ], ...;
  fhir:DeviceDefinition.owner [ Reference(Organization) ]; # 0..1 Organization responsible for device
  fhir:DeviceDefinition.contact [ ContactPoint ], ... ; # 0..* Details for human/organization for support
  fhir:DeviceDefinition.url [ uri ]; # 0..1 Network address to contact device
  fhir:DeviceDefinition.onlineInformation [ uri ]; # 0..1 Access to on-line information
  fhir:DeviceDefinition.note [ Annotation ], ... ; # 0..* Device notes and comments
  fhir:DeviceDefinition.quantity [ Quantity ]; # 0..1 The quantity of the device present in the packaging (e.g. the
    number of devices present in a pack, or the number of devices in the same package of the medicinal product)
  fhir:DeviceDefinition.parentDevice [ Reference(DeviceDefinition) ]; # 0..1 The parent device it can be part of
  fhir:DeviceDefinition.material [ # 0..* A substance used to create the material(s) of which the device is made
    fhir:DeviceDefinition.material.substance [ CodeableConcept ]; # 1..1 The substance
    fhir:DeviceDefinition.material.alternate [ boolean ]; # 0..1 Indicates an alternative material of the device
    fhir:DeviceDefinition.material.allergenicIndicator [ boolean ]; # 0..1 Whether the substance is a known or suspected
     allergen
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.15.4.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceDefinition.deviceName.type | The type of name the device is referred by. | Required | DeviceNameType
DeviceDefinition.type | Type of device e.g. according to official classification. | Example | FHIRDeviceTypes
DeviceDefinition.safety | | Example | DeviceSafety

8.15.5 Notes:

8.15.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | The identifier of the component | DeviceDefinition.identifier
parent | reference | The parent DeviceDefinition resource | DeviceDefinition.parentDevice (DeviceDefinition)
type | token | The device component type | DeviceDefinition.type

8.15.7 Resource DeviceDefinition - Examples

URL: https://hl7.org/fhir/R4/devicedefinition-examples.html

8.15.7 Resource DeviceDefinition - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
General DeviceDefinition example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.15.8 Resource DeviceDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/devicedefinition-definitions.html

8.15.8 Resource DeviceDefinition - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the DeviceDefinition resource.

DeviceDefinition
Element Id | DeviceDefinition
Definition | The characteristics, operational status and capabilities of a medical-related component of a medical device.
Cardinality | 0..*
Type | DomainResource
Comments | For the initial scope, this DeviceDefinition resource is only applicable to describe a single node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard and that does not represent a metric. Examples for such a node are MDS, VMD, or Channel.
DeviceDefinition.identifier
Element Id | DeviceDefinition.identifier
Definition | Unique instance identifiers assigned to a device by the software, manufacturers, other organizations or owners. For example: handle ID.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
DeviceDefinition.udiDeviceIdentifier
Element Id | DeviceDefinition.udiDeviceIdentifier
Definition | Unique device identifier (UDI) assigned to device label or package. Note that the Device may include multiple udiCarriers as it either may include just the udiCarrier for the jurisdiction it is sold, or for multiple jurisdictions it could have been sold.
Cardinality | 0..*
DeviceDefinition.udiDeviceIdentifier.deviceIdentifier
Element Id | DeviceDefinition.udiDeviceIdentifier.deviceIdentifier
Definition | The identifier that is to be associated with every Device that references this DeviceDefintiion for the issuer and jurisdication porvided in the DeviceDefinition.udiDeviceIdentifier.
Cardinality | 1..1
Type | string
DeviceDefinition.udiDeviceIdentifier.issuer
Element Id | DeviceDefinition.udiDeviceIdentifier.issuer
Definition | The organization that assigns the identifier algorithm.
Cardinality | 1..1
Type | uri
DeviceDefinition.udiDeviceIdentifier.jurisdiction
Element Id | DeviceDefinition.udiDeviceIdentifier.jurisdiction
Definition | The jurisdiction to which the deviceIdentifier applies.
Cardinality | 1..1
Type | uri
DeviceDefinition.manufacturer[x]
Element Id | DeviceDefinition.manufacturer[x]
Definition | A name of the manufacturer.
Cardinality | 0..1
Type | string|Reference(Organization)
[x] Note | See Choice of Data Types for further information about how to use [x]
DeviceDefinition.deviceName
Element Id | DeviceDefinition.deviceName
Definition | A name given to the device to identify it.
Cardinality | 0..*
DeviceDefinition.deviceName.name
Element Id | DeviceDefinition.deviceName.name
Definition | The name of the device.
Cardinality | 1..1
Type | string
Alternate Names | Σ
DeviceDefinition.deviceName.type
Element Id | DeviceDefinition.deviceName.type
Definition | The type of deviceName. UDILabelName | UserFriendlyName | PatientReportedName | ManufactureDeviceName | ModelName.
Cardinality | 1..1
Terminology Binding | DeviceNameType (Required)
Type | code
DeviceDefinition.modelNumber

Element Id | DeviceDefinition.modelNumber
Definition | The model number for the device.
Cardinality | 0..1
Type | string
DeviceDefinition.type
Element Id | DeviceDefinition.type
Definition | What kind of device or device system this is.
Cardinality | 0..1
Terminology Binding | FHIR Device Types (Example)
Type | CodeableConcept
DeviceDefinition.specialization
Element Id | DeviceDefinition.specialization
Definition | The capabilities supported on a device, the standards to which the device conforms for a particular purpose, and used for the communication.
Cardinality | 0..*
DeviceDefinition.specialization.systemType
Element Id | DeviceDefinition.specialization.systemType
Definition | The standard that is used to operate and communicate.
Cardinality | 1..1
Type | string
Alternate Names | Σ
DeviceDefinition.specialization.version
Element Id | DeviceDefinition.specialization.version
Definition | The version of the standard that is used to operate and communicate.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
DeviceDefinition.version
Element Id | DeviceDefinition.version
Definition | The available versions of the device, e.g., software versions.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..*
Type | string
DeviceDefinition.safety
Element Id | DeviceDefinition.safety
Definition | Safety characteristics of the device.
Cardinality | 0..*
Terminology Binding | Device safety (Example)
Type | CodeableConcept
Summary | true
DeviceDefinition.shelfLifeStorage
Element Id | DeviceDefinition.shelfLifeStorage
Definition | Shelf Life and storage information.
Cardinality | 0..*
Type | ProductShelfLife
DeviceDefinition.physicalCharacteristics
Element Id | DeviceDefinition.physicalCharacteristics
Definition | Dimensions, color etc.
Cardinality | 0..1
Type | ProdCharacteristic
DeviceDefinition.languageCode
Element Id | DeviceDefinition.languageCode
Definition | Language code for the human-readable text strings produced by the device (all supported).
Cardinality | 0..*
Type | CodeableConcept
DeviceDefinition.capability
Element Id | DeviceDefinition.capability
Definition | Device capabilities.
Cardinality | 0..*
DeviceDefinition.capability.type
Element Id | DeviceDefinition.capability.type
Definition | Type of capability.
Cardinality | 1..1
Type | CodeableConcept
DeviceDefinition.capability.description
Element Id | DeviceDefinition.capability.description
Definition | Description of capability.
Cardinality | 0..*
Type | CodeableConcept
DeviceDefinition.property
Element Id | DeviceDefinition.property
Definition | The actual configuration settings of a device as it actually operates, e.g., regulation status, time properties.

Cardinality | 0..*
DeviceDefinition.property.type
Element Id | DeviceDefinition.property.type
Definition | Code that specifies the property DeviceDefinitionPropetyCode (Extensible).
Cardinality | 1..1
Type | CodeableConcept
DeviceDefinition.property.valueQuantity
Element Id | DeviceDefinition.property.valueQuantity
Definition | Property value as a quantity.
Cardinality | 0..*
Type | Quantity
DeviceDefinition.property.valueCode
Element Id | DeviceDefinition.property.valueCode
Definition | Property value as a code, e.g., NTP4 (synced to NTP).
Cardinality | 0..*
Type | CodeableConcept
DeviceDefinition.owner
Element Id | DeviceDefinition.owner
Definition | An organization that is responsible for the provision and ongoing maintenance of the device.
Cardinality | 0..1
Type | Reference(Organization)
DeviceDefinition.contact
Element Id | DeviceDefinition.contact
Definition | Contact details for an organization or a particular human that is responsible for the device.
Cardinality | 0..*
Type | ContactPoint
Comments | used for troubleshooting etc.
DeviceDefinition.url
Element Id | DeviceDefinition.url
Definition | A network address on which the device may be contacted directly.
Cardinality | 0..1
Type | uri
Comments | If the device is running a FHIR server, the network address should be the Base URL from which a conformance statement may be retrieved.
DeviceDefinition.onlineInformation
Element Id | DeviceDefinition.onlineInformation
Definition | Access to on-line information about the device.
Cardinality | 0..1
Type | uri
DeviceDefinition.note
Element Id | DeviceDefinition.note
Definition | Descriptive information, usage information or implantation information that is not captured in an existing element.
Cardinality | 0..*
Type | Annotation
DeviceDefinition.quantity
Element Id | DeviceDefinition.quantity
Definition | The quantity of the device present in the packaging (e.g. the number of devices present in a pack, or the number of devices in the same package of the medicinal product).
Cardinality | 0..1
Type | Quantity
DeviceDefinition.parentDevice
Element Id | DeviceDefinition.parentDevice
Definition | The parent device it can be part of.
Cardinality | 0..1
Type | Reference(DeviceDefinition)
Summary | true
DeviceDefinition.material
Element Id | DeviceDefinition.material
Definition | A substance used to create the material(s) of which the device is made.
Cardinality | 0..*
DeviceDefinition.material.substance
Element Id | DeviceDefinition.material.substance
Definition | The substance.
Cardinality | 1..1
Type | CodeableConcept
DeviceDefinition.material.alternate
Element Id | DeviceDefinition.material.alternate
Definition | Indicates an alternative material of the device.
Cardinality | 0..1
Type | boolean
DeviceDefinition.material.allergenicIndicator
Element Id | DeviceDefinition.material.allergenicIndicator

Definition | Whether the substance is a known or suspected allergen.
Cardinality | 0..1
Type | boolean

8.15.9 Resource DeviceDefinition - Mappings

URL: https://hl7.org/fhir/R4/devicedefinition-mappings.html

8.15.9 Resource DeviceDefinition - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the devicedefinition resource (see Mappings to Other Standards for further information & status).

8.15.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DeviceDefinition
identifier | FiveWs.identifier
manufacturer[x] | FiveWs.what[x]
type | FiveWs.what[x]
modelNumber | FiveWs.what[x]
version | FiveWs.what[x]
owner | FiveWs.source
contact | FiveWs.source
url | FiveWs.where[x]

8.15.9.2 RIM Mapping (http://hl7.org/v3)

DeviceDefinition
manufacturer[x] | .playedRole[typeCode=MANU].scoper.name
type | .playedRole[typeCode=MANU].code
modelNumber | .softwareName (included as part)
owner | .playedRole[typeCode=OWN].scoper
contact | .scopedRole[typeCode=CON].player
url | .telecom
note | .text

8.15.10 Resource DeviceDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/devicedefinition-profiles.html

8.15.10 Resource DeviceDefinition - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the DeviceDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.15.10.1 Profiles

No Profiles defined for this resource

8.15.10.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.15.10.3 Search Extensions

No Search Extensions defined for this resource

8.15.11 JSON Schema for DeviceDefinition

URL: https://hl7.org/fhir/R4/devicedefinition.schema.json.html

8.15.11 JSON Schema for DeviceDefinition

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for DeviceDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DeviceDefinition",
  "$ref": "#/definitions/DeviceDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DeviceDefinition": {
      "description": "The characteristics, operational status and capabilities of a medical-related component of a
    medical device.",
      "properties": {
        "resourceType": {
          "description": "This is a DeviceDefinition resource",
          "const": "DeviceDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
```

```
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique instance identifiers assigned to a device by the software, manufacturers, other
organizations or owners. For example: handle ID.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "udiDeviceIdentifier": {
      "description": "Unique device identifier (UDI) assigned to device label or package.  Note that the Device may
include multiple udiCarriers as it either may include just the udiCarrier for the jurisdiction it is sold, or for
multiple jurisdictions it could have been sold.",
      "items": {
        "$ref": "#/definitions/DeviceDefinition_UdiDeviceIdentifier"
      },
      "type": "array"
    },
    "manufacturerString": {
      "description": "A name of the manufacturer.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_manufacturerString": {
      "description": "Extensions for manufacturerString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "manufacturerReference": {
      "description": "A name of the manufacturer.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "deviceName": {
      "description": "A name given to the device to identify it.",
      "items": {
        "$ref": "#/definitions/DeviceDefinition_DeviceName"
      },
      "type": "array"
    },
    "modelNumber": {
      "description": "The model number for the device.",
      "$ref": "#/definitions/string"
    },
    "_modelNumber": {
      "description": "Extensions for modelNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "What kind of device or device system this is.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "specialization": {
      "description": "The capabilities supported on a  device, the standards to which the device conforms for a
particular purpose, and used for the communication.",
      "items": {
        "$ref": "#/definitions/DeviceDefinition_Specialization"
      },
      "type": "array"
    },
    "version": {
      "description": "The available versions of the device, e.g., software versions.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_version": {
      "description": "Extensions for version",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "safety": {
      "description": "Safety characteristics of the device.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "shelfLifeStorage": {
      "description": "Shelf Life and storage information.",
      "items": {
```

```
        "$ref": "ProductShelfLife.schema.json#/definitions/ProductShelfLife"
      },
      "type": "array"
    },
    "physicalCharacteristics": {
      "description": "Dimensions, color etc.",
      "$ref": "ProdCharacteristic.schema.json#/definitions/ProdCharacteristic"
    },
    "languageCode": {
      "description": "Language code for the human-readable text strings produced by the device (all supported).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "capability": {
      "description": "Device capabilities.",
      "items": {
        "$ref": "#/definitions/DeviceDefinition_Capability"
      },
      "type": "array"
    },
    "property": {
      "description": "The actual configuration settings of a device as it actually operates, e.g., regulation
status, time properties.",
      "items": {
        "$ref": "#/definitions/DeviceDefinition_Property"
      },
      "type": "array"
    },
    "owner": {
      "description": "An organization that is responsible for the provision and ongoing maintenance of the device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "contact": {
      "description": "Contact details for an organization or a particular human that is responsible for the
device.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "url": {
      "description": "A network address on which the device may be contacted directly.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "onlineInformation": {
      "description": "Access to on-line information about the device.",
      "$ref": "#/definitions/uri"
    },
    "_onlineInformation": {
      "description": "Extensions for onlineInformation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "Descriptive information, usage information or implantation information that is not captured in
 an existing element.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "quantity": {
      "description": "The quantity of the device present in the packaging (e.g. the number of devices present in a
pack, or the number of devices in the same package of the medicinal product).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "parentDevice": {
      "description": "The parent device it can be part of.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "material": {
      "description": "A substance used to create the material(s) of which the device is made.",
      "items": {
        "$ref": "#/definitions/DeviceDefinition_Material"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"DeviceDefinition_UdiDeviceIdentifier": {
```

```
  "description": "The characteristics, operational status and capabilities of a medical-related component of a
medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "deviceIdentifier": {
      "description": "The identifier that is to be associated with every Device that references this
DeviceDefintiion for the issuer and jurisdication porvided in the DeviceDefinition.udiDeviceIdentifier.",
      "$ref": "#/definitions/string"
    },
    "_deviceIdentifier": {
      "description": "Extensions for deviceIdentifier",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "issuer": {
      "description": "The organization that assigns the identifier algorithm.",
      "$ref": "#/definitions/uri"
    },
    "_issuer": {
      "description": "Extensions for issuer",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "jurisdiction": {
      "description": "The jurisdiction to which the deviceIdentifier applies.",
      "$ref": "#/definitions/uri"
    },
    "_jurisdiction": {
      "description": "Extensions for jurisdiction",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"DeviceDefinition_DeviceName": {
  "description": "The characteristics, operational status and capabilities of a medical-related component of a
medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The name of the device.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of deviceName.\nUDILabelName | UserFriendlyName | PatientReportedName |
ManufactureDeviceName | ModelName.",
      "enum": [
        "udi-label-name",
        "user-friendly-name",
        "patient-reported-name",
        "manufacturer-name",
        "model-name",
        "other"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"DeviceDefinition_Specialization": {
  "description": "The characteristics, operational status and capabilities of a medical-related component of a
medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "systemType": {
      "description": "The standard that is used to operate and communicate.",
      "$ref": "#/definitions/string"
    },
    "_systemType": {
      "description": "Extensions for systemType",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The version of the standard that is used to operate and communicate.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"DeviceDefinition_Capability": {
  "description": "The characteristics, operational status and capabilities of a medical-related component of a
medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
```

```
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of capability.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "Description of capability.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"DeviceDefinition_Property": {
  "description": "The characteristics, operational status and capabilities of a medical-related component of a
medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Code that specifies the property DeviceDefinitionPropetyCode (Extensible).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueQuantity": {
      "description": "Property value as a quantity.",
      "items": {
        "$ref": "Quantity.schema.json#/definitions/Quantity"
      },
      "type": "array"
    },
    "valueCode": {
```

```
        "description": "Property value as a code, e.g., NTP4 (synced to NTP).",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "type"
    ]
  },
  "DeviceDefinition_Material": {
    "description": "The characteristics, operational status and capabilities of a medical-related component of a
  medical device.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
  value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
  definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the element and that modifies the understanding of the element in which it is contained and/or the understanding of
  the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
  use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
   of the definition of the extension. Applications processing a resource are required to check for modifier
  extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
  (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "substance": {
        "description": "The substance.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "alternate": {
        "description": "Indicates an alternative material of the device.",
        "$ref": "#/definitions/boolean"
      },
      "_alternate": {
        "description": "Extensions for alternate",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "allergenicIndicator": {
        "description": "Whether the substance is a known or suspected allergen.",
        "$ref": "#/definitions/boolean"
      },
      "_allergenicIndicator": {
        "description": "Extensions for allergenicIndicator",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false,
    "required": [
      "substance"
    ]
  }
 }
}
```


8.15.12 Resource DeviceDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/devicedefinition-dependencies.html

8.15.12 Resource DeviceDefinition - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for DeviceDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
DeviceDefinition | | DomainResource

identifier | 0..* | Identifier
udiDeviceIdentifier | 0..* | BackboneElement
deviceIdentifier | 1..1 | string
issuer | 1..1 | uri
jurisdiction | 1..1 | uri
manufacturer[x] | 0..1
manufacturerString | | string
manufacturerReference | | Reference(Organization
deviceName | 0..* | BackboneElement
name | 1..1 | string
type | 1..1 | code
modelNumber | 0..1 | string
type | 0..1 | CodeableConcept
specialization | 0..* | BackboneElement
systemType | 1..1 | string
version | 0..1 | string
version | 0..* | string
safety | 0..* | CodeableConcept
shelfLifeStorage | 0..* | ProductShelfLife | Type Error: (ProductShelfLife = FMM5-Draft vs. Element = FMM0-Trial Use)
physicalCharacteristics | 0..1 | ProdCharacteristic | Type Error: (ProdCharacteristic = FMM5-Draft vs. Element = FMM0-Trial Use)
languageCode | 0..* | CodeableConcept
capability | 0..* | BackboneElement
type | 1..1 | CodeableConcept
description | 0..* | CodeableConcept
property | 0..* | BackboneElement
type | 1..1 | CodeableConcept
valueQuantity | 0..* | Quantity
valueCode | 0..* | CodeableConcept
owner | 0..1 | Reference(Organization)
contact | 0..* | ContactPoint
url | 0..1 | uri
onlineInformation | 0..1 | uri
note | 0..* | Annotation
quantity | 0..1 | Quantity
parentDevice | 0..1 | Reference(DeviceDefinition)
material | 0..* | BackboneElement
substance | 1..1 | CodeableConcept
alternate | 0..1 | boolean
allergenicIndicator | 0..1 | boolean

8.16 Resource DeviceMetric - Content

URL: https://hl7.org/fhir/R4/devicemetric.html
Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Describes a measurement, calculation or setting capability of a medical device.

8.16.1 Scope and Usage

The DeviceMetric resource describes mandatory static properties that characterize a direct or derived, quantitative or qualitative biosignal measurement, setting, or calculation produced by a medical device. The DeviceMetric resource can also be used to describe the non-static but highly relevant properties to the metric such as metric status, metric last calibration time and type, measurement mode, color, reference link to the parent DeviceComponent to where it belongs, and any capabilities that the metric offers (for example: setting the metric label).

Note:

For the initial scope, this DeviceMetric resource is only applicable to describe a single metric node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard.

8.16.2 Boundaries and Relationships

There are two related resources

• Device - The physical device that this DeviceMetric belongs to.

This resource is referenced by Media and Observation

8.16.3 Resource Content

• Structure

• UML

- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceMetric | ΣTU | | DomainResource | Measurement, calculation or setting capability of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Instance identifier
type | Σ | 1..1 | CodeableConcept | Identity of metric, for example Heart Rate or PEEP Setting Device Metric and Component Types (Preferred)
unit | Σ | 0..1 | CodeableConcept | Unit of Measure for the Metric Device Metric and Component Types (Preferred)
source | Σ | 0..1 | Reference(Device) | Describes the link to the source Device
parent | Σ | 0..1 | Reference(Device) | Describes the link to the parent Device
operationalStatus | Σ | 0..1 | code | on | off | standby | entered-in-error DeviceMetricOperationalStatus (Required)
color | Σ | 0..1 | code | black | red | green | yellow | blue | magenta | cyan | white DeviceMetricColor (Required)
category | Σ | 1..1 | code | measurement | setting | calculation | unspecified DeviceMetricCategory (Required)
measurementPeriod | Σ | 0..1 | Timing | Describes the measurement repetition time
calibration | Σ | 0..* | BackboneElement | Describes the calibrations that have been performed or that are required to be performed
type | Σ | 0..1 | code | unspecified | offset | gain | two-point DeviceMetricCalibrationType (Required)
state | Σ | 0..1 | code | not-calibrated | calibration-required | calibrated | unspecified DeviceMetricCalibrationState (Required)
time | Σ | 0..1 | instant | Describes the time last calibration has been performed
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceMetric xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <type><!-- 1..1 CodeableConcept Identity of metric, for example Heart Rate or PEEP Setting --></type>
 <unit><!-- 0..1 CodeableConcept Unit of Measure for the Metric --></unit>
 <source><!-- 0..1 Reference(Device) Describes the link to the source Device --></source>
 <parent><!-- 0..1 Reference(Device) Describes the link to the parent Device --></parent>
 <operationalStatus value="[code]"/><!-- 0..1 on | off | standby | entered-in-error -->
 <color value="[code]"/><!-- 0..1 black | red | green | yellow | blue | magenta | cyan | white -->
 <category value="[code]"/><!-- 1..1 measurement | setting | calculation | unspecified -->
 <measurementPeriod><!-- 0..1 Timing Describes the measurement repetition time --></measurementPeriod>
 <calibration>  <!-- 0..* Describes the calibrations that have been performed or that are required to be performed -->
  <type value="[code]"/><!-- 0..1 unspecified | offset | gain | two-point -->
  <state value="[code]"/><!-- 0..1 not-calibrated | calibration-required | calibrated | unspecified -->
  <time value="[instant]"/><!-- 0..1 Describes the time last calibration has been performed -->
 </calibration>
</DeviceMetric>
```

JSON Template

```
{
  "resourceType" : "DeviceMetric",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "type" : { CodeableConcept }, // R!  Identity of metric, for example Heart Rate or PEEP Setting
  "unit" : { CodeableConcept }, // Unit of Measure for the Metric
  "source" : { Reference(Device) }, // Describes the link to the source Device
  "parent" : { Reference(Device) }, // Describes the link to the parent Device
  "operationalStatus" : "<code>", // on | off | standby | entered-in-error
  "color" : "<code>", // black | red | green | yellow | blue | magenta | cyan | white
  "category" : "<code>", // R!  measurement | setting | calculation | unspecified
  "measurementPeriod" : { Timing }, // Describes the measurement repetition time
  "calibration" : [{ // Describes the calibrations that have been performed or that are required to be performed
    "type" : "<code>", // unspecified | offset | gain | two-point
    "state" : "<code>", // not-calibrated | calibration-required | calibrated | unspecified
    "time" : "<instant>" // Describes the time last calibration has been performed
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceMetric;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceMetric.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:DeviceMetric.type [ CodeableConcept ]; # 1..1 Identity of metric, for example Heart Rate or PEEP Setting
  fhir:DeviceMetric.unit [ CodeableConcept ]; # 0..1 Unit of Measure for the Metric
  fhir:DeviceMetric.source [ Reference(Device) ]; # 0..1 Describes the link to the source Device
```

```
  fhir:DeviceMetric.parent [ Reference(Device) ]; # 0..1 Describes the link to the parent Device
  fhir:DeviceMetric.operationalStatus [ code ]; # 0..1 on | off | standby | entered-in-error
  fhir:DeviceMetric.color [ code ]; # 0..1 black | red | green | yellow | blue | magenta | cyan | white
  fhir:DeviceMetric.category [ code ]; # 1..1 measurement | setting | calculation | unspecified
  fhir:DeviceMetric.measurementPeriod [ Timing ]; # 0..1 Describes the measurement repetition time
  fhir:DeviceMetric.calibration [ # 0..* Describes the calibrations that have been performed or that are required to be
    performed
    fhir:DeviceMetric.calibration.type [ code ]; # 0..1 unspecified | offset | gain | two-point
    fhir:DeviceMetric.calibration.state [ code ]; # 0..1 not-calibrated | calibration-required | calibrated |
    unspecified
    fhir:DeviceMetric.calibration.time [ instant ]; # 0..1 Describes the time last calibration has been performed
  ], ...;
]
```

Changes since R3

DeviceMetric

DeviceMetric.identifier | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

DeviceMetric.parent | Type Reference: Added Target Type Device Type Reference: Removed Target Type DeviceComponent

DeviceMetric.operationalStatus | Change value set from http://hl7.org/fhir/ValueSet/metric-operational-status to http://hl7.org/fhir/ValueSet/metric-operational-status|4.0.1

DeviceMetric.color | Change value set from http://hl7.org/fhir/ValueSet/metric-color to http://hl7.org/fhir/ValueSet/metric-color|4.0.1

DeviceMetric.category | Change value set from http://hl7.org/fhir/ValueSet/metric-category to http://hl7.org/fhir/ValueSet/metric-category|4.0.1

DeviceMetric.calibration.type | Change value set from http://hl7.org/fhir/ValueSet/metric-calibration-type to http://hl7.org/fhir/ValueSet/metric-calibration-type|4.0.1

DeviceMetric.calibration.state | Change value set from http://hl7.org/fhir/ValueSet/metric-calibration-state to http://hl7.org/fhir/ValueSet/metric-calibration-state|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

DeviceMetric | ΣTU | | DomainResource | Measurement, calculation or setting capability of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Instance identifier

type | Σ | 1..1 | CodeableConcept | Identity of metric, for example Heart Rate or PEEP Setting Device Metric and Component Types (Preferred)

unit | Σ | 0..1 | CodeableConcept | Unit of Measure for the Metric Device Metric and Component Types (Preferred)

source | Σ | 0..1 | Reference(Device) | Describes the link to the source Device

parent | Σ | 0..1 | Reference(Device) | Describes the link to the parent Device

operationalStatus | Σ | 0..1 | code | on | off | standby | entered-in-error DeviceMetricOperationalStatus (Required)

color | Σ | 0..1 | code | black | red | green | yellow | blue | magenta | cyan | white DeviceMetricColor (Required)

category | Σ | 1..1 | code | measurement | setting | calculation | unspecified DeviceMetricCategory (Required)

measurementPeriod | Σ | 0..1 | Timing | Describes the measurement repetition time

calibration | Σ | 0..* | BackboneElement | Describes the calibrations that have been performed or that are required to be performed

type | Σ | 0..1 | code | unspecified | offset | gain | two-point DeviceMetricCalibrationType (Required)

state | Σ | 0..1 | code | not-calibrated | calibration-required | calibrated | unspecified DeviceMetricCalibrationState (Required)

time | Σ | 0..1 | instant | Describes the time last calibration has been performed

Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceMetric xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <type><!-- 1..1 CodeableConcept Identity of metric, for example Heart Rate or PEEP Setting --></type>
 <unit><!-- 0..1 CodeableConcept Unit of Measure for the Metric --></unit>
 <source><!-- 0..1 Reference(Device) Describes the link to the source Device --></source>
 <parent><!-- 0..1 Reference(Device) Describes the link to the parent Device --></parent>
 <operationalStatus value="[code]"/><!-- 0..1 on | off | standby | entered-in-error -->
 <color value="[code]"/><!-- 0..1 black | red | green | yellow | blue | magenta | cyan | white -->
 <category value="[code]"/><!-- 1..1 measurement | setting | calculation | unspecified -->
 <measurementPeriod><!-- 0..1 Timing Describes the measurement repetition time --></measurementPeriod>
 <calibration>  <!-- 0..* Describes the calibrations that have been performed or that are required to be performed -->
  <type value="[code]"/><!-- 0..1 unspecified | offset | gain | two-point -->
  <state value="[code]"/><!-- 0..1 not-calibrated | calibration-required | calibrated | unspecified -->
  <time value="[instant]"/><!-- 0..1 Describes the time last calibration has been performed -->
 </calibration>
</DeviceMetric>
```

JSON Template

```
{
  "resourceType" : "DeviceMetric",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
```

```
  "identifier" : [{ Identifier }], // Instance identifier
  "type" : { CodeableConcept }, // R!  Identity of metric, for example Heart Rate or PEEP Setting
  "unit" : { CodeableConcept }, // Unit of Measure for the Metric
  "source" : { Reference(Device) }, // Describes the link to the source Device
  "parent" : { Reference(Device) }, // Describes the link to the parent Device
  "operationalStatus" : "<code>", // on | off | standby | entered-in-error
  "color" : "<code>", // black | red | green | yellow | blue | magenta | cyan | white
  "category" : "<code>", // R!  measurement | setting | calculation | unspecified
  "measurementPeriod" : { Timing }, // Describes the measurement repetition time
  "calibration" : [{ // Describes the calibrations that have been performed or that are required to be performed
    "type" : "<code>", // unspecified | offset | gain | two-point
    "state" : "<code>", // not-calibrated | calibration-required | calibrated | unspecified
    "time" : "<instant>" // Describes the time last calibration has been performed
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceMetric;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceMetric.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:DeviceMetric.type [ CodeableConcept ]; # 1..1 Identity of metric, for example Heart Rate or PEEP Setting
  fhir:DeviceMetric.unit [ CodeableConcept ]; # 0..1 Unit of Measure for the Metric
  fhir:DeviceMetric.source [ Reference(Device) ]; # 0..1 Describes the link to the source Device
  fhir:DeviceMetric.parent [ Reference(Device) ]; # 0..1 Describes the link to the parent Device
  fhir:DeviceMetric.operationalStatus [ code ]; # 0..1 on | off | standby | entered-in-error
  fhir:DeviceMetric.color [ code ]; # 0..1 black | red | green | yellow | blue | magenta | cyan | white
  fhir:DeviceMetric.category [ code ]; # 1..1 measurement | setting | calculation | unspecified
  fhir:DeviceMetric.measurementPeriod [ Timing ]; # 0..1 Describes the measurement repetition time
  fhir:DeviceMetric.calibration [ # 0..* Describes the calibrations that have been performed or that are required to be
    performed
    fhir:DeviceMetric.calibration.type [ code ]; # 0..1 unspecified | offset | gain | two-point
    fhir:DeviceMetric.calibration.state [ code ]; # 0..1 not-calibrated | calibration-required | calibrated |
    unspecified
    fhir:DeviceMetric.calibration.time [ instant ]; # 0..1 Describes the time last calibration has been performed
  ], ...;
]
```

Changes since Release 3

DeviceMetric

DeviceMetric.identifier | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

DeviceMetric.parent | Type Reference: Added Target Type Device Type Reference: Removed Target Type DeviceComponent

DeviceMetric.operationalStatus | Change value set from http://hl7.org/fhir/ValueSet/metric-operational-status to http://hl7.org/fhir/ValueSet/metric-operational-status|4.0.1

DeviceMetric.color | Change value set from http://hl7.org/fhir/ValueSet/metric-color to http://hl7.org/fhir/ValueSet/metric-color|4.0.1

DeviceMetric.category | Change value set from http://hl7.org/fhir/ValueSet/metric-category to http://hl7.org/fhir/ValueSet/metric-category|4.0.1

DeviceMetric.calibration.type | Change value set from http://hl7.org/fhir/ValueSet/metric-calibration-type to http://hl7.org/fhir/ValueSet/metric-calibration-type|4.0.1

DeviceMetric.calibration.state | Change value set from http://hl7.org/fhir/ValueSet/metric-calibration-state to http://hl7.org/fhir/ValueSet/metric-calibration-state|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.16.3.1 Terminology Bindings

Path | Definition | Type | Reference

DeviceMetric.type | Describes the metric type. | Preferred | DeviceMetricAndComponentTypes

DeviceMetric.unit | Describes the unit of the metric. | Preferred | DeviceMetricAndComponentTypes

DeviceMetric.operationalStatus | Describes the operational status of the DeviceMetric. | Required | DeviceMetricOperationalStatus

DeviceMetric.color | Describes the typical color of representation. | Required | DeviceMetricColor

DeviceMetric.category | Describes the category of the metric. | Required | DeviceMetricCategory

DeviceMetric.calibration.type | Describes the type of a metric calibration. | Required | DeviceMetricCalibrationType

DeviceMetric.calibration.state | Describes the state of a metric calibration. | Required | DeviceMetricCalibrationState

8.16.4 Notes:

• The correct codes for the metric types are registered in the RTM Management service , but this is not required. See Terminology Systems for the correct representation of these codes in a Coding data type.

8.16.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The category of the metric | DeviceMetric.category
identifier | token | The identifier of the metric | DeviceMetric.identifier
parent | reference | The parent DeviceMetric resource | DeviceMetric.parent (Device)
source | reference | The device resource | DeviceMetric.source (Device)
type | token | The component type | DeviceMetric.type

8.16.6 Resource DeviceMetric - Examples

URL: https://hl7.org/fhir/R4/devicemetric-examples.html

8.16.6 Resource DeviceMetric - Examples

Health Care Devices Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
General DeviceMetric example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.16.7 Resource DeviceMetric - Detailed Descriptions

URL: https://hl7.org/fhir/R4/devicemetric-definitions.html

8.16.7 Resource DeviceMetric - Detailed Descriptions

Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the DeviceMetric resource.

DeviceMetric
Element Id | DeviceMetric
Definition | Describes a measurement, calculation or setting capability of a medical device.
Cardinality | 0..*
Type | DomainResource
Summary | true
Comments | For the initial scope, this DeviceMetric resource is only applicable to describe a single metric node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard.
DeviceMetric.identifier
Element Id | DeviceMetric.identifier
Definition | Unique instance identifiers assigned to a device by the device or gateway software, manufacturers, other organizations or owners. For example: handle ID.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Comments | For identifiers assigned to a device by the device or gateway software, the system element of the identifier should be set to the unique identifier of the device.
DeviceMetric.type
Element Id | DeviceMetric.type
Definition | Describes the type of the metric. For example: Heart Rate, PEEP Setting, etc.
Cardinality | 1..1
Terminology Binding | Device Metric and Component Types (Preferred)
Type | CodeableConcept
Summary | true
Comments | DeviceMetric.type can be referred to either IEEE 11073-10101 or LOINC.
DeviceMetric.unit
Element Id | DeviceMetric.unit
Definition | Describes the unit that an observed value determined for this metric will have. For example: Percent, Seconds, etc.
Cardinality | 0..1
Terminology Binding | Device Metric and Component Types (Preferred)
Type | CodeableConcept
Summary | true
Comments | DeviceMetric.unit can refer to either UCUM or preferable a RTMMS coding system.
DeviceMetric.source
Element Id | DeviceMetric.source
Definition | Describes the link to the Device that this DeviceMetric belongs to and that contains administrative device information such as manufacturer, serial number, etc.
Cardinality | 0..1

Type | Reference(Device)
Summary | true
DeviceMetric.parent
Element Id | DeviceMetric.parent
Definition | Describes the link to the Device that this DeviceMetric belongs to and that provide information about the location of this DeviceMetric in the containment structure of the parent Device. An example would be a Device that represents a Channel. This reference can be used by a client application to distinguish DeviceMetrics that have the same type, but should be interpreted based on their containment location.
Cardinality | 0..1
Type | Reference(Device)
Summary | true
DeviceMetric.operationalStatus
Element Id | DeviceMetric.operationalStatus
Definition | Indicates current operational state of the device. For example: On, Off, Standby, etc.
Cardinality | 0..1
Terminology Binding | DeviceMetricOperationalStatus (Required)
Type | code
Summary | true
DeviceMetric.color
Element Id | DeviceMetric.color
Definition | Describes the color representation for the metric. This is often used to aid clinicians to track and identify parameter types by color. In practice, consider a Patient Monitor that has ECG/HR and Pleth for example; the parameters are displayed in different characteristic colors, such as HR-blue, BP-green, and PR and SpO2- magenta.
Cardinality | 0..1
Terminology Binding | DeviceMetricColor (Required)
Type | code
Summary | true
DeviceMetric.category
Element Id | DeviceMetric.category
Definition | Indicates the category of the observation generation process. A DeviceMetric can be for example a setting, measurement, or calculation.
Cardinality | 1..1
Terminology Binding | DeviceMetricCategory (Required)
Type | code
Summary | true
DeviceMetric.measurementPeriod
Element Id | DeviceMetric.measurementPeriod
Definition | Describes the measurement repetition time. This is not necessarily the same as the update period. The measurement repetition time can range from milliseconds up to hours. An example for a measurement repetition time in the range of milliseconds is the sampling rate of an ECG. An example for a measurement repetition time in the range of hours is a NIBP that is triggered automatically every hour. The update period may be different than the measurement repetition time, if the device does not update the published observed value with the same frequency as it was measured.
Cardinality | 0..1
Type | Timing
Summary | true
DeviceMetric.calibration
Element Id | DeviceMetric.calibration
Definition | Describes the calibrations that have been performed or that are required to be performed.
Cardinality | 0..*
Summary | true
DeviceMetric.calibration.type
Element Id | DeviceMetric.calibration.type
Definition | Describes the type of the calibration method.
Cardinality | 0..1
Terminology Binding | DeviceMetricCalibrationType (Required)
Type | code
Summary | true
DeviceMetric.calibration.state
Element Id | DeviceMetric.calibration.state
Definition | Describes the state of the calibration.
Cardinality | 0..1
Terminology Binding | DeviceMetricCalibrationState (Required)
Type | code
Summary | true
DeviceMetric.calibration.time
Element Id | DeviceMetric.calibration.time
Definition | Describes the time last calibration has been performed.
Cardinality | 0..1
Type | instant
Summary | true

8.16.8 Resource DeviceMetric - Mappings

URL: https://hl7.org/fhir/R4/devicemetric-mappings.html

8.16.8 Resource DeviceMetric - Mappings

Health Care Devices Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Mappings for the devicemetric resource (see Mappings to Other Standards for further information & status).

8.16.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DeviceMetric
identifier | FiveWs.identifier
type | FiveWs.class
category | FiveWs.class

8.16.9 Resource DeviceMetric - Extensions & Profiles

URL: https://hl7.org/fhir/R4/devicemetric-profiles.html

8.16.9 Resource DeviceMetric - Extensions & Profiles

Health Care Devices Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the DeviceMetric resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.16.9.1 Profiles

No Profiles defined for this resource

8.16.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.16.9.3 Search Extensions

No Search Extensions defined for this resource

8.16.10 Resource DeviceMetric - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/devicemetric-version-maps.html
No validation errors - all conversions are clean

8.16.10 Resource DeviceMetric - R3/R4 Conversions

Health Care Devices Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for DeviceMetric.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.16.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DeviceMetric3to4" = "R3 to R4 Conversions for DeviceMetric"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DeviceMetric" alias DeviceMetricR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceMetric" alias DeviceMetric as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group DeviceMetric(source src : DeviceMetricR3, target tgt : DeviceMetric) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.type -> tgt.type;
  src.unit -> tgt.unit;
  src.source -> tgt.source;
  src.parent -> tgt.parent;
  src.operationalStatus -> tgt.operationalStatus;
  src.color -> tgt.color;
  src.category -> tgt.category;
  src.measurementPeriod -> tgt.measurementPeriod;
  src.calibration as vs0 -> tgt.calibration as vt0 then {
    vs0.type -> vt0.type;
    vs0.state -> vt0.state;
    vs0.time -> vt0.time;
  };
}
```

8.16.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceMetric4to3" = "R4 to R3 Conversion for DeviceMetric"

uses "http://hl7.org/fhir/StructureDefinition/DeviceMetric" alias DeviceMetric as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DeviceMetric" alias DeviceMetricR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DeviceMetric(source src : DeviceMetric, target tgt : DeviceMetricR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.type -> tgt.type;
  src.unit -> tgt.unit;
  src.source -> tgt.source;
  src.parent -> tgt.parent;
  src.operationalStatus -> tgt.operationalStatus;
  src.color -> tgt.color;
  src.category -> tgt.category;
  src.measurementPeriod -> tgt.measurementPeriod;
  src.calibration as vs0 -> tgt.calibration as vt0 then {
    vs0.type -> vt0.type;
    vs0.state -> vt0.state;
    vs0.time -> vt0.time;
  };
}
```

8.16.10.3 R4 Validation Errors

8.16.11 JSON Schema for DeviceMetric

URL: https://hl7.org/fhir/R4/devicemetric.schema.json.html

8.16.11 JSON Schema for DeviceMetric

Health Care Devices Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for DeviceMetric, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DeviceMetric",
  "$ref": "#/definitions/DeviceMetric",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DeviceMetric": {
      "description": "Describes a measurement, calculation or setting capability of a medical device.",
      "properties": {
        "resourceType": {
          "description": "This is a DeviceMetric resource",
          "const": "DeviceMetric"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
```

```
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique instance identifiers assigned to a device by the device or gateway software,
manufacturers, other organizations or owners. For example: handle ID.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "Describes the type of the metric. For example: Heart Rate, PEEP Setting, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "unit": {
      "description": "Describes the unit that an observed value determined for this metric will have. For example:
Percent, Seconds, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "source": {
      "description": "Describes the link to the  Device that this DeviceMetric belongs to and that contains
administrative device information such as manufacturer, serial number, etc.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "parent": {
      "description": "Describes the link to the  Device that this DeviceMetric belongs to and that provide
information about the location of this DeviceMetric in the containment structure of the parent Device. An example
would be a Device that represents a Channel. This reference can be used by a client application to distinguish
DeviceMetrics that have the same type, but should be interpreted based on their containment location.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "operationalStatus": {
      "description": "Indicates current operational state of the device. For example: On, Off, Standby, etc.",
      "enum": [
        "on",
        "off",
        "standby",
        "entered-in-error"
      ]
    },
    "_operationalStatus": {
      "description": "Extensions for operationalStatus",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "color": {
      "description": "Describes the color representation for the metric. This is often used to aid clinicians to
track and identify parameter types by color. In practice, consider a Patient Monitor that has ECG/HR and Pleth for
example; the parameters are displayed in different characteristic colors, such as HR-blue, BP-green, and PR and
SpO2- magenta.",
      "enum": [
        "black",
        "red",
        "green",
        "yellow",
        "blue",
        "magenta",
        "cyan",
        "white"
      ]
    },
    "_color": {
      "description": "Extensions for color",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Indicates the category of the observation generation process. A DeviceMetric can be for
example a setting, measurement, or calculation.",
      "enum": [
        "measurement",
        "setting",
        "calculation",
        "unspecified"
      ]
    },
    "_category": {
      "description": "Extensions for category",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "measurementPeriod": {
      "description": "Describes the measurement repetition time. This is not necessarily the same as the update
period. The measurement repetition time can range from milliseconds up to hours. An example for a measurement
repetition time in the range of milliseconds is the sampling rate of an ECG. An example for a measurement repetition
 time in the range of hours is a NIBP that is triggered automatically every hour. The update period may be different
 than the measurement repetition time, if the device does not update the published observed value with the same
frequency as it was measured.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "calibration": {
      "description": "Describes the calibrations that have been performed or that are required to be performed.",
      "items": {
        "$ref": "#/definitions/DeviceMetric_Calibration"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type",
    "resourceType"
  ]
},
"DeviceMetric_Calibration": {
  "description": "Describes a measurement, calculation or setting capability of a medical device.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
        "type": {
          "description": "Describes the type of the calibration method.",
          "enum": [
            "unspecified",
            "offset",
            "gain",
            "two-point"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "state": {
          "description": "Describes the state of the calibration.",
          "enum": [
            "not-calibrated",
            "calibration-required",
            "calibrated",
            "unspecified"
          ]
        },
        "_state": {
          "description": "Extensions for state",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "time": {
          "description": "Describes the time last calibration has been performed.",
          "$ref": "#/definitions/instant"
        },
        "_time": {
          "description": "Extensions for time",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

8.16.12 Resource DeviceMetric - Maturity Dependency View

URL: https://hl7.org/fhir/R4/devicemetric-dependencies.html

8.16.12 Resource DeviceMetric - Maturity Dependency View

Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for DeviceMetric FMM level 1

Name | Card. | Type | Dependency Analysis
DeviceMetric | | DomainResource
identifier | 0..* | Identifier
type | 1..1 | CodeableConcept
unit | 0..1 | CodeableConcept
source | 0..1 | Reference(Device)
parent | 0..1 | Reference(Device)
operationalStatus | 0..1 | code
color | 0..1 | code
category | 1..1 | code
measurementPeriod | 0..1 | Timing
calibration | 0..* | BackboneElement
type | 0..1 | code
state | 0..1 | code
time | 0..1 | instant

8.16.13 Value Set Sc-valueset-metric-operational-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-metric-operational-status.html

8.16.13 Value Set Sc-valueset-metric-operational-status - Status Code Mapping

Health Care Devices Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.16.14 DeviceMetricOperationalStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-metric-operational-status)

Mapping from http://hl7.org/fhir/ValueSet/metric-operational-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Describes the operational status of the DeviceMetric."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
on | equivalent | active
standby | equivalent | suspended
off | equivalent | inactive

8.17 Resource Substance - Content

URL: https://hl7.org/fhir/R4/substance.html
Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

A homogeneous material with a definite composition.

8.17.1 Scope and Usage

This resource allows for a material to be represented. The resource can be used to represent either a kind of a substance - e.g. a formulation commonly used for treating patients, or it can be used to describe a particular package of a known substance (e.g. bottle, jar, packet).

The composition of the material can be specified in terms of a mix of other materials, including with precise amounts if required.

8.17.2 Boundaries and Relationships

A medication is a substance that is packaged and used as an administered medication. The medication resource uses the substance resource to represent the actual ingredients of a medication.

This resource is referenced by ActivityDefinition, AdverseEvent, BiologicallyDerivedProduct, CarePlan, ChargeItem, ChargeItemDefinition, Contract, Group, Medication, MedicationKnowledge, MedicinalProductContraindication, MedicinalProductIndication, MedicinalProductInteraction, Procedure, Specimen, SpecimenDefinition, itself, SubstanceSpecification, SupplyDelivery and SupplyRequest

8.17.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Substance | TU | | DomainResource | A homogeneous material with a definite composition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error FHIRSubstanceStatus (Required)
category | Σ | 0..* | CodeableConcept | What class/type of substance this is Substance Category Codes (Extensible)
code | Σ | 1..1 | CodeableConcept | What substance this is Substance Code (Example)
description | Σ | 0..1 | string | Textual description of the substance, comments
instance | Σ | 0..* | BackboneElement | If this describes a specific package/container of the substance
identifier | Σ | 0..1 | Identifier | Identifier of the package/container
expiry | Σ | 0..1 | dateTime | When no longer valid to use
quantity | Σ | 0..1 | SimpleQuantity | Amount of substance in the package
ingredient | Σ | 0..* | BackboneElement | Composition information about the substance
quantity | Σ | 0..1 | Ratio | Optional amount (concentration)
substance[x] | Σ | 1..1 | | A component of the substance Substance Code (Example)
substanceCodeableConcept | | | CodeableConcept
substanceReference | | | Reference(Substance)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Substance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept What class/type of substance this is --></category>
 <code><!-- 1..1 CodeableConcept What substance this is --></code>
 <description value="[string]"/><!-- 0..1 Textual description of the substance, comments -->
 <instance>  <!-- 0..* If this describes a specific package/container of the substance -->
  <identifier><!-- 0..1 Identifier Identifier of the package/container --></identifier>
  <expiry value="[dateTime]"/><!-- 0..1 When no longer valid to use -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of substance in the package --></quantity>
 </instance>
 <ingredient>  <!-- 0..* Composition information about the substance -->
  <quantity><!-- 0..1 Ratio Optional amount (concentration) --></quantity>
  <substance[x]><!-- 1..1 CodeableConcept|Reference(Substance) A component of the substance --></substance[x]>
 </ingredient>
</Substance>
```

JSON Template

```
{
  "resourceType" : "Substance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "status" : "<code>", // active | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // What class/type of substance this is
  "code" : { CodeableConcept }, // R!  What substance this is
  "description" : "<string>", // Textual description of the substance, comments
  "instance" : [{ // If this describes a specific package/container of the substance
    "identifier" : { Identifier }, // Identifier of the package/container
    "expiry" : "<dateTime>", // When no longer valid to use
    "quantity" : { Quantity(SimpleQuantity) } // Amount of substance in the package
  }],
  "ingredient" : [{ // Composition information about the substance
    "quantity" : { Ratio }, // Optional amount (concentration)
    // substance[x]: A component of the substance. One of these 2:
    "substanceCodeableConcept" : { CodeableConcept }
    "substanceReference" : { Reference(Substance) }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Substance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Substance.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Substance.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Substance.category [ CodeableConcept ], ... ; # 0..* What class/type of substance this is
  fhir:Substance.code [ CodeableConcept ]; # 1..1 What substance this is
  fhir:Substance.description [ string ]; # 0..1 Textual description of the substance, comments
  fhir:Substance.instance [ # 0..* If this describes a specific package/container of the substance
    fhir:Substance.instance.identifier [ Identifier ]; # 0..1 Identifier of the package/container
    fhir:Substance.instance.expiry [ dateTime ]; # 0..1 When no longer valid to use
    fhir:Substance.instance.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of substance in the package
  ], ...;
  fhir:Substance.ingredient [ # 0..* Composition information about the substance
    fhir:Substance.ingredient.quantity [ Ratio ]; # 0..1 Optional amount (concentration)
    # Substance.ingredient.substance[x] : 1..1 A component of the substance. One of these 2
      fhir:Substance.ingredient.substanceCodeableConcept [ CodeableConcept ]
      fhir:Substance.ingredient.substanceReference [ Reference(Substance) ]
  ], ...;
]
```

Changes since R3

Substance

Substance.status | Change value set from http://hl7.org/fhir/ValueSet/substance-status to http://hl7.org/fhir/ValueSet/substance-status|4.0.1 Now marked as Modifier

Substance.category | Change code system for extensibly bound codes from "http://hl7.org/fhir/substance-category" to "http://terminology.hl7.org/CodeSystem/substance-category"

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Substance | TU | | DomainResource | A homogeneous material with a definite composition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error FHIRSubstanceStatus (Required)
category | Σ | 0..* | CodeableConcept | What class/type of substance this is Substance Category Codes (Extensible)
code | Σ | 1..1 | CodeableConcept | What substance this is Substance Code (Example)
description | Σ | 0..1 | string | Textual description of the substance, comments
instance | Σ | 0..* | BackboneElement | If this describes a specific package/container of the substance
identifier | Σ | 0..1 | Identifier | Identifier of the package/container
expiry | Σ | 0..1 | dateTime | When no longer valid to use
quantity | Σ | 0..1 | SimpleQuantity | Amount of substance in the package
ingredient | Σ | 0..* | BackboneElement | Composition information about the substance
quantity | Σ | 0..1 | Ratio | Optional amount (concentration)
substance[x] | Σ | 1..1 | | A component of the substance Substance Code (Example)
substanceCodeableConcept | | | CodeableConcept
substanceReference | | | Reference(Substance)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Substance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept What class/type of substance this is --></category>
 <code><!-- 1..1 CodeableConcept What substance this is --></code>
 <description value="[string]"/><!-- 0..1 Textual description of the substance, comments -->
 <instance>  <!-- 0..* If this describes a specific package/container of the substance -->
  <identifier><!-- 0..1 Identifier Identifier of the package/container --></identifier>
  <expiry value="[dateTime]"/><!-- 0..1 When no longer valid to use -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of substance in the package --></quantity>
 </instance>
 <ingredient>  <!-- 0..* Composition information about the substance -->
  <quantity><!-- 0..1 Ratio Optional amount (concentration) --></quantity>
  <substance[x]><!-- 1..1 CodeableConcept|Reference(Substance) A component of the substance --></substance[x]>
 </ingredient>
</Substance>
```

JSON Template

```
{
  "resourceType" : "Substance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "status" : "<code>", // active | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // What class/type of substance this is
  "code" : { CodeableConcept }, // R!  What substance this is
  "description" : "<string>", // Textual description of the substance, comments
  "instance" : [{ // If this describes a specific package/container of the substance
    "identifier" : { Identifier }, // Identifier of the package/container
    "expiry" : "<dateTime>", // When no longer valid to use
    "quantity" : { Quantity(SimpleQuantity) } // Amount of substance in the package
  }],
  "ingredient" : [{ // Composition information about the substance
    "quantity" : { Ratio }, // Optional amount (concentration)
    // substance[x]: A component of the substance. One of these 2:
    "substanceCodeableConcept" : { CodeableConcept }
    "substanceReference" : { Reference(Substance) }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Substance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Substance.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Substance.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Substance.category [ CodeableConcept ], ... ; # 0..* What class/type of substance this is
  fhir:Substance.code [ CodeableConcept ]; # 1..1 What substance this is
  fhir:Substance.description [ string ]; # 0..1 Textual description of the substance, comments
  fhir:Substance.instance [ # 0..* If this describes a specific package/container of the substance
    fhir:Substance.instance.identifier [ Identifier ]; # 0..1 Identifier of the package/container
    fhir:Substance.instance.expiry [ dateTime ]; # 0..1 When no longer valid to use
    fhir:Substance.instance.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of substance in the package
  ], ...;
  fhir:Substance.ingredient [ # 0..* Composition information about the substance
    fhir:Substance.ingredient.quantity [ Ratio ]; # 0..1 Optional amount (concentration)
    # Substance.ingredient.substance[x] : 1..1 A component of the substance. One of these 2
```

```
      fhir:Substance.ingredient.substanceCodeableConcept [ CodeableConcept ]
      fhir:Substance.ingredient.substanceReference [ Reference(Substance) ]
  ], ...;
]
```

Changes since Release 3

Substance
Substance.status | Change value set from http://hl7.org/fhir/ValueSet/substance-status to http://hl7.org/fhir/ValueSet/substance-status|4.0.1 Now marked as Modifier
Substance.category | Change code system for extensibly bound codes from "http://hl7.org/fhir/substance-category" to "http://terminology.hl7.org/CodeSystem/substance-category"

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.17.3.1 Terminology Bindings

Path | Definition | Type | Reference
Substance.status | A code to indicate if the substance is actively used. | Required | FHIRSubstanceStatus
Substance.category | Category or classification of substance. | Extensible | SubstanceCategoryCodes
Substance.code | Substance codes. | Example | SubstanceCode
Substance.ingredient.substance[x] | Substance Ingredient codes. | Example | SubstanceCode

8.17.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The category of the substance | Substance.category
code | token | The code of the substance or ingredient | Substance.code | (Substance.ingredient.substance as CodeableConcept)
container-identifier | token | Identifier of the package/container | Substance.instance.identifier
expiry | date | Expiry date of package or container of substance | Substance.instance.expiry
identifier | token | Unique identifier for the substance | Substance.identifier
quantity | quantity | Amount of substance in the package | Substance.instance.quantity
status | token | active | inactive | entered-in-error | Substance.status
substance-reference | reference | A component of the substance | (Substance.ingredient.substance as Reference) (Substance)

8.17.5 Resource Substance - Examples

URL: https://hl7.org/fhir/R4/substance-examples.html

8.17.5 Resource Substance - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
General Example | example | XML | JSON | Turtle
Dust Mite Example | f201 | XML | JSON | Turtle
Staphylococcus Example | f202 | XML | JSON | Turtle
Potassium Example | f203 | XML | JSON | Turtle
Silver Nitrate Product Example | f204 | XML | JSON | Turtle
Amoxicillin-Clavulanate Example | f205 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.17.6 Resource Substance - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substance-definitions.html

8.17.6 Resource Substance - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Substance resource.

Substance
Element Id | Substance
Definition | A homogeneous material with a definite composition.
Cardinality | 0..*
Type | DomainResource

Substance.identifier
Element Id | Substance.identifier
Definition | Unique identifier for the substance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Comments | This identifier is associated with the kind of substance in contrast to the Substance.instance.identifier which is associated with the package/container.
Substance.status
Element Id | Substance.status
Definition | A code to indicate if the substance is actively used.
Cardinality | 0..1
Terminology Binding | FHIRSubstanceStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Substance.category
Element Id | Substance.category
Definition | A code that classifies the general type of substance. This is used for searching, sorting and display purposes.
Cardinality | 0..*
Terminology Binding | Substance Category Codes (Extensible)
Type | CodeableConcept
Summary | true
Comments | The level of granularity is defined by the category concepts in the value set. More fine-grained filtering can be performed using the metadata and/or terminology hierarchy in Substance.code.
Substance.code
Element Id | Substance.code
Definition | A code (or set of codes) that identify this substance.
Cardinality | 1..1
Terminology Binding | Substance Code (Example)
Type | CodeableConcept
Summary | true
Comments | This could be a reference to an externally defined code. It could also be a locally assigned code (e.g. a formulary), optionally with translations to the standard drug codes.
Substance.description
Element Id | Substance.description
Definition | A description of the substance - its appearance, handling requirements, and other usage notes.
Cardinality | 0..1
Type | string
Summary | true
Substance.instance
Element Id | Substance.instance
Definition | Substance may be used to describe a kind of substance, or a specific package/container of the substance: an instance.
Cardinality | 0..*
Meaning if Missing | If this element is not present, then the substance resource describes a kind of substance
Summary | true
Substance.instance.identifier
Element Id | Substance.instance.identifier
Definition | Identifier associated with the package/container (usually a label affixed directly).
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
Substance.instance.expiry
Element Id | Substance.instance.expiry
Definition | When the substance is no longer valid to use. For some substances, a single arbitrary date is used for expiry.
Cardinality | 0..1
Type | dateTime
Summary | true
Substance.instance.quantity
Element Id | Substance.instance.quantity
Definition | The amount of the substance.
Cardinality | 0..1
Type | SimpleQuantity
Summary | true
Substance.ingredient
Element Id | Substance.ingredient
Definition | A substance can be composed of other substances.
Cardinality | 0..*
Summary | true
Substance.ingredient.quantity

Element Id | Substance.ingredient.quantity
Definition | The amount of the ingredient in the substance - a concentration ratio.
Cardinality | 0..1
Type | Ratio
Summary | true
Substance.ingredient.substance[x]
Element Id | Substance.ingredient.substance[x]
Definition | Another substance that is a component of this substance.
Cardinality | 1..1
Terminology Binding | Substance Code (Example)
Type | CodeableConcept|Reference(Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true

8.17.7 Resource Substance - Mappings

URL: https://hl7.org/fhir/R4/substance-mappings.html

8.17.7 Resource Substance - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Ontological RIM Mapping

Mappings for the substance resource (see Mappings to Other Standards for further information & status).

8.17.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Substance
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
code | FiveWs.what[x]

8.17.7.2 RIM Mapping (http://hl7.org/v3)

Substance | Material
identifier | .id
status | .statusCode
category | .code
code | .code
description | .player.desc
instance | .player.determinerCode
identifier | .id
expiry | .expiryTime
quantity | .player.quantity
ingredient | .player.scopesRole[classCode=INGR]
quantity | .quantity
substance[x] | .player

8.17.7.3 Ontological RIM Mapping (http://hl7.org/orim)

Substance | fhir:Substance rdfs:subClassOf rim:Entity, [a owl:Restriction; owl:onProperty rim:Act.classCode; owl:allValuesFrom cs:EntityClass\#MAT].
identifier
status
category
code | fhir:Substance.code rdfs:subPropertyOf rim:Entity.code
description
instance
identifier
expiry
quantity
ingredient
quantity
substance[x]

8.17.8 Resource Substance - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substance-profiles.html

8.17.8 Resource Substance - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Substance resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.17.8.1 Profiles

No Profiles defined for this resource

8.17.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

observation-reagent | reagent | for Observation HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.17.8.3 Search Extensions

No Search Extensions defined for this resource

8.17.9 Resource Substance - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/substance-version-maps.html

No validation errors - all conversions are clean

8.17.9 Resource Substance - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Substance.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

8.17.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Substance3to4" = "R3 to R4 Conversions for Substance"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Substance" alias SubstanceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Substance" alias Substance as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Substance(source src : SubstanceR3, target tgt : Substance) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.instance as vs0 -> tgt.instance as vt0 then instance(vs0, vt0);
  src.ingredient as vs0 -> tgt.ingredient as vt0 then ingredient(vs0, vt0);
}

group instance(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.expiry -> tgt.expiry;
  src.quantity -> tgt.quantity;
}

group ingredient(source src, target tgt) extends BackboneElement {
  src.quantity -> tgt.quantity;
  src.substance : CodeableConcept as vs -> tgt.substance = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.substance : Reference as vs -> tgt.substance = create('Reference') as vt then Reference(vs, vt);
}
```

8.17.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Substance4to3" = "R4 to R3 Conversion for Substance"
```

```
uses "http://hl7.org/fhir/StructureDefinition/Substance" alias Substance as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Substance" alias SubstanceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Substance(source src : Substance, target tgt : SubstanceR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.instance as vs0 -> tgt.instance as vt0 then instance(vs0, vt0);
  src.ingredient as vs0 -> tgt.ingredient as vt0 then ingredient(vs0, vt0);
}

group instance(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.expiry -> tgt.expiry;
  src.quantity -> tgt.quantity;
}

group ingredient(source src, target tgt) extends BackboneElement {
  src.quantity -> tgt.quantity;
  src.substance : CodeableConcept as vs -> tgt.substance = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.substance : Reference as vs -> tgt.substance = create('Reference') as vt then Reference(vs, vt);
}
```

8.17.9.3 R4 Validation Errors

8.17.10 JSON Schema for Substance

URL: https://hl7.org/fhir/R4/substance.schema.json.html

8.17.10 JSON Schema for Substance

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Substance, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Substance",
  "$ref": "#/definitions/Substance",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Substance": {
      "description": "A homogeneous material with a definite composition.",
      "properties": {
        "resourceType": {
          "description": "This is a Substance resource",
          "const": "Substance"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
```

```
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique identifier for the substance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "A code to indicate if the substance is actively used.",
      "enum": [
        "active",
        "inactive",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "A code that classifies the general type of substance.  This is used  for searching, sorting
and display purposes.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "code": {
      "description": "A code (or set of codes) that identify this substance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "A description of the substance - its appearance, handling requirements, and other usage
notes.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "instance": {
      "description": "Substance may be used to describe a kind of substance, or a specific package/container of the
substance: an instance.",
      "items": {
        "$ref": "#/definitions/Substance_Instance"
      },
      "type": "array"
    },
    "ingredient": {
      "description": "A substance can be composed of other substances.",
      "items": {
        "$ref": "#/definitions/Substance_Ingredient"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
```

```
  "required": [
    "code",
    "resourceType"
  ]
},
"Substance_Instance": {
  "description": "A homogeneous material with a definite composition.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier associated with the package/container (usually a label affixed directly).",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "expiry": {
      "description": "When the substance is no longer valid to use. For some substances, a single arbitrary date is
used for expiry.",
      "$ref": "#/definitions/dateTime"
    },
    "_expiry": {
      "description": "Extensions for expiry",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "quantity": {
      "description": "The amount of the substance.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    }
  },
  "additionalProperties": false
},
"Substance_Ingredient": {
  "description": "A homogeneous material with a definite composition.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
        "quantity": {
          "description": "The amount of the ingredient in the substance - a concentration ratio.",
          "$ref": "Ratio.schema.json#/definitions/Ratio"
        },
        "substanceCodeableConcept": {
          "description": "Another substance that is a component of this substance.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "substanceReference": {
          "description": "Another substance that is a component of this substance.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    }
  }
}
```

8.17.11 Resource Substance - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substance-dependencies.html

8.17.11 Resource Substance - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Substance FMM level 2

Name | Card. | Type | Dependency Analysis
Substance | | DomainResource
identifier | 0..* | Identifier
status | 0..1 | code
category | 0..* | CodeableConcept
code | 1..1 | CodeableConcept
description | 0..1 | string
instance | 0..* | BackboneElement
identifier | 0..1 | Identifier
expiry | 0..1 | dateTime
quantity | 0..1 | SimpleQuantity
ingredient | 0..* | BackboneElement
quantity | 0..1 | Ratio
substance[x] | 1..1
substanceCodeableConcept | | CodeableConcept
substanceReference | | Reference(Substance

8.17.12 Value Set Sc-valueset-substance-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-substance-status.html

8.17.12 Value Set Sc-valueset-substance-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.17.13 FHIRSubstanceStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-substance-status)

Mapping from http://hl7.org/fhir/ValueSet/substance-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "A code to indicate if the substance is actively used."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
active | equivalent | active
inactive | equivalent | inactive

8.18 Resource Person - Content

URL: https://hl7.org/fhir/R4/person.html
Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Demographics and administrative information about a person independent of a specific health-related context.

8.18.1 Scope and Usage

The Person resource serves as a linkage resource that may provide a reference set of common demographics for an individual (human or animal) across multiple roles. This linkage can be direct to role-specific FHIR resources (Patient, Practitioner and RelatedPerson) residing on the same or possibly distinct FHIR systems/applications, or indirectly through the use of business identifiers.

The Person resource may be used in many situations/contexts, including:

• A set of demographics that can be used to co-ordinate the maintenance of this de-normalized information across practitioners, patients and/or related persons e.g. link known resources of the different types together within a system

• A state/network based Master Person Index e.g. A National Identifier Index or a Network membership/subscriber list

• A central register that links patient resources from multiple servers, indicating they all correspond to the same individual e.g. Within a large organization with many systems to be able to link the various records, without having to modify the source information with external links

• A support for access monitoring software that is able to assert what practitioner, patient and related person records correspond to the same human being to assist in detecting improper querying. e.g. asserting that a Patient A is the same individual as a RelatedPerson B who is related to Patient C, who is also Practitioner D. As such, monitoring for Practitioner D's accesses to Patient A's records should be carefully scrutinized.

• Local record(s) on a mobile device listing links to remote server patient resources e.g. A mobile phone application storing references to the hospitals they have access to

Note: The Person resource is an advanced feature. Many systems don't have a way to relate information across resource types, or systems, especially from Patient to Practitioner, and therefore might not implement this functionality. Some do have ways of relating the Patient and RelatedPerson resource types. FHIR is also able to do this without the Person resource, utilizing the Patient.link property (as shown in the Mother and newborn relationship example)

8.18.2 Boundaries and Relationships

Person resources may have references to Patient, RelatedPerson and/or Practitioner resources. These linkages assert that all the records relate to the same individual.

Person instances are never directly referenced as actors (authors, subjects, performers, etc.) Individual actors are always identified as either Patient, Practitioner or RelatedPerson, depending on the role of the individual when undertaking the action. Therefore this resource SHALL NOT be referenced by any other clinical or administrative resources. E.g. it can be the focus of an operation or message or included in a Bundle, but cannot be a member of a List or Group, the subject of an Observation, or other similar non-infrastructural usage.

In some use cases the linkages will be indirect, and through business Identifiers, such as in the case of the various types of Master Person Indexes, where the records are not there for the purpose of healthcare, but simply describing a population of people, and has other purposes and not just to represent a population patients.

This resource is referenced by itself

8.18.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name	Flags	Card.	Type	Description & Constraints
Person	TU		DomainResource	A generic person record Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier		0..*	Identifier	A human identifier for this person
name	Σ	0..*	HumanName	A name associated with the person
telecom	Σ	0..*	ContactPoint	A contact detail for the person
gender	Σ	0..1	code	male | female | other | unknown AdministrativeGender (Required)
birthDate	Σ	0..1	date	The date on which the person was born
address		0..*	Address	One or more addresses for the person
photo		0..1	Attachment	Image of the person
managingOrganization	Σ	0..1	Reference(Organization)	The organization that is the custodian of the person record

active | ?!Σ | 0..1 | boolean | This person's record is in active use
link | | 0..* | BackboneElement | Link to a resource that concerns the same actual person
target | | 1..1 | Reference(Patient | Practitioner | RelatedPerson | Person) | The resource to which this actual person is associated
assurance | | 0..1 | code | level1 | level2 | level3 | level4 IdentityAssuranceLevel (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Person xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the person was born -->
 <address><!-- 0..* Address One or more addresses for the person --></address>
 <photo><!-- 0..1 Attachment Image of the person --></photo>
 <managingOrganization><!-- 0..1 Reference(Organization) The organization that is the custodian of the person record
    --></managingOrganization>
 <active value="[boolean]"/><!-- 0..1 This person's record is in active use -->
 <link>  <!-- 0..* Link to a resource that concerns the same actual person -->
  <target><!-- 1..1 Reference(Patient|Practitioner|RelatedPerson|Person) The resource to which this actual person is
    associated --></target>
  <assurance value="[code]"/><!-- 0..1 level1 | level2 | level3 | level4 -->
 </link>
</Person>
```

JSON Template

```
{
  "resourceType" : "Person",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the person was born
  "address" : [{ Address }], // One or more addresses for the person
  "photo" : { Attachment }, // Image of the person
  "managingOrganization" : { Reference(Organization) }, // The organization that is the custodian of the person record
  "active" : <boolean>, // This person's record is in active use
  "link" : [{ // Link to a resource that concerns the same actual person
    "target" : { Reference(Patient|Practitioner|RelatedPerson|Person) }, // R!  The resource to which this actual person
     is associated
    "assurance" : "<code>" // level1 | level2 | level3 | level4
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Person;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Person.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:Person.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:Person.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:Person.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Person.birthDate [ date ]; # 0..1 The date on which the person was born
  fhir:Person.address [ Address ], ... ; # 0..* One or more addresses for the person
  fhir:Person.photo [ Attachment ]; # 0..1 Image of the person
  fhir:Person.managingOrganization [ Reference(Organization) ]; # 0..1 The organization that is the custodian of the
    person record
  fhir:Person.active [ boolean ]; # 0..1 This person's record is in active use
  fhir:Person.link [ # 0..* Link to a resource that concerns the same actual person
    fhir:Person.link.target [ Reference(Patient|Practitioner|RelatedPerson|Person) ]; # 1..1 The resource to which this
    actual person is associated
    fhir:Person.link.assurance [ code ]; # 0..1 level1 | level2 | level3 | level4
  ], ...;
]
```

Changes since R3

Person
Person.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
Person.link.assurance | Change value set from http://hl7.org/fhir/ValueSet/identity-assuranceLevel to http://hl7.org/fhir/ValueSet/identity-assuranceLevel|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 5 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Person | TU | | DomainResource | A generic person record Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | A human identifier for this person
name | Σ | 0..* | HumanName | A name associated with the person
telecom | Σ | 0..* | ContactPoint | A contact detail for the person
gender | Σ | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | Σ | 0..1 | date | The date on which the person was born
address | | 0..* | Address | One or more addresses for the person
photo | | 0..1 | Attachment | Image of the person
managingOrganization | Σ | 0..1 | Reference(Organization) | The organization that is the custodian of the person record
active | ?!Σ | 0..1 | boolean | This person's record is in active use
link | | 0..* | BackboneElement | Link to a resource that concerns the same actual person
target | | 1..1 | Reference(Patient | Practitioner | RelatedPerson | Person) | The resource to which this actual person is associated
assurance | | 0..1 | code | level1 | level2 | level3 | level4 IdentityAssuranceLevel (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Person xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the person was born -->
 <address><!-- 0..* Address One or more addresses for the person --></address>
 <photo><!-- 0..1 Attachment Image of the person --></photo>
 <managingOrganization><!-- 0..1 Reference(Organization) The organization that is the custodian of the person record
    --></managingOrganization>
 <active value="[boolean]"/><!-- 0..1 This person's record is in active use -->
 <link>  <!-- 0..* Link to a resource that concerns the same actual person -->
  <target><!-- 1..1 Reference(Patient|Practitioner|RelatedPerson|Person) The resource to which this actual person is
    associated --></target>
  <assurance value="[code]"/><!-- 0..1 level1 | level2 | level3 | level4 -->
 </link>
</Person>
```

JSON Template

```
{
  "resourceType" : "Person",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the person was born
  "address" : [{ Address }], // One or more addresses for the person
  "photo" : { Attachment }, // Image of the person
  "managingOrganization" : { Reference(Organization) }, // The organization that is the custodian of the person record
  "active" : <boolean>, // This person's record is in active use
  "link" : [{ // Link to a resource that concerns the same actual person
    "target" : { Reference(Patient|Practitioner|RelatedPerson|Person) }, // R!  The resource to which this actual person
     is associated
    "assurance" : "<code>" // level1 | level2 | level3 | level4
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Person;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Person.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:Person.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:Person.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:Person.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Person.birthDate [ date ]; # 0..1 The date on which the person was born
  fhir:Person.address [ Address ], ... ; # 0..* One or more addresses for the person
  fhir:Person.photo [ Attachment ]; # 0..1 Image of the person
  fhir:Person.managingOrganization [ Reference(Organization) ]; # 0..1 The organization that is the custodian of the
    person record
  fhir:Person.active [ boolean ]; # 0..1 This person's record is in active use
  fhir:Person.link [ # 0..* Link to a resource that concerns the same actual person
```

```
    fhir:Person.link.target [ Reference(Patient|Practitioner|RelatedPerson|Person) ]; # 1..1 The resource to which this
    actual person is associated
    fhir:Person.link.assurance [ code ]; # 0..1 level1 | level2 | level3 | level4
  ], ...;
]
```

Changes since Release 3

Person

Person.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Person.link.assurance | Change value set from http://hl7.org/fhir/ValueSet/identity-assuranceLevel to http://hl7.org/fhir/ValueSet/identity-assuranceLevel|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 5 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.18.3.1 Terminology Bindings

Path | Definition | Type | Reference

Person.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender

Person.link.assurance | The level of confidence that this link represents the same actual person, based on NIST Authentication Levels. | Required | IdentityAssuranceLevel

8.18.4 De-normalized Data

Most of the properties of the Person resource are replicated within the other resources that they are shared with. This is intentional and highlights the loose coupling of the resources.

Not many systems implement a shared Person record, and as such the values DO become out of sync with each other. The inclusion of this resource does permit a capability for systems to identify other instances of this actual person's data via a centralized registry that can assist in keeping things up to date.

8.18.5 Person and Linking

The link element is used to relate resources under a common person record. This element supports two primary scenarios where other resources refer to the same person resource.

The link element cannot be used to link to RelatedPerson entries, so we can use a Person resource to relate these elements together identifying them as the same individual.

8.18.6 Cross-Domain Patient Directory

In a data sharing network, finding the location of patient records across different systems is a necessary pre-requisite for accessing external patient data. Using the link element, systems associate patient resources from different organizations. The assuranceLevel associated with the link provides a way for a system to qualify its confidence in the asserted link. For example, a relationship from the person to a patient using a probabilistic matching algorithm may be represented using a link with an assurance level of level1, while a relationship established using a government-issued photo ID may be created with an assurance level of level3.

8.18.7 Cross-Domain Provider Directory

Similarly, providers working in multiple healthcare service settings may be linked across different organizations using the link element. The various practitioner resources can be related using a common person resource with a link for each of the practitioner resources located in other organizations.

8.18.8 Person Master Index

Many national and state programs maintain a central register of people, and their key Identifiers. These systems do not store any health related information, they are purely a validated set data, often provided by government departments. These may contain identifiers for patients, practitioners from various departments for various purposes. Access to these systems typically requires some form of professional role or permission, such as those in healthcare, law enforcement, or other internal government uses.

These may also operate outside of government in the private sector, such as associations or networks that a person may be involved with and may work in the same was as they do inside government. The usage of the data into the healthcare environment is not the only use of the register.

Note: This style of system may use the Person resource without any FHIR references to Patient or Practitioner resources. In these environments the Master Index is likely to have a master identifier that performs this logical linking. This style of usage has fewer security implications, as it doesn’t disclose where the information is being used.

8.18.9 Client Portal

Client Portals provide consumer access to a window of their data locked up in healthcare systems. In many cases these systems are externally integrated and do not have access to the legacy CIS/PAS systems.

In contrast to most systems where a user has access to multiple patients depending on their roles and permissions, a client portal provides a consumer with direct access to their data (with permissions applied). This can also include not just their data (via the Person.link Patient entries), but also to information that they have entered, or were involved with (via the Person.link RelatedPerson entries), such as observations that they entered. With appropriate permissions/consent applied, the user could have access to other Patient records linked via a RelatedPerson.

8.18.10 Security and Privacy Considerations

The Person resource may contain identifying and demographic information about an individual, therefore it must be protected as Identifying Information.

The Person resource may contain references to the individual's other various Patient, Practitioner, and RelatedPerson resources, where the referenced record might be to very sensitive information. The access to the linked sensitive Resource may be protected, but the link itself may expose information. For example where a Practitioner and Patient resource are linked through a Person resource, inspecting the Patient URL could reveal that the Practitioner has a record at a sensitive medical health clinic (e.g. Drug Rehabilitation Clinic).

When implementing the Person resource consideration should be given to the security and privacy of this information. Some mitigations to these risks include higher access control protections, filling out the Person resource as sparsely as practical for the given use-cases, and masking some data elements for certain users of the information. Additional security guidance can be found on the security page.

8.18.11 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
address | string | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Person.address | 3 Resources
address-city | string | A city specified in an address | Person.address.city | 3 Resources
address-country | string | A country specified in an address | Person.address.country | 3 Resources
address-postalcode | string | A postal code specified in an address | Person.address.postalCode | 3 Resources
address-state | string | A state specified in an address | Person.address.state | 3 Resources
address-use | token | A use code specified in an address | Person.address.use | 3 Resources
birthdate | date | The person's date of birth | Person.birthDate | 2 Resources
email | token | A value in an email contact | Person.telecom.where(system='email') | 4 Resources
gender | token | The gender of the person | Person.gender | 3 Resources
identifier | token | A person Identifier | Person.identifier
link | reference | Any link has this Patient, Person, RelatedPerson or Practitioner reference | Person.link.target (Practitioner, Patient, Person, RelatedPerson)
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Person.name
organization | reference | The organization at which this person record is being managed | Person.managingOrganization (Organization)
patient | reference | The Person links to this Patient | Person.link.target.where(resolve() is Patient) (Patient)
phone | token | A value in a phone contact | Person.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of name using some kind of phonetic matching algorithm | Person.name | 3 Resources
practitioner | reference | The Person links to this Practitioner | Person.link.target.where(resolve() is Practitioner) (Practitioner)
relatedperson | reference | The Person links to this RelatedPerson | Person.link.target.where(resolve() is RelatedPerson) (RelatedPerson)
telecom | token | The value in any kind of contact | Person.telecom | 4 Resources

8.18.12 Resource Person - Examples

URL: https://hl7.org/fhir/R4/person-examples.html

8.18.12 Resource Person - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Person Example | example | XML | JSON | Turtle
Grahame's person resource | grahame | XML | JSON | Turtle
Patient Portal Example | pp | XML | JSON | Turtle
Provider Directory Example | pd | XML | JSON | Turtle
Same person from RelatedPerson | f002 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.18.13 Resource Person - Detailed Descriptions

URL: https://hl7.org/fhir/R4/person-definitions.html

8.18.13 Resource Person - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Person resource.

Person
Element Id | Person
Definition | Demographics and administrative information about a person independent of a specific health-related context.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to track persons potentially across multiple roles.
Comments | The Person resource does justice to person registries that keep track of persons regardless of their role. The Person resource is also a primary resource to point to for people acting in a particular role such as SubjectofCare, Practitioner, and Agent. Very few attributes are specific to any role and so Person is kept lean. Most attributes are expected to be tied to the role the Person plays rather than the Person himself. Examples of that are Guardian (SubjectofCare), ContactParty (SubjectOfCare, Practitioner), and multipleBirthInd (SubjectofCare).
Person.identifier
Element Id | Person.identifier
Definition | Identifier for a person within a particular scope.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | People are known by a variety of ids. Some institutions maintain several, and most collect identifiers for exchange with other organizations concerning the person. Examples are national person identifier and local identifier.
Person.name
Element Id | Person.name
Definition | A name associated with the person.
Cardinality | 0..*
Type | HumanName
Requirements | Need to be able to track the person by multiple names. Examples are your official name and a partner name.
Summary | true
Comments | Person may have multiple names with different uses or applicable periods.
Person.telecom
Element Id | Person.telecom
Definition | A contact detail for the person, e.g. a telephone number or an email address.
Cardinality | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Summary | true
Comments | Person may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently and also to help with identification.
Person.gender
Element Id | Person.gender
Definition | Administrative Gender.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed for identification of the person, in combination with (at least) name and birth date. Gender of person drives many clinical processes.
Summary | true
Comments | The gender might not match the biological sex as determined by genetics, or the individual's preferred identification. Note that for both humans and particularly animals, there are other legitimate possibilities than M and F, though a clear majority of systems and contexts only support M and F.
Person.birthDate
Element Id | Person.birthDate
Definition | The birth date for the person.
Cardinality | 0..1
Type | date
Requirements | Age of person drives many clinical processes, and is often used in performing identification of the person. Times are not included so as to not confuse things with potential timezone issues.
Summary | true
Comments | At least an estimated year should be provided as a guess if the real DOB is unknown.
Person.address
Element Id | Person.address
Definition | One or more addresses for the person.
Cardinality | 0..*

Type | Address
Requirements | May need to keep track of person's addresses for contacting, billing or reporting requirements and also to help with identification.
Comments | Person may have multiple addresses with different uses or applicable periods.
Person.photo
Element Id | Person.photo
Definition | An image that can be displayed as a thumbnail of the person to enhance the identification of the individual.
Cardinality | 0..1
Type | Attachment
Person.managingOrganization
Element Id | Person.managingOrganization
Definition | The organization that is the custodian of the person record.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | Need to know who recognizes this person record, manages and updates it.
Summary | true
Person.active
Element Id | Person.active
Definition | Whether this person's record is in active use.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)
Requirements | Need to be able to mark a person record as not to be used because it was created in error.
Summary | true
Person.link
Element Id | Person.link
Definition | Link to a resource that concerns the same actual person.
Cardinality | 0..*
Person.link.target
Element Id | Person.link.target
Definition | The resource to which this actual person is associated.
Cardinality | 1..1
Type | Reference(Patient | Practitioner | RelatedPerson | Person)
Person.link.assurance
Element Id | Person.link.assurance
Definition | Level of assurance that this link is associated with the target resource.
Cardinality | 0..1
Terminology Binding | IdentityAssuranceLevel (Required)
Type | code

8.18.14 Resource Person - Mappings

URL: https://hl7.org/fhir/R4/person-mappings.html

8.18.14 Resource Person - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the person resource (see Mappings to Other Standards for further information & status).

8.18.14.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Person
identifier | FiveWs.identifier
active | FiveWs.status

8.18.14.2 HL7 v2 Mapping (http://hl7.org/v2)

Person
identifier | PID-3
name | PID-5, PID-9
telecom | PID-13, PID-14
gender | PID-8
birthDate | PID-7
address | PID-11
photo | OBX-5 - needs a profile
managingOrganization

active
link
target
assurance

8.18.14.3 RIM Mapping (http://hl7.org/v3)

Person | Person(classCode="PSN" and determinerCode="INST" and quantity="1")
identifier | .plays:Role(classCode='IDENT').id
name | ./name
telecom | ./telecom
gender | player[classCode=PSN|ANM and determinerCode=INSTANCE]/administrativeGender
birthDate | ./birthTime
address | ./addr
photo | player[classCode=PSN|ANM and determinerCode=INSTANCE]/desc
managingOrganization | scoper
active | statusCode
link | outboundLink
target | id
assurance | priorityNumber?

8.18.15 Resource Person - Extensions & Profiles

URL: https://hl7.org/fhir/R4/person-profiles.html

8.18.15 Resource Person - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Person resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.18.15.1 Profiles

No Profiles defined for this resource

8.18.15.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

cqf-initiatingPerson | initiatingPerson | for Clinical Reasoning Extensions
cqf-receivingPerson | receivingPerson | for Clinical Reasoning Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.18.15.3 Search Extensions

No Search Extensions defined for this resource

8.18.16 Resource Person - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/person-version-maps.html

No validation errors - all conversions are clean

8.18.16 Resource Person - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Person.

Functional status for this map: 5 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.18.16.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Person3to4" = "R3 to R4 Conversions for Person"
```

```
uses "http://hl7.org/fhir/3.0/StructureDefinition/Person" alias PersonR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Person" alias Person as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Person(source src : PersonR3, target tgt : Person) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.address -> tgt.address;
  src.photo -> tgt.photo;
  src.managingOrganization -> tgt.managingOrganization;
  src.active -> tgt.active;
  src.link as vs0 -> tgt.link as vt0 then {
    vs0.target -> vt0.target;
    vs0.assurance -> vt0.assurance;
  };
}
```

8.18.16.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Person4to3" = "R4 to R3 Conversion for Person"

uses "http://hl7.org/fhir/StructureDefinition/Person" alias Person as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Person" alias PersonR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Person(source src : Person, target tgt : PersonR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.name -> tgt.name;
  src.telecom -> tgt.telecom;
  src.gender -> tgt.gender;
  src.birthDate -> tgt.birthDate;
  src.address -> tgt.address;
  src.photo -> tgt.photo;
  src.managingOrganization -> tgt.managingOrganization;
  src.active -> tgt.active;
  src.link as vs0 -> tgt.link as vt0 then {
    vs0.target -> vt0.target;
    vs0.assurance -> vt0.assurance;
  };
}
```

8.18.16.3 R4 Validation Errors

8.18.17 JSON Schema for Person

URL: https://hl7.org/fhir/R4/person.schema.json.html

8.18.17 JSON Schema for Person

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Person, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Person",
  "$ref": "#/definitions/Person",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Person": {
      "description": "Demographics and administrative information about a person independent of a specific health-
    related context.",
      "properties": {
        "resourceType": {
          "description": "This is a Person resource",
          "const": "Person"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
```

```
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for a person within a particular scope.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "name": {
      "description": "A name associated with the person.",
      "items": {
        "$ref": "HumanName.schema.json#/definitions/HumanName"
      },
      "type": "array"
    },
    "telecom": {
      "description": "A contact detail for the person, e.g. a telephone number or an email address.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "gender": {
      "description": "Administrative Gender.",
      "enum": [
        "male",
        "female",
        "other",
        "unknown"
      ]
    },
    "_gender": {
      "description": "Extensions for gender",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "birthDate": {
      "description": "The birth date for the person.",
      "$ref": "#/definitions/date"
    },
```

```
    "_birthDate": {
      "description": "Extensions for birthDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "address": {
      "description": "One or more addresses for the person.",
      "items": {
        "$ref": "Address.schema.json#/definitions/Address"
      },
      "type": "array"
    },
    "photo": {
      "description": "An image that can be displayed as a thumbnail of the person to enhance the identification of
the individual.",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "managingOrganization": {
      "description": "The organization that is the custodian of the person record.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "active": {
      "description": "Whether this person\u0027s record is in active use.",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "link": {
      "description": "Link to a resource that concerns the same actual person.",
      "items": {
        "$ref": "#/definitions/Person_Link"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Person_Link": {
  "description": "Demographics and administrative information about a person independent of a specific health-
related context.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "target": {
      "description": "The resource to which this actual person is associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "assurance": {
      "description": "Level of assurance that this link is associated with the target resource.",
      "enum": [
        "level1",
        "level2",
        "level3",
        "level4"
      ]
    },
    "_assurance": {
      "description": "Extensions for assurance",
```

```
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "target"
      ]
    }
  }
}
```

8.18.18 Resource Person - Maturity Dependency View

URL: https://hl7.org/fhir/R4/person-dependencies.html

8.18.18 Resource Person - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Dependency Graph for Person FMM level 2

Name | Card. | Type | Dependency Analysis
Person | | DomainResource
identifier | 0..* | Identifier
name | 0..* | HumanName
telecom | 0..* | ContactPoint
gender | 0..1 | code
birthDate | 0..1 | date
address | 0..* | Address
photo | 0..1 | Attachment
managingOrganization | 0..1 | Reference(Organization)
active | 0..1 | boolean
link | 0..* | BackboneElement
target | 1..1 | Reference(Patient | Practitioner | RelatedPerson | Person)
assurance | 0..1 | code

8.19 Resource Linkage - Content

URL: https://hl7.org/fhir/R4/linkage.html
Patient Care Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Practitioner

Identifies two or more records (resource instances) that refer to the same real-world "occurrence".

8.19.1 Scope and Usage

This resource allows the assertion of linkages between multiple resource instances (generally of the same type) that are referring to the same underlying business objects. For example, multiple Condition records that refer to the same underlying problem/issue for a Patient; multiple AllergyIntolerance records that refer to the same reaction susceptibility; multiple Patient, Practitioner and/or RelatedPerson records that refer to the same human being or animal.

FHIR supports a process for sharing electronic records. It is common for multiple records to exist that deal with the same real-world phenomenon. This can result from information being captured by different systems, information being captured within a single system by different users (either deliberately to represent distinct perspectives or accidentally when a new record is created rather than updating an existing record). These multiple records may be referred to as "duplicate" records, but in practice they aren't often actually "duplicate" in that the data represented (and the history of the records) will be at least somewhat distinct.

In some cases, the solution after identifying duplicates is to deprecate one of the records (e.g. by changing the status to "Entered in Error") and to move all relevant information to the surviving record. In other cases, the resource may support the ability to establish a linkage directly between the resources. However, in some cases, both records may need to survive or there might be a desire to have both resources continue to be maintained, perhaps because the resources live on different servers or have different 'owners'. The Linkage resource is intended to satisfy this use-case.

8.19.2 Boundaries and Relationships

The Person resource should be used to link a person independent of a specific health-related context. Linkage can be used for all other use cases.

8.19.3 Resource Content

- Structure
- UML
- XML
- JSON

- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Linkage | ΣITU | | DomainResource | Links records for 'same' item + Rule: Must have at least two items Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
active | Σ | 0..1 | boolean | Whether this linkage assertion is active or not
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Who is responsible for linkages
item | ΣI | 1..* | BackboneElement | Item to be linked
type | Σ | 1..1 | code | source | alternate | historical LinkageType (Required)
resource | Σ | 1..1 | Reference(Any) | Resource being linked
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Linkage xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <active value="[boolean]"/><!-- 0..1 Whether this linkage assertion is active or not -->
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Who is responsible for linkages --></author>
 <item>  <!--  1..* Item to be linked -->
  <type value="[code]"/><!-- 1..1 source | alternate | historical -->
  <resource><!-- 1..1 Reference(Any) Resource being linked --></resource>
 </item>
</Linkage>
```

JSON Template

```
{
  "resourceType" : "Linkage",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "active" : <boolean>, // Whether this linkage assertion is active or not
  "author" : { Reference(Practitioner|PractitionerRole|Organization) }, // Who is responsible for linkages
  "item" : [{ // C? R!  Item to be linked
    "type" : "<code>", // R!  source | alternate | historical
    "resource" : { Reference(Any) } // R!  Resource being linked
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Linkage;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Linkage.active [ boolean ]; # 0..1 Whether this linkage assertion is active or not
  fhir:Linkage.author [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Who is responsible for linkages
  fhir:Linkage.item [ # 1..* Item to be linked
    fhir:Linkage.item.type [ code ]; # 1..1 source | alternate | historical
    fhir:Linkage.item.resource [ Reference(Any) ]; # 1..1 Resource being linked
  ], ...;
]
```

Changes since R3

Linkage
Linkage.active | Default Value "true" removed
Linkage.author | Type Reference: Added Target Type PractitionerRole
Linkage.item.type | Change value set from http://hl7.org/fhir/ValueSet/linkage-type to http://hl7.org/fhir/ValueSet/linkage-type|4.0.1
Linkage.item.resource | Type changed from Reference to Reference(Resource)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Linkage | ΣITU | | DomainResource | Links records for 'same' item + Rule: Must have at least two items Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
active | Σ | 0..1 | boolean | Whether this linkage assertion is active or not
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Who is responsible for linkages
item | ΣI | 1..* | BackboneElement | Item to be linked
type | Σ | 1..1 | code | source | alternate | historical LinkageType (Required)
resource | Σ | 1..1 | Reference(Any) | Resource being linked

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Linkage xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <active value="[boolean]"/><!-- 0..1 Whether this linkage assertion is active or not -->
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Who is responsible for linkages --></author>
 <item>  <!--  1..* Item to be linked -->
  <type value="[code]"/><!-- 1..1 source | alternate | historical -->
  <resource><!-- 1..1 Reference(Any) Resource being linked --></resource>
 </item>
</Linkage>
```

JSON Template

```
{
  "resourceType" : "Linkage",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "active" : <boolean>, // Whether this linkage assertion is active or not
  "author" : { Reference(Practitioner|PractitionerRole|Organization) }, // Who is responsible for linkages
  "item" : [{ // C? R!  Item to be linked
    "type" : "<code>", // R!  source | alternate | historical
    "resource" : { Reference(Any) } // R!  Resource being linked
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Linkage;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Linkage.active [ boolean ]; # 0..1 Whether this linkage assertion is active or not
  fhir:Linkage.author [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Who is responsible for linkages
  fhir:Linkage.item [ # 1..* Item to be linked
    fhir:Linkage.item.type [ code ]; # 1..1 source | alternate | historical
    fhir:Linkage.item.resource [ Reference(Any) ]; # 1..1 Resource being linked
  ], ...;
]
```

Changes since Release 3

Linkage
Linkage.active | Default Value "true" removed
Linkage.author | Type Reference: Added Target Type PractitionerRole
Linkage.item.type | Change value set from http://hl7.org/fhir/ValueSet/linkage-type to http://hl7.org/fhir/ValueSet/linkage-type|4.0.1
Linkage.item.resource | Type changed from Reference to Reference(Resource)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.19.3.1 Terminology Bindings

Path | Definition | Type | Reference
Linkage.item.type | Used to distinguish different roles a resource can play within a set of linked resources. | Required | LinkageType

8.19.3.2 Constraints

id | Level | Location | Description | Expression
lnk-1 | Rule | (base) | Must have at least two items | item.count()>1

8.19.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | Author of the Linkage | Linkage.author (Practitioner, Organization, PractitionerRole)
item | reference | Matches on any item in the Linkage | Linkage.item.resource (Any)
source | reference | Matches on any item in the Linkage with a type of 'source' | Linkage.item.resource (Any)

8.19.5 Resource Linkage - Examples

URL: https://hl7.org/fhir/R4/linkage-examples.html

8.19.5 Resource Linkage - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Practitioner

Example Name | id | Format
Example of linkage | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.19.6 Resource Linkage - Detailed Descriptions

URL: https://hl7.org/fhir/R4/linkage-definitions.html

8.19.6 Resource Linkage - Detailed Descriptions

Patient Care Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Practitioner

Detailed Descriptions for the elements in the Linkage resource.

Linkage
Element Id | Linkage
Definition | Identifies two or more records (resource instances) that refer to the same real-world "occurrence".
Cardinality | 0..*
Type | DomainResource
Summary | true
Invariants | Defined on this element lnk-1 Rule Must have at least two items item.count()>1
Defined on this element
lnk-1 | Rule | Must have at least two items | item.count()>1
Linkage.active
Element Id | Linkage.active
Definition | Indicates whether the asserted set of linkages are considered to be "in effect".
Cardinality | 0..1
Type | boolean
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a linkage record as not to be used because it was created in error.
Summary | true
Comments | If false, any asserted linkages should not be considered current/relevant/applicable.
Linkage.author
Element Id | Linkage.author
Definition | Identifies the user or organization responsible for asserting the linkages as well as the user or organization who establishes the context in which the nature of each linkage is evaluated.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Requirements | "Primary" for the same set of linked resources may be different from different user's/organization's perspectives.
Summary | true
Linkage.item
Element Id | Linkage.item
Definition | Identifies which record considered as the reference to the same real-world occurrence as well as how the items should be evaluated within the collection of linked items.
Cardinality | 1..*
Summary | true
Invariants | Affect this element lnk-1 Rule Must have at least two items item.count()>1
Affect this element
lnk-1 | Rule | Must have at least two items | item.count()>1
Linkage.item.type
Element Id | Linkage.item.type
Definition | Distinguishes which item is "source of truth" (if any) and which items are no longer considered to be current representations.
Cardinality | 1..1
Terminology Binding | LinkageType (Required)
Type | code
Summary | true
Linkage.item.resource
Element Id | Linkage.item.resource
Definition | The resource instance being linked as part of the group.
Cardinality | 1..1
Type | Reference(Any)
Summary | true

8.19.7 Resource Linkage - Mappings

URL: https://hl7.org/fhir/R4/linkage-mappings.html

8.19.7 Resource Linkage - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Practitioner

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the linkage resource (see Mappings to Other Standards for further information & status).

8.19.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Linkage
active | FiveWs.status
author | FiveWs.author

8.19.7.2 RIM Mapping (http://hl7.org/v3)

Linkage | N/A - RIM doesn't know how to do this

8.19.8 Resource Linkage - Extensions & Profiles

URL: https://hl7.org/fhir/R4/linkage-profiles.html

8.19.8 Resource Linkage - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Practitioner

This table lists profiles and extensions for the Linkage resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.19.8.1 Profiles

No Profiles defined for this resource

8.19.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.19.8.3 Search Extensions

No Search Extensions defined for this resource

8.19.9 Resource Linkage - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/linkage-version-maps.html

No validation errors - all conversions are clean

8.19.9 Resource Linkage - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Practitioner

R3 : R4 Conversion maps for Linkage.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.19.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Linkage3to4" = "R3 to R4 Conversions for Linkage"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Linkage" alias LinkageR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Linkage" alias Linkage as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Linkage(source src : LinkageR3, target tgt : Linkage) extends DomainResource <<type+>> {
  src.active -> tgt.active;
  src.author -> tgt.author;
  src.item as vs0 -> tgt.item as vt0 then linkage(vs0, vt0);
}

group linkage(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.resource -> tgt.resource;
}
```

8.19.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Linkage4to3" = "R4 to R3 Conversion for Linkage"

uses "http://hl7.org/fhir/StructureDefinition/Linkage" alias Linkage as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Linkage" alias LinkageR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Linkage(source src : Linkage, target tgt : LinkageR3) extends DomainResource <<type+>> {
  src.active -> tgt.active;
  src.author -> tgt.author;
  src.item as vs0 -> tgt.item as vt0 then linkage(vs0, vt0);
}

group linkage(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.resource -> tgt.resource;
}
```

8.19.9.3 R4 Validation Errors

8.19.10 JSON Schema for Linkage

URL: https://hl7.org/fhir/R4/linkage.schema.json.html

8.19.10 JSON Schema for Linkage

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Practitioner

This is a representation of the json schema for Linkage, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Linkage",
  "$ref": "#/definitions/Linkage",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Linkage": {
      "description": "Identifies two or more records (resource instances) that refer to the same real-world
    \"occurrence\".",
      "properties": {
        "resourceType": {
          "description": "This is a Linkage resource",
          "const": "Linkage"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
```

```
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "active": {
      "description": "Indicates whether the asserted set of linkages are considered to be \"in effect\".",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "Identifies the user or organization responsible for asserting the linkages as well as the user or organization who establishes the context in which the nature of each linkage is evaluated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "item": {
      "description": "Identifies which record considered as the reference to the same real-world occurrence as well as how the items should be evaluated within the collection of linked items.",
      "items": {
        "$ref": "#/definitions/Linkage_Item"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "item",
    "resourceType"
  ]
},
"Linkage_Item": {
  "description": "Identifies two or more records (resource instances) that refer to the same real-world \"occurrence\".",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "type": {
          "description": "Distinguishes which item is \"source of truth\" (if any) and which items are no longer
    considered to be current representations.",
          "enum": [
            "source",
            "alternate",
            "historical"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "resource": {
          "description": "The resource instance being linked as part of the group.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "resource"
      ]
    }
  }
}
```


8.19.11 Resource Linkage - Maturity Dependency View

URL: https://hl7.org/fhir/R4/linkage-dependencies.html

8.19.11 Resource Linkage - Maturity Dependency View

Patient Care Work Group | Maturity Level: 0 | Trial Use | Compartments: Practitioner

Dependency Graph for Linkage FMM level 0

Name	Card.	Type	Dependency Analysis
Linkage			DomainResource
active	0..1	boolean	
author	0..1	Reference(Practitioner | PractitionerRole | Organization)	
item	1..*	BackboneElement	
type	1..1	code	
resource	1..1	Reference(Any)	

8.20 Resource Communication - Content

URL: https://hl7.org/fhir/R4/communication.html

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An occurrence of information being transmitted; e.g. an alert that was sent to a responsible provider, a public health agency that was notified about a reportable condition.

8.20.1 Scope and Usage

Communication is one of the event resources in the FHIR workflow specification.

This resource is a record of a communication even if it is planned or has failed. A communication is a conveyance of information from one entity, a sender, to another entity, a receiver. The sender and receivers may be patients, practitioners, related persons, organizations, or devices. Communication use cases include:

• A reminder or alert delivered to a responsible provider

• A recorded notification from the nurse to the on-call physician (or any other specified person) that a patient's temperature exceeds a value

• A notification to a public health agency of a patient presenting with a communicable disease reportable to the public health agency

• Patient educational material sent by a provider to a patient

• Unable to deliver lab results to ordering physician

Non-patient specific communication use cases may include:

• A nurse call from a hall bathroom

• Advisory for battery service from a pump

8.20.2 Boundaries and Relationships

This resource is a record of a communication that has occurred. It does not represent the actual flow of communication. While AuditEvent can track electronic disclosures of information, it cannot track conversations, phone calls, letters and other interactions that are not system-to-system. And even for system-to-system communications, the specific end recipients might not be known. Furthermore, AuditEvents are not considered to be "part" of the patient record, while Communication instances are. The Communication resource is not used as a general audit mechanism to track every disclosure of every record. Rather, it is used when a clinician or other user wants to ensure a record of a particular communication is itself maintained as part of the reviewable health record.

Flag resources represent a continuous ongoing "communication" alerting anyone dealing with the patient of certain precautions to take or issues to be aware of. The flags are continuously present as an ongoing reminder. This is distinct from Communication where there is a specific intended sender and receiver and the information is delivered only once.

Communication and Encounter

The Communication is about the transfer of information (which might or might not occur as part of an encounter), while Encounter is about the coming together (in person or virtually) of a Patient with a Practitioner. Communication does not deal with the duration of a call, it represents the fact that information was transferred at a particular point in time.

The phone calls involving the Patient should be handled using Encounter. Phone calls not involving the patient (e.g. between practitioners or practitioner to relative) that are tracked for billing or other purposes can use Communication to represent the information transferred but are not ideal to represent the call itself. A better mechanism for handling such calls will be explored in a future release.

This resource is referenced by itself

8.20.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Communication | TU | | DomainResource | A record of information transmitted from a sender to a receiver Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(Any) | Request fulfilled by this communication
partOf | Σ | 0..* | Reference(Any) | Part of this action
inResponseTo | | 0..* | Reference(Communication) | Reply to
status | ?!Σ | 1..1 | code | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown EventStatus (Required)
statusReason | Σ | 0..1 | CodeableConcept | Reason for current status CommunicationNotDoneReason (Example)
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | Σ | 0..1 | Reference(Patient | Group) | Focus of message
topic | | 0..1 | CodeableConcept | Description of the purpose/content CommunicationTopic (Example)
about | | 0..* | Reference(Any) | Resources that pertain to this communication
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
sent | | 0..1 | dateTime | When sent
received | | 0..1 | dateTime | When received
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService) | Message recipient
sender | | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService) | Message sender

reasonCode | Σ | 0..* | CodeableConcept | Indication for message SNOMED CT Clinical Findings (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why was communication done?
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
note | | 0..* | Annotation | Comments made about the communication
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Communication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this communication --></basedOn>
 <partOf><!-- 0..* Reference(Any) Part of this action --></partOf>
 <inResponseTo><!-- 0..* Reference(Communication) Reply to --></inResponseTo>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..* CodeableConcept Message category --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <topic><!-- 0..1 CodeableConcept Description of the purpose/content --></topic>
 <about><!-- 0..* Reference(Any) Resources that pertain to this communication --></about>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <sent value="[dateTime]"/><!-- 0..1 When sent -->
 <received value="[dateTime]"/><!-- 0..1 When received -->
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) Message recipient --></recipient>
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) Message sender --></sender>
 <reasonCode><!-- 0..* CodeableConcept Indication for message --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why was communication done? --></reasonReference>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <note><!-- 0..* Annotation Comments made about the communication --></note>
</Communication>
```

JSON Template

```
{
  "resourceType" : "Communication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "instantiatesCanonical" : [{ canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this communication
  "partOf" : [{ Reference(Any) }], // Part of this action
  "inResponseTo" : [{ Reference(Communication) }], // Reply to
  "status" : "<code>", // R!  preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error |
    unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "category" : [{ CodeableConcept }], // Message category
  "priority" : "<code>", // routine | urgent | asap | stat
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "topic" : { CodeableConcept }, // Description of the purpose/content
  "about" : [{ Reference(Any) }], // Resources that pertain to this communication
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "sent" : "<dateTime>", // When sent
  "received" : "<dateTime>", // When received
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) }], // Message recipient
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) }, // Message sender
  "reasonCode" : [{ CodeableConcept }], // Indication for message
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why was communication done?
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  "note" : [{ Annotation }] // Comments made about the communication
```

```
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Communication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Communication.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Communication.instantiatesCanonical [
    canonical(PlanDefinition|ActivityDefinition|Measure|OperationDefinition|Questionnaire) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:Communication.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:Communication.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this communication
  fhir:Communication.partOf [ Reference(Any) ], ... ; # 0..* Part of this action
  fhir:Communication.inResponseTo [ Reference(Communication) ], ... ; # 0..* Reply to
  fhir:Communication.status [ code ]; # 1..1 preparation | in-progress | not-done | on-hold | stopped | completed |
    entered-in-error | unknown
  fhir:Communication.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Communication.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:Communication.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:Communication.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:Communication.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:Communication.topic [ CodeableConcept ]; # 0..1 Description of the purpose/content
  fhir:Communication.about [ Reference(Any) ], ... ; # 0..* Resources that pertain to this communication
  fhir:Communication.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:Communication.sent [ dateTime ]; # 0..1 When sent
  fhir:Communication.received [ dateTime ]; # 0..1 When received
  fhir:Communication.recipient [
    Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|Group|
  CareTeam|HealthcareService) ], ... ; # 0..* Message recipient
  fhir:Communication.sender [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|
  HealthcareService) ]; # 0..1 Message sender
  fhir:Communication.reasonCode [ CodeableConcept ], ... ; # 0..* Indication for message
  fhir:Communication.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Why was communication done?
  fhir:Communication.payload [ # 0..* Message payload
    # Communication.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:Communication.payload.contentString [ string ]
      fhir:Communication.payload.contentAttachment [ Attachment ]
      fhir:Communication.payload.contentReference [ Reference(Any) ]
  ], ...;
  fhir:Communication.note [ Annotation ], ... ; # 0..* Comments made about the communication
]
```

Changes since R3

Communication
Communication.instantiatesCanonical | Added Element
Communication.instantiatesUri | Added Element
Communication.inResponseTo | Added Element
Communication.status | Change value set from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/event-status|4.0.1
Communication.statusReason | Added Element
Communication.priority | Added Element
Communication.topic | Max Cardinality changed from * to 1 Type changed from Reference(Resource) to CodeableConcept
Communication.about | Added Element
Communication.encounter | Added Element
Communication.recipient | Type Reference: Added Target Types PractitionerRole, CareTeam, HealthcareService
Communication.sender | Type Reference: Added Target Types PractitionerRole, HealthcareService
Communication.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference
Communication.definition | deleted
Communication.notDone | deleted
Communication.notDoneReason | deleted
Communication.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Communication | TU | | DomainResource | A record of information transmitted from a sender to a receiver Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(Any) | Request fulfilled by this communication

partOf | Σ | 0..* | Reference(Any) | Part of this action
inResponseTo | | 0..* | Reference(Communication) | Reply to
status | ?!Σ | 1..1 | code | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown EventStatus (Required)
statusReason | Σ | 0..1 | CodeableConcept | Reason for current status CommunicationNotDoneReason (Example)
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | Σ | 0..1 | Reference(Patient | Group) | Focus of message
topic | | 0..1 | CodeableConcept | Description of the purpose/content CommunicationTopic (Example)
about | | 0..* | Reference(Any) | Resources that pertain to this communication
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
sent | | 0..1 | dateTime | When sent
received | | 0..1 | dateTime | When received
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService) | Message recipient
sender | | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService) | Message sender
reasonCode | Σ | 0..* | CodeableConcept | Indication for message SNOMED CT Clinical Findings (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why was communication done?
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
note | | 0..* | Annotation | Comments made about the communication
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Communication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this communication --></basedOn>
 <partOf><!-- 0..* Reference(Any) Part of this action --></partOf>
 <inResponseTo><!-- 0..* Reference(Communication) Reply to --></inResponseTo>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..* CodeableConcept Message category --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <topic><!-- 0..1 CodeableConcept Description of the purpose/content --></topic>
 <about><!-- 0..* Reference(Any) Resources that pertain to this communication --></about>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <sent value="[dateTime]"/><!-- 0..1 When sent -->
 <received value="[dateTime]"/><!-- 0..1 When received -->
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) Message recipient --></recipient>
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) Message sender --></sender>
 <reasonCode><!-- 0..* CodeableConcept Indication for message --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why was communication done? --></reasonReference>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <note><!-- 0..* Annotation Comments made about the communication --></note>
</Communication>
```

JSON Template

```
{
  "resourceType" : "Communication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "instantiatesCanonical" : [{ canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this communication
  "partOf" : [{ Reference(Any) }], // Part of this action
  "inResponseTo" : [{ Reference(Communication) }], // Reply to
  "status" : "<code>", // R!  preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error |
    unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "category" : [{ CodeableConcept }], // Message category
```

```
  "priority" : "<code>", // routine | urgent | asap | stat
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "topic" : { CodeableConcept }, // Description of the purpose/content
  "about" : [{ Reference(Any) }], // Resources that pertain to this communication
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "sent" : "<dateTime>", // When sent
  "received" : "<dateTime>", // When received
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) }], // Message recipient
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) }, // Message sender
  "reasonCode" : [{ CodeableConcept }], // Indication for message
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why was communication done?
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  "note" : [{ Annotation }] // Comments made about the communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Communication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Communication.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Communication.instantiatesCanonical [
    canonical(PlanDefinition|ActivityDefinition|Measure|OperationDefinition|Questionnaire) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:Communication.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:Communication.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this communication
  fhir:Communication.partOf [ Reference(Any) ], ... ; # 0..* Part of this action
  fhir:Communication.inResponseTo [ Reference(Communication) ], ... ; # 0..* Reply to
  fhir:Communication.status [ code ]; # 1..1 preparation | in-progress | not-done | on-hold | stopped | completed |
    entered-in-error | unknown
  fhir:Communication.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Communication.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:Communication.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:Communication.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:Communication.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:Communication.topic [ CodeableConcept ]; # 0..1 Description of the purpose/content
  fhir:Communication.about [ Reference(Any) ], ... ; # 0..* Resources that pertain to this communication
  fhir:Communication.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:Communication.sent [ dateTime ]; # 0..1 When sent
  fhir:Communication.received [ dateTime ]; # 0..1 When received
  fhir:Communication.recipient [
    Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|Group|
  CareTeam|HealthcareService) ], ... ; # 0..* Message recipient
  fhir:Communication.sender [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|
  HealthcareService) ]; # 0..1 Message sender
  fhir:Communication.reasonCode [ CodeableConcept ], ... ; # 0..* Indication for message
  fhir:Communication.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Why was communication done?
  fhir:Communication.payload [ # 0..* Message payload
    # Communication.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:Communication.payload.contentString [ string ]
      fhir:Communication.payload.contentAttachment [ Attachment ]
      fhir:Communication.payload.contentReference [ Reference(Any) ]
  ], ...;
  fhir:Communication.note [ Annotation ], ... ; # 0..* Comments made about the communication
]
```

Changes since Release 3

Communication
Communication.instantiatesCanonical | Added Element
Communication.instantiatesUri | Added Element
Communication.inResponseTo | Added Element
Communication.status | Change value set from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/event-status|4.0.1
Communication.statusReason | Added Element
Communication.priority | Added Element
Communication.topic | Max Cardinality changed from * to 1 Type changed from Reference(Resource) to CodeableConcept
Communication.about | Added Element
Communication.encounter | Added Element
Communication.recipient | Type Reference: Added Target Types PractitionerRole, CareTeam, HealthcareService
Communication.sender | Type Reference: Added Target Types PractitionerRole, HealthcareService
Communication.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference

Communication.definition | deleted
Communication.notDone | deleted
Communication.notDoneReason | deleted
Communication.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.20.3.1 Terminology Bindings

Path | Definition | Type | Reference
Communication.status | The status of the communication. | Required | EventStatus
Communication.statusReason | Codes for the reason why a communication did not happen. | Example | CommunicationNotDoneReason
Communication.category | Codes for general categories of communications such as alerts, instructions, etc. | Example | CommunicationCategory
Communication.priority | Codes indicating the relative importance of a communication. | Required | RequestPriority
Communication.medium | Codes for communication mediums such as phone, fax, email, in person, etc. | Example | v3.ParticipationMode
Communication.topic | Codes describing the purpose or content of the communication. | Example | CommunicationTopic
Communication.reasonCode | Codes for describing reasons for the occurrence of a communication. | Example | SNOMEDCTClinicalFindings

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

8.20.3.2 Communication.sender and Communication.recipient

Communication.sender allows Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService and Communication.recipient allows Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService - but it is not unusual to have a communication target - even a defined one - where it is unknown what kind of role the person is playing.

If the communication is to or from an individual whose role is not known (practitioner, patient or related person) - for example, only email address is captured in the system - then RelatedPerson should be used by default.

8.20.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | Request fulfilled by this communication | Communication.basedOn (Any)
category | token | Message category | Communication.category
encounter | reference | Encounter created as part of | Communication.encounter (Encounter)
identifier | token | Unique identifier | Communication.identifier
instantiates-canonical | reference | Instantiates FHIR protocol or definition | Communication.instantiatesCanonical (Questionnaire, Measure, PlanDefinition, OperationDefinition, ActivityDefinition)
instantiates-uri | uri | Instantiates external protocol or definition | Communication.instantiatesUri
medium | token | A channel of communication | Communication.medium
part-of | reference | Part of this action | Communication.partOf (Any)
patient | reference | Focus of message | Communication.subject.where(resolve() is Patient) (Patient)
received | date | When received | Communication.received
recipient | reference | Message recipient | Communication.recipient (Practitioner, Group, Organization, CareTeam, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
sender | reference | Message sender | Communication.sender (Practitioner, Organization, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
sent | date | When sent | Communication.sent
status | token | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown | Communication.status
subject | reference | Focus of message | Communication.subject (Group, Patient)

8.20.5 Resource Communication - Examples

URL: https://hl7.org/fhir/R4/communication-examples.html

8.20.5 Resource Communication - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An introduction to this page as well as proper examples are to be submitted shortly.

Example Name | id | Format
An example of an alert message | example | XML | JSON | Turtle
An example of a response to a request for additional information | fm-solicited | XML | JSON | Turtle
An example of an unsolicited attachment - provision of additional information | fm-attachment | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.20.6 Resource Communication - Detailed Descriptions

URL: https://hl7.org/fhir/R4/communication-definitions.html

8.20.6 Resource Communication - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Communication resource.

Communication
Element Id | Communication
Definition | An occurrence of information being transmitted; e.g. an alert that was sent to a responsible provider, a public health agency that was notified about a reportable condition.
Cardinality | 0..*
Type | DomainResource
Communication.identifier
Element Id | Communication.identifier
Definition | Business identifiers assigned to this communication by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the communication as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
Communication.instantiatesCanonical
Element Id | Communication.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this Communication.
Cardinality | 0..*
Type | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire)
Summary | true
Communication.instantiatesUri
Element Id | Communication.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this Communication.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
Communication.basedOn
Element Id | Communication.basedOn
Definition | An order, proposal or plan fulfilled in whole or in part by this Communication.
Cardinality | 0..*
Type | Reference(Any)
Alternate Names | fulfills
Summary | true
Comments | This must point to some sort of a 'Request' resource, such as CarePlan, CommunicationRequest, ServiceRequest, MedicationRequest, etc.
Communication.partOf
Element Id | Communication.partOf
Definition | Part of this action.
Cardinality | 0..*
Type | Reference(Any)
Alternate Names | container
Summary | true
Communication.inResponseTo
Element Id | Communication.inResponseTo
Definition | Prior communication that this communication is in response to.
Cardinality | 0..*

Type | Reference(Communication)
Communication.status
Element Id | Communication.status
Definition | The status of the transmission.
Cardinality | 1..1
Terminology Binding | EventStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes aborted and entered-in-error that mark the communication as not currently valid.
Communication.statusReason
Element Id | Communication.statusReason
Definition | Captures the reason for the current state of the Communication.
Cardinality | 0..1
Terminology Binding | CommunicationNotDoneReason (Example)
Type | CodeableConcept
Alternate Names | Suspended Reason; Cancelled Reason
Summary | true
Comments | This is generally only used for "exception" statuses such as "not-done", "suspended" or "aborted". The reason for performing the event at all is captured in reasonCode, not here.
Communication.category
Element Id | Communication.category
Definition | The type of message conveyed such as alert, notification, reminder, instruction, etc.
Cardinality | 0..*
Terminology Binding | CommunicationCategory (Example)
Type | CodeableConcept
Comments | There may be multiple axes of categorization and one communication may serve multiple purposes.
Communication.priority
Element Id | Communication.priority
Definition | Characterizes how quickly the planned or in progress communication must be addressed. Includes concepts such as stat, urgent, routine.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, this communication should be treated with normal priority
Summary | true
Comments | Used to prioritize workflow (such as which communication to read first) when the communication is planned or in progress.
Communication.medium
Element Id | Communication.medium
Definition | A channel that was used for this communication (e.g. email, fax).
Cardinality | 0..*
Terminology Binding | v3 Code System ParticipationMode (Example)
Type | CodeableConcept
Communication.subject
Element Id | Communication.subject
Definition | The patient or group that was the focus of this communication.
Cardinality | 0..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
Communication.topic
Element Id | Communication.topic
Definition | Description of the purpose/content, similar to a subject line in an email.
Cardinality | 0..1
Terminology Binding | CommunicationTopic (Example)
Type | CodeableConcept
Comments | Communication.topic.text can be used without any codings.
Communication.about
Element Id | Communication.about
Definition | Other resources that pertain to this communication and to which this communication should be associated.
Cardinality | 0..*
Type | Reference(Any)
Comments | Don't use Communication.about element when a more specific element exists, such as basedOn or reasonReference.
Communication.encounter
Element Id | Communication.encounter
Definition | The Encounter during which this Communication was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true

Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter.
Communication.sent
Element Id | Communication.sent
Definition | The time when this communication was sent.
Cardinality | 0..1
Type | dateTime
Communication.received
Element Id | Communication.received
Definition | The time when this communication arrived at the destination.
Cardinality | 0..1
Type | dateTime
Communication.recipient
Element Id | Communication.recipient
Definition | The entity (e.g. person, organization, clinical information system, care team or device) which was the target of the communication. If receipts need to be tracked by an individual, a separate resource instance will need to be created for each recipient. Multiple recipient communications are intended where either receipts are not tracked (e.g. a mass mail-out) or a receipt is captured in aggregate (all emails confirmed received by a particular time).
Cardinality | 0..*
Type | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService)
Communication.sender
Element Id | Communication.sender
Definition | The entity (e.g. person, organization, clinical information system, or device) which was the source of the communication.
Cardinality | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService)
Communication.reasonCode
Element Id | Communication.reasonCode
Definition | The reason or justification for the communication.
Cardinality | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be captured using reasonCode.text.
Communication.reasonReference
Element Id | Communication.reasonReference
Definition | Indicates another resource whose existence justifies this communication.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
Summary | true
Communication.payload
Element Id | Communication.payload
Definition | Text, attachment(s), or resource(s) that was communicated to the recipient.
Cardinality | 0..*
Communication.payload.content[x]
Element Id | Communication.payload.content[x]
Definition | A communicated content (or for multi-part communications, one portion of the communication).
Cardinality | 1..1
Type | string|Attachment|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Communication.note
Element Id | Communication.note
Definition | Additional notes or commentary about the communication by the sender, receiver or other interested parties.
Cardinality | 0..*
Type | Annotation

8.20.7 Resource Communication - Mappings

URL: https://hl7.org/fhir/R4/communication-mappings.html

8.20.7 Resource Communication - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the communication resource (see Mappings to Other Standards for further information & status).

8.20.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Communication
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
priority | FiveWs.grade
subject | FiveWs.subject[x]
topic | FiveWs.context
about | FiveWs.context
encounter | FiveWs.context
sent | FiveWs.done[x]
received | FiveWs.done[x]
recipient | FiveWs.actor
sender | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

8.20.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Communication | Event
identifier | Event.identifier
instantiatesCanonical | Event.instantiatesCanonical
instantiatesUri | Event.instantiatesUri
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
statusReason | Event.statusReason
subject | Event.subject
encounter | Event.context
sent | Event.occurrence[x] {Invariant: maps to period.start}
received | Event.occurrence[x] {Invariant: maps to period.end}
recipient | Event.performer.actor
sender | Event.performer.actor
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note

8.20.7.3 RIM Mapping (http://hl7.org/v3)

Communication
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target

8.20.8 Resource Communication - Extensions & Profiles

URL: https://hl7.org/fhir/R4/communication-profiles.html

8.20.8 Resource Communication - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Communication resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.20.8.1 Profiles

No Profiles defined for this resource

8.20.8.2 Extensions

communication-media | Communication | media | for Communication HL7 Extensions
workflow-episodeOfCare | Communication | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions

relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.20.8.3 Search Extensions

No Search Extensions defined for this resource

8.20.9 Resource Communication - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/communication-version-maps.html

No validation errors - all conversions are clean

8.20.9 Resource Communication - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Communication.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

8.20.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Communication3to4" = "R3 to R4 Conversions for Communication"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Communication" alias CommunicationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Communication" alias Communication as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Communication(source src : CommunicationR3, target tgt : Communication) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.instantiatesCanonical;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.status where src.notDone.not() -> tgt.status;
  src.notDone where src.notDone -> tgt.status = 'not-done';
  src.notDoneReason -> tgt.statusReason;
  src.category -> tgt.category;
  src.medium -> tgt.medium;
  src.subject -> tgt.subject;
  src.recipient -> tgt.recipient;
  src.topic as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Communication.topic',  vt.value = v;
  src.context -> tgt.encounter;
  src.sent -> tgt.sent;
  src.received -> tgt.received;
  src.sender -> tgt.sender;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.payload as s -> tgt.payload as t then payload(s, t);
  src.note -> tgt.note;
}

group payload(source src, target tgt) extends BackboneElement {
  src.content : string as s -> tgt.content = create('string') as t then string(s, t);
  src.content : Attachment as s -> tgt.content = create('Attachment') as t then Attachment(s, t);
  src.content : Reference as s -> tgt.content = create('Reference') as t then Reference(s, t);
}
```

8.20.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Communication4to3" = "R4 to R3 Conversion for Communication"

uses "http://hl7.org/fhir/StructureDefinition/Communication" alias Communication as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Communication" alias CommunicationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Communication(source src : CommunicationR3, target tgt : Communication) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.definition;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.status where value != 'not-done' -> tgt.status;
  src.status as v where value = 'not-done' ->  tgt.status = 'completed',  tgt.notDone = v;
  src.statusReason where src.status = 'not-done' -> tgt.notDoneReason;
  src.category -> tgt.category;
  src.medium -> tgt.medium;
  src.subject -> tgt.subject;
  src.recipient -> tgt.recipient;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Communication.topic' then {
    e.value -> tgt.topic;
  };
```

```
  src.encounter -> tgt.context;
  src.sent -> tgt.sent;
  src.received -> tgt.received;
  src.sender -> tgt.sender;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.payload as s -> tgt.payload as t then payload(s, t);
  src.note -> tgt.note;
}

group payload(source src, target tgt) extends BackboneElement {
  src.content : string as s -> tgt.content = create('string') as t then string(s, t);
  src.content : Attachment as s -> tgt.content = create('Attachment') as t then Attachment(s, t);
  src.content : Reference as s -> tgt.content = create('Reference') as t then Reference(s, t);
}
```

8.20.9.3 R4 Validation Errors

8.20.10 Communication HL7 Extensions

URL: https://hl7.org/fhir/R4/communication-extensions.html

Defines common extensions used with or related to the Communication resource

8.20.10 Communication HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

8.20.10.1 Content

Extensions:

communication-media | media : It contains enriched media representation of the alert message, such as a voice recording. This may be used, for example for compliance with jurisdictional accessibility requirements, literacy issues, or translations of the unstructured text content in other languages.

8.20.11 JSON Schema for Communication

URL: https://hl7.org/fhir/R4/communication.schema.json.html

8.20.11 JSON Schema for Communication

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Communication, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Communication",
  "$ref": "#/definitions/Communication",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Communication": {
      "description": "An occurrence of information being transmitted; e.g. an alert that was sent to a responsible
    provider, a public health agency that was notified about a reportable condition.",
      "properties": {
        "resourceType": {
          "description": "This is a Communication resource",
          "const": "Communication"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this communication by the performer or other systems which
remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is
adhered to in whole or in part by this Communication.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, orderset or other definition
 that is adhered to in whole or in part by this Communication.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "An order, proposal or plan fulfilled in whole or in part by this Communication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "Part of this action.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "inResponseTo": {
      "description": "Prior communication that this communication is in response to.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
```

```
      },
      "type": "array"
    },
    "status": {
      "description": "The status of the transmission.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Captures the reason for the current state of the Communication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "The type of message conveyed such as alert, notification, reminder, instruction, etc.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "priority": {
      "description": "Characterizes how quickly the planned or in progress communication must be addressed. Includes
 concepts such as stat, urgent, routine.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "medium": {
      "description": "A channel that was used for this communication (e.g. email, fax).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "subject": {
      "description": "The patient or group that was the focus of this communication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "topic": {
      "description": "Description of the purpose/content, similar to a subject line in an email.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "about": {
      "description": "Other resources that pertain to this communication and to which this communication should be
associated.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "encounter": {
      "description": "The Encounter during which this Communication was created or to which the creation of this
record is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "sent": {
      "description": "The time when this communication was sent.",
      "$ref": "#/definitions/dateTime"
    },
    "_sent": {
      "description": "Extensions for sent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "received": {
      "description": "The time when this communication arrived at the destination.",
      "$ref": "#/definitions/dateTime"
    },
    "_received": {
      "description": "Extensions for received",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recipient": {
      "description": "The entity (e.g. person, organization, clinical information system, care team or device) which
 was the target of the communication. If receipts need to be tracked by an individual, a separate resource instance
will need to be created for each recipient.  Multiple recipient communications are intended where either receipts
are not tracked (e.g. a mass mail-out) or a receipt is captured in aggregate (all emails confirmed received by a
particular time).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "sender": {
      "description": "The entity (e.g. person, organization, clinical information system, or device) which was the
source of the communication.",
```

```
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "The reason or justification for the communication.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates another resource whose existence justifies this communication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "payload": {
      "description": "Text, attachment(s), or resource(s) that was communicated to the recipient.",
      "items": {
        "$ref": "#/definitions/Communication_Payload"
      },
      "type": "array"
    },
    "note": {
      "description": "Additional notes or commentary about the communication by the sender, receiver or other
interested parties.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Communication_Payload": {
  "description": "An occurrence of information being transmitted; e.g. an alert that was sent to a responsible
provider, a public health agency that was notified about a reportable condition.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "contentString": {
      "description": "A communicated content (or for multi-part communications, one portion of the communication).",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_contentString": {
      "description": "Extensions for contentString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contentAttachment": {
      "description": "A communicated content (or for multi-part communications, one portion of the communication).",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "contentReference": {
      "description": "A communicated content (or for multi-part communications, one portion of the communication).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
```

```
    }
  }
}
```

8.20.12 Resource Communication - Maturity Dependency View

URL: https://hl7.org/fhir/R4/communication-dependencies.html

8.20.12 Resource Communication - Maturity Dependency View

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for Communication FMM level 2

Name | Card. | Type | Dependency Analysis
Communication | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire)
instantiatesUri | 0..* | uri
basedOn | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
partOf | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
inResponseTo | 0..* | Reference(Communication)
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
category | 0..* | CodeableConcept
priority | 0..1 | code
medium | 0..* | CodeableConcept
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
topic | 0..1 | CodeableConcept
about | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
sent | 0..1 | dateTime
received | 0..1 | dateTime
recipient | 0..* | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
sender | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
payload | 0..* | BackboneElement
content[x] | 1..1
contentString | | string
contentAttachment | | Attachment
contentReference | | Reference(Any
note | 0..* | Annotation

8.20.13 Value Set Sc-valueset-event-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-event-status.html

8.20.13 Value Set Sc-valueset-event-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.20.14 EventStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-event-status)

Mapping from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes identifying the lifecycle stage of an event."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
preparation | equivalent | planned
in-progress | equivalent | active
on-hold | equivalent | suspended
stopped | equivalent | failed
completed | equivalent | complete
not-done | equivalent | abandoned
unknown | equivalent | unknown

8.21 Resource CommunicationRequest - Content

URL: https://hl7.org/fhir/R4/communicationrequest.html
Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A request to convey information; e.g. the CDS system proposes that an alert be sent to a responsible provider, the CDS system proposes that the public health agency be notified about a reportable condition.

8.21.1 Scope and Usage

CommunicationRequest is one of the request resources in the FHIR workflow specification.

This resource is a record of a request for a communication to be performed. A communication is a conveyance of information from one entity, a sender, to another entity, a receiver. The requester requests the sender to send the payload to the recipient. The sender and receivers may be patients, practitioners, related persons, organizations, and devices. Uses of communication request include:

• A computer-based decision-support system requesting a reminder or alert be delivered to a responsible provider

• A physician requesting notification from the nurse if a patient's temperature exceeds a value

• A monitoring system or a provider requesting a staff member or department to notify a public health agency of a patient presenting with a communicable disease reportable to the public health agency

• A computer-based decision-support system proposes to send educational material to a patient

8.21.2 Boundaries and Relationships

This resource is a record of a request. It does not represent the actual flow of communication.

The use of CommunicationRequest excludes requests for referrals and requests for therapy or counseling which would be handled by the ServiceRequest resource. The performance of a CommunicationRequest may result in a Communication resource.

This resource is referenced by CarePlan, ClaimResponse and itself

8.21.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
CommunicationRequest | TU | | DomainResource | A request for information to be sent to a receiver Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier
basedOn | Σ | 0..* | Reference(Any) | Fulfills plan or proposal
replaces | Σ | 0..* | Reference(CommunicationRequest) | Request(s) replaced by this request
groupIdentifier | Σ | 0..1 | Identifier | Composite request this is part of
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if request is prohibiting action
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | | 0..1 | Reference(Patient | Group) | Focus of message
about | | 0..* | Reference(Any) | Resources that pertain to this communication request
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
occurrence[x] | Σ | 0..1 | | When scheduled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
authoredOn | Σ | 0..1 | dateTime | When request transitioned to being actionable
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/what is requesting service

recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService) | Message recipient

sender | Σ | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService) | Message sender

reasonCode | Σ | 0..* | CodeableConcept | Why is communication needed? v3 Code System ActReason (Example)

reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why is communication needed?

note | | 0..* | Annotation | Comments made about communication request

Documentation for this format

UML Diagram (Legend)

XML Template

```
<CommunicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <basedOn><!-- 0..* Reference(Any) Fulfills plan or proposal --></basedOn>
 <replaces><!-- 0..* Reference(CommunicationRequest) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..* CodeableConcept Message category --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if request is prohibiting action -->
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <about><!-- 0..* Reference(Any) Resources that pertain to this communication request --></about>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <occurrence[x]><!-- 0..1 dateTime|Period When scheduled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request transitioned to being actionable -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Who/what is requesting service --></requester>
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) Message recipient --></recipient>
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) Message sender --></sender>
 <reasonCode><!-- 0..* CodeableConcept Why is communication needed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why is communication needed? --></reasonReference>
 <note><!-- 0..* Annotation Comments made about communication request --></note>
</CommunicationRequest>
```

JSON Template

```
{
  "resourceType" : "CommunicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "basedOn" : [{ Reference(Any) }], // Fulfills plan or proposal
  "replaces" : [{ Reference(CommunicationRequest) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "category" : [{ CodeableConcept }], // Message category
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if request is prohibiting action
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "about" : [{ Reference(Any) }], // Resources that pertain to this communication request
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  // occurrence[x]: When scheduled. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "authoredOn" : "<dateTime>", // When request transitioned to being actionable
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Who/what is requesting service
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) }], // Message recipient
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) }, // Message sender
  "reasonCode" : [{ CodeableConcept }], // Why is communication needed?
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why is communication needed?
  "note" : [{ Annotation }] // Comments made about communication request
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CommunicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CommunicationRequest.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:CommunicationRequest.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan or proposal
  fhir:CommunicationRequest.replaces [ Reference(CommunicationRequest) ], ... ; # 0..* Request(s) replaced by this
    request
  fhir:CommunicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:CommunicationRequest.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error |
    unknown
  fhir:CommunicationRequest.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:CommunicationRequest.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:CommunicationRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:CommunicationRequest.doNotPerform [ boolean ]; # 0..1 True if request is prohibiting action
  fhir:CommunicationRequest.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:CommunicationRequest.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:CommunicationRequest.about [ Reference(Any) ], ... ; # 0..* Resources that pertain to this communication request
  fhir:CommunicationRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:CommunicationRequest.payload [ # 0..* Message payload
    # CommunicationRequest.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:CommunicationRequest.payload.contentString [ string ]
      fhir:CommunicationRequest.payload.contentAttachment [ Attachment ]
      fhir:CommunicationRequest.payload.contentReference [ Reference(Any) ]
  ], ...;
  # CommunicationRequest.occurrence[x] : 0..1 When scheduled. One of these 2
    fhir:CommunicationRequest.occurrenceDateTime [ dateTime ]
    fhir:CommunicationRequest.occurrencePeriod [ Period ]
  fhir:CommunicationRequest.authoredOn [ dateTime ]; # 0..1 When request transitioned to being actionable
  fhir:CommunicationRequest.requester [
    Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device) ]; # 0..1 Who/what is requesting
    service
  fhir:CommunicationRequest.recipient [
    Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|Group|
  CareTeam|HealthcareService) ], ... ; # 0..* Message recipient
  fhir:CommunicationRequest.sender [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|
  HealthcareService) ]; # 0..1 Message sender
  fhir:CommunicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Why is communication needed?
  fhir:CommunicationRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ...
     ; # 0..* Why is communication needed?
  fhir:CommunicationRequest.note [ Annotation ], ... ; # 0..* Comments made about communication request
]
```

Changes since R3

CommunicationRequest

CommunicationRequest.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1

CommunicationRequest.statusReason | Added Element

CommunicationRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1

CommunicationRequest.doNotPerform | Added Element

CommunicationRequest.about | Added Element

CommunicationRequest.encounter | Added Element

CommunicationRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)

CommunicationRequest.recipient | Type Reference: Added Target Types PractitionerRole, HealthcareService

CommunicationRequest.sender | Type Reference: Added Target Types PractitionerRole, HealthcareService

CommunicationRequest.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference

CommunicationRequest.topic | deleted

CommunicationRequest.context | deleted

CommunicationRequest.requester.agent | deleted

CommunicationRequest.requester.onBehalfOf | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

CommunicationRequest | TU | | DomainResource | A request for information to be sent to a receiver Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Unique identifier

basedOn | Σ | 0..* | Reference(Any) | Fulfills plan or proposal

replaces | Σ | 0..* | Reference(CommunicationRequest) | Request(s) replaced by this request

groupIdentifier | Σ | 0..1 | Identifier | Composite request this is part of

status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)

statusReason | | 0..1 | CodeableConcept | Reason for current status

category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if request is prohibiting action
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | | 0..1 | Reference(Patient | Group) | Focus of message
about | | 0..* | Reference(Any) | Resources that pertain to this communication request
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
occurrence[x] | Σ | 0..1 | | When scheduled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
authoredOn | Σ | 0..1 | dateTime | When request transitioned to being actionable
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/what is requesting service
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService) | Message recipient
sender | Σ | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService) | Message sender
reasonCode | Σ | 0..* | CodeableConcept | Why is communication needed? v3 Code System ActReason (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why is communication needed?
note | | 0..* | Annotation | Comments made about communication request
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CommunicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <basedOn><!-- 0..* Reference(Any) Fulfills plan or proposal --></basedOn>
 <replaces><!-- 0..* Reference(CommunicationRequest) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..* CodeableConcept Message category --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if request is prohibiting action -->
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <about><!-- 0..* Reference(Any) Resources that pertain to this communication request --></about>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <occurrence[x]><!-- 0..1 dateTime|Period When scheduled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request transitioned to being actionable -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Who/what is requesting service --></requester>
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) Message recipient --></recipient>
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) Message sender --></sender>
 <reasonCode><!-- 0..* CodeableConcept Why is communication needed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why is communication needed? --></reasonReference>
 <note><!-- 0..* Annotation Comments made about communication request --></note>
</CommunicationRequest>
```

JSON Template

```
{
  "resourceType" : "CommunicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "basedOn" : [{ Reference(Any) }], // Fulfills plan or proposal
  "replaces" : [{ Reference(CommunicationRequest) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "category" : [{ CodeableConcept }], // Message category
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if request is prohibiting action
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "about" : [{ Reference(Any) }], // Resources that pertain to this communication request
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "payload" : [{ // Message payload
```

```
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  // occurrence[x]: When scheduled. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "authoredOn" : "<dateTime>", // When request transitioned to being actionable
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Who/what is requesting service
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService) }], // Message recipient
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson|HealthcareService) }, // Message sender
  "reasonCode" : [{ CodeableConcept }], // Why is communication needed?
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why is communication needed?
  "note" : [{ Annotation }] // Comments made about communication request
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CommunicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CommunicationRequest.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:CommunicationRequest.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan or proposal
  fhir:CommunicationRequest.replaces [ Reference(CommunicationRequest) ], ... ; # 0..* Request(s) replaced by this
    request
  fhir:CommunicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:CommunicationRequest.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error |
    unknown
  fhir:CommunicationRequest.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:CommunicationRequest.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:CommunicationRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:CommunicationRequest.doNotPerform [ boolean ]; # 0..1 True if request is prohibiting action
  fhir:CommunicationRequest.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:CommunicationRequest.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:CommunicationRequest.about [ Reference(Any) ], ... ; # 0..* Resources that pertain to this communication request
  fhir:CommunicationRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:CommunicationRequest.payload [ # 0..* Message payload
    # CommunicationRequest.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:CommunicationRequest.payload.contentString [ string ]
      fhir:CommunicationRequest.payload.contentAttachment [ Attachment ]
      fhir:CommunicationRequest.payload.contentReference [ Reference(Any) ]
  ], ...;
  # CommunicationRequest.occurrence[x] : 0..1 When scheduled. One of these 2
    fhir:CommunicationRequest.occurrenceDateTime [ dateTime ]
    fhir:CommunicationRequest.occurrencePeriod [ Period ]
  fhir:CommunicationRequest.authoredOn [ dateTime ]; # 0..1 When request transitioned to being actionable
  fhir:CommunicationRequest.requester [
    Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device) ]; # 0..1 Who/what is requesting
    service
  fhir:CommunicationRequest.recipient [
    Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|Group|
  CareTeam|HealthcareService) ], ... ; # 0..* Message recipient
  fhir:CommunicationRequest.sender [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|
  HealthcareService) ]; # 0..1 Message sender
  fhir:CommunicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Why is communication needed?
  fhir:CommunicationRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ...
     ; # 0..* Why is communication needed?
  fhir:CommunicationRequest.note [ Annotation ], ... ; # 0..* Comments made about communication request
]
```

Changes since Release 3

CommunicationRequest

CommunicationRequest.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1

CommunicationRequest.statusReason | Added Element

CommunicationRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1

CommunicationRequest.doNotPerform | Added Element

CommunicationRequest.about | Added Element

CommunicationRequest.encounter | Added Element

CommunicationRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)

CommunicationRequest.recipient | Type Reference: Added Target Types PractitionerRole, HealthcareService

CommunicationRequest.sender | Type Reference: Added Target Types PractitionerRole, HealthcareService

CommunicationRequest.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference

CommunicationRequest.topic | deleted

CommunicationRequest.context | deleted
CommunicationRequest.requester.agent | deleted
CommunicationRequest.requester.onBehalfOf | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.21.3.1 Terminology Bindings

Path | Definition | Type | Reference
CommunicationRequest.status | The status of the communication request. | Required | RequestStatus
CommunicationRequest.statusReason | Codes identifying the reason for the current state of a request. | Unknown | No details provided yet
CommunicationRequest.category | Codes for general categories of communications such as alerts, instruction, etc. | Example | CommunicationCategory
CommunicationRequest.priority | Codes indicating the relative importance of a communication request. | Required | RequestPriority
CommunicationRequest.medium | Codes for communication mediums such as phone, fax, email, in person, etc. | Example | v3.ParticipationMode
CommunicationRequest.reasonCode | Codes for describing reasons for the occurrence of a communication. | Example | v3.ActReason

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

8.21.3.2 CommunicationRequest.sender and CommunicationRequest.recepient

CommunicationRequest.sender allows Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService and CommunicationRequest.recipient allows Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService - but it is not unusual to have a communication target - even a defined one - where it is unknown what kind of role the person is playing.

If the communication request is to or from an individual, whose role is not known (practitioner, patient or related person), - for example, only email address is captured in the system; then RelatedPerson should be used by default.

8.21.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored | date | When request transitioned to being actionable | CommunicationRequest.authoredOn
based-on | reference | Fulfills plan or proposal | CommunicationRequest.basedOn (Any)
category | token | Message category | CommunicationRequest.category
encounter | reference | Encounter created as part of | CommunicationRequest.encounter (Encounter)
group-identifier | token | Composite request this is part of | CommunicationRequest.groupIdentifier
identifier | token | Unique identifier | CommunicationRequest.identifier
medium | token | A channel of communication | CommunicationRequest.medium
occurrence | date | When scheduled | (CommunicationRequest.occurrence as dateTime)
patient | reference | Focus of message | CommunicationRequest.subject.where(resolve() is Patient) (Patient)
priority | token | routine | urgent | asap | stat | CommunicationRequest.priority
recipient | reference | Message recipient | CommunicationRequest.recipient (Practitioner, Group, Organization, CareTeam, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
replaces | reference | Request(s) replaced by this request | CommunicationRequest.replaces (CommunicationRequest)
requester | reference | Who/what is requesting service | CommunicationRequest.requester (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
sender | reference | Message sender | CommunicationRequest.sender (Practitioner, Organization, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
status | token | draft | active | on-hold | revoked | completed | entered-in-error | unknown | CommunicationRequest.status
subject | reference | Focus of message | CommunicationRequest.subject (Group, Patient)

8.21.5 Resource CommunicationRequest - Examples

URL: https://hl7.org/fhir/R4/communicationrequest-examples.html

8.21.5 Resource CommunicationRequest - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An introduction to this page as well as proper examples are to be submitted shortly.

Example Name | id | Format
An example of an alert message | example | XML | JSON | Turtle
An example of a request for additional information | fm-solicit | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.21.6 Resource CommunicationRequest - Detailed Descriptions

URL: https://hl7.org/fhir/R4/communicationrequest-definitions.html

8.21.6 Resource CommunicationRequest - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the CommunicationRequest resource.

CommunicationRequest
Element Id | CommunicationRequest
Definition | A request to convey information; e.g. the CDS system proposes that an alert be sent to a responsible provider, the CDS system proposes that the public health agency be notified about a reportable condition.
Cardinality | 0..*
Type | DomainResource
CommunicationRequest.identifier
Element Id | CommunicationRequest.identifier
Definition | Business identifiers assigned to this communication request by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the communication request as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
CommunicationRequest.basedOn
Element Id | CommunicationRequest.basedOn
Definition | A plan or proposal that is fulfilled in whole or in part by this request.
Cardinality | 0..*
Type | Reference(Any)
Requirements | Allows tracing of authorization for the request and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
Summary | true
CommunicationRequest.replaces
Element Id | CommunicationRequest.replaces
Definition | Completed or terminated request(s) whose function is taken by this new request.
Cardinality | 0..*
Type | Reference(CommunicationRequest)
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple requests.
Alternate Names | supersedes; prior
Summary | true
Comments | The replacement could be because the initial request was immediately rejected (due to an issue) or because the previous request was completed, but the need for the action described by the request remains ongoing.
CommunicationRequest.groupIdentifier
Element Id | CommunicationRequest.groupIdentifier
Definition | A shared identifier common to all requests that were authorized more or less simultaneously by a single author, representing the identifier of the requisition, prescription or similar form.
Cardinality | 0..1
Type | Identifier
Alternate Names | grouperId
Summary | true
Comments | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
CommunicationRequest.status
Element Id | CommunicationRequest.status
Definition | The status of the proposal or order.
Cardinality | 1..1
Terminology Binding | RequestStatus (Required)
Type | code

Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid.)
Summary | true
CommunicationRequest.statusReason
Element Id | CommunicationRequest.statusReason
Definition | Captures the reason for the current state of the CommunicationRequest.
Cardinality | 0..1
Terminology Binding | CommunicationRequestStatusReason:
Type | CodeableConcept
Alternate Names | Suspended Reason; Cancelled Reason
Comments | This is generally only used for "exception" statuses such as "suspended" or "cancelled". The reason why the CommunicationRequest was created at all is captured in reasonCode, not here. [distinct reason codes for different statuses can be enforced using invariants if they are universal bindings].
CommunicationRequest.category
Element Id | CommunicationRequest.category
Definition | The type of message to be sent such as alert, notification, reminder, instruction, etc.
Cardinality | 0..*
Terminology Binding | CommunicationCategory (Example)
Type | CodeableConcept
Comments | There may be multiple axes of categorization and one communication request may serve multiple purposes.
CommunicationRequest.priority
Element Id | CommunicationRequest.priority
Definition | Characterizes how quickly the proposed act must be initiated. Includes concepts such as stat, urgent, routine.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Summary | true
CommunicationRequest.doNotPerform
Element Id | CommunicationRequest.doNotPerform
Definition | If true indicates that the CommunicationRequest is asking for the specified action to not occur.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: If true this element negates the specified action. For Example, instead of a request for a communication, it is a request for the communication to not occur.)
Meaning if Missing | If do not perform is not specified, the request is a positive request e.g. "do perform"
Alternate Names | prohibited
Summary | true
Comments | The attributes provided with the request qualify what is not to be done.
CommunicationRequest.medium
Element Id | CommunicationRequest.medium
Definition | A channel that was used for this communication (e.g. email, fax).
Cardinality | 0..*
Terminology Binding | v3 Code System ParticipationMode (Example)
Type | CodeableConcept
CommunicationRequest.subject
Element Id | CommunicationRequest.subject
Definition | The patient or group that is the focus of this communication request.
Cardinality | 0..1
Type | Reference(Patient | Group)
Alternate Names | patient
CommunicationRequest.about
Element Id | CommunicationRequest.about
Definition | Other resources that pertain to this communication request and to which this communication request should be associated.
Cardinality | 0..*
Type | Reference(Any)
Comments | Don't use CommunicationRequest.about element when a more specific element exists, such as basedOn, reasonReference, or replaces.
CommunicationRequest.encounter
Element Id | CommunicationRequest.encounter
Definition | The Encounter during which this CommunicationRequest was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter.
CommunicationRequest.payload
Element Id | CommunicationRequest.payload
Definition | Text, attachment(s), or resource(s) to be communicated to the recipient.
Cardinality | 0..*
CommunicationRequest.payload.content[x]

Element Id | CommunicationRequest.payload.content[x]
Definition | The communicated content (or for multi-part communications, one portion of the communication).
Cardinality | 1..1
Type | string|Attachment|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
CommunicationRequest.occurrence[x]
Element Id | CommunicationRequest.occurrence[x]
Definition | The time when this communication is to occur.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | timing
Summary | true
CommunicationRequest.authoredOn
Element Id | CommunicationRequest.authoredOn
Definition | For draft requests, indicates the date of initial creation. For requests with other statuses, indicates the date of activation.
Cardinality | 0..1
Type | dateTime
Alternate Names | createdOn
Summary | true
CommunicationRequest.requester
Element Id | CommunicationRequest.requester
Definition | The device, individual, or organization who initiated the request and has responsibility for its activation.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
Alternate Names | author
Summary | true
CommunicationRequest.recipient
Element Id | CommunicationRequest.recipient
Definition | The entity (e.g. person, organization, clinical information system, device, group, or care team) which is the intended target of the communication.
Cardinality | 0..*
Type | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService)
CommunicationRequest.sender
Element Id | CommunicationRequest.sender
Definition | The entity (e.g. person, organization, clinical information system, or device) which is to be the source of the communication.
Cardinality | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService)
Summary | true
CommunicationRequest.reasonCode
Element Id | CommunicationRequest.reasonCode
Definition | Describes why the request is being made in coded or textual form.
Cardinality | 0..*
Terminology Binding | v3 Code System ActReason (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be captured using reasonCode.text.
CommunicationRequest.reasonReference
Element Id | CommunicationRequest.reasonReference
Definition | Indicates another resource whose existence justifies this request.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
Summary | true
CommunicationRequest.note
Element Id | CommunicationRequest.note
Definition | Comments made about the request by the requester, sender, recipient, subject or other participants.
Cardinality | 0..*
Type | Annotation

8.21.7 Resource CommunicationRequest - Mappings

URL: https://hl7.org/fhir/R4/communicationrequest-mappings.html

8.21.7 Resource CommunicationRequest - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the communicationrequest resource (see Mappings to Other Standards for further information & status).

8.21.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

CommunicationRequest
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
priority | FiveWs.grade
subject | FiveWs.subject[x]
about | FiveWs.context
encounter | FiveWs.context
occurrence[x] | FiveWs.planned
authoredOn | FiveWs.recorded
requester | FiveWs.author
recipient | FiveWs.actor
sender | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

8.21.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

CommunicationRequest | Request
identifier | Request.identifier
basedOn | Request.basedOn
replaces | Request.replaces
groupIdentifier | Request.groupIdentifier
status | Request.status
statusReason | Request.statusReason
priority | Request.priority
doNotPerform | Request.doNotPerform
subject | Request.subject
about | Request.supportingInfo
encounter | Request.context
content[x] | Request.note
occurrence[x] | Request.occurrence[x]
authoredOn | Request.authoredOn
requester | Request.requester
recipient | Request.performer
sender | Request.performer
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
note | Request.note

8.21.7.3 RIM Mapping (http://hl7.org/v3)

CommunicationRequest
basedOn | .outboundRelationship[typeCode=FLFS].target
replaces | .outboundRelationship[typeCode=RPLC].target
groupIdentifier | .inboundRelationship(typeCode=COMP].source[moodCode=INT].identifier
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN].reasonCOde
requester | .participation[typeCode=AUT].role
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]

8.21.8 Resource CommunicationRequest - Extensions & Profiles

URL: https://hl7.org/fhir/R4/communicationrequest-profiles.html

8.21.8 Resource CommunicationRequest - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the CommunicationRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.21.8.1 Profiles

No Profiles defined for this resource

8.21.8.2 Extensions

communicationrequest-initiatingLocation | CommunicationRequest | initiatingLocation | for CommunicationRequest HL7 Extensions
request-relevantHistory | CommunicationRequest | relevantHistory | for Request Pattern HL7 Extensions
workflow-episodeOfCare | CommunicationRequest | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-instantiatesCanonical | CommunicationRequest | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | CommunicationRequest | instantiatesUri | for Workflow Pattern HL7 Extensions
workflow-supportingInfo | CommunicationRequest | supportingInfo | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.21.8.3 Search Extensions

No Search Extensions defined for this resource

8.21.9 Resource CommunicationRequest - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/communicationrequest-version-maps.html

No validation errors - all conversions are clean

8.21.9 Resource CommunicationRequest - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for CommunicationRequest.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

8.21.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CommunicationRequest3to4" = "R3 to R4 Conversions for CommunicationRequest"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CommunicationRequest" alias CommunicationRequestR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CommunicationRequest" alias CommunicationRequest as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CommunicationRequest(source src : CommunicationRequestR3, target tgt : CommunicationRequest) extends
    DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.replaces -> tgt.replaces;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.medium -> tgt.medium;
  src.subject -> tgt.subject;
  src.recipient -> tgt.recipient;
  src.topic -> tgt.about;
  src.context -> tgt.encounter;
  src.payload as s -> tgt.payload as t then payload(s, t);
  src.occurrence : dateTime as s -> tgt.occurrence = create('dateTime') as t then dateTime(s, t);
  src.occurrence : Period as s -> tgt.occurrence = create('Period') as t then Period(s, t);
  src.authoredOn -> tgt.authoredOn;
  src.sender -> tgt.sender;
  src.requester as vs0 -> tgt then agent(vs0, tgt);
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
}

group agent(source src, target tgt) {
  src.agent -> tgt.requester;
  src.onBehalfOf as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CommunicationRequest.requester.onBehalfOf',  vt.value = v;
```

```
}

group payload(source src, target tgt) extends BackboneElement {
  src.content : string as s -> tgt.content = create('string') as t then string(s, t);
  src.content : Attachment as s -> tgt.content = create('Attachment') as t then Attachment(s, t);
  src.content : Reference as s -> tgt.content = create('Reference') as t then Reference(s, t);
}
```

8.21.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CommunicationRequest4to3" = "R4 to R3 Conversion for CommunicationRequest"

uses "http://hl7.org/fhir/StructureDefinition/CommunicationRequest" alias CommunicationRequest as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CommunicationRequest" alias CommunicationRequestR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group CommunicationRequest(source src : CommunicationRequest, target tgt : CommunicationRequestR3) extends
    DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.replaces -> tgt.replaces;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.medium -> tgt.medium;
  src.subject -> tgt.subject;
  src.recipient -> tgt.recipient;
  src.about -> tgt.topic;
  src.encounter -> tgt.context;
  src.payload as s -> tgt.payload as t then payload(s, t);
  src.occurrence : dateTime as s -> tgt.occurrence = create('dateTime') as t then dateTime(s, t);
  src.occurrence : Period as s -> tgt.occurrence = create('Period') as t then Period(s, t);
  src.authoredOn -> tgt.authoredOn;
  src.sender -> tgt.sender;
  src where requester.exists() or extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CommunicationRequest.requester.onBehalfOf').exists() -> tgt.requester as vt0 then agent(src, vt0) "requester";
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
}

group agent(source src, target tgt) {
  src.requester -> tgt.agent;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CommunicationRequest.requester.onBehalfOf' then {
    e.value -> tgt.onBehalfOf;
  };
}

group payload(source src, target tgt) extends BackboneElement {
  src.content : string as s -> tgt.content = create('string') as t then string(s, t);
  src.content : Attachment as s -> tgt.content = create('Attachment') as t then Attachment(s, t);
  src.content : Reference as s -> tgt.content = create('Reference') as t then Reference(s, t);
}
```

8.21.9.3 R4 Validation Errors

8.21.10 CommunicationRequest HL7 Extensions

URL: https://hl7.org/fhir/R4/communicationrequest-extensions.html

Defines common extensions used with or related to the CommunicationRequest resource

8.21.10 CommunicationRequest HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

8.21.10.1 Content

Extensions:

communicationrequest-initiatingLocation | initiatingLocation : Location where the information being requested to be communicated happened.

8.21.11 JSON Schema for CommunicationRequest

URL: https://hl7.org/fhir/R4/communicationrequest.schema.json.html

8.21.11 JSON Schema for CommunicationRequest

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for CommunicationRequest, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CommunicationRequest",
  "$ref": "#/definitions/CommunicationRequest",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CommunicationRequest": {
      "description": "A request to convey information; e.g. the CDS system proposes that an alert be sent to a
    responsible provider, the CDS system proposes that the public health agency be notified about a reportable
    condition.",
      "properties": {
        "resourceType": {
          "description": "This is a CommunicationRequest resource",
          "const": "CommunicationRequest"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "Business identifiers assigned to this communication request by the performer or other systems
    which remain constant as the resource is updated and propagates from server to server.",
          "items": {
            "$ref": "Identifier.schema.json#/definitions/Identifier"
          },
          "type": "array"
```

```
    },
    "basedOn": {
      "description": "A plan or proposal that is fulfilled in whole or in part by this request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "replaces": {
      "description": "Completed or terminated request(s) whose function is taken by this new request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "groupIdentifier": {
      "description": "A shared identifier common to all requests that were authorized more or less simultaneously by
 a single author, representing the identifier of the requisition, prescription or similar form.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "status": {
      "description": "The status of the proposal or order.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Captures the reason for the current state of the CommunicationRequest.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "The type of message to be sent such as alert, notification, reminder, instruction, etc.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "priority": {
      "description": "Characterizes how quickly the proposed act must be initiated. Includes concepts such as stat,
urgent, routine.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doNotPerform": {
      "description": "If true indicates that the CommunicationRequest is asking for the specified action to *not*
occur.",
      "$ref": "#/definitions/boolean"
    },
    "_doNotPerform": {
      "description": "Extensions for doNotPerform",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "medium": {
      "description": "A channel that was used for this communication (e.g. email, fax).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "subject": {
      "description": "The patient or group that is the focus of this communication request.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "about": {
      "description": "Other resources that pertain to this communication request and to which this communication
request should be associated.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "encounter": {
      "description": "The Encounter during which this CommunicationRequest was created or to which the creation of
this record is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "payload": {
      "description": "Text, attachment(s), or resource(s) to be communicated to the recipient.",
      "items": {
        "$ref": "#/definitions/CommunicationRequest_Payload"
      },
      "type": "array"
    },
    "occurrenceDateTime": {
```

```
      "description": "The time when this communication is to occur.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "The time when this communication is to occur.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "authoredOn": {
      "description": "For draft requests, indicates the date of initial creation.  For requests with other statuses,
 indicates the date of activation.",
      "$ref": "#/definitions/dateTime"
    },
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requester": {
      "description": "The device, individual, or organization who initiated the request and has responsibility for
its activation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "recipient": {
      "description": "The entity (e.g. person, organization, clinical information system, device, group, or care
team) which is the intended target of the communication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "sender": {
      "description": "The entity (e.g. person, organization, clinical information system, or device) which is to be
the source of the communication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "Describes why the request is being made in coded or textual form.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates another resource whose existence justifies this request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Comments made about the request by the requester, sender, recipient, subject or other
participants.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"CommunicationRequest_Payload": {
  "description": "A request to convey information; e.g. the CDS system proposes that an alert be sent to a
responsible provider, the CDS system proposes that the public health agency be notified about a reportable
condition.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "contentString": {
      "description": "The communicated content (or for multi-part communications, one portion of the
    communication).",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_contentString": {
      "description": "Extensions for contentString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contentAttachment": {
      "description": "The communicated content (or for multi-part communications, one portion of the
    communication).",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "contentReference": {
      "description": "The communicated content (or for multi-part communications, one portion of the
    communication).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
 }
 }
}
```


8.21.12 Resource CommunicationRequest - Maturity Dependency View

URL: https://hl7.org/fhir/R4/communicationrequest-dependencies.html

8.21.12 Resource CommunicationRequest - Maturity Dependency View

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for CommunicationRequest FMM level 2

Name | Card. | Type | Dependency Analysis
CommunicationRequest | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
replaces | 0..* | Reference(CommunicationRequest)
groupIdentifier | 0..1 | Identifier
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
category | 0..* | CodeableConcept
priority | 0..1 | code
doNotPerform | 0..1 | boolean
medium | 0..* | CodeableConcept
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
about | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
payload | 0..* | BackboneElement
content[x] | 1..1
contentString | | string
contentAttachment | | Attachment
contentReference | | Reference(Any
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
authoredOn | 0..1 | dateTime
requester | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
recipient | 0..* | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | Group | CareTeam | HealthcareService) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
sender | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson | HealthcareService)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
note | 0..* | Annotation

8.22 Resource ResearchStudy - Content

URL: https://hl7.org/fhir/R4/researchstudy.html
Biomedical Research and Regulation Work Group | Maturity Level: 1 | Trial Use | Security Category: Business | Compartments: Practitioner

A process where a researcher or organization plans and then executes a series of steps intended to increase the field of healthcare-related knowledge. This includes studies of safety, efficacy, comparative effectiveness and other information about medications, devices, therapies and other interventional and investigative techniques. A ResearchStudy involves the gathering of information about human or animal subjects.

8.22.1 Scope and Usage

This resource is an definition resource from a FHIR workflow perspective - see Workflow.

A research study is a scientific way to improve or develop new methods of health care. Studies are designed to answer specific questions on how to prevent, diagnose, or treat diseases and disorders. The ResearchStudy resource describes essential information about the study, including the purpose, objective, sponsor, investigator, therapy, condition being studied, schedule of activities, and other key items.

Uses for the ResearchStudy resource include:

- Registering a clinical trial in trial registry such as clinicaltrials.gov
- Communicating study protocol information between study stakeholders
- Supporting study set up of sites, forms
- Grouping Observation resources by study
- Setting up a Schedule of Activities for the study and its subjects

This resource is referenced by UsageContext, AdverseEvent, MedicinalProduct, itself and ResearchSubject

8.22.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchStudy | TU | | DomainResource | Investigation to increase healthcare-related patient-independent knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for study
title | Σ | 0..1 | string | Name for this study
protocol | Σ | 0..* | Reference(PlanDefinition) | Steps followed in executing study
partOf | Σ | 0..* | Reference(ResearchStudy) | Part of larger study
status | ?!Σ | 1..1 | code | active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-accrual-and-intervention | withdrawn ResearchStudyStatus (Required)
primaryPurposeType | Σ | 0..1 | CodeableConcept | treatment | prevention | diagnostic | supportive-care | screening | health-services-research | basic-science | device-feasibility ResearchStudyPrimaryPurposeType (Extensible)
phase | Σ | 0..1 | CodeableConcept | n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 | phase-2-phase-3 | phase-3 | phase-4 ResearchStudyPhase (Example)
category | Σ | 0..* | CodeableConcept | Classifications for the study
focus | Σ | 0..* | CodeableConcept | Drugs, devices, etc. under study
condition | Σ | 0..* | CodeableConcept | Condition being studied Condition/Problem/Diagnosis Codes (Example)
contact | Σ | 0..* | ContactDetail | Contact details for the study
relatedArtifact | | 0..* | RelatedArtifact | References and dependencies
keyword | Σ | 0..* | CodeableConcept | Used to search for the study
location | Σ | 0..* | CodeableConcept | Geographic region(s) for study Jurisdiction (Extensible)
description | | 0..1 | markdown | What this is study doing
enrollment | Σ | 0..* | Reference(Group) | Inclusion & exclusion criteria
period | Σ | 0..1 | Period | When the study began and ended
sponsor | Σ | 0..1 | Reference(Organization) | Organization that initiates and is legally responsible for the study
principalInvestigator | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Researcher who oversees multiple aspects of the study
site | Σ | 0..* | Reference(Location) | Facility where study activities are conducted
reasonStopped | Σ | 0..1 | CodeableConcept | accrual-goal-met | closed-due-to-toxicity | closed-due-to-lack-of-study-progress | temporarily-closed-per-study-design ResearchStudyReasonStopped (Example)
note | | 0..* | Annotation | Comments made about the study

arm | | 0..* | BackboneElement | Defined path through the study for a subject
name | | 1..1 | string | Label for study arm
type | | 0..1 | CodeableConcept | Categorization of study arm
description | | 0..1 | string | Short explanation of study path
objective | | 0..* | BackboneElement | A goal for the study
name | | 0..1 | string | Label for the objective
type | | 0..1 | CodeableConcept | primary | secondary | exploratory ResearchStudyObjectiveType (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for study --></identifier>
 <title value="[string]"/><!-- 0..1 Name for this study -->
 <protocol><!-- 0..* Reference(PlanDefinition) Steps followed in executing study --></protocol>
 <partOf><!-- 0..* Reference(ResearchStudy) Part of larger study --></partOf>
 <status value="[code]"/><!-- 1..1 active | administratively-completed | approved | closed-to-accrual | closed-to-
    accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-
    to-accrual-and-intervention | withdrawn -->
 <primaryPurposeType><!-- 0..1 CodeableConcept treatment | prevention | diagnostic | supportive-care | screening |
    health-services-research | basic-science | device-feasibility --></primaryPurposeType>
 <phase><!-- 0..1 CodeableConcept n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 | phase-2-phase-3 | phase-3
    | phase-4 --></phase>
 <category><!-- 0..* CodeableConcept Classifications for the study --></category>
 <focus><!-- 0..* CodeableConcept Drugs, devices, etc. under study --></focus>
 <condition><!-- 0..* CodeableConcept Condition being studied --></condition>
 <contact><!-- 0..* ContactDetail Contact details for the study --></contact>
 <relatedArtifact><!-- 0..* RelatedArtifact References and dependencies --></relatedArtifact>
 <keyword><!-- 0..* CodeableConcept Used to search for the study --></keyword>
 <location><!-- 0..* CodeableConcept Geographic region(s) for study --></location>
 <description value="[markdown]"/><!-- 0..1 What this is study doing -->
 <enrollment><!-- 0..* Reference(Group) Inclusion & exclusion criteria --></enrollment>
 <period><!-- 0..1 Period When the study began and ended --></period>
 <sponsor><!-- 0..1 Reference(Organization) Organization that initiates and is legally responsible for the study
    --></sponsor>
 <principalInvestigator><!-- 0..1 Reference(Practitioner|PractitionerRole) Researcher who oversees multiple aspects of
    the study --></principalInvestigator>
 <site><!-- 0..* Reference(Location) Facility where study activities are conducted --></site>
 <reasonStopped><!-- 0..1 CodeableConcept accrual-goal-met | closed-due-to-toxicity | closed-due-to-lack-of-study-
    progress | temporarily-closed-per-study-design --></reasonStopped>
 <note><!-- 0..* Annotation Comments made about the study --></note>
 <arm>  <!-- 0..* Defined path through the study for a subject -->
  <name value="[string]"/><!-- 1..1 Label for study arm -->
  <type><!-- 0..1 CodeableConcept Categorization of study arm --></type>
  <description value="[string]"/><!-- 0..1 Short explanation of study path -->
 </arm>
 <objective>  <!-- 0..* A goal for the study -->
  <name value="[string]"/><!-- 0..1 Label for the objective -->
  <type><!-- 0..1 CodeableConcept primary | secondary | exploratory --></type>
 </objective>
</ResearchStudy>
```

JSON Template

```
{
  "resourceType" : "ResearchStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for study
  "title" : "<string>", // Name for this study
  "protocol" : [{ Reference(PlanDefinition) }], // Steps followed in executing study
  "partOf" : [{ Reference(ResearchStudy) }], // Part of larger study
  "status" : "<code>", // R!  active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-
    and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-
    accrual-and-intervention | withdrawn
  "primaryPurposeType" : { CodeableConcept }, // treatment | prevention | diagnostic | supportive-care | screening |
    health-services-research | basic-science | device-feasibility
  "phase" : { CodeableConcept }, // n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 | phase-2-phase-3 |
    phase-3 | phase-4
  "category" : [{ CodeableConcept }], // Classifications for the study
  "focus" : [{ CodeableConcept }], // Drugs, devices, etc. under study
  "condition" : [{ CodeableConcept }], // Condition being studied
  "contact" : [{ ContactDetail }], // Contact details for the study
  "relatedArtifact" : [{ RelatedArtifact }], // References and dependencies
  "keyword" : [{ CodeableConcept }], // Used to search for the study
  "location" : [{ CodeableConcept }], // Geographic region(s) for study
  "description" : "<markdown>", // What this is study doing
  "enrollment" : [{ Reference(Group) }], // Inclusion & exclusion criteria
  "period" : { Period }, // When the study began and ended
  "sponsor" : { Reference(Organization) }, // Organization that initiates and is legally responsible for the study
  "principalInvestigator" : { Reference(Practitioner|PractitionerRole) }, // Researcher who oversees multiple aspects of
     the study
  "site" : [{ Reference(Location) }], // Facility where study activities are conducted
  "reasonStopped" : { CodeableConcept }, // accrual-goal-met | closed-due-to-toxicity | closed-due-to-lack-of-study-
    progress | temporarily-closed-per-study-design
```

```
  "note" : [{ Annotation }], // Comments made about the study
  "arm" : [{ // Defined path through the study for a subject
    "name" : "<string>", // R!  Label for study arm
    "type" : { CodeableConcept }, // Categorization of study arm
    "description" : "<string>" // Short explanation of study path
  }],
  "objective" : [{ // A goal for the study
    "name" : "<string>", // Label for the objective
    "type" : { CodeableConcept } // primary | secondary | exploratory
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchStudy.identifier [ Identifier ], ... ; # 0..* Business Identifier for study
  fhir:ResearchStudy.title [ string ]; # 0..1 Name for this study
  fhir:ResearchStudy.protocol [ Reference(PlanDefinition) ], ... ; # 0..* Steps followed in executing study
  fhir:ResearchStudy.partOf [ Reference(ResearchStudy) ], ... ; # 0..* Part of larger study
  fhir:ResearchStudy.status [ code ]; # 1..1 active | administratively-completed | approved | closed-to-accrual |
    closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual |
    temporarily-closed-to-accrual-and-intervention | withdrawn
  fhir:ResearchStudy.primaryPurposeType [ CodeableConcept ]; # 0..1 treatment | prevention | diagnostic | supportive-
    care | screening | health-services-research | basic-science | device-feasibility
  fhir:ResearchStudy.phase [ CodeableConcept ]; # 0..1 n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 |
    phase-2-phase-3 | phase-3 | phase-4
  fhir:ResearchStudy.category [ CodeableConcept ], ... ; # 0..* Classifications for the study
  fhir:ResearchStudy.focus [ CodeableConcept ], ... ; # 0..* Drugs, devices, etc. under study
  fhir:ResearchStudy.condition [ CodeableConcept ], ... ; # 0..* Condition being studied
  fhir:ResearchStudy.contact [ ContactDetail ], ... ; # 0..* Contact details for the study
  fhir:ResearchStudy.relatedArtifact [ RelatedArtifact ], ... ; # 0..* References and dependencies
  fhir:ResearchStudy.keyword [ CodeableConcept ], ... ; # 0..* Used to search for the study
  fhir:ResearchStudy.location [ CodeableConcept ], ... ; # 0..* Geographic region(s) for study
  fhir:ResearchStudy.description [ markdown ]; # 0..1 What this is study doing
  fhir:ResearchStudy.enrollment [ Reference(Group) ], ... ; # 0..* Inclusion & exclusion criteria
  fhir:ResearchStudy.period [ Period ]; # 0..1 When the study began and ended
  fhir:ResearchStudy.sponsor [ Reference(Organization) ]; # 0..1 Organization that initiates and is legally responsible
    for the study
  fhir:ResearchStudy.principalInvestigator [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Researcher who oversees
    multiple aspects of the study
  fhir:ResearchStudy.site [ Reference(Location) ], ... ; # 0..* Facility where study activities are conducted
  fhir:ResearchStudy.reasonStopped [ CodeableConcept ]; # 0..1 accrual-goal-met | closed-due-to-toxicity | closed-due-
    to-lack-of-study-progress | temporarily-closed-per-study-design
  fhir:ResearchStudy.note [ Annotation ], ... ; # 0..* Comments made about the study
  fhir:ResearchStudy.arm [ # 0..* Defined path through the study for a subject
    fhir:ResearchStudy.arm.name [ string ]; # 1..1 Label for study arm
    fhir:ResearchStudy.arm.type [ CodeableConcept ]; # 0..1 Categorization of study arm
    fhir:ResearchStudy.arm.description [ string ]; # 0..1 Short explanation of study path
  ], ...;
  fhir:ResearchStudy.objective [ # 0..* A goal for the study
    fhir:ResearchStudy.objective.name [ string ]; # 0..1 Label for the objective
    fhir:ResearchStudy.objective.type [ CodeableConcept ]; # 0..1 primary | secondary | exploratory
  ], ...;
]
```

Changes since R3

ResearchStudy
ResearchStudy.status | Change value set from http://hl7.org/fhir/ValueSet/research-study-status to http://hl7.org/fhir/ValueSet/research-study-status|4.0.1
ResearchStudy.primaryPurposeType | Added Element
ResearchStudy.phase | Added Element
ResearchStudy.condition | Added Element
ResearchStudy.location | Added Element
ResearchStudy.principalInvestigator | Type Reference: Added Target Type PractitionerRole
ResearchStudy.arm.type | Added Element
ResearchStudy.objective | Added Element
ResearchStudy.objective.name | Added Element
ResearchStudy.objective.type | Added Element
ResearchStudy.jurisdiction | deleted
ResearchStudy.arm.code | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

ResearchStudy | TU | | DomainResource | Investigation to increase healthcare-related patient-independent knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for study
title | Σ | 0..1 | string | Name for this study
protocol | Σ | 0..* | Reference(PlanDefinition) | Steps followed in executing study
partOf | Σ | 0..* | Reference(ResearchStudy) | Part of larger study
status | ?!Σ | 1..1 | code | active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-accrual-and-intervention | withdrawn ResearchStudyStatus (Required)
primaryPurposeType | Σ | 0..1 | CodeableConcept | treatment | prevention | diagnostic | supportive-care | screening | health-services-research | basic-science | device-feasibility ResearchStudyPrimaryPurposeType (Extensible)
phase | Σ | 0..1 | CodeableConcept | n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 | phase-2-phase-3 | phase-3 | phase-4 ResearchStudyPhase (Example)
category | Σ | 0..* | CodeableConcept | Classifications for the study
focus | Σ | 0..* | CodeableConcept | Drugs, devices, etc. under study
condition | Σ | 0..* | CodeableConcept | Condition being studied Condition/Problem/Diagnosis Codes (Example)
contact | Σ | 0..* | ContactDetail | Contact details for the study
relatedArtifact | | 0..* | RelatedArtifact | References and dependencies
keyword | Σ | 0..* | CodeableConcept | Used to search for the study
location | Σ | 0..* | CodeableConcept | Geographic region(s) for study Jurisdiction (Extensible)
description | | 0..1 | markdown | What this is study doing
enrollment | Σ | 0..* | Reference(Group) | Inclusion & exclusion criteria
period | Σ | 0..1 | Period | When the study began and ended
sponsor | Σ | 0..1 | Reference(Organization) | Organization that initiates and is legally responsible for the study
principalInvestigator | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Researcher who oversees multiple aspects of the study
site | Σ | 0..* | Reference(Location) | Facility where study activities are conducted
reasonStopped | Σ | 0..1 | CodeableConcept | accrual-goal-met | closed-due-to-toxicity | closed-due-to-lack-of-study-progress | temporarily-closed-per-study-design ResearchStudyReasonStopped (Example)
note | | 0..* | Annotation | Comments made about the study
arm | | 0..* | BackboneElement | Defined path through the study for a subject
name | | 1..1 | string | Label for study arm
type | | 0..1 | CodeableConcept | Categorization of study arm
description | | 0..1 | string | Short explanation of study path
objective | | 0..* | BackboneElement | A goal for the study
name | | 0..1 | string | Label for the objective
type | | 0..1 | CodeableConcept | primary | secondary | exploratory ResearchStudyObjectiveType (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for study --></identifier>
 <title value="[string]"/><!-- 0..1 Name for this study -->
 <protocol><!-- 0..* Reference(PlanDefinition) Steps followed in executing study --></protocol>
 <partOf><!-- 0..* Reference(ResearchStudy) Part of larger study --></partOf>
 <status value="[code]"/><!-- 1..1 active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-accrual-and-intervention | withdrawn -->
 <primaryPurposeType><!-- 0..1 CodeableConcept treatment | prevention | diagnostic | supportive-care | screening | health-services-research | basic-science | device-feasibility --></primaryPurposeType>
 <phase><!-- 0..1 CodeableConcept n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 | phase-2-phase-3 | phase-3 | phase-4 --></phase>
 <category><!-- 0..* CodeableConcept Classifications for the study --></category>
 <focus><!-- 0..* CodeableConcept Drugs, devices, etc. under study --></focus>
 <condition><!-- 0..* CodeableConcept Condition being studied --></condition>
 <contact><!-- 0..* ContactDetail Contact details for the study --></contact>
 <relatedArtifact><!-- 0..* RelatedArtifact References and dependencies --></relatedArtifact>
 <keyword><!-- 0..* CodeableConcept Used to search for the study --></keyword>
 <location><!-- 0..* CodeableConcept Geographic region(s) for study --></location>
 <description value="[markdown]"/><!-- 0..1 What this is study doing -->
 <enrollment><!-- 0..* Reference(Group) Inclusion & exclusion criteria --></enrollment>
 <period><!-- 0..1 Period When the study began and ended --></period>
 <sponsor><!-- 0..1 Reference(Organization) Organization that initiates and is legally responsible for the study --></sponsor>
 <principalInvestigator><!-- 0..1 Reference(Practitioner|PractitionerRole) Researcher who oversees multiple aspects of the study --></principalInvestigator>
 <site><!-- 0..* Reference(Location) Facility where study activities are conducted --></site>
 <reasonStopped><!-- 0..1 CodeableConcept accrual-goal-met | closed-due-to-toxicity | closed-due-to-lack-of-study-progress | temporarily-closed-per-study-design --></reasonStopped>
 <note><!-- 0..* Annotation Comments made about the study --></note>
 <arm>  <!-- 0..* Defined path through the study for a subject -->
  <name value="[string]"/><!-- 1..1 Label for study arm -->
  <type><!-- 0..1 CodeableConcept Categorization of study arm --></type>
  <description value="[string]"/><!-- 0..1 Short explanation of study path -->
 </arm>
 <objective>  <!-- 0..* A goal for the study -->
  <name value="[string]"/><!-- 0..1 Label for the objective -->
  <type><!-- 0..1 CodeableConcept primary | secondary | exploratory --></type>
```

```
 </objective>
</ResearchStudy>
```

JSON Template

```
{
  "resourceType" : "ResearchStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for study
  "title" : "<string>", // Name for this study
  "protocol" : [{ Reference(PlanDefinition) }], // Steps followed in executing study
  "partOf" : [{ Reference(ResearchStudy) }], // Part of larger study
  "status" : "<code>", // R!  active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-
    and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-
    accrual-and-intervention | withdrawn
  "primaryPurposeType" : { CodeableConcept }, // treatment | prevention | diagnostic | supportive-care | screening |
    health-services-research | basic-science | device-feasibility
  "phase" : { CodeableConcept }, // n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 | phase-2-phase-3 |
    phase-3 | phase-4
  "category" : [{ CodeableConcept }], // Classifications for the study
  "focus" : [{ CodeableConcept }], // Drugs, devices, etc. under study
  "condition" : [{ CodeableConcept }], // Condition being studied
  "contact" : [{ ContactDetail }], // Contact details for the study
  "relatedArtifact" : [{ RelatedArtifact }], // References and dependencies
  "keyword" : [{ CodeableConcept }], // Used to search for the study
  "location" : [{ CodeableConcept }], // Geographic region(s) for study
  "description" : "<markdown>", // What this is study doing
  "enrollment" : [{ Reference(Group) }], // Inclusion & exclusion criteria
  "period" : { Period }, // When the study began and ended
  "sponsor" : { Reference(Organization) }, // Organization that initiates and is legally responsible for the study
  "principalInvestigator" : { Reference(Practitioner|PractitionerRole) }, // Researcher who oversees multiple aspects of
     the study
  "site" : [{ Reference(Location) }], // Facility where study activities are conducted
  "reasonStopped" : { CodeableConcept }, // accrual-goal-met | closed-due-to-toxicity | closed-due-to-lack-of-study-
    progress | temporarily-closed-per-study-design
  "note" : [{ Annotation }], // Comments made about the study
  "arm" : [{ // Defined path through the study for a subject
    "name" : "<string>", // R!  Label for study arm
    "type" : { CodeableConcept }, // Categorization of study arm
    "description" : "<string>" // Short explanation of study path
  }],
  "objective" : [{ // A goal for the study
    "name" : "<string>", // Label for the objective
    "type" : { CodeableConcept } // primary | secondary | exploratory
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchStudy.identifier [ Identifier ], ... ; # 0..* Business Identifier for study
  fhir:ResearchStudy.title [ string ]; # 0..1 Name for this study
  fhir:ResearchStudy.protocol [ Reference(PlanDefinition) ], ... ; # 0..* Steps followed in executing study
  fhir:ResearchStudy.partOf [ Reference(ResearchStudy) ], ... ; # 0..* Part of larger study
  fhir:ResearchStudy.status [ code ]; # 1..1 active | administratively-completed | approved | closed-to-accrual |
    closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual |
    temporarily-closed-to-accrual-and-intervention | withdrawn
  fhir:ResearchStudy.primaryPurposeType [ CodeableConcept ]; # 0..1 treatment | prevention | diagnostic | supportive-
    care | screening | health-services-research | basic-science | device-feasibility
  fhir:ResearchStudy.phase [ CodeableConcept ]; # 0..1 n-a | early-phase-1 | phase-1 | phase-1-phase-2 | phase-2 |
    phase-2-phase-3 | phase-3 | phase-4
  fhir:ResearchStudy.category [ CodeableConcept ], ... ; # 0..* Classifications for the study
  fhir:ResearchStudy.focus [ CodeableConcept ], ... ; # 0..* Drugs, devices, etc. under study
  fhir:ResearchStudy.condition [ CodeableConcept ], ... ; # 0..* Condition being studied
  fhir:ResearchStudy.contact [ ContactDetail ], ... ; # 0..* Contact details for the study
  fhir:ResearchStudy.relatedArtifact [ RelatedArtifact ], ... ; # 0..* References and dependencies
  fhir:ResearchStudy.keyword [ CodeableConcept ], ... ; # 0..* Used to search for the study
  fhir:ResearchStudy.location [ CodeableConcept ], ... ; # 0..* Geographic region(s) for study
  fhir:ResearchStudy.description [ markdown ]; # 0..1 What this is study doing
  fhir:ResearchStudy.enrollment [ Reference(Group) ], ... ; # 0..* Inclusion & exclusion criteria
  fhir:ResearchStudy.period [ Period ]; # 0..1 When the study began and ended
  fhir:ResearchStudy.sponsor [ Reference(Organization) ]; # 0..1 Organization that initiates and is legally responsible
    for the study
  fhir:ResearchStudy.principalInvestigator [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Researcher who oversees
    multiple aspects of the study
  fhir:ResearchStudy.site [ Reference(Location) ], ... ; # 0..* Facility where study activities are conducted
  fhir:ResearchStudy.reasonStopped [ CodeableConcept ]; # 0..1 accrual-goal-met | closed-due-to-toxicity | closed-due-
    to-lack-of-study-progress | temporarily-closed-per-study-design
  fhir:ResearchStudy.note [ Annotation ], ... ; # 0..* Comments made about the study
  fhir:ResearchStudy.arm [ # 0..* Defined path through the study for a subject
    fhir:ResearchStudy.arm.name [ string ]; # 1..1 Label for study arm
    fhir:ResearchStudy.arm.type [ CodeableConcept ]; # 0..1 Categorization of study arm
    fhir:ResearchStudy.arm.description [ string ]; # 0..1 Short explanation of study path
```

```
  ], ...;
  fhir:ResearchStudy.objective [ # 0..* A goal for the study
    fhir:ResearchStudy.objective.name [ string ]; # 0..1 Label for the objective
    fhir:ResearchStudy.objective.type [ CodeableConcept ]; # 0..1 primary | secondary | exploratory
  ], ...;
]
```

Changes since Release 3

ResearchStudy
ResearchStudy.status | Change value set from http://hl7.org/fhir/ValueSet/research-study-status to http://hl7.org/fhir/ValueSet/research-study-status|4.0.1
ResearchStudy.primaryPurposeType | Added Element
ResearchStudy.phase | Added Element
ResearchStudy.condition | Added Element
ResearchStudy.location | Added Element
ResearchStudy.principalInvestigator | Type Reference: Added Target Type PractitionerRole
ResearchStudy.arm.type | Added Element
ResearchStudy.objective | Added Element
ResearchStudy.objective.name | Added Element
ResearchStudy.objective.type | Added Element
ResearchStudy.jurisdiction | deleted
ResearchStudy.arm.code | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.22.2.1 Terminology Bindings

Path | Definition | Type | Reference
ResearchStudy.status | Codes that convey the current status of the research study. | Required | ResearchStudyStatus
ResearchStudy.primaryPurposeType | Codes for the main intent of the study. | Extensible | ResearchStudyPrimaryPurposeType
ResearchStudy.phase | Codes for the stage in the progression of a therapy from initial experimental use in humans in clinical trials to post-market evaluation. | Example | ResearchStudyPhase
ResearchStudy.category | Codes that describe the type of research study. E.g. Study phase, Interventional/Observational, blinding type, etc. | Unknown | No details provided yet
ResearchStudy.focus | Codes for medications, devices and other interventions. | Unknown | No details provided yet
ResearchStudy.condition | Identification of the condition or diagnosis. | Example | Condition/Problem/DiagnosisCodes
ResearchStudy.keyword | Words associated with the study that may be useful in discovery. | Unknown | No details provided yet
ResearchStudy.location | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
ResearchStudy.reasonStopped | Codes for why the study ended prematurely. | Example | ResearchStudyReasonStopped
ResearchStudy.objective.type | Codes for the kind of study objective. | Preferred | ResearchStudyObjectiveType

8.22.3 ResearchStudy State Machine

The following diagram reflects the "typical" state machine for ResearchStudy.

8.22.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | Classifications for the study | ResearchStudy.category
date | date | When the study began and ended | ResearchStudy.period
focus | token | Drugs, devices, etc. under study | ResearchStudy.focus
identifier | token | Business Identifier for study | ResearchStudy.identifier
keyword | token | Used to search for the study | ResearchStudy.keyword
location | token | Geographic region(s) for study | ResearchStudy.location
partof | reference | Part of larger study | ResearchStudy.partOf (ResearchStudy)
principalinvestigator | reference | Researcher who oversees multiple aspects of the study | ResearchStudy.principalInvestigator (Practitioner, PractitionerRole)
protocol | reference | Steps followed in executing study | ResearchStudy.protocol (PlanDefinition)
site | reference | Facility where study activities are conducted | ResearchStudy.site (Location)
sponsor | reference | Organization that initiates and is legally responsible for the study | ResearchStudy.sponsor (Organization)
status | token | active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-accrual-and-intervention | withdrawn | ResearchStudy.status
title | string | Name for this study | ResearchStudy.title

8.22.5 Resource ResearchStudy - Examples

URL: https://hl7.org/fhir/R4/researchstudy-examples.html

8.22.5 Resource ResearchStudy - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Practitioner

Example Name | id | Format
Placeholder example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.22.6 Resource ResearchStudy - Detailed Descriptions

URL: https://hl7.org/fhir/R4/researchstudy-definitions.html

8.22.6 Resource ResearchStudy - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 1 | Trial Use | Security Category: Business | Compartments: Practitioner

Detailed Descriptions for the elements in the ResearchStudy resource.

ResearchStudy
Element Id | ResearchStudy
Definition | A process where a researcher or organization plans and then executes a series of steps intended to increase the field of healthcare-related knowledge. This includes studies of safety, efficacy, comparative effectiveness and other information about medications, devices, therapies and other interventional and investigative techniques. A ResearchStudy involves the gathering of information about human or animal subjects.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Clinical Trial; Study
Comments | Need to make sure we encompass public health studies.
ResearchStudy.identifier
Element Id | ResearchStudy.identifier
Definition | Identifiers assigned to this research study by the sponsor or other systems.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the research study as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
ResearchStudy.title
Element Id | ResearchStudy.title
Definition | A short, descriptive user-friendly label for the study.
Cardinality | 0..1
Type | string
Summary | true
ResearchStudy.protocol
Element Id | ResearchStudy.protocol
Definition | The set of steps expected to be performed as part of the execution of the study.
Cardinality | 0..*
Type | Reference(PlanDefinition)
Summary | true
ResearchStudy.partOf
Element Id | ResearchStudy.partOf
Definition | A larger research study of which this particular study is a component or step.
Cardinality | 0..*
Type | Reference(ResearchStudy)
Requirements | Allows breaking a study into components (e.g. by study site) each with their own PI, status, enrollment, etc.
Alternate Names | container
Summary | true
ResearchStudy.status
Element Id | ResearchStudy.status
Definition | The current state of the study.
Cardinality | 1..1
Terminology Binding | ResearchStudyStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
ResearchStudy.primaryPurposeType
Element Id | ResearchStudy.primaryPurposeType
Definition | The type of study based upon the intent of the study's activities. A classification of the intent of the study.
Cardinality | 0..1

Terminology Binding | ResearchStudyPrimaryPurposeType (Extensible)
Type | CodeableConcept
Summary | true
ResearchStudy.phase
Element Id | ResearchStudy.phase
Definition | The stage in the progression of a therapy from initial experimental use in humans in clinical trials to post-market evaluation.
Cardinality | 0..1
Terminology Binding | ResearchStudyPhase (Example)
Type | CodeableConcept
Summary | true
ResearchStudy.category
Element Id | ResearchStudy.category
Definition | Codes categorizing the type of study such as investigational vs. observational, type of blinding, type of randomization, safety vs. efficacy, etc.
Cardinality | 0..*
Terminology Binding | ResearchStudyCategory:
Type | CodeableConcept
Summary | true
ResearchStudy.focus
Element Id | ResearchStudy.focus
Definition | The medication(s), food(s), therapy(ies), device(s) or other concerns or interventions that the study is seeking to gain more information about.
Cardinality | 0..*
Terminology Binding | ResearchStudyFocus:
Type | CodeableConcept
Summary | true
ResearchStudy.condition
Element Id | ResearchStudy.condition
Definition | The condition that is the focus of the study. For example, In a study to examine risk factors for Lupus, might have as an inclusion criterion "healthy volunteer", but the target condition code would be a Lupus SNOMED code.
Cardinality | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Summary | true
ResearchStudy.contact
Element Id | ResearchStudy.contact
Definition | Contact details to assist a user in learning more about or engaging with the study.
Cardinality | 0..*
Type | ContactDetail
Summary | true
ResearchStudy.relatedArtifact
Element Id | ResearchStudy.relatedArtifact
Definition | Citations, references and other related documents.
Cardinality | 0..*
Type | RelatedArtifact
ResearchStudy.keyword
Element Id | ResearchStudy.keyword
Definition | Key terms to aid in searching for or filtering the study.
Cardinality | 0..*
Terminology Binding | ResearchStudyKeyword:
Type | CodeableConcept
Summary | true
ResearchStudy.location
Element Id | ResearchStudy.location
Definition | Indicates a country, state or other region where the study is taking place.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
ResearchStudy.description
Element Id | ResearchStudy.description
Definition | A full description of how the study is being conducted.
Cardinality | 0..1
Type | markdown
ResearchStudy.enrollment
Element Id | ResearchStudy.enrollment
Definition | Reference to a Group that defines the criteria for and quantity of subjects participating in the study. E.g. " 200 female Europeans between the ages of 20 and 45 with early onset diabetes".
Cardinality | 0..*
Type | Reference(Group)
Alternate Names | patient
Summary | true

Comments | The Group referenced should not generally enumerate specific subjects. Subjects will be linked to the study using the ResearchSubject resource.
ResearchStudy.period
Element Id | ResearchStudy.period
Definition | Identifies the start date and the expected (or actual, depending on status) end date for the study.
Cardinality | 0..1
Type | Period
Alternate Names | timing
Summary | true
ResearchStudy.sponsor
Element Id | ResearchStudy.sponsor
Definition | An organization that initiates the investigation and is legally responsible for the study.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
ResearchStudy.principalInvestigator
Element Id | ResearchStudy.principalInvestigator
Definition | A researcher in a study who oversees multiple aspects of the study, such as concept development, protocol writing, protocol submission for IRB approval, participant recruitment, informed consent, data collection, analysis, interpretation and presentation.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Summary | true
ResearchStudy.site
Element Id | ResearchStudy.site
Definition | A facility in which study activities are conducted.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
ResearchStudy.reasonStopped
Element Id | ResearchStudy.reasonStopped
Definition | A description and/or code explaining the premature termination of the study.
Cardinality | 0..1
Terminology Binding | ResearchStudyReasonStopped (Example)
Type | CodeableConcept
Summary | true
ResearchStudy.note
Element Id | ResearchStudy.note
Definition | Comments made about the study by the performer, subject or other participants.
Cardinality | 0..*
Type | Annotation
ResearchStudy.arm
Element Id | ResearchStudy.arm
Definition | Describes an expected sequence of events for one of the participants of a study. E.g. Exposure to drug A, wash-out, exposure to drug B, wash-out, follow-up.
Cardinality | 0..*
ResearchStudy.arm.name
Element Id | ResearchStudy.arm.name
Definition | Unique, human-readable label for this arm of the study.
Cardinality | 1..1
Type | string
ResearchStudy.arm.type
Element Id | ResearchStudy.arm.type
Definition | Categorization of study arm, e.g. experimental, active comparator, placebo comparater.
Cardinality | 0..1
Type | CodeableConcept
ResearchStudy.arm.description
Element Id | ResearchStudy.arm.description
Definition | A succinct description of the path through the study that would be followed by a subject adhering to this arm.
Cardinality | 0..1
Type | string
ResearchStudy.objective
Element Id | ResearchStudy.objective
Definition | A goal that the study is aiming to achieve in terms of a scientific question to be answered by the analysis of data collected during the study.
Cardinality | 0..*
ResearchStudy.objective.name
Element Id | ResearchStudy.objective.name
Definition | Unique, human-readable label for this objective of the study.
Cardinality | 0..1
Type | string
ResearchStudy.objective.type
Element Id | ResearchStudy.objective.type

Definition | The kind of study objective.
Cardinality | 0..1
Terminology Binding | ResearchStudyObjectiveType (Preferred)
Type | CodeableConcept

8.22.7 Resource ResearchStudy - Mappings

URL: https://hl7.org/fhir/R4/researchstudy-mappings.html

8.22.7 Resource ResearchStudy - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Practitioner

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

ClinicalTrials.gov Mapping

BRIDG 5.1 Mapping

Mappings for the researchstudy resource (see Mappings to Other Standards for further information & status).

8.22.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ResearchStudy
identifier | FiveWs.identifier
status | FiveWs.status
focus | FiveWs.what[x]
condition | FiveWs.what[x]
enrollment | FiveWs.subject[x]
period | FiveWs.done[x]
sponsor | FiveWs.actor
principalInvestigator | FiveWs.actor
site | FiveWs.where[x]
reasonStopped | FiveWs.why[x]

8.22.7.2 HL7 v2 Mapping (http://hl7.org/v2)

ResearchStudy | OBX
identifier
title
protocol
partOf
status
primaryPurposeType
phase
category
focus
condition
contact
relatedArtifact
keyword
location
description
enrollment
period
sponsor
principalInvestigator
site
reasonStopped
note
arm
name
type
description
objective
name
type

8.22.7.3 RIM Mapping (http://hl7.org/v3)

ResearchStudy | Observation[classCode=CLNTRL, moodCode=EVN]
identifier | .identifier

protocol | .outboundRelationship[typeCode=DEFN].target
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
status | .status
enrollment | .participation[typeCode=SBJ].role
period | .effectiveTime
sponsor | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]

8.22.7.4 ClinicalTrials.gov Mapping (http://clinicaltrials.gov)

ResearchStudy
identifier | Organization's Unique Protocol ID; Secondary IDs
title | Brief Title
protocol | Interventions; Arm/Intervention Cross-Reference
partOf
status
primaryPurposeType
phase | Study Phase
category | Study Type; Primary Purpose; Interventional Model; Masking; Allocation; study Classification; Observational Study Model; Time Perspective; Biospecimen Retention
focus | Focus of Study
condition | Conditions
contact | Central Contact; Central Contact Backup; Overall Study Officials; Facility Contact & Backup (if Study tied to only one location); Investigators
relatedArtifact | Citations, Links, Available Study Data/Documents
keyword | Keywords
location
description | Detailed Description
enrollment | Enrollment; Number of Groups/Cohorts (= number of "enrollment" instances); Groups (Group/Cohort Label = Group.name; Group/Cohort Description = Group.description); Eligibility (Group criteria)
period | Study Start Date (start); Study Completion Date (end)
sponsor | Sponsor
principalInvestigator | Principal Investigator
site | Facility
reasonStopped | Why Study Stopped
note
arm | Arms
name | Arm Label
type | Arm Type
description | Arm Description
objective
name
type

8.22.7.5 BRIDG 5.1 Mapping (https://bridgmodel.nci.nih.gov)

ResearchStudy | ObservationalStudyProtocolVersion; Study; StudyProtocolVersion
identifier | No BRIDG mapping (although there is DocumentVersion.identifier > StudyProtocolDocumentVersion, this is arguably a different semantic)
title | No BRIDG mapping (although there is StudyProtocolDocumentVersionPublicTitle this is arguably a different semantic: title of a document vs title of a research study)
protocol | StudyProtocolDocumentVersionPublicTitle.name
partOf | CompanionStudyRelationship
status | StudyOverallStatus.code
primaryPurposeType | StudyProtocolVersion.primaryPurposeTypeCode
phase | StudyProtocolVersion.phaseCode
category | InterventionalStudyProtocolVersion; InterventionalStudyProtocol.allocationCode; InterventionalStudyProtocol.blindedRoleCode; InterventionalStudyProtocol.blindingSchemaCode; InterventionalStudyProtocol.controlTypeCode
focus | InterventionalStudyProtocol.interventionTypeCode
condition | StudyCondition.code
contact | PointOfContact > Project > Research Project > Study > StudyProtocol > StudyProtocolVersion
relatedArtifact | StudyProtocolDocument, StudyProtocolDocumentVersion
keyword | DocumentVersion.keywordCode; DocumentVersion.keywordText
location | StudyProtocolVersion.participatingLocationCode
description | StudyProtocolDocumentVersion.publicDescription
enrollment | InterventionalStudyProtocol.interventionGroupQuantity
period | ProjectConduct.dateRange; StudyProtocolVersion.plannedDuration
sponsor | RegulatoryApplicationSponsor; StudyLegalSponsor; StudyProtocolVersion.responsiblePartyCode
principalInvestigator | StudyInvestigator
site | PlannedStudySite; StudySite
reasonStopped | StudyOverallStatus.studyStoppedReasonCode
note | No BRIDG mapping
arm | Arm

name | Arm.name
type | Arm.typeCode
description | Arm.description
objective | StudyObjective
name | No BRIDG mapping (BRIDG has a description, but not a name)
type | StudyObjective.typeCode

8.22.8 Resource ResearchStudy - Extensions & Profiles

URL: https://hl7.org/fhir/R4/researchstudy-profiles.html

8.22.8 Resource ResearchStudy - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Practitioner

This table lists profiles and extensions for the ResearchStudy resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.22.8.1 Profiles

No Profiles defined for this resource

8.22.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-researchStudy | researchStudy | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.22.8.3 Search Extensions

No Search Extensions defined for this resource

8.22.9 Resource ResearchStudy - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/researchstudy-version-maps.html

No validation errors - all conversions are clean

8.22.9 Resource ResearchStudy - R3/R4 Conversions

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Practitioner

R3 : R4 Conversion maps for ResearchStudy.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.22.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ResearchStudy3to4" = "R3 to R4 Conversions for ResearchStudy"

conceptmap "ResourceStudyStatus" {
  prefix s = "http://hl7.org/fhir/research-study-status"
  prefix t = "http://hl7.org/fhir/research-study-status"

  s:"'draft'" - t:"'in-review'"
  s:"'in-progress'" - t:"'active'"
  s:"'suspended'" - t:"'disapproved'"
  s:"'stopped'" - t:"'withdrawn'"
  s:"'completed'" - t:"'administratively-completed'"
  s:"'entered-in-error'" - t:"'withdrawn'"
}

uses "http://hl7.org/fhir/3.0/StructureDefinition/ResearchStudy" alias ResearchStudyR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ResearchStudy" alias ResearchStudy as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group ResearchStudy(source src : ResearchStudyR3, target tgt : ResearchStudy) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.title -> tgt.title;
  src.protocol -> tgt.protocol;
  src.partOf -> tgt.partOf;
  src.status as v -> tgt.status = translate(v, '#ResourceStudyStatus', 'code');
  src.category -> tgt.category;
  src.focus -> tgt.focus;
  src.contact -> tgt.contact;
  src.relatedArtifact -> tgt.relatedArtifact;
  src.keyword -> tgt.keyword;
  src.location -> tgt.location;
  src.jurisdiction as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ResearchStudy.jurisdiction',  vt.value = v;
  src.description -> tgt.description;
  src.enrollment -> tgt.enrollment;
  src.period -> tgt.period;
  src.sponsor -> tgt.sponsor;
  src.principalInvestigator -> tgt.principalInvestigator;
  src.site -> tgt.site;
  src.reasonStopped -> tgt.reasonStopped;
  src.note -> tgt.note;
  src.arm as s -> tgt.arm as t then arm(s, t);
}

group arm(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.code -> tgt.type;
  src.description -> tgt.description;
}
```

8.22.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ResearchStudy4to3" = "R4 to R3 Conversion for ResearchStudy"

conceptmap "ResourceStudyStatus" {
  prefix s = "http://hl7.org/fhir/research-study-status"
  prefix t = "http://hl7.org/fhir/research-study-status"

  s:"'active'" - t:"'in-progress'"
  s:"'administratively-completed'" - t:"'completed'"
  s:"'approved'" - t:"'in-progress'"
  s:"'closed-to-accrual'" - t:"'suspended'"
  s:"'closed-to-accrual-and-intervention'" - t:"'suspended'"
  s:"'completed'" - t:"'completed'"
  s:"'disapproved'" - t:"'stopped'"
  s:"'in-review'" - t:"'draft'"
  s:"'temporarily-closed-to-accrual'" - t:"'suspended'"
  s:"'temporarily-closed-to-accrual-and-intervention'" - t:"'suspended'"
  s:"'withdrawn  '" - t:"'stopped'"
}

uses "http://hl7.org/fhir/StructureDefinition/ResearchStudy" alias ResearchStudy as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ResearchStudy" alias ResearchStudyR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ResearchStudy(source src : ResearchStudyR3, target tgt : ResearchStudy) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.title -> tgt.title;
  src.protocol -> tgt.protocol;
  src.partOf -> tgt.partOf;
  src.status as v -> tgt.status = translate(v, '#ResourceStudyStatus', 'code');
  src.category -> tgt.category;
  src.focus -> tgt.focus;
  src.contact -> tgt.contact;
  src.relatedArtifact -> tgt.relatedArtifact;
  src.keyword -> tgt.keyword;
  src.location -> tgt.location;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-ResearchStudy.jurisdiction' then
     {
    e.value -> tgt.jurisdiction;
  };
  src.description -> tgt.description;
  src.enrollment -> tgt.enrollment;
  src.period -> tgt.period;
  src.sponsor -> tgt.sponsor;
  src.principalInvestigator -> tgt.principalInvestigator;
  src.site -> tgt.site;
  src.reasonStopped -> tgt.reasonStopped;
  src.note -> tgt.note;
  src.arm as s -> tgt.arm as t then arm(s, t);
}

group arm(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.type -> tgt.code;
  src.description -> tgt.description;
}
```

8.22.9.3 R4 Validation Errors

8.22.10 JSON Schema for ResearchStudy

URL: https://hl7.org/fhir/R4/researchstudy.schema.json.html

8.22.10 JSON Schema for ResearchStudy

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Practitioner

This is a representation of the json schema for ResearchStudy, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ResearchStudy",
  "$ref": "#/definitions/ResearchStudy",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ResearchStudy": {
      "description": "A process where a researcher or organization plans and then executes a series of steps intended to
     increase the field of healthcare-related knowledge.  This includes studies of safety, efficacy, comparative
    effectiveness and other information about medications, devices, therapies and other interventional and investigative
     techniques.  A ResearchStudy involves the gathering of information about human or animal subjects.",
      "properties": {
        "resourceType": {
          "description": "This is a ResearchStudy resource",
          "const": "ResearchStudy"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
```

```
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this research study by the sponsor or other systems.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "title": {
      "description": "A short, descriptive user-friendly label for the study.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "protocol": {
      "description": "The set of steps expected to be performed as part of the execution of the study.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger research study of which this particular study is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the study.",
      "enum": [
        "active",
        "administratively-completed",
        "approved",
        "closed-to-accrual",
        "closed-to-accrual-and-intervention",
        "completed",
        "disapproved",
        "in-review",
        "temporarily-closed-to-accrual",
        "temporarily-closed-to-accrual-and-intervention",
        "withdrawn"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "primaryPurposeType": {
      "description": "The type of study based upon the intent of the study\u0027s activities. A classification of
the intent of the study.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "phase": {
      "description": "The stage in the progression of a therapy from initial experimental use in humans in clinical
trials to post-market evaluation.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "Codes categorizing the type of study such as investigational vs. observational, type of
blinding, type of randomization, safety vs. efficacy, etc.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "focus": {
      "description": "The medication(s), food(s), therapy(ies), device(s) or other concerns or interventions that
the study is seeking to gain more information about.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "condition": {
      "description": "The condition that is the focus of the study.  For example, In a study to examine risk factors
 for Lupus, might have as an inclusion criterion \"healthy volunteer\", but the target condition code would be a
Lupus SNOMED code.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
      },
      "type": "array"
    },
    "contact": {
      "description": "Contact details to assist a user in learning more about or engaging with the study.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Citations, references and other related documents.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "keyword": {
      "description": "Key terms to aid in searching for or filtering the study.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "location": {
      "description": "Indicates a country, state or other region where the study is taking place.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "description": {
      "description": "A full description of how the study is being conducted.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "enrollment": {
      "description": "Reference to a Group that defines the criteria for and quantity of subjects participating in
the study.  E.g. \" 200 female Europeans between the ages of 20 and 45 with early onset diabetes\".",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "period": {
      "description": "Identifies the start date and the expected (or actual, depending on status) end date for the
study.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "sponsor": {
      "description": "An organization that initiates the investigation and is legally responsible for the study.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "principalInvestigator": {
      "description": "A researcher in a study who oversees multiple aspects of the study, such as concept
development, protocol writing, protocol submission for IRB approval, participant recruitment, informed consent, data
 collection, analysis, interpretation and presentation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "site": {
      "description": "A facility in which study activities are conducted.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "reasonStopped": {
      "description": "A description and/or code explaining the premature termination of the study.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "note": {
      "description": "Comments made about the study by the performer, subject or other participants.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "arm": {
      "description": "Describes an expected sequence of events for one of the participants of a study.  E.g.
Exposure to drug A, wash-out, exposure to drug B, wash-out, follow-up.",
      "items": {
        "$ref": "#/definitions/ResearchStudy_Arm"
      },
      "type": "array"
    },
    "objective": {
```

```
      "description": "A goal that the study is aiming to achieve in terms of a scientific question to be answered by
 the analysis of data collected during the study.",
      "items": {
        "$ref": "#/definitions/ResearchStudy_Objective"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ResearchStudy_Arm": {
  "description": "A process where a researcher or organization plans and then executes a series of steps intended to
 increase the field of healthcare-related knowledge.  This includes studies of safety, efficacy, comparative
effectiveness and other information about medications, devices, therapies and other interventional and investigative
 techniques.  A ResearchStudy involves the gathering of information about human or animal subjects.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Unique, human-readable label for this arm of the study.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Categorization of study arm, e.g. experimental, active comparator, placebo comparater.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "A succinct description of the path through the study that would be followed by a subject
adhering to this arm.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ResearchStudy_Objective": {
  "description": "A process where a researcher or organization plans and then executes a series of steps intended to
 increase the field of healthcare-related knowledge.  This includes studies of safety, efficacy, comparative
effectiveness and other information about medications, devices, therapies and other interventional and investigative
 techniques.  A ResearchStudy involves the gathering of information about human or animal subjects.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
```

```
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
   the element and that modifies the understanding of the element in which it is contained and/or the understanding of
   the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
   use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
   extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
    of the definition of the extension. Applications processing a resource are required to check for modifier
   extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
   (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "name": {
        "description": "Unique, human-readable label for this objective of the study.",
        "$ref": "#/definitions/string"
      },
      "_name": {
        "description": "Extensions for name",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": {
        "description": "The kind of study objective.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      }
    },
    "additionalProperties": false
  }
 }
}
```


8.22.11 Resource ResearchStudy - Maturity Dependency View

URL: https://hl7.org/fhir/R4/researchstudy-dependencies.html

8.22.11 Resource ResearchStudy - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 1 | Trial Use | Compartments: Practitioner

Dependency Graph for ResearchStudy FMM level 1

Name | Card. | Type | Dependency Analysis
ResearchStudy | | DomainResource
identifier | 0..* | Identifier
title | 0..1 | string
protocol | 0..* | Reference(PlanDefinition)
partOf | 0..* | Reference(ResearchStudy)
status | 1..1 | code
primaryPurposeType | 0..1 | CodeableConcept
phase | 0..1 | CodeableConcept
category | 0..* | CodeableConcept
focus | 0..* | CodeableConcept
condition | 0..* | CodeableConcept
contact | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
keyword | 0..* | CodeableConcept
location | 0..* | CodeableConcept
description | 0..1 | markdown
enrollment | 0..* | Reference(Group)
period | 0..1 | Period
sponsor | 0..1 | Reference(Organization)
principalInvestigator | 0..1 | Reference(Practitioner | PractitionerRole)
site | 0..* | Reference(Location)
reasonStopped | 0..1 | CodeableConcept
note | 0..* | Annotation
arm | 0..* | BackboneElement
name | 1..1 | string
type | 0..1 | CodeableConcept
description | 0..1 | string
objective | 0..* | BackboneElement
name | 0..1 | string
type | 0..1 | CodeableConcept

8.22.12 Value Set Sc-valueset-research-study-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-research-study-status.html

8.22.12 Value Set Sc-valueset-research-study-status - Status Code Mapping

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.22.13 ResearchStudyStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-research-study-status)

Mapping from http://hl7.org/fhir/ValueSet/research-study-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes that convey the current status of the research study."

Source Code | Equivalence | Destination Code
in-review | equivalent | draft
disapproved | equivalent | declined
approved | equivalent | accepted
active, closed-to-accrual, closed-to-accrual-and-intervention | equivalent | active
temporarily-closed-to-accrual, temporarily-closed-to-accrual-and-intervention | equivalent | suspended
completed, administratively-completed | equivalent | complete
withdrawn | equivalent | abandoned

8.23 Resource ResearchSubject - Content

URL: https://hl7.org/fhir/R4/researchsubject.html
Biomedical Research and Regulation Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

A physical entity which is the primary unit of operational and/or administrative interest in a study.

8.23.1 Scope and Usage

A research subject is the entity of interest in a research study, typically a human being or an animal, but can also be a device, group of humans or animals, or a tissue sample. The ResearchSubject resource describes essential information about the subject, including the gender, birthdate, study, study arm, consent status, and other key items. Human research subjects are usually not traceable to a particular person to protect the subject’s privacy.

The ResearchSubject resource is used in the following research activities:

- Consent and registration to a research study
- Exposure to test therapy
- Assessment of test therapy results
- Reporting of adverse events

8.23.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchSubject | TU | | DomainResource | Physical entity which is the primary unit of interest in the study Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for research subject in a study
status | ?!Σ | 1..1 | code | candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn ResearchSubjectStatus (Required)
period | Σ | 0..1 | Period | Start and end of participation
study | Σ | 1..1 | Reference(ResearchStudy) | Study subject is part of
individual | Σ | 1..1 | Reference(Patient) | Who is part of study
assignedArm | | 0..1 | string | What path should be followed

actualArm | | 0..1 | string | What path was followed
consent | | 0..1 | Reference(Consent) | Agreement to participate in study
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchSubject xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for research subject in a study --></identifier>
 <status value="[code]"/><!-- 1..1 candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study
     | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn -->
 <period><!-- 0..1 Period Start and end of participation --></period>
 <study><!-- 1..1 Reference(ResearchStudy) Study subject is part of --></study>
 <individual><!-- 1..1 Reference(Patient) Who is part of study --></individual>
 <assignedArm value="[string]"/><!-- 0..1 What path should be followed -->
 <actualArm value="[string]"/><!-- 0..1 What path was followed -->
 <consent><!-- 0..1 Reference(Consent) Agreement to participate in study --></consent>
</ResearchSubject>
```

JSON Template

```
{
  "resourceType" : "ResearchSubject",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for research subject in a study
  "status" : "<code>", // R!  candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study |
    on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn
  "period" : { Period }, // Start and end of participation
  "study" : { Reference(ResearchStudy) }, // R!  Study subject is part of
  "individual" : { Reference(Patient) }, // R!  Who is part of study
  "assignedArm" : "<string>", // What path should be followed
  "actualArm" : "<string>", // What path was followed
  "consent" : { Reference(Consent) } // Agreement to participate in study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchSubject;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchSubject.identifier [ Identifier ], ... ; # 0..* Business Identifier for research subject in a study
  fhir:ResearchSubject.status [ code ]; # 1..1 candidate | eligible | follow-up | ineligible | not-registered | off-
    study | on-study | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening
     | withdrawn
  fhir:ResearchSubject.period [ Period ]; # 0..1 Start and end of participation
  fhir:ResearchSubject.study [ Reference(ResearchStudy) ]; # 1..1 Study subject is part of
  fhir:ResearchSubject.individual [ Reference(Patient) ]; # 1..1 Who is part of study
  fhir:ResearchSubject.assignedArm [ string ]; # 0..1 What path should be followed
  fhir:ResearchSubject.actualArm [ string ]; # 0..1 What path was followed
  fhir:ResearchSubject.consent [ Reference(Consent) ]; # 0..1 Agreement to participate in study
]
```

Changes since R3

ResearchSubject
ResearchSubject.identifier | Max Cardinality changed from 1 to *
ResearchSubject.status | Change value set from http://hl7.org/fhir/ValueSet/research-subject-status to
http://hl7.org/fhir/ValueSet/research-subject-status|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchSubject | TU | | DomainResource | Physical entity which is the primary unit of interest in the study Elements defined in
Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for research subject in a study
status | ?!Σ | 1..1 | code | candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study | on-study-intervention |
on-study-observation | pending-on-study | potential-candidate | screening | withdrawn ResearchSubjectStatus (Required)
period | Σ | 0..1 | Period | Start and end of participation
study | Σ | 1..1 | Reference(ResearchStudy) | Study subject is part of
individual | Σ | 1..1 | Reference(Patient) | Who is part of study
assignedArm | | 0..1 | string | What path should be followed
actualArm | | 0..1 | string | What path was followed
consent | | 0..1 | Reference(Consent) | Agreement to participate in study
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchSubject xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for research subject in a study --></identifier>
 <status value="[code]"/><!-- 1..1 candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study
     | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn -->
 <period><!-- 0..1 Period Start and end of participation --></period>
 <study><!-- 1..1 Reference(ResearchStudy) Study subject is part of --></study>
 <individual><!-- 1..1 Reference(Patient) Who is part of study --></individual>
 <assignedArm value="[string]"/><!-- 0..1 What path should be followed -->
 <actualArm value="[string]"/><!-- 0..1 What path was followed -->
 <consent><!-- 0..1 Reference(Consent) Agreement to participate in study --></consent>
</ResearchSubject>
```

JSON Template

```
{
  "resourceType" : "ResearchSubject",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for research subject in a study
  "status" : "<code>", // R!  candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study |
    on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn
  "period" : { Period }, // Start and end of participation
  "study" : { Reference(ResearchStudy) }, // R!  Study subject is part of
  "individual" : { Reference(Patient) }, // R!  Who is part of study
  "assignedArm" : "<string>", // What path should be followed
  "actualArm" : "<string>", // What path was followed
  "consent" : { Reference(Consent) } // Agreement to participate in study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchSubject;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchSubject.identifier [ Identifier ], ... ; # 0..* Business Identifier for research subject in a study
  fhir:ResearchSubject.status [ code ]; # 1..1 candidate | eligible | follow-up | ineligible | not-registered | off-
    study | on-study | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening
     | withdrawn
  fhir:ResearchSubject.period [ Period ]; # 0..1 Start and end of participation
  fhir:ResearchSubject.study [ Reference(ResearchStudy) ]; # 1..1 Study subject is part of
  fhir:ResearchSubject.individual [ Reference(Patient) ]; # 1..1 Who is part of study
  fhir:ResearchSubject.assignedArm [ string ]; # 0..1 What path should be followed
  fhir:ResearchSubject.actualArm [ string ]; # 0..1 What path was followed
  fhir:ResearchSubject.consent [ Reference(Consent) ]; # 0..1 Agreement to participate in study
]
```

Changes since Release 3

ResearchSubject

ResearchSubject.identifier | Max Cardinality changed from 1 to *

ResearchSubject.status | Change value set from http://hl7.org/fhir/ValueSet/research-subject-status to http://hl7.org/fhir/ValueSet/research-subject-status|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.23.2.1 Terminology Bindings

Path | Definition | Type | Reference

ResearchSubject.status | Indicates the progression of a study subject through a study. | Required | ResearchSubjectStatus

8.23.3 ResearchSubject State Machine

The following diagram reflects the "typical" state machine for ResearchSubject.

8.23.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

date | date | Start and end of participation | ResearchSubject.period

identifier | token | Business Identifier for research subject in a study | ResearchSubject.identifier
individual | reference | Who is part of study | ResearchSubject.individual (Patient)
patient | reference | Who is part of study | ResearchSubject.individual (Patient)
status | token | candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn | ResearchSubject.status
study | reference | Study subject is part of | ResearchSubject.study (ResearchStudy)

8.23.5 Resource ResearchSubject - Examples

URL: https://hl7.org/fhir/R4/researchsubject-examples.html

8.23.5 Resource ResearchSubject - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
Placeholder example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.23.6 Resource ResearchSubject - Detailed Descriptions

URL: https://hl7.org/fhir/R4/researchsubject-definitions.html

8.23.6 Resource ResearchSubject - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the ResearchSubject resource.

ResearchSubject
Element Id | ResearchSubject
Definition | A physical entity which is the primary unit of operational and/or administrative interest in a study.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Study Subject
Comments | Need to make sure we encompass public health studies.
ResearchSubject.identifier
Element Id | ResearchSubject.identifier
Definition | Identifiers assigned to this research subject for a study.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
ResearchSubject.status
Element Id | ResearchSubject.status
Definition | The current state of the subject.
Cardinality | 1..1
Terminology Binding | ResearchSubjectStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
To Do | Should this be a common code system for all events?
ResearchSubject.period
Element Id | ResearchSubject.period
Definition | The dates the subject began and ended their participation in the study.
Cardinality | 0..1
Type | Period
Alternate Names | timing
Summary | true
To Do | .effectiveTime.
ResearchSubject.study
Element Id | ResearchSubject.study
Definition | Reference to the study the subject is participating in.
Cardinality | 1..1
Type | Reference(ResearchStudy)
Summary | true
ResearchSubject.individual
Element Id | ResearchSubject.individual
Definition | The record of the person or animal who is involved in the study.
Cardinality | 1..1
Type | Reference(Patient)

Summary | true
ResearchSubject.assignedArm
Element Id | ResearchSubject.assignedArm
Definition | The name of the arm in the study the subject is expected to follow as part of this study.
Cardinality | 0..1
Type | string
ResearchSubject.actualArm
Element Id | ResearchSubject.actualArm
Definition | The name of the arm in the study the subject actually followed as part of this study.
Cardinality | 0..1
Type | string
ResearchSubject.consent
Element Id | ResearchSubject.consent
Definition | A record of the patient's informed agreement to participate in the study.
Cardinality | 0..1
Type | Reference(Consent)

8.23.7 Resource ResearchSubject - Mappings

URL: https://hl7.org/fhir/R4/researchsubject-mappings.html

8.23.7 Resource ResearchSubject - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

BRIDG 5.1 Mapping

Mappings for the researchsubject resource (see Mappings to Other Standards for further information & status).

8.23.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ResearchSubject
identifier | FiveWs.identifier
status | FiveWs.status
period | FiveWs.planned

8.23.7.2 HL7 v2 Mapping (http://hl7.org/v2)

ResearchSubject | OBX
identifier
status
period
study
individual
assignedArm
actualArm
consent

8.23.7.3 RIM Mapping (http://hl7.org/v3)

ResearchSubject | Observation[classCode=CLNTRL, moodCode=EVN]
identifier | .identifier
status | .status

8.23.7.4 BRIDG 5.1 Mapping (https://bridgmodel.nci.nih.gov)

ResearchSubject | StudySubject
identifier | StudySubject > Subject.identifier
status | StudySubject.statusCode
period | PerformedStudySubjectMilestone.studyReferenceDateRange
study | StudySubjectProtocolVersionRelationship
individual | StudySubject
assignedArm | Arm > ExperimentalUnit > BiologicEntity > Subject > StudySubject
actualArm | Arm > ExperimentalUnit > BiologicEntity > Subject > StudySubject
consent | ResearchStudy.consent is related to PerformedStudySubjectMilestone.informedConsentIndicator in that the informedConsentIndicator can be derived from the data in the Consent resource.

8.23.8 Resource ResearchSubject - Extensions & Profiles

URL: https://hl7.org/fhir/R4/researchsubject-profiles.html

8.23.8 Resource ResearchSubject - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the ResearchSubject resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.23.8.1 Profiles

No Profiles defined for this resource

8.23.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.23.8.3 Search Extensions

No Search Extensions defined for this resource

8.23.9 Resource ResearchSubject - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/researchsubject-version-maps.html

No validation errors - all conversions are clean

8.23.9 Resource ResearchSubject - R3/R4 Conversions

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for ResearchSubject.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.23.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ResearchSubject3to4" = "R3 to R4 Conversions for ResearchSubject"

conceptmap "ResourceSubjectStatus" {
  prefix s = "http://hl7.org/fhir/research-study-status"
  prefix t = "http://hl7.org/fhir/research-study-status"

  s:"'candidate'" - t:"'candidate'"
  s:"'eligible'" - t:"'candidate'"
  s:"'follow-up'" - t:"'completed'"
  s:"'ineligible'" - t:"'candidate'"
  s:"'not-registered'" - t:"'candidate'"
  s:"'off-study'" - t:"'completed'"
  s:"'on-study'" - t:"'enrolled'"
  s:"'on-study-intervention'" - t:"'active'"
  s:"'on-study-observation'" - t:"'suspended'"
  s:"'pending-on-study'" - t:"'enrolled'"
  s:"'potential-candidate'" - t:"'candidate'"
  s:"'screening'" - t:"'candidate'"
  s:"'withdrawn'" - t:"'withdrawn'"
}

uses "http://hl7.org/fhir/3.0/StructureDefinition/ResearchSubject" alias ResearchSubjectR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ResearchSubject" alias ResearchSubject as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ResearchSubject(source src : ResearchSubjectR3, target tgt : ResearchSubject) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status as v -> tgt.status = translate(v, '#ResourceSubjectStatus', 'code');
  src.status -> tgt.status;
```

```
  src.period -> tgt.period;
  src.study -> tgt.study;
  src.individual -> tgt.individual;
  src.assignedArm -> tgt.assignedArm;
  src.actualArm -> tgt.actualArm;
  src.consent -> tgt.consent;
}
```

8.23.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ResearchSubject4to3" = "R4 to R3 Conversion for ResearchSubject"

conceptmap "ResourceSubjectStatus" {
  prefix s = "http://hl7.org/fhir/research-study-status"
  prefix t = "http://hl7.org/fhir/research-study-status"

  s:"'candidate'" - t:"'candidate'"
  s:"'enrolled'" - t:"'on-study'"
  s:"'active'" - t:"'on-study-intervention'"
  s:"'suspended'" - t:"'on-study-observation'"
  s:"'withdrawn'" - t:"'withdrawn'"
  s:"'completed'" - t:"'off-study'"
}

uses "http://hl7.org/fhir/StructureDefinition/ResearchSubject" alias ResearchSubject as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ResearchSubject" alias ResearchSubjectR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ResearchSubject(source src : ResearchSubjectR3, target tgt : ResearchSubject) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status as v -> tgt.status = translate(v, '#ResourceSubjectStatus', 'code');
  src.status -> tgt.status;
  src.period -> tgt.period;
  src.study -> tgt.study;
  src.individual -> tgt.individual;
  src.assignedArm -> tgt.assignedArm;
  src.actualArm -> tgt.actualArm;
  src.consent -> tgt.consent;
}
```

8.23.9.3 R4 Validation Errors

8.23.10 JSON Schema for ResearchSubject

URL: https://hl7.org/fhir/R4/researchsubject.schema.json.html

8.23.10 JSON Schema for ResearchSubject

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for ResearchSubject, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ResearchSubject",
  "$ref": "#/definitions/ResearchSubject",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ResearchSubject": {
      "description": "A physical entity which is the primary unit of operational and/or administrative interest in a
    study.",
      "properties": {
        "resourceType": {
          "description": "This is a ResearchSubject resource",
          "const": "ResearchSubject"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
```

```
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this research subject for a study.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the subject.",
      "enum": [
        "candidate",
        "eligible",
        "follow-up",
        "ineligible",
        "not-registered",
        "off-study",
        "on-study",
        "on-study-intervention",
        "on-study-observation",
        "pending-on-study",
        "potential-candidate",
        "screening",
        "withdrawn"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The dates the subject began and ended their participation in the study.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "study": {
      "description": "Reference to the study the subject is participating in.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "individual": {
      "description": "The record of the person or animal who is involved in the study.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "assignedArm": {
      "description": "The name of the arm in the study the subject is expected to follow as part of this study.",
      "$ref": "#/definitions/string"
```

```
        },
        "_assignedArm": {
          "description": "Extensions for assignedArm",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "actualArm": {
          "description": "The name of the arm in the study the subject actually followed as part of this study.",
          "$ref": "#/definitions/string"
        },
        "_actualArm": {
          "description": "Extensions for actualArm",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "consent": {
          "description": "A record of the patient\u0027s informed agreement to participate in the study.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "study",
        "individual",
        "resourceType"
      ]
    }
  }
}
```


8.23.11 Resource ResearchSubject - Maturity Dependency View

URL: https://hl7.org/fhir/R4/researchsubject-dependencies.html

8.23.11 Resource ResearchSubject - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Dependency Graph for ResearchSubject FMM level 1

Name | Card. | Type | Dependency Analysis
ResearchSubject | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
period | 0..1 | Period
study | 1..1 | Reference(ResearchStudy)
individual | 1..1 | Reference(Patient)
assignedArm | 0..1 | string
actualArm | 0..1 | string
consent | 0..1 | Reference(Consent)

8.23.12 Value Set Sc-valueset-research-subject-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-research-subject-status.html

8.23.12 Value Set Sc-valueset-research-subject-status - Status Code Mapping

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.23.13 ResearchSubjectStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-research-subject-status)

Mapping from http://hl7.org/fhir/ValueSet/research-subject-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Indicates the progression of a study subject through a study."

Source Code | Equivalence | Destination Code
candidate, screening, eligible, potential-candidate | equivalent | draft
ineligible | equivalent | declined
on-study, on-study-intervention, on-study-observation | equivalent | active
pending-on-study | equivalent | suspended
withdrawn | equivalent | failed
follow-up | equivalent | replaced
off-study | equivalent | complete

not-registered | equivalent | abandoned

8.24 Resource ChargeItem - Content

URL: https://hl7.org/fhir/R4/chargeitem.html
Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

The resource ChargeItem describes the provision of healthcare provider products for a certain patient, therefore referring not only to the product, but containing in addition details of the provision, like date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the billing process and internal cost allocation.

8.24.1 Scope and Usage

Tracking Financial information is vital in Patient Administration and Finance systems in most Healthcare Organizations. The resource ChargeItem describes the charge for provision of healthcare provider products for a certain patient, therefore referring not only to the product, but containing in addition details of the provision, like date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the billing process and internal cost allocation. They are created as soon as the products are planned or provisioned, references to Encounters and/or Accounts can be maintained in a later process step.

The target of ChargeItem.definition may provide information on the Charge code such as pricing and inclusion/exclusion rules as well as factors that apply under certain conditions. In many cases however this information may have been drawn from sources outside of FHIR depending on the distribution format of the code catalogue. The ChargeItem assumes that such information is either implicitly known by the communicating systems or explicitly shared through the ChargeItem.definition. Therefore explicit pricing information is not shared within the ChargeItem resource. Also, the systems posting the ChargeItems are not expected to apply the rules associated with the charge codes as they may not know the whole context of the patient/encounter to evaluate such rules. It lies within the responsibility of a billing engine, to collect the ChargeItems in the context of an Account or Encounter at a certain point in time (e.g. discharge of the patient) and to evaluate the associated rules resulting in some of the ChargeItems to be set to the status "not billable" in case the rules exclude them from being billed, or to create financial transactions according to base price and factors. Additional references to Encounter/EpisodeOfCare, Patient/Group and Services provide further context to help billing systems determine the appropriate account and establish the clinical/financial context to evaluate the rules associated with the charge codes.

8.24.2 Boundaries and Relationships

This resource is not an actual financial transaction (such as an item on an invoice or any concise monetary amount being transferred from one Account to another) but is the base administrative data that may be used by a billing engine to create the financial transactions based on rules, factors and base prices associated with the charge code.

Unlike the Financial Transaction the ChargeItem primarily describes the provision, whereas the Financial Transaction documents cash flow. Therefore, the Financial Transaction results from ChargeItems created via the subsequent billing- or cost allocation process.

The actual financial transaction resulting from the evaluation of these rules against the clinical and financial context may be represented in formats appropriate to the financial realm. These are considered out of scope for the FHIR Standard, as they are not specific to the healthcare domain. The FHIR Claim resource does contain line items, and this ChargeItem resource provides the source material for the billing engine to create the items on the claim (which may be different due to business rules).

This resource is referenced by itself and Invoice

8.24.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ChargeItem | TU | | DomainResource | Item containing charge code(s) associated with the provision of healthcare provider products Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Business Identifier for item
definitionUri | | 0..* | uri | Defining information about the code of this charge item
definitionCanonical | | 0..* | canonical(ChargeItemDefinition) | Resource defining the code of this ChargeItem
status | ?!Σ | 1..1 | code | planned | billable | not-billable | aborted | billed | entered-in-error | unknown ChargeItemStatus (Required)
partOf | | 0..* | Reference(ChargeItem) | Part of referenced ChargeItem
code | Σ | 1..1 | CodeableConcept | A code that identifies the charge, like a billing code ChargeItemCode (Example)
subject | Σ | 1..1 | Reference(Patient | Group) | Individual service was done for/to
context | Σ | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
occurrence[x] | Σ | 0..1 | | When the charged service was applied
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
performer | | 0..* | BackboneElement | Who performed charged service
function | | 0..1 | CodeableConcept | What type of performance was done Procedure Performer Role Codes (Example)
actor | | 1..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | Individual who was performing
performingOrganization | | 0..1 | Reference(Organization) | Organization providing the charged service
requestingOrganization | | 0..1 | Reference(Organization) | Organization requesting the charged service
costCenter | | 0..1 | Reference(Organization) | Organization that has ownership of the (potential, future) revenue
quantity | Σ | 0..1 | Quantity | Quantity of which the charge item has been serviced
bodysite | Σ | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)
factorOverride | | 0..1 | decimal | Factor overriding the associated rules
priceOverride | | 0..1 | Money | Price overriding the associated rules
overrideReason | | 0..1 | string | Reason for overriding the list price/factor
enterer | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson) | Individual who was entering
enteredDate | Σ | 0..1 | dateTime | Date the charge item was entered
reason | | 0..* | CodeableConcept | Why was the charged service rendered? ICD-10 Codes (Example)
service | | 0..* | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery) | Which rendered service is being charged?
product[x] | | 0..1 | | Product charged FHIR Device Types (Example)
productReference | | | Reference(Device | Medication | Substance)
productCodeableConcept | | | CodeableConcept
account | Σ | 0..* | Reference(Account) | Account to place this charge
note | | 0..* | Annotation | Comments made about the ChargeItem
supportingInformation | | 0..* | Reference(Any) | Further information supporting this charge
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ChargeItem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for item --></identifier>
 <definitionUri value="[uri]"/><!-- 0..* Defining information about the code of this charge item -->
 <definitionCanonical><!-- 0..* canonical(ChargeItemDefinition) Resource defining the code of this ChargeItem
    --></definitionCanonical>
 <status value="[code]"/><!-- 1..1 planned | billable | not-billable | aborted | billed | entered-in-error | unknown -->
 <partOf><!-- 0..* Reference(ChargeItem) Part of referenced ChargeItem --></partOf>
 <code><!-- 1..1 CodeableConcept A code that identifies the charge, like a billing code --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Individual service was done for/to --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the charged service was applied --></occurrence[x]>
 <performer>  <!-- 0..* Who performed charged service -->
  <function><!-- 0..1 CodeableConcept What type of performance was done --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|CareTeam|
    Patient|Device|RelatedPerson) Individual who was performing --></actor>
 </performer>
 <performingOrganization><!-- 0..1 Reference(Organization) Organization providing the charged service
    --></performingOrganization>
 <requestingOrganization><!-- 0..1 Reference(Organization) Organization requesting the charged service
    --></requestingOrganization>
 <costCenter><!-- 0..1 Reference(Organization) Organization that has ownership of the (potential, future) revenue
    --></costCenter>
 <quantity><!-- 0..1 Quantity Quantity of which the charge item has been serviced --></quantity>
 <bodysite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodysite>
 <factorOverride value="[decimal]"/><!-- 0..1 Factor overriding the associated rules -->
 <priceOverride><!-- 0..1 Money Price overriding the associated rules --></priceOverride>
 <overrideReason value="[string]"/><!-- 0..1 Reason for overriding the list price/factor -->
 <enterer><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|Patient|
   Device|RelatedPerson) Individual who was entering --></enterer>
 <enteredDate value="[dateTime]"/><!-- 0..1 Date the charge item was entered -->
 <reason><!-- 0..* CodeableConcept Why was the charged  service rendered? --></reason>
 <service><!-- 0..* Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) Which rendered service is being charged? --></service>
 <product[x]><!-- 0..1 Reference(Device|Medication|Substance)|CodeableConcept Product charged --></product[x]>
 <account><!-- 0..* Reference(Account) Account to place this charge --></account>
 <note><!-- 0..* Annotation Comments made about the ChargeItem --></note>
```

```
 <supportingInformation><!-- 0..* Reference(Any) Further information supporting this charge --></supportingInformation>
</ChargeItem>
```

JSON Template

```
{
  "resourceType" : "ChargeItem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for item
  "definitionUri" : ["<uri>"], // Defining information about the code of this charge item
  "definitionCanonical" : [{ canonical(ChargeItemDefinition) }], // Resource defining the code of this ChargeItem
  "status" : "<code>", // R!  planned | billable | not-billable | aborted | billed | entered-in-error | unknown
  "partOf" : [{ Reference(ChargeItem) }], // Part of referenced ChargeItem
  "code" : { CodeableConcept }, // R!  A code that identifies the charge, like a billing code
  "subject" : { Reference(Patient|Group) }, // R!  Individual service was done for/to
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  // occurrence[x]: When the charged service was applied. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "performer" : [{ // Who performed charged service
    "function" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner|PractitionerRole|Organization|CareTeam|
    Patient|Device|RelatedPerson) } // R!  Individual who was performing
  }],
  "performingOrganization" : { Reference(Organization) }, // Organization providing the charged service
  "requestingOrganization" : { Reference(Organization) }, // Organization requesting the charged service
  "costCenter" : { Reference(Organization) }, // Organization that has ownership of the (potential, future) revenue
  "quantity" : { Quantity }, // Quantity of which the charge item has been serviced
  "bodysite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "factorOverride" : <decimal>, // Factor overriding the associated rules
  "priceOverride" : { Money }, // Price overriding the associated rules
  "overrideReason" : "<string>", // Reason for overriding the list price/factor
  "enterer" : { Reference(Practitioner|PractitionerRole|Organization|Patient|
   Device|RelatedPerson) }, // Individual who was entering
  "enteredDate" : "<dateTime>", // Date the charge item was entered
  "reason" : [{ CodeableConcept }], // Why was the charged  service rendered?
  "service" : [{ Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) }], // Which rendered service is being charged?
  // product[x]: Product charged. One of these 2:
  "productReference" : { Reference(Device|Medication|Substance) },
  "productCodeableConcept" : { CodeableConcept },
  "account" : [{ Reference(Account) }], // Account to place this charge
  "note" : [{ Annotation }], // Comments made about the ChargeItem
  "supportingInformation" : [{ Reference(Any) }] // Further information supporting this charge
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ChargeItem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ChargeItem.identifier [ Identifier ], ... ; # 0..* Business Identifier for item
  fhir:ChargeItem.definitionUri [ uri ], ... ; # 0..* Defining information about the code of this charge item
  fhir:ChargeItem.definitionCanonical [ canonical(ChargeItemDefinition) ], ... ; # 0..* Resource defining the code of
    this ChargeItem
  fhir:ChargeItem.status [ code ]; # 1..1 planned | billable | not-billable | aborted | billed | entered-in-error |
    unknown
  fhir:ChargeItem.partOf [ Reference(ChargeItem) ], ... ; # 0..* Part of referenced ChargeItem
  fhir:ChargeItem.code [ CodeableConcept ]; # 1..1 A code that identifies the charge, like a billing code
  fhir:ChargeItem.subject [ Reference(Patient|Group) ]; # 1..1 Individual service was done for/to
  fhir:ChargeItem.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with event
  # ChargeItem.occurrence[x] : 0..1 When the charged service was applied. One of these 3
    fhir:ChargeItem.occurrenceDateTime [ dateTime ]
    fhir:ChargeItem.occurrencePeriod [ Period ]
    fhir:ChargeItem.occurrenceTiming [ Timing ]
  fhir:ChargeItem.performer [ # 0..* Who performed charged service
    fhir:ChargeItem.performer.function [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:ChargeItem.performer.actor [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson) ]; # 1..1 Individual who
     was performing
  ], ...;
  fhir:ChargeItem.performingOrganization [ Reference(Organization) ]; # 0..1 Organization providing the charged service
  fhir:ChargeItem.requestingOrganization [ Reference(Organization) ]; # 0..1 Organization requesting the charged service
  fhir:ChargeItem.costCenter [ Reference(Organization) ]; # 0..1 Organization that has ownership of the (potential,
    future) revenue
  fhir:ChargeItem.quantity [ Quantity ]; # 0..1 Quantity of which the charge item has been serviced
  fhir:ChargeItem.bodysite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:ChargeItem.factorOverride [ decimal ]; # 0..1 Factor overriding the associated rules
  fhir:ChargeItem.priceOverride [ Money ]; # 0..1 Price overriding the associated rules
  fhir:ChargeItem.overrideReason [ string ]; # 0..1 Reason for overriding the list price/factor
  fhir:ChargeItem.enterer [ Reference(Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson) ]; # 0..1
     Individual who was entering
  fhir:ChargeItem.enteredDate [ dateTime ]; # 0..1 Date the charge item was entered
```

```
  fhir:ChargeItem.reason [ CodeableConcept ], ... ; # 0..* Why was the charged  service rendered?
  fhir:ChargeItem.service [ Reference(DiagnosticReport|ImagingStudy|Immunization|MedicationAdministration|
  MedicationDispense|Observation|Procedure|SupplyDelivery) ], ... ; # 0..* Which rendered service is being charged?
  # ChargeItem.product[x] : 0..1 Product charged. One of these 2
    fhir:ChargeItem.productReference [ Reference(Device|Medication|Substance) ]
    fhir:ChargeItem.productCodeableConcept [ CodeableConcept ]
  fhir:ChargeItem.account [ Reference(Account) ], ... ; # 0..* Account to place this charge
  fhir:ChargeItem.note [ Annotation ], ... ; # 0..* Comments made about the ChargeItem
  fhir:ChargeItem.supportingInformation [ Reference(Any) ], ... ; # 0..* Further information supporting this charge
]
```

Changes since R3

ChargeItem
ChargeItem.identifier | Max Cardinality changed from 1 to *
ChargeItem.definitionUri | Added Element
ChargeItem.definitionCanonical | Added Element
ChargeItem.status | Change value set from http://hl7.org/fhir/ValueSet/chargeitem-status to http://hl7.org/fhir/ValueSet/chargeitem-status|4.0.1
ChargeItem.performer | Added Element
ChargeItem.performer.function | Added Element
ChargeItem.performer.actor | Added Mandatory Element
ChargeItem.costCenter | Added Element
ChargeItem.enterer | Type Reference: Added Target Type PractitionerRole
ChargeItem.product[x] | Added Element
ChargeItem.definition | deleted
ChargeItem.participant | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
ChargeItem | TU | | DomainResource | Item containing charge code(s) associated with the provision of healthcare provider products Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for item
definitionUri | | 0..* | uri | Defining information about the code of this charge item
definitionCanonical | | 0..* | canonical(ChargeItemDefinition) | Resource defining the code of this ChargeItem
status | ?!Σ | 1..1 | code | planned | billable | not-billable | aborted | billed | entered-in-error | unknown ChargeItemStatus (Required)
partOf | | 0..* | Reference(ChargeItem) | Part of referenced ChargeItem
code | Σ | 1..1 | CodeableConcept | A code that identifies the charge, like a billing code ChargeItemCode (Example)
subject | Σ | 1..1 | Reference(Patient | Group) | Individual service was done for/to
context | Σ | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
occurrence[x] | Σ | 0..1 | | When the charged service was applied
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
performer | | 0..* | BackboneElement | Who performed charged service
function | | 0..1 | CodeableConcept | What type of performance was done Procedure Performer Role Codes (Example)
actor | | 1..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | Individual who was performing
performingOrganization | | 0..1 | Reference(Organization) | Organization providing the charged service
requestingOrganization | | 0..1 | Reference(Organization) | Organization requesting the charged service
costCenter | | 0..1 | Reference(Organization) | Organization that has ownership of the (potential, future) revenue
quantity | Σ | 0..1 | Quantity | Quantity of which the charge item has been serviced
bodysite | Σ | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)
factorOverride | | 0..1 | decimal | Factor overriding the associated rules
priceOverride | | 0..1 | Money | Price overriding the associated rules
overrideReason | | 0..1 | string | Reason for overriding the list price/factor
enterer | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson) | Individual who was entering
enteredDate | Σ | 0..1 | dateTime | Date the charge item was entered
reason | | 0..* | CodeableConcept | Why was the charged service rendered? ICD-10 Codes (Example)
service | | 0..* | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery) | Which rendered service is being charged?
product[x] | | 0..1 | | Product charged FHIR Device Types (Example)
productReference | | | Reference(Device | Medication | Substance)
productCodeableConcept | | | CodeableConcept
account | Σ | 0..* | Reference(Account) | Account to place this charge
note | | 0..* | Annotation | Comments made about the ChargeItem
supportingInformation | | 0..* | Reference(Any) | Further information supporting this charge
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ChargeItem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for item --></identifier>
 <definitionUri value="[uri]"/><!-- 0..* Defining information about the code of this charge item -->
 <definitionCanonical><!-- 0..* canonical(ChargeItemDefinition) Resource defining the code of this ChargeItem
    --></definitionCanonical>
 <status value="[code]"/><!-- 1..1 planned | billable | not-billable | aborted | billed | entered-in-error | unknown -->
 <partOf><!-- 0..* Reference(ChargeItem) Part of referenced ChargeItem --></partOf>
 <code><!-- 1..1 CodeableConcept A code that identifies the charge, like a billing code --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Individual service was done for/to --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the charged service was applied --></occurrence[x]>
 <performer>  <!-- 0..* Who performed charged service -->
  <function><!-- 0..1 CodeableConcept What type of performance was done --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|CareTeam|
    Patient|Device|RelatedPerson) Individual who was performing --></actor>
 </performer>
 <performingOrganization><!-- 0..1 Reference(Organization) Organization providing the charged service
    --></performingOrganization>
 <requestingOrganization><!-- 0..1 Reference(Organization) Organization requesting the charged service
    --></requestingOrganization>
 <costCenter><!-- 0..1 Reference(Organization) Organization that has ownership of the (potential, future) revenue
    --></costCenter>
 <quantity><!-- 0..1 Quantity Quantity of which the charge item has been serviced --></quantity>
 <bodysite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodysite>
 <factorOverride value="[decimal]"/><!-- 0..1 Factor overriding the associated rules -->
 <priceOverride><!-- 0..1 Money Price overriding the associated rules --></priceOverride>
 <overrideReason value="[string]"/><!-- 0..1 Reason for overriding the list price/factor -->
 <enterer><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|Patient|
   Device|RelatedPerson) Individual who was entering --></enterer>
 <enteredDate value="[dateTime]"/><!-- 0..1 Date the charge item was entered -->
 <reason><!-- 0..* CodeableConcept Why was the charged  service rendered? --></reason>
 <service><!-- 0..* Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) Which rendered service is being charged? --></service>
 <product[x]><!-- 0..1 Reference(Device|Medication|Substance)|CodeableConcept Product charged --></product[x]>
 <account><!-- 0..* Reference(Account) Account to place this charge --></account>
 <note><!-- 0..* Annotation Comments made about the ChargeItem --></note>
 <supportingInformation><!-- 0..* Reference(Any) Further information supporting this charge --></supportingInformation>
</ChargeItem>
```

JSON Template

```
{
  "resourceType" : "ChargeItem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for item
  "definitionUri" : ["<uri>"], // Defining information about the code of this charge item
  "definitionCanonical" : [{ canonical(ChargeItemDefinition) }], // Resource defining the code of this ChargeItem
  "status" : "<code>", // R!  planned | billable | not-billable | aborted | billed | entered-in-error | unknown
  "partOf" : [{ Reference(ChargeItem) }], // Part of referenced ChargeItem
  "code" : { CodeableConcept }, // R!  A code that identifies the charge, like a billing code
  "subject" : { Reference(Patient|Group) }, // R!  Individual service was done for/to
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  // occurrence[x]: When the charged service was applied. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "performer" : [{ // Who performed charged service
    "function" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner|PractitionerRole|Organization|CareTeam|
    Patient|Device|RelatedPerson) } // R!  Individual who was performing
  }],
  "performingOrganization" : { Reference(Organization) }, // Organization providing the charged service
  "requestingOrganization" : { Reference(Organization) }, // Organization requesting the charged service
  "costCenter" : { Reference(Organization) }, // Organization that has ownership of the (potential, future) revenue
  "quantity" : { Quantity }, // Quantity of which the charge item has been serviced
  "bodysite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "factorOverride" : <decimal>, // Factor overriding the associated rules
  "priceOverride" : { Money }, // Price overriding the associated rules
  "overrideReason" : "<string>", // Reason for overriding the list price/factor
  "enterer" : { Reference(Practitioner|PractitionerRole|Organization|Patient|
   Device|RelatedPerson) }, // Individual who was entering
  "enteredDate" : "<dateTime>", // Date the charge item was entered
  "reason" : [{ CodeableConcept }], // Why was the charged  service rendered?
  "service" : [{ Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) }], // Which rendered service is being charged?
  // product[x]: Product charged. One of these 2:
  "productReference" : { Reference(Device|Medication|Substance) },
  "productCodeableConcept" : { CodeableConcept },
  "account" : [{ Reference(Account) }], // Account to place this charge
  "note" : [{ Annotation }], // Comments made about the ChargeItem
  "supportingInformation" : [{ Reference(Any) }] // Further information supporting this charge
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ChargeItem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ChargeItem.identifier [ Identifier ], ... ; # 0..* Business Identifier for item
  fhir:ChargeItem.definitionUri [ uri ], ... ; # 0..* Defining information about the code of this charge item
  fhir:ChargeItem.definitionCanonical [ canonical(ChargeItemDefinition) ], ... ; # 0..* Resource defining the code of
    this ChargeItem
  fhir:ChargeItem.status [ code ]; # 1..1 planned | billable | not-billable | aborted | billed | entered-in-error |
    unknown
  fhir:ChargeItem.partOf [ Reference(ChargeItem) ], ... ; # 0..* Part of referenced ChargeItem
  fhir:ChargeItem.code [ CodeableConcept ]; # 1..1 A code that identifies the charge, like a billing code
  fhir:ChargeItem.subject [ Reference(Patient|Group) ]; # 1..1 Individual service was done for/to
  fhir:ChargeItem.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with event
  # ChargeItem.occurrence[x] : 0..1 When the charged service was applied. One of these 3
    fhir:ChargeItem.occurrenceDateTime [ dateTime ]
    fhir:ChargeItem.occurrencePeriod [ Period ]
    fhir:ChargeItem.occurrenceTiming [ Timing ]
  fhir:ChargeItem.performer [ # 0..* Who performed charged service
    fhir:ChargeItem.performer.function [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:ChargeItem.performer.actor [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson) ]; # 1..1 Individual who
     was performing
  ], ...;
  fhir:ChargeItem.performingOrganization [ Reference(Organization) ]; # 0..1 Organization providing the charged service
  fhir:ChargeItem.requestingOrganization [ Reference(Organization) ]; # 0..1 Organization requesting the charged service
  fhir:ChargeItem.costCenter [ Reference(Organization) ]; # 0..1 Organization that has ownership of the (potential,
    future) revenue
  fhir:ChargeItem.quantity [ Quantity ]; # 0..1 Quantity of which the charge item has been serviced
  fhir:ChargeItem.bodysite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:ChargeItem.factorOverride [ decimal ]; # 0..1 Factor overriding the associated rules
  fhir:ChargeItem.priceOverride [ Money ]; # 0..1 Price overriding the associated rules
  fhir:ChargeItem.overrideReason [ string ]; # 0..1 Reason for overriding the list price/factor
  fhir:ChargeItem.enterer [ Reference(Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson) ]; # 0..1
     Individual who was entering
  fhir:ChargeItem.enteredDate [ dateTime ]; # 0..1 Date the charge item was entered
  fhir:ChargeItem.reason [ CodeableConcept ], ... ; # 0..* Why was the charged  service rendered?
  fhir:ChargeItem.service [ Reference(DiagnosticReport|ImagingStudy|Immunization|MedicationAdministration|
  MedicationDispense|Observation|Procedure|SupplyDelivery) ], ... ; # 0..* Which rendered service is being charged?
  # ChargeItem.product[x] : 0..1 Product charged. One of these 2
    fhir:ChargeItem.productReference [ Reference(Device|Medication|Substance) ]
    fhir:ChargeItem.productCodeableConcept [ CodeableConcept ]
  fhir:ChargeItem.account [ Reference(Account) ], ... ; # 0..* Account to place this charge
  fhir:ChargeItem.note [ Annotation ], ... ; # 0..* Comments made about the ChargeItem
  fhir:ChargeItem.supportingInformation [ Reference(Any) ], ... ; # 0..* Further information supporting this charge
]
```

Changes since Release 3

ChargeItem
ChargeItem.identifier | Max Cardinality changed from 1 to *
ChargeItem.definitionUri | Added Element
ChargeItem.definitionCanonical | Added Element
ChargeItem.status | Change value set from http://hl7.org/fhir/ValueSet/chargeitem-status to http://hl7.org/fhir/ValueSet/chargeitem-status|4.0.1
ChargeItem.performer | Added Element
ChargeItem.performer.function | Added Element
ChargeItem.performer.actor | Added Mandatory Element
ChargeItem.costCenter | Added Element
ChargeItem.enterer | Type Reference: Added Target Type PractitionerRole
ChargeItem.product[x] | Added Element
ChargeItem.definition | deleted
ChargeItem.participant | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.24.3.1 Terminology Bindings

Path | Definition | Type | Reference
ChargeItem.status | Codes identifying the lifecycle stage of a ChargeItem. | Required | ChargeItemStatus
ChargeItem.code | Example set of codes that can be used for billing purposes. | Example | ChargeItemCode
ChargeItem.performer.function | Codes describing the types of functional roles performers can take on when performing events. | Example | ProcedurePerformerRoleCodes

ChargeItem.bodysite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures
ChargeItem.reason | Example binding for reason. | Example | ICD-10Codes
ChargeItem.product[x] | Example binding for product type. | Example | FHIRDeviceTypes

8.24.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
account | reference | Account to place this charge | ChargeItem.account (Account)
code | token | A code that identifies the charge, like a billing code | ChargeItem.code
context | reference | Encounter / Episode associated with event | ChargeItem.context (EpisodeOfCare, Encounter)
entered-date | date | Date the charge item was entered | ChargeItem.enteredDate
enterer | reference | Individual who was entering | ChargeItem.enterer (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
factor-override | number | Factor overriding the associated rules | ChargeItem.factorOverride
identifier | token | Business Identifier for item | ChargeItem.identifier
occurrence | date | When the charged service was applied | ChargeItem.occurrence
patient | reference | Individual service was done for/to | ChargeItem.subject.where(resolve() is Patient) (Patient)
performer-actor | reference | Individual who was performing | ChargeItem.performer.actor (Practitioner, Organization, CareTeam, Device, Patient, PractitionerRole, RelatedPerson)
performer-function | token | What type of performance was done | ChargeItem.performer.function
performing-organization | reference | Organization providing the charged service | ChargeItem.performingOrganization (Organization)
price-override | quantity | Price overriding the associated rules | ChargeItem.priceOverride
quantity | quantity | Quantity of which the charge item has been serviced | ChargeItem.quantity
requesting-organization | reference | Organization requesting the charged service | ChargeItem.requestingOrganization (Organization)
service | reference | Which rendered service is being charged? | ChargeItem.service (Immunization, MedicationDispense, SupplyDelivery, Observation, DiagnosticReport, ImagingStudy, MedicationAdministration, Procedure)
subject | reference | Individual service was done for/to | ChargeItem.subject (Group, Patient)

8.24.5 Resource ChargeItem - Examples

URL: https://hl7.org/fhir/R4/chargeitem-examples.html

8.24.5 Resource ChargeItem - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Example of chargeitem | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.24.6 Resource ChargeItem - Detailed Descriptions

URL: https://hl7.org/fhir/R4/chargeitem-definitions.html

8.24.6 Resource ChargeItem - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the ChargeItem resource.

ChargeItem
Element Id | ChargeItem
Definition | The resource ChargeItem describes the provision of healthcare provider products for a certain patient, therefore referring not only to the product, but containing in addition details of the provision, like date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the billing process and internal cost allocation.
Cardinality | 0..*
Type | DomainResource
ChargeItem.identifier
Element Id | ChargeItem.identifier
Definition | Identifiers assigned to this event performer or other systems.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the charge Item as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
ChargeItem.definitionUri
Element Id | ChargeItem.definitionUri

Definition | References the (external) source of pricing information, rules of application for the code this ChargeItem uses.
Cardinality | 0..*
Type | uri
Summary | false
ChargeItem.definitionCanonical
Element Id | ChargeItem.definitionCanonical
Definition | References the source of pricing information, rules of application for the code this ChargeItem uses.
Cardinality | 0..*
Type | canonical(ChargeItemDefinition)
Summary | false
ChargeItem.status
Element Id | ChargeItem.status
Definition | The current state of the ChargeItem.
Cardinality | 1..1
Terminology Binding | ChargeItemStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | Unknown does not represent "other" - one of the defined statuses must apply. Unknown is used when the authoring system is not sure what the current status is. This element is labeled as a modifier because the status contains the code entered-in-error that marks the charge item as not currently valid.
ChargeItem.partOf
Element Id | ChargeItem.partOf
Definition | ChargeItems can be grouped to larger ChargeItems covering the whole set.
Cardinality | 0..*
Type | Reference(ChargeItem)
Requirements | E.g. Drug administration as part of a procedure, procedure as part of observation, etc.
Alternate Names | container
Summary | false
ChargeItem.code
Element Id | ChargeItem.code
Definition | A code that identifies the charge, like a billing code.
Cardinality | 1..1
Terminology Binding | ChargeItemCode (Example)
Type | CodeableConcept
Alternate Names | type
Summary | true
ChargeItem.subject
Element Id | ChargeItem.subject
Definition | The individual or set of individuals the action is being or was performed on.
Cardinality | 1..1
Type | Reference(Patient | Group)
Requirements | Links the event to the Patient context.
Alternate Names | patient
Summary | true
ChargeItem.context
Element Id | ChargeItem.context
Definition | The encounter or episode of care that establishes the context for this event.
Cardinality | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Requirements | Links the request to the Encounter context.
Alternate Names | encounter
Summary | true
ChargeItem.occurrence[x]
Element Id | ChargeItem.occurrence[x]
Definition | Date/time(s) or duration when the charged service was applied.
Cardinality | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | timing
Summary | true
Comments | The list of types may be constrained as appropriate for the type of charge item.
ChargeItem.performer
Element Id | ChargeItem.performer
Definition | Indicates who or what performed or participated in the charged service.
Cardinality | 0..*
Summary | false
ChargeItem.performer.function
Element Id | ChargeItem.performer.function
Definition | Describes the type of performance or participation(e.g. primary surgeon, anesthesiologiest, etc.).
Cardinality | 0..1

Terminology Binding | Procedure Performer Role Codes (Example)
Type | CodeableConcept
Summary | false
ChargeItem.performer.actor
Element Id | ChargeItem.performer.actor
Definition | The device, practitioner, etc. who performed or participated in the service.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson)
Summary | false
ChargeItem.performingOrganization
Element Id | ChargeItem.performingOrganization
Definition | The organization requesting the service.
Cardinality | 0..1
Type | Reference(Organization)
Summary | false
Comments | Practitioners and Devices can be associated with multiple organizations. It has to be made clear, on behalf of which Organization the services have been rendered.
ChargeItem.requestingOrganization
Element Id | ChargeItem.requestingOrganization
Definition | The organization performing the service.
Cardinality | 0..1
Type | Reference(Organization)
Summary | false
Comments | The rendered Service might not be associated with a Request. This property indicates which Organization requested the services to be rendered. (In many cases, this may just be the Department associated with the Encounter.location).
ChargeItem.costCenter
Element Id | ChargeItem.costCenter
Definition | The financial cost center permits the tracking of charge attribution.
Cardinality | 0..1
Type | Reference(Organization)
Summary | false
Comments | The costCenter could either be given as a reference to an Organization(Role) resource or as the identifier of the cost center determined by Reference.identifier.value and Reference.identifier.system, depending on use case requirements.
ChargeItem.quantity
Element Id | ChargeItem.quantity
Definition | Quantity of which the charge item has been serviced.
Cardinality | 0..1
Type | Quantity
Summary | true
Comments | In many cases this may just be a value, if the underlying units are implicit in the definition of the charge item code.
ChargeItem.bodysite
Element Id | ChargeItem.bodysite
Definition | The anatomical location where the related service has been applied.
Cardinality | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Comments | Only used if not implicit in code found in Condition.code. If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension bodySite. May be a summary code, or a reference to a very precise definition of the location, or both.
ChargeItem.factorOverride
Element Id | ChargeItem.factorOverride
Definition | Factor overriding the factor determined by the rules associated with the code.
Cardinality | 0..1
Type | decimal
Summary | false
Comments | There is no reason to carry the factor in the instance of a ChargeItem unless special circumstances require a manual override. The factors are usually defined by a set of rules in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overridden.
ChargeItem.priceOverride
Element Id | ChargeItem.priceOverride
Definition | Total price of the charge overriding the list price associated with the code.
Cardinality | 0..1
Type | Money
Summary | false
Comments | There is no reason to carry the price in the instance of a ChargeItem unless circumstances require a manual override. The list prices or are usually defined in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overridden.
ChargeItem.overrideReason
Element Id | ChargeItem.overrideReason
Definition | If the list price or the rule-based factor associated with the code is overridden, this attribute can capture a text to indicate the reason for this action.
Cardinality | 0..1

Type | string
Summary | false
Comments | Derived Profiles may choose to add invariants requiring this field to be populated if either priceOverride or factorOverride have been filled.
ChargeItem.enterer
Element Id | ChargeItem.enterer
Definition | The device, practitioner, etc. who entered the charge item.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson)
Summary | true
Comments | The enterer is also the person considered responsible for factor/price overrides if applicable.
ChargeItem.enteredDate
Element Id | ChargeItem.enteredDate
Definition | Date the charge item was entered.
Cardinality | 0..1
Type | dateTime
Summary | true
Comments | The actual date when the service associated with the charge has been rendered is captured in occurrence[x].
ChargeItem.reason
Element Id | ChargeItem.reason
Definition | Describes why the event occurred in coded or textual form.
Cardinality | 0..*
Terminology Binding | ICD-10 Codes (Example)
Type | CodeableConcept
Summary | false
Comments | If the application of the charge item requires a reason to be given, it can be captured here. Textual reasons can be captured using reasonCode.text.
ChargeItem.service
Element Id | ChargeItem.service
Definition | Indicated the rendered service that caused this charge.
Cardinality | 0..*
Type | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery)
Summary | false
ChargeItem.product[x]
Element Id | ChargeItem.product[x]
Definition | Identifies the device, food, drug or other product being charged either by type code or reference to an instance.
Cardinality | 0..1
Terminology Binding | FHIR Device Types (Example)
Type | Reference(Device | Medication | Substance)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | false
ChargeItem.account
Element Id | ChargeItem.account
Definition | Account into which this ChargeItems belongs.
Cardinality | 0..*
Type | Reference(Account)
Summary | true
Comments | Systems posting the ChargeItems might not always be able to determine, which accounts the Items need to be places into. It is up to the postprocessing Financial System to apply internal rules to decide based on the Encounter/EpisodeOfCare/Patient/Coverage context and the type of ChargeItem, which Account is appropriate.
ChargeItem.note
Element Id | ChargeItem.note
Definition | Comments made about the event by the performer, subject or other participants.
Cardinality | 0..*
Type | Annotation
Summary | false
ChargeItem.supportingInformation
Element Id | ChargeItem.supportingInformation
Definition | Further information supporting this charge.
Cardinality | 0..*
Type | Reference(Any)
Summary | false

8.24.7 Resource ChargeItem - Mappings

URL: https://hl7.org/fhir/R4/chargeitem-mappings.html

8.24.7 Resource ChargeItem - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the chargeitem resource (see Mappings to Other Standards for further information & status).

8.24.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ChargeItem
identifier | FiveWs.identifier
status | FiveWs.status
code | FiveWs.what[x]
subject | FiveWs.subject[x]
context | FiveWs.context
occurrence[x] | FiveWs.done[x]
actor | FiveWs.actor
enterer | FiveWs.actor
reason | FiveWs.why[x]
service | FiveWs.why[x]

8.24.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ChargeItem | Event
identifier | Event.identifier
definitionUri | Event.instantiates
definitionCanonical | Event.instantiates
status | Event.status
partOf | Event.partOf
code | Event.code
subject | Event.subject
context | Event.context
occurrence[x] | Event.occurrence[x]
performer | Event.performer
function | Event.performer.function
actor | Event.performer.actor
performingOrganization | Event.performer.actor
reason | Event.reasonCode
service | Event.reasonReference
note | Event.note
supportingInformation | Event.supportingInfo

8.24.7.3 HL7 v2 Mapping (http://hl7.org/v2)

ChargeItem | FT1
identifier | FT1.2
definitionUri | Varies by domain
definitionCanonical | Varies by domain
status | Varies by domain
partOf | Varies by domain
code | FT1.7
subject | PAT in proximity to FT1 segment
context | PV1 in proximity to FT1 segment
occurrence[x] | Varies by domain
performer | EVN.5 or by domain
function | Varies by domain
actor | EVN.5 or by domain
performingOrganization | N/A
requestingOrganization | N/A
costCenter | FT1.13
quantity | FT1.10
bodysite
factorOverride | FT1.26
priceOverride | FT1.22
overrideReason
enterer | EVN.5 or by domain
enteredDate
reason | EVN.4 or by domain
service | EVN.4 or by domain
product[x]
account
note | NTE

supportingInformation

8.24.7.4 RIM Mapping (http://hl7.org/v3)

ChargeItem | Act[moodCode=EVN]
identifier | .identifier
definitionUri | .outboundRelationship[typeCode=DEFN].target
definitionCanonical | .outboundRelationship[typeCode=DEFN].target
status | .status
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
code | .code
subject | .participation[typeCode=SBJ].role
context | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
occurrence[x] | .effectiveTime
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
function | .participation.functionCode
actor | .player
performingOrganization | .scoper
requestingOrganization | .scoper
bodysite | .targetBodySiteCode
enterer | .player
reason | .reasonCode
service | .outboundRelationship[typeCode=RSON].target
product[x] | .participation[typeCode=CSM].role
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]

8.24.8 Resource ChargeItem - Extensions & Profiles

URL: https://hl7.org/fhir/R4/chargeitem-profiles.html

8.24.8 Resource ChargeItem - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the ChargeItem resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.24.8.1 Profiles

No Profiles defined for this resource

8.24.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.24.8.3 Search Extensions

No Search Extensions defined for this resource

8.24.9 Resource ChargeItem - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/chargeitem-version-maps.html

No validation errors - all conversions are clean

8.24.9 Resource ChargeItem - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for ChargeItem.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.24.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ChargeItem3to4" = "R3 to R4 Conversions for ChargeItem"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ChargeItem" alias ChargeItemR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ChargeItem" alias ChargeItem as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ChargeItem(source src : ChargeItemR3, target tgt : ChargeItem) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.definitionUri;
  src.status -> tgt.status;
  src.partOf -> tgt.partOf;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.context -> tgt.context;
  src.occurrence -> tgt.occurrence;
  src.participant as s -> tgt.performer as t then ChargeItemPerformer(s, t);
  src.performingOrganization -> tgt.performingOrganization;
  src.requestingOrganization -> tgt.requestingOrganization;
  src.costCenter -> tgt.costCenter;
  src.quantity -> tgt.quantity;
  src.bodysite -> tgt.bodysite;
  src.factorOverride -> tgt.factorOverride;
  src.priceOverride -> tgt.priceOverride;
  src.overrideReason -> tgt.overrideReason;
  src.enterer -> tgt.enterer;
  src.enteredDate -> tgt.enteredDate;
  src.reason -> tgt.reason;
  src.service -> tgt.service;
  src.product -> tgt.product;
  src.account -> tgt.account;
  src.note -> tgt.note;
  src.supportingInformation -> tgt.supportingInformation;
}

group ChargeItemPerformer(source src, target tgt) extends BackboneElement {
  src.role -> tgt.function;
  src.actor -> tgt.actor;
}
```

8.24.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ChargeItem4to3" = "R4 to R3 Conversion for ChargeItem"

uses "http://hl7.org/fhir/StructureDefinition/ChargeItem" alias ChargeItem as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ChargeItem" alias ChargeItemR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ChargeItem(source src : ChargeItemR3, target tgt : ChargeItem) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.definitionUri -> tgt.definition;
  src.status -> tgt.status;
  src.partOf -> tgt.partOf;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.context -> tgt.context;
  src.occurrence -> tgt.occurrence;
  src.performer as s -> tgt.participant as t then ChargeItemPerformer(s, t);
  src.performingOrganization -> tgt.performingOrganization;
  src.requestingOrganization -> tgt.requestingOrganization;
  src.costCenter -> tgt.costCenter;
  src.quantity -> tgt.quantity;
  src.bodysite -> tgt.bodysite;
  src.factorOverride -> tgt.factorOverride;
  src.priceOverride -> tgt.priceOverride;
  src.overrideReason -> tgt.overrideReason;
  src.enterer -> tgt.enterer;
  src.enteredDate -> tgt.enteredDate;
  src.reason -> tgt.reason;
  src.service -> tgt.service;
  src.product -> tgt.product;
  src.account -> tgt.account;
  src.note -> tgt.note;
  src.supportingInformation -> tgt.supportingInformation;
}

group ChargeItemPerformer(source src, target tgt) extends BackboneElement {
  src.function -> tgt.role;
  src.actor -> tgt.actor;
}
```

8.24.9.3 R4 Validation Errors

8.24.10 JSON Schema for ChargeItem

URL: https://hl7.org/fhir/R4/chargeitem.schema.json.html

8.24.10 JSON Schema for ChargeItem

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for ChargeItem, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ChargeItem",
  "$ref": "#/definitions/ChargeItem",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ChargeItem": {
      "description": "The resource ChargeItem describes the provision of healthcare provider products for a certain
    patient, therefore referring not only to the product, but containing in addition details of the provision, like
    date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the
    billing process and internal cost allocation.",
      "properties": {
        "resourceType": {
          "description": "This is a ChargeItem resource",
          "const": "ChargeItem"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
```

```
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this event performer or other systems.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "definitionUri": {
      "description": "References the (external) source of pricing information, rules of application for the code
this ChargeItem uses.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_definitionUri": {
      "description": "Extensions for definitionUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "definitionCanonical": {
      "description": "References the source of pricing information, rules of application for the code this
ChargeItem uses.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the ChargeItem.",
      "enum": [
        "planned",
        "billable",
        "not-billable",
        "aborted",
        "billed",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "partOf": {
      "description": "ChargeItems can be grouped to larger ChargeItems covering the whole set.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "code": {
      "description": "A code that identifies the charge, like a billing code.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The individual or set of individuals the action is being or was performed on.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "context": {
      "description": "The encounter or episode of care that establishes the context for this event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "occurrenceDateTime": {
      "description": "Date/time(s) or duration when the charged service was applied.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "Date/time(s) or duration when the charged service was applied.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "occurrenceTiming": {
      "description": "Date/time(s) or duration when the charged service was applied.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "performer": {
```

```
      "description": "Indicates who or what performed or participated in the charged service.",
      "items": {
        "$ref": "#/definitions/ChargeItem_Performer"
      },
      "type": "array"
    },
    "performingOrganization": {
      "description": "The organization requesting the service.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "requestingOrganization": {
      "description": "The organization performing the service.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "costCenter": {
      "description": "The financial cost center permits the tracking of charge attribution.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "quantity": {
      "description": "Quantity of which the charge item has been serviced.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "bodysite": {
      "description": "The anatomical location where the related service has been applied.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "factorOverride": {
      "description": "Factor overriding the factor determined by the rules associated with the code.",
      "$ref": "#/definitions/decimal"
    },
    "_factorOverride": {
      "description": "Extensions for factorOverride",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priceOverride": {
      "description": "Total price of the charge overriding the list price associated with the code.",
      "$ref": "Money.schema.json#/definitions/Money"
    },
    "overrideReason": {
      "description": "If the list price or the rule-based factor associated with the code is overridden, this
attribute can capture a text to indicate the  reason for this action.",
      "$ref": "#/definitions/string"
    },
    "_overrideReason": {
      "description": "Extensions for overrideReason",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "enterer": {
      "description": "The device, practitioner, etc. who entered the charge item.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "enteredDate": {
      "description": "Date the charge item was entered.",
      "$ref": "#/definitions/dateTime"
    },
    "_enteredDate": {
      "description": "Extensions for enteredDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reason": {
      "description": "Describes why the event occurred in coded or textual form.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "service": {
      "description": "Indicated the rendered service that caused this charge.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "productReference": {
      "description": "Identifies the device, food, drug or other product being charged either by type code or
reference to an instance.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "productCodeableConcept": {
      "description": "Identifies the device, food, drug or other product being charged either by type code or
reference to an instance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "account": {
      "description": "Account into which this ChargeItems belongs.",
      "items": {
```

```
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "note": {
          "description": "Comments made about the event by the performer, subject or other participants.",
          "items": {
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        },
        "supportingInformation": {
          "description": "Further information supporting this charge.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "code",
        "subject",
        "resourceType"
      ]
    },
    "ChargeItem_Performer": {
      "description": "The resource ChargeItem describes the provision of healthcare provider products for a certain
    patient, therefore referring not only to the product, but containing in addition details of the provision, like
    date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the
    billing process and internal cost allocation.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "function": {
          "description": "Describes the type of performance or participation(e.g. primary surgeon, anesthesiologiest,
    etc.).",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "actor": {
          "description": "The device, practitioner, etc. who performed or participated in the service.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "actor"
      ]
    }
  }
}
```


8.24.11 Resource ChargeItem - Maturity Dependency View

URL: https://hl7.org/fhir/R4/chargeitem-dependencies.html

8.24.11 Resource ChargeItem - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for ChargeItem FMM level 0

Name | Card. | Type | Dependency Analysis
ChargeItem | | DomainResource
identifier | 0..* | Identifier
definitionUri | 0..* | uri
definitionCanonical | 0..* | canonical(ChargeItemDefinition)
status | 1..1 | code
partOf | 0..* | Reference(ChargeItem)
code | 1..1 | CodeableConcept
subject | 1..1 | Reference(Patient | Group)
context | 0..1 | Reference(Encounter | EpisodeOfCare)
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
occurrenceTiming | | Timing
performer | 0..* | BackboneElement
function | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson)
performingOrganization | 0..1 | Reference(Organization)
requestingOrganization | 0..1 | Reference(Organization)
costCenter | 0..1 | Reference(Organization)
quantity | 0..1 | Quantity
bodysite | 0..* | CodeableConcept
factorOverride | 0..1 | decimal
priceOverride | 0..1 | Money
overrideReason | 0..1 | string
enterer | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson)
enteredDate | 0..1 | dateTime
reason | 0..* | CodeableConcept
service | 0..* | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery)
product[x] | 0..1
productReference | | Reference(Device | Medication | Substance
productCodeableConcept | | CodeableConcept
account | 0..* | Reference(Account)
note | 0..* | Annotation
supportingInformation | 0..* | Reference(Any)

8.24.12 Value Set Sc-valueset-chargeitem-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-chargeitem-status.html

8.24.12 Value Set Sc-valueset-chargeitem-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.24.13 ChargeItemStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-chargeitem-status)

Mapping from http://hl7.org/fhir/ValueSet/chargeitem-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes identifying the lifecycle stage of a ChargeItem."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
planned | equivalent | planned
not-billable | equivalent | declined
billable | equivalent | active
aborted | equivalent | failed
billed | equivalent | complete
unknown | equivalent | unknown

8.25 Resource ChargeItemDefinition - Content

URL: https://hl7.org/fhir/R4/chargeitemdefinition.html
Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The ChargeItemDefinition resource provides the properties that apply to the (billing) codes necessary to calculate costs and prices. The properties may differ largely depending on type and realm, therefore this resource gives only a rough structure and requires profiling for each type of billing code system.

8.25.1 Scope and Usage

The ChargeItemDefinition holds all relevant billing information pertaining to a billing code or a product.

Many billing systems have rules associated with the code which may define whether a billing code is applicable in a specific context or whether certain billing details (surcharges, discounts) apply. ChargeItemDefinitions may also be used to hold pricing information pertaining to only one specific instance of a product (e.g. Device, Medication or Substance) which may be a requirement for custom made devices or medications or other products where prices vary by instance. Rules for applicability may include

- a billing code is only applicable in an outpatient setting
- a surcharge that applies only to services provided outside regular business hours
- a discount that applies only to patients up to the age of 14
- a billing code that may only be charged once per Encounter
- a billing code that can only be charged in combination with another code from the same billing system

A billing engine will apply the ChargeItemDefinitions to all ChargeItems within a specific Account in order to select all billable items an acquire the pricing details to create an Invoice. However, ChargeItems may be validated against a ChargeItemDefinition at any point in time, using the $apply-Operation

8.25.2 Boundaries and Relationships

While the ChargeItem resource represents the occurrence of a billing code (for services rendered or products supplied to a specific patient), the ChargeItemDefinition represents the properties and rules that a billing engine needs to determine the pricing and applicability of a billing code in the context of a specific setting.

Billing Codes may be defined in a CodeSystem Resource, however, each of these codes - if it is used in a financial context - can have a large number of business rules and properties associated with it, that lay outside the scope of CodeSystem and may change in time independently from the underlying Code's validity period. Even though CodeSystem can associate a concept with any number of additional properties, such a construct would be very hard to maintain. This is why the ChargItemDefinition has been introduced. It allows to maintain the attributes of each billing code independently and has a wider scope to also include business rules.

In most use cases, billing codes will be both defined as semantical concepts in a CodeSystem and as a financial artifact in ChargeItemDefinition.

ChargeItemDefinitions that pertain to codes from the same CodeSystem may be aggregated using the Catalog profile for Composition.

Resources representing products that can be sold or billed (Device, Medication, Substance...), do not specify financial details for the product instance. Most of the time, such products are not billed seperately but are a part of a generic billing code that covers both the product and the service rendered. In some occasions however, product prices may vary depending on type or manufactuer or even on instance (e.g. custom made products). For these cases, ChargeItemDefinitions may be created to specify the financial details for a product kind, type or instance.

This resource is referenced by ChargeItem and itself

8.25.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ChargeItemDefinition | ITU | | DomainResource | Definition of properties and rules about how the price and the applicability of a ChargeItem can be determined + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this charge item definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the charge item definition
version | Σ | 0..1 | string | Business version of the charge item definition
title | Σ | 0..1 | string | Name for this charge item definition (human friendly)

derivedFromUri | Σ | 0..* | uri | Underlying externally-defined charge item definition
partOf | Σ | 0..* | canonical(ChargeItemDefinition) | A larger definition of which this particular definition is a component or step
replaces | Σ | 0..* | canonical(ChargeItemDefinition) | Completed or terminated request(s) whose function is taken by this new request
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the charge item definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for charge item definition (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the charge item definition was approved by publisher
lastReviewDate | | 0..1 | date | When the charge item definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the charge item definition is expected to be used
code | Σ | 0..1 | CodeableConcept | Billing codes or product types this definition applies to ChargeItemCode (Example)
instance | | 0..* | Reference(Medication | Substance | Device) | Instances this definition applies to
applicability | | 0..* | BackboneElement | Whether or not the billing code is applicable
description | | 0..1 | string | Natural language description of the condition
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | Boolean-valued expression
propertyGroup | | 0..* | BackboneElement | Group of properties which are applicable under the same conditions
applicability | | 0..* | see applicability | Conditions under which the priceComponent is applicable
priceComponent | | 0..* | BackboneElement | Components of total line item price
type | | 1..1 | code | base | surcharge | deduction | discount | tax | informational InvoicePriceComponentType (Required)
code | | 0..1 | CodeableConcept | Code identifying the specific component
factor | | 0..1 | decimal | Factor used for calculating this component
amount | | 0..1 | Money | Monetary amount associated with this component
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ChargeItemDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this charge item definition, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the charge item definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the charge item definition -->
 <title value="[string]"/><!-- 0..1 Name for this charge item definition (human friendly) -->
 <derivedFromUri value="[uri]"/><!-- 0..* Underlying externally-defined charge item definition -->
 <partOf><!-- 0..* canonical(ChargeItemDefinition) A larger definition of which this particular definition is a
    component or step --></partOf>
 <replaces><!-- 0..* canonical(ChargeItemDefinition) Completed or terminated request(s) whose function is taken by this
    new request --></replaces>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the charge item definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for charge item definition (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the charge item definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the charge item definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the charge item definition is expected to be used --></effectivePeriod>
 <code><!-- 0..1 CodeableConcept Billing codes or product types this definition applies to --></code>
 <instance><!-- 0..* Reference(Medication|Substance|Device) Instances this definition applies to --></instance>
 <applicability>  <!-- 0..* Whether or not the billing code is applicable -->
  <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
  <language value="[string]"/><!-- 0..1 Language of the expression -->
  <expression value="[string]"/><!-- 0..1 Boolean-valued expression -->
 </applicability>
 <propertyGroup>  <!-- 0..* Group of properties which are applicable under the same conditions -->
  <applicability><!-- 0..* Content as for ChargeItemDefinition.applicability Conditions under which the priceComponent
     is applicable --></applicability>
  <priceComponent>  <!-- 0..* Components of total line item price -->
   <type value="[code]"/><!-- 1..1 base | surcharge | deduction | discount | tax | informational -->
   <code><!-- 0..1 CodeableConcept Code identifying the specific component --></code>
   <factor value="[decimal]"/><!-- 0..1 Factor used for calculating this component -->
   <amount><!-- 0..1 Money Monetary amount associated with this component --></amount>
  </priceComponent>
 </propertyGroup>
</ChargeItemDefinition>
```

JSON Template

```
{
  "resourceType" : "ChargeItemDefinition",
```

```
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this charge item definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the charge item definition
  "version" : "<string>", // Business version of the charge item definition
  "title" : "<string>", // Name for this charge item definition (human friendly)
  "derivedFromUri" : ["<uri>"], // Underlying externally-defined charge item definition
  "partOf" : [{ canonical(ChargeItemDefinition) }], // A larger definition of which this particular definition is a
    component or step
  "replaces" : [{ canonical(ChargeItemDefinition) }], // Completed or terminated request(s) whose function is taken by
    this new request
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the charge item definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for charge item definition (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the charge item definition was approved by publisher
  "lastReviewDate" : "<date>", // When the charge item definition was last reviewed
  "effectivePeriod" : { Period }, // When the charge item definition is expected to be used
  "code" : { CodeableConcept }, // Billing codes or product types this definition applies to
  "instance" : [{ Reference(Medication|Substance|Device) }], // Instances this definition applies to
  "applicability" : [{ // Whether or not the billing code is applicable
    "description" : "<string>", // Natural language description of the condition
    "language" : "<string>", // Language of the expression
    "expression" : "<string>" // Boolean-valued expression
  }],
  "propertyGroup" : [{ // Group of properties which are applicable under the same conditions
    "applicability" : [{ Content as for ChargeItemDefinition.applicability }], // Conditions under which the
    priceComponent is applicable
    "priceComponent" : [{ // Components of total line item price
      "type" : "<code>", // R!  base | surcharge | deduction | discount | tax | informational
      "code" : { CodeableConcept }, // Code identifying the specific component
      "factor" : <decimal>, // Factor used for calculating this component
      "amount" : { Money } // Monetary amount associated with this component
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ChargeItemDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ChargeItemDefinition.url [ uri ]; # 1..1 Canonical identifier for this charge item definition, represented as a
    URI (globally unique)
  fhir:ChargeItemDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the charge item definition
  fhir:ChargeItemDefinition.version [ string ]; # 0..1 Business version of the charge item definition
  fhir:ChargeItemDefinition.title [ string ]; # 0..1 Name for this charge item definition (human friendly)
  fhir:ChargeItemDefinition.derivedFromUri [ uri ], ... ; # 0..* Underlying externally-defined charge item definition
  fhir:ChargeItemDefinition.partOf [ canonical(ChargeItemDefinition) ], ... ; # 0..* A larger definition of which this
    particular definition is a component or step
  fhir:ChargeItemDefinition.replaces [ canonical(ChargeItemDefinition) ], ... ; # 0..* Completed or terminated
    request(s) whose function is taken by this new request
  fhir:ChargeItemDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ChargeItemDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ChargeItemDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ChargeItemDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ChargeItemDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ChargeItemDefinition.description [ markdown ]; # 0..1 Natural language description of the charge item definition
  fhir:ChargeItemDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:ChargeItemDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for charge item
    definition (if applicable)
  fhir:ChargeItemDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ChargeItemDefinition.approvalDate [ date ]; # 0..1 When the charge item definition was approved by publisher
  fhir:ChargeItemDefinition.lastReviewDate [ date ]; # 0..1 When the charge item definition was last reviewed
  fhir:ChargeItemDefinition.effectivePeriod [ Period ]; # 0..1 When the charge item definition is expected to be used
  fhir:ChargeItemDefinition.code [ CodeableConcept ]; # 0..1 Billing codes or product types this definition applies to
  fhir:ChargeItemDefinition.instance [ Reference(Medication|Substance|Device) ], ... ; # 0..* Instances this definition
    applies to
  fhir:ChargeItemDefinition.applicability [ # 0..* Whether or not the billing code is applicable
    fhir:ChargeItemDefinition.applicability.description [ string ]; # 0..1 Natural language description of the condition
    fhir:ChargeItemDefinition.applicability.language [ string ]; # 0..1 Language of the expression
    fhir:ChargeItemDefinition.applicability.expression [ string ]; # 0..1 Boolean-valued expression
  ], ...;
  fhir:ChargeItemDefinition.propertyGroup [ # 0..* Group of properties which are applicable under the same conditions
    fhir:ChargeItemDefinition.propertyGroup.applicability [ See ChargeItemDefinition.applicability ], ... ; # 0..*
    Conditions under which the priceComponent is applicable
    fhir:ChargeItemDefinition.propertyGroup.priceComponent [ # 0..* Components of total line item price
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.type [ code ]; # 1..1 base | surcharge | deduction |
    discount | tax | informational
```

```
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.code [ CodeableConcept ]; # 0..1 Code identifying the
    specific component
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.factor [ decimal ]; # 0..1 Factor used for calculating this
     component
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.amount [ Money ]; # 0..1 Monetary amount associated with
    this component
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
ChargeItemDefinition | ITU | | DomainResource | Definition of properties and rules about how the price and the applicability of a ChargeItem can be determined + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this charge item definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the charge item definition
version | Σ | 0..1 | string | Business version of the charge item definition
title | Σ | 0..1 | string | Name for this charge item definition (human friendly)
derivedFromUri | Σ | 0..* | uri | Underlying externally-defined charge item definition
partOf | Σ | 0..* | canonical(ChargeItemDefinition) | A larger definition of which this particular definition is a component or step
replaces | Σ | 0..* | canonical(ChargeItemDefinition) | Completed or terminated request(s) whose function is taken by this new request
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the charge item definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for charge item definition (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the charge item definition was approved by publisher
lastReviewDate | | 0..1 | date | When the charge item definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the charge item definition is expected to be used
code | Σ | 0..1 | CodeableConcept | Billing codes or product types this definition applies to ChargeItemCode (Example)
instance | | 0..* | Reference(Medication | Substance | Device) | Instances this definition applies to
applicability | | 0..* | BackboneElement | Whether or not the billing code is applicable
description | | 0..1 | string | Natural language description of the condition
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | Boolean-valued expression
propertyGroup | | 0..* | BackboneElement | Group of properties which are applicable under the same conditions
applicability | | 0..* | see applicability | Conditions under which the priceComponent is applicable
priceComponent | | 0..* | BackboneElement | Components of total line item price
type | | 1..1 | code | base | surcharge | deduction | discount | tax | informational InvoicePriceComponentType (Required)
code | | 0..1 | CodeableConcept | Code identifying the specific component
factor | | 0..1 | decimal | Factor used for calculating this component
amount | | 0..1 | Money | Monetary amount associated with this component
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ChargeItemDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this charge item definition, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the charge item definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the charge item definition -->
 <title value="[string]"/><!-- 0..1 Name for this charge item definition (human friendly) -->
 <derivedFromUri value="[uri]"/><!-- 0..* Underlying externally-defined charge item definition -->
 <partOf><!-- 0..* canonical(ChargeItemDefinition) A larger definition of which this particular definition is a
    component or step --></partOf>
 <replaces><!-- 0..* canonical(ChargeItemDefinition) Completed or terminated request(s) whose function is taken by this
    new request --></replaces>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the charge item definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for charge item definition (if applicable)
```

```
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the charge item definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the charge item definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the charge item definition is expected to be used --></effectivePeriod>
 <code><!-- 0..1 CodeableConcept Billing codes or product types this definition applies to --></code>
 <instance><!-- 0..* Reference(Medication|Substance|Device) Instances this definition applies to --></instance>
 <applicability>  <!-- 0..* Whether or not the billing code is applicable -->
  <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
  <language value="[string]"/><!-- 0..1 Language of the expression -->
  <expression value="[string]"/><!-- 0..1 Boolean-valued expression -->
 </applicability>
 <propertyGroup>  <!-- 0..* Group of properties which are applicable under the same conditions -->
  <applicability><!-- 0..* Content as for ChargeItemDefinition.applicability Conditions under which the priceComponent
    is applicable --></applicability>
  <priceComponent>  <!-- 0..* Components of total line item price -->
   <type value="[code]"/><!-- 1..1 base | surcharge | deduction | discount | tax | informational -->
   <code><!-- 0..1 CodeableConcept Code identifying the specific component --></code>
   <factor value="[decimal]"/><!-- 0..1 Factor used for calculating this component -->
   <amount><!-- 0..1 Money Monetary amount associated with this component --></amount>
  </priceComponent>
 </propertyGroup>
</ChargeItemDefinition>
```

JSON Template

```
{
  "resourceType" : "ChargeItemDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this charge item definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the charge item definition
  "version" : "<string>", // Business version of the charge item definition
  "title" : "<string>", // Name for this charge item definition (human friendly)
  "derivedFromUri" : ["<uri>"], // Underlying externally-defined charge item definition
  "partOf" : [{ canonical(ChargeItemDefinition) }], // A larger definition of which this particular definition is a
    component or step
  "replaces" : [{ canonical(ChargeItemDefinition) }], // Completed or terminated request(s) whose function is taken by
    this new request
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the charge item definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for charge item definition (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the charge item definition was approved by publisher
  "lastReviewDate" : "<date>", // When the charge item definition was last reviewed
  "effectivePeriod" : { Period }, // When the charge item definition is expected to be used
  "code" : { CodeableConcept }, // Billing codes or product types this definition applies to
  "instance" : [{ Reference(Medication|Substance|Device) }], // Instances this definition applies to
  "applicability" : [{ // Whether or not the billing code is applicable
    "description" : "<string>", // Natural language description of the condition
    "language" : "<string>", // Language of the expression
    "expression" : "<string>" // Boolean-valued expression
  }],
  "propertyGroup" : [{ // Group of properties which are applicable under the same conditions
    "applicability" : [{ Content as for ChargeItemDefinition.applicability }], // Conditions under which the
    priceComponent is applicable
    "priceComponent" : [{ // Components of total line item price
      "type" : "<code>", // R!  base | surcharge | deduction | discount | tax | informational
      "code" : { CodeableConcept }, // Code identifying the specific component
      "factor" : <decimal>, // Factor used for calculating this component
      "amount" : { Money } // Monetary amount associated with this component
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ChargeItemDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ChargeItemDefinition.url [ uri ]; # 1..1 Canonical identifier for this charge item definition, represented as a
    URI (globally unique)
  fhir:ChargeItemDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the charge item definition
  fhir:ChargeItemDefinition.version [ string ]; # 0..1 Business version of the charge item definition
  fhir:ChargeItemDefinition.title [ string ]; # 0..1 Name for this charge item definition (human friendly)
  fhir:ChargeItemDefinition.derivedFromUri [ uri ], ... ; # 0..* Underlying externally-defined charge item definition
  fhir:ChargeItemDefinition.partOf [ canonical(ChargeItemDefinition) ], ... ; # 0..* A larger definition of which this
    particular definition is a component or step
  fhir:ChargeItemDefinition.replaces [ canonical(ChargeItemDefinition) ], ... ; # 0..* Completed or terminated
    request(s) whose function is taken by this new request
  fhir:ChargeItemDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
```

```
  fhir:ChargeItemDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ChargeItemDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ChargeItemDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ChargeItemDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ChargeItemDefinition.description [ markdown ]; # 0..1 Natural language description of the charge item definition
  fhir:ChargeItemDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:ChargeItemDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for charge item
    definition (if applicable)
  fhir:ChargeItemDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ChargeItemDefinition.approvalDate [ date ]; # 0..1 When the charge item definition was approved by publisher
  fhir:ChargeItemDefinition.lastReviewDate [ date ]; # 0..1 When the charge item definition was last reviewed
  fhir:ChargeItemDefinition.effectivePeriod [ Period ]; # 0..1 When the charge item definition is expected to be used
  fhir:ChargeItemDefinition.code [ CodeableConcept ]; # 0..1 Billing codes or product types this definition applies to
  fhir:ChargeItemDefinition.instance [ Reference(Medication|Substance|Device) ], ... ; # 0..* Instances this definition
    applies to
  fhir:ChargeItemDefinition.applicability [ # 0..* Whether or not the billing code is applicable
    fhir:ChargeItemDefinition.applicability.description [ string ]; # 0..1 Natural language description of the condition
    fhir:ChargeItemDefinition.applicability.language [ string ]; # 0..1 Language of the expression
    fhir:ChargeItemDefinition.applicability.expression [ string ]; # 0..1 Boolean-valued expression
  ], ...;
  fhir:ChargeItemDefinition.propertyGroup [ # 0..* Group of properties which are applicable under the same conditions
    fhir:ChargeItemDefinition.propertyGroup.applicability [ See ChargeItemDefinition.applicability ], ... ; # 0..*
    Conditions under which the priceComponent is applicable
    fhir:ChargeItemDefinition.propertyGroup.priceComponent [ # 0..* Components of total line item price
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.type [ code ]; # 1..1 base | surcharge | deduction |
    discount | tax | informational
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.code [ CodeableConcept ]; # 0..1 Code identifying the
    specific component
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.factor [ decimal ]; # 0..1 Factor used for calculating this
     component
      fhir:ChargeItemDefinition.propertyGroup.priceComponent.amount [ Money ]; # 0..1 Monetary amount associated with
    this component
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.25.3.1 Terminology Bindings

Path	Definition	Type	Reference
ChargeItemDefinition.status	The lifecycle status of an artifact.	Required	PublicationStatus
ChargeItemDefinition.jurisdiction	Countries and regions within which this artifact is targeted for use.	Extensible	Jurisdiction ValueSet
ChargeItemDefinition.code	Billing Code defined by this ChargeItemDefinition.	Example	ChargeItemCode
ChargeItemDefinition.propertyGroup.priceComponent.type	Codes indicating the kind of the price component.	Required	InvoicePriceComponentType

8.25.3.2 Constraints

id	Level	Location	Description	Expression
cid-0	Warning	(base)	Name should be usable as an identifier for the module by machine processing applications such as code generation	name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

8.25.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
context	token	A use context assigned to the charge item definition	(ChargeItemDefinition.useContext.value as CodeableConcept)	
context-quantity	quantity	A quantity- or range-valued use context assigned to the charge item definition	(ChargeItemDefinition.useContext.value as Quantity) | (ChargeItemDefinition.useContext.value as Range)	
context-type	token	A type of use context assigned to the charge item definition	ChargeItemDefinition.useContext.code	
context-type-quantity	composite	A use context type and quantity- or range-based value assigned to the charge item definition	On ChargeItemDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)	
context-type-value	composite	A use context type and value assigned to the charge item definition	On ChargeItemDefinition.useContext: context-type: code context: value.as(CodeableConcept)	
date	date	The charge item definition publication date	ChargeItemDefinition.date	
description	string	The description of the charge item definition	ChargeItemDefinition.description	
effective	date	The time during which the charge item definition is intended to be in use	ChargeItemDefinition.effectivePeriod	
identifier	token	External identifier for the charge item definition	ChargeItemDefinition.identifier	
jurisdiction	token	Intended jurisdiction for the charge item definition	ChargeItemDefinition.jurisdiction	
publisher	string	Name of the publisher of the charge item definition	ChargeItemDefinition.publisher	
status	token	The current status of the charge item definition	ChargeItemDefinition.status	

title | string | The human-friendly name of the charge item definition | ChargeItemDefinition.title
url | uri | The uri that identifies the charge item definition | ChargeItemDefinition.url
version | token | The business version of the charge item definition | ChargeItemDefinition.version

8.25.5 Resource ChargeItemDefinition - Examples

URL: https://hl7.org/fhir/R4/chargeitemdefinition-examples.html

8.25.5 Resource ChargeItemDefinition - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
example definition of EMB code 30110 (from a german billing system) | ebm | XML | JSON | Turtle
instance-specific billing information for a device with time-dependent tax rate | device | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.25.6 Resource ChargeItemDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/chargeitemdefinition-definitions.html

8.25.6 Resource ChargeItemDefinition - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ChargeItemDefinition resource.

ChargeItemDefinition
Element Id | ChargeItemDefinition
Definition | The ChargeItemDefinition resource provides the properties that apply to the (billing) codes necessary to calculate costs and prices. The properties may differ largely depending on type and realm, therefore this resource gives only a rough structure and requires profiling for each type of billing code system.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element cid-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
cid-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ChargeItemDefinition.url
Element Id | ChargeItemDefinition.url
Definition | An absolute URI that is used to identify this charge item definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this charge item definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the charge item definition is stored on different servers.
Cardinality | 1..1
Type | uri
Requirements | Allows the charge item definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ChargeItemDefinition.identifier
Element Id | ChargeItemDefinition.identifier
Definition | A formal identifier that is used to identify this charge item definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this charge item definition outside of FHIR, where it is not possible to use the logical URI.
ChargeItemDefinition.version
Element Id | ChargeItemDefinition.version
Definition | The identifier that is used to identify this version of the charge item definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the charge item definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active assets.

Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different charge item definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the charge item definition with the format [url]|[version].
ChargeItemDefinition.title
Element Id | ChargeItemDefinition.title
Definition | A short, descriptive, user-friendly title for the charge item definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ChargeItemDefinition.derivedFromUri
Element Id | ChargeItemDefinition.derivedFromUri
Definition | The URL pointing to an externally-defined charge item definition that is adhered to in whole or in part by this definition.
Cardinality | 0..*
Type | uri
Summary | true
ChargeItemDefinition.partOf
Element Id | ChargeItemDefinition.partOf
Definition | A larger definition of which this particular definition is a component or step.
Cardinality | 0..*
Type | canonical(ChargeItemDefinition)
Alternate Names | container
Summary | true
ChargeItemDefinition.replaces
Element Id | ChargeItemDefinition.replaces
Definition | As new versions of a protocol or guideline are defined, allows identification of what versions are replaced by a new instance.
Cardinality | 0..*
Type | canonical(ChargeItemDefinition)
Alternate Names | supersedes; prior; renewed order
Summary | true
ChargeItemDefinition.status
Element Id | ChargeItemDefinition.status
Definition | The current state of the ChargeItemDefinition.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of charge item definitions that are appropriate for use versus not.
ChargeItemDefinition.experimental
Element Id | ChargeItemDefinition.experimental
Definition | A Boolean value to indicate that this charge item definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level charge item definition.
Summary | true
Comments | Allows filtering of charge item definitions that are appropriate for use versus not.
ChargeItemDefinition.date
Element Id | ChargeItemDefinition.date
Definition | The date (and optionally time) when the charge item definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the charge item definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the charge item definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ChargeItemDefinition.publisher
Element Id | ChargeItemDefinition.publisher
Definition | The name of the organization or individual that published the charge item definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the charge item definition. May also allow for contact.

Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the charge item definition is the organization or individual primarily responsible for the maintenance and upkeep of the charge item definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the charge item definition. This item SHOULD be populated unless the information is available from context.
ChargeItemDefinition.contact
Element Id | ChargeItemDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ChargeItemDefinition.description
Element Id | ChargeItemDefinition.description
Definition | A free text natural language description of the charge item definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the charge item definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the charge item definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the charge item definition is presumed to be the predominant language in the place the charge item definition was created).
ChargeItemDefinition.useContext
Element Id | ChargeItemDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate charge item definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ChargeItemDefinition.jurisdiction
Element Id | ChargeItemDefinition.jurisdiction
Definition | A legal or geographic region in which the charge item definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the charge item definition to be used in jurisdictions other than those for which it was originally designed or intended.
ChargeItemDefinition.copyright
Element Id | ChargeItemDefinition.copyright
Definition | A copyright statement relating to the charge item definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the charge item definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the charge item definition and/or its content.
Alternate Names | License; Restrictions
ChargeItemDefinition.approvalDate
Element Id | ChargeItemDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
ChargeItemDefinition.lastReviewDate
Element Id | ChargeItemDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
ChargeItemDefinition.effectivePeriod
Element Id | ChargeItemDefinition.effectivePeriod
Definition | The period during which the charge item definition content was or is planned to be in active use.
Cardinality | 0..1
Type | Period

Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the charge item definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a charge item definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
ChargeItemDefinition.code
Element Id | ChargeItemDefinition.code
Definition | The defined billing details in this resource pertain to the given billing code.
Cardinality | 0..1
Terminology Binding | ChargeItemCode (Example)
Type | CodeableConcept
Summary | true
ChargeItemDefinition.instance
Element Id | ChargeItemDefinition.instance
Definition | The defined billing details in this resource pertain to the given product instance(s).
Cardinality | 0..*
Type | Reference(Medication | Substance | Device)
Comments | In case of highly customized, individually produced or fitted devices/substances, the pricing information may be different for each instance of the product. This reference links pricing details to specific product instances.
ChargeItemDefinition.applicability
Element Id | ChargeItemDefinition.applicability
Definition | Expressions that describe applicability criteria for the billing code.
Cardinality | 0..*
Comments | The applicability conditions can be used to ascertain whether a billing item is allowed in a specific context. E.g. some billing codes may only be applicable in out-patient settings, only to male/female patients or only to children.
ChargeItemDefinition.applicability.description
Element Id | ChargeItemDefinition.applicability.description
Definition | A brief, natural language description of the condition that effectively communicates the intended semantics.
Cardinality | 0..1
Type | string
ChargeItemDefinition.applicability.language
Element Id | ChargeItemDefinition.applicability.language
Definition | The media type of the language for the expression, e.g. "text/cql" for Clinical Query Language expressions or "text/fhirpath" for FHIRPath expressions.
Cardinality | 0..1
Type | string
ChargeItemDefinition.applicability.expression
Element Id | ChargeItemDefinition.applicability.expression
Definition | An expression that returns true or false, indicating whether the condition is satisfied. When using FHIRPath expressions, the %context environment variable must be replaced at runtime with the ChargeItem resource to which this definition is applied.
Cardinality | 0..1
Type | string
Comments | Please note that FHIRPath Expressions can only be evaluated in the scope of the current ChargeItem resource to which this definition is being applied. FHIRPath expressions can traverse into other resources linked from the ChargeItem resource, however, testing rules such as that a billing code may be billed only once per encounter need a wider scope. In such scenarios, CQL may be the appropriate choice.
ChargeItemDefinition.propertyGroup
Element Id | ChargeItemDefinition.propertyGroup
Definition | Group of properties which are applicable under the same conditions. If no applicability rules are established for the group, then all properties always apply.
Cardinality | 0..*
ChargeItemDefinition.propertyGroup.applicability
Element Id | ChargeItemDefinition.propertyGroup.applicability
Definition | Expressions that describe applicability criteria for the priceComponent.
Cardinality | 0..*
Type | See ChargeItemDefinition.applicability
Comments | The applicability conditions can be used to ascertain whether a billing item is allowed in a specific context. E.g. some billing codes may only be applicable in out-patient settings, only to male/female patients or only to children.
ChargeItemDefinition.propertyGroup.priceComponent
Element Id | ChargeItemDefinition.propertyGroup.priceComponent
Definition | The price for a ChargeItem may be calculated as a base price with surcharges/deductions that apply in certain conditions. A ChargeItemDefinition resource that defines the prices, factors and conditions that apply to a billing code is currently under development. The priceComponent element can be used to offer transparency to the recipient of the Invoice of how the prices have been calculated.
Cardinality | 0..*
ChargeItemDefinition.propertyGroup.priceComponent.type
Element Id | ChargeItemDefinition.propertyGroup.priceComponent.type
Definition | This code identifies the type of the component.
Cardinality | 1..1
Terminology Binding | InvoicePriceComponentType (Required)
Type | code
ChargeItemDefinition.propertyGroup.priceComponent.code

Element Id | ChargeItemDefinition.propertyGroup.priceComponent.code
Definition | A code that identifies the component. Codes may be used to differentiate between kinds of taxes, surcharges, discounts etc.
Cardinality | 0..1
Type | CodeableConcept
ChargeItemDefinition.propertyGroup.priceComponent.factor
Element Id | ChargeItemDefinition.propertyGroup.priceComponent.factor
Definition | The factor that has been applied on the base price for calculating this component.
Cardinality | 0..1
Type | decimal
ChargeItemDefinition.propertyGroup.priceComponent.amount
Element Id | ChargeItemDefinition.propertyGroup.priceComponent.amount
Definition | The amount calculated for this component.
Cardinality | 0..1
Type | Money

8.25.7 Resource ChargeItemDefinition - Mappings

URL: https://hl7.org/fhir/R4/chargeitemdefinition-mappings.html

8.25.7 Resource ChargeItemDefinition - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the chargeitemdefinition resource (see Mappings to Other Standards for further information & status).

8.25.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ChargeItemDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness

8.25.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ChargeItemDefinition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status {different ValueSet}
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher {as string instead of CodeableConcept}
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

8.25.7.3 RIM Mapping (http://hl7.org/v3)

ChargeItemDefinition | N/A

8.25.8 Resource ChargeItemDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/chargeitemdefinition-profiles.html

8.25.8 Resource ChargeItemDefinition - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ChargeItemDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.25.8.1 Profiles

No Profiles defined for this resource

8.25.8.2 Extensions

replaces | ChargeItemDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.25.8.3 Search Extensions

No Search Extensions defined for this resource

8.25.9 Resource ChargeItemDefinition - Operations

URL: https://hl7.org/fhir/R4/chargeitemdefinition-operations.html

8.25.9 Resource ChargeItemDefinition - Operations

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

This resource has 1 operation operations associated with it:

$apply | Apply

For more information about operations, including how they are invoked, see Operations.

8.25.9.1 Operation $apply on ChargeItemDefinition

URL: https://hl7.org/fhir/R4/chargeitemdefinition-operation-apply.html

8.25.9.1 Operation $apply on ChargeItemDefinition

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Compartments: N/A

The apply operation applies a definition in a specific context.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ChargeItemDefinition-apply
```

Formal Definition (as a OperationDefinition).

URL: [base]/ChargeItemDefinition/[id]/$apply

This is not an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
chargeItem | 1..1 | Reference(ChargeItem) | | | The ChargeItem on which the definition is to ba applies
account | 0..1 | Reference(Account) | | | The account in context, if any
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | Any | | | The resource that is the result of applying the definition Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The result of invoking this operation is a resource of the type specified by the activity definition, with all the definitions resolved as appropriate for the type of resource. Any dynamicValue elements will be evaluated (in the order in which they appear in the resource) and the results applied to the returned resource. If the ActivityDefinition includes library references, those libraries will be available to the evaluated expressions. If those libraries have parameters, those parameters will be bound by name to the parameters given to the operation. For a more detailed description, refer to the ActivityDefinition resource

8.25.9.1.1 Examples

Request: Apply a ChargeItemDefinitionDefinition to an account

```
GET [base]/ChargeItemDefinition/ebm-1100/$apply?account=1234567
```

Response: Returned on successful application

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p> Error applying ChargeItemDefinition: Billing Code SHALL NOT be charged for in-patient encounters</p>
    </div>
  </text>
  <issue>
    <severity value="error"/>
    <code value="business-rule"/>
    <details>
      <text value="Billing Code SHALL NOT be charged for in-patient encounters"/>
    </details>
  </issue>
</OperationOutcome>
```

For more information about operations, including how they are invoked, see Operations.

8.25.10 JSON Schema for ChargeItemDefinition

URL: https://hl7.org/fhir/R4/chargeitemdefinition.schema.json.html

8.25.10 JSON Schema for ChargeItemDefinition

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ChargeItemDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ChargeItemDefinition",
  "$ref": "#/definitions/ChargeItemDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ChargeItemDefinition": {
      "description": "The ChargeItemDefinition resource provides the properties that apply to the (billing) codes
    necessary to calculate costs and prices. The properties may differ largely depending on type and realm, therefore
    this resource gives only a rough structure and requires profiling for each type of billing code system.",
      "properties": {
        "resourceType": {
          "description": "This is a ChargeItemDefinition resource",
          "const": "ChargeItemDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
```

```
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this charge item definition when it is referenced in
a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this charge item definition is (or
will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the charge
item definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this charge item definition when it is
represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the charge item definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the charge item
definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
 a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence. To provide a version consistent with the Decision Support Service specification, use the format
Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision
Support Service specification. Note that a version is required for non-experimental active assets.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the charge item definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "derivedFromUri": {
      "description": "The URL pointing to an externally-defined charge item definition that is adhered to in whole
or in part by this definition.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_derivedFromUri": {
      "description": "Extensions for derivedFromUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
```

```
    },
    "partOf": {
      "description": "A larger definition of which this particular definition is a component or step.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "replaces": {
      "description": "As new versions of a protocol or guideline are defined, allows identification of what versions
 are replaced by a new instance.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the ChargeItemDefinition.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this charge item definition is authored for testing purposes
(or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the charge item definition was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the charge item definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the charge item definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the charge item definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate charge item definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the charge item definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
```

```
    "copyright": {
      "description": "A copyright statement relating to the charge item definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the charge item definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the charge item definition content was or is planned to be in active
use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "code": {
      "description": "The defined billing details in this resource pertain to the given billing code.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "instance": {
      "description": "The defined billing details in this resource pertain to the given product instance(s).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "applicability": {
      "description": "Expressions that describe applicability criteria for the billing code.",
      "items": {
        "$ref": "#/definitions/ChargeItemDefinition_Applicability"
      },
      "type": "array"
    },
    "propertyGroup": {
      "description": "Group of properties which are applicable under the same conditions. If no applicability rules
are established for the group, then all properties always apply.",
      "items": {
        "$ref": "#/definitions/ChargeItemDefinition_PropertyGroup"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ChargeItemDefinition_Applicability": {
  "description": "The ChargeItemDefinition resource provides the properties that apply to the (billing) codes
necessary to calculate costs and prices. The properties may differ largely depending on type and realm, therefore
this resource gives only a rough structure and requires profiling for each type of billing code system.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
```

```
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "A brief, natural language description of the condition that effectively communicates the
intended semantics.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The media type of the language for the expression, e.g. \"text/cql\" for Clinical Query
Language expressions or \"text/fhirpath\" for FHIRPath expressions.",
      "$ref": "#/definitions/string"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expression": {
      "description": "An expression that returns true or false, indicating whether the condition is satisfied. When
using FHIRPath expressions, the %context environment variable must be replaced at runtime with the ChargeItem
resource to which this definition is applied.",
      "$ref": "#/definitions/string"
    },
    "_expression": {
      "description": "Extensions for expression",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ChargeItemDefinition_PropertyGroup": {
  "description": "The ChargeItemDefinition resource provides the properties that apply to the (billing) codes
necessary to calculate costs and prices. The properties may differ largely depending on type and realm, therefore
this resource gives only a rough structure and requires profiling for each type of billing code system.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "applicability": {
      "description": "Expressions that describe applicability criteria for the priceComponent.",
      "items": {
        "$ref": "#/definitions/ChargeItemDefinition_Applicability"
      },
      "type": "array"
    },
    "priceComponent": {
      "description": "The price for a ChargeItem may be calculated as a base price with surcharges/deductions that
apply in certain conditions. A ChargeItemDefinition resource that defines the prices, factors and conditions that
apply to a billing code is currently under development. The priceComponent element can be used to offer transparency
 to the recipient of the Invoice of how the prices have been calculated.",
      "items": {
        "$ref": "#/definitions/ChargeItemDefinition_PriceComponent"
      },
```

```
        "type": "array"
      }
    },
    "additionalProperties": false
  },
  "ChargeItemDefinition_PriceComponent": {
    "description": "The ChargeItemDefinition resource provides the properties that apply to the (billing) codes
  necessary to calculate costs and prices. The properties may differ largely depending on type and realm, therefore
  this resource gives only a rough structure and requires profiling for each type of billing code system.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
  value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
  definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the element and that modifies the understanding of the element in which it is contained and/or the understanding of
  the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
  use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
   of the definition of the extension. Applications processing a resource are required to check for modifier
  extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
  (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "type": {
        "description": "This code identifies the type of the component.",
        "$ref": "#/definitions/code"
      },
      "_type": {
        "description": "Extensions for type",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "code": {
        "description": "A code that identifies the component. Codes may be used to differentiate between kinds of
  taxes, surcharges, discounts etc.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "factor": {
        "description": "The factor that has been applied on the base price for calculating this component.",
        "$ref": "#/definitions/decimal"
      },
      "_factor": {
        "description": "Extensions for factor",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "amount": {
        "description": "The amount calculated for this component.",
        "$ref": "Money.schema.json#/definitions/Money"
      }
    },
    "additionalProperties": false
  }
 }
}
```


8.25.11 Resource ChargeItemDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/chargeitemdefinition-dependencies.html

8.25.11 Resource ChargeItemDefinition - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ChargeItemDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
ChargeItemDefinition | | DomainResource
url | 1..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string

title | 0..1 | string
derivedFromUri | 0..* | uri
partOf | 0..* | canonical(ChargeItemDefinition)
replaces | 0..* | canonical(ChargeItemDefinition)
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
code | 0..1 | CodeableConcept
instance | 0..* | Reference(Medication | Substance | Device)
applicability | 0..* | BackboneElement
description | 0..1 | string
language | 0..1 | string
expression | 0..1 | string
propertyGroup | 0..* | BackboneElement
applicability | 0..* | see applicability
priceComponent | 0..* | BackboneElement
type | 1..1 | code
code | 0..1 | CodeableConcept
factor | 0..1 | decimal
amount | 0..1 | Money

8.26 Resource ObservationDefinition - Content

URL: https://hl7.org/fhir/R4/observationdefinition.html
Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Set of definitional characteristics for a kind of observation or measurement produced or consumed by an orderable health care service.

8.26.1 Scope and Usage

Note that this content is preliminary has not undergone proper review by the appropriate Workgroups.

An instance of ObservationDefinition represents the definitional aspects of a kind of observation. This resource is used in a catalog of products or services for healthcare. It provides the definition of observations, which are used as input or produced as outcome of these products and services. The focus of the definitional aspects of ObservationDefinition is not on the action of performing observations of that kind. Rather, it is on the outcome of the observations. An instance of ObservationDefinition represents a set of constraints applicable to the value produced by observations of this kind. This set of constraints may have two utilities: 1) It guides the entry of the result of an observation of the kind defined. 2) It helps the consumer of the results of observations of that kind, to interpret properly these results.

For instance, in a catalog of laboratory in vitro diagnostic services, the ObservationDefinition resources describe the observations that should be provided by the orderer alongside with an ordered service (questions asked at order entry or at specimen collection, prior results) as well as the observations that will be reported once the service has been performed.

In a catalog of medications, ObservationDefinition resources can be used to describe observations the result thereof need to be checked against the prescription of a particular medicinal product (therapeutic range, units ...)

In the catalog of tests of an in vitro diagnostic device, ObservationDefinition resources can be used to characterize (absolute range, unit, decimal precision, critical values ...) the observations that will be produced by the IVD tests performed by the device.

ObservationDefinition may be used to characterize observations from any discipline.

8.26.2 Boundaries and Relationships

This resource relates to these other resource:.

• Resource ObservationDefinition is referenced by resource CatalogEntry. This enables items of catalogs, representing healthcare services or products, to declare the observations, which are consumed or produced by these items.

• Observation.basedOn references ObservationDefinition resource.

This resource is referenced by ActivityDefinition, CatalogEntry, MedicationKnowledge and MedicinalProductInteraction

8.26.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ObservationDefinition | TU | | DomainResource | Definition of an observation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
category | Σ | 0..* | CodeableConcept | Category of observation Observation Category Codes (Example)
code | Σ | 1..1 | CodeableConcept | Type of observation (code / type) LOINC Codes (Example)
identifier | Σ | 0..* | Identifier | Business identifier for this ObservationDefinition instance
permittedDataType | | 0..* | code | Quantity | CodeableConcept | string | boolean | integer | Range | Ratio | SampledData | time | dateTime | Period ObservationDataType (Required)
multipleResultsAllowed | | 0..1 | boolean | Multiple results allowed
method | | 0..1 | CodeableConcept | Method used to produce the observation Observation Methods (Example)
preferredReportName | | 0..1 | string | Preferred report name
quantitativeDetails | | 0..1 | BackboneElement | Characteristics of quantitative results
customaryUnit | | 0..1 | CodeableConcept | Customary unit for quantitative results UCUM Codes (Extensible)
unit | | 0..1 | CodeableConcept | SI unit for quantitative results UCUM Codes (Extensible)
conversionFactor | | 0..1 | decimal | SI to Customary unit conversion factor
decimalPrecision | | 0..1 | integer | Decimal precision of observation quantitative results
qualifiedInterval | | 0..* | BackboneElement | Qualified range for continuous and ordinal observation results
category | | 0..1 | code | reference | critical | absolute ObservationRangeCategory (Required)
range | | 0..1 | Range | The interval itself, for continuous or ordinal observations
context | | 0..1 | CodeableConcept | Range context qualifier Observation Reference Range Meaning Codes (Extensible)
appliesTo | | 0..* | CodeableConcept | Targetted population of the range Observation Reference Range Applies To Codes (Example)
gender | | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
age | | 0..1 | Range | Applicable age range, if relevant
gestationalAge | | 0..1 | Range | Applicable gestational age range, if relevant
condition | | 0..1 | string | Condition associated with the reference range
validCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of valid coded values for the observations conforming to this ObservationDefinition
normalCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of normal coded values for the observations conforming to this ObservationDefinition
abnormalCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of abnormal coded values for the observations conforming to this ObservationDefinition
criticalCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of critical coded values for the observations conforming to this ObservationDefinition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ObservationDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <category><!-- 0..* CodeableConcept Category of observation --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type) --></code>
 <identifier><!-- 0..* Identifier Business identifier for this ObservationDefinition instance --></identifier>
 <permittedDataType value="[code]"/><!-- 0..* Quantity | CodeableConcept | string | boolean | integer | Range | Ratio |
    SampledData | time | dateTime | Period -->
 <multipleResultsAllowed value="[boolean]"/><!-- 0..1 Multiple results allowed -->
 <method><!-- 0..1 CodeableConcept Method used to produce the observation --></method>
 <preferredReportName value="[string]"/><!-- 0..1 Preferred report name -->
 <quantitativeDetails>  <!-- 0..1 Characteristics of quantitative results -->
  <customaryUnit><!-- 0..1 CodeableConcept Customary unit for quantitative results --></customaryUnit>
  <unit><!-- 0..1 CodeableConcept SI unit for quantitative results --></unit>
  <conversionFactor value="[decimal]"/><!-- 0..1 SI to Customary unit conversion factor -->
  <decimalPrecision value="[integer]"/><!-- 0..1 Decimal precision of observation quantitative results -->
 </quantitativeDetails>
 <qualifiedInterval>  <!-- 0..* Qualified range for continuous and ordinal observation results -->
  <category value="[code]"/><!-- 0..1 reference | critical | absolute -->
  <range><!-- 0..1 Range The interval itself, for continuous or ordinal observations --></range>
  <context><!-- 0..1 CodeableConcept Range context qualifier --></context>
  <appliesTo><!-- 0..* CodeableConcept Targetted population of the range --></appliesTo>
  <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
  <age><!-- 0..1 Range Applicable age range, if relevant --></age>
  <gestationalAge><!-- 0..1 Range Applicable gestational age range, if relevant --></gestationalAge>
```

```
  <condition value="[string]"/><!-- 0..1 Condition associated with the reference range -->
 </qualifiedInterval>
 <validCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of valid coded values for the observations conforming to
    this ObservationDefinition --></validCodedValueSet>
 <normalCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of normal coded values for the observations conforming to
    this ObservationDefinition --></normalCodedValueSet>
 <abnormalCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of abnormal coded values for the observations conforming
     to this ObservationDefinition --></abnormalCodedValueSet>
 <criticalCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of critical coded values for the observations conforming
     to this ObservationDefinition --></criticalCodedValueSet>
</ObservationDefinition>
```

JSON Template

```
{
  "resourceType" : "ObservationDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "category" : [{ CodeableConcept }], // Category of observation
  "code" : { CodeableConcept }, // R!  Type of observation (code / type)
  "identifier" : [{ Identifier }], // Business identifier for this ObservationDefinition instance
  "permittedDataType" : ["<code>"], // Quantity | CodeableConcept | string | boolean | integer | Range | Ratio |
    SampledData | time | dateTime | Period
  "multipleResultsAllowed" : <boolean>, // Multiple results allowed
  "method" : { CodeableConcept }, // Method used to produce the observation
  "preferredReportName" : "<string>", // Preferred report name
  "quantitativeDetails" : { // Characteristics of quantitative results
    "customaryUnit" : { CodeableConcept }, // Customary unit for quantitative results
    "unit" : { CodeableConcept }, // SI unit for quantitative results
    "conversionFactor" : <decimal>, // SI to Customary unit conversion factor
    "decimalPrecision" : <integer> // Decimal precision of observation quantitative results
  },
  "qualifiedInterval" : [{ // Qualified range for continuous and ordinal observation results
    "category" : "<code>", // reference | critical | absolute
    "range" : { Range }, // The interval itself, for continuous or ordinal observations
    "context" : { CodeableConcept }, // Range context qualifier
    "appliesTo" : [{ CodeableConcept }], // Targetted population of the range
    "gender" : "<code>", // male | female | other | unknown
    "age" : { Range }, // Applicable age range, if relevant
    "gestationalAge" : { Range }, // Applicable gestational age range, if relevant
    "condition" : "<string>" // Condition associated with the reference range
  }],
  "validCodedValueSet" : { Reference(ValueSet) }, // Value set of valid coded values for the observations conforming to
    this ObservationDefinition
  "normalCodedValueSet" : { Reference(ValueSet) }, // Value set of normal coded values for the observations conforming
    to this ObservationDefinition
  "abnormalCodedValueSet" : { Reference(ValueSet) }, // Value set of abnormal coded values for the observations
    conforming to this ObservationDefinition
  "criticalCodedValueSet" : { Reference(ValueSet) } // Value set of critical coded values for the observations
    conforming to this ObservationDefinition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ObservationDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ObservationDefinition.category [ CodeableConcept ], ... ; # 0..* Category of observation
  fhir:ObservationDefinition.code [ CodeableConcept ]; # 1..1 Type of observation (code / type)
  fhir:ObservationDefinition.identifier [ Identifier ], ... ; # 0..* Business identifier for this ObservationDefinition
    instance
  fhir:ObservationDefinition.permittedDataType [ code ], ... ; # 0..* Quantity | CodeableConcept | string | boolean |
    integer | Range | Ratio | SampledData | time | dateTime | Period
  fhir:ObservationDefinition.multipleResultsAllowed [ boolean ]; # 0..1 Multiple results allowed
  fhir:ObservationDefinition.method [ CodeableConcept ]; # 0..1 Method used to produce the observation
  fhir:ObservationDefinition.preferredReportName [ string ]; # 0..1 Preferred report name
  fhir:ObservationDefinition.quantitativeDetails [ # 0..1 Characteristics of quantitative results
    fhir:ObservationDefinition.quantitativeDetails.customaryUnit [ CodeableConcept ]; # 0..1 Customary unit for
    quantitative results
    fhir:ObservationDefinition.quantitativeDetails.unit [ CodeableConcept ]; # 0..1 SI unit for quantitative results
    fhir:ObservationDefinition.quantitativeDetails.conversionFactor [ decimal ]; # 0..1 SI to Customary unit conversion
    factor
    fhir:ObservationDefinition.quantitativeDetails.decimalPrecision [ integer ]; # 0..1 Decimal precision of observation
     quantitative results
  ];
  fhir:ObservationDefinition.qualifiedInterval [ # 0..* Qualified range for continuous and ordinal observation results
    fhir:ObservationDefinition.qualifiedInterval.category [ code ]; # 0..1 reference | critical | absolute
    fhir:ObservationDefinition.qualifiedInterval.range [ Range ]; # 0..1 The interval itself, for continuous or ordinal
    observations
    fhir:ObservationDefinition.qualifiedInterval.context [ CodeableConcept ]; # 0..1 Range context qualifier
    fhir:ObservationDefinition.qualifiedInterval.appliesTo [ CodeableConcept ], ... ; # 0..* Targetted population of the
     range
    fhir:ObservationDefinition.qualifiedInterval.gender [ code ]; # 0..1 male | female | other | unknown
    fhir:ObservationDefinition.qualifiedInterval.age [ Range ]; # 0..1 Applicable age range, if relevant
    fhir:ObservationDefinition.qualifiedInterval.gestationalAge [ Range ]; # 0..1 Applicable gestational age range, if
    relevant
```

```
    fhir:ObservationDefinition.qualifiedInterval.condition [ string ]; # 0..1 Condition associated with the reference
      range
  ], ...;
  fhir:ObservationDefinition.validCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of valid coded values for the
    observations conforming to this ObservationDefinition
  fhir:ObservationDefinition.normalCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of normal coded values for
    the observations conforming to this ObservationDefinition
  fhir:ObservationDefinition.abnormalCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of abnormal coded values
    for the observations conforming to this ObservationDefinition
  fhir:ObservationDefinition.criticalCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of critical coded values
    for the observations conforming to this ObservationDefinition
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ObservationDefinition | TU | | DomainResource | Definition of an observation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

category | Σ | 0..* | CodeableConcept | Category of observation Observation Category Codes (Example)

code | Σ | 1..1 | CodeableConcept | Type of observation (code / type) LOINC Codes (Example)

identifier | Σ | 0..* | Identifier | Business identifier for this ObservationDefinition instance

permittedDataType | | 0..* | code | Quantity | CodeableConcept | string | boolean | integer | Range | Ratio | SampledData | time | dateTime | Period ObservationDataType (Required)

multipleResultsAllowed | | 0..1 | boolean | Multiple results allowed

method | | 0..1 | CodeableConcept | Method used to produce the observation Observation Methods (Example)

preferredReportName | | 0..1 | string | Preferred report name

quantitativeDetails | | 0..1 | BackboneElement | Characteristics of quantitative results

customaryUnit | | 0..1 | CodeableConcept | Customary unit for quantitative results UCUM Codes (Extensible)

unit | | 0..1 | CodeableConcept | SI unit for quantitative results UCUM Codes (Extensible)

conversionFactor | | 0..1 | decimal | SI to Customary unit conversion factor

decimalPrecision | | 0..1 | integer | Decimal precision of observation quantitative results

qualifiedInterval | | 0..* | BackboneElement | Qualified range for continuous and ordinal observation results

category | | 0..1 | code | reference | critical | absolute ObservationRangeCategory (Required)

range | | 0..1 | Range | The interval itself, for continuous or ordinal observations

context | | 0..1 | CodeableConcept | Range context qualifier Observation Reference Range Meaning Codes (Extensible)

appliesTo | | 0..* | CodeableConcept | Targetted population of the range Observation Reference Range Applies To Codes (Example)

gender | | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

age | | 0..1 | Range | Applicable age range, if relevant

gestationalAge | | 0..1 | Range | Applicable gestational age range, if relevant

condition | | 0..1 | string | Condition associated with the reference range

validCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of valid coded values for the observations conforming to this ObservationDefinition

normalCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of normal coded values for the observations conforming to this ObservationDefinition

abnormalCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of abnormal coded values for the observations conforming to this ObservationDefinition

criticalCodedValueSet | | 0..1 | Reference(ValueSet) | Value set of critical coded values for the observations conforming to this ObservationDefinition

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ObservationDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <category><!-- 0..* CodeableConcept Category of observation --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type) --></code>
 <identifier><!-- 0..* Identifier Business identifier for this ObservationDefinition instance --></identifier>
 <permittedDataType value="[code]"/><!-- 0..* Quantity | CodeableConcept | string | boolean | integer | Range | Ratio |
    SampledData | time | dateTime | Period -->
 <multipleResultsAllowed value="[boolean]"/><!-- 0..1 Multiple results allowed -->
 <method><!-- 0..1 CodeableConcept Method used to produce the observation --></method>
 <preferredReportName value="[string]"/><!-- 0..1 Preferred report name -->
 <quantitativeDetails>  <!-- 0..1 Characteristics of quantitative results -->
  <customaryUnit><!-- 0..1 CodeableConcept Customary unit for quantitative results --></customaryUnit>
  <unit><!-- 0..1 CodeableConcept SI unit for quantitative results --></unit>
  <conversionFactor value="[decimal]"/><!-- 0..1 SI to Customary unit conversion factor -->
  <decimalPrecision value="[integer]"/><!-- 0..1 Decimal precision of observation quantitative results -->
 </quantitativeDetails>
 <qualifiedInterval>  <!-- 0..* Qualified range for continuous and ordinal observation results -->
  <category value="[code]"/><!-- 0..1 reference | critical | absolute -->
  <range><!-- 0..1 Range The interval itself, for continuous or ordinal observations --></range>
  <context><!-- 0..1 CodeableConcept Range context qualifier --></context>
  <appliesTo><!-- 0..* CodeableConcept Targetted population of the range --></appliesTo>
  <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
  <age><!-- 0..1 Range Applicable age range, if relevant --></age>
  <gestationalAge><!-- 0..1 Range Applicable gestational age range, if relevant --></gestationalAge>
```

```
  <condition value="[string]"/><!-- 0..1 Condition associated with the reference range -->
 </qualifiedInterval>
 <validCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of valid coded values for the observations conforming to
    this ObservationDefinition --></validCodedValueSet>
 <normalCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of normal coded values for the observations conforming to
    this ObservationDefinition --></normalCodedValueSet>
 <abnormalCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of abnormal coded values for the observations conforming
     to this ObservationDefinition --></abnormalCodedValueSet>
 <criticalCodedValueSet><!-- 0..1 Reference(ValueSet) Value set of critical coded values for the observations conforming
     to this ObservationDefinition --></criticalCodedValueSet>
</ObservationDefinition>
```

JSON Template

```
{
  "resourceType" : "ObservationDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "category" : [{ CodeableConcept }], // Category of observation
  "code" : { CodeableConcept }, // R!  Type of observation (code / type)
  "identifier" : [{ Identifier }], // Business identifier for this ObservationDefinition instance
  "permittedDataType" : ["<code>"], // Quantity | CodeableConcept | string | boolean | integer | Range | Ratio |
    SampledData | time | dateTime | Period
  "multipleResultsAllowed" : <boolean>, // Multiple results allowed
  "method" : { CodeableConcept }, // Method used to produce the observation
  "preferredReportName" : "<string>", // Preferred report name
  "quantitativeDetails" : { // Characteristics of quantitative results
    "customaryUnit" : { CodeableConcept }, // Customary unit for quantitative results
    "unit" : { CodeableConcept }, // SI unit for quantitative results
    "conversionFactor" : <decimal>, // SI to Customary unit conversion factor
    "decimalPrecision" : <integer> // Decimal precision of observation quantitative results
  },
  "qualifiedInterval" : [{ // Qualified range for continuous and ordinal observation results
    "category" : "<code>", // reference | critical | absolute
    "range" : { Range }, // The interval itself, for continuous or ordinal observations
    "context" : { CodeableConcept }, // Range context qualifier
    "appliesTo" : [{ CodeableConcept }], // Targetted population of the range
    "gender" : "<code>", // male | female | other | unknown
    "age" : { Range }, // Applicable age range, if relevant
    "gestationalAge" : { Range }, // Applicable gestational age range, if relevant
    "condition" : "<string>" // Condition associated with the reference range
  }],
  "validCodedValueSet" : { Reference(ValueSet) }, // Value set of valid coded values for the observations conforming to
    this ObservationDefinition
  "normalCodedValueSet" : { Reference(ValueSet) }, // Value set of normal coded values for the observations conforming
    to this ObservationDefinition
  "abnormalCodedValueSet" : { Reference(ValueSet) }, // Value set of abnormal coded values for the observations
    conforming to this ObservationDefinition
  "criticalCodedValueSet" : { Reference(ValueSet) } // Value set of critical coded values for the observations
    conforming to this ObservationDefinition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ObservationDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ObservationDefinition.category [ CodeableConcept ], ... ; # 0..* Category of observation
  fhir:ObservationDefinition.code [ CodeableConcept ]; # 1..1 Type of observation (code / type)
  fhir:ObservationDefinition.identifier [ Identifier ], ... ; # 0..* Business identifier for this ObservationDefinition
    instance
  fhir:ObservationDefinition.permittedDataType [ code ], ... ; # 0..* Quantity | CodeableConcept | string | boolean |
    integer | Range | Ratio | SampledData | time | dateTime | Period
  fhir:ObservationDefinition.multipleResultsAllowed [ boolean ]; # 0..1 Multiple results allowed
  fhir:ObservationDefinition.method [ CodeableConcept ]; # 0..1 Method used to produce the observation
  fhir:ObservationDefinition.preferredReportName [ string ]; # 0..1 Preferred report name
  fhir:ObservationDefinition.quantitativeDetails [ # 0..1 Characteristics of quantitative results
    fhir:ObservationDefinition.quantitativeDetails.customaryUnit [ CodeableConcept ]; # 0..1 Customary unit for
    quantitative results
    fhir:ObservationDefinition.quantitativeDetails.unit [ CodeableConcept ]; # 0..1 SI unit for quantitative results
    fhir:ObservationDefinition.quantitativeDetails.conversionFactor [ decimal ]; # 0..1 SI to Customary unit conversion
    factor
    fhir:ObservationDefinition.quantitativeDetails.decimalPrecision [ integer ]; # 0..1 Decimal precision of observation
     quantitative results
  ];
  fhir:ObservationDefinition.qualifiedInterval [ # 0..* Qualified range for continuous and ordinal observation results
    fhir:ObservationDefinition.qualifiedInterval.category [ code ]; # 0..1 reference | critical | absolute
    fhir:ObservationDefinition.qualifiedInterval.range [ Range ]; # 0..1 The interval itself, for continuous or ordinal
    observations
    fhir:ObservationDefinition.qualifiedInterval.context [ CodeableConcept ]; # 0..1 Range context qualifier
    fhir:ObservationDefinition.qualifiedInterval.appliesTo [ CodeableConcept ], ... ; # 0..* Targetted population of the
     range
    fhir:ObservationDefinition.qualifiedInterval.gender [ code ]; # 0..1 male | female | other | unknown
    fhir:ObservationDefinition.qualifiedInterval.age [ Range ]; # 0..1 Applicable age range, if relevant
    fhir:ObservationDefinition.qualifiedInterval.gestationalAge [ Range ]; # 0..1 Applicable gestational age range, if
    relevant
```

```
    fhir:ObservationDefinition.qualifiedInterval.condition [ string ]; # 0..1 Condition associated with the reference
      range
  ], ...;
  fhir:ObservationDefinition.validCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of valid coded values for the
    observations conforming to this ObservationDefinition
  fhir:ObservationDefinition.normalCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of normal coded values for
    the observations conforming to this ObservationDefinition
  fhir:ObservationDefinition.abnormalCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of abnormal coded values
    for the observations conforming to this ObservationDefinition
  fhir:ObservationDefinition.criticalCodedValueSet [ Reference(ValueSet) ]; # 0..1 Value set of critical coded values
    for the observations conforming to this ObservationDefinition
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.26.3.1 Terminology Bindings

Path | Definition | Type | Reference
ObservationDefinition.category | Codes for high level observation categories. | Example | ObservationCategoryCodes
ObservationDefinition.code | Codes identifying names of simple observations. | Example | LOINCCodes
ObservationDefinition.permittedDataType | Permitted data type for observation value. | Required | ObservationDataType
ObservationDefinition.method | Methods for simple observations. | Example | ObservationMethods
ObservationDefinition.quantitativeDetails.customaryUnit ObservationDefinition.quantitativeDetails.unit | Codes identifying units of measure. | Extensible | UCUMCodes
ObservationDefinition.qualifiedInterval.category | Codes identifying the category of observation range. | Required | ObservationRangeCategory
ObservationDefinition.qualifiedInterval.context | Code identifying the health context of a range. | Extensible | ObservationReferenceRangeMeaningCodes
ObservationDefinition.qualifiedInterval.appliesTo | Codes identifying the population the reference range applies to. | Example | ObservationReferenceRangeAppliesToCodes
ObservationDefinition.qualifiedInterval.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender

8.26.4 Resource ObservationDefinition - Examples

URL: https://hl7.org/fhir/R4/observationdefinition-examples.html

8.26.4 Resource ObservationDefinition - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of observationdefinition | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.26.5 Resource ObservationDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/observationdefinition-definitions.html

8.26.5 Resource ObservationDefinition - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ObservationDefinition resource.

ObservationDefinition
Element Id | ObservationDefinition
Definition | Set of definitional characteristics for a kind of observation or measurement produced or consumed by an orderable health care service.
Cardinality | 0..*
Type | DomainResource
Requirements | In a catalog of health-related services that use or produce observations and measurements, this resource describes the expected characteristics of these observation / measurements.
Comments | An instance of this resource informs the consumer of a health-related service (such as a lab diagnostic test or panel) about how the observations used or produced by this service will look like.
ObservationDefinition.category
Element Id | ObservationDefinition.category
Definition | A code that classifies the general type of observation.
Cardinality | 0..*
Terminology Binding | Observation Category Codes (Example)
Type | CodeableConcept

Requirements | Used for filtering what kinds of observations are retrieved and displayed.
Alternate Names | Class of observation
Summary | true
Comments | This element allows various categorization schemes based on the owner’s definition of the category and effectively multiple categories can be used for one instance of ObservationDefinition. The level of granularity is defined by the category concepts in the value set.
ObservationDefinition.code
Element Id | ObservationDefinition.code
Definition | Describes what will be observed. Sometimes this is called the observation "name".
Cardinality | 1..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Requirements | Knowing what kind of observation is being made is essential to understanding the observation.
Alternate Names | Name
Summary | true
ObservationDefinition.identifier
Element Id | ObservationDefinition.identifier
Definition | A unique identifier assigned to this ObservationDefinition artifact.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
ObservationDefinition.permittedDataType
Element Id | ObservationDefinition.permittedDataType
Definition | The data types allowed for the value element of the instance observations conforming to this ObservationDefinition.
Cardinality | 0..*
Terminology Binding | ObservationDataType (Required)
Type | code
ObservationDefinition.multipleResultsAllowed
Element Id | ObservationDefinition.multipleResultsAllowed
Definition | Multiple results allowed for observations conforming to this ObservationDefinition.
Cardinality | 0..1
Type | boolean
Comments | An example of observation allowing multiple results is "bacteria identified by culture". Conversely, the measurement of a potassium level allows a single result.
ObservationDefinition.method
Element Id | ObservationDefinition.method
Definition | The method or technique used to perform the observation.
Cardinality | 0..1
Terminology Binding | Observation Methods (Example)
Type | CodeableConcept
Requirements | In some cases, method can impact results.
Comments | Only used if not implicit in observation code.
ObservationDefinition.preferredReportName
Element Id | ObservationDefinition.preferredReportName
Definition | The preferred name to be used when reporting the results of observations conforming to this ObservationDefinition.
Cardinality | 0..1
Type | string
ObservationDefinition.quantitativeDetails
Element Id | ObservationDefinition.quantitativeDetails
Definition | Characteristics for quantitative results of this observation.
Cardinality | 0..1
ObservationDefinition.quantitativeDetails.customaryUnit
Element Id | ObservationDefinition.quantitativeDetails.customaryUnit
Definition | Customary unit used to report quantitative results of observations conforming to this ObservationDefinition.
Cardinality | 0..1
Terminology Binding | UCUM Codes (Extensible)
Type | CodeableConcept
ObservationDefinition.quantitativeDetails.unit
Element Id | ObservationDefinition.quantitativeDetails.unit
Definition | SI unit used to report quantitative results of observations conforming to this ObservationDefinition.
Cardinality | 0..1
Terminology Binding | UCUM Codes (Extensible)
Type | CodeableConcept
ObservationDefinition.quantitativeDetails.conversionFactor
Element Id | ObservationDefinition.quantitativeDetails.conversionFactor
Definition | Factor for converting value expressed with SI unit to value expressed with customary unit.
Cardinality | 0..1
Type | decimal
ObservationDefinition.quantitativeDetails.decimalPrecision
Element Id | ObservationDefinition.quantitativeDetails.decimalPrecision
Definition | Number of digits after decimal separator when the results of such observations are of type Quantity.

Cardinality | 0..1
Type | integer
ObservationDefinition.qualifiedInterval
Element Id | ObservationDefinition.qualifiedInterval
Definition | Multiple ranges of results qualified by different contexts for ordinal or continuous observations conforming to this ObservationDefinition.
Cardinality | 0..*
ObservationDefinition.qualifiedInterval.category
Element Id | ObservationDefinition.qualifiedInterval.category
Definition | The category of interval of values for continuous or ordinal observations conforming to this ObservationDefinition.
Cardinality | 0..1
Terminology Binding | ObservationRangeCategory (Required)
Type | code
ObservationDefinition.qualifiedInterval.range
Element Id | ObservationDefinition.qualifiedInterval.range
Definition | The low and high values determining the interval. There may be only one of the two.
Cardinality | 0..1
Type | Range
Requirements | The unit may be not relevant for ordinal values. In case it is there, it is the same as quantitativeDetails.unit.
ObservationDefinition.qualifiedInterval.context
Element Id | ObservationDefinition.qualifiedInterval.context
Definition | Codes to indicate the health context the range applies to. For example, the normal or therapeutic range.
Cardinality | 0..1
Terminology Binding | Observation Reference Range Meaning Codes (Extensible)
Type | CodeableConcept
ObservationDefinition.qualifiedInterval.appliesTo
Element Id | ObservationDefinition.qualifiedInterval.appliesTo
Definition | Codes to indicate the target population this reference range applies to.
Cardinality | 0..*
Terminology Binding | Observation Reference Range Applies To Codes (Example)
Type | CodeableConcept
Comments | If this element is not present then the global population is assumed.
ObservationDefinition.qualifiedInterval.gender
Element Id | ObservationDefinition.qualifiedInterval.gender
Definition | Sex of the population the range applies to.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
ObservationDefinition.qualifiedInterval.age
Element Id | ObservationDefinition.qualifiedInterval.age
Definition | The age at which this reference range is applicable. This is a neonatal age (e.g. number of weeks at term) if the meaning says so.
Cardinality | 0..1
Type | Range
Comments | Some analytes vary greatly over age.
ObservationDefinition.qualifiedInterval.gestationalAge
Element Id | ObservationDefinition.qualifiedInterval.gestationalAge
Definition | The gestational age to which this reference range is applicable, in the context of pregnancy.
Cardinality | 0..1
Type | Range
ObservationDefinition.qualifiedInterval.condition
Element Id | ObservationDefinition.qualifiedInterval.condition
Definition | Text based condition for which the reference range is valid.
Cardinality | 0..1
Type | string
ObservationDefinition.validCodedValueSet
Element Id | ObservationDefinition.validCodedValueSet
Definition | The set of valid coded results for the observations conforming to this ObservationDefinition.
Cardinality | 0..1
Type | Reference(ValueSet)
ObservationDefinition.normalCodedValueSet
Element Id | ObservationDefinition.normalCodedValueSet
Definition | The set of normal coded results for the observations conforming to this ObservationDefinition.
Cardinality | 0..1
Type | Reference(ValueSet)
ObservationDefinition.abnormalCodedValueSet
Element Id | ObservationDefinition.abnormalCodedValueSet
Definition | The set of abnormal coded results for the observation conforming to this ObservationDefinition.
Cardinality | 0..1
Type | Reference(ValueSet)
ObservationDefinition.criticalCodedValueSet
Element Id | ObservationDefinition.criticalCodedValueSet

Definition | The set of critical coded results for the observation conforming to this ObservationDefinition.
Cardinality | 0..1
Type | Reference(ValueSet)

8.26.6 Resource ObservationDefinition - Mappings

URL: https://hl7.org/fhir/R4/observationdefinition-mappings.html

8.26.6 Resource ObservationDefinition - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the observationdefinition resource (see Mappings to Other Standards for further information & status).

8.26.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ObservationDefinition
category | FiveWs.class
code | FiveWs.what[x]

8.26.6.2 HL7 v2 Mapping (http://hl7.org/v2)

ObservationDefinition | OM2 OM3 OMC
category
code | OM1-2 OMC-4
identifier
permittedDataType | OM1-3
multipleResultsAllowed
method | OM1-14
preferredReportName | OM1-9
quantitativeDetails | OM2
customaryUnit | OM2-2
unit | OM2-4
conversionFactor | OM2-5
decimalPrecision | OM2-3
qualifiedInterval | OM2-6
category | OM-2
range | OM-2
context | n.a.
appliesTo | RFR.6
gender | RFR.2
age | RFR.3
gestationalAge | RFR.4
condition | RFR.7
validCodedValueSet | OM3-3
normalCodedValueSet | OM3-4
abnormalCodedValueSet | OM3-5
criticalCodedValueSet | OM3-6

8.26.6.3 RIM Mapping (http://hl7.org/v3)

ObservationDefinition | Observation[classCode=OBS, moodCode=DEF]
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="DEF"].code
code | code
method | methodCode

8.26.7 Resource ObservationDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/observationdefinition-profiles.html

8.26.7 Resource ObservationDefinition - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ObservationDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.26.7.1 Profiles

No Profiles defined for this resource

8.26.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.26.7.3 Search Extensions

No Search Extensions defined for this resource

8.26.8 JSON Schema for ObservationDefinition

URL: https://hl7.org/fhir/R4/observationdefinition.schema.json.html

8.26.8 JSON Schema for ObservationDefinition

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ObservationDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ObservationDefinition",
  "$ref": "#/definitions/ObservationDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ObservationDefinition": {
      "description": "Set of definitional characteristics for a kind of observation or measurement produced or consumed
    by an orderable health care service.",
      "properties": {
        "resourceType": {
          "description": "This is a ObservationDefinition resource",
          "const": "ObservationDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
```

```
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "category": {
      "description": "A code that classifies the general type of observation.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "code": {
      "description": "Describes what will be observed. Sometimes this is called the observation \"name\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "identifier": {
      "description": "A unique identifier assigned to this ObservationDefinition artifact.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "permittedDataType": {
      "description": "The data types allowed for the value element of the instance observations conforming to this
ObservationDefinition.",
      "items": {
        "enum": [
          "Quantity",
          "CodeableConcept",
          "string",
          "boolean",
          "integer",
          "Range",
          "Ratio",
          "SampledData",
          "time",
          "dateTime",
          "Period"
        ]
      },
      "type": "array"
    },
    "_permittedDataType": {
      "description": "Extensions for permittedDataType",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "multipleResultsAllowed": {
      "description": "Multiple results allowed for observations conforming to this ObservationDefinition.",
      "$ref": "#/definitions/boolean"
    },
    "_multipleResultsAllowed": {
      "description": "Extensions for multipleResultsAllowed",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "method": {
      "description": "The method or technique used to perform the observation.",
```

```
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "preferredReportName": {
      "description": "The preferred name to be used when reporting the results of observations conforming to this
ObservationDefinition.",
      "$ref": "#/definitions/string"
    },
    "_preferredReportName": {
      "description": "Extensions for preferredReportName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "quantitativeDetails": {
      "description": "Characteristics for quantitative results of this observation.",
      "$ref": "#/definitions/ObservationDefinition_QuantitativeDetails"
    },
    "qualifiedInterval": {
      "description": "Multiple  ranges of results qualified by different contexts for ordinal or continuous
observations conforming to this ObservationDefinition.",
      "items": {
        "$ref": "#/definitions/ObservationDefinition_QualifiedInterval"
      },
      "type": "array"
    },
    "validCodedValueSet": {
      "description": "The set of valid coded results for the observations  conforming to this
ObservationDefinition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "normalCodedValueSet": {
      "description": "The set of normal coded results for the observations conforming to this
ObservationDefinition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "abnormalCodedValueSet": {
      "description": "The set of abnormal coded results for the observation conforming to this
ObservationDefinition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "criticalCodedValueSet": {
      "description": "The set of critical coded results for the observation conforming to this
ObservationDefinition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "code",
    "resourceType"
  ]
},
"ObservationDefinition_QuantitativeDetails": {
  "description": "Set of definitional characteristics for a kind of observation or measurement produced or consumed
by an orderable health care service.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "customaryUnit": {
      "description": "Customary unit used to report quantitative results of observations conforming to this
ObservationDefinition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "unit": {
```

```
      "description": "SI unit used to report quantitative results of observations conforming to this
ObservationDefinition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "conversionFactor": {
      "description": "Factor for converting value expressed with SI unit to value expressed with customary unit.",
      "$ref": "#/definitions/decimal"
    },
    "_conversionFactor": {
      "description": "Extensions for conversionFactor",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "decimalPrecision": {
      "description": "Number of digits after decimal separator when the results of such observations are of type
Quantity.",
      "$ref": "#/definitions/integer"
    },
    "_decimalPrecision": {
      "description": "Extensions for decimalPrecision",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ObservationDefinition_QualifiedInterval": {
  "description": "Set of definitional characteristics for a kind of observation or measurement produced or consumed
by an orderable health care service.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "category": {
      "description": "The category of interval of values for continuous or ordinal observations conforming to this
ObservationDefinition.",
      "enum": [
        "reference",
        "critical",
        "absolute"
      ]
    },
    "_category": {
      "description": "Extensions for category",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "range": {
      "description": "The low and high values determining the interval. There may be only one of the two.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "context": {
      "description": "Codes to indicate the health context the range applies to. For example, the normal or
therapeutic range.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "appliesTo": {
      "description": "Codes to indicate the target population this reference range applies to.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "gender": {
      "description": "Sex of the population the range applies to.",
      "enum": [
```

```
            "male",
            "female",
            "other",
            "unknown"
          ]
        },
        "_gender": {
          "description": "Extensions for gender",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "age": {
          "description": "The age at which this reference range is applicable. This is a neonatal age (e.g. number of
    weeks at term) if the meaning says so.",
          "$ref": "Range.schema.json#/definitions/Range"
        },
        "gestationalAge": {
          "description": "The gestational age to which this reference range is applicable, in the context of
    pregnancy.",
          "$ref": "Range.schema.json#/definitions/Range"
        },
        "condition": {
          "description": "Text based condition for which the reference range is valid.",
          "$ref": "#/definitions/string"
        },
        "_condition": {
          "description": "Extensions for condition",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

8.26.9 Resource ObservationDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/observationdefinition-dependencies.html

8.26.9 Resource ObservationDefinition - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ObservationDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
ObservationDefinition | | DomainResource
category | 0..* | CodeableConcept
code | 1..1 | CodeableConcept
identifier | 0..* | Identifier
permittedDataType | 0..* | code
multipleResultsAllowed | 0..1 | boolean
method | 0..1 | CodeableConcept
preferredReportName | 0..1 | string
quantitativeDetails | 0..1 | BackboneElement
customaryUnit | 0..1 | CodeableConcept
unit | 0..1 | CodeableConcept
conversionFactor | 0..1 | decimal
decimalPrecision | 0..1 | integer
qualifiedInterval | 0..* | BackboneElement
category | 0..1 | code
range | 0..1 | Range
context | 0..1 | CodeableConcept
appliesTo | 0..* | CodeableConcept
gender | 0..1 | code
age | 0..1 | Range
gestationalAge | 0..1 | Range
condition | 0..1 | string
validCodedValueSet | 0..1 | Reference(ValueSet)
normalCodedValueSet | 0..1 | Reference(ValueSet)
abnormalCodedValueSet | 0..1 | Reference(ValueSet)
criticalCodedValueSet | 0..1 | Reference(ValueSet)

8.28 Resource CatalogEntry - Content

URL: https://hl7.org/fhir/R4/catalogentry.html
Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Catalog entries are wrappers that contextualize items included in a catalog.

8.28.1 Scope and Usage

The CatalogEntry resource (formally known as EntryDefinition) allows additional information specification for items that may be found in the following types of clinical collections, including but not limited to:

- A pharmacy formulary
- A laboratory compendium
- A radiology compendium of exams that may be ordered
- A dietary nutrition order catalog
- A supply charge master
- A CDS catalog of knowledge artifacts

The Catalog profile of the Composition resource is used to organize catalog entries.

Catalogs are composed of nestable sections which contain catalog entries. Catalog entries wrap the items that are orderable or selectable in a catalog and any other required supporting entries. Examples of information provided by CatalogEntry includes whether the status of the item in the catalog is "active", "inactive", "legacy", or "informative" or whether the item is an orderable item in the context of this catalog (or catalog section). the relation between the item and other items ("contains", "includes", "requires"). Catalog entries also support relationships to other catalog entries such as whether a product is now replaced by another product in a product catalog. Please refer to the Boundaries and Relationships part of this document for more information on the topic.

This CatalogEntry resource wraps and contextualizes definitional items that make up a catalog such as the following resources:

8.28.2 Boundaries and Relationships

- Medication
- Device
- Activity Definition (including laboratory service definitions)
- Specimen Definition
- ...

It is important to note that CatalogEntry shall not specify attributes that are pertinent to the item that it wraps if those attributes hold true regardless of the catalog context. For instance, a catalog entry should generally not provide attributes to describe the hierarchical relationship of an item to other items (e.g., drug class or other such ontological relationships) since that information holds outside of the context of a catalog and thus should reside in the referenced item itself. A catalog entry may, however, specify a status of a given item in the context of the catalog such as whether that item is orderable a given institution or context. Catalog entries can also be related to one another. Here again, it is important to note that only relationships that hold within the context of a catalog and not generally for the item referenced by the catalog entry should be considered. For instance, in the context of a given laboratory service compendium, a laboratory order may be preferred over other laboratory orders. The other laboratory orders in this use case may thus be related to the preferred test using an 'is-replaced-by' relationship and assigned a new status of 'retired'. Other definitional relationships such as the specimen or expected input and output observation definitions associated with a particular laboratory test, relationships that generally hold outside of a laboratory catalog context, should reside at the level of the definitional resource itself. That is, laboratory tests often state the required input observation specifications, the format of the expected observation output, and the attributes of the required specimen and such relationships are probably best described outside of the CatalogEntry within the referenced item.

This resource is referenced by itself

8.28.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
CatalogEntry	TU		DomainResource	An entry in a catalog Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Unique identifier of the catalog item

type | | 0..1 | CodeableConcept | The type of item - medication, device, service, protocol or other
orderable | Σ | 1..1 | boolean | Whether the entry represents an orderable item
referencedItem | Σ | 1..1 | Reference(Medication | Device | Organization | Practitioner | PractitionerRole | HealthcareService | ActivityDefinition | PlanDefinition | SpecimenDefinition | ObservationDefinition | Binary) | The item that is being defined
additionalIdentifier | | 0..* | Identifier | Any additional identifier(s) for the catalog item, in the same granularity or concept
classification | | 0..* | CodeableConcept | Classification (category or class) of the item entry
status | | 0..1 | code | draft | active | retired | unknown PublicationStatus (Required)
validityPeriod | | 0..1 | Period | The time period in which this catalog entry is expected to be active
validTo | | 0..1 | dateTime | The date until which this catalog entry is expected to be active
lastUpdated | | 0..1 | dateTime | When was this catalog last updated
additionalCharacteristic | | 0..* | CodeableConcept | Additional characteristics of the catalog entry
additionalClassification | | 0..* | CodeableConcept | Additional classification of the catalog entry
relatedEntry | | 0..* | BackboneElement | An item that this catalog entry is related to
relationtype | | 1..1 | code | triggers | is-replaced-by CatalogEntryRelationType (Required)
item | | 1..1 | Reference(CatalogEntry) | The reference to the related item
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CatalogEntry xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier of the catalog item --></identifier>
 <type><!-- 0..1 CodeableConcept The type of item - medication, device, service, protocol or other --></type>
 <orderable value="[boolean]"/><!-- 1..1 Whether the entry represents an orderable item -->
 <referencedItem><!-- 1..1 Reference(Medication|Device|Organization|Practitioner|
   PractitionerRole|HealthcareService|ActivityDefinition|PlanDefinition|
   SpecimenDefinition|ObservationDefinition|Binary) The item that is being defined --></referencedItem>
 <additionalIdentifier><!-- 0..* Identifier Any additional identifier(s) for the catalog item, in the same granularity
    or concept --></additionalIdentifier>
 <classification><!-- 0..* CodeableConcept Classification (category or class) of the item entry --></classification>
 <status value="[code]"/><!-- 0..1 draft | active | retired | unknown -->
 <validityPeriod><!-- 0..1 Period The time period in which this catalog entry is expected to be active
    --></validityPeriod>
 <validTo value="[dateTime]"/><!-- 0..1 The date until which this catalog entry is expected to be active -->
 <lastUpdated value="[dateTime]"/><!-- 0..1 When was this catalog last updated -->
 <additionalCharacteristic><!-- 0..* CodeableConcept Additional characteristics of the catalog entry
    --></additionalCharacteristic>
 <additionalClassification><!-- 0..* CodeableConcept Additional classification of the catalog entry
    --></additionalClassification>
 <relatedEntry>  <!-- 0..* An item that this catalog entry is related to -->
  <relationtype value="[code]"/><!-- 1..1 triggers | is-replaced-by -->
  <item><!-- 1..1 Reference(CatalogEntry) The reference to the related item --></item>
 </relatedEntry>
</CatalogEntry>
```

JSON Template

```
{
  "resourceType" : "CatalogEntry",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier of the catalog item
  "type" : { CodeableConcept }, // The type of item - medication, device, service, protocol or other
  "orderable" : <boolean>, // R!  Whether the entry represents an orderable item
  "referencedItem" : { Reference(Medication|Device|Organization|Practitioner|
   PractitionerRole|HealthcareService|ActivityDefinition|PlanDefinition|
   SpecimenDefinition|ObservationDefinition|Binary) }, // R!  The item that is being defined
  "additionalIdentifier" : [{ Identifier }], // Any additional identifier(s) for the catalog item, in the same
    granularity or concept
  "classification" : [{ CodeableConcept }], // Classification (category or class) of the item entry
  "status" : "<code>", // draft | active | retired | unknown
  "validityPeriod" : { Period }, // The time period in which this catalog entry is expected to be active
  "validTo" : "<dateTime>", // The date until which this catalog entry is expected to be active
  "lastUpdated" : "<dateTime>", // When was this catalog last updated
  "additionalCharacteristic" : [{ CodeableConcept }], // Additional characteristics of the catalog entry
  "additionalClassification" : [{ CodeableConcept }], // Additional classification of the catalog entry
  "relatedEntry" : [{ // An item that this catalog entry is related to
    "relationtype" : "<code>", // R!  triggers | is-replaced-by
    "item" : { Reference(CatalogEntry) } // R!  The reference to the related item
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CatalogEntry;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CatalogEntry.identifier [ Identifier ], ... ; # 0..* Unique identifier of the catalog item
  fhir:CatalogEntry.type [ CodeableConcept ]; # 0..1 The type of item - medication, device, service, protocol or other
  fhir:CatalogEntry.orderable [ boolean ]; # 1..1 Whether the entry represents an orderable item
```

```
  fhir:CatalogEntry.referencedItem [
    Reference(Medication|Device|Organization|Practitioner|PractitionerRole|HealthcareService|
  ActivityDefinition|PlanDefinition|SpecimenDefinition|ObservationDefinition|
  Binary) ]; # 1..1 The item that is being defined
  fhir:CatalogEntry.additionalIdentifier [ Identifier ], ... ; # 0..* Any additional identifier(s) for the catalog item,
     in the same granularity or concept
  fhir:CatalogEntry.classification [ CodeableConcept ], ... ; # 0..* Classification (category or class) of the item
    entry
  fhir:CatalogEntry.status [ code ]; # 0..1 draft | active | retired | unknown
  fhir:CatalogEntry.validityPeriod [ Period ]; # 0..1 The time period in which this catalog entry is expected to be
    active
  fhir:CatalogEntry.validTo [ dateTime ]; # 0..1 The date until which this catalog entry is expected to be active
  fhir:CatalogEntry.lastUpdated [ dateTime ]; # 0..1 When was this catalog last updated
  fhir:CatalogEntry.additionalCharacteristic [ CodeableConcept ], ... ; # 0..* Additional characteristics of the catalog
     entry
  fhir:CatalogEntry.additionalClassification [ CodeableConcept ], ... ; # 0..* Additional classification of the catalog
    entry
  fhir:CatalogEntry.relatedEntry [ # 0..* An item that this catalog entry is related to
    fhir:CatalogEntry.relatedEntry.relationtype [ code ]; # 1..1 triggers | is-replaced-by
    fhir:CatalogEntry.relatedEntry.item [ Reference(CatalogEntry) ]; # 1..1 The reference to the related item
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
CatalogEntry	TU		DomainResource	An entry in a catalog Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Unique identifier of the catalog item
type		0..1	CodeableConcept	The type of item - medication, device, service, protocol or other
orderable	Σ	1..1	boolean	Whether the entry represents an orderable item
referencedItem	Σ	1..1	Reference(Medication | Device | Organization | Practitioner | PractitionerRole | HealthcareService | ActivityDefinition | PlanDefinition | SpecimenDefinition | ObservationDefinition | Binary)	The item that is being defined
additionalIdentifier		0..*	Identifier	Any additional identifier(s) for the catalog item, in the same granularity or concept
classification		0..*	CodeableConcept	Classification (category or class) of the item entry
status		0..1	code | draft | active | retired | unknown PublicationStatus (Required)	
validityPeriod		0..1	Period	The time period in which this catalog entry is expected to be active
validTo		0..1	dateTime	The date until which this catalog entry is expected to be active
lastUpdated		0..1	dateTime	When was this catalog last updated
additionalCharacteristic		0..*	CodeableConcept	Additional characteristics of the catalog entry
additionalClassification		0..*	CodeableConcept	Additional classification of the catalog entry
relatedEntry		0..*	BackboneElement	An item that this catalog entry is related to
relationtype		1..1	code | triggers | is-replaced-by CatalogEntryRelationType (Required)	
item		1..1	Reference(CatalogEntry)	The reference to the related item

Documentation for this format

UML Diagram (Legend)

XML Template

```
<CatalogEntry xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier of the catalog item --></identifier>
 <type><!-- 0..1 CodeableConcept The type of item - medication, device, service, protocol or other --></type>
 <orderable value="[boolean]"/><!-- 1..1 Whether the entry represents an orderable item -->
 <referencedItem><!-- 1..1 Reference(Medication|Device|Organization|Practitioner|
   PractitionerRole|HealthcareService|ActivityDefinition|PlanDefinition|
   SpecimenDefinition|ObservationDefinition|Binary) The item that is being defined --></referencedItem>
 <additionalIdentifier><!-- 0..* Identifier Any additional identifier(s) for the catalog item, in the same granularity
    or concept --></additionalIdentifier>
 <classification><!-- 0..* CodeableConcept Classification (category or class) of the item entry --></classification>
 <status value="[code]"/><!-- 0..1 draft | active | retired | unknown -->
 <validityPeriod><!-- 0..1 Period The time period in which this catalog entry is expected to be active
    --></validityPeriod>
 <validTo value="[dateTime]"/><!-- 0..1 The date until which this catalog entry is expected to be active -->
 <lastUpdated value="[dateTime]"/><!-- 0..1 When was this catalog last updated -->
 <additionalCharacteristic><!-- 0..* CodeableConcept Additional characteristics of the catalog entry
    --></additionalCharacteristic>
 <additionalClassification><!-- 0..* CodeableConcept Additional classification of the catalog entry
    --></additionalClassification>
 <relatedEntry>  <!-- 0..* An item that this catalog entry is related to -->
  <relationtype value="[code]"/><!-- 1..1 triggers | is-replaced-by -->
  <item><!-- 1..1 Reference(CatalogEntry) The reference to the related item --></item>
 </relatedEntry>
</CatalogEntry>
```

JSON Template

```
{
  "resourceType" : "CatalogEntry",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier of the catalog item
  "type" : { CodeableConcept }, // The type of item - medication, device, service, protocol or other
  "orderable" : <boolean>, // R!  Whether the entry represents an orderable item
  "referencedItem" : { Reference(Medication|Device|Organization|Practitioner|
   PractitionerRole|HealthcareService|ActivityDefinition|PlanDefinition|
   SpecimenDefinition|ObservationDefinition|Binary) }, // R!  The item that is being defined
  "additionalIdentifier" : [{ Identifier }], // Any additional identifier(s) for the catalog item, in the same
    granularity or concept
  "classification" : [{ CodeableConcept }], // Classification (category or class) of the item entry
  "status" : "<code>", // draft | active | retired | unknown
  "validityPeriod" : { Period }, // The time period in which this catalog entry is expected to be active
  "validTo" : "<dateTime>", // The date until which this catalog entry is expected to be active
  "lastUpdated" : "<dateTime>", // When was this catalog last updated
  "additionalCharacteristic" : [{ CodeableConcept }], // Additional characteristics of the catalog entry
  "additionalClassification" : [{ CodeableConcept }], // Additional classification of the catalog entry
  "relatedEntry" : [{ // An item that this catalog entry is related to
    "relationtype" : "<code>", // R!  triggers | is-replaced-by
    "item" : { Reference(CatalogEntry) } // R!  The reference to the related item
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CatalogEntry;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CatalogEntry.identifier [ Identifier ], ... ; # 0..* Unique identifier of the catalog item
  fhir:CatalogEntry.type [ CodeableConcept ]; # 0..1 The type of item - medication, device, service, protocol or other
  fhir:CatalogEntry.orderable [ boolean ]; # 1..1 Whether the entry represents an orderable item
  fhir:CatalogEntry.referencedItem [
    Reference(Medication|Device|Organization|Practitioner|PractitionerRole|HealthcareService|
  ActivityDefinition|PlanDefinition|SpecimenDefinition|ObservationDefinition|
  Binary) ]; # 1..1 The item that is being defined
  fhir:CatalogEntry.additionalIdentifier [ Identifier ], ... ; # 0..* Any additional identifier(s) for the catalog item,
     in the same granularity or concept
  fhir:CatalogEntry.classification [ CodeableConcept ], ... ; # 0..* Classification (category or class) of the item
    entry
  fhir:CatalogEntry.status [ code ]; # 0..1 draft | active | retired | unknown
  fhir:CatalogEntry.validityPeriod [ Period ]; # 0..1 The time period in which this catalog entry is expected to be
    active
  fhir:CatalogEntry.validTo [ dateTime ]; # 0..1 The date until which this catalog entry is expected to be active
  fhir:CatalogEntry.lastUpdated [ dateTime ]; # 0..1 When was this catalog last updated
  fhir:CatalogEntry.additionalCharacteristic [ CodeableConcept ], ... ; # 0..* Additional characteristics of the catalog
     entry
  fhir:CatalogEntry.additionalClassification [ CodeableConcept ], ... ; # 0..* Additional classification of the catalog
    entry
  fhir:CatalogEntry.relatedEntry [ # 0..* An item that this catalog entry is related to
    fhir:CatalogEntry.relatedEntry.relationtype [ code ]; # 1..1 triggers | is-replaced-by
    fhir:CatalogEntry.relatedEntry.item [ Reference(CatalogEntry) ]; # 1..1 The reference to the related item
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.28.3.1 Terminology Bindings

Path	Definition	Type	Reference
CatalogEntry.status	The lifecycle status of an artifact.	Required	PublicationStatus
CatalogEntry.relatedEntry.relationtype	The type of relations between entries.	Required	CatalogEntryRelationType

8.28.4 Resource CatalogEntry - Examples

URL: https://hl7.org/fhir/R4/catalogentry-examples.html

8.28.4 Resource CatalogEntry - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name	id	Format
Example of catalogentry	example	XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.28.5 Resource CatalogEntry - Detailed Descriptions

URL: https://hl7.org/fhir/R4/catalogentry-definitions.html

8.28.5 Resource CatalogEntry - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the CatalogEntry resource.

CatalogEntry
Element Id | CatalogEntry
Definition | Catalog entries are wrappers that contextualize items included in a catalog.
Cardinality | 0..*
Type | DomainResource
CatalogEntry.identifier
Element Id | CatalogEntry.identifier
Definition | Used in supporting different identifiers for the same product, e.g. manufacturer code and retailer code.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
CatalogEntry.type
Element Id | CatalogEntry.type
Definition | The type of item - medication, device, service, protocol or other.
Cardinality | 0..1
Type | CodeableConcept
CatalogEntry.orderable
Element Id | CatalogEntry.orderable
Definition | Whether the entry represents an orderable item.
Cardinality | 1..1
Type | boolean
Summary | true
CatalogEntry.referencedItem
Element Id | CatalogEntry.referencedItem
Definition | The item in a catalog or definition.
Cardinality | 1..1
Type | Reference(Medication | Device | Organization | Practitioner | PractitionerRole | HealthcareService | ActivityDefinition | PlanDefinition | SpecimenDefinition | ObservationDefinition | Binary)
Summary | true
CatalogEntry.additionalIdentifier
Element Id | CatalogEntry.additionalIdentifier
Definition | Used in supporting related concepts, e.g. NDC to RxNorm.
Cardinality | 0..*
Type | Identifier
CatalogEntry.classification
Element Id | CatalogEntry.classification
Definition | Classes of devices, or ATC for medication.
Cardinality | 0..*
Type | CodeableConcept
CatalogEntry.status
Element Id | CatalogEntry.status
Definition | Used to support catalog exchange even for unsupported products, e.g. getting list of medications even if not prescribable.
Cardinality | 0..1
Terminology Binding | PublicationStatus (Required)
Type | code
CatalogEntry.validityPeriod
Element Id | CatalogEntry.validityPeriod
Definition | The time period in which this catalog entry is expected to be active.
Cardinality | 0..1
Type | Period
CatalogEntry.validTo
Element Id | CatalogEntry.validTo
Definition | The date until which this catalog entry is expected to be active.
Cardinality | 0..1
Type | dateTime
CatalogEntry.lastUpdated
Element Id | CatalogEntry.lastUpdated
Definition | Typically date of issue is different from the beginning of the validity. This can be used to see when an item was last updated.
Cardinality | 0..1
Type | dateTime

Comments | Perhaps not needed - if we use fhir resource metadata.
CatalogEntry.additionalCharacteristic
Element Id | CatalogEntry.additionalCharacteristic
Definition | Used for examplefor Out of Formulary, or any specifics.
Cardinality | 0..*
Type | CodeableConcept
CatalogEntry.additionalClassification
Element Id | CatalogEntry.additionalClassification
Definition | User for example for ATC classification, or.
Cardinality | 0..*
Type | CodeableConcept
CatalogEntry.relatedEntry
Element Id | CatalogEntry.relatedEntry
Definition | Used for example, to point to a substance, or to a device used to administer a medication.
Cardinality | 0..*
CatalogEntry.relatedEntry.relationtype
Element Id | CatalogEntry.relatedEntry.relationtype
Definition | The type of relation to the related item: child, parent, packageContent, containerPackage, usedIn, uses, requires, etc.
Cardinality | 1..1
Terminology Binding | CatalogEntryRelationType (Required)
Type | code
CatalogEntry.relatedEntry.item
Element Id | CatalogEntry.relatedEntry.item
Definition | The reference to the related item.
Cardinality | 1..1
Type | Reference(CatalogEntry)

8.28.6 Resource CatalogEntry - Mappings

URL: https://hl7.org/fhir/R4/catalogentry-mappings.html

8.28.6 Resource CatalogEntry - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings for the catalogentry resource (see Mappings to Other Standards for further information & status).

8.28.7 Resource CatalogEntry - Extensions & Profiles

URL: https://hl7.org/fhir/R4/catalogentry-profiles.html

8.28.7 Resource CatalogEntry - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the CatalogEntry resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.28.7.1 Profiles

No Profiles defined for this resource

8.28.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.28.7.3 Search Extensions

No Search Extensions defined for this resource

8.28.8 JSON Schema for CatalogEntry

URL: https://hl7.org/fhir/R4/catalogentry.schema.json.html

8.28.8 JSON Schema for CatalogEntry

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for CatalogEntry, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CatalogEntry",
  "$ref": "#/definitions/CatalogEntry",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CatalogEntry": {
      "description": "Catalog entries are wrappers that contextualize items included in a catalog.",
      "properties": {
        "resourceType": {
          "description": "This is a CatalogEntry resource",
          "const": "CatalogEntry"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
```

```
      "type": "array"
    },
    "identifier": {
      "description": "Used in supporting different identifiers for the same product, e.g. manufacturer code and
retailer code.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of item - medication, device, service, protocol or other.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "orderable": {
      "description": "Whether the entry represents an orderable item.",
      "$ref": "#/definitions/boolean"
    },
    "_orderable": {
      "description": "Extensions for orderable",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "referencedItem": {
      "description": "The item in a catalog or definition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "additionalIdentifier": {
      "description": "Used in supporting related concepts, e.g. NDC to RxNorm.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "classification": {
      "description": "Classes of devices, or ATC for medication.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "status": {
      "description": "Used to support catalog exchange even for unsupported products, e.g. getting list of
medications even if not prescribable.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "validityPeriod": {
      "description": "The time period in which this catalog entry is expected to be active.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "validTo": {
      "description": "The date until which this catalog entry is expected to be active.",
      "$ref": "#/definitions/dateTime"
    },
    "_validTo": {
      "description": "Extensions for validTo",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastUpdated": {
      "description": "Typically date of issue is different from the beginning of the validity. This can be used to
see when an item was last updated.",
      "$ref": "#/definitions/dateTime"
    },
    "_lastUpdated": {
      "description": "Extensions for lastUpdated",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "additionalCharacteristic": {
      "description": "Used for examplefor Out of Formulary, or any specifics.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "additionalClassification": {
      "description": "User for example for ATC classification, or.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
```

```
        "relatedEntry": {
          "description": "Used for example, to point to a substance, or to a device used to administer a medication.",
          "items": {
            "$ref": "#/definitions/CatalogEntry_RelatedEntry"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType",
        "referencedItem"
      ]
    },
    "CatalogEntry_RelatedEntry": {
      "description": "Catalog entries are wrappers that contextualize items included in a catalog.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "relationtype": {
          "description": "The type of relation to the related item: child, parent, packageContent, containerPackage,
    usedIn, uses, requires, etc.",
          "enum": [
            "triggers",
            "is-replaced-by"
          ]
        },
        "_relationtype": {
          "description": "Extensions for relationtype",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "item": {
          "description": "The reference to the related item.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "item"
      ]
    }
  }
}
```


8.28.9 Resource CatalogEntry - Maturity Dependency View

URL: https://hl7.org/fhir/R4/catalogentry-dependencies.html

8.28.9 Resource CatalogEntry - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for CatalogEntry FMM level 0

Name | Card. | Type | Dependency Analysis
CatalogEntry | | DomainResource
identifier | 0..* | Identifier
type | 0..1 | CodeableConcept
orderable | 1..1 | boolean

referencedItem | 1..1 | Reference(Medication | Device | Organization | Practitioner | PractitionerRole | HealthcareService | ActivityDefinition | PlanDefinition | SpecimenDefinition | ObservationDefinition | Binary)
additionalIdentifier | 0..* | Identifier
classification | 0..* | CodeableConcept
status | 0..1 | code
validityPeriod | 0..1 | Period
validTo | 0..1 | dateTime
lastUpdated | 0..1 | dateTime
additionalCharacteristic | 0..* | CodeableConcept
additionalClassification | 0..* | CodeableConcept
relatedEntry | 0..* | BackboneElement
relationtype | 1..1 | code
item | 1..1 | Reference(CatalogEntry)

8.28.10 Value Set Sc-valueset-publication-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-publication-status.html

8.28.10 Value Set Sc-valueset-publication-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.28.11 PublicationStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-publication-status)

Mapping from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The lifecycle status of an artifact."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
draft | equivalent | draft
active | equivalent | active
retired | equivalent | inactive
unknown | equivalent | unknown

8.29 Resource Invoice - Content

URL: https://hl7.org/fhir/R4/invoice.html
Financial Management Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Invoice containing collected ChargeItems from an Account with calculated individual and total price for Billing purpose.

8.29.1 Scope and Usage

Tracking Financial information is vital in Patient Administration and Finance systems in most Healthcare Organizations. An invoice is a financial document issued by a healthcare provider to a patient or a payer indicating the goods and services (ChargeItems) performed with their quantities and prices.

Example implementations:

• Any solution that tracks billing information and needs to issue invoices

• Private Insurance Providers who want to deliver structured information to patients to increase cost transparency

• Patient apps that want to include information on the amount and reason of the charged costs for the healthcare services a patient received

8.29.2 Boundaries and Relationships

The existing Claim resource is constricted to use cases where Claims are sent to insurances for reimbursement, in a message-like style. It neither references ChargeItems nor Accounts.

Competing invoicing standards such as EDIFACT or X12 are aiming at inter-organizational exchange and do not offer human readable representations or traceable links to services rendered which is a requirement for invoicing towards patients. This resource is used where existing ChargeItems in an Account are collected to create an Invoice to be sent out to Individuals or Organizations in a structured way. The invoice is the hospital's view, whereas the Claim is the payer's view on the performed services.

Since the individual ChargeItem resources that make up the line items of the Invoice do not carry any pricing information, ChargeItemDefinitions will have to be applied to the ChargeItems in order to calculate the price components for the Invoice depending on the context of the ChargeItem

8.29.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Invoice | TU | | DomainResource | Invoice containing ChargeItems from an Account Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for item
status | ?!Σ | 1..1 | code | draft | issued | balanced | cancelled | entered-in-error InvoiceStatus (Required)
cancelledReason | | 0..1 | string | Reason for cancellation of this Invoice
type | Σ | 0..1 | CodeableConcept | Type of Invoice
subject | Σ | 0..1 | Reference(Patient | Group) | Recipient(s) of goods and services
recipient | Σ | 0..1 | Reference(Organization | Patient | RelatedPerson) | Recipient of this invoice
date | Σ | 0..1 | dateTime | Invoice date / posting date
participant | | 0..* | BackboneElement | Participant in creation of this Invoice
role | | 0..1 | CodeableConcept | Type of involvement in creation of this Invoice
actor | | 1..1 | Reference(Practitioner | Organization | Patient | PractitionerRole | Device | RelatedPerson) | Individual who was involved
issuer | | 0..1 | Reference(Organization) | Issuing Organization of Invoice
account | | 0..1 | Reference(Account) | Account that is being balanced
lineItem | | 0..* | BackboneElement | Line items of this Invoice
sequence | | 0..1 | positiveInt | Sequence number of line item
chargeItem[x] | | 1..1 | | Reference to ChargeItem containing details of this line item or an inline billing code
chargeItemReference | | | Reference(ChargeItem)
chargeItemCodeableConcept | | | CodeableConcept
priceComponent | | 0..* | BackboneElement | Components of total line item price
type | | 1..1 | code | base | surcharge | deduction | discount | tax | informational InvoicePriceComponentType (Required)
code | | 0..1 | CodeableConcept | Code identifying the specific component
factor | | 0..1 | decimal | Factor used for calculating this component
amount | | 0..1 | Money | Monetary amount associated with this component
totalPriceComponent | | 0..* | see priceComponent | Components of Invoice total
totalNet | Σ | 0..1 | Money | Net total of this Invoice
totalGross | Σ | 0..1 | Money | Gross total of this Invoice
paymentTerms | | 0..1 | markdown | Payment details
note | | 0..* | Annotation | Comments made about the invoice
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Invoice xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for item --></identifier>
 <status value="[code]"/><!-- 1..1 draft | issued | balanced | cancelled | entered-in-error -->
 <cancelledReason value="[string]"/><!-- 0..1 Reason for cancellation of this Invoice -->
 <type><!-- 0..1 CodeableConcept Type of Invoice --></type>
 <subject><!-- 0..1 Reference(Patient|Group) Recipient(s) of goods and services --></subject>
 <recipient><!-- 0..1 Reference(Organization|Patient|RelatedPerson) Recipient of this invoice --></recipient>
 <date value="[dateTime]"/><!-- 0..1 Invoice date / posting date -->
 <participant>  <!-- 0..* Participant in creation of this Invoice -->
  <role><!-- 0..1 CodeableConcept Type of involvement in creation of this Invoice --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|PractitionerRole|
    Device|RelatedPerson) Individual who was involved --></actor>
 </participant>
 <issuer><!-- 0..1 Reference(Organization) Issuing Organization of Invoice --></issuer>
 <account><!-- 0..1 Reference(Account) Account that is being balanced --></account>
 <lineItem>  <!-- 0..* Line items of this Invoice -->
  <sequence value="[positiveInt]"/><!-- 0..1 Sequence number of line item -->
  <chargeItem[x]><!-- 1..1 Reference(ChargeItem)|CodeableConcept Reference to ChargeItem containing details of this line
     item or an inline billing code --></chargeItem[x]>
  <priceComponent>  <!-- 0..* Components of total line item price -->
   <type value="[code]"/><!-- 1..1 base | surcharge | deduction | discount | tax | informational -->
   <code><!-- 0..1 CodeableConcept Code identifying the specific component --></code>
   <factor value="[decimal]"/><!-- 0..1 Factor used for calculating this component -->
```

```
   <amount><!-- 0..1 Money Monetary amount associated with this component --></amount>
  </priceComponent>
 </lineItem>
 <totalPriceComponent><!-- 0..* Content as for Invoice.lineItem.priceComponent Components of Invoice total
    --></totalPriceComponent>
 <totalNet><!-- 0..1 Money Net total of this Invoice --></totalNet>
 <totalGross><!-- 0..1 Money Gross total of this Invoice --></totalGross>
 <paymentTerms value="[markdown]"/><!-- 0..1 Payment details -->
 <note><!-- 0..* Annotation Comments made about the invoice --></note>
</Invoice>
```

JSON Template

```
{
  "resourceType" : "Invoice",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for item
  "status" : "<code>", // R!  draft | issued | balanced | cancelled | entered-in-error
  "cancelledReason" : "<string>", // Reason for cancellation of this Invoice
  "type" : { CodeableConcept }, // Type of Invoice
  "subject" : { Reference(Patient|Group) }, // Recipient(s) of goods and services
  "recipient" : { Reference(Organization|Patient|RelatedPerson) }, // Recipient of this invoice
  "date" : "<dateTime>", // Invoice date / posting date
  "participant" : [{ // Participant in creation of this Invoice
    "role" : { CodeableConcept }, // Type of involvement in creation of this Invoice
    "actor" : { Reference(Practitioner|Organization|Patient|PractitionerRole|
    Device|RelatedPerson) } // R!  Individual who was involved
  }],
  "issuer" : { Reference(Organization) }, // Issuing Organization of Invoice
  "account" : { Reference(Account) }, // Account that is being balanced
  "lineItem" : [{ // Line items of this Invoice
    "sequence" : "<positiveInt>", // Sequence number of line item
    // chargeItem[x]:
      Reference to ChargeItem containing details of this line item or an inline billing code. One of these 2:
    "chargeItemReference" : { Reference(ChargeItem) },
    "chargeItemCodeableConcept" : { CodeableConcept },
    "priceComponent" : [{ // Components of total line item price
      "type" : "<code>", // R!  base | surcharge | deduction | discount | tax | informational
      "code" : { CodeableConcept }, // Code identifying the specific component
      "factor" : <decimal>, // Factor used for calculating this component
      "amount" : { Money } // Monetary amount associated with this component
    }]
  }],
  "totalPriceComponent" : [{ Content as for Invoice.lineItem.priceComponent }], // Components of Invoice total
  "totalNet" : { Money }, // Net total of this Invoice
  "totalGross" : { Money }, // Gross total of this Invoice
  "paymentTerms" : "<markdown>", // Payment details
  "note" : [{ Annotation }] // Comments made about the invoice
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Invoice;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Invoice.identifier [ Identifier ], ... ; # 0..* Business Identifier for item
  fhir:Invoice.status [ code ]; # 1..1 draft | issued | balanced | cancelled | entered-in-error
  fhir:Invoice.cancelledReason [ string ]; # 0..1 Reason for cancellation of this Invoice
  fhir:Invoice.type [ CodeableConcept ]; # 0..1 Type of Invoice
  fhir:Invoice.subject [ Reference(Patient|Group) ]; # 0..1 Recipient(s) of goods and services
  fhir:Invoice.recipient [ Reference(Organization|Patient|RelatedPerson) ]; # 0..1 Recipient of this invoice
  fhir:Invoice.date [ dateTime ]; # 0..1 Invoice date / posting date
  fhir:Invoice.participant [ # 0..* Participant in creation of this Invoice
    fhir:Invoice.participant.role [ CodeableConcept ]; # 0..1 Type of involvement in creation of this Invoice
    fhir:Invoice.participant.actor [ Reference(Practitioner|Organization|Patient|PractitionerRole|Device|RelatedPerson)
    ]; # 1..1 Individual who was involved
  ], ...;
  fhir:Invoice.issuer [ Reference(Organization) ]; # 0..1 Issuing Organization of Invoice
  fhir:Invoice.account [ Reference(Account) ]; # 0..1 Account that is being balanced
  fhir:Invoice.lineItem [ # 0..* Line items of this Invoice
    fhir:Invoice.lineItem.sequence [ positiveInt ]; # 0..1 Sequence number of line item
    # Invoice.lineItem.chargeItem[x] : 1..1 Reference to ChargeItem containing details of this line item or an inline
    billing code. One of these 2
      fhir:Invoice.lineItem.chargeItemReference [ Reference(ChargeItem) ]
      fhir:Invoice.lineItem.chargeItemCodeableConcept [ CodeableConcept ]
    fhir:Invoice.lineItem.priceComponent [ # 0..* Components of total line item price
      fhir:Invoice.lineItem.priceComponent.type [ code ]; # 1..1 base | surcharge | deduction | discount | tax |
    informational
      fhir:Invoice.lineItem.priceComponent.code [ CodeableConcept ]; # 0..1 Code identifying the specific component
      fhir:Invoice.lineItem.priceComponent.factor [ decimal ]; # 0..1 Factor used for calculating this component
      fhir:Invoice.lineItem.priceComponent.amount [ Money ]; # 0..1 Monetary amount associated with this component
    ], ...;
  ], ...;
  fhir:Invoice.totalPriceComponent [ See Invoice.lineItem.priceComponent ], ... ; # 0..* Components of Invoice total
  fhir:Invoice.totalNet [ Money ]; # 0..1 Net total of this Invoice
```

```
  fhir:Invoice.totalGross [ Money ]; # 0..1 Gross total of this Invoice
  fhir:Invoice.paymentTerms [ markdown ]; # 0..1 Payment details
  fhir:Invoice.note [ Annotation ], ... ; # 0..* Comments made about the invoice
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
Invoice | TU | | DomainResource | Invoice containing ChargeItems from an Account Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for item
status | ?!Σ | 1..1 | code | draft | issued | balanced | cancelled | entered-in-error InvoiceStatus (Required)
cancelledReason | | 0..1 | string | Reason for cancellation of this Invoice
type | Σ | 0..1 | CodeableConcept | Type of Invoice
subject | Σ | 0..1 | Reference(Patient | Group) | Recipient(s) of goods and services
recipient | Σ | 0..1 | Reference(Organization | Patient | RelatedPerson) | Recipient of this invoice
date | Σ | 0..1 | dateTime | Invoice date / posting date
participant | | 0..* | BackboneElement | Participant in creation of this Invoice
role | | 0..1 | CodeableConcept | Type of involvement in creation of this Invoice
actor | | 1..1 | Reference(Practitioner | Organization | Patient | PractitionerRole | Device | RelatedPerson) | Individual who was involved
issuer | | 0..1 | Reference(Organization) | Issuing Organization of Invoice
account | | 0..1 | Reference(Account) | Account that is being balanced
lineItem | | 0..* | BackboneElement | Line items of this Invoice
sequence | | 0..1 | positiveInt | Sequence number of line item
chargeItem[x] | | 1..1 | | Reference to ChargeItem containing details of this line item or an inline billing code
chargeItemReference | | | Reference(ChargeItem)
chargeItemCodeableConcept | | | CodeableConcept
priceComponent | | 0..* | BackboneElement | Components of total line item price
type | | 1..1 | code | base | surcharge | deduction | discount | tax | informational InvoicePriceComponentType (Required)
code | | 0..1 | CodeableConcept | Code identifying the specific component
factor | | 0..1 | decimal | Factor used for calculating this component
amount | | 0..1 | Money | Monetary amount associated with this component
totalPriceComponent | | 0..* | see priceComponent | Components of Invoice total
totalNet | Σ | 0..1 | Money | Net total of this Invoice
totalGross | Σ | 0..1 | Money | Gross total of this Invoice
paymentTerms | | 0..1 | markdown | Payment details
note | | 0..* | Annotation | Comments made about the invoice
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Invoice xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for item --></identifier>
 <status value="[code]"/><!-- 1..1 draft | issued | balanced | cancelled | entered-in-error -->
 <cancelledReason value="[string]"/><!-- 0..1 Reason for cancellation of this Invoice -->
 <type><!-- 0..1 CodeableConcept Type of Invoice --></type>
 <subject><!-- 0..1 Reference(Patient|Group) Recipient(s) of goods and services --></subject>
 <recipient><!-- 0..1 Reference(Organization|Patient|RelatedPerson) Recipient of this invoice --></recipient>
 <date value="[dateTime]"/><!-- 0..1 Invoice date / posting date -->
 <participant>  <!-- 0..* Participant in creation of this Invoice -->
  <role><!-- 0..1 CodeableConcept Type of involvement in creation of this Invoice --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|PractitionerRole|
    Device|RelatedPerson) Individual who was involved --></actor>
 </participant>
 <issuer><!-- 0..1 Reference(Organization) Issuing Organization of Invoice --></issuer>
 <account><!-- 0..1 Reference(Account) Account that is being balanced --></account>
 <lineItem>  <!-- 0..* Line items of this Invoice -->
  <sequence value="[positiveInt]"/><!-- 0..1 Sequence number of line item -->
  <chargeItem[x]><!-- 1..1 Reference(ChargeItem)|CodeableConcept Reference to ChargeItem containing details of this line
     item or an inline billing code --></chargeItem[x]>
  <priceComponent>  <!-- 0..* Components of total line item price -->
   <type value="[code]"/><!-- 1..1 base | surcharge | deduction | discount | tax | informational -->
   <code><!-- 0..1 CodeableConcept Code identifying the specific component --></code>
   <factor value="[decimal]"/><!-- 0..1 Factor used for calculating this component -->
   <amount><!-- 0..1 Money Monetary amount associated with this component --></amount>
  </priceComponent>
 </lineItem>
 <totalPriceComponent><!-- 0..* Content as for Invoice.lineItem.priceComponent Components of Invoice total
    --></totalPriceComponent>
 <totalNet><!-- 0..1 Money Net total of this Invoice --></totalNet>
 <totalGross><!-- 0..1 Money Gross total of this Invoice --></totalGross>
 <paymentTerms value="[markdown]"/><!-- 0..1 Payment details -->
 <note><!-- 0..* Annotation Comments made about the invoice --></note>
</Invoice>
```

JSON Template

```
{
  "resourceType" : "Invoice",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for item
  "status" : "<code>", // R!  draft | issued | balanced | cancelled | entered-in-error
  "cancelledReason" : "<string>", // Reason for cancellation of this Invoice
  "type" : { CodeableConcept }, // Type of Invoice
  "subject" : { Reference(Patient|Group) }, // Recipient(s) of goods and services
  "recipient" : { Reference(Organization|Patient|RelatedPerson) }, // Recipient of this invoice
  "date" : "<dateTime>", // Invoice date / posting date
  "participant" : [{ // Participant in creation of this Invoice
    "role" : { CodeableConcept }, // Type of involvement in creation of this Invoice
    "actor" : { Reference(Practitioner|Organization|Patient|PractitionerRole|
    Device|RelatedPerson) } // R!  Individual who was involved
  }],
  "issuer" : { Reference(Organization) }, // Issuing Organization of Invoice
  "account" : { Reference(Account) }, // Account that is being balanced
  "lineItem" : [{ // Line items of this Invoice
    "sequence" : "<positiveInt>", // Sequence number of line item
    // chargeItem[x]:
      Reference to ChargeItem containing details of this line item or an inline billing code. One of these 2:
    "chargeItemReference" : { Reference(ChargeItem) },
    "chargeItemCodeableConcept" : { CodeableConcept },
    "priceComponent" : [{ // Components of total line item price
      "type" : "<code>", // R!  base | surcharge | deduction | discount | tax | informational
      "code" : { CodeableConcept }, // Code identifying the specific component
      "factor" : <decimal>, // Factor used for calculating this component
      "amount" : { Money } // Monetary amount associated with this component
    }]
  }],
  "totalPriceComponent" : [{ Content as for Invoice.lineItem.priceComponent }], // Components of Invoice total
  "totalNet" : { Money }, // Net total of this Invoice
  "totalGross" : { Money }, // Gross total of this Invoice
  "paymentTerms" : "<markdown>", // Payment details
  "note" : [{ Annotation }] // Comments made about the invoice
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Invoice;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Invoice.identifier [ Identifier ], ... ; # 0..* Business Identifier for item
  fhir:Invoice.status [ code ]; # 1..1 draft | issued | balanced | cancelled | entered-in-error
  fhir:Invoice.cancelledReason [ string ]; # 0..1 Reason for cancellation of this Invoice
  fhir:Invoice.type [ CodeableConcept ]; # 0..1 Type of Invoice
  fhir:Invoice.subject [ Reference(Patient|Group) ]; # 0..1 Recipient(s) of goods and services
  fhir:Invoice.recipient [ Reference(Organization|Patient|RelatedPerson) ]; # 0..1 Recipient of this invoice
  fhir:Invoice.date [ dateTime ]; # 0..1 Invoice date / posting date
  fhir:Invoice.participant [ # 0..* Participant in creation of this Invoice
    fhir:Invoice.participant.role [ CodeableConcept ]; # 0..1 Type of involvement in creation of this Invoice
    fhir:Invoice.participant.actor [ Reference(Practitioner|Organization|Patient|PractitionerRole|Device|RelatedPerson)
    ]; # 1..1 Individual who was involved
  ], ...;
  fhir:Invoice.issuer [ Reference(Organization) ]; # 0..1 Issuing Organization of Invoice
  fhir:Invoice.account [ Reference(Account) ]; # 0..1 Account that is being balanced
  fhir:Invoice.lineItem [ # 0..* Line items of this Invoice
    fhir:Invoice.lineItem.sequence [ positiveInt ]; # 0..1 Sequence number of line item
    # Invoice.lineItem.chargeItem[x] : 1..1 Reference to ChargeItem containing details of this line item or an inline
    billing code. One of these 2
      fhir:Invoice.lineItem.chargeItemReference [ Reference(ChargeItem) ]
      fhir:Invoice.lineItem.chargeItemCodeableConcept [ CodeableConcept ]
    fhir:Invoice.lineItem.priceComponent [ # 0..* Components of total line item price
      fhir:Invoice.lineItem.priceComponent.type [ code ]; # 1..1 base | surcharge | deduction | discount | tax |
    informational
      fhir:Invoice.lineItem.priceComponent.code [ CodeableConcept ]; # 0..1 Code identifying the specific component
      fhir:Invoice.lineItem.priceComponent.factor [ decimal ]; # 0..1 Factor used for calculating this component
      fhir:Invoice.lineItem.priceComponent.amount [ Money ]; # 0..1 Monetary amount associated with this component
    ], ...;
  ], ...;
  fhir:Invoice.totalPriceComponent [ See Invoice.lineItem.priceComponent ], ... ; # 0..* Components of Invoice total
  fhir:Invoice.totalNet [ Money ]; # 0..1 Net total of this Invoice
  fhir:Invoice.totalGross [ Money ]; # 0..1 Gross total of this Invoice
  fhir:Invoice.paymentTerms [ markdown ]; # 0..1 Payment details
  fhir:Invoice.note [ Annotation ], ... ; # 0..* Comments made about the invoice
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.29.3.1 Terminology Bindings

Path | Definition | Type | Reference
Invoice.status | Codes identifying the lifecycle stage of an Invoice. | Required | InvoiceStatus
Invoice.lineItem.priceComponent.type | Codes indicating the kind of the price component. | Required | InvoicePriceComponentType

8.29.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
account | reference | Account that is being balanced | Invoice.account (Account)
date | date | Invoice date / posting date | Invoice.date
identifier | token | Business Identifier for item | Invoice.identifier
issuer | reference | Issuing Organization of Invoice | Invoice.issuer (Organization)
participant | reference | Individual who was involved | Invoice.participant.actor (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
participant-role | token | Type of involvement in creation of this Invoice | Invoice.participant.role
patient | reference | Recipient(s) of goods and services | Invoice.subject.where(resolve() is Patient) (Patient)
recipient | reference | Recipient of this invoice | Invoice.recipient (Organization, Patient, RelatedPerson)
status | token | draft | issued | balanced | cancelled | entered-in-error | Invoice.status
subject | reference | Recipient(s) of goods and services | Invoice.subject (Group, Patient)
totalgross | quantity | Gross total of this Invoice | Invoice.totalGross
totalnet | quantity | Net total of this Invoice | Invoice.totalNet
type | token | Type of Invoice | Invoice.type

8.29.5 Resource Invoice - Examples

URL: https://hl7.org/fhir/R4/invoice-examples.html

8.29.5 Resource Invoice - Examples

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Example of invoice | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.29.6 Resource Invoice - Detailed Descriptions

URL: https://hl7.org/fhir/R4/invoice-definitions.html

8.29.6 Resource Invoice - Detailed Descriptions

Financial Management Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Invoice resource.

Invoice
Element Id | Invoice
Definition | Invoice containing collected ChargeItems from an Account with calculated individual and total price for Billing purpose.
Cardinality | 0..*
Type | DomainResource
Invoice.identifier
Element Id | Invoice.identifier
Definition | Identifier of this Invoice, often used for reference in correspondence about this invoice or for tracking of payments.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows Identification of this Invoice instance.
Summary | true
Invoice.status
Element Id | Invoice.status
Definition | The current state of the Invoice.
Cardinality | 1..1
Terminology Binding | InvoiceStatus (Required)
Type | code

Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Invoice.cancelledReason
Element Id | Invoice.cancelledReason
Definition | In case of Invoice cancellation a reason must be given (entered in error, superseded by corrected invoice etc.).
Cardinality | 0..1
Type | string
Summary | false
Comments | Derived Profiles may choose to add invariants requiring this field to be populated if either priceOverride or factorOverride have been filled.
Invoice.type
Element Id | Invoice.type
Definition | Type of Invoice depending on domain, realm an usage (e.g. internal/external, dental, preliminary).
Cardinality | 0..1
Type | CodeableConcept
Alternate Names | type
Summary | true
Invoice.subject
Element Id | Invoice.subject
Definition | The individual or set of individuals receiving the goods and services billed in this invoice.
Cardinality | 0..1
Type | Reference(Patient | Group)
Requirements | Links the event to the Patient context.
Alternate Names | patient
Summary | true
Invoice.recipient
Element Id | Invoice.recipient
Definition | The individual or Organization responsible for balancing of this invoice.
Cardinality | 0..1
Type | Reference(Organization | Patient | RelatedPerson)
Summary | true
Invoice.date
Element Id | Invoice.date
Definition | Date/time(s) of when this Invoice was posted.
Cardinality | 0..1
Type | dateTime
Summary | true
Comments | The list of types may be constrained as appropriate for the type of charge item.
Invoice.participant
Element Id | Invoice.participant
Definition | Indicates who or what performed or participated in the charged service.
Cardinality | 0..*
Summary | false
Invoice.participant.role
Element Id | Invoice.participant.role
Definition | Describes the type of involvement (e.g. transcriptionist, creator etc.). If the invoice has been created automatically, the Participant may be a billing engine or another kind of device.
Cardinality | 0..1
Type | CodeableConcept
Summary | false
Invoice.participant.actor
Element Id | Invoice.participant.actor
Definition | The device, practitioner, etc. who performed or participated in the service.
Cardinality | 1..1
Type | Reference(Practitioner | Organization | Patient | PractitionerRole | Device | RelatedPerson)
Summary | false
Invoice.issuer
Element Id | Invoice.issuer
Definition | The organizationissuing the Invoice.
Cardinality | 0..1
Type | Reference(Organization)
Summary | false
Comments | Practitioners and Devices can be associated with multiple organizations. It has to be made clear, on behalf of which Organization the services have been rendered.
Invoice.account
Element Id | Invoice.account
Definition | Account which is supposed to be balanced with this Invoice.
Cardinality | 0..1
Type | Reference(Account)
Summary | false
Comments | Systems posting the ChargeItems might not always be able to determine, which accounts the Items need to be places into. It is up to the potprocessing Financial System to apply internal rules to decide based on the

Encounter/EpisodeOfCare/Patient/Coverage context and the type of ChargeItem, which Account is appropriate.
Invoice.lineItem
Element Id | Invoice.lineItem
Definition | Each line item represents one charge for goods and services rendered. Details such as date, code and amount are found in the referenced ChargeItem resource.
Cardinality | 0..*
Summary | false
Invoice.lineItem.sequence
Element Id | Invoice.lineItem.sequence
Definition | Sequence in which the items appear on the invoice.
Cardinality | 0..1
Type | positiveInt
Summary | false
Invoice.lineItem.chargeItem[x]
Element Id | Invoice.lineItem.chargeItem[x]
Definition | The ChargeItem contains information such as the billing code, date, amount etc. If no further details are required for the lineItem, inline billing codes can be added using the CodeableConcept data type instead of the Reference.
Cardinality | 1..1
Type | Reference(ChargeItem)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | false
Invoice.lineItem.priceComponent
Element Id | Invoice.lineItem.priceComponent
Definition | The price for a ChargeItem may be calculated as a base price with surcharges/deductions that apply in certain conditions. A ChargeItemDefinition resource that defines the prices, factors and conditions that apply to a billing code is currently under development. The priceComponent element can be used to offer transparency to the recipient of the Invoice as to how the prices have been calculated.
Cardinality | 0..*
Summary | false
Invoice.lineItem.priceComponent.type
Element Id | Invoice.lineItem.priceComponent.type
Definition | This code identifies the type of the component.
Cardinality | 1..1
Terminology Binding | InvoicePriceComponentType (Required)
Type | code
Summary | false
Invoice.lineItem.priceComponent.code
Element Id | Invoice.lineItem.priceComponent.code
Definition | A code that identifies the component. Codes may be used to differentiate between kinds of taxes, surcharges, discounts etc.
Cardinality | 0..1
Type | CodeableConcept
Summary | false
Invoice.lineItem.priceComponent.factor
Element Id | Invoice.lineItem.priceComponent.factor
Definition | The factor that has been applied on the base price for calculating this component.
Cardinality | 0..1
Type | decimal
Summary | false
Comments | There is no reason to carry the price in the instance of a ChargeItem unless circumstances require a manual override. The list prices or are usually defined in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overridden.
Invoice.lineItem.priceComponent.amount
Element Id | Invoice.lineItem.priceComponent.amount
Definition | The amount calculated for this component.
Cardinality | 0..1
Type | Money
Summary | false
Comments | There is no reason to carry the price in the instance of a ChargeItem unless circumstances require a manual override. The list prices or are usually defined in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overridden.
Invoice.totalPriceComponent
Element Id | Invoice.totalPriceComponent
Definition | The total amount for the Invoice may be calculated as the sum of the line items with surcharges/deductions that apply in certain conditions. The priceComponent element can be used to offer transparency to the recipient of the Invoice of how the total price was calculated.
Cardinality | 0..*
Type | See Invoice.lineItem.priceComponent
Summary | false
Invoice.totalNet
Element Id | Invoice.totalNet
Definition | Invoice total , taxes excluded.
Cardinality | 0..1

Type | Money
Summary | true
Comments | There is no reason to carry the price in the instance of a ChargeItem unless circumstances require a manual override. The list prices or are usually defined in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overridden.
Invoice.totalGross
Element Id | Invoice.totalGross
Definition | Invoice total, tax included.
Cardinality | 0..1
Type | Money
Summary | true
Comments | There is no reason to carry the price in the instance of a ChargeItem unless circumstances require a manual override. The list prices or are usually defined in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overridden.
Invoice.paymentTerms
Element Id | Invoice.paymentTerms
Definition | Payment details such as banking details, period of payment, deductibles, methods of payment.
Cardinality | 0..1
Type | markdown
Summary | false
Comments | Derived Profiles may chose to add invariants requiring this field to be populated if either priceOverride or factorOverride have been filled.
Invoice.note
Element Id | Invoice.note
Definition | Comments made about the invoice by the issuer, subject, or other participants.
Cardinality | 0..*
Type | Annotation
Summary | false

8.29.7 Resource Invoice - Mappings

URL: https://hl7.org/fhir/R4/invoice-mappings.html

8.29.7 Resource Invoice - Mappings

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the invoice resource (see Mappings to Other Standards for further information & status).

8.29.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Invoice
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.what[x]
subject | FiveWs.subject[x]
date | FiveWs.done[x]
actor | FiveWs.actor

8.29.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Invoice
type | Event.code
subject | Event.subject
recipient | Event.context
type | Event.code
code | Event.code
note | Event.note

8.29.7.3 HL7 v2 Mapping (http://hl7.org/v2)

Invoice | FT1
identifier | FT1.2
status | Varies by domain
cancelledReason
type | FT1.7
subject | PAT in proximity to FT1 segment
recipient | PV1 in proximity to FT1 segment

date | Varies by domain
participant | EVN.5 or by domain
role | Varies by domain
actor | EVN.5 or by domain
issuer | N/A
account
lineItem | EVN.5 or by domain
sequence
chargeItem[x] | EVN.5 or by domain
priceComponent | EVN.5 or by domain
type | FT1.7
code | FT1.7
factor | FT1.22
amount | FT1.22
totalPriceComponent | EVN.5 or by domain
totalNet | FT1.22
totalGross | FT1.22
paymentTerms
note | NTE

8.29.7.4 RIM Mapping (http://hl7.org/v3)

Invoice | Act[moodCode=EVN]
identifier | .identifier
status | .status
type | .code
subject | .participation[typeCode=SBJ].role
recipient | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
date | .effectiveTime
participant | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
role | .participation.functionCode
actor | .player
issuer | .scoper
lineItem | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
chargeItem[x] | .player
priceComponent | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
type | .code
code | .code
totalPriceComponent | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]

8.29.8 Resource Invoice - Extensions & Profiles

URL: https://hl7.org/fhir/R4/invoice-profiles.html

8.29.8 Resource Invoice - Extensions & Profiles

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Invoice resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.29.8.1 Profiles

No Profiles defined for this resource

8.29.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.29.8.3 Search Extensions

No Search Extensions defined for this resource

8.29.9 JSON Schema for Invoice

URL: https://hl7.org/fhir/R4/invoice.schema.json.html

8.29.9 JSON Schema for Invoice

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Invoice, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Invoice",
  "$ref": "#/definitions/Invoice",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Invoice": {
      "description": "Invoice containing collected ChargeItems from an Account with calculated individual and total
    price for Billing purpose.",
      "properties": {
        "resourceType": {
          "description": "This is a Invoice resource",
          "const": "Invoice"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
```

```
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier of this Invoice, often used for reference in correspondence about this invoice or
for tracking of payments.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the Invoice.",
      "enum": [
        "draft",
        "issued",
        "balanced",
        "cancelled",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cancelledReason": {
      "description": "In case of Invoice cancellation a reason must be given (entered in error, superseded by
corrected invoice etc.).",
      "$ref": "#/definitions/string"
    },
    "_cancelledReason": {
      "description": "Extensions for cancelledReason",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Type of Invoice depending on domain, realm an usage (e.g. internal/external, dental,
preliminary).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The individual or set of individuals receiving the goods and services billed in this
invoice.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "recipient": {
      "description": "The individual or Organization responsible for balancing of this invoice.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "Date/time(s) of when this Invoice was posted.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "participant": {
      "description": "Indicates who or what performed or participated in the charged service.",
      "items": {
        "$ref": "#/definitions/Invoice_Participant"
      },
      "type": "array"
    },
    "issuer": {
      "description": "The organizationissuing the Invoice.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "account": {
      "description": "Account which is supposed to be balanced with this Invoice.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "lineItem": {
      "description": "Each line item represents one charge for goods and services rendered. Details such as date,
code and amount are found in the referenced ChargeItem resource.",
      "items": {
        "$ref": "#/definitions/Invoice_LineItem"
      },
      "type": "array"
    },
    "totalPriceComponent": {
      "description": "The total amount for the Invoice may be calculated as the sum of the line items with
surcharges/deductions that apply in certain conditions.  The priceComponent element can be used to offer
transparency to the recipient of the Invoice of how the total price was calculated.",
      "items": {
```

```
        "$ref": "#/definitions/Invoice_PriceComponent"
      },
      "type": "array"
    },
    "totalNet": {
      "description": "Invoice total , taxes excluded.",
      "$ref": "Money.schema.json#/definitions/Money"
    },
    "totalGross": {
      "description": "Invoice total, tax included.",
      "$ref": "Money.schema.json#/definitions/Money"
    },
    "paymentTerms": {
      "description": "Payment details such as banking details, period of payment, deductibles, methods of payment.",
      "$ref": "#/definitions/markdown"
    },
    "_paymentTerms": {
      "description": "Extensions for paymentTerms",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "Comments made about the invoice by the issuer, subject, or other participants.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Invoice_Participant": {
  "description": "Invoice containing collected ChargeItems from an Account with calculated individual and total
price for Billing purpose.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "role": {
      "description": "Describes the type of involvement (e.g. transcriptionist, creator etc.). If the invoice has
been created automatically, the Participant may be a billing engine or another kind of device.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "actor": {
      "description": "The device, practitioner, etc. who performed or participated in the service.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "actor"
  ]
},
"Invoice_LineItem": {
  "description": "Invoice containing collected ChargeItems from an Account with calculated individual and total
price for Billing purpose.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
```

```
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "sequence": {
      "description": "Sequence in which the items appear on the invoice.",
      "$ref": "#/definitions/positiveInt"
    },
    "_sequence": {
      "description": "Extensions for sequence",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "chargeItemReference": {
      "description": "The ChargeItem contains information such as the billing code, date, amount etc. If no further
details are required for the lineItem, inline billing codes can be added using the CodeableConcept data type instead
 of the Reference.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "chargeItemCodeableConcept": {
      "description": "The ChargeItem contains information such as the billing code, date, amount etc. If no further
details are required for the lineItem, inline billing codes can be added using the CodeableConcept data type instead
 of the Reference.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "priceComponent": {
      "description": "The price for a ChargeItem may be calculated as a base price with surcharges/deductions that
apply in certain conditions. A ChargeItemDefinition resource that defines the prices, factors and conditions that
apply to a billing code is currently under development. The priceComponent element can be used to offer transparency
 to the recipient of the Invoice as to how the prices have been calculated.",
      "items": {
        "$ref": "#/definitions/Invoice_PriceComponent"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"Invoice_PriceComponent": {
  "description": "Invoice containing collected ChargeItems from an Account with calculated individual and total
price for Billing purpose.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
```

```
          "type": "array"
        },
        "type": {
          "description": "This code identifies the type of the component.",
          "enum": [
            "base",
            "surcharge",
            "deduction",
            "discount",
            "tax",
            "informational"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "code": {
          "description": "A code that identifies the component. Codes may be used to differentiate between kinds of
    taxes, surcharges, discounts etc.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "factor": {
          "description": "The factor that has been applied on the base price for calculating this component.",
          "$ref": "#/definitions/decimal"
        },
        "_factor": {
          "description": "Extensions for factor",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "amount": {
          "description": "The amount calculated for this component.",
          "$ref": "Money.schema.json#/definitions/Money"
        }
      },
      "additionalProperties": false
    }
  }
}
```


8.29.10 Resource Invoice - Maturity Dependency View

URL: https://hl7.org/fhir/R4/invoice-dependencies.html

8.29.10 Resource Invoice - Maturity Dependency View

Financial Management Work Group | Maturity Level: 0 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Dependency Graph for Invoice FMM level 0

Name | Card. | Type | Dependency Analysis
Invoice | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
cancelledReason | 0..1 | string
type | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group)
recipient | 0..1 | Reference(Organization | Patient | RelatedPerson)
date | 0..1 | dateTime
participant | 0..* | BackboneElement
role | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | Organization | Patient | PractitionerRole | Device | RelatedPerson)
issuer | 0..1 | Reference(Organization)
account | 0..1 | Reference(Account)
lineItem | 0..* | BackboneElement
sequence | 0..1 | positiveInt
chargeItem[x] | 1..1
chargeItemReference | | Reference(ChargeItem
chargeItemCodeableConcept | | CodeableConcept
priceComponent | 0..* | BackboneElement
type | 1..1 | code
code | 0..1 | CodeableConcept
factor | 0..1 | decimal
amount | 0..1 | Money
totalPriceComponent | 0..* | see priceComponent
totalNet | 0..1 | Money
totalGross | 0..1 | Money
paymentTerms | 0..1 | markdown
note | 0..* | Annotation

8.29.11 Value Set Sc-valueset-invoice-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-invoice-status.html

8.29.11 Value Set Sc-valueset-invoice-status - Status Code Mapping

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.29.12 InvoiceStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-invoice-status)

Mapping from http://hl7.org/fhir/ValueSet/invoice-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes identifying the lifecycle stage of an Invoice."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
draft | equivalent | draft
issued | equivalent | active
cancelled | equivalent | failed
balanced | equivalent | complete

8.30 Resource InsurancePlan - Content

URL: https://hl7.org/fhir/R4/insuranceplan.html

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Details of a Health Insurance product/plan provided by an organization.

8.30.1 Scope and Usage

A product is a discrete package of health insurance coverage benefits that are offered under a particular network type. A given payer’s products typically differ by network type and/or coverage benefits. A plan pairs the health insurance coverage benefits under a product with the particular cost sharing structure offered to a consumer. A given product may comprise multiple plans.

InsurancePlan describes a health insurance offering comprised of a list of covered benefits (i.e. the product), costs associated with those benefits (i.e. the plan), and additional information about the offering, such as who it is owned and administered by, a coverage area, contact information, etc.

8.30.2 Boundaries and Relationships

TBD.

This resource is referenced by UsageContext

8.30.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
InsurancePlan | ITU | | DomainResource | Details of a Health Insurance product/plan provided by an organization + Rule: The organization SHALL at least have a name or an idendtifier, and possibly more than one Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ΣI | 0..* | Identifier | Business Identifier for Product
status | ?!Σ | 0..1 | code | draft | active | retired | unknown PublicationStatus (Required)
type | Σ | 0..* | CodeableConcept | Kind of product Insurance plan type (Example)
name | ΣI | 0..1 | string | Official name
alias | | 0..* | string | Alternate names
period | | 0..1 | Period | When the product is available
ownedBy | Σ | 0..1 | Reference(Organization) | Plan issuer

administeredBy | Σ | 0..1 | Reference(Organization) | Product administrator
coverageArea | Σ | 0..* | Reference(Location) | Where product applies
contact | | 0..* | BackboneElement | Contact for the product
purpose | | 0..1 | CodeableConcept | The type of contact Contact entity type (Extensible)
name | | 0..1 | HumanName | A name associated with the contact
telecom | | 0..* | ContactPoint | Contact details (telephone, email, etc.) for a contact
address | | 0..1 | Address | Visiting or postal addresses for the contact
endpoint | | 0..* | Reference(Endpoint) | Technical endpoint
network | | 0..* | Reference(Organization) | What networks are Included
coverage | | 0..* | BackboneElement | Coverage details
type | | 1..1 | CodeableConcept | Type of coverage
network | | 0..* | Reference(Organization) | What networks provide coverage
benefit | | 1..* | BackboneElement | List of benefits
type | | 1..1 | CodeableConcept | Type of benefit
requirement | | 0..1 | string | Referral requirements
limit | | 0..* | BackboneElement | Benefit limits
value | | 0..1 | Quantity | Maximum value allowed
code | | 0..1 | CodeableConcept | Benefit limit details
plan | | 0..* | BackboneElement | Plan details
identifier | Σ | 0..* | Identifier | Business Identifier for Product
type | | 0..1 | CodeableConcept | Type of plan
coverageArea | Σ | 0..* | Reference(Location) | Where product applies
network | | 0..* | Reference(Organization) | What networks provide coverage
generalCost | | 0..* | BackboneElement | Overall costs
type | | 0..1 | CodeableConcept | Type of cost
groupSize | | 0..1 | positiveInt | Number of enrollees
cost | | 0..1 | Money | Cost value
comment | | 0..1 | string | Additional cost information
specificCost | | 0..* | BackboneElement | Specific costs
category | | 1..1 | CodeableConcept | General category of benefit
benefit | | 0..* | BackboneElement | Benefits list
type | | 1..1 | CodeableConcept | Type of specific benefit
cost | | 0..* | BackboneElement | List of the costs
type | | 1..1 | CodeableConcept | Type of cost
applicability | | 0..1 | CodeableConcept | in-network | out-of-network | other Benefit cost applicability (Required)
qualifiers | | 0..* | CodeableConcept | Additional information about the cost
value | | 0..1 | Quantity | The actual cost value
Documentation for this format

UML Diagram (Legend)

XML Template

```
<InsurancePlan xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!--  0..* Identifier Business Identifier for Product --></identifier>
 <status value="[code]"/><!-- 0..1 draft | active | retired | unknown -->
 <type><!-- 0..* CodeableConcept Kind of product --></type>
 <name value="[string]"/><!--  0..1 Official name -->
 <alias value="[string]"/><!-- 0..* Alternate names -->
 <period><!-- 0..1 Period When the product is available --></period>
 <ownedBy><!-- 0..1 Reference(Organization) Plan issuer --></ownedBy>
 <administeredBy><!-- 0..1 Reference(Organization) Product administrator --></administeredBy>
 <coverageArea><!-- 0..* Reference(Location) Where product applies --></coverageArea>
 <contact>  <!-- 0..* Contact for the product -->
  <purpose><!-- 0..1 CodeableConcept The type of contact --></purpose>
  <name><!-- 0..1 HumanName A name associated with the contact --></name>
  <telecom><!-- 0..* ContactPoint Contact details (telephone, email, etc.)  for a contact --></telecom>
  <address><!-- 0..1 Address Visiting or postal addresses for the contact --></address>
 </contact>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoint --></endpoint>
 <network><!-- 0..* Reference(Organization) What networks are Included --></network>
 <coverage>  <!-- 0..* Coverage details -->
  <type><!-- 1..1 CodeableConcept Type of coverage --></type>
  <network><!-- 0..* Reference(Organization) What networks provide coverage --></network>
  <benefit>  <!-- 1..* List of benefits -->
   <type><!-- 1..1 CodeableConcept Type of benefit --></type>
   <requirement value="[string]"/><!-- 0..1 Referral requirements -->
   <limit>  <!-- 0..* Benefit limits -->
    <value><!-- 0..1 Quantity Maximum value allowed --></value>
    <code><!-- 0..1 CodeableConcept Benefit limit details --></code>
   </limit>
  </benefit>
 </coverage>
 <plan>  <!-- 0..* Plan details -->
  <identifier><!-- 0..* Identifier Business Identifier for Product --></identifier>
  <type><!-- 0..1 CodeableConcept Type of plan --></type>
  <coverageArea><!-- 0..* Reference(Location) Where product applies --></coverageArea>
  <network><!-- 0..* Reference(Organization) What networks provide coverage --></network>
  <generalCost>  <!-- 0..* Overall costs -->
```

```
   <type><!-- 0..1 CodeableConcept Type of cost --></type>
   <groupSize value="[positiveInt]"/><!-- 0..1 Number of enrollees -->
   <cost><!-- 0..1 Money Cost value --></cost>
   <comment value="[string]"/><!-- 0..1 Additional cost information -->
  </generalCost>
  <specificCost>  <!-- 0..* Specific costs -->
   <category><!-- 1..1 CodeableConcept General category of benefit --></category>
   <benefit>  <!-- 0..* Benefits list -->
    <type><!-- 1..1 CodeableConcept Type of specific benefit --></type>
    <cost>  <!-- 0..* List of the costs -->
     <type><!-- 1..1 CodeableConcept Type of cost --></type>
     <applicability><!-- 0..1 CodeableConcept in-network | out-of-network | other --></applicability>
     <qualifiers><!-- 0..* CodeableConcept Additional information about the cost --></qualifiers>
     <value><!-- 0..1 Quantity The actual cost value --></value>
    </cost>
   </benefit>
  </specificCost>
 </plan>
</InsurancePlan>
```

JSON Template

```
{
  "resourceType" : "InsurancePlan",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // C? Business Identifier for Product
  "status" : "<code>", // draft | active | retired | unknown
  "type" : [{ CodeableConcept }], // Kind of product
  "name" : "<string>", // C? Official name
  "alias" : ["<string>"], // Alternate names
  "period" : { Period }, // When the product is available
  "ownedBy" : { Reference(Organization) }, // Plan issuer
  "administeredBy" : { Reference(Organization) }, // Product administrator
  "coverageArea" : [{ Reference(Location) }], // Where product applies
  "contact" : [{ // Contact for the product
    "purpose" : { CodeableConcept }, // The type of contact
    "name" : { HumanName }, // A name associated with the contact
    "telecom" : [{ ContactPoint }], // Contact details (telephone, email, etc.)  for a contact
    "address" : { Address } // Visiting or postal addresses for the contact
  }],
  "endpoint" : [{ Reference(Endpoint) }], // Technical endpoint
  "network" : [{ Reference(Organization) }], // What networks are Included
  "coverage" : [{ // Coverage details
    "type" : { CodeableConcept }, // R!  Type of coverage
    "network" : [{ Reference(Organization) }], // What networks provide coverage
    "benefit" : [{ // R!  List of benefits
      "type" : { CodeableConcept }, // R!  Type of benefit
      "requirement" : "<string>", // Referral requirements
      "limit" : [{ // Benefit limits
        "value" : { Quantity }, // Maximum value allowed
        "code" : { CodeableConcept } // Benefit limit details
      }]
    }]
  }],
  "plan" : [{ // Plan details
    "identifier" : [{ Identifier }], // Business Identifier for Product
    "type" : { CodeableConcept }, // Type of plan
    "coverageArea" : [{ Reference(Location) }], // Where product applies
    "network" : [{ Reference(Organization) }], // What networks provide coverage
    "generalCost" : [{ // Overall costs
      "type" : { CodeableConcept }, // Type of cost
      "groupSize" : "<positiveInt>", // Number of enrollees
      "cost" : { Money }, // Cost value
      "comment" : "<string>" // Additional cost information
    }],
    "specificCost" : [{ // Specific costs
      "category" : { CodeableConcept }, // R!  General category of benefit
      "benefit" : [{ // Benefits list
        "type" : { CodeableConcept }, // R!  Type of specific benefit
        "cost" : [{ // List of the costs
          "type" : { CodeableConcept }, // R!  Type of cost
          "applicability" : { CodeableConcept }, // in-network | out-of-network | other
          "qualifiers" : [{ CodeableConcept }], // Additional information about the cost
          "value" : { Quantity } // The actual cost value
        }]
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:InsurancePlan;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
```

```
  fhir:InsurancePlan.identifier [ Identifier ], ... ; # 0..* Business Identifier for Product
  fhir:InsurancePlan.status [ code ]; # 0..1 draft | active | retired | unknown
  fhir:InsurancePlan.type [ CodeableConcept ], ... ; # 0..* Kind of product
  fhir:InsurancePlan.name [ string ]; # 0..1 Official name
  fhir:InsurancePlan.alias [ string ], ... ; # 0..* Alternate names
  fhir:InsurancePlan.period [ Period ]; # 0..1 When the product is available
  fhir:InsurancePlan.ownedBy [ Reference(Organization) ]; # 0..1 Plan issuer
  fhir:InsurancePlan.administeredBy [ Reference(Organization) ]; # 0..1 Product administrator
  fhir:InsurancePlan.coverageArea [ Reference(Location) ], ... ; # 0..* Where product applies
  fhir:InsurancePlan.contact [ # 0..* Contact for the product
    fhir:InsurancePlan.contact.purpose [ CodeableConcept ]; # 0..1 The type of contact
    fhir:InsurancePlan.contact.name [ HumanName ]; # 0..1 A name associated with the contact
    fhir:InsurancePlan.contact.telecom [ ContactPoint ], ... ; # 0..* Contact details (telephone, email, etc.)  for a
    contact
    fhir:InsurancePlan.contact.address [ Address ]; # 0..1 Visiting or postal addresses for the contact
  ], ...;
  fhir:InsurancePlan.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoint
  fhir:InsurancePlan.network [ Reference(Organization) ], ... ; # 0..* What networks are Included
  fhir:InsurancePlan.coverage [ # 0..* Coverage details
    fhir:InsurancePlan.coverage.type [ CodeableConcept ]; # 1..1 Type of coverage
    fhir:InsurancePlan.coverage.network [ Reference(Organization) ], ... ; # 0..* What networks provide coverage
    fhir:InsurancePlan.coverage.benefit [ # 1..* List of benefits
      fhir:InsurancePlan.coverage.benefit.type [ CodeableConcept ]; # 1..1 Type of benefit
      fhir:InsurancePlan.coverage.benefit.requirement [ string ]; # 0..1 Referral requirements
      fhir:InsurancePlan.coverage.benefit.limit [ # 0..* Benefit limits
        fhir:InsurancePlan.coverage.benefit.limit.value [ Quantity ]; # 0..1 Maximum value allowed
        fhir:InsurancePlan.coverage.benefit.limit.code [ CodeableConcept ]; # 0..1 Benefit limit details
      ], ...;
    ], ...;
  ], ...;
  fhir:InsurancePlan.plan [ # 0..* Plan details
    fhir:InsurancePlan.plan.identifier [ Identifier ], ... ; # 0..* Business Identifier for Product
    fhir:InsurancePlan.plan.type [ CodeableConcept ]; # 0..1 Type of plan
    fhir:InsurancePlan.plan.coverageArea [ Reference(Location) ], ... ; # 0..* Where product applies
    fhir:InsurancePlan.plan.network [ Reference(Organization) ], ... ; # 0..* What networks provide coverage
    fhir:InsurancePlan.plan.generalCost [ # 0..* Overall costs
      fhir:InsurancePlan.plan.generalCost.type [ CodeableConcept ]; # 0..1 Type of cost
      fhir:InsurancePlan.plan.generalCost.groupSize [ positiveInt ]; # 0..1 Number of enrollees
      fhir:InsurancePlan.plan.generalCost.cost [ Money ]; # 0..1 Cost value
      fhir:InsurancePlan.plan.generalCost.comment [ string ]; # 0..1 Additional cost information
    ], ...;
    fhir:InsurancePlan.plan.specificCost [ # 0..* Specific costs
      fhir:InsurancePlan.plan.specificCost.category [ CodeableConcept ]; # 1..1 General category of benefit
      fhir:InsurancePlan.plan.specificCost.benefit [ # 0..* Benefits list
        fhir:InsurancePlan.plan.specificCost.benefit.type [ CodeableConcept ]; # 1..1 Type of specific benefit
        fhir:InsurancePlan.plan.specificCost.benefit.cost [ # 0..* List of the costs
          fhir:InsurancePlan.plan.specificCost.benefit.cost.type [ CodeableConcept ]; # 1..1 Type of cost
          fhir:InsurancePlan.plan.specificCost.benefit.cost.applicability [ CodeableConcept ]; # 0..1 in-network | out-
    of-network | other
          fhir:InsurancePlan.plan.specificCost.benefit.cost.qualifiers [ CodeableConcept ], ... ; # 0..* Additional
    information about the cost
          fhir:InsurancePlan.plan.specificCost.benefit.cost.value [ Quantity ]; # 0..1 The actual cost value
        ], ...;
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
InsurancePlan	ITU		DomainResource	Details of a Health Insurance product/plan provided by an organization + Rule: The organization SHALL at least have a name or an identtifier, and possibly more than one Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	ΣI	0..*	Identifier	Business Identifier for Product
status	?!Σ	0..1	code	draft | active | retired | unknown PublicationStatus (Required)
type	Σ	0..*	CodeableConcept	Kind of product Insurance plan type (Example)
name	ΣI	0..1	string	Official name
alias		0..*	string	Alternate names
period		0..1	Period	When the product is available
ownedBy	Σ	0..1	Reference(Organization)	Plan issuer
administeredBy	Σ	0..1	Reference(Organization)	Product administrator
coverageArea	Σ	0..*	Reference(Location)	Where product applies
contact		0..*	BackboneElement	Contact for the product
purpose		0..1	CodeableConcept	The type of contact Contact entity type (Extensible)
name		0..1	HumanName	A name associated with the contact
telecom		0..*	ContactPoint	Contact details (telephone, email, etc.) for a contact
address		0..1	Address	Visiting or postal addresses for the contact
endpoint		0..*	Reference(Endpoint)	Technical endpoint
network		0..*	Reference(Organization)	What networks are Included

coverage | | 0..* | BackboneElement | Coverage details
type | | 1..1 | CodeableConcept | Type of coverage
network | | 0..* | Reference(Organization) | What networks provide coverage
benefit | | 1..* | BackboneElement | List of benefits
type | | 1..1 | CodeableConcept | Type of benefit
requirement | | 0..1 | string | Referral requirements
limit | | 0..* | BackboneElement | Benefit limits
value | | 0..1 | Quantity | Maximum value allowed
code | | 0..1 | CodeableConcept | Benefit limit details
plan | | 0..* | BackboneElement | Plan details
identifier | Σ | 0..* | Identifier | Business Identifier for Product
type | | 0..1 | CodeableConcept | Type of plan
coverageArea | Σ | 0..* | Reference(Location) | Where product applies
network | | 0..* | Reference(Organization) | What networks provide coverage
generalCost | | 0..* | BackboneElement | Overall costs
type | | 0..1 | CodeableConcept | Type of cost
groupSize | | 0..1 | positiveInt | Number of enrollees
cost | | 0..1 | Money | Cost value
comment | | 0..1 | string | Additional cost information
specificCost | | 0..* | BackboneElement | Specific costs
category | | 1..1 | CodeableConcept | General category of benefit
benefit | | 0..* | BackboneElement | Benefits list
type | | 1..1 | CodeableConcept | Type of specific benefit
cost | | 0..* | BackboneElement | List of the costs
type | | 1..1 | CodeableConcept | Type of cost
applicability | | 0..1 | CodeableConcept | in-network | out-of-network | other Benefit cost applicability (Required)
qualifiers | | 0..* | CodeableConcept | Additional information about the cost
value | | 0..1 | Quantity | The actual cost value
Documentation for this format

UML Diagram (Legend)

XML Template

```
<InsurancePlan xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!--  0..* Identifier Business Identifier for Product --></identifier>
 <status value="[code]"/><!-- 0..1 draft | active | retired | unknown -->
 <type><!-- 0..* CodeableConcept Kind of product --></type>
 <name value="[string]"/><!--  0..1 Official name -->
 <alias value="[string]"/><!-- 0..* Alternate names -->
 <period><!-- 0..1 Period When the product is available --></period>
 <ownedBy><!-- 0..1 Reference(Organization) Plan issuer --></ownedBy>
 <administeredBy><!-- 0..1 Reference(Organization) Product administrator --></administeredBy>
 <coverageArea><!-- 0..* Reference(Location) Where product applies --></coverageArea>
 <contact>  <!-- 0..* Contact for the product -->
  <purpose><!-- 0..1 CodeableConcept The type of contact --></purpose>
  <name><!-- 0..1 HumanName A name associated with the contact --></name>
  <telecom><!-- 0..* ContactPoint Contact details (telephone, email, etc.)  for a contact --></telecom>
  <address><!-- 0..1 Address Visiting or postal addresses for the contact --></address>
 </contact>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoint --></endpoint>
 <network><!-- 0..* Reference(Organization) What networks are Included --></network>
 <coverage>  <!-- 0..* Coverage details -->
  <type><!-- 1..1 CodeableConcept Type of coverage --></type>
  <network><!-- 0..* Reference(Organization) What networks provide coverage --></network>
  <benefit>  <!-- 1..* List of benefits -->
   <type><!-- 1..1 CodeableConcept Type of benefit --></type>
   <requirement value="[string]"/><!-- 0..1 Referral requirements -->
   <limit>  <!-- 0..* Benefit limits -->
    <value><!-- 0..1 Quantity Maximum value allowed --></value>
    <code><!-- 0..1 CodeableConcept Benefit limit details --></code>
   </limit>
  </benefit>
 </coverage>
 <plan>  <!-- 0..* Plan details -->
  <identifier><!-- 0..* Identifier Business Identifier for Product --></identifier>
  <type><!-- 0..1 CodeableConcept Type of plan --></type>
  <coverageArea><!-- 0..* Reference(Location) Where product applies --></coverageArea>
  <network><!-- 0..* Reference(Organization) What networks provide coverage --></network>
  <generalCost>  <!-- 0..* Overall costs -->
   <type><!-- 0..1 CodeableConcept Type of cost --></type>
   <groupSize value="[positiveInt]"/><!-- 0..1 Number of enrollees -->
   <cost><!-- 0..1 Money Cost value --></cost>
   <comment value="[string]"/><!-- 0..1 Additional cost information -->
  </generalCost>
  <specificCost>  <!-- 0..* Specific costs -->
   <category><!-- 1..1 CodeableConcept General category of benefit --></category>
   <benefit>  <!-- 0..* Benefits list -->
    <type><!-- 1..1 CodeableConcept Type of specific benefit --></type>
    <cost>  <!-- 0..* List of the costs -->
     <type><!-- 1..1 CodeableConcept Type of cost --></type>
     <applicability><!-- 0..1 CodeableConcept in-network | out-of-network | other --></applicability>
```

```
     <qualifiers><!-- 0..* CodeableConcept Additional information about the cost --></qualifiers>
     <value><!-- 0..1 Quantity The actual cost value --></value>
    </cost>
   </benefit>
  </specificCost>
 </plan>
</InsurancePlan>
```

JSON Template

```
{
  "resourceType" : "InsurancePlan",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // C? Business Identifier for Product
  "status" : "<code>", // draft | active | retired | unknown
  "type" : [{ CodeableConcept }], // Kind of product
  "name" : "<string>", // C? Official name
  "alias" : ["<string>"], // Alternate names
  "period" : { Period }, // When the product is available
  "ownedBy" : { Reference(Organization) }, // Plan issuer
  "administeredBy" : { Reference(Organization) }, // Product administrator
  "coverageArea" : [{ Reference(Location) }], // Where product applies
  "contact" : [{ // Contact for the product
    "purpose" : { CodeableConcept }, // The type of contact
    "name" : { HumanName }, // A name associated with the contact
    "telecom" : [{ ContactPoint }], // Contact details (telephone, email, etc.)  for a contact
    "address" : { Address } // Visiting or postal addresses for the contact
  }],
  "endpoint" : [{ Reference(Endpoint) }], // Technical endpoint
  "network" : [{ Reference(Organization) }], // What networks are Included
  "coverage" : [{ // Coverage details
    "type" : { CodeableConcept }, // R!  Type of coverage
    "network" : [{ Reference(Organization) }], // What networks provide coverage
    "benefit" : [{ // R!  List of benefits
      "type" : { CodeableConcept }, // R!  Type of benefit
      "requirement" : "<string>", // Referral requirements
      "limit" : [{ // Benefit limits
        "value" : { Quantity }, // Maximum value allowed
        "code" : { CodeableConcept } // Benefit limit details
      }]
    }]
  }],
  "plan" : [{ // Plan details
    "identifier" : [{ Identifier }], // Business Identifier for Product
    "type" : { CodeableConcept }, // Type of plan
    "coverageArea" : [{ Reference(Location) }], // Where product applies
    "network" : [{ Reference(Organization) }], // What networks provide coverage
    "generalCost" : [{ // Overall costs
      "type" : { CodeableConcept }, // Type of cost
      "groupSize" : "<positiveInt>", // Number of enrollees
      "cost" : { Money }, // Cost value
      "comment" : "<string>" // Additional cost information
    }],
    "specificCost" : [{ // Specific costs
      "category" : { CodeableConcept }, // R!  General category of benefit
      "benefit" : [{ // Benefits list
        "type" : { CodeableConcept }, // R!  Type of specific benefit
        "cost" : [{ // List of the costs
          "type" : { CodeableConcept }, // R!  Type of cost
          "applicability" : { CodeableConcept }, // in-network | out-of-network | other
          "qualifiers" : [{ CodeableConcept }], // Additional information about the cost
          "value" : { Quantity } // The actual cost value
        }]
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:InsurancePlan;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:InsurancePlan.identifier [ Identifier ], ... ; # 0..* Business Identifier for Product
  fhir:InsurancePlan.status [ code ]; # 0..1 draft | active | retired | unknown
  fhir:InsurancePlan.type [ CodeableConcept ], ... ; # 0..* Kind of product
  fhir:InsurancePlan.name [ string ]; # 0..1 Official name
  fhir:InsurancePlan.alias [ string ], ... ; # 0..* Alternate names
  fhir:InsurancePlan.period [ Period ]; # 0..1 When the product is available
  fhir:InsurancePlan.ownedBy [ Reference(Organization) ]; # 0..1 Plan issuer
  fhir:InsurancePlan.administeredBy [ Reference(Organization) ]; # 0..1 Product administrator
  fhir:InsurancePlan.coverageArea [ Reference(Location) ], ... ; # 0..* Where product applies
  fhir:InsurancePlan.contact [ # 0..* Contact for the product
    fhir:InsurancePlan.contact.purpose [ CodeableConcept ]; # 0..1 The type of contact
    fhir:InsurancePlan.contact.name [ HumanName ]; # 0..1 A name associated with the contact
```

```
    fhir:InsurancePlan.contact.telecom [ ContactPoint ], ... ; # 0..* Contact details (telephone, email, etc.)  for a
    contact
    fhir:InsurancePlan.contact.address [ Address ]; # 0..1 Visiting or postal addresses for the contact
  ], ...;
  fhir:InsurancePlan.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoint
  fhir:InsurancePlan.network [ Reference(Organization) ], ... ; # 0..* What networks are Included
  fhir:InsurancePlan.coverage [ # 0..* Coverage details
    fhir:InsurancePlan.coverage.type [ CodeableConcept ]; # 1..1 Type of coverage
    fhir:InsurancePlan.coverage.network [ Reference(Organization) ], ... ; # 0..* What networks provide coverage
    fhir:InsurancePlan.coverage.benefit [ # 1..* List of benefits
      fhir:InsurancePlan.coverage.benefit.type [ CodeableConcept ]; # 1..1 Type of benefit
      fhir:InsurancePlan.coverage.benefit.requirement [ string ]; # 0..1 Referral requirements
      fhir:InsurancePlan.coverage.benefit.limit [ # 0..* Benefit limits
        fhir:InsurancePlan.coverage.benefit.limit.value [ Quantity ]; # 0..1 Maximum value allowed
        fhir:InsurancePlan.coverage.benefit.limit.code [ CodeableConcept ]; # 0..1 Benefit limit details
      ], ...;
    ], ...;
  ], ...;
  fhir:InsurancePlan.plan [ # 0..* Plan details
    fhir:InsurancePlan.plan.identifier [ Identifier ], ... ; # 0..* Business Identifier for Product
    fhir:InsurancePlan.plan.type [ CodeableConcept ]; # 0..1 Type of plan
    fhir:InsurancePlan.plan.coverageArea [ Reference(Location) ], ... ; # 0..* Where product applies
    fhir:InsurancePlan.plan.network [ Reference(Organization) ], ... ; # 0..* What networks provide coverage
    fhir:InsurancePlan.plan.generalCost [ # 0..* Overall costs
      fhir:InsurancePlan.plan.generalCost.type [ CodeableConcept ]; # 0..1 Type of cost
      fhir:InsurancePlan.plan.generalCost.groupSize [ positiveInt ]; # 0..1 Number of enrollees
      fhir:InsurancePlan.plan.generalCost.cost [ Money ]; # 0..1 Cost value
      fhir:InsurancePlan.plan.generalCost.comment [ string ]; # 0..1 Additional cost information
    ], ...;
    fhir:InsurancePlan.plan.specificCost [ # 0..* Specific costs
      fhir:InsurancePlan.plan.specificCost.category [ CodeableConcept ]; # 1..1 General category of benefit
      fhir:InsurancePlan.plan.specificCost.benefit [ # 0..* Benefits list
        fhir:InsurancePlan.plan.specificCost.benefit.type [ CodeableConcept ]; # 1..1 Type of specific benefit
        fhir:InsurancePlan.plan.specificCost.benefit.cost [ # 0..* List of the costs
          fhir:InsurancePlan.plan.specificCost.benefit.cost.type [ CodeableConcept ]; # 1..1 Type of cost
          fhir:InsurancePlan.plan.specificCost.benefit.cost.applicability [ CodeableConcept ]; # 0..1 in-network | out-
    of-network | other
          fhir:InsurancePlan.plan.specificCost.benefit.cost.qualifiers [ CodeableConcept ], ... ; # 0..* Additional
    information about the cost
          fhir:InsurancePlan.plan.specificCost.benefit.cost.value [ Quantity ]; # 0..1 The actual cost value
        ], ...;
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.30.3.1 Terminology Bindings

Path | Definition | Type | Reference
InsurancePlan.status | The lifecycle status of an artifact. | Required | PublicationStatus
InsurancePlan.type | Used to categorize the product/plan. | Example | InsurancePlanType
InsurancePlan.contact.purpose | The purpose for which you would contact a contact party. | Extensible | ContactEntityType
InsurancePlan.plan.specificCost.benefit.cost.applicability | Whether the cost applies to in-network or out-of-network providers. | Required | BenefitCostApplicability

8.30.3.2 Constraints

id | Level | Location | Description | Expression
ipn-1 | Rule | (base) | The organization SHALL at least have a name or an idendtifier, and possibly more than one | (identifier.count() + name.count()) > 0

8.30.4 Notes:

8.30.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
address | string | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | InsurancePlan.contact.address
address-city | string | A city specified in an address | InsurancePlan.contact.address.city
address-country | string | A country specified in an address | InsurancePlan.contact.address.country
address-postalcode | string | A postal code specified in an address | InsurancePlan.contact.address.postalCode
address-state | string | A state specified in an address | InsurancePlan.contact.address.state
address-use | token | A use code specified in an address | InsurancePlan.contact.address.use
administered-by | reference | Product administrator | InsurancePlan.administeredBy (Organization)

endpoint | reference | Technical endpoint | InsurancePlan.endpoint (Endpoint)
identifier | token | Any identifier for the organization (not the accreditation issuer's identifier) | InsurancePlan.identifier
name | string | A portion of the organization's name or alias | name | alias
owned-by | reference | An organization of which this organization forms a part | InsurancePlan.ownedBy (Organization)
phonetic | string | A portion of the organization's name using some kind of phonetic matching algorithm | InsurancePlan.name
status | token | Is the Organization record active | InsurancePlan.status
type | token | A code for the type of organization | InsurancePlan.type

8.30.6 Resource InsurancePlan - Examples

URL: https://hl7.org/fhir/R4/insuranceplan-examples.html

8.30.6 Resource InsurancePlan - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
A general insurance plan example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.30.7 Resource InsurancePlan - Detailed Descriptions

URL: https://hl7.org/fhir/R4/insuranceplan-definitions.html

8.30.7 Resource InsurancePlan - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the InsurancePlan resource.

InsurancePlan
Element Id | InsurancePlan
Definition | Details of a Health Insurance product/plan provided by an organization.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element ipn-1 Rule The organization SHALL at least have a name or an idendtifier, and possibly more than one (identifier.count() + name.count()) > 0
Defined on this element
ipn-1 | Rule | The organization SHALL at least have a name or an idendtifier, and possibly more than one | (identifier.count() + name.count()) > 0
InsurancePlan.identifier
Element Id | InsurancePlan.identifier
Definition | Business identifiers assigned to this health insurance product which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Invariants | Affect this element ipn-1 Rule The organization SHALL at least have a name or an idendtifier, and possibly more than one (identifier.count() + name.count()) > 0
Affect this element
ipn-1 | Rule | The organization SHALL at least have a name or an idendtifier, and possibly more than one | (identifier.count() + name.count()) > 0
InsurancePlan.status
Element Id | InsurancePlan.status
Definition | The current state of the health insurance product.
Cardinality | 0..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains codes which means that the resource should not be treated as valid)
Meaning if Missing | Default interpretation is active.
Requirements | Need a flag to indicate a record is no longer to be used and should generally be hidden for the user in the UI.
Summary | true
InsurancePlan.type
Element Id | InsurancePlan.type
Definition | The kind of health insurance product.
Cardinality | 0..*
Terminology Binding | Insurance plan type (Example)
Type | CodeableConcept
Summary | true
InsurancePlan.name

Element Id | InsurancePlan.name
Definition | Official name of the health insurance product as designated by the owner.
Cardinality | 0..1
Type | string
Summary | true
Comments | If the name of the product/plan changes, consider putting the old name in the alias column so that it can still be located through searches.
Invariants | Affect this element ipn-1 Rule The organization SHALL at least have a name or an idendtifier, and possibly more than one (identifier.count() + name.count()) > 0
Affect this element
ipn-1 | Rule | The organization SHALL at least have a name or an idendtifier, and possibly more than one | (identifier.count() + name.count()) > 0
InsurancePlan.alias
Element Id | InsurancePlan.alias
Definition | A list of alternate names that the product is known as, or was known as in the past.
Cardinality | 0..*
Type | string
Requirements | Over time products/plans go through many changes and can be known by different names. For searching knowing previous names that the product/plan was known by can be very useful.
Comments | There are no dates associated with the alias/historic names, as this is not intended to track when names were used, but to assist in searching so that older names can still result in identifying the product/plan.
InsurancePlan.period
Element Id | InsurancePlan.period
Definition | The period of time that the health insurance product is available.
Cardinality | 0..1
Type | Period
InsurancePlan.ownedBy
Element Id | InsurancePlan.ownedBy
Definition | The entity that is providing the health insurance product and underwriting the risk. This is typically an insurance carriers, other third-party payers, or health plan sponsors comonly referred to as 'payers'.
Cardinality | 0..1
Type | Reference(Organization)
Alternate Names | Payer
Summary | true
InsurancePlan.administeredBy
Element Id | InsurancePlan.administeredBy
Definition | An organization which administer other services such as underwriting, customer service and/or claims processing on behalf of the health insurance product owner.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
InsurancePlan.coverageArea
Element Id | InsurancePlan.coverageArea
Definition | The geographic region in which a health insurance product's benefits apply.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
InsurancePlan.contact
Element Id | InsurancePlan.contact
Definition | The contact for the health insurance product for a certain purpose.
Cardinality | 0..*
Requirements | Need to keep track of assigned contact points within bigger organization.
Comments | Where multiple contacts for the same purpose are provided there is a standard extension that can be used to determine which one is the preferred contact to use.
InsurancePlan.contact.purpose
Element Id | InsurancePlan.contact.purpose
Definition | Indicates a purpose for which the contact can be reached.
Cardinality | 0..1
Terminology Binding | Contact entity type (Extensible)
Type | CodeableConcept
Requirements | Need to distinguish between multiple contact persons.
InsurancePlan.contact.name
Element Id | InsurancePlan.contact.name
Definition | A name associated with the contact.
Cardinality | 0..1
Type | HumanName
Requirements | Need to be able to track the person by name.
InsurancePlan.contact.telecom
Element Id | InsurancePlan.contact.telecom
Definition | A contact detail (e.g. a telephone number or an email address) by which the party may be contacted.
Cardinality | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.

InsurancePlan.contact.address
Element Id | InsurancePlan.contact.address
Definition | Visiting or postal addresses for the contact.
Cardinality | 0..1
Type | Address
Requirements | May need to keep track of a contact party's address for contacting, billing or reporting requirements.
InsurancePlan.endpoint
Element Id | InsurancePlan.endpoint
Definition | The technical endpoints providing access to services operated for the health insurance product.
Cardinality | 0..*
Type | Reference(Endpoint)
Requirements | Organizations have multiple systems that provide various services and need to be able to define the technical connection details for how to connect to them, and for what purpose.
InsurancePlan.network
Element Id | InsurancePlan.network
Definition | Reference to the network included in the health insurance product.
Cardinality | 0..*
Type | Reference(Organization)
Comments | Networks are represented as a hierarchy of organization resources.
InsurancePlan.coverage
Element Id | InsurancePlan.coverage
Definition | Details about the coverage offered by the insurance product.
Cardinality | 0..*
InsurancePlan.coverage.type
Element Id | InsurancePlan.coverage.type
Definition | Type of coverage (Medical; Dental; Mental Health; Substance Abuse; Vision; Drug; Short Term; Long Term Care; Hospice; Home Health).
Cardinality | 1..1
Type | CodeableConcept
InsurancePlan.coverage.network
Element Id | InsurancePlan.coverage.network
Definition | Reference to the network that providing the type of coverage.
Cardinality | 0..*
Type | Reference(Organization)
Comments | Networks are represented as a hierarchy of organization resources.
InsurancePlan.coverage.benefit
Element Id | InsurancePlan.coverage.benefit
Definition | Specific benefits under this type of coverage.
Cardinality | 1..*
InsurancePlan.coverage.benefit.type
Element Id | InsurancePlan.coverage.benefit.type
Definition | Type of benefit (primary care; speciality care; inpatient; outpatient).
Cardinality | 1..1
Type | CodeableConcept
InsurancePlan.coverage.benefit.requirement
Element Id | InsurancePlan.coverage.benefit.requirement
Definition | The referral requirements to have access/coverage for this benefit.
Cardinality | 0..1
Type | string
InsurancePlan.coverage.benefit.limit
Element Id | InsurancePlan.coverage.benefit.limit
Definition | The specific limits on the benefit.
Cardinality | 0..*
InsurancePlan.coverage.benefit.limit.value
Element Id | InsurancePlan.coverage.benefit.limit.value
Definition | The maximum amount of a service item a plan will pay for a covered benefit. For examples. wellness visits, or eyeglasses.
Cardinality | 0..1
Type | Quantity
Comments | May also be called “eligible expense,” “payment allowance,” or “negotiated rate.”.
InsurancePlan.coverage.benefit.limit.code
Element Id | InsurancePlan.coverage.benefit.limit.code
Definition | The specific limit on the benefit.
Cardinality | 0..1
Type | CodeableConcept
Comments | Use CodeableConcept.text element if the data is free (uncoded) text.
InsurancePlan.plan
Element Id | InsurancePlan.plan
Definition | Details about an insurance plan.
Cardinality | 0..*
InsurancePlan.plan.identifier
Element Id | InsurancePlan.plan.identifier

Definition | Business identifiers assigned to this health insurance plan which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
InsurancePlan.plan.type
Element Id | InsurancePlan.plan.type
Definition | Type of plan. For example, "Platinum" or "High Deductable".
Cardinality | 0..1
Type | CodeableConcept
InsurancePlan.plan.coverageArea
Element Id | InsurancePlan.plan.coverageArea
Definition | The geographic region in which a health insurance plan's benefits apply.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
InsurancePlan.plan.network
Element Id | InsurancePlan.plan.network
Definition | Reference to the network that providing the type of coverage.
Cardinality | 0..*
Type | Reference(Organization)
Comments | Networks are represented as a hierarchy of organization resources.
InsurancePlan.plan.generalCost
Element Id | InsurancePlan.plan.generalCost
Definition | Overall costs associated with the plan.
Cardinality | 0..*
InsurancePlan.plan.generalCost.type
Element Id | InsurancePlan.plan.generalCost.type
Definition | Type of cost.
Cardinality | 0..1
Type | CodeableConcept
InsurancePlan.plan.generalCost.groupSize
Element Id | InsurancePlan.plan.generalCost.groupSize
Definition | Number of participants enrolled in the plan.
Cardinality | 0..1
Type | positiveInt
InsurancePlan.plan.generalCost.cost
Element Id | InsurancePlan.plan.generalCost.cost
Definition | Value of the cost.
Cardinality | 0..1
Type | Money
InsurancePlan.plan.generalCost.comment
Element Id | InsurancePlan.plan.generalCost.comment
Definition | Additional information about the general costs associated with this plan.
Cardinality | 0..1
Type | string
InsurancePlan.plan.specificCost
Element Id | InsurancePlan.plan.specificCost
Definition | Costs associated with the coverage provided by the product.
Cardinality | 0..*
InsurancePlan.plan.specificCost.category
Element Id | InsurancePlan.plan.specificCost.category
Definition | General category of benefit (Medical; Dental; Vision; Drug; Mental Health; Substance Abuse; Hospice, Home Health).
Cardinality | 1..1
Type | CodeableConcept
InsurancePlan.plan.specificCost.benefit
Element Id | InsurancePlan.plan.specificCost.benefit
Definition | List of the specific benefits under this category of benefit.
Cardinality | 0..*
InsurancePlan.plan.specificCost.benefit.type
Element Id | InsurancePlan.plan.specificCost.benefit.type
Definition | Type of specific benefit (preventative; primary care office visit; speciality office visit; hospitalization; emergency room; urgent care).
Cardinality | 1..1
Type | CodeableConcept
InsurancePlan.plan.specificCost.benefit.cost
Element Id | InsurancePlan.plan.specificCost.benefit.cost
Definition | List of the costs associated with a specific benefit.
Cardinality | 0..*
InsurancePlan.plan.specificCost.benefit.cost.type
Element Id | InsurancePlan.plan.specificCost.benefit.cost.type

Definition | Type of cost (copay; individual cap; family cap; coinsurance; deductible).
Cardinality | 1..1
Type | CodeableConcept
InsurancePlan.plan.specificCost.benefit.cost.applicability
Element Id | InsurancePlan.plan.specificCost.benefit.cost.applicability
Definition | Whether the cost applies to in-network or out-of-network providers (in-network; out-of-network; other).
Cardinality | 0..1
Terminology Binding | Benefit cost applicability (Required)
Type | CodeableConcept
InsurancePlan.plan.specificCost.benefit.cost.qualifiers
Element Id | InsurancePlan.plan.specificCost.benefit.cost.qualifiers
Definition | Additional information about the cost, such as information about funding sources (e.g. HSA, HRA, FSA, RRA).
Cardinality | 0..*
Type | CodeableConcept
InsurancePlan.plan.specificCost.benefit.cost.value
Element Id | InsurancePlan.plan.specificCost.benefit.cost.value
Definition | The actual cost value. (some of the costs may be represented as percentages rather than currency, e.g. 10% coinsurance).
Cardinality | 0..1
Type | Quantity

8.30.8 Resource InsurancePlan - Mappings

URL: https://hl7.org/fhir/R4/insuranceplan-mappings.html

8.30.8 Resource InsurancePlan - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the insuranceplan resource (see Mappings to Other Standards for further information & status).

8.30.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

InsurancePlan
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class
identifier | FiveWs.identifier

8.30.8.2 RIM Mapping (http://hl7.org/v3)

InsurancePlan
identifier | .scopes[Role](classCode=IDENT)
status | .status
type | .code
name | .name
alias | .name
contact | .contactParty
purpose | ./type
name | ./name
telecom | ./telecom
address | ./addr
endpoint | n/a
identifier | .scopes[Role](classCode=IDENT)

8.30.9 Resource InsurancePlan - Extensions & Profiles

URL: https://hl7.org/fhir/R4/insuranceplan-profiles.html

8.30.9 Resource InsurancePlan - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the InsurancePlan resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.30.9.1 Profiles

No Profiles defined for this resource

8.30.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.30.9.3 Search Extensions

No Search Extensions defined for this resource

8.30.10 JSON Schema for InsurancePlan

URL: https://hl7.org/fhir/R4/insuranceplan.schema.json.html

8.30.10 JSON Schema for InsurancePlan

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for InsurancePlan, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/InsurancePlan",
  "$ref": "#/definitions/InsurancePlan",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "InsurancePlan": {
      "description": "Details of a Health Insurance product/plan provided by an organization.",
      "properties": {
        "resourceType": {
          "description": "This is a InsurancePlan resource",
          "const": "InsurancePlan"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
```

```
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this health insurance product which remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the health insurance product.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The kind of health insurance product.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "name": {
      "description": "Official name of the health insurance product as designated by the owner.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "alias": {
      "description": "A list of alternate names that the product is known as, or was known as in the past.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_alias": {
      "description": "Extensions for alias",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "period": {
      "description": "The period of time that the health insurance product is available.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "ownedBy": {
      "description": "The entity that is providing  the health insurance product and underwriting the risk.  This is typically an insurance carriers, other third-party payers, or health plan sponsors comonly referred to as \u0027payers\u0027.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "administeredBy": {
```

```
      "description": "An organization which administer other services such as underwriting, customer service and/or
claims processing on behalf of the health insurance product owner.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "coverageArea": {
      "description": "The geographic region in which a health insurance product\u0027s benefits apply.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "contact": {
      "description": "The contact for the health insurance product for a certain purpose.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Contact"
      },
      "type": "array"
    },
    "endpoint": {
      "description": "The technical endpoints providing access to services operated for the health insurance
product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "network": {
      "description": "Reference to the network included in the health insurance product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "coverage": {
      "description": "Details about the coverage offered by the insurance product.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Coverage"
      },
      "type": "array"
    },
    "plan": {
      "description": "Details about an insurance plan.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Plan"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"InsurancePlan_Contact": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Indicates a purpose for which the contact can be reached.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
```

```
    "name": {
      "description": "A name associated with the contact.",
      "$ref": "HumanName.schema.json#/definitions/HumanName"
    },
    "telecom": {
      "description": "A contact detail (e.g. a telephone number or an email address) by which the party may be
contacted.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "address": {
      "description": "Visiting or postal addresses for the contact.",
      "$ref": "Address.schema.json#/definitions/Address"
    }
  },
  "additionalProperties": false
},
"InsurancePlan_Coverage": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of coverage  (Medical; Dental; Mental Health; Substance Abuse; Vision; Drug; Short Term;
Long Term Care; Hospice; Home Health).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "network": {
      "description": "Reference to the network that providing the type of coverage.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "benefit": {
      "description": "Specific benefits under this type of coverage.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Benefit"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type",
    "benefit"
  ]
},
"InsurancePlan_Benefit": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of benefit (primary care; speciality care; inpatient; outpatient).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "requirement": {
      "description": "The referral requirements to have access/coverage for this benefit.",
      "$ref": "#/definitions/string"
    },
    "_requirement": {
      "description": "Extensions for requirement",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "limit": {
      "description": "The specific limits on the benefit.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Limit"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"InsurancePlan_Limit": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "value": {
      "description": "The maximum amount of a service item a plan will pay for a covered benefit.  For examples.
wellness visits, or eyeglasses.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "code": {
      "description": "The specific limit on the benefit.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
```

```
"InsurancePlan_Plan": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this health insurance plan which remain constant as the
resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of plan. For example, \"Platinum\" or \"High Deductable\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "coverageArea": {
      "description": "The geographic region in which a health insurance plan\u0027s benefits apply.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "network": {
      "description": "Reference to the network that providing the type of coverage.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "generalCost": {
      "description": "Overall costs associated with the plan.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_GeneralCost"
      },
      "type": "array"
    },
    "specificCost": {
      "description": "Costs associated with the coverage provided by the product.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_SpecificCost"
      },
      "type": "array"
    }

  },
  "additionalProperties": false
},
"InsurancePlan_GeneralCost": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of cost.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "groupSize": {
      "description": "Number of participants enrolled in the plan.",
      "$ref": "#/definitions/positiveInt"
    },
    "_groupSize": {
      "description": "Extensions for groupSize",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cost": {
      "description": "Value of the cost.",
      "$ref": "Money.schema.json#/definitions/Money"
    },
    "comment": {
      "description": "Additional information about the general costs associated with this plan.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"InsurancePlan_SpecificCost": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "category": {
      "description": "General category of benefit (Medical; Dental; Vision; Drug; Mental Health; Substance Abuse;
Hospice, Home Health).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "benefit": {
      "description": "List of the specific benefits under this category of benefit.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Benefit1"
      },
      "type": "array"
    }
```

```
  },
  "additionalProperties": false,
  "required": [
    "category"
  ]
},
"InsurancePlan_Benefit1": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of specific benefit (preventative; primary care office visit; speciality office visit;
hospitalization; emergency room; urgent care).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "cost": {
      "description": "List of the costs associated with a specific benefit.",
      "items": {
        "$ref": "#/definitions/InsurancePlan_Cost"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"InsurancePlan_Cost": {
  "description": "Details of a Health Insurance product/plan provided by an organization.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
```

```
          "description": "Type of cost (copay; individual cap; family cap; coinsurance; deductible).",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "applicability": {
          "description": "Whether the cost applies to in-network or out-of-network providers (in-network; out-of-
    network; other).",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "qualifiers": {
          "description": "Additional information about the cost, such as information about funding sources (e.g. HSA,
    HRA, FSA, RRA).",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "value": {
          "description": "The actual cost value. (some of the costs may be represented as percentages rather than
    currency, e.g. 10% coinsurance).",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        }
      },
      "additionalProperties": false,
      "required": [
        "type"
      ]
    }
  }
}
```


8.30.11 Resource InsurancePlan - Maturity Dependency View

URL: https://hl7.org/fhir/R4/insuranceplan-dependencies.html

8.30.11 Resource InsurancePlan - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for InsurancePlan FMM level 0

Name | Card. | Type | Dependency Analysis
InsurancePlan | | DomainResource
identifier | 0..* | Identifier
status | 0..1 | code
type | 0..* | CodeableConcept
name | 0..1 | string
alias | 0..* | string
period | 0..1 | Period
ownedBy | 0..1 | Reference(Organization)
administeredBy | 0..1 | Reference(Organization)
coverageArea | 0..* | Reference(Location)
contact | 0..* | BackboneElement
purpose | 0..1 | CodeableConcept
name | 0..1 | HumanName
telecom | 0..* | ContactPoint
address | 0..1 | Address
endpoint | 0..* | Reference(Endpoint)
network | 0..* | Reference(Organization)
coverage | 0..* | BackboneElement
type | 1..1 | CodeableConcept
network | 0..* | Reference(Organization)
benefit | 1..* | BackboneElement
type | 1..1 | CodeableConcept
requirement | 0..1 | string
limit | 0..* | BackboneElement
value | 0..1 | Quantity
code | 0..1 | CodeableConcept
plan | 0..* | BackboneElement
identifier | 0..* | Identifier
type | 0..1 | CodeableConcept
coverageArea | 0..* | Reference(Location)
network | 0..* | Reference(Organization)
generalCost | 0..* | BackboneElement
type | 0..1 | CodeableConcept
groupSize | 0..1 | positiveInt
cost | 0..1 | Money
comment | 0..1 | string
specificCost | 0..* | BackboneElement
category | 1..1 | CodeableConcept

benefit | 0..* | BackboneElement
type | 1..1 | CodeableConcept
cost | 0..* | BackboneElement
type | 1..1 | CodeableConcept
applicability | 0..1 | CodeableConcept
qualifiers | 0..* | CodeableConcept
value | 0..1 | Quantity

8.31 Resource OrganizationAffiliation - Content

URL: https://hl7.org/fhir/R4/organizationaffiliation.html
Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Defines an affiliation/assotiation/relationship between 2 distinct oganizations, that is not a part-of relationship/sub-division relationship.

8.31.1 Scope and Usage

A relationship between 2 organizations over a period of time, where the entities are separate business entities. The relationship can optionally include details of locations/services from the participating organization.

The OrganizationAffiliation enables defining non-hierarchical relationships between organizations. For example:

• One organization may provide services to another organization e.g. An agency service providing casual staff, a radiology service, a diagnostic lab, catering services, community care services etc.

• Two or more organizations may form a partnership or joint venture

• An organization may be a member of an association, but not owned by it e.g. a hospital is a member the American Hospital Association, a hospital is a member of a health information exchange network

8.31.2 Boundaries and Relationships

The Organization.partOf is used to form a hierarchical relationships within an organization which eventually resolves to a single organization. Each child in the tree is a subdivision of the parent. The OrganizationAffiliation is used to describe the relationship between two distinct organizations. It does not require a hierarchical relationship. A resource instance is for a singular linkage between 2 organizations (it does not contain all members), to create multiple affiliations, create additional resources for the other relationship between other organizations. Each instance contains its own period, and optionally links to specific related services/locations that are available. These referenced locations/services should be associated with the participating organization, and are available to the primary organization.

This resource should not be used when the affiliates are part of a single organization.

8.31.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
OrganizationAffiliation | TU | | DomainResource | Defines an affiliation/assotiation/relationship between 2 distinct oganizations, that is not a part-of relationship/sub-division relationship Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifiers that are specific to this role
active | Σ | 0..1 | boolean | Whether this organization affiliation record is in active use
period | Σ | 0..1 | Period | The period during which the participatingOrganization is affiliated with the primary organization
organization | Σ | 0..1 | Reference(Organization) | Organization where the role is available
participatingOrganization | Σ | 0..1 | Reference(Organization) | Organization that provides/performs the role (e.g. providing services or is a member of)
network | Σ | 0..* | Reference(Organization) | Health insurance provider network in which the participatingOrganization provides the role's services (if defined) at the indicated locations (if defined)
code | Σ | 0..* | CodeableConcept | Definition of the role the participatingOrganization plays Organization Affiliation Role (Example)
specialty | Σ | 0..* | CodeableConcept | Specific specialty of the participatingOrganization in the context of the role Practice Setting Code Value Set (Preferred)
location | Σ | 0..* | Reference(Location) | The location(s) at which the role occurs

healthcareService | | 0..* | Reference(HealthcareService) | Healthcare services provided through the role
telecom | Σ | 0..* | ContactPoint | Contact details at the participatingOrganization relevant to this Affiliation
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for this role
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OrganizationAffiliation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifiers that are specific to this role --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this organization affiliation record is in active use -->
 <period><!-- 0..1 Period The period during which the participatingOrganization is affiliated with the primary
    organization --></period>
 <organization><!-- 0..1 Reference(Organization) Organization where the role is available --></organization>
 <participatingOrganization><!-- 0..1 Reference(Organization) Organization that provides/performs the role (e.g.
    providing services or is a member of) --></participatingOrganization>
 <network><!-- 0..* Reference(Organization) Health insurance provider network in which the participatingOrganization
    provides the role's services (if defined) at the indicated locations (if defined) --></network>
 <code><!-- 0..* CodeableConcept Definition of the role the participatingOrganization plays --></code>
 <specialty><!-- 0..* CodeableConcept Specific specialty of the participatingOrganization in the context of the role
    --></specialty>
 <location><!-- 0..* Reference(Location) The location(s) at which the role occurs --></location>
 <healthcareService><!-- 0..* Reference(HealthcareService) Healthcare services provided through the role
    --></healthcareService>
 <telecom><!-- 0..* ContactPoint Contact details at the participatingOrganization relevant to this Affiliation
    --></telecom>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for this role
    --></endpoint>
</OrganizationAffiliation>
```

JSON Template

```
{
  "resourceType" : "OrganizationAffiliation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifiers that are specific to this role
  "active" : <boolean>, // Whether this organization affiliation record is in active use
  "period" : { Period }, // The period during which the participatingOrganization is affiliated with the primary
    organization
  "organization" : { Reference(Organization) }, // Organization where the role is available
  "participatingOrganization" : { Reference(Organization) }, // Organization that provides/performs the role (e.g.
    providing services or is a member of)
  "network" : [{ Reference(Organization) }], // Health insurance provider network in which the participatingOrganization
     provides the role's services (if defined) at the indicated locations (if defined)
  "code" : [{ CodeableConcept }], // Definition of the role the participatingOrganization plays
  "specialty" : [{ CodeableConcept }], // Specific specialty of the participatingOrganization in the context of the role
  "location" : [{ Reference(Location) }], // The location(s) at which the role occurs
  "healthcareService" : [{ Reference(HealthcareService) }], // Healthcare services provided through the role
  "telecom" : [{ ContactPoint }], // Contact details at the participatingOrganization relevant to this Affiliation
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for this role
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OrganizationAffiliation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OrganizationAffiliation.identifier [ Identifier ], ... ; # 0..* Business identifiers that are specific to this
    role
  fhir:OrganizationAffiliation.active [ boolean ]; # 0..1 Whether this organization affiliation record is in active use
  fhir:OrganizationAffiliation.period [ Period ]; # 0..1 The period during which the participatingOrganization is
    affiliated with the primary organization
  fhir:OrganizationAffiliation.organization [ Reference(Organization) ]; # 0..1 Organization where the role is available
  fhir:OrganizationAffiliation.participatingOrganization [ Reference(Organization) ]; # 0..1 Organization that
    provides/performs the role (e.g. providing services or is a member of)
  fhir:OrganizationAffiliation.network [ Reference(Organization) ], ... ; # 0..* Health insurance provider network in
    which the participatingOrganization provides the role's services (if defined) at the indicated locations (if
    defined)
  fhir:OrganizationAffiliation.code [ CodeableConcept ], ... ; # 0..* Definition of the role the
    participatingOrganization plays
  fhir:OrganizationAffiliation.specialty [ CodeableConcept ], ... ; # 0..* Specific specialty of the
    participatingOrganization in the context of the role
  fhir:OrganizationAffiliation.location [ Reference(Location) ], ... ; # 0..* The location(s) at which the role occurs
  fhir:OrganizationAffiliation.healthcareService [ Reference(HealthcareService) ], ... ; # 0..* Healthcare services
    provided through the role
  fhir:OrganizationAffiliation.telecom [ ContactPoint ], ... ; # 0..* Contact details at the participatingOrganization
    relevant to this Affiliation
  fhir:OrganizationAffiliation.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to
    services operated for this role
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
OrganizationAffiliation | TU | | DomainResource | Defines an affiliation/assotiation/relationship between 2 distinct oganizations, that is not a part-of relationship/sub-division relationship Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifiers that are specific to this role
active | Σ | 0..1 | boolean | Whether this organization affiliation record is in active use
period | Σ | 0..1 | Period | The period during which the participatingOrganization is affiliated with the primary organization
organization | Σ | 0..1 | Reference(Organization) | Organization where the role is available
participatingOrganization | Σ | 0..1 | Reference(Organization) | Organization that provides/performs the role (e.g. providing services or is a member of)
network | Σ | 0..* | Reference(Organization) | Health insurance provider network in which the participatingOrganization provides the role's services (if defined) at the indicated locations (if defined)
code | Σ | 0..* | CodeableConcept | Definition of the role the participatingOrganization plays Organization Affiliation Role (Example)
specialty | Σ | 0..* | CodeableConcept | Specific specialty of the participatingOrganization in the context of the role Practice Setting Code Value Set (Preferred)
location | Σ | 0..* | Reference(Location) | The location(s) at which the role occurs
healthcareService | | 0..* | Reference(HealthcareService) | Healthcare services provided through the role
telecom | Σ | 0..* | ContactPoint | Contact details at the participatingOrganization relevant to this Affiliation
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for this role
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OrganizationAffiliation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifiers that are specific to this role --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this organization affiliation record is in active use -->
 <period><!-- 0..1 Period The period during which the participatingOrganization is affiliated with the primary
    organization --></period>
 <organization><!-- 0..1 Reference(Organization) Organization where the role is available --></organization>
 <participatingOrganization><!-- 0..1 Reference(Organization) Organization that provides/performs the role (e.g.
    providing services or is a member of) --></participatingOrganization>
 <network><!-- 0..* Reference(Organization) Health insurance provider network in which the participatingOrganization
    provides the role's services (if defined) at the indicated locations (if defined) --></network>
 <code><!-- 0..* CodeableConcept Definition of the role the participatingOrganization plays --></code>
 <specialty><!-- 0..* CodeableConcept Specific specialty of the participatingOrganization in the context of the role
    --></specialty>
 <location><!-- 0..* Reference(Location) The location(s) at which the role occurs --></location>
 <healthcareService><!-- 0..* Reference(HealthcareService) Healthcare services provided through the role
    --></healthcareService>
 <telecom><!-- 0..* ContactPoint Contact details at the participatingOrganization relevant to this Affiliation
    --></telecom>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for this role
    --></endpoint>
</OrganizationAffiliation>
```

JSON Template

```
{
  "resourceType" : "OrganizationAffiliation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifiers that are specific to this role
  "active" : <boolean>, // Whether this organization affiliation record is in active use
  "period" : { Period }, // The period during which the participatingOrganization is affiliated with the primary
    organization
  "organization" : { Reference(Organization) }, // Organization where the role is available
  "participatingOrganization" : { Reference(Organization) }, // Organization that provides/performs the role (e.g.
    providing services or is a member of)
  "network" : [{ Reference(Organization) }], // Health insurance provider network in which the participatingOrganization
     provides the role's services (if defined) at the indicated locations (if defined)
  "code" : [{ CodeableConcept }], // Definition of the role the participatingOrganization plays
  "specialty" : [{ CodeableConcept }], // Specific specialty of the participatingOrganization in the context of the role
  "location" : [{ Reference(Location) }], // The location(s) at which the role occurs
  "healthcareService" : [{ Reference(HealthcareService) }], // Healthcare services provided through the role
  "telecom" : [{ ContactPoint }], // Contact details at the participatingOrganization relevant to this Affiliation
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for this role
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OrganizationAffiliation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OrganizationAffiliation.identifier [ Identifier ], ... ; # 0..* Business identifiers that are specific to this
```

```
    role
  fhir:OrganizationAffiliation.active [ boolean ]; # 0..1 Whether this organization affiliation record is in active use
  fhir:OrganizationAffiliation.period [ Period ]; # 0..1 The period during which the participatingOrganization is
    affiliated with the primary organization
  fhir:OrganizationAffiliation.organization [ Reference(Organization) ]; # 0..1 Organization where the role is available
  fhir:OrganizationAffiliation.participatingOrganization [ Reference(Organization) ]; # 0..1 Organization that
    provides/performs the role (e.g. providing services or is a member of)
  fhir:OrganizationAffiliation.network [ Reference(Organization) ], ... ; # 0..* Health insurance provider network in
    which the participatingOrganization provides the role's services (if defined) at the indicated locations (if
    defined)
  fhir:OrganizationAffiliation.code [ CodeableConcept ], ... ; # 0..* Definition of the role the
    participatingOrganization plays
  fhir:OrganizationAffiliation.specialty [ CodeableConcept ], ... ; # 0..* Specific specialty of the
    participatingOrganization in the context of the role
  fhir:OrganizationAffiliation.location [ Reference(Location) ], ... ; # 0..* The location(s) at which the role occurs
  fhir:OrganizationAffiliation.healthcareService [ Reference(HealthcareService) ], ... ; # 0..* Healthcare services
    provided through the role
  fhir:OrganizationAffiliation.telecom [ ContactPoint ], ... ; # 0..* Contact details at the participatingOrganization
    relevant to this Affiliation
  fhir:OrganizationAffiliation.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to
    services operated for this role
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.31.3.1 Terminology Bindings

Path | Definition | Type | Reference
OrganizationAffiliation.code | The role the participating organization providing services to the primary organization. | Example | OrganizationAffiliationRole
OrganizationAffiliation.specialty | Specific specialty associated with the participating organization. | Preferred | PracticeSettingCodeValueSet

8.31.4 Notes:

• Multiple instances of an OrganizationAffilitiation may exist between the same 2 organizations if a specific combination of the other related properties is required, or has a different period of usage. Some systems record a collection of service values for a single location, others record the single service and the list of locations it is available. Both are acceptable options for representing this data.

• If the HealthcareService contains more locations than specified in the Affilition, then only those in the Affiliation are relevant in the OrganizationAffiliation, it cannot introduce new locations not covered by the HealthcareService. Conversely if no Locations are specified, then it can be assumed that all locations are available.

• For some types of affiliation the HealthcareService might not be relevant and be empty, and location could be included.

8.31.5 Joint Ventures

With a joint venture between multiple organizations, a seperate Organization is created which represents the Venture. Each Organization in the venture is created through a seperate OrganizationAffiliation resource with the telecom details provided in each being the contact point to call for enquiries relating to their affiliation in the venture.

8.31.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Whether this organization affiliation record is in active use | OrganizationAffiliation.active
date | date | The period during which the participatingOrganization is affiliated with the primary organization | OrganizationAffiliation.period
email | token | A value in an email contact | OrganizationAffiliation.telecom.where(system='email')
endpoint | reference | Technical endpoints providing access to services operated for this role | OrganizationAffiliation.endpoint (Endpoint)
identifier | token | An organization affiliation's Identifier | OrganizationAffiliation.identifier
location | reference | The location(s) at which the role occurs | OrganizationAffiliation.location (Location)
network | reference | Health insurance provider network in which the participatingOrganization provides the role's services (if defined) at the indicated locations (if defined) | OrganizationAffiliation.network (Organization)
participating-organization | reference | The organization that provides services to the primary organization | OrganizationAffiliation.participatingOrganization (Organization)
phone | token | A value in a phone contact | OrganizationAffiliation.telecom.where(system='phone')

primary-organization | reference | The organization that receives the services from the participating organization | OrganizationAffiliation.organization (Organization)
role | token | Definition of the role the participatingOrganization plays | OrganizationAffiliation.code
service | reference | Healthcare services provided through the role | OrganizationAffiliation.healthcareService (HealthcareService)
specialty | token | Specific specialty of the participatingOrganization in the context of the role | OrganizationAffiliation.specialty
telecom | token | The value in any kind of contact | OrganizationAffiliation.telecom

8.31.7 Resource OrganizationAffiliation - Examples

URL: https://hl7.org/fhir/R4/organizationaffiliation-examples.html

8.31.7 Resource OrganizationAffiliation - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
HL7 Payer Network services for Acme Hospital | example | XML | JSON | Turtle
Hospital member | orgrole2 | XML | JSON | Turtle
Provider of rehabilitation services | orgrole1 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.31.8 Resource OrganizationAffiliation - Detailed Descriptions

URL: https://hl7.org/fhir/R4/organizationaffiliation-definitions.html

8.31.8 Resource OrganizationAffiliation - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the OrganizationAffiliation resource.

OrganizationAffiliation
Element Id | OrganizationAffiliation
Definition | Defines an affiliation/assotiation/relationship between 2 distinct oganizations, that is not a part-of relationship/sub-division relationship.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to define relationships between organizations that are not sub-divisions of the same organization (part-of relationships).
OrganizationAffiliation.identifier
Element Id | OrganizationAffiliation.identifier
Definition | Business identifiers that are specific to this role.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the agent.
Summary | true
OrganizationAffiliation.active
Element Id | OrganizationAffiliation.active
Definition | Whether this organization affiliation record is in active use.
Cardinality | 0..1
Type | boolean
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark an organization affiliation record as not to be used because it was created in error, or otherwise no longer in active use.
Summary | true
Comments | If this value is false, you may refer to the period to see when the role was in active use. If there is no period specified, no inference can be made about when it was active.
OrganizationAffiliation.period
Element Id | OrganizationAffiliation.period
Definition | The period during which the participatingOrganization is affiliated with the primary organization.
Cardinality | 0..1
Type | Period
Requirements | An affiliation may be created ahead of time and only be active for a defined period of time. Once the affiliation is complete, the fact that it existed may still be required.
Summary | true
OrganizationAffiliation.organization
Element Id | OrganizationAffiliation.organization
Definition | Organization where the role is available (primary organization/has members).
Cardinality | 0..1
Type | Reference(Organization)
Summary | true

OrganizationAffiliation.participatingOrganization
Element Id | OrganizationAffiliation.participatingOrganization
Definition | The Participating Organization provides/performs the role(s) defined by the code to the Primary Organization (e.g. providing services or is a member of).
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
OrganizationAffiliation.network
Element Id | OrganizationAffiliation.network
Definition | Health insurance provider network in which the participatingOrganization provides the role's services (if defined) at the indicated locations (if defined).
Cardinality | 0..*
Type | Reference(Organization)
Summary | true
OrganizationAffiliation.code
Element Id | OrganizationAffiliation.code
Definition | Definition of the role the participatingOrganization plays in the association.
Cardinality | 0..*
Terminology Binding | Organization Affiliation Role (Example)
Type | CodeableConcept
Summary | true
OrganizationAffiliation.specialty
Element Id | OrganizationAffiliation.specialty
Definition | Specific specialty of the participatingOrganization in the context of the role.
Cardinality | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
OrganizationAffiliation.location
Element Id | OrganizationAffiliation.location
Definition | The location(s) at which the role occurs.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
OrganizationAffiliation.healthcareService
Element Id | OrganizationAffiliation.healthcareService
Definition | Healthcare services provided through the role.
Cardinality | 0..*
Type | Reference(HealthcareService)
OrganizationAffiliation.telecom
Element Id | OrganizationAffiliation.telecom
Definition | Contact details at the participatingOrganization relevant to this Affiliation.
Cardinality | 0..*
Type | ContactPoint
Requirements | Sometimes a participating organization provides dedicated contacts to handle enquiries specifically for the role provided to the affiliated organization.
Summary | true
OrganizationAffiliation.endpoint
Element Id | OrganizationAffiliation.endpoint
Definition | Technical endpoints providing access to services operated for this role.
Cardinality | 0..*
Type | Reference(Endpoint)

8.31.9 Resource OrganizationAffiliation - Mappings

URL: https://hl7.org/fhir/R4/organizationaffiliation-mappings.html

8.31.9 Resource OrganizationAffiliation - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the organizationaffiliation resource (see Mappings to Other Standards for further information & status).

8.31.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

OrganizationAffiliation
identifier | FiveWs.identifier
active | FiveWs.status

period | FiveWs.done[x]
location | FiveWs.where[x]

8.31.9.2 RIM Mapping (http://hl7.org/v3)

OrganizationAffiliation | Role
identifier | .id
active | .statusCode
period | .performance[@typeCode <= 'PPRF'].ActDefinitionOrEvent.effectiveTime
organization | .scoper
participatingOrganization | .player
code | .code
specialty | .player.HealthCareProvider[@classCode = 'PROV'].code
location | .performance.ActDefinitionOrEvent.ServiceDeliveryLocation[@classCode = 'SDLOC']
healthcareService | .player.QualifiedEntity[@classCode = 'QUAL'].code
telecom | .telecom
endpoint | n/a

8.31.10 Resource OrganizationAffiliation - Extensions & Profiles

URL: https://hl7.org/fhir/R4/organizationaffiliation-profiles.html

8.31.10 Resource OrganizationAffiliation - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the OrganizationAffiliation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.31.10.1 Profiles

No Profiles defined for this resource

8.31.10.2 Extensions

organizationaffiliation-primaryInd | OrganizationAffiliation.specialty | primaryInd | for OrganizationAffiliation HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.31.10.3 Search Extensions

No Search Extensions defined for this resource

8.31.11 OrganizationAffiliation HL7 Extensions

URL: https://hl7.org/fhir/R4/organizationaffiliation-extensions.html

Defines common extensions used with or related to the OrganizationAffiliation resource

8.31.11 OrganizationAffiliation HL7 Extensions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative

8.31.11.1 Content

Extensions:
organizationaffiliation-primaryInd | primaryInd : Flag indicating if the specialty is the primary specialty of the provider. Normally, a practitioner will have one primary specialty, but in some cases more than one can be primary.

8.31.12 JSON Schema for OrganizationAffiliation

URL: https://hl7.org/fhir/R4/organizationaffiliation.schema.json.html

8.31.12 JSON Schema for OrganizationAffiliation

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for OrganizationAffiliation, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/OrganizationAffiliation",
  "$ref": "#/definitions/OrganizationAffiliation",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "OrganizationAffiliation": {
      "description": "Defines an affiliation/assotiation/relationship between 2 distinct oganizations, that is not a
    part-of relationship/sub-division relationship.",
      "properties": {
        "resourceType": {
          "description": "This is a OrganizationAffiliation resource",
          "const": "OrganizationAffiliation"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
```

```
      "identifier": {
        "description": "Business identifiers that are specific to this role.",
        "items": {
          "$ref": "Identifier.schema.json#/definitions/Identifier"
        },
        "type": "array"
      },
      "active": {
        "description": "Whether this organization affiliation record is in active use.",
        "$ref": "#/definitions/boolean"
      },
      "_active": {
        "description": "Extensions for active",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "period": {
        "description": "The period during which the participatingOrganization is affiliated with the primary
  organization.",
        "$ref": "Period.schema.json#/definitions/Period"
      },
      "organization": {
        "description": "Organization where the role is available (primary organization/has members).",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "participatingOrganization": {
        "description": "The Participating Organization provides/performs the role(s) defined by the code to the
  Primary Organization (e.g. providing services or is a member of).",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "network": {
        "description": "Health insurance provider network in which the participatingOrganization provides the
  role\u0027s services (if defined) at the indicated locations (if defined).",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "code": {
        "description": "Definition of the role the participatingOrganization plays in the association.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "specialty": {
        "description": "Specific specialty of the participatingOrganization in the context of the role.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "location": {
        "description": "The location(s) at which the role occurs.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "healthcareService": {
        "description": "Healthcare services provided through the role.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "telecom": {
        "description": "Contact details at the participatingOrganization relevant to this Affiliation.",
        "items": {
          "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
        },
        "type": "array"
      },
      "endpoint": {
        "description": "Technical endpoints providing access to services operated for this role.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "resourceType"
    ]
  }
 }
}
```

8.31.13 Resource OrganizationAffiliation - Maturity Dependency View

URL: https://hl7.org/fhir/R4/organizationaffiliation-dependencies.html

8.31.13 Resource OrganizationAffiliation - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for OrganizationAffiliation FMM level 0

Name | Card. | Type | Dependency Analysis
OrganizationAffiliation | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
period | 0..1 | Period
organization | 0..1 | Reference(Organization)
participatingOrganization | 0..1 | Reference(Organization)
network | 0..* | Reference(Organization)
code | 0..* | CodeableConcept
specialty | 0..* | CodeableConcept
location | 0..* | Reference(Location)
healthcareService | 0..* | Reference(HealthcareService)
telecom | 0..* | ContactPoint
endpoint | 0..* | Reference(Endpoint)

8.32 Resource VerificationResult - Content

URL: https://hl7.org/fhir/R4/verificationresult.html
Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Describes validation requirements, source(s), status and dates for one or more elements.

8.32.1 Scope and Usage

The VerificationResult can be used where content (such as found in a directory) is aggregated between systems, and the details of this process needs to be recorded, to be able to determine the likely accuracy of the content.

8.32.2 Boundaries and Relationships

TBD.

8.32.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
VerificationResult | TU | | DomainResource | Describes validation requirements, source(s), status and dates for one or more elements Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
target | Σ | 0..* | Reference(Any) | A resource that was validated
targetLocation | Σ | 0..* | string | The fhirpath location(s) within the resource that was validated
need | Σ | 0..1 | CodeableConcept | none | initial | periodic Need (Preferred)
status | Σ | 1..1 | code | attested | validated | in-process | req-revalid | val-fail | reval-fail Status (Required)
statusDate | Σ | 0..1 | dateTime | When the validation status was updated
validationType | Σ | 0..1 | CodeableConcept | nothing | primary | multiple Validation-type (Preferred)
validationProcess | Σ | 0..* | CodeableConcept | The primary process by which the target is validated (edit check; value set; primary source; multiple sources; standalone; in context) Validation-process (Example)
frequency | | 0..1 | Timing | Frequency of revalidation
lastPerformed | | 0..1 | dateTime | The date/time validation was last completed (including failed validations)
nextScheduled | | 0..1 | date | The date when target is next validated, if appropriate
failureAction | Σ | 0..1 | CodeableConcept | fatal | warn | rec-only | none Failure-action (Preferred)
primarySource | | 0..* | BackboneElement | Information about the primary source(s) involved in validation
who | | 0..1 | Reference(Organization | Practitioner | PractitionerRole) | Reference to the primary source

type | Σ | 0..* | CodeableConcept | Type of primary source (License Board; Primary Education; Continuing Education; Postal Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source) Primary-source-type (Example)
communicationMethod | Σ | 0..* | CodeableConcept | Method for exchanging information with the primary source VerificationResult Communication Method (Example)
validationStatus | | 0..1 | CodeableConcept | successful | failed | unknown Validation-status (Preferred)
validationDate | | 0..1 | dateTime | When the target was validated against the primary source
canPushUpdates | Σ | 0..1 | CodeableConcept | yes | no | undetermined Can-push-updates (Preferred)
pushTypeAvailable | | 0..* | CodeableConcept | specific | any | source Push-type-available (Preferred)
attestation | | 0..1 | BackboneElement | Information about the entity attesting to information
who | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | The individual or organization attesting to information
onBehalfOf | Σ | 0..1 | Reference(Organization | Practitioner | PractitionerRole) | When the who is asserting on behalf of another (organization or individual)
communicationMethod | Σ | 0..1 | CodeableConcept | The method by which attested information was submitted/retrieved VerificationResult Communication Method (Example)
date | Σ | 0..1 | date | The date the information was attested to
sourceIdentityCertificate | | 0..1 | string | A digital identity certificate associated with the attestation source
proxyIdentityCertificate | | 0..1 | string | A digital identity certificate associated with the proxy entity submitting attested information on behalf of the attestation source
proxySignature | | 0..1 | Signature | Proxy signature
sourceSignature | | 0..1 | Signature | Attester signature
validator | | 0..* | BackboneElement | Information about the entity validating information
organization | | 1..1 | Reference(Organization) | Reference to the organization validating information
identityCertificate | | 0..1 | string | A digital identity certificate associated with the validator
attestationSignature | | 0..1 | Signature | Validator signature
Documentation for this format

UML Diagram (Legend)

XML Template

```
<VerificationResult xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <target><!-- 0..* Reference(Any) A resource that was validated --></target>
 <targetLocation value="[string]"/><!-- 0..* The fhirpath location(s) within the resource that was validated -->
 <need><!-- 0..1 CodeableConcept none | initial | periodic --></need>
 <status value="[code]"/><!-- 1..1 attested | validated | in-process | req-revalid | val-fail | reval-fail -->
 <statusDate value="[dateTime]"/><!-- 0..1 When the validation status was updated -->
 <validationType><!-- 0..1 CodeableConcept nothing | primary | multiple --></validationType>
 <validationProcess><!-- 0..* CodeableConcept The primary process by which the target is validated (edit check; value
    set; primary source; multiple sources; standalone; in context) --></validationProcess>
 <frequency><!-- 0..1 Timing Frequency of revalidation --></frequency>
 <lastPerformed value="[dateTime]"/><!-- 0..1 The date/time validation was last completed (including failed validations)
     -->
 <nextScheduled value="[date]"/><!-- 0..1 The date when target is next validated, if appropriate -->
 <failureAction><!-- 0..1 CodeableConcept fatal | warn | rec-only | none --></failureAction>
 <primarySource>  <!-- 0..* Information about the primary source(s) involved in validation -->
  <who><!-- 0..1 Reference(Organization|Practitioner|PractitionerRole) Reference to the primary source --></who>
  <type><!-- 0..* CodeableConcept Type of primary source (License Board; Primary Education; Continuing Education; Postal
     Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source) --></type>
  <communicationMethod><!-- 0..* CodeableConcept Method for exchanging information with the primary source
    --></communicationMethod>
  <validationStatus><!-- 0..1 CodeableConcept successful | failed | unknown --></validationStatus>
  <validationDate value="[dateTime]"/><!-- 0..1 When the target was validated against the primary source -->
  <canPushUpdates><!-- 0..1 CodeableConcept yes | no | undetermined --></canPushUpdates>
  <pushTypeAvailable><!-- 0..* CodeableConcept specific | any | source --></pushTypeAvailable>
 </primarySource>
 <attestation>  <!-- 0..1 Information about the entity attesting to information -->
  <who><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) The individual or organization attesting to
    information --></who>
  <onBehalfOf><!-- 0..1 Reference(Organization|Practitioner|PractitionerRole) When the who is asserting on behalf of
    another (organization or individual) --></onBehalfOf>
  <communicationMethod><!-- 0..1 CodeableConcept The method by which attested information was submitted/retrieved
    --></communicationMethod>
  <date value="[date]"/><!-- 0..1 The date the information was attested to -->
  <sourceIdentityCertificate value="[string]"/><!-- 0..1 A digital identity certificate associated with the attestation
    source -->
  <proxyIdentityCertificate value="[string]"/><!-- 0..1 A digital identity certificate associated with the proxy entity
    submitting attested information on behalf of the attestation source -->
  <proxySignature><!-- 0..1 Signature Proxy signature --></proxySignature>
  <sourceSignature><!-- 0..1 Signature Attester signature --></sourceSignature>
 </attestation>
 <validator>  <!-- 0..* Information about the entity validating information -->
  <organization><!-- 1..1 Reference(Organization) Reference to the organization validating information
    --></organization>
  <identityCertificate value="[string]"/><!-- 0..1 A digital identity certificate associated with the validator -->
  <attestationSignature><!-- 0..1 Signature Validator signature --></attestationSignature>
 </validator>
</VerificationResult>
```

JSON Template

```
{
  "resourceType" : "VerificationResult",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "target" : [{ Reference(Any) }], // A resource that was validated
  "targetLocation" : ["<string>"], // The fhirpath location(s) within the resource that was validated
  "need" : { CodeableConcept }, // none | initial | periodic
  "status" : "<code>", // R!  attested | validated | in-process | req-revalid | val-fail | reval-fail
  "statusDate" : "<dateTime>", // When the validation status was updated
  "validationType" : { CodeableConcept }, // nothing | primary | multiple
  "validationProcess" : [{ CodeableConcept }], // The primary process by which the target is validated (edit check;
    value set; primary source; multiple sources; standalone; in context)
  "frequency" : { Timing }, // Frequency of revalidation
  "lastPerformed" : "<dateTime>", // The date/time validation was last completed (including failed validations)
  "nextScheduled" : "<date>", // The date when target is next validated, if appropriate
  "failureAction" : { CodeableConcept }, // fatal | warn | rec-only | none
  "primarySource" : [{ // Information about the primary source(s) involved in validation
    "who" : { Reference(Organization|Practitioner|PractitionerRole) }, // Reference to the primary source
    "type" : [{ CodeableConcept }], // Type of primary source (License Board; Primary Education; Continuing Education;
    Postal Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source)
    "communicationMethod" : [{ CodeableConcept }], // Method for exchanging information with the primary source
    "validationStatus" : { CodeableConcept }, // successful | failed | unknown
    "validationDate" : "<dateTime>", // When the target was validated against the primary source
    "canPushUpdates" : { CodeableConcept }, // yes | no | undetermined
    "pushTypeAvailable" : [{ CodeableConcept }] // specific | any | source
  }],
  "attestation" : { // Information about the entity attesting to information
    "who" : { Reference(Practitioner|PractitionerRole|Organization) }, // The individual or organization attesting to
    information
    "onBehalfOf" : { Reference(Organization|Practitioner|PractitionerRole) }, // When the who is asserting on behalf of
    another (organization or individual)
    "communicationMethod" : { CodeableConcept }, // The method by which attested information was submitted/retrieved
    "date" : "<date>", // The date the information was attested to
    "sourceIdentityCertificate" : "<string>", // A digital identity certificate associated with the attestation source
    "proxyIdentityCertificate" : "<string>", // A digital identity certificate associated with the proxy entity
    submitting attested information on behalf of the attestation source
    "proxySignature" : { Signature }, // Proxy signature
    "sourceSignature" : { Signature } // Attester signature
  },
  "validator" : [{ // Information about the entity validating information
    "organization" : { Reference(Organization) }, // R!  Reference to the organization validating information
    "identityCertificate" : "<string>", // A digital identity certificate associated with the validator
    "attestationSignature" : { Signature } // Validator signature
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:VerificationResult;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:VerificationResult.target [ Reference(Any) ], ... ; # 0..* A resource that was validated
  fhir:VerificationResult.targetLocation [ string ], ... ; # 0..* The fhirpath location(s) within the resource that was
    validated
  fhir:VerificationResult.need [ CodeableConcept ]; # 0..1 none | initial | periodic
  fhir:VerificationResult.status [ code ]; # 1..1 attested | validated | in-process | req-revalid | val-fail | reval-
    fail
  fhir:VerificationResult.statusDate [ dateTime ]; # 0..1 When the validation status was updated
  fhir:VerificationResult.validationType [ CodeableConcept ]; # 0..1 nothing | primary | multiple
  fhir:VerificationResult.validationProcess [ CodeableConcept ], ... ; # 0..* The primary process by which the target is
     validated (edit check; value set; primary source; multiple sources; standalone; in context)
  fhir:VerificationResult.frequency [ Timing ]; # 0..1 Frequency of revalidation
  fhir:VerificationResult.lastPerformed [ dateTime ]; # 0..1 The date/time validation was last completed (including
    failed validations)
  fhir:VerificationResult.nextScheduled [ date ]; # 0..1 The date when target is next validated, if appropriate
  fhir:VerificationResult.failureAction [ CodeableConcept ]; # 0..1 fatal | warn | rec-only | none
  fhir:VerificationResult.primarySource [ # 0..* Information about the primary source(s) involved in validation
    fhir:VerificationResult.primarySource.who [ Reference(Organization|Practitioner|PractitionerRole) ]; # 0..1
    Reference to the primary source
    fhir:VerificationResult.primarySource.type [ CodeableConcept ], ... ; # 0..* Type of primary source (License Board;
    Primary Education; Continuing Education; Postal Service; Relationship owner; Registration Authority; legal source;
    issuing source; authoritative source)
    fhir:VerificationResult.primarySource.communicationMethod [ CodeableConcept ], ... ; # 0..* Method for exchanging
    information with the primary source
    fhir:VerificationResult.primarySource.validationStatus [ CodeableConcept ]; # 0..1 successful | failed | unknown
    fhir:VerificationResult.primarySource.validationDate [ dateTime ]; # 0..1 When the target was validated against the
    primary source
    fhir:VerificationResult.primarySource.canPushUpdates [ CodeableConcept ]; # 0..1 yes | no | undetermined
    fhir:VerificationResult.primarySource.pushTypeAvailable [ CodeableConcept ], ... ; # 0..* specific | any | source
  ], ...;
  fhir:VerificationResult.attestation [ # 0..1 Information about the entity attesting to information
    fhir:VerificationResult.attestation.who [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 The
    individual or organization attesting to information
    fhir:VerificationResult.attestation.onBehalfOf [ Reference(Organization|Practitioner|PractitionerRole) ]; # 0..1
    When the who is asserting on behalf of another (organization or individual)
    fhir:VerificationResult.attestation.communicationMethod [ CodeableConcept ]; # 0..1 The method by which attested
    information was submitted/retrieved
    fhir:VerificationResult.attestation.date [ date ]; # 0..1 The date the information was attested to
```

```
    fhir:VerificationResult.attestation.sourceIdentityCertificate [ string ]; # 0..1 A digital identity certificate
    associated with the attestation source
    fhir:VerificationResult.attestation.proxyIdentityCertificate [ string ]; # 0..1 A digital identity certificate
    associated with the proxy entity submitting attested information on behalf of the attestation source
    fhir:VerificationResult.attestation.proxySignature [ Signature ]; # 0..1 Proxy signature
    fhir:VerificationResult.attestation.sourceSignature [ Signature ]; # 0..1 Attester signature
  ];
  fhir:VerificationResult.validator [ # 0..* Information about the entity validating information
    fhir:VerificationResult.validator.organization [ Reference(Organization) ]; # 1..1 Reference to the organization
    validating information
    fhir:VerificationResult.validator.identityCertificate [ string ]; # 0..1 A digital identity certificate associated
    with the validator
    fhir:VerificationResult.validator.attestationSignature [ Signature ]; # 0..1 Validator signature
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

VerificationResult | TU | | DomainResource | Describes validation requirements, source(s), status and dates for one or more elements Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

target | Σ | 0..* | Reference(Any) | A resource that was validated

targetLocation | Σ | 0..* | string | The fhirpath location(s) within the resource that was validated

need | Σ | 0..1 | CodeableConcept | none | initial | periodic Need (Preferred)

status | Σ | 1..1 | code | attested | validated | in-process | req-revalid | val-fail | reval-fail Status (Required)

statusDate | Σ | 0..1 | dateTime | When the validation status was updated

validationType | Σ | 0..1 | CodeableConcept | nothing | primary | multiple Validation-type (Preferred)

validationProcess | Σ | 0..* | CodeableConcept | The primary process by which the target is validated (edit check; value set; primary source; multiple sources; standalone; in context) Validation-process (Example)

frequency | | 0..1 | Timing | Frequency of revalidation

lastPerformed | | 0..1 | dateTime | The date/time validation was last completed (including failed validations)

nextScheduled | | 0..1 | date | The date when target is next validated, if appropriate

failureAction | Σ | 0..1 | CodeableConcept | fatal | warn | rec-only | none Failure-action (Preferred)

primarySource | | 0..* | BackboneElement | Information about the primary source(s) involved in validation

who | | 0..1 | Reference(Organization | Practitioner | PractitionerRole) | Reference to the primary source

type | Σ | 0..* | CodeableConcept | Type of primary source (License Board; Primary Education; Continuing Education; Postal Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source) Primary-source-type (Example)

communicationMethod | Σ | 0..* | CodeableConcept | Method for exchanging information with the primary source VerificationResult Communication Method (Example)

validationStatus | | 0..1 | CodeableConcept | successful | failed | unknown Validation-status (Preferred)

validationDate | | 0..1 | dateTime | When the target was validated against the primary source

canPushUpdates | Σ | 0..1 | CodeableConcept | yes | no | undetermined Can-push-updates (Preferred)

pushTypeAvailable | | 0..* | CodeableConcept | specific | any | source Push-type-available (Preferred)

attestation | | 0..1 | BackboneElement | Information about the entity attesting to information

who | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | The individual or organization attesting to information

onBehalfOf | Σ | 0..1 | Reference(Organization | Practitioner | PractitionerRole) | When the who is asserting on behalf of another (organization or individual)

communicationMethod | Σ | 0..1 | CodeableConcept | The method by which attested information was submitted/retrieved VerificationResult Communication Method (Example)

date | Σ | 0..1 | date | The date the information was attested to

sourceIdentityCertificate | | 0..1 | string | A digital identity certificate associated with the attestation source

proxyIdentityCertificate | | 0..1 | string | A digital identity certificate associated with the proxy entity submitting attested information on behalf of the attestation source

proxySignature | | 0..1 | Signature | Proxy signature

sourceSignature | | 0..1 | Signature | Attester signature

validator | | 0..* | BackboneElement | Information about the entity validating information

organization | | 1..1 | Reference(Organization) | Reference to the organization validating information

identityCertificate | | 0..1 | string | A digital identity certificate associated with the validator

attestationSignature | | 0..1 | Signature | Validator signature

Documentation for this format

UML Diagram (Legend)

XML Template

```
<VerificationResult xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <target><!-- 0..* Reference(Any) A resource that was validated --></target>
 <targetLocation value="[string]"/><!-- 0..* The fhirpath location(s) within the resource that was validated -->
 <need><!-- 0..1 CodeableConcept none | initial | periodic --></need>
 <status value="[code]"/><!-- 1..1 attested | validated | in-process | req-revalid | val-fail | reval-fail -->
 <statusDate value="[dateTime]"/><!-- 0..1 When the validation status was updated -->
 <validationType><!-- 0..1 CodeableConcept nothing | primary | multiple --></validationType>
 <validationProcess><!-- 0..* CodeableConcept The primary process by which the target is validated (edit check; value
    set; primary source; multiple sources; standalone; in context) --></validationProcess>
```

```
 <frequency><!-- 0..1 Timing Frequency of revalidation --></frequency>
 <lastPerformed value="[dateTime]"/><!-- 0..1 The date/time validation was last completed (including failed validations)
     -->
 <nextScheduled value="[date]"/><!-- 0..1 The date when target is next validated, if appropriate -->
 <failureAction><!-- 0..1 CodeableConcept fatal | warn | rec-only | none --></failureAction>
 <primarySource>  <!-- 0..* Information about the primary source(s) involved in validation -->
  <who><!-- 0..1 Reference(Organization|Practitioner|PractitionerRole) Reference to the primary source --></who>
  <type><!-- 0..* CodeableConcept Type of primary source (License Board; Primary Education; Continuing Education; Postal
     Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source) --></type>
  <communicationMethod><!-- 0..* CodeableConcept Method for exchanging information with the primary source
    --></communicationMethod>
  <validationStatus><!-- 0..1 CodeableConcept successful | failed | unknown --></validationStatus>
  <validationDate value="[dateTime]"/><!-- 0..1 When the target was validated against the primary source -->
  <canPushUpdates><!-- 0..1 CodeableConcept yes | no | undetermined --></canPushUpdates>
  <pushTypeAvailable><!-- 0..* CodeableConcept specific | any | source --></pushTypeAvailable>
 </primarySource>
 <attestation>  <!-- 0..1 Information about the entity attesting to information -->
  <who><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) The individual or organization attesting to
    information --></who>
  <onBehalfOf><!-- 0..1 Reference(Organization|Practitioner|PractitionerRole) When the who is asserting on behalf of
    another (organization or individual) --></onBehalfOf>
  <communicationMethod><!-- 0..1 CodeableConcept The method by which attested information was submitted/retrieved
    --></communicationMethod>
  <date value="[date]"/><!-- 0..1 The date the information was attested to -->
  <sourceIdentityCertificate value="[string]"/><!-- 0..1 A digital identity certificate associated with the attestation
    source -->
  <proxyIdentityCertificate value="[string]"/><!-- 0..1 A digital identity certificate associated with the proxy entity
    submitting attested information on behalf of the attestation source -->
  <proxySignature><!-- 0..1 Signature Proxy signature --></proxySignature>
  <sourceSignature><!-- 0..1 Signature Attester signature --></sourceSignature>
 </attestation>
 <validator>  <!-- 0..* Information about the entity validating information -->
  <organization><!-- 1..1 Reference(Organization) Reference to the organization validating information
    --></organization>
  <identityCertificate value="[string]"/><!-- 0..1 A digital identity certificate associated with the validator -->
  <attestationSignature><!-- 0..1 Signature Validator signature --></attestationSignature>
 </validator>
</VerificationResult>
```

JSON Template

```
{
  "resourceType" : "VerificationResult",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "target" : [{ Reference(Any) }], // A resource that was validated
  "targetLocation" : ["<string>"], // The fhirpath location(s) within the resource that was validated
  "need" : { CodeableConcept }, // none | initial | periodic
  "status" : "<code>", // R!  attested | validated | in-process | req-revalid | val-fail | reval-fail
  "statusDate" : "<dateTime>", // When the validation status was updated
  "validationType" : { CodeableConcept }, // nothing | primary | multiple
  "validationProcess" : [{ CodeableConcept }], // The primary process by which the target is validated (edit check;
    value set; primary source; multiple sources; standalone; in context)
  "frequency" : { Timing }, // Frequency of revalidation
  "lastPerformed" : "<dateTime>", // The date/time validation was last completed (including failed validations)
  "nextScheduled" : "<date>", // The date when target is next validated, if appropriate
  "failureAction" : { CodeableConcept }, // fatal | warn | rec-only | none
  "primarySource" : [{ // Information about the primary source(s) involved in validation
    "who" : { Reference(Organization|Practitioner|PractitionerRole) }, // Reference to the primary source
    "type" : [{ CodeableConcept }], // Type of primary source (License Board; Primary Education; Continuing Education;
    Postal Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source)
    "communicationMethod" : [{ CodeableConcept }], // Method for exchanging information with the primary source
    "validationStatus" : { CodeableConcept }, // successful | failed | unknown
    "validationDate" : "<dateTime>", // When the target was validated against the primary source
    "canPushUpdates" : { CodeableConcept }, // yes | no | undetermined
    "pushTypeAvailable" : [{ CodeableConcept }] // specific | any | source
  }],
  "attestation" : { // Information about the entity attesting to information
    "who" : { Reference(Practitioner|PractitionerRole|Organization) }, // The individual or organization attesting to
    information
    "onBehalfOf" : { Reference(Organization|Practitioner|PractitionerRole) }, // When the who is asserting on behalf of
    another (organization or individual)
    "communicationMethod" : { CodeableConcept }, // The method by which attested information was submitted/retrieved
    "date" : "<date>", // The date the information was attested to
    "sourceIdentityCertificate" : "<string>", // A digital identity certificate associated with the attestation source
    "proxyIdentityCertificate" : "<string>", // A digital identity certificate associated with the proxy entity
    submitting attested information on behalf of the attestation source
    "proxySignature" : { Signature }, // Proxy signature
    "sourceSignature" : { Signature } // Attester signature
  },
  "validator" : [{ // Information about the entity validating information
    "organization" : { Reference(Organization) }, // R!  Reference to the organization validating information
    "identityCertificate" : "<string>", // A digital identity certificate associated with the validator
    "attestationSignature" : { Signature } // Validator signature
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:VerificationResult;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:VerificationResult.target [ Reference(Any) ], ... ; # 0..* A resource that was validated
  fhir:VerificationResult.targetLocation [ string ], ... ; # 0..* The fhirpath location(s) within the resource that was
    validated
  fhir:VerificationResult.need [ CodeableConcept ]; # 0..1 none | initial | periodic
  fhir:VerificationResult.status [ code ]; # 1..1 attested | validated | in-process | req-revalid | val-fail | reval-
    fail
  fhir:VerificationResult.statusDate [ dateTime ]; # 0..1 When the validation status was updated
  fhir:VerificationResult.validationType [ CodeableConcept ]; # 0..1 nothing | primary | multiple
  fhir:VerificationResult.validationProcess [ CodeableConcept ], ... ; # 0..* The primary process by which the target is
     validated (edit check; value set; primary source; multiple sources; standalone; in context)
  fhir:VerificationResult.frequency [ Timing ]; # 0..1 Frequency of revalidation
  fhir:VerificationResult.lastPerformed [ dateTime ]; # 0..1 The date/time validation was last completed (including
    failed validations)
  fhir:VerificationResult.nextScheduled [ date ]; # 0..1 The date when target is next validated, if appropriate
  fhir:VerificationResult.failureAction [ CodeableConcept ]; # 0..1 fatal | warn | rec-only | none
  fhir:VerificationResult.primarySource [ # 0..* Information about the primary source(s) involved in validation
    fhir:VerificationResult.primarySource.who [ Reference(Organization|Practitioner|PractitionerRole) ]; # 0..1
    Reference to the primary source
    fhir:VerificationResult.primarySource.type [ CodeableConcept ], ... ; # 0..* Type of primary source (License Board;
    Primary Education; Continuing Education; Postal Service; Relationship owner; Registration Authority; legal source;
    issuing source; authoritative source)
    fhir:VerificationResult.primarySource.communicationMethod [ CodeableConcept ], ... ; # 0..* Method for exchanging
    information with the primary source
    fhir:VerificationResult.primarySource.validationStatus [ CodeableConcept ]; # 0..1 successful | failed | unknown
    fhir:VerificationResult.primarySource.validationDate [ dateTime ]; # 0..1 When the target was validated against the
    primary source
    fhir:VerificationResult.primarySource.canPushUpdates [ CodeableConcept ]; # 0..1 yes | no | undetermined
    fhir:VerificationResult.primarySource.pushTypeAvailable [ CodeableConcept ], ... ; # 0..* specific | any | source
  ], ...;
  fhir:VerificationResult.attestation [ # 0..1 Information about the entity attesting to information
    fhir:VerificationResult.attestation.who [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 The
    individual or organization attesting to information
    fhir:VerificationResult.attestation.onBehalfOf [ Reference(Organization|Practitioner|PractitionerRole) ]; # 0..1
    When the who is asserting on behalf of another (organization or individual)
    fhir:VerificationResult.attestation.communicationMethod [ CodeableConcept ]; # 0..1 The method by which attested
    information was submitted/retrieved
    fhir:VerificationResult.attestation.date [ date ]; # 0..1 The date the information was attested to
    fhir:VerificationResult.attestation.sourceIdentityCertificate [ string ]; # 0..1 A digital identity certificate
    associated with the attestation source
    fhir:VerificationResult.attestation.proxyIdentityCertificate [ string ]; # 0..1 A digital identity certificate
    associated with the proxy entity submitting attested information on behalf of the attestation source
    fhir:VerificationResult.attestation.proxySignature [ Signature ]; # 0..1 Proxy signature
    fhir:VerificationResult.attestation.sourceSignature [ Signature ]; # 0..1 Attester signature
  ];
  fhir:VerificationResult.validator [ # 0..* Information about the entity validating information
    fhir:VerificationResult.validator.organization [ Reference(Organization) ]; # 1..1 Reference to the organization
    validating information
    fhir:VerificationResult.validator.identityCertificate [ string ]; # 0..1 A digital identity certificate associated
    with the validator
    fhir:VerificationResult.validator.attestationSignature [ Signature ]; # 0..1 Validator signature
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

8.32.3.1 Terminology Bindings

Path | Definition | Type | Reference
VerificationResult.need | The frequency with which the target must be validated. | Preferred | need
VerificationResult.status | The validation status of the target. | Required | status
VerificationResult.validationType | What the target is validated against. | Preferred | validation-type
VerificationResult.validationProcess | The primary process by which the target is validated. | Example | validation-process
VerificationResult.failureAction | The result if validation fails. | Preferred | failure-action
VerificationResult.primarySource.type | Type of the validation primary source. | Example | primary-source-type
VerificationResult.primarySource.communicationMethod VerificationResult.attestation.communicationMethod | Method for communicating with the data source (manual; API; Push). | Example | verificationresult-communication-method
VerificationResult.primarySource.validationStatus | Status of the validation of the target against the primary source. | Preferred | validation-status
VerificationResult.primarySource.canPushUpdates | Ability of the primary source to push updates/alerts. | Preferred | can-push-updates
VerificationResult.primarySource.pushTypeAvailable | Type of alerts/updates the primary source can send. | Preferred | push-type-available

8.32.4 Notes:

8.32.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
target | reference | A resource that was validated | VerificationResult.target (Any)

8.32.6 Resource VerificationResult - Examples

URL: https://hl7.org/fhir/R4/verificationresult-examples.html

8.32.6 Resource VerificationResult - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
A general verification result example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.32.7 Resource VerificationResult - Detailed Descriptions

URL: https://hl7.org/fhir/R4/verificationresult-definitions.html

8.32.7 Resource VerificationResult - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the VerificationResult resource.

VerificationResult
Element Id | VerificationResult
Definition | Describes validation requirements, source(s), status and dates for one or more elements.
Cardinality | 0..*
Type | DomainResource
VerificationResult.target
Element Id | VerificationResult.target
Definition | A resource that was validated.
Cardinality | 0..*
Type | Reference(Any)
Summary | true
VerificationResult.targetLocation
Element Id | VerificationResult.targetLocation
Definition | The fhirpath location(s) within the resource that was validated.
Cardinality | 0..*
Type | string
Summary | true
VerificationResult.need
Element Id | VerificationResult.need
Definition | The frequency with which the target must be validated (none; initial; periodic).
Cardinality | 0..1
Terminology Binding | Need (Preferred)
Type | CodeableConcept
Summary | true
VerificationResult.status
Element Id | VerificationResult.status
Definition | The validation status of the target (attested; validated; in process; requires revalidation; validation failed; revalidation failed).
Cardinality | 1..1
Terminology Binding | Status (Required)
Type | code
Summary | true
VerificationResult.statusDate
Element Id | VerificationResult.statusDate
Definition | When the validation status was updated.
Cardinality | 0..1
Type | dateTime
Summary | true
VerificationResult.validationType
Element Id | VerificationResult.validationType
Definition | What the target is validated against (nothing; primary source; multiple sources).
Cardinality | 0..1

Terminology Binding | Validation-type (Preferred)
Type | CodeableConcept
Summary | true
VerificationResult.validationProcess
Element Id | VerificationResult.validationProcess
Definition | The primary process by which the target is validated (edit check; value set; primary source; multiple sources; standalone; in context).
Cardinality | 0..*
Terminology Binding | Validation-process (Example)
Type | CodeableConcept
Summary | true
VerificationResult.frequency
Element Id | VerificationResult.frequency
Definition | Frequency of revalidation.
Cardinality | 0..1
Type | Timing
VerificationResult.lastPerformed
Element Id | VerificationResult.lastPerformed
Definition | The date/time validation was last completed (including failed validations).
Cardinality | 0..1
Type | dateTime
VerificationResult.nextScheduled
Element Id | VerificationResult.nextScheduled
Definition | The date when target is next validated, if appropriate.
Cardinality | 0..1
Type | date
VerificationResult.failureAction
Element Id | VerificationResult.failureAction
Definition | The result if validation fails (fatal; warning; record only; none).
Cardinality | 0..1
Terminology Binding | Failure-action (Preferred)
Type | CodeableConcept
Summary | true
VerificationResult.primarySource
Element Id | VerificationResult.primarySource
Definition | Information about the primary source(s) involved in validation.
Cardinality | 0..*
VerificationResult.primarySource.who
Element Id | VerificationResult.primarySource.who
Definition | Reference to the primary source.
Cardinality | 0..1
Type | Reference(Organization | Practitioner | PractitionerRole)
VerificationResult.primarySource.type
Element Id | VerificationResult.primarySource.type
Definition | Type of primary source (License Board; Primary Education; Continuing Education; Postal Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source).
Cardinality | 0..*
Terminology Binding | Primary-source-type (Example)
Type | CodeableConcept
Summary | true
VerificationResult.primarySource.communicationMethod
Element Id | VerificationResult.primarySource.communicationMethod
Definition | Method for communicating with the primary source (manual; API; Push).
Cardinality | 0..*
Terminology Binding | VerificationResult Communication Method (Example)
Type | CodeableConcept
Summary | true
VerificationResult.primarySource.validationStatus
Element Id | VerificationResult.primarySource.validationStatus
Definition | Status of the validation of the target against the primary source (successful; failed; unknown).
Cardinality | 0..1
Terminology Binding | Validation-status (Preferred)
Type | CodeableConcept
VerificationResult.primarySource.validationDate
Element Id | VerificationResult.primarySource.validationDate
Definition | When the target was validated against the primary source.
Cardinality | 0..1
Type | dateTime
VerificationResult.primarySource.canPushUpdates
Element Id | VerificationResult.primarySource.canPushUpdates
Definition | Ability of the primary source to push updates/alerts (yes; no; undetermined).
Cardinality | 0..1

Terminology Binding | Can-push-updates (Preferred)
Type | CodeableConcept
Summary | true
VerificationResult.primarySource.pushTypeAvailable
Element Id | VerificationResult.primarySource.pushTypeAvailable
Definition | Type of alerts/updates the primary source can send (specific requested changes; any changes; as defined by source).
Cardinality | 0..*
Terminology Binding | Push-type-available (Preferred)
Type | CodeableConcept
VerificationResult.attestation
Element Id | VerificationResult.attestation
Definition | Information about the entity attesting to information.
Cardinality | 0..1
VerificationResult.attestation.who
Element Id | VerificationResult.attestation.who
Definition | The individual or organization attesting to information.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Summary | true
VerificationResult.attestation.onBehalfOf
Element Id | VerificationResult.attestation.onBehalfOf
Definition | When the who is asserting on behalf of another (organization or individual).
Cardinality | 0..1
Type | Reference(Organization | Practitioner | PractitionerRole)
Summary | true
VerificationResult.attestation.communicationMethod
Element Id | VerificationResult.attestation.communicationMethod
Definition | The method by which attested information was submitted/retrieved (manual; API; Push).
Cardinality | 0..1
Terminology Binding | VerificationResult Communication Method (Example)
Type | CodeableConcept
Summary | true
VerificationResult.attestation.date
Element Id | VerificationResult.attestation.date
Definition | The date the information was attested to.
Cardinality | 0..1
Type | date
Summary | true
VerificationResult.attestation.sourceIdentityCertificate
Element Id | VerificationResult.attestation.sourceIdentityCertificate
Definition | A digital identity certificate associated with the attestation source.
Cardinality | 0..1
Type | string
VerificationResult.attestation.proxyIdentityCertificate
Element Id | VerificationResult.attestation.proxyIdentityCertificate
Definition | A digital identity certificate associated with the proxy entity submitting attested information on behalf of the attestation source.
Cardinality | 0..1
Type | string
VerificationResult.attestation.proxySignature
Element Id | VerificationResult.attestation.proxySignature
Definition | Signed assertion by the proxy entity indicating that they have the right to submit attested information on behalf of the attestation source.
Cardinality | 0..1
Type | Signature
VerificationResult.attestation.sourceSignature
Element Id | VerificationResult.attestation.sourceSignature
Definition | Signed assertion by the attestation source that they have attested to the information.
Cardinality | 0..1
Type | Signature
VerificationResult.validator
Element Id | VerificationResult.validator
Definition | Information about the entity validating information.
Cardinality | 0..*
VerificationResult.validator.organization
Element Id | VerificationResult.validator.organization
Definition | Reference to the organization validating information.
Cardinality | 1..1
Type | Reference(Organization)
VerificationResult.validator.identityCertificate
Element Id | VerificationResult.validator.identityCertificate
Definition | A digital identity certificate associated with the validator.

Cardinality | 0..1
Type | string
VerificationResult.validator.attestationSignature
Element Id | VerificationResult.validator.attestationSignature
Definition | Signed assertion by the validator that they have validated the information.
Cardinality | 0..1
Type | Signature

8.32.8 Resource VerificationResult - Mappings

URL: https://hl7.org/fhir/R4/verificationresult-mappings.html

8.32.8 Resource VerificationResult - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Mappings for the verificationresult resource (see Mappings to Other Standards for further information & status).

8.32.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

VerificationResult
who | FiveWs.who

8.32.9 Resource VerificationResult - Extensions & Profiles

URL: https://hl7.org/fhir/R4/verificationresult-profiles.html

8.32.9 Resource VerificationResult - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the VerificationResult resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

8.32.9.1 Profiles

No Profiles defined for this resource

8.32.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

8.32.9.3 Search Extensions

No Search Extensions defined for this resource

8.32.10 JSON Schema for VerificationResult

URL: https://hl7.org/fhir/R4/verificationresult.schema.json.html

8.32.10 JSON Schema for VerificationResult

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for VerificationResult, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
```

```
"id": "http://hl7.org/fhir/json-schema/VerificationResult",
"$ref": "#/definitions/VerificationResult",
"description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
"definitions": {
  "VerificationResult": {
    "description": "Describes validation requirements, source(s), status and dates for one or more elements.",
    "properties": {
      "resourceType": {
        "description": "This is a VerificationResult resource",
        "const": "VerificationResult"
      },
      "id": {
        "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
   never changes.",
        "$ref": "id.schema.json#/definitions/id"
      },
      "meta": {
        "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
  Changes to the content might not always be associated with version changes to the resource.",
        "$ref": "Meta.schema.json#/definitions/Meta"
      },
      "implicitRules": {
        "description": "A reference to a set of rules that were followed when the resource was constructed, and which
  must be understood when processing the content. Often, this is a reference to an implementation guide that defines
  the special rules along with other profiles etc.",
        "$ref": "#/definitions/uri"
      },
      "_implicitRules": {
        "description": "Extensions for implicitRules",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "language": {
        "description": "The base language in which the resource is written.",
        "$ref": "#/definitions/code"
      },
      "_language": {
        "description": "Extensions for language",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "text": {
        "description": "A human-readable narrative that contains a summary of the resource and can be used to
  represent the content of the resource to a human. The narrative need not encode all the structured data, but is
  required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
  Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
        "$ref": "Narrative.schema.json#/definitions/Narrative"
      },
      "contained": {
        "description": "These resources do not have an independent existence apart from the resource that contains
  them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
        "items": {
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
   definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource and that modifies the understanding of the element that contains it and/or the understanding of the
  containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
  of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
  met as part of the definition of the extension. Applications processing a resource are required to check for
  modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
  DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "target": {
        "description": "A resource that was validated.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "targetLocation": {
        "description": "The fhirpath location(s) within the resource that was validated.",
        "items": {
          "$ref": "#/definitions/string"
```

```
      },
      "type": "array"
    },
    "_targetLocation": {
      "description": "Extensions for targetLocation",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "need": {
      "description": "The frequency with which the target must be validated (none; initial; periodic).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "The validation status of the target (attested; validated; in process; requires revalidation;
validation failed; revalidation failed).",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusDate": {
      "description": "When the validation status was updated.",
      "$ref": "#/definitions/dateTime"
    },
    "_statusDate": {
      "description": "Extensions for statusDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "validationType": {
      "description": "What the target is validated against (nothing; primary source; multiple sources).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "validationProcess": {
      "description": "The primary process by which the target is validated (edit check; value set; primary source;
multiple sources; standalone; in context).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "frequency": {
      "description": "Frequency of revalidation.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "lastPerformed": {
      "description": "The date/time validation was last completed (including failed validations).",
      "$ref": "#/definitions/dateTime"
    },
    "_lastPerformed": {
      "description": "Extensions for lastPerformed",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "nextScheduled": {
      "description": "The date when target is next validated, if appropriate.",
      "$ref": "#/definitions/date"
    },
    "_nextScheduled": {
      "description": "Extensions for nextScheduled",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "failureAction": {
      "description": "The result if validation fails (fatal; warning; record only; none).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "primarySource": {
      "description": "Information about the primary source(s) involved in validation.",
      "items": {
        "$ref": "#/definitions/VerificationResult_PrimarySource"
      },
      "type": "array"
    },
    "attestation": {
      "description": "Information about the entity attesting to information.",
      "$ref": "#/definitions/VerificationResult_Attestation"
    },
    "validator": {
      "description": "Information about the entity validating information.",
      "items": {
        "$ref": "#/definitions/VerificationResult_Validator"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
```

```
  ]
},
"VerificationResult_PrimarySource": {
  "description": "Describes validation requirements, source(s), status and dates for one or more elements.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "who": {
      "description": "Reference to the primary source.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "type": {
      "description": "Type of primary source (License Board; Primary Education; Continuing Education; Postal
Service; Relationship owner; Registration Authority; legal source; issuing source; authoritative source).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "communicationMethod": {
      "description": "Method for communicating with the primary source (manual; API; Push).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "validationStatus": {
      "description": "Status of the validation of the target against the primary source (successful; failed;
unknown).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "validationDate": {
      "description": "When the target was validated against the primary source.",
      "$ref": "#/definitions/dateTime"
    },
    "_validationDate": {
      "description": "Extensions for validationDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "canPushUpdates": {
      "description": "Ability of the primary source to push updates/alerts (yes; no; undetermined).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "pushTypeAvailable": {
      "description": "Type of alerts/updates the primary source can send (specific requested changes; any changes;
as defined by source).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"VerificationResult_Attestation": {
  "description": "Describes validation requirements, source(s), status and dates for one or more elements.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
```

```
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "who": {
      "description": "The individual or organization attesting to information.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onBehalfOf": {
      "description": "When the who is asserting on behalf of another (organization or individual).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "communicationMethod": {
      "description": "The method by which attested information was submitted/retrieved (manual; API; Push).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "date": {
      "description": "The date the information was attested to.",
      "$ref": "#/definitions/date"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sourceIdentityCertificate": {
      "description": "A digital identity certificate associated with the attestation source.",
      "$ref": "#/definitions/string"
    },
    "_sourceIdentityCertificate": {
      "description": "Extensions for sourceIdentityCertificate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "proxyIdentityCertificate": {
      "description": "A digital identity certificate associated with the proxy entity submitting attested
information on behalf of the attestation source.",
      "$ref": "#/definitions/string"
    },
    "_proxyIdentityCertificate": {
      "description": "Extensions for proxyIdentityCertificate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "proxySignature": {
      "description": "Signed assertion by the proxy entity indicating that they have the right to submit attested
information on behalf of the attestation source.",
      "$ref": "Signature.schema.json#/definitions/Signature"
    },
    "sourceSignature": {
      "description": "Signed assertion by the attestation source that they have attested to the information.",
      "$ref": "Signature.schema.json#/definitions/Signature"
    }
  },
  "additionalProperties": false
},
"VerificationResult_Validator": {
  "description": "Describes validation requirements, source(s), status and dates for one or more elements.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
```

```
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "organization": {
        "description": "Reference to the organization validating information.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "identityCertificate": {
        "description": "A digital identity certificate associated with the validator.",
        "$ref": "#/definitions/string"
      },
      "_identityCertificate": {
        "description": "Extensions for identityCertificate",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "attestationSignature": {
        "description": "Signed assertion by the validator that they have validated the information.",
        "$ref": "Signature.schema.json#/definitions/Signature"
      }
    },
    "additionalProperties": false,
    "required": [
      "organization"
    ]
  }
 }
}
```


8.32.11 Resource VerificationResult - Maturity Dependency View

URL: https://hl7.org/fhir/R4/verificationresult-dependencies.html

8.32.11 Resource VerificationResult - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for VerificationResult FMM level 0

Name | Card. | Type | Dependency Analysis
VerificationResult | | DomainResource
target | 0..* | Reference(Any)
targetLocation | 0..* | string
need | 0..1 | CodeableConcept
status | 1..1 | code
statusDate | 0..1 | dateTime
validationType | 0..1 | CodeableConcept
validationProcess | 0..* | CodeableConcept
frequency | 0..1 | Timing
lastPerformed | 0..1 | dateTime
nextScheduled | 0..1 | date
failureAction | 0..1 | CodeableConcept
primarySource | 0..* | BackboneElement
who | 0..1 | Reference(Organization | Practitioner | PractitionerRole)
type | 0..* | CodeableConcept
communicationMethod | 0..* | CodeableConcept
validationStatus | 0..1 | CodeableConcept
validationDate | 0..1 | dateTime
canPushUpdates | 0..1 | CodeableConcept
pushTypeAvailable | 0..* | CodeableConcept
attestation | 0..1 | BackboneElement
who | 0..1 | Reference(Practitioner | PractitionerRole | Organization)
onBehalfOf | 0..1 | Reference(Organization | Practitioner | PractitionerRole)
communicationMethod | 0..1 | CodeableConcept
date | 0..1 | date
sourceIdentityCertificate | 0..1 | string
proxyIdentityCertificate | 0..1 | string
proxySignature | 0..1 | Signature

sourceSignature | 0..1 | Signature
validator | 0..* | BackboneElement
organization | 1..1 | Reference(Organization)
identityCertificate | 0..1 | string
attestationSignature | 0..1 | Signature

8.32.12 Value Set Sc-valueset-verificationresult-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-verificationresult-status.html

8.32.12 Value Set Sc-valueset-verificationresult-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.32.13 statusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-verificationresult-status)

Mapping from http://hl7.org/fhir/ValueSet/verificationresult-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 06/06/2018 12:06:02 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The validation status of the target"

Source Code | Equivalence | Destination Code
attested | equivalent | draft
validated | equivalent | accepted
in-process | equivalent | active
val-fail, reval-fail | equivalent | failed
req-revalid (Requires revalidation) | equivalent | replaced

8.32.14 Value Set Sc-valueset-verificationresult-validation-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-verificationresult-validation-status.html

8.32.14 Value Set Sc-valueset-verificationresult-validation-status - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

8.32.15 validation-statusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-verificationresult-validation-status)

Mapping from http://hl7.org/fhir/ValueSet/verificationresult-validation-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 06/06/2018 12:06:02 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Status of the validation of the target against the primary source"

Source Code | Equivalence | Destination Code
successful | equivalent | active
failed | equivalent | failed
unknown | equivalent | unknown